

State of Rhode Island Hazard Mitigation Plan

February 2024



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List of Commonly Used Acronyms

Acronym	Meaning
AEL	Annualized Earthquake Loss
AELR	Annualized Earthquake Loss Ratio
ASCE	American Society of Civil Engineers
BFE	Base Flood Elevation
BRIC	Building Resilient Infrastructure and Communities
CDC	Centers for Disease Control and Prevention
CFR	Code of Federal Regulations
CRMC	Coastal Resources Management Council
CRS	Community Rating System
DMA	Disaster Mitigation Act
DOA	Department of Administration
EC4	Executive Climate Change Coordinating Council
FEMA	Federal Emergency Management Agency
FEMA GO	FEMA Grant Outcomes
FIRMs	Flood Insurance Rate Maps
FMA	Flood Mitigation Assistance
GIS	Geographic Information System
HHPD	Rehabilitation Of High Hazard Potential Dam Grant Program
HMA	Hazard Mitigation Assistance
HMGP	Hazard Mitigation Grant Program
HMGP	Hazard Mitigation Grant Program
HAZUS	Hazards-United States
IBC	International Building Code
LEPC	Local Emergency Planning Committee
NCEI	National Centers for Environmental Information
NFIP	National Flood Insurance Program
NOAA	National Oceanic and Atmospheric Administration
NRI	National Risk Index
NWS	National Weather Service
PDA	Preliminary Damage Assessment
RIDEM	Rhode Island Department of Environmental Management
RIDOH	Rhode Island Department of Health
RIDOT	Rhode Island Department of Transportation
RIEMA	Rhode Island Emergency Management Agency
RIGIS	Rhode Island Geographical Information System
RIIB	Rhode Island Infrastructure Bank
Risk MAP	Risk Mapping, Assessment, and Planning
RL	Repetitive Loss
SFHA	Special Flood Hazard Area
SHMO	State Hazard Mitigation Officer
SHMP	State of Rhode Island Hazard Mitigation Plan
SIHMC	State Interagency Hazard Mitigation Committee
SRL	Severe Repetitive Loss
STAPLEE	Social, Technical, Administrative, Political, Legal, Economic, and Environmental
STIP	State Transportation Improvement Program
URI CRC	University of Rhode Island Coastal Resources Center
USACE	U.S. Army Corps of Engineers
USDA	U.S. Department of Agriculture
USGS	United States Geologic Survey
WUI	Wildland/Urban Interface

Section 1 – Introduction, Assurances, and Adoption

1.1 Introduction

Hazard mitigation planning refers to the process of identifying, assessing, and implementing strategies to reduce or eliminate the long-term risks and vulnerabilities associated with natural and human-made hazards. The goal is to minimize the impact of disasters and enhance community resilience. Hazard mitigation planning provides communities with a roadmap to aid in the creation and revision of policies and procedures, and the use of available resources, to provide long-term, tangible benefits. A well-designed hazard mitigation plan provides communities with realistic actions that can be taken to reduce potential vulnerability and exposure to identified hazards.

This State of Rhode Island Hazard Mitigation Plan (SHMP) was prepared to provide sustained actions to reduce or eliminate risk to the people and infrastructure of the State from the effects of natural, technological, and human caused hazards. This plan documents the planning process and identifies applicable hazards, vulnerabilities, and hazard mitigation strategies. This plan will serve to direct available state resources towards creating policies and actions that provide long-term benefits to the community. Additionally, state officials can refer to the plan when making decisions regarding regulations and ordinances, granting permits, and in funding capital improvements and other community initiatives.

Specifically, this hazard mitigation plan was developed to:

- Update the 2019 SHMP
- Build for a safer future for all citizens
- Foster cooperation for planning and resiliency
- Identify, prioritize, and mitigate against hazards
- Assist with sensible and effective planning and budgeting
- Educate citizens about hazards, mitigation, and preparedness
- Comply with relevant federal requirements

This plan has been designed to be a living document, a document that will evolve to reflect changes, correct any omissions, and constantly strive to ensure the safety of Rhode Island’s citizens.

1.2 Assurances

In an effort to reduce natural disaster losses, the United States Congress passed the Disaster Mitigation Act of 2000 (DMA 2000) in order to amend the Robert T. Stafford Disaster Relief and Emergency Assistance Act (Stafford Act). DMA 2000 amended the Stafford Act by repealing the previous Mitigation Planning section (409) and replacing it with a new Mitigation Planning section (322). Section 322 of the DMA makes the development of a hazard mitigation plan a specific eligibility requirement for any local government applying for Federal mitigation grant funds. This SHMP was prepared to meet the requirements of the DMA 2000, as defined in regulations set forth by the Interim Final Rule (44 Code of Federal Regulations (CFR) Part 201.4).

The State of Rhode Island certifies that it will comply with all applicable Federal statutes and regulations during the periods for which it receives grant funding, in compliance with 44 CFR 13.11(c), and will amend its plan whenever necessary to reflect changes in State or Federal laws and statutes as required in 44 CFR 13.11(d).

This hazard mitigation plan was prepared to comply with all relevant requirements of the Robert T. Stafford Disaster Relief and Emergency Assistance Act of 1988, as amended by the Disaster Mitigation Act of 2000. This plan complies with all the relevant requirements of:

- Code of Federal Regulations (44 CFR) pertaining to hazard mitigation planning
- Federal Emergency Management Agency (FEMA) planning directives and guidelines
- Interim final, and final rules pertaining to hazard mitigation planning and grant funding
- Relevant presidential directives
- Office of Management and Budget circulars

- Any additional and relevant federal government documents, guidelines, and rules.

1.3 Authorities

The SHMP relies on the authorities given to the State of Rhode Island by its citizens and encoded in state law. This plan is intended to be consistent with all policies and procedures that govern activities related to the mitigation programming and planning. In all cases of primacy, State of Rhode Island laws, statutes, and policies will supersede the provisions of the plan.

1.4 Plan Adoption

State of Rhode Island Governor Daniel McKee has formally adopted the 2024 Rhode Island SHMP. Adoption of this plan demonstrates Rhode Island's commitment to continuing to make a safer future for all citizens. Both adoption documents and FEMA Region I approval documentation may be found in Appendix A.

Administration and oversight of the hazard mitigation program is the responsibility of the Rhode Island Emergency Management Agency (RIEMA) Planning Branch. The plan will be reviewed annually and will be updated every five years, or as required by changing hazard mitigation regulations or guidelines.

Section 2 – Documentation of the Planning Process

2.1 Planning Process

The process established for this planning effort is based on the Disaster Mitigation Act of 2000 planning and update requirements and the FEMA associated guidance for state hazard mitigation plans. To accomplish this, the following planning process methodology was followed:

- Inform, invite, and involve other mitigation plan stakeholders throughout the state, including federal agencies, state agencies, regional groups, businesses, non-profits, underserved communities, and local emergency management organizations.
- Conduct a thorough review of all relevant current and historic planning efforts.
- Collect data on all related state plans and initiatives, local plans' hazard risk, local plans' mitigation strategies and actions, state owned facilities, flood plains, Repetitive Loss/Severe Repetitive Loss properties, hazard events, on-going and completed mitigation actions, and mitigation program changes since the development of the previous plan.
- Conduct a review of all related and relevant state and local plans for integration and incorporation.
- Develop the planning and project management process, including methodology, review procedures, details about plan development changes, interagency coordination, planning integration, and the organization and contribution of stakeholders.
- Develop and update the profile of the State of Rhode Island.
- Complete a risk and vulnerability assessment using a Geographic Information System (GIS) driven approach using data from the FEMA and other federal and state agency resources. Analyses were conducted at the state level, county by county, of state-owned facilities, and county by county drawing on local assessments.
- Develop a comprehensive mitigation strategy effectively addressing the State of Rhode Island's hazards and mitigation program objectives. This included identifying state and local capabilities, reviewing pre and post disaster policies and programs, identifying objectives and goals, identifying mitigation actions and projects, and assessing mitigation actions and projects.
- Determination and implementation of a plan maintenance cycle, including a timeline for plan upgrades and improvements.
- The petition of the Governor's Office of the State of Rhode Island for a letter of formal plan adoption.
- Submission of the plan to FEMA for review and approval.

2.2 Hazard Mitigation Planning Equity

Planning equity refers to the principle of fairness and justice in planning and development processes. It emphasizes the equitable distribution of resources, opportunities, and benefits among all members of a community, particularly those who have historically been marginalized or disadvantaged. The concept of planning equity recognizes that planning decisions can have significant impacts on different groups of people and aims to ensure that these decisions promote social justice and inclusivity. It involves addressing spatial inequalities, such as disparities in access to housing, transportation, public services, green spaces, and employment opportunities.

Planning equity entails involving diverse stakeholders in decision-making processes, including community members, advocacy groups, and underrepresented populations. It seeks to empower marginalized communities by giving them a voice in shaping the development and planning policies that directly affect their lives.

Planning equity and hazard mitigation planning are closely related, as both aim to create more resilient and inclusive communities. As part of this planning effort, the following intersections were considered between planning equity and hazard mitigation planning:

- Vulnerability assessment: Planning equity recognizes that certain communities, particularly marginalized and disadvantaged populations, may be more vulnerable to hazards due to social, economic, and environmental factors. When conducting a vulnerability assessment as part of hazard mitigation planning, it is important to consider equity issues and identify areas or groups that may experience disproportionate impacts.

- Engaging marginalized communities: Planning equity emphasizes the inclusion and participation of diverse stakeholders, including marginalized communities, in decision-making processes. In hazard mitigation planning it is crucial to engage these communities to understand their unique needs, concerns, and perspectives regarding hazards.
- Addressing social disparities: Hazard mitigation planning can help address social disparities by considering the unequal distribution of resources and opportunities in the context of hazards. This can involve implementing mitigation measures that specifically target vulnerable populations, such as affordable housing in safer areas or improved access to emergency services and transportation for underserved communities.
- Equitable distribution of resources: Planning equity promotes the equitable distribution of resources, and this principle can be applied to hazard mitigation planning. It involves ensuring that mitigation measures and investments are allocated fairly, with consideration given to communities that have historically received less attention or investment. This can help reduce existing disparities and enhance the resilience of marginalized communities.

By integrating planning equity into hazard mitigation planning, it becomes possible to develop strategies and actions that not only reduce the risks associated with hazards but also promote social justice, inclusivity, and resilience for all members of the community. As part of this effort, Rhode Island is committed to making all hazard mitigation communications and meetings accessible, including translation services, hearing and visual accommodations, and meetings in accessible locations.

2.3 2024 Plan Update

In undertaking this planning effort, the State of Rhode Island determined that wide variances in planning format and data do not allow for effective continuous planning. To provide planning continuity every effort was made during this plan update to adhere as closely as possible to elements of the 2019 SHMP. As such, the level of analysis and detail included in this risk assessment is cumulative, allowing the state to have a robust base to further mold and improve its mitigation strategy over the next five years.

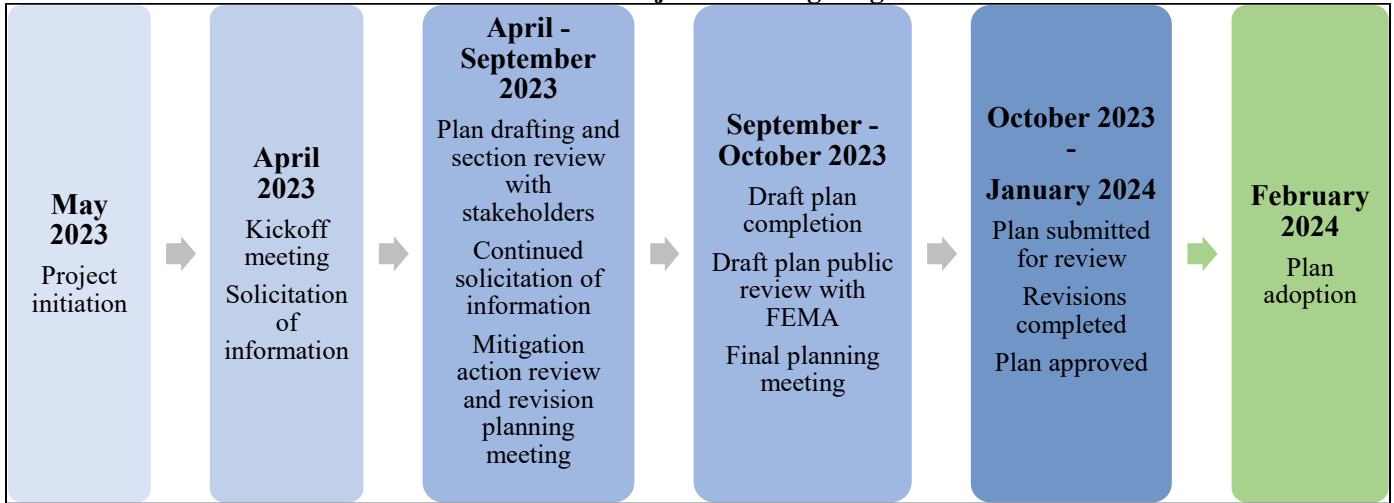
As part of this planning effort, each section of the previous mitigation plan was reviewed and revised based on current and available data. The plan was reviewed and revised against the following elements:

- Compliance with the current regulatory environment
- Completeness of data
- Correctness of data
- Capability differentials
- Current state environment

Based on the above criteria, each section of the 2019 SHMP was reviewed and revised as required. In addition to data revisions, the format and sequencing of the previous plan was updated for ease of use and plan clarity. Additionally, during this process, and after a thorough review and discussion with all stakeholders, it was determined that the priorities of the State of Rhode Island in relation to hazard mitigation planning have not changed during the five years of the previous planning cycle.

The SHMP review and revision process began in May 2023, with the first public meeting held in April 2023. The following chart indicates the planning stages completed as part of this process:

Chart 1: Project Planning Stages



A FEMA State Plan Review Guide (April 2023) detailing how all FEMA planning requirements have been met and addressed by this SHMP may be found in Appendix B.

2.4 State Interagency Mitigation Planning Committee

Project initiation began with the selection of a State Interagency Hazard Mitigation Committee (SIHMC). Members of the SIHMC were chosen by RIEMA from a cross section of state agencies and planning partners. From project inception to completion, the SIHMC was notified at each major plan development milestone through a combination of meetings and electronic communication.

In general, all SIHMC members were asked to participate in the following ways:

- Attend and participate in meetings
- Assist with the collection of data
- Assure the accuracy and completeness of data
- Assist with the revision and development of mitigation actions
- Review planning elements and drafts
- Integrate hazard mitigation planning elements with other planning mechanisms

As an additional responsibility as part of the SIHMC, RIEMA members helped establish project operating procedures and timelines, and assisted with the establishment of project milestones.

The following table represents members of the SIHMC:

Table 1: SIHMC Members

Agency	Representative	Title
Rhode Island Coastal Resources Management Council	Bruce Lofgren	Coastal Policy Advisor
Rhode Island Coastal Resources Management Council	Laura Miguel	Deputy Director
Rhode Island Commission of Deaf and Hard of Hearing	Michelle Cline	Executive Director
Rhode Island Commission of Deaf and Hard of Hearing	Michael Baer	Director, Emergency and Public Communication Access Program
Rhode Island Department of Children, Youth, and Families	Pam Leary	Operations Manager
Rhode Island Office of Healthy Aging	Michael Szylin	Associate Director
Rhode Island Department of Health	Nicholas Larmore	Program Support Specialist

Table 1: SIHMC Members

Agency	Representative	Title
Rhode Island Department of Housing	Mike Tondra	Director of Community Planning
Rhode Island Department of Transportation	Steve Cascione	PSO/Meteorologist
Rhode Island Department of Transportation	Christos Xenophontos	Assistant Director for Admin SVS
Rhode Island Department of Transportation	Stephen Kut	Program Analyst
Rhode Island Department of Transportation	Pamela Cotter	Division Administrator for Planning
Rhode Island National Guard	David Lemont	Lieutenant Colonel
Rhode Island Public Transit Authority	Joe Philbin	Chief of Security and Operations
Rhode Island Public Utilities Commission	Thomas Kogut	Associate Administrator
Rhode Island College	Kevin Fitta	Assistant Vice President
Rhode Island Commission of Deaf and Hard of Hearing	Michelle Cline	Director
Rhode Island Commission of Deaf and Hard of Hearing	Michale Baer	Staff Person
RIEMA	Rae-Anne Culp	Mitigation Planning Supervisor
RIEMA	Melinda Hopkins	Planning Branch Chief
RIEMA	Melissa Carden	Public Information Officer
RIEMA	Caterina Massidda	Supervising GOS Specialist
RIEMA	Morgan Reilly	National Flood Insurance Program Coordinator
Rhode Island State Police	Nicholas Rivello	Sergeant
Rhode Island Division of Statewide Planning	Joshua O'Neill, AICP	Supervising Planner
Rhode Island Turnpike and Bridge Authority	Jeffrey Wiggin	Assistant Project Manager

Repeated outreach efforts were made to identified equity partners, including the National Association for the Advancement of Colored People Providence, the Rhode Island Commission for Human Rights, the Rhode Island Developmental Disabilities Council, and the Rhode Island Housing Resources Commission, extending opportunities to have a representative on the SIHMC. Additionally, the Narragansett Tribe, a Federally recognized sovereign nation within Rhode Island was invited to participate in the planning process. Despite repeated outreach attempts no response was received from the Narragansett Tribe.

2.5 Stakeholders

RIEMA acknowledges that effective hazard mitigation planning should involve a diverse group of stakeholders, including government agencies, private sector entities, private non-profit organizations, quasi-governmental authorities, and special districts. The coordination and cooperation of these stakeholders assists with all aspects of plan development, including:

- Data collection
- Risk analysis
- High and Significant Hazard dam information
- Statewide capability assessment
- Mitigation action review, revision, and development
- Plan implementation

All stakeholders were invited to attend any and all SIHMC scheduled meetings. If unable to attend, stakeholders were kept involved in the planning process through:

- Email or phone updates as to planning progress
- Review and comment opportunities on planning elements

RIEMA would like to thank the following entities for their participation in the planning effort. Their contribution has not only substantially improved the quality of this plan but has increased the safety and wellbeing of the citizens of the State of Rhode Island.

Table 2: Stakeholders

Rhode Island Coastal Resources Management Council (CRMC)
Rhode Island Department of Administration
Rhode Island Department of Business Administration
Rhode Island Building Code Commission
Rhode Island Department of Children, Youths, and Families
Rhode Island Department of Corrections
Rhode Island Department of Elderly Affairs
Rhode Island Department of Environmental Management (RIDEM)
Rhode Island Department of Health (RIDOH)
Rhode Island Department of Housing
Rhode Island Department of Human Services
Rhode Island Department of Transportation (RIDOT)
Rhode Island Department of Public Safety
Rhode Island Department of Revenue
Rhode Island Infrastructure Bank*
Rhode Island Fire Marshall's Office
Rhode Island Historical Preservation and Heritage Commission
Rhode Island Office of Information Technology
Rhode Island Office of the Governor
Rhode Island Public Transit Authority*
Rhode Island Public Utilities Commission
RIEMA
Rhode Island Division of Statewide Planning
Rhode Island State Police
Rhode Island National Guard
Rhode Island Narragansett Bay Commission*
Rhode Island Turnpike and Bridge Authority*
Rhode Island Water Resources Board
PPL Rhode Island Energy*
Quonset Development Corporation*
University of Rhode Island Natural Resources and Environmental Management

*: Private sector entity, private non-profit organization, quasi-governmental authority, or special districts.

2.6 Community Outreach

As part of the overall planning process, the community was provided with numerous opportunities to contribute and comment on the creation and adoption of the plan. These opportunities included:

- Advertised meeting invitations
- Comment period upon completion of draft plan
- Online survey

Experience has indicated that public meetings, no matter how well advertised, generally do not generate either participation or interest in the planning process. Even so, three open meetings were held at an easily accessible community location, the Warwick Public Library. To help generate community interest and participation, a parallel online outreach strategy was undertaken. An online SHMP survey was created, the Rhode Island Hazard Mitigation Plan Update Survey ([https:// publicinput.com/g8177](https://publicinput.com/g8177)). This online survey portal allowed community members to provide feedback and input on the SHMP update using a series of guided questions and open comment fields. Over 100 community members commented through this survey, and these comments are both incorporated in this SHMP and are included in Appendix C.

Input from the general public provided the SIHMC with a clearer understanding of local concerns, increased the likelihood of citizen buy-in concerning proposed mitigation actions, and provided elected officials with a guide and tool to set regional ordinances and regulations. Additionally, as citizens were made more aware of potential hazards and the local process to mitigation against their impacts, it was believed that they would take a stronger role in making their homes, neighborhoods, schools, and businesses safer from the potential effects of natural hazards.

2.7 Planning Meetings

Three in-person meetings were conducted for the 2024 SHMP update. All of the meetings were held in a publicly accessible location and advertised as open to the public. These meetings were conducted to discuss the mitigation planning process as well as gain public support and input for the plan update. The following is a brief synopsis of those meetings.

- **SHMP Update Kick-Off and Public Information Meeting – April 19, 2023:** RIEMA hosted a kick-off meeting for the SIHMC, stakeholders, and the public. Prior to the meeting, a public announcement was published on the Secretary of State’s Website and advertised in the Warwick Public Library. At the meeting, SIHMC members, plan stakeholders, and the public were invited to voice any concerns, ask questions, and provide input on the mitigation plan update. Additionally, BOLDplanning worked with SIHMC members and plan stakeholders to collect contact information, hazard history, facility information, and other pertinent information.
- **SHMP Plan Review, Capability Review, and Mitigation Strategy Review Meeting - September 7, 2023:** RIEMA hosted a mid-term planning meeting for the SIHMC, stakeholders, and the public. Prior to the meeting, a public announcement was published on the Secretary of State’s Website and advertised in the Warwick Public Library. Attendees met to review and revise, as necessary, the state’s hazards list and vulnerability assessment. SIHMC members also reviewed the proposed and revised mitigation strategy to ensure it was in-line with the current planning environment. Finally, each mitigation action item was reviewed, and a determination was made if the action was still applicable and achievable if not already completed.
- **SHMP Update Final Review Meeting – November 7, 2023:** RIEMA hosted a public final plan review meeting for the SIHMC, stakeholders, and the public. Prior to the meeting, a public announcement was published on the Secretary of State’s Website and advertised in the Warwick Public Library. At the meeting, SIHMC members, plan stakeholders, and the public were invited to voice any concerns, ask questions, and provide input on the mitigation plan update. Additionally, members of the public were invited to review a draft copy of the SHMP update posted to RIEMA’s website for two weeks prior to the final meeting, and prior to its submission to FEMA Region I.

Additionally, there were monthly situation reports and planning calls with the State Hazard Mitigation Officer (SHMO) to provide updates concerning the phases of plan development and to solicit plan review recommendations.

2.8 Technical Resources

The SIHMC employed a variety of technical resources in its plan development. These technical resources were instrumental in completing an accurate vulnerability and risk assessment, and include:

- **BOLDplanning Inc.:** With over 20 years of experience in hazard mitigation planning, BOLDplanning was the principal plan writer.

- **Environmental Systems Research Institute ArcGIS v10:** Assisted with the development of maps for this plan, along with the Hazards-United States (Hazus) models.
- **FEMA Digital Flood Insurance Rate Maps (DFRIM):** FEMA’s National Flood Hazard Layer data was instrumental in mapping floodplain locations and estimating potential flood impacts and loss estimates.
- **FEMA National Risk Index (NRI):** An online mapping application that identifies communities most at risk of natural hazards. The mapping service visualizes natural hazard risk metrics and includes data about expected annual losses from natural hazards, social vulnerability, and community resilience. The NRI's interactive web maps are at the county and Census tract level and made available via GIS services for custom analyses.
- **National Oceanic and Atmospheric Administration (NOAA)/National Centers for Environmental Information (NCEI):** Weather data and historical events were primarily provided by NCEI.
- **U.S. Army Corps of Engineers (USACE):** Sea level data, flood data, project data.
- **U.S. Department of Agriculture (USDA):** Drought and agricultural data.
- **U.S. Geological Survey:** Geologic hazard occurrence and probability data.
- **National Weather Service:** Storm event occurrence and probability data.
- **RIDEM:** Dam safety program, including location and condition reports, and hazardous material data.
- **RIDOH:** Climate change, pandemic preparedness and response and drinking water quality information.
- **RIDOT:** Critical infrastructure data.
- **Rhode Island State Police:** Cyber security information.
- **State Building Code Commission:** Building code compliance data.
- **CRMC:** State coastal and estuarine data.
- **CRMC:** Vulnerability assessments for coastal hazards and flood loss data

2.9 Hazard Mitigation Planning State Integration

This SHMP is available to all state agencies and departments to reference when seeking information and guidance on state mitigation goals and objectives. The general information in this plan is also intended for use by interested local governments, universities, businesses, and private associations, in addition to state and federal departments and agencies.

The hazard mitigation plan is an overarching document that is both comprised of, and contributes to, various state plans. In creating this SHMP, the following planning documents were extensively consulted and reviewed. In turn, when each of these plans is updated, they will be measured against the contents of the SHMP.

- **Rhode Island Comprehensive Emergency Management Plan (CEMP):** A primary task in fulfilling RIEMA’s mission is the development and maintenance of the CEMP. The purpose of the CEMP is to establish the overall framework for integration and coordination of emergency management activities of government, volunteer organizations, and the private sector. The CEMP is designed to provide guidance and a flexible framework that the State of Rhode Island may utilize to prevent, prepare for, respond to, recover from, and mitigate potential hazards identified in the Rhode Island Hazard Identification and Risk Assessment and Threat Hazard Identification and Risk Assessment. The provisions of the CEMP are promulgated under the authority of the Rhode Island General Laws Emergency Management Act, Chapter 30-15, as amended. The CEMP was developed using the planning process outlined in the FEMA Comprehensive Preparedness Guide (CPG-101).
- **The State Guide Plan:** The State Plan Guide is Rhode Island’s centralized and integrated long-range planning document. The State Guide Plan is not a single document but a collection of plans that have been adopted over many years. The following list details the elements of the State Plan Guide utilized, and informed by, the SHMP:
 - o Economic Development Plan (December 2014)
 - o Energy 2035, Rhode Island State Energy Plan
 - o Protecting Our Legacy of Buildings, Places, and Culture: An Historic Preservation Plan for Rhode Island (October 2021) - Element 210
 - o Rhode Island Five Year Strategic Housing Plan (2010) – Element 423
 - o State Housing Plan (March 2000) – Element 421
 - o Land Use 2025: Rhode Island's State Land Use Policies and Plan (April 13, 2006) – Element 121

- o Forest Resources Management Plan (March 10, 2005) – Element 161
- o Urban and Community Forestry Plan (May 13, 1999) – Element 156
- o Ocean State Outdoors: Rhode Island's Comprehensive Outdoor Recreation Plan (2019) – Element 152
- o A Greener Path: Greenspace and Greenways for Rhode Island's Future (Nov. 10, 1994) – Element 155
- o Rhode Island Comprehensive Solid Waste Management Plan (May 14, 2015)
- o State Airport System Plan (September 15, 2011) – Element 640
- o Moving Forward Rhode Island 2040, Long Range Transportation Plan (LRTP) (December 10, 2020) - Element 611
- o Rhode Island State Rail Plan 2014 - Element 661
- o Rhode Island Waterborne Passenger Transportation Plan (August 1998) - Element 651
- o Rhode Island Water 2030 (June 14, 2012) – Element 721
- o Water Quality 2035, RI Water Quality Management Plan (October 13, 2016)
- **2021 Act on Climate:** Signed into Rhode Island law, the Act sets enforceable climate emissions reduction mandates and updates the 2014 Resilient Rhode Island Act.
- **2022 Lead by Example Executive Order:** Directs any state agency with 15 or more employees to collaborate on lowering state energy consumption.
- **Resilient Rhody: The Statewide Climate Resilience Action Strategy, July 2018:** The State's climate adaptation strategy which addresses the impacts of climate change on critical infrastructure and utilities, natural systems, emergency preparedness, community health, and community resilience.
- **Rhode Island Pandemic Response and Recovery Plans**
- **State Emergency Response Commission plans for implementing the Emergency Planning and Community Right-to-Know Act of 1986**
- **Municipal Resilience Program:** Provides direct support to cities and towns to complete a municipal-driven workshop process that brings together climate change information and local knowledge to identify top hazards, current challenges, and community strengths.

Each of these planning documents utilized the respective iteration of the SHMP to inform and frame their formation.

State-level mitigation is inherently integrated into a host of federal programs and initiatives. Utilizing federal grant programs, the state and its local jurisdictions have accomplished numerous quality mitigation activities, negating some of the adverse effects associated with hazards. The following list illustrates federal programs integrated and referenced in the State of Rhode Island's mitigation efforts.

- **Hazard Mitigation Grant Program (HMGP) and HMGP Fire:** HMGP grants assist in implementing long-term hazard mitigation measures following Presidential disaster declarations, including fire declarations. Funding is available to implement projects in accordance with State, Tribal, and local priorities.
- **Building Resilient Infrastructure and Communities (BRIC):** BRIC supports states, local communities, tribes, and territories as they undertake hazard mitigation projects, reducing the risks they face from disasters and natural hazards. The BRIC program guiding principles are supporting communities through capability- and capacity-building; encouraging and enabling innovation; promoting partnerships; enabling large projects; maintaining flexibility; and providing consistency. Working in coordination with BRIC, the National Mitigation Investment Strategy is intended to provide a national, whole-community approach to investments in mitigation activities and risk management.
- **Flood Mitigation Assistance (FMA) Grant Program:** FMA is a competitive grant program that provides funding to states, local communities, federally recognized tribes, and territories. Funds can be used for projects that reduce or eliminate the risk of repetitive flood damage. FEMA chooses projects based on the recipients' ranking of the project and the eligibility and cost-effectiveness of the project. FEMA requires state, local, tribal, and territorial governments to develop and adopt hazard mitigation plans as a condition for receiving certain types of non-emergency disaster assistance, including funding for hazard mitigation assistance projects.
- **National Flood Insurance Program (NFIP):** The NFIP aims to reduce the impact of flooding on private and public structures. It does so by providing affordable insurance to property owners, renters, and businesses and by encouraging communities to adopt and enforce floodplain management regulations. Overall, the program

reduces the socio-economic impact of disasters by promoting the purchase and retention of general risk insurance and flood insurance.

- **Community Rating System (CRS):** Provides incentives to communities for the work they do managing their floodplains. Eligible communities that qualify for this voluntary program go above the minimum NFIP requirements and can offer their citizens discounted flood insurance in both Special Flood Hazard Areas (SFHAs) and non-SFHAs.

Plan integration is crucial for creating a cohesive and coordinated approach to address various aspects of hazard mitigation. While stakeholders will utilize their internal procedures for plan integration, the following represent commonly utilized methods:

- **Cross-Referencing:** Identify and cross-reference relevant sections of different plans and policies. This involves explicitly noting connections between the goals, strategies, and actions outlined in one plan with those in others.
- **Consistency Checks:** Conduct consistency checks to ensure that the language, objectives, and strategies in different plans and policies align with each other. This involves reviewing documents side by side to identify any conflicting or contradictory elements that need to be addressed.
- **Joint Planning Committees:** Establish joint planning committees or task forces that involve representatives from different departments or agencies responsible for various plans (for example, the SIHMC). These committees facilitate communication, collaboration, and the coordination of planning efforts across sectors.
- **Collaborative Workshops and Meetings:** Organize collaborative workshops and meetings to bring together stakeholders involved in different planning processes (as seen in the planning meetings for the SHMP). These forums provide an opportunity for stakeholders to share information, discuss common goals, and identify areas of overlap or potential conflict.
- **Alignment with State and Regional Plans:** Ensure that local plans align with broader regional and state plans. This involves considering regional and state priorities and incorporating them into local planning efforts to create a harmonized approach to development.
- **Data Sharing and Analysis:** Share relevant data among planning efforts and conduct joint data analysis. This helps in creating a common understanding of the challenges and opportunities, facilitating evidence-based decision-making across different plans.
- **Unified Implementation Strategies:** This involves identifying common actions and initiatives that contribute to the achievement of multiple goals outlined in various plans.

Rhode Island has a good working relationship with both local jurisdictions and FEMA and has historically had successes in integrating our state plan with both local plans and FEMA mitigation programs. As always, challenges exist due to the day to day demands of the working environment, including scheduling conflicts, budget restrictions, and staffing changes and shortages.

2.10 Local Hazard Mitigation Plan Integration

Information from the Rhode Island SHMP is often used by local jurisdictions to incorporate into their hazard mitigation plans. This information includes hazard identification and risk assessment, goals and objectives, local capabilities, and mitigation initiatives. Conversely, the SHMP includes the integration of all available data from Rhode Island's local Hazard Mitigation Plans. Local Hazard Mitigation Plans were utilized for the following information:

- Identified hazards
- Vulnerability and risk assessment methodologies
- Jurisdictional capabilities
- Mitigation goals
- Identified mitigation actions, both proposed and completed

This information was incorporated throughout this SHMP, when possible, as both minor and major data variances were inherent in the local plans.

Section 3 – State Profile and Development Trends

3.1 Introduction

Data concerning development trends and conditions is of great importance in determining statewide risk and vulnerability to identified hazards, especially in locations which are susceptible to identified hazards. In general, any increase in population or development in hazard susceptible areas tends to increase both the risk and the vulnerability to that hazard. As such, the information presented in this chapter details relevant population and building statistics for the State of Rhode Island. This data will then be used to determine and refine potential hazard vulnerability in succeeding sections.

3.2 Statewide Population Data

The State of Rhode Island has seen small scale population growth over the 20-year period from 2000 to 2020, as indicated by data collected from the United State Census Bureau. The following table, and associated chart, presents population data for the State of Rhode Island and its five counties:

Table 3: Rhode Island Population Data, 2010 - 2020

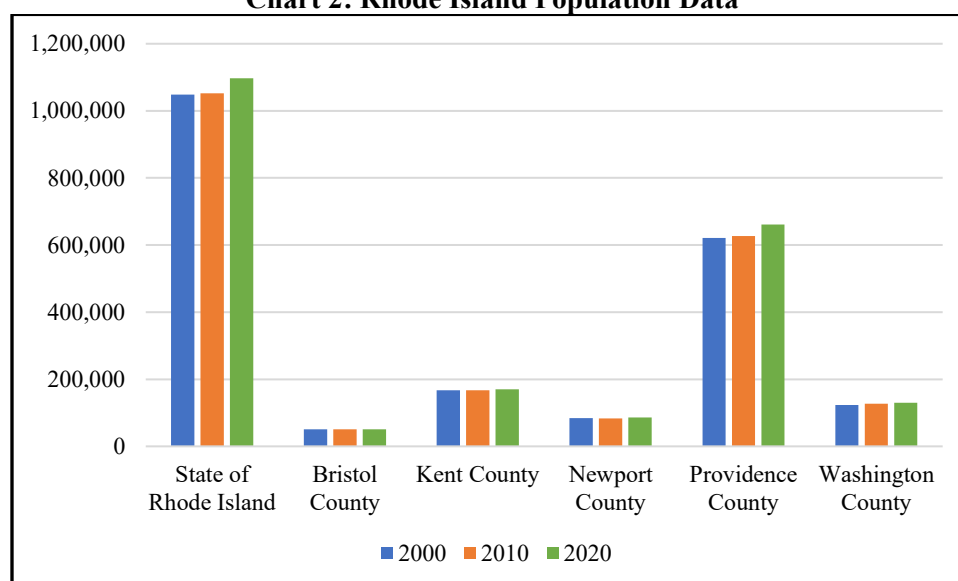
Jurisdiction	Population			Percentage Population Change 2000-2020	Total Land Area (Sq. Mi.)	Population Density
	2000	2010	2020			
State of Rhode Island	1,048,319	1,052,567	1,097,379	4.7%	1,034	1,061
Bristol County	50,648	50,501	50,793	0.3%	24	2,116
Barrington	16,819	16,501	16,099	-4.3%	8.2	1,963
Bristol	22,496	23,189	22,070	-1.9%	10.1	2,185
Warren	11,360	10,811	10,476	-7.8%	6.2	1,690
Kent County	167,090	167,235	170,363	2.0%	169	1,008
Coventry	33,638	35,040	34,747	3.3%	59.5	584
East Greenwich	12,948	13,187	13,098	1.2%	16.5	794
Warwick	85,808	82,643	81,043	-5.6%	35.5	2,283
West Greenwich	5,115	5,974	6,297	23.1%	50.6	124
West Warwick	29,581	29,358	28,937	-2.2%	7.9	3,663
Newport County	84,433	82,852	85,643	1.4%	102	840
Jamestown	5,622	5,436	5,494	-2.3%	9.7	566
Little Compton	3,593	3,508	3,496	-2.7%	20.9	167
Middletown	17,334	16,362	15,982	-7.8%	13.0	1,229
Newport	26,475	24,957	24,697	-6.7%	7.7	3,207
Portsmouth	17,149	17,316	17,332	1.1%	23.2	747
Tiverton	15,260	15,674	15,730	3.1%	29.4	535
Providence County	621,602	626,667	660,741	6.3%	410	1,612
Burrillville	15,796	15,971	16,684	5.6%	55.6	300
Central Falls	18,928	19,391	19,437	2.7%	1.2	16,198
Cranston	79,269	80,424	81,252	2.5%	28.5	2,851
Cumberland	31,840	33,291	34,980	9.9%	26.4	1,325
East Providence	48,688	47,491	47,442	-2.6%	13.4	3,540
Foster	4,274	4,556	4,722	10.5%	50.4	94
Glocester	9,948	9,805	10,202	2.6%	54.8	186
Johnston	28,219	28,752	29,326	3.9%	23.7	1,237
Lincoln	20,898	21,120	21,793	4.3%	18.4	1,184
North Providence	32,411	32,244	32,532	0.4%	5.7	5,707
North Smithfield	10,618	11,768	12,478	17.5%	24.0	520
Pawtucket	72,958	71,204	71,821	-1.6%	8.7	8,255
Providence	173,618	178,162	179,472	3.4%	18.5	9,701
Scituate	10,324	10,337	10,669	3.3%	48.7	219

Table 3: Rhode Island Population Data, 2010 - 2020

Jurisdiction	Population			Percentage Population Change 2000-2020	Total Land Area (Sq. Mi.)	Population Density
	2000	2010	2020			
Smithfield	20,589	21,477	21,735	5.6%	26.6	817
Woonsocket	43,224	41,682	41,616	-3.7%	7.7	5,405
Washington County	123,546	126,979	129,839	5.1%	329	395
Charlestown	7,859	7,876	7,832	-0.3%	36.8	213
Exeter	6,045	6,537	6,711	11.0%	57.7	116
Hopkinton	7,836	8,168	8,110	3.5%	43.0	189
Narragansett	16,361	16,034	15,451	-5.6%	14.1	1,096
New Shoreham	1,010	914	871	-13.8%	9.1	96
North Kingstown	26,326	26,611	26,289	-0.1%	43.6	603
Richmond	7,222	7,672	7,723	6.9%	40.6	190
South Kingstown	27,921	30,226	30,651	9.8%	57.1	537
Westerly	22,966	22,949	22,501	-2.0%	30.1	748

Source: US Census Bureau

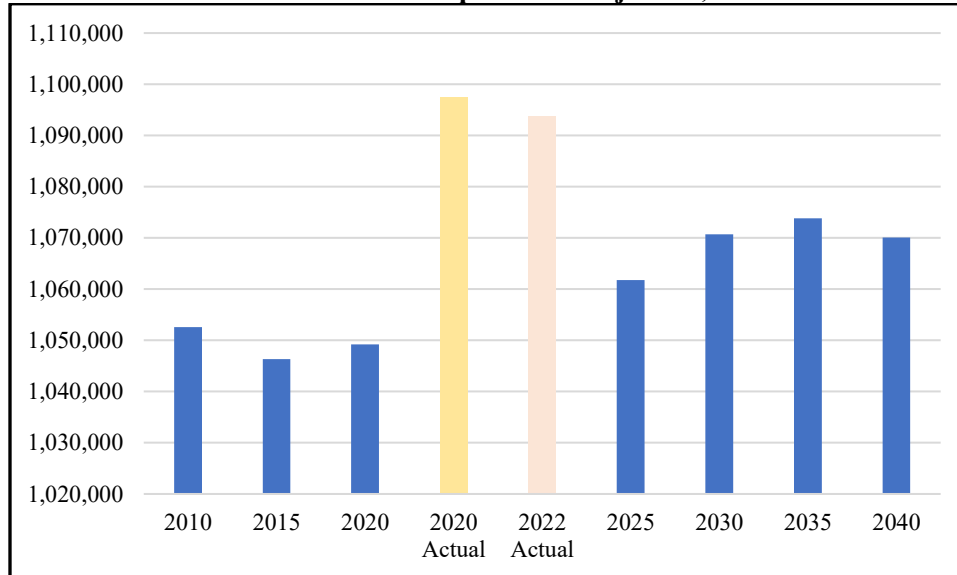
Chart 2: Rhode Island Population Data



Source: US Census Bureau

The following chart, using data from the April 2013 Rhode Island Statewide Planning Program report “Rhode Island Population Projections 2010-2040” indicates population projects for Rhode Island. As indicated by the overlay of actual state populations for both 2020 and 2022, the data presented in this report, despite using a robust set of parameters, underestimates actual population figures. However, the population trends, either increasing or decreasing, are useful for predicting any large population fluctuations.

Chart 3: Rhode Island Population Projection, 2010-2040



Source: US Census Bureau and April 2013 Rhode Island Statewide Planning Program

3.3 Statewide Vulnerable Population Data

As a subset of the population data, Rhode Island has socially vulnerable and at-risk populations, populations that may have difficulty with medical issues, poverty, and extremes in age. Several principles may be considered when discussing potentially at-risk populations, including:

- Not all people who are considered at risk are at risk
- Outward appearance does not necessarily mark a person as at risk
- The hazard event will, in many cases, affect at risk population in differing ways

In Rhode Island, the Department of Human Services takes the lead on socially vulnerable and at-risk populations and works with community partners and resources to deliver critical benefits, support, and services to families, adults, children, older adults, individuals with disabilities and veterans. Additionally, the Rhode Island Department of Children, Youth, and Families and the Office of Housing and Community Development provide valuable services to unserved and at-risk communities.

The National Response Framework defines at risk populations as "populations whose members may have additional needs before, during, and after an incident in functional areas, including but not limited to: maintaining independence, communication, transportation, supervision, and medical care."

The following tables, and associated charts, present information on potential at risk populations within Rhode Island:

Table 4: Rhode Island Population Under Five, 2000 - 2020

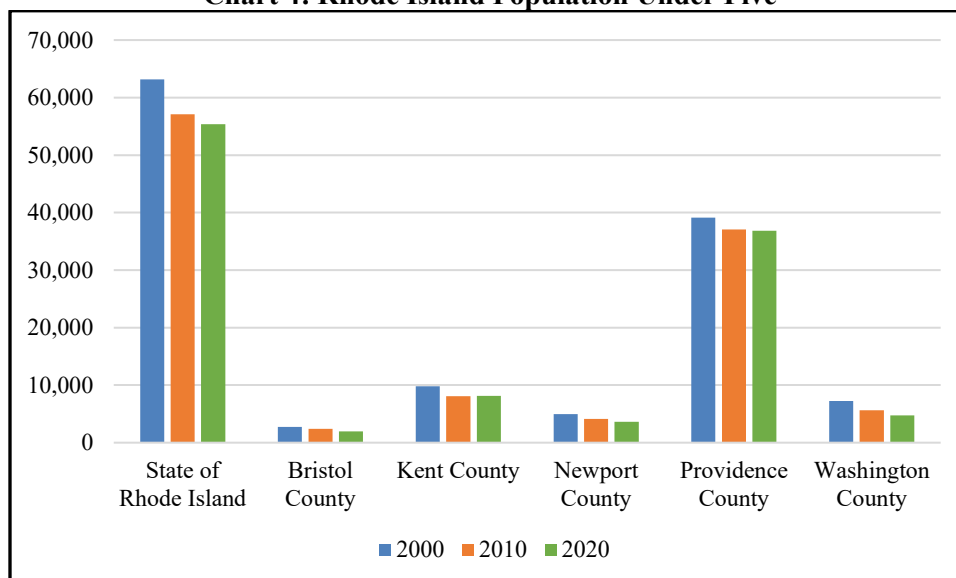
Jurisdiction	2000	2010	2020	Percentage Change 2000 - 2020
State of Rhode Island	63,135	57,112	55,370	-12.3%
Bristol County	2,729	2,388	1,976	-27.6%
Barrington	983	896	974	-0.9%
Bristol	1,129	933	718	-36.4%
Warren	572	586	296	-48.3%
Kent County	9,829	8,059	8,148	-17.1%
Coventry	2,057	1,686	1,487	-27.7%
East Greenwich	759	575	764	0.7%
Warwick	4,730	3,528	3,831	-19.0%
West Greenwich	416	308	189	-54.6%
West Warwick	1,937	1,902	1,724	-11.0%

Table 4: Rhode Island Population Under Five, 2000 - 2020

Jurisdiction	2000	2010	2020	Percentage Change 2000 - 2020
Newport County	4,958	4,140	3,641	-26.6%
Jamestown	224	176	184	-17.9%
Little Compton	177	135	22	-87.6%
Middletown	1,193	1,003	814	-31.8%
Newport	1,595	1,013	1,318	-17.4%
Portsmouth	1,065	972	551	-48.3%
Tiverton	783	783	676	-13.7%
Providence County	39,120	37,068	36,846	-5.8%
Burrillville	895	597	1,092	22.0%
Central Falls	1,540	2,117	1,673	8.6%
Cranston	4,188	4,182	4,670	11.5%
Cumberland	2,008	1,789	1,979	-1.4%
East Providence	2,593	2,430	2,350	-9.4%
Foster	202	277	273	35.1%
Glocester	574	487	543	-5.4%
Johnston	1,589	1,502	1,231	-22.5%
Lincoln	1,137	887	1,382	21.5%
North Providence	1,433	1,580	1,536	7.2%
North Smithfield	548	422	332	-39.4%
Pawtucket	4,877	3,761	4,149	-14.9%
Providence	11,987	11,852	11,017	-8.1%
Scituate	615	346	255	-58.5%
Smithfield	864	820	598	-30.8%
Woonsocket	3,274	3,465	3,281	0.2%
Washington County	7,260	5,650	4,760	-34.4%
Charlestown	450	290	202	-55.1%
Exeter	380	210	329	-13.4%
Hopkinton	503	422	306	-39.2%
Narragansett	636	662	562	-11.6%
New Shoreham	59	10	19	-67.8%
North Kingstown	1,760	1,439	1,356	-23.0%
Richmond	574	416	303	-47.2%
South Kingstown	1,467	1,201	953	-35.0%
Westerly	1,362	1,262	749	-45.0%

Source: US Census Bureau

Chart 4: Rhode Island Population Under Five



Source: US Census Bureau

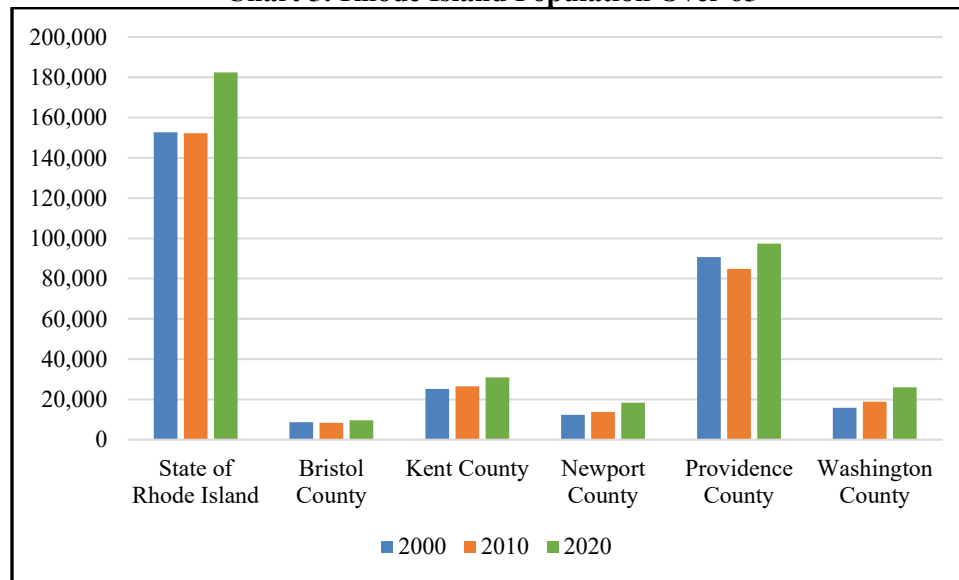
Table 5: Rhode Island Population Over 65, 2000 - 2020

Jurisdiction	2000	2010	2020	Percentage Change 2000 - 2020
State of Rhode Island	152,719	152,183	182,486	19.5%
Bristol County	8,747	8,372	9,674	10.6%
Barrington	2,471	2,427	2,655	7.4%
Bristol	4,009	4,039	4,488	11.9%
Warren	2,048	1,906	2,531	23.6%
Kent County	25,222	26,437	30,970	22.8%
Coventry	4,508	4,908	5,831	29.3%
East Greenwich	1,751	2,056	2,279	30.2%
Warwick	14,582	15,896	16,968	16.4%
West Greenwich	314	532	960	205.7%
West Warwick	4,179	3,924	4,932	18.0%
Newport County	12,281	13,819	18,390	49.7%
Jamestown	844	863	1,701	101.5%
Little Compton	635	688	1,183	86.3%
Middletown	2,534	2,646	3,155	24.5%
Newport	3,422	3,481	4,608	34.7%
Portsmouth	2,351	3,021	4,247	80.6%
Tiverton	2,510	2,961	3,496	39.3%
Providence County	90,659	84,768	97,381	7.4%
Burrillville	1,799	1,796	2,525	40.4%
Central Falls	2,147	1,665	1,499	-30.2%
Cranston	13,820	11,559	13,332	-3.5%
Cumberland	5,136	5,500	6,364	23.9%
East Providence	9,163	8,493	9,154	-0.1%
Foster	449	553	708	57.7%
Glocester	914	1,092	1,673	83.0%
Johnston	5,318	5,571	6,260	17.7%
Lincoln	3,420	3,458	4,387	28.3%
North Providence	6,458	5,876	6,183	-4.3%
North Smithfield	1,952	2,132	2,603	33.4%
Pawtucket	10,800	9,836	9,176	-15.0%

Table 5: Rhode Island Population Over 65, 2000 - 2020

Jurisdiction	2000	2010	2020	Percentage Change 2000 - 2020
Providence	18,351	15,377	20,740	13.0%
Scituate	1,240	1,344	2,360	90.3%
Smithfield	3,391	3,583	4,748	40.0%
Woonsocket	6,547	6,507	5,669	-13.4%
Washington County	15,766	18,797	26,071	65.4%
Charlestown	1,124	1,292	1,876	66.9%
Exeter	590	680	1,269	115.1%
Hopkinton	866	1,155	1,306	50.8%
Narragansett	2,146	2,643	3,593	67.4%
New Shoreham	176	209	189	7.4%
North Kingstown	3,064	3,294	5,216	70.2%
Richmond	508	626	1,088	114.2%
South Kingstown	3,299	4,121	6,330	91.9%
Westerly	3,883	3,963	5,204	34.0%

Source: US Census Bureau

Chart 5: Rhode Island Population Over 65

Source: US Census Bureau

The parameters used to define poverty in the census were refined in 2010, causing a discrepancy in data. As such, only data from 2010 and 2020 were used.

Table 6: Rhode Island Population Below 100% of the Poverty Level, 2010 - 2020

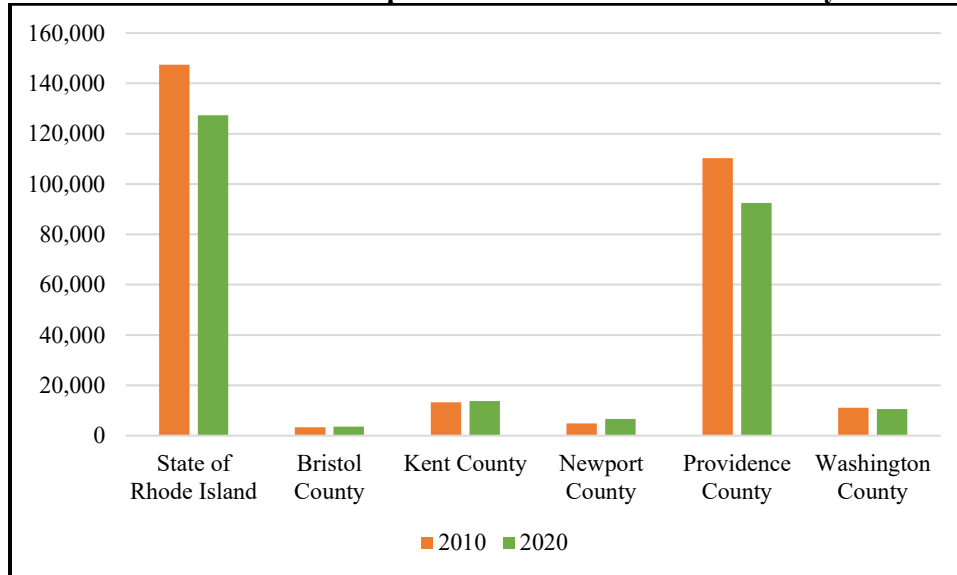
Jurisdiction	2010	2020	Percentage Change 2010 - 2020
State of Rhode Island	147,359	127,296	-2.4%
Bristol County	3,283	3,606	0.6%
Barrington	429	869	2.8%
Bristol	1,925	1,611	-1.0%
Warren	962	964	0.3%
Kent County	13,212	13,799	0.2%
Coventry	2,348	3,266	2.7%
East Greenwich	580	616	0.3%
Warwick	6,611	5,835	-0.8%
West Greenwich	412	107	-5.2%

Table 6: Rhode Island Population Below 100% of the Poverty Level, 2010 - 2020

Jurisdiction	2010	2020	Percentage Change 2010 - 2020
West Warwick	3,464	3,501	0.3%
Newport County	4,805	6,595	1.9%
Jamestown	511	187	-6.0%
Little Compton	102	147	1.3%
Middletown	1,227	991	-1.3%
Newport	2,546	3,556	4.2%
Portsmouth	745	867	0.7%
Tiverton	987	1,038	0.3%
Providence County	110,293	92,504	-3.6%
Burrillville	1,533	1,101	-3.0%
Central Falls	4,925	5,812	4.5%
Cranston	6,756	6,013	-1.0%
Cumberland	1,731	2,064	0.7%
East Providence	4,417	4,554	0.3%
Foster	91	255	3.4%
Glocester	422	541	1.0%
Johnston	2,588	2,082	-1.9%
Lincoln	1,140	1,526	1.6%
North Providence	3,192	3,741	1.6%
North Smithfield	506	898	2.9%
Pawtucket	12,674	8,331	-6.2%
Providence	46,857	41,279	-3.3%
Scituate	538	363	-1.8%
Smithfield	859	782	-0.4%
Woonsocket	9,253	8,739	-1.2%
Washington County	11,047	10,647	-0.5%
Charlestown	315	587	3.5%
Exeter	170	262	1.3%
Hopkinton	180	470	3.6%
Narragansett	2,758	2,982	2.1%
New Shoreham	102	77	-2.4%
North Kingstown	1,490	2,313	3.2%
Richmond	376	100	-3.6%
South Kingstown	2,388	1,655	-2.5%
Westerly	1,675	1,710	0.3%

Source: US Census Bureau

Note: The parameters used to define poverty were refined causing a discrepancy in data. As such, only data from 2010 and 2020 were used.

Chart 6: Rhode Island Population Below 100% of the Poverty Level

Source: US Census Bureau

Data concerning citizens with a disability is detailed the U.S. Census QuickFacts and is reported as an average in a date range rather than as a yearly statistic. The following table details the percentage of the population under the age of 65 with a disability for the period 2018-2022:

Table 7: Rhode Island Average Percentage of Population with a Disability, 2018 - 2022

Jurisdiction	2018-2022
State of Rhode Island	9.80%
Bristol County	6.80%
Barrington	4.30%
Bristol	7.10%
Warren	10.40%
Kent County	10.60%
Coventry	11.10%
East Greenwich	6.70%
Warwick	11.30%
West Greenwich	8.30%
West Warwick	10.80%
Newport County	7.50%
Jamestown	5.30%
Little Compton	-
Middletown	6.40%
Newport	10.20%
Portsmouth	7.00%
Tiverton	8.30%
Providence County	10.60%
Burrillville	10.60%
Central Falls	13.80%
Cranston	8.40%
Cumberland	7.70%
East Providence	10.30%
Foster	-
Glocester	9.60%
Johnston	9.10%

Table 7: Rhode Island Average Percentage of Population with a Disability, 2018 - 2022

Jurisdiction	2018-2022
Lincoln	8.00%
North Providence	10.90%
North Smithfield	10.60%
Pawtucket	11.30%
Providence	10.80%
Scituate	5.60%
Smithfield	6.50%
Woonsocket	18.90%
Washington County	6.70%
Charlestown	5.60%
Exeter	10.10%
Hopkinton	9.40%
Narragansett	4.80%
New Shoreham	-
North Kingstown	5.50%
Richmond	5.20%
South Kingstown	6.50%
Westerly	8.50%

Source: US Census Bureau

Note: QuickFacts only reports data for jurisdictions greater than 5,000 persons

Using data from the Centers for Disease Control and Prevention (CDC)/Agency for Toxic Substances and Disease Registry Social Vulnerability Index FEMA’s NRI creates and maps a Social Vulnerability score. In this context, social vulnerability is the susceptibility of social groups to the adverse impacts of natural hazards, including disproportionate death, injury, loss, or disruption of livelihood. This score represents the relative level of a community’s social vulnerability compared to all other communities at the same level. A qualitative rating that describes the community in comparison to all other communities at the same level, ranging from “Very Low” to “Very High” is used quantify Social Vulnerability. Census tracts with the social vulnerability score highest qualify for designation as a community disaster resilience zone. Census tracts designated as a community disaster resilience zone may receive special technical assistance, planning assistance, and a 90% federal funding match (as opposed to the standard 75% federal match) for mitigation projects.

Community resilience is the ability of a community to prepare for anticipated natural hazards, adapt to changing conditions, and withstand and recover rapidly from disruptions. Factors that are considered when calculating community resilience include governance, infrastructure, education, and other capabilities that help communities deal with hazards on their own. As a consequence reduction risk component of the NRI, a community resilience score and rating represent the relative level of a community’s resilience compared to all other communities at the same level. A community resilience score is inversely proportional to a community’s risk.

Data concerning both social vulnerability and community resilience is reported by census tract. As many State of Rhode Island jurisdictions encompass multiple tracts, an average of tract ratings was completed to determine the overall jurisdictional rating. The following table details both social vulnerability and community resilience by jurisdiction for the State of Rhode Island:

Table 8: State of Rhode Island Jurisdiction Social Vulnerability and Community Resilience Ratings

County	Jurisdiction	Social Vulnerability Rating	Community Resilience Rating
Bristol	Bristol County	Very Low	Very High
	Barrington	Very Low	Very High
	Bristol	Very Low	Very High
	Warren	Relatively Moderate	Very High
Kent	Kent County	Relatively Low	Very High

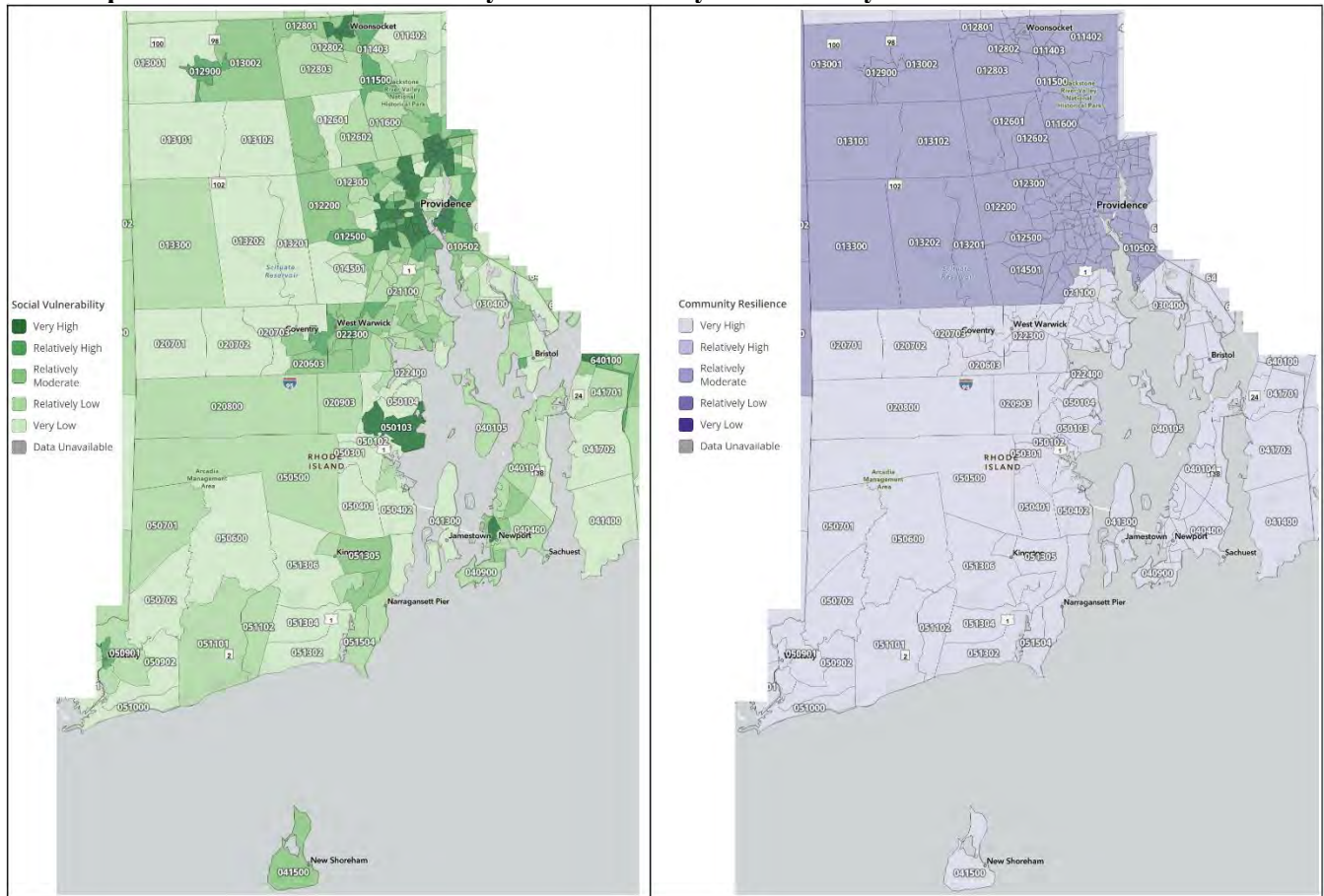
Table 8: State of Rhode Island Jurisdiction Social Vulnerability and Community Resilience Ratings

County	Jurisdiction	Social Vulnerability Rating	Community Resilience Rating
	Coventry	Relatively Low	Very High
	East Greenwich	Relatively Low	Very High
	Warwick	Relatively Low	Very High
	West Greenwich	Relatively Low	Very High
	West Warwick	Relatively Moderate	Very High
Newport	Newport County	Relatively Low	Very High
	Jamestown	Very Low	Very High
	Little Compton	Very Low	Very High
	Middletown	Relatively Low	Very High
	Newport	Relatively Low	Very High
	Portsmouth	Relatively Low	Very High
	Tiverton	Relatively Low	Very High
Providence	Providence County	Relatively High	Relatively High
	Burrillville	Relatively Moderate	Relatively Moderate
	Central Falls	Very High	Relatively Moderate
	Cranston	Relatively Moderate	Relatively Moderate
	Cumberland	Relatively Low	Relatively Moderate
	East Providence	Relatively Low	Relatively Moderate
	Foster	Relatively Low	Relatively Moderate
	Glocester	Very Low	Relatively Moderate
	Johnson	Relatively Low	Relatively Moderate
	Lincoln	Relatively Moderate	Relatively Moderate
	North Providence	Relatively Moderate	Relatively Moderate
	North Smithfield	Relatively Low	Relatively Moderate
	Pawtucket	Relatively High	Relatively Moderate
	Providence	Relatively High	Relatively Moderate
	Scituate	Very Low	Relatively Moderate
	Smithfield	Relatively Low	Relatively Moderate
	Woonsocket	Relatively High	Relatively Moderate
Washington	Washington County	Very Low	Very High
	Charlestown	Relatively Low	Very High
	Exeter	Relatively Low	Very High
	Hopkinton	Relatively Low	Very High
	Narragansett	Relatively Low	Very High
	New Shoreham	Relatively Moderate	Very High
	North Kingstown	Very Low	Very High
	Richmond	Very Low	Very High
	South Kingstown	Relatively Low	Very High
	Westerly	Relatively Low	Very High

Source: FEMA NRI

The following maps indicate the FEMA NRI social vulnerability and community resilience by census tract for the state of Rhode Island.

Map 1: FEMA Social Vulnerability and Community Resilience by Census Tract for Rhode Island



Source: FEMA NRI

Based on the analysis of census data concerning vulnerable populations and the FEMA NRI data, the following jurisdictions have been determined to be potentially the most vulnerable to the impacts of hazards due to social factors:

Table 9: Most Socially Vulnerable Jurisdictions

Jurisdiction	Social Vulnerability Rating	Community Resilience Score
Central Falls	Very High	Relatively Moderate
Pawtucket	Relatively High	Relatively Moderate
Providence	Relatively High	Relatively Moderate
Woonsocket	Relatively High	Relatively Moderate
Burrillville	Relatively Moderate	Relatively Moderate
Cranston	Relatively Moderate	Relatively Moderate
Lincoln	Relatively Moderate	Relatively Moderate
North Providence	Relatively Moderate	Relatively Moderate

Source: FEMA NRI

3.4 Statewide Housing Data

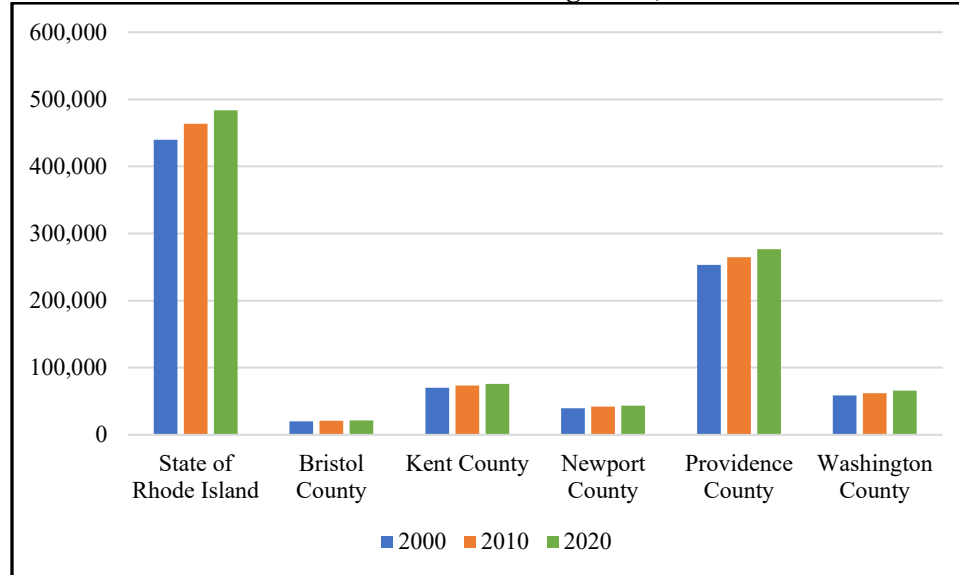
Closely tracking population data, but tending to lag population changes, housing data is a good indicator of changing State demographics and growth. The State of Rhode Island has seen a growth in housing units over the 20-year period from 2000 to 2020, as indicated by data collected from the United State Census Bureau.

The following table, and associated chart, present housing unit data for the State of Rhode Island and its five counties:

Table 10: Rhode Island Housing Units 2000 - 2020

Jurisdiction	2000	2010	2020
Bristol County	19,881	20,850	21,504
Kent County	70,365	73,701	76,084
Newport County	39,561	41,796	43,422
Providence County	253,214	264,835	276,770
Washington County	58,816	62,206	65,694
State of Rhode Island	439,837	463,416	483,474

Source: US Census Bureau

Chart 7: Rhode Island Housing Units, 2000-2020

Source: US Census Bureau

2020 Census data indicates that Rhode Island has approximately 441,274 occupied housing units out of the 483,377 available housing units. This equals a vacancy rate of 9%.

3.5 Statewide Vulnerable Housing Data

Of particular concern when considering housing data is manufactured buildings (often referred to as mobile homes). Data from the NOAA National Severe Storms Laboratory reports that people living in mobile homes are especially at risk for injury and death as even anchored mobile homes can be seriously damaged when winds gust over 80 miles per hour. Additionally, study data from Michigan State University reported that the two biggest factors related to wind event fatalities were housing quality (measured by mobile homes as a proportion of housing units) and income level. When a tornadic wind strikes, a county with double the number of mobile homes as a proportion of all homes will experience 62% more fatalities than a county with fewer mobile homes, according to the study data. The following indicates the percentage of mobile homes for each Rhode Island County:

Table 11: Rhode Island 2020 Census Mobile Home Data

Jurisdiction	Number of Mobile Homes	Percentage Of Housing Stock as Mobile Homes
Bristol County	0	0%
Kent County	902	1.2%
Newport County	701	1.6%
Providence County	1,059	0.4%
Washington County	793	1.2%
State of Rhode Island	3,455	0.7%

Source: United States Census Bureau

A household is considered cost burdened if it spends 30% or more of its income on housing costs, with cost burdens being an indication of rents increasing faster than incomes. The data on cost burdens is collected over a five-year period

in Housing and Urban Development’s Comprehensive Housing Affordability Strategy. Data for Rhode Island was available for 2009 to 2019, with the time lag on the data collection and calculation meaning these numbers are not reflective of recent rapid rent increases that occurred during the COVID-19 pandemic. According to the most recent 2019 data, there are 156,000 cost burdened renters in Rhode Island, up from 149,000 in 2009. This trend indicates declining affordability.

Table 12: Rhode Island Cost Burdened Renters and Homeowners

Jurisdiction	Single Family	Two or More Family
Barrington	64%	24%
Bristol	40%	27%
Burrillville	31%	18%
Central Falls	52%	45%
Charlestown	43%	23%
Coventry	48%	22%
Cranston	46%	29%
Cumberland	54%	21%
East Greenwich	44%	24%
East Providence	49%	25%
Exeter	73%	20%
Foster	26%	25%
Glocester	14%	29%
Hopkinton	42%	22%
Jamestown	47%	24%
Johnston	46%	25%
Lincoln	42%	22%
Little Compton	30%	28%
Middletown	45%	29%
Narragansett	65%	31%
New Shoreham	47%	31%
Newport	11%	49%
North Kingstown	45%	19%
North Providence	45%	29%
North Smithfield	38%	30%
Pawtucket	46%	32%
Portsmouth	54%	22%
Providence	47%	35%
Richmond	11%	24%
Scituate	45%	19%
Smithfield	53%	22%
South Kingstown	39%	24%
Tiverton	37%	27%
Warren	52%	20%
Warwick	50%	28%
West Greenwich	47%	24%
West Warwick	42%	27%
Westerly	47%	31%
Woonsocket	50%	28%
State Total	47%	27%

Source: Rhode Island Department of Housing 2022 Integrated Housing Report

Additionally, the age of a structure could have a significant impact on its resilience to a hazard as an older structure may be less resilient due to the following factors:

- Building Codes and Standards: Older buildings may not have been constructed to meet modern building codes, which often incorporate lessons learned from previous disasters. Newer buildings are typically designed to be more resilient to known hazards.
- Construction Materials and Techniques: Older buildings may have been constructed using materials and techniques that are less resilient to hazards. Modern construction materials and methods, such as reinforced concrete and steel frames, are often more resilient to hazards.

The following table indicates the construction date of all housing units, by city, for the State of Rhode Island:

Table 13: Age of Housing Units in Rhode Island by City

Jurisdiction	2020+	2010-2019	2000-2009	1990-1999	1980-1989	1970-1979	1960-1969	1950-1959	1940-1949	1939 and later
Barrington	0.0%	2.2%	3.5%	5.9%	5.3%	8.8%	16.1%	25.7%	7.6%	25.0%
Bristol	0.0%	2.7%	7.1%	7.5%	15.8%	11.2%	12.4%	10.9%	6.6%	25.9%
Burrillville	0.3%	2.2%	6.9%	6.7%	15.9%	15.6%	8.5%	9.2%	7.5%	27.0%
Central Falls	0.0%	2.5%	4.1%	6.4%	7.5%	6.0%	7.5%	6.6%	8.9%	50.4%
Charlestown	0.5%	2.5%	13.4%	14.0%	19.3%	13.8%	9.5%	12.5%	3.2%	11.3%
Coventry	0.3%	3.0%	6.3%	10.0%	12.6%	17.4%	16.9%	13.7%	4.6%	15.2%
Cranston	0.1%	1.9%	4.2%	5.9%	10.6%	10.5%	12.7%	16.7%	7.6%	29.8%
Cumberland	0.0%	4.5%	8.3%	9.8%	13.2%	13.5%	21.5%	14.0%	3.8%	11.5%
East Greenwich	0.0%	3.8%	11.1%	14.6%	11.2%	15.3%	15.6%	7.4%	4.0%	16.9%
East Providence	0.1%	2.6%	3.7%	4.0%	8.4%	10.3%	15.7%	17.5%	8.1%	29.5%
Exeter	0.6%	3.0%	12.0%	16.3%	22.3%	15.7%	10.1%	5.5%	4.8%	9.7%
Foster	0.0%	3.5%	12.3%	9.8%	15.5%	17.6%	7.2%	7.2%	9.3%	17.6%
Glocester	0.0%	5.2%	8.0%	8.2%	15.0%	20.0%	11.0%	8.3%	6.9%	17.4%
Hopkinton	0.0%	5.9%	13.3%	12.2%	9.2%	11.6%	8.3%	5.3%	9.7%	24.4%
Jamestown	0.6%	7.4%	13.2%	7.6%	16.9%	15.2%	9.6%	6.5%	5.3%	17.8%
Johnston	0.0%	1.4%	3.9%	11.8%	16.0%	14.7%	14.2%	16.6%	5.7%	15.6%
Lincoln	0.0%	4.5%	4.9%	14.5%	8.2%	13.0%	11.1%	13.8%	8.2%	21.8%
Little Compton	0.0%	6.5%	10.5%	8.3%	10.6%	8.7%	8.3%	6.7%	12.3%	28.2%
Middletown	0.3%	2.9%	5.5%	11.6%	14.6%	12.9%	15.7%	13.9%	8.5%	14.2%
Narragansett	0.7%	2.4%	10.4%	8.4%	15.8%	20.4%	15.0%	12.4%	3.4%	11.1%
New Shoreham	0.0%	4.3%	9.7%	11.7%	23.8%	12.2%	9.0%	3.5%	3.0%	22.8%
Newport	0.2%	1.6%	3.5%	3.9%	7.0%	7.1%	8.6%	6.8%	6.1%	55.2%
North Kingstown	0.0%	4.9%	7.3%	12.0%	12.9%	14.2%	16.7%	9.8%	5.0%	17.1%
North Providence	0.0%	1.0%	2.8%	6.5%	23.4%	17.3%	12.5%	14.0%	7.0%	15.5%
North Smithfield	0.1%	3.2%	10.9%	10.3%	13.7%	9.8%	11.8%	12.3%	5.7%	22.2%
Pawtucket	0.1%	1.2%	3.1%	4.2%	5.9%	9.1%	9.9%	13.4%	9.8%	43.3%
Portsmouth	0.0%	4.9%	10.1%	8.9%	14.8%	13.2%	14.6%	10.8%	7.8%	15.0%
Providence	0.1%	1.2%	4.6%	4.6%	5.7%	7.9%	5.4%	7.1%	5.7%	57.7%
Richmond	0.0%	5.1%	10.4%	20.2%	17.7%	22.1%	8.8%	2.1%	0.0%	13.5%
Scituate	0.0%	2.5%	4.8%	8.3%	16.5%	11.4%	10.9%	13.1%	4.4%	28.1%
Smithfield	0.0%	2.8%	7.7%	10.4%	20.2%	14.7%	13.6%	10.5%	4.2%	15.9%
South Kingstown	0.0%	4.7%	12.5%	12.2%	15.0%	15.7%	6.6%	10.2%	4.5%	18.6%
Tiverton	0.0%	5.0%	11.8%	9.1%	13.5%	10.1%	8.8%	15.6%	6.1%	19.9%
Warren	0.0%	1.5%	3.6%	3.9%	11.8%	12.6%	9.6%	11.0%	4.6%	41.4%
Warwick	0.1%	1.9%	3.0%	6.0%	11.1%	13.5%	15.3%	20.2%	7.8%	21.1%
Westerly	0.0%	7.2%	18.9%	21.3%	20.6%	12.8%	9.7%	3.8%	0.0%	5.7%
West Greenwich	0.0%	0.4%	4.5%	9.4%	15.8%	15.2%	7.7%	11.0%	4.8%	31.2%
West Warwick	0.1%	2.9%	8.4%	10.8%	16.2%	13.1%	10.2%	9.8%	3.8%	24.8%
Woonsocket	0.1%	1.1%	4.1%	3.7%	5.1%	14.1%	10.4%	10.2%	7.7%	43.5%
State Total	0.3%	2.4%	5.8%	7.5%	11.3%	12.1%	11.3%	12.1%	6.4%	31.0%

Source: Rhode Island Department of Housing 2022 Integrated Housing Report

For simplification, those ages were combined into a county level summary. The following table indicates the construction date of all housing units, by county, for the State of Rhode Island:

Table 14: Age of Housing Units in Rhode Island by County

Jurisdiction	2020+	2010-2019	2000-2009	1990-1999	1980-1989	1970-1979	1960-1969	1950-1959	1940-1949	1939 and later
Bristol County	0.0%	2.1%	4.7%	5.8%	11.0%	10.9%	12.7%	15.9%	6.3%	30.8%
Kent County	0.1%	3.3%	8.8%	12.3%	14.3%	14.8%	13.0%	11.2%	4.2%	18.0%
Newport County	0.2%	4.7%	9.1%	8.2%	12.9%	11.2%	10.9%	10.1%	7.7%	25.1%
Providence County	0.1%	2.6%	5.9%	7.8%	12.6%	12.8%	11.5%	11.9%	6.9%	27.9%
Washington County	0.2%	4.0%	10.8%	13.1%	16.9%	15.4%	10.5%	7.9%	4.2%	17.0%
State Total	0.3%	2.4%	5.8%	7.5%	11.3%	12.1%	11.3%	12.1%	6.4%	31.0%

Source: Rhode Island Department of Housing 2022 Integrated Housing Report

3.6 Structure Valuation Data

FEMA's Hazus is a nationally standardized risk modeling methodology that uses GIS-based data to identify areas with high risk for natural hazards. Hazus also details the number of buildings and the replacement value of those buildings within the defined area. The following data, from Hazus, indicates valuation (excluding contents) and the percentage of buildings identified as residential properties, by county, for Rhode Island:

Table 15: Rhode Island Structure Valuation

Jurisdiction	Valuation	Percentage Residential
State of Rhode Island	\$194,765,170,000	81.9%
Bristol County	\$15,398,714,000	87.5%
Barrington	\$6,232,480,408	-
Bristol	\$3,890,801,005	-
Warren	\$5,240,338,072	-
Kent County	\$30,632,473,000	87.5%
Coventry	\$5,276,105,592	-
East Greenwich	\$4,005,683,167	-
Warwick	\$14,930,920,436	-
West Greenwich	\$1,489,197,257	-
West Warwick	\$4,748,713,731	-
Newport County	\$18,922,865,000	87.1%
Jamestown	\$1,620,049,528	-
Little Compton	\$1,186,875,686	-
Middletown	\$3,612,874,026	-
Newport	\$5,310,694,435	-
Portsmouth	\$4,106,831,450	-
Tiverton	\$3,088,906,867	-
Providence County	\$99,990,882,000	58.6%
Burrillville	\$2,656,042,158	-
Central Falls	\$1,982,535,924	-
Cranston	\$12,512,812,083	-
Cumberland	\$6,299,110,619	-
East Providence	\$7,790,136,843	-
Foster	\$1,013,244,033	-
Glocester	\$1,954,456,625	-
Johnston	\$4,506,757,792	-
Lincoln	\$5,131,234,662	-
North Providence	\$4,947,271,223	-
North Smithfield	\$2,879,772,507	-

Table 15: Rhode Island Structure Valuation

Jurisdiction	Valuation	Percentage Residential
Pawtucket	\$10,409,990,144	-
Providence	\$25,504,876,133	-
Scituate	\$2,075,370,757	-
Smithfield	\$4,432,256,778	-
Woonsocket	\$5,804,799,090	-
Washington County	\$29,910,236,000	88.9%
Charlestown	\$2,369,173,046	-
Exeter	\$1,304,512,062	-
Hopkinton	\$1,449,289,753	-
Narragansett	\$2,740,655,158	-
New Shoreham	\$812,384,726	-
North Kingstown	\$6,403,427,187	-
Richmond	\$1,619,011,575	-
South Kingstown	\$7,644,732,887	-
Westerly	\$5,571,489,563	-

Source: FEMA

3.7 State and Jurisdictional Development

All current and future development for both the State of Rhode Island and local jurisdictions is potentially vulnerable to the hazards identified in this plan. However, the State of Rhode Island and local jurisdictions have taken steps to reduce the potential impacts through the utilization of building codes and comprehensive plans, which can help restrict development in hazard prone areas. Building codes are evenly and universally enforced throughout the State of Rhode Island, and the application of building codes plays a crucial role in enhancing hazard resilience by promoting the construction of structures that can withstand and mitigate the impacts of hazards. A comprehensive plan outlines the long-term vision and goals for the development of a city or municipality. It serves as a strategic guide for future growth, land use, infrastructure, and community development. Comprehensive plans are typically created through a collaborative process involving local government officials, city planners, residents, and various stakeholders. A key component of a comprehensive plan is land use planning, which defines how land will be used, including residential, commercial, industrial, recreational, and green spaces.

All building and infrastructure construction and improvements play a crucial role in hazard mitigation by enhancing a community's resilience and ability to withstand, respond to, and recover from various hazards. The relationship between building and infrastructure construction and improvement and hazard mitigation can be understood through several key aspects:

- **Enhanced Structural Resilience:** Upgrading and fortifying critical infrastructure, such as buildings, bridges, and utilities, can improve their structural resilience to withstand the forces of natural disasters. Reinforced structures are less likely to suffer damage or fail during hazardous events.
- **Upgraded Utility Systems:** Modernizing and reinforcing utility systems, such as water, sewer, and electrical networks, can minimize disruptions during hazardous events. Upgraded utility systems are less prone to damage, reducing the risk of service outages and improving the community's ability to recover quickly.
- **Flood Control and Stormwater Management Infrastructure** improvements related to flood control and stormwater management can mitigate the impact of flooding. These measures help channel water away from vulnerable areas, protecting lives, property, and critical infrastructure.
- **Climate-Resilient Building and Infrastructure:** With the increasing frequency and intensity of climate-related hazards, building and infrastructure improvement can involve incorporating climate-resilient design principles. This includes constructing buildings and infrastructure that can withstand extreme weather events, rising sea levels, and other climate-related challenges.

The future vulnerability of both the State of Rhode Island and local jurisdictions to these identified hazards is influenced by numerous factors, and the relationship between jurisdictional growth and potential vulnerability is complex. While population growth and urbanization can bring about economic development and improved and more resilient infrastructure, increased growth can also result in an increased vulnerability to identified hazards, including:

- Higher population density in urban areas can mean that more people and assets are concentrated in a limited space. This concentration can amplify the impacts of disaster events, leading to greater loss of life and property.
- Urban expansion often involves changes in land use, such as deforestation or alteration of natural drainage patterns. These changes can influence the community's vulnerability to hazards like floods.
- Urbanization may result in the reduction of green spaces and natural buffers, which can contribute to increased vulnerability to certain hazards. Green spaces can act as natural mitigation measures.
- Growth may lead to economic disparities within a community. Vulnerable populations with limited resources may be disproportionately affected by natural disasters, as they may lack the means to prepare for, respond to, or recover from such events.
- The growth of communities can coincide with the broader impacts of climate change, such as more frequent and intense extreme weather events. This can further increase the vulnerability of communities to natural hazards.

Please note that for all of the following, projects marked with an Asterix denote jurisdictions indicated as most threatened by the hazard being mitigated against.

The Rhode Island Infrastructure Bank (RIIB) is the state's central hub for financing infrastructure improvements, including:

- Water and wastewater
- Road and bridge
- Energy efficiency and renewable energy
- Brownfield remediation

The following recent projects have received funding from RIIB to help improve the infrastructure within the State of Rhode Island:

- City of Warren's Municipal Wastewater Treatment Facility*: Project upgraded wastewater treatment process to reduce nitrogen discharge levels and better protect the health of the bay. Warren's upgraded wastewater treatment facility now releases close to zero nitrogen into the harbor, is more energy efficient, and is more resilient to the impacts of storms, flooding, and the effects of climate change.
- City of Woonsocket's Drinking Water Plant: The new state-of-the-art water treatment plant is providing safe, clean drinking water to residents while improving the Blackstone River's water quality. By moving operations out of the flood plain, the new plant has increased resilience to future storm events. The new facility is the largest public works project in the City of Woonsocket's history.
- Pascoag Utility District: The existing energy infrastructure was insufficient to meet customer needs during peak demand periods. This project upgraded infrastructure with a battery storage facility.
- Boombridge Road Bridge: This bridge, closed since 2008 due to safety concerns, and the road servicing it underwent reconditioning and resurfacing.

In addition, as part of the RIIB's Municipal Resilience Program, the following projects have received funding:

- **Grantee: Barrington ***
Project Name: Resilience Improvements Projects: Walker Farm
Award: \$201,000
Walker Farm is a 48.5-acre Town-owned site comprising ecologically significant upland, salt marsh, and ponds habitats located along the Barrington River. The site has an active recreational and agricultural component and

hosts a municipal services facility. Rather than attempt to block floodwaters and impending sea level rise, the project adapts the site to these conditions by enhancing natural ecosystem functions and relocating recreation areas. Specifically, the project will reconfigure the shoreline to increase stormwater filtration, wave buffering, and flood storage; restore native plant buffers and salt marsh to stabilize the shoreline; and move recreation areas landward to provide areas for floodable open space as needed.

- **Grantee: Portsmouth***

Project Name: Stormwater Inland Flooding Climate Adaptation Project

Award: \$199,000

Portsmouth will use action grant funding to redesign and improve drainage infrastructure at the entrance to the Common Fence Point neighborhood. Period street flooding can compromise the only emergency evacuation route for the neighborhood, creating a public safety concern for the roughly 700 households in the area. To improve drainage and redirect stormwater runoff, Portsmouth will install two new rain gardens, clean existing wetlands, rehabilitate catch basins, and re-contour the roadside along Anthony Road.

- **Grantee: Portsmouth**

Project Name: Melville Dam Rehabilitation

Award: \$140,000

The Town of Portsmouth used action grant funding to help offset the costs of completing ongoing rehabilitation of the Melville Dam in anticipation of the increased intensity and frequency of major storms due to climate change. The earthen dam performs a critical role in flood control for downstream economic and environmental resources. In addition, the dam protects a wetlands environment situated within a popular recreation area. This project was the final step in the Portsmouth Department of Public Works' effort to remove excessive vegetation growth that was impeding regularly required dam inspections. Funding supported efforts to remove stumps, roots and vegetation; re-grade the existing slope; and apply loam and hydroseed as a final stabilization measure. The Melville Dam Rehabilitation project was completed in April 2021.

- **Grantee: Warren***

Project Name: Stormwater Mitigation at Public Access Points

Award: \$156,000

This project will allow the Town of Warren to incorporate natural stormwater mitigation practices at several public rights-of-ways that terminate at the edge of waterways. Currently, many of the rights-of-ways have hard surfaces that create unnecessary stormwater runoff and erosion and contribute to increased levels of pollutants entering waterways. The Town will replace pavement with loam and vegetation and install filter fabric and rip rap. The Town previously worked with Save The Bay, Fuss and O'Neill, and the Rhode Island School of Design to develop conceptual designs for these rights-of-ways as part of a coastal adaptation pilot program.

- **Grantee: Westerly***

Project Name: Old Canal Street Pump Station Flood Protection Wall

Award: \$191,000

The Old Canal Street Pump Station Flood Protection Wall project involves improvements to the Westerly wastewater treatment system that will limit discharge quantities, limit overtopping by floodwaters, and improve water quality in the Pawcatuck River. The Town of Westerly will install a concrete footing and flood protection wall around the perimeter of the pump station located at 20 Canal Street. Without this project, the station is vulnerable to flooding during extreme weather. Flooding of the station would cause tremendous strain to the system and potentially lead to a pump failure, triggering a Sanitary Sewer Overflow of raw sewage flowing directly into the Pawcatuck River.

- **Grantee: Westerly***

Project Name: Springbrook Road Rain Garden and South/West Fairway Infiltration

Basin Award: \$87,936

The Town of Westerly will use action grant funding to implement two projects to address stormwater runoff and water quality issues. The first project entails the installation of rain gardens on Springbrook Road near the entrance to Springbrook Elementary School. The rain gardens will capture and infiltrate stormwater runoff, which will improve the water quality of Pawcatuck River and mitigate safety issues that occur when stormwater runoff collects and freezes alongside the road. The second project will install an infiltration basin on the

Winnapaug golf course near the intersection of South and West Fairway. The basin will slow the movement of road drainage discharges to the golf course and into Winnapaug Pond.

- **Grantee: Bristol***

Project Name: Watershed Restoration at the Bristol Golf Course, Phase II

Award: \$222,863

This project will address downstream flooding and water quality and habitat degradation within two watersheds resulting from historical alterations to wetlands and two streams at the Bristol Golf Course. Today, the golf course provides few natural wetland benefits and precipitation runs unchecked to impoundments and watercourses on the property and eventually flows downstream. Permitted work includes restoring ponds and impoundments within the golf course, removing fill and culverts and “daylighting” streams, and restoring a buffer of native shrubs, trees and grasses between the active course and wetland resource areas. Work will increase flood storage capacity and enhance functions of freshwater wetland habitat within the property as well as address concerns for downstream flooding in both the Towns of Bristol and Warren and improve resiliency and asset protection.

- **Grantee: Little Compton***

Project Name: Stormwater Management Program

Award: \$164,000

The Town of Little Compton will use action grant funding to implement three projects intended to mitigate stormwater runoff and flooding on town roads and public spaces. The South Shore Beach Parking Area project includes the installation of a sediment forebay and dry swale to capture and treat stormwater runoff, provide sediment capture, and encourage infiltration. The Town Way project will install a drainage retention area to collect and retain stormwater along a public right-of-way and help limit erosion of the roadway, reduce the flow of sediments into the ocean, and improve a parking area with low-impact design. The John Dyer Road project will install a detention area for storm flow drainage to reduce the flow of stormwater and sediments across roadways, into neighboring properties, and into the Westport River.

- **Grantee: Newport***

Project Name: Almy Pond/Spouting Rock Drive Restoration

Award: \$180,990

This project will remove unnecessary infrastructure at Almy Pond, one of the most distressed bodies of water in the state and restore the surrounding meadow and marsh. Specifically, the City of Newport will remove approximately 25,200 square feet of pavement from Spouting Rock Drive and its associated catch basins, which were built in the 1990s to service an illegal subdivision. Removal will expand the drainage buffer around the pond, improve local water quality, and allow the pond to act as a more effective buffer against stronger and more frequent storms anticipated as a result of climate change.

- **Grantee: Pawtucket and Central Falls***

Project Name: Pawtucket-Central Falls Transit Oriented Development District Stormwater and Streetscape Improvements Award: \$400,000

The Cities of Pawtucket and Central Falls will use action grant funding to improve and upgrade aging infrastructure in the joint Transit Oriented Development District around the Pawtucket-Central Falls train station and transit hub opening in mid-2022. Improvements include new street trees, stormwater improvements with additional green infrastructure, new sidewalks, bicycle-friendly infrastructure, and new crosswalks. Both cities are particularly vulnerable to high precipitation events and heat waves due to high levels of impervious surfaces (86% of the Transit Oriented Development District is impervious) and aging stormwater infrastructure with limited capacity. The planned stormwater and streetscape improvements will make the area more accessible to pedestrians and cyclists while also improving air quality, reducing the heat island effect, and mitigating flooding issues.

- **Grantee: Warwick***

Project Name: Oakland Beach Nature Based Resiliency Enhancements

Award: \$225,000

This project involves the construction of a bioretention stormwater system and coastal embankment restoration in the seaside community of Oakland Beach. The bioretention system is a nature-based solution that will help

reduce flooding and minimize bacterial pollution in Greenwich Bay by intercepting and treating stormwater runoff before it discharges into coastal waters. The coastal embankment restoration will help reduce and prevent erosion of the existing beach area, increase wildlife habitat, and enhance the overall aesthetics of the area. This project is part of a larger effort supported by local, state, and federal agencies to incorporate nature-based resilience solutions along the coastline of Oakland Beach that will increase the public's enjoyment of the shoreline by expanding public access and recreation opportunities.

- **Grantee: Woonsocket**

Project Name: Blackstone River Vision Report Implementation

Award: \$150,000

This project will implement three green infrastructure projects identified in the Blackstone River Vision Report. The projects will divert stormwater from the City's gray infrastructure and reduce flow rates and pollutant loads of stormwater entering the Blackstone River, an impaired waterway. The proposed projects are (1) vegetated swales at River's Edge Park, (2) vegetated swales and rain gardens at Bernon Memorial Park, and (3) permeable paving, stormwater planters, gravel and sand filtration, and underground water storage at City Hall's parking lot.

- **Grantee: Woonsocket**

Project Name: Iron Rock Brook at Mendon Road

Award: \$150,000

This project will address recurring flooding at Iron Rock Brook and Mendon Road. Currently, an undersized culvert results in flooding at a residential property and across the state-owned Mendon Road during heavy rain events. This is a public safety concern as vehicles attempt to pass the flood section and a public health concern as floodwaters can infiltrate and overflow the local sewer system, causing untreated sewage to be introduced into the environment. The City of Woonsocket will increase culvert capacity so more stormwater can be managed. If feasible, a nature-based, green infrastructure solution will also be included upstream to help capture stormwater and reduce overall demand on the culvert.

- **Grantee: Barrington***

Project Name: Shoreline Adaptation & Resilience Project

Award: \$249,521

The Town of Barrington will use grant funding to support four key resilience actions: shoreline adaptation, upgradient tree trench Best Management Practices, a municipal tree nursery, and community engagement. Projects will take place at six locations. At Woodbine Avenue, Adams Avenue, Watson Avenue, Clark Road, and Water Way, the Town will remove invasive plant material and excess asphalt; install new inlet structures, sediment forebays, riprap, infiltration basins, and native plantings; and enhance visitor experience through restoring pedestrian public shoreline access and placing green infrastructure educational signage. In addition, upland from these coastal sites, corresponding tree box filters will be placed to treat stormwater and promote infiltration ahead of interaction with each coastal area. At the sixth site, the Municipal Tree Nursery, which provides native trees to support tree box filters across Town, fencing will be placed to protect growing trees from deer and tree species will be selected for resilience to climate change impacts. The six projects, which will address flooding, water quality, and high temperatures, will be supported by Town community engagement efforts regarding stormwater, green infrastructure, and tree planting.

- **Grantee: Bristol***

Project Name: Improving Coastal Resiliency at Independence Park

Award: \$461,500

The northern end of Independence Park provides public access to Narragansett Bay and includes a public boat ramp, a dinghy dock, kayak racks, and an entrance to the East Bay Bike Path. However, this area currently has oversized impervious surface area, uncontrolled stormwater runoff, and severe erosion, creating both flooding and water quality concerns. This project will remove approximately 4,000 square feet of unneeded impervious pavement at the waterfront, provide green infrastructure Best Management Practices including bioretention areas, porous or grass pavers, and other infiltration techniques to treat stormwater, and provide an improved landscaped area and safer public access between the East Bay Bike Path, the parking area, and the Bay.

- **Grantee: Central Falls & Pawtucket**

Project Name: Pine Street North: Green Stormwater Parklets

Award: \$678,600

This project builds upon a past grant funded project, which aims to bring stormwater infrastructure and improved pedestrian and bike access along the Pine Street corridor in the Pawtucket-Central Falls Transit Oriented Development District. This corridor serves as a major connection between the two municipalities and to the Pawtucket Central Falls train station opening in late 2022. The Towns will use this new round of funding to create two parklets, one in each city, designed to manage stormwater as well as to provide public gathering areas along the Pine Street corridor. These parklets, located at Conant Street and Rand Street, will include permeable pavers, bioretention planters, furnishings, and shade structures to invite residents and workers to enjoy these outdoor spaces. The Rand Street parklet will serve as a gateway to the Transit Oriented Development District, and the Conant Street parklet will be adjacent to new housing opportunities, including a building being rehabbed by Shri Service Corps to include both affordable housing units and health and wellness services to children, the elderly, and people with disabilities.

- **Grantee: Cumberland**

Project Name: Valley Falls/Lonsdale (Urban Forestry Municipal Resilience Project)

Award: \$250,000

The Town of Cumberland will plant one hundred ninety-six (196) trees at twenty-six (26) locations throughout the Valley Falls and Lonsdale neighborhoods. These neighborhoods are densely populated and have been identified as need areas through American Forests' Tree Equity Score Analyzer, which evaluates data on tree canopy cover, surface temperature, and demographics such as income, age, race, and health to determine where lack of trees is contributing to economic and social disparity. This tree planting will restore the area's urban forest and, in doing so, will provide stormwater management and cooling benefits.

- **Grantee: Cumberland**

Project Name: Industrial Road, Largest Commercial District, Chronic Flooding - Install a Stormwater Drainage System

Award: \$500,000

Industrial Road, Cumberland's largest commercial district, experiences chronic flooding. In this area, twenty-nine of the Town's largest employers have experienced business loss and disruption with increasing frequency as flooding has increased with climate change impacts, and vehicles have become trapped during flash flood events. This project will enhance Cumberland's stormwater management system along Industrial Road, using a combination of green and gray stormwater management techniques. The existing undersized and outdated stormwater infrastructure will be replaced with updated structures, and roadside plantings, absorbent gardens, and tree filters will be installed along the roadway to further address flooding and water quality concerns.

- **Grantee: East Providence***

Project Name: Stormwater Infiltration and Flood Protection in East Providence

Award: \$225,000

The City of East Providence will implement three climate resiliency projects that together will improve stormwater infiltration, erosion control, and water quality at two public parks, and will strengthen flood protection at a city pumping station. The Beach Road stormwater improvement project will install two catch basins, six rows of infiltration chambers, and native plantings where the roadway ends at the coast, preventing runoff and erosion as well as improving public access to the water. The Silver Street pump station project will install flood proof doors, remove an underground storage tank, and replace it with a new above ground tank to house the generator (raised above the 100-year flood elevation), which will protect this critical sewage pumping station from future floods and hazardous weather events. The Willet Pond improvement project will remove 2,700 square feet of impervious area and replace the remaining 11,880 square feet of the existing parking area and driveway with crushed stone and pervious pavement, which will improve stormwater infiltration and water quality at this public recreation area.

- Grantee: Newport***
 Project Name: Pine Street Community Resiliency Project through Pavement Removal, Stormwater Management and Public Access Enhancement
 Award: \$58,982
 Pine Street ends at the southern edge of Battery Park, overlooking Newport Harbor and the Point neighborhood. Currently, stormwater runoff from Pine Street flows down to Blue Rocks, a public access site to Narragansett Bay, which has been hardened with asphalt. This project would remove this impervious pavement, replacing it with a permeable walking path, landscaping, infiltration swales, and updated sidewalks and curbing at its entrance. The project will not only improve public access to the waterfront, but will also provide erosion control, water quality, and stormwater management benefits.
- Grantee: North Kingstown***
 Project Name: Roger Williams Drive End of Road Retrofit
 Award: \$24,217
 At the end of Roger Williams Drive, which extends toward Wickford Cove, the Town will mitigate the impacts of the stormwater flow by removing unnecessary pavement and installing grassy areas, a sediment forebay, and a sand filter. These measures will address erosion and provide stormwater management benefits on site by slowing runoff, increasing infiltration, and catching sediment before it discharges into the cove. The existing scrub shrubbery will remain in place as a natural buffer and infiltration area.
- Grantee: North Kingstown***
 Project Name: Wickford Waterfront Improvements
 Award: \$647,365
 This project will implement low impact stormwater management with enhanced green infrastructure to adapt to changing coastal conditions, mitigate stormwater runoff, and address high tide flooding at the municipal waterfront parking lot, known as the Brown Street Parking Lot, in Wickford Village. These improvements are part of a multi-tiered approach to expand recreational, cultural, and civic opportunities in Wickford, as well as to address climate change impacts including sea level rise and increasing storm surges. The MRP Action Grant will support elevating the lowest segments of the Brown Street Parking Lot that are susceptible to flooding, removing existing impervious pavement, and installing pervious pavers around the exterior of the parking lot. The project will also include the installation of tide gates and bioretention basins. These additions will minimize nuisance flooding in the area and reduce pollutants entering the harbor.
- Grantee: Portsmouth***
 Project Name: Stormwater Inland Flooding Climate Adaptation Projects II
 Award: \$140,000
 The Town of Portsmouth will build off of a previous MRP Action Grant award to continue providing stormwater management projects that address increased intensity and frequency of major storm events due to climate change. Two projects will be implemented. At Riverside Street, the Town will install an underground stormwater injection facility below the roadway in an area subject to constant flooding in major storms. Stormwater will be captured through catch basins and piped into a galley-type structure or gravel trench system designed to improve stormwater infiltration. At Founders Brook, dredging will take place to widen, deepen, and enhance 560 of natural watercourse, removing silt and vegetation buildup without hardening or channelizing the Brook, with the aim of reducing flooding at nearby Boyd's Lane.
- Grantee: Providence***
 Project Name: Mashapaug Watershed Water Quality Improvements
 Award: \$470,000
 Mashapaug Pond is an impaired waterway that is the main feed for the Roger Williams (RW) Park ponds and contributes to Phosphorus loading to these ponds. The combination of increased flooding due to climate change (which brings more Phosphorus to the Mashapaug and Park ponds), as well as increased heat (which allows algae that use nutrients like Phosphorus to grow faster), is leading to more frequent blue-green algal blooms in these ponds and public health hazards. With this grant, Providence's Parks Department will install a "treatment train" in line on the pipe at the Roosevelt Lake outlet, including a jelly fish filter, a bio-char platform with a wet vegetated swale, and additional adjacent green infrastructure elements to handle overflow in larger rain

events. Not only will this installation address the Phosphorus contamination, but it will also allow for testing of innovative stormwater management methods to add to the RI Stormwater Manual and Rules. Testing ports will be installed before and after each treatment element to test the effectiveness of each installation. This site will also serve as a training site for RIDOT maintenance personnel on these innovative stormwater treatment techniques.

- **Grantee: Providence**

Project Name: Emergency Generators for Davey Lopes Recreation Center and Elmwood

Community Center

Award: \$375,000

The City of Providence will install emergency generators at two (2) City-owned community centers: Davey Lopes Recreation center and Elmwood Community center. The generators will keep the centers operational and functional during times of adverse weather hazards/events which have been exacerbated by climate change. With the generator installations, these centers will be able to provide shelter during storms and extreme heat and cold events, creating an additional level of public safety measures for the City's constituents who regularly use, and live near, these centers. Public engagement has highlighted that these centers will significantly benefit from these generators as they serve disadvantaged populations, with both centers serving thousands of families who reside in a majority low-moderate income area within Providence. In addition, as a part of this MRP Action Grant, Providence will also begin the process of developing these centers as Resilience Hubs, which will provide shelter and resources to meet community needs during extreme events, provide information and engagement on the topic of climate resilience, and transition to clean-energy systems in line with City-wide resilience goals.

- **Grantee: Tiverton***

Project Name: Fogland Beach and Conservation Area Stormwater Mitigation and Restoration

Project Award: \$228,425

This project addresses longstanding challenges at Fogland Beach, Tiverton's largest and most important recreational beach, and Fogland Conservation Area, a relatively intact and healthy salt marsh threatened by stormwater runoff and sea level rise. The entire project area is bisected by Three Rod Way, a town roadway that serves over twenty year-round and summer homes located on Fogland Point. This project combines several elements to mitigate stormwater runoff along Fogland Road and Three Rod Way, restore natural hydrology of the 7+ acre Fogland Marsh, relocate existing infrastructure, and create natural sand dunes to provide storm surge resilience to the area. The Town will install infiltration areas and swales, install a trench drain under Three Rod Way, conduct regrading to restore the hydrology at the skating pond, expand the dunes and further stabilize with beach grass, excavate runnels in the marsh to restore nature hydrology and allow impounded water to drain, and relocate the existing ticket booth to a less flood prone area.

- **Grantee: Warren***

Project Name: Fern Drive Drainage Improvements

Award: \$94,485

This project will improve the drainage infrastructure and stormwater management within the cul-de-sac on Fern Drive by installing green infrastructure Best Management Practices (BMPs). The Town of Warren will install three bioretention tree filters to mitigate flooding on the abutting residential properties on the cul-de-sac, as well as to provide water quality treatment and reduce the flow of runoff to the Town's drainage network.

- **Grantee: Westerly***

Project Name: Greening Downtown Westerly - A Main Street Renaissance

Award: \$500,000

This project will install nature-based stormwater infrastructure at several locations along Main Street in Westerly, serving as a nature-based solutions demonstration project that will achieve the goal of revitalization of Downtown Westerly while also improving the water quality of the adjacent Pawcatuck River, reducing flooding, reducing heat, and increasing green space. Installations will include tree planters, rain gardens, parking lot infiltration medians, planted streetscape buffers, bioswales, stormwater infiltration planters, pervious pavers, and pervious parking.

The RI READY program is a statewide industrial site readiness initiative designed to prepare sites to achieve pre-permitted status. This program provides access to an array of services designed to create an inventory of pre-permitted properties ready for industrial development throughout the State. State investment is recaptured to the extent possible to make the initiative self-sustaining and extend the life of available funds.

The RI READY program leverages the Quonset Development Corporation, a quasi-state agency which partners with both Rhode Island Commerce and municipalities and/or private parties to spur development activity throughout the state. Site-readiness awards are made in two categories, site-specific improvements and municipal assistance. Site-specific improvements include site planning and pre-development activities and project improvements. Projects intended to catalyze growth in advanced industry clusters, regardless of project size, are prioritized. Municipal Assistance awards are for developing projects that will lead to further growth and support the municipality's development goals.

In 2023 over \$650,000 in Site Readiness Awards were given to eight major projects across Rhode Island. These projects are:

- Bowen's Wharf (Newport)* – \$35,700 to investigate its vulnerability to flooding. Funding will support long-term sustainability by hiring a local engineering firm to do research and fieldwork.
- House of the Rising Star LLC (Providence) – \$125,000 to assist in the revitalization of the historic, 3-story mill at 239 Oak Street in Providence. Funding will go towards building and to sidewalk and street renovations.
- JR Hess Co., Inc. (Cranston) – \$125,000 to assist in the construction of a new warehouse that will allow the company to expand and contribute to the growing pharma and biotech growth in its surrounding area.
- Manton Development Partners, LLC (Providence) – \$125,000 to assist in the redevelopment of the historic St. Teresa site for neighborhood scale commercial uses and parking with a solar canopy.
- Rise Above All, LLC (Pawtucket) – \$75,000 to support the predevelopment costs for a \$61,000,000 mixed-use development located adjacent to the Pawtucket Central Falls Transit Hub. The proposed project will transform a blighted contaminated property and will be home to 180 housing units including 52 designated for affordable housing.
- The Coletta Group LLC (Providence) – \$100,000 to aid in the development of the 2.5-acre Swan Business Park campus in Providence.
- The Compost Plant L3C (TCP) (Providence) – \$51,200 to support the development of an organics pre-processing facility including required design and engineering proposals, and relevant planning analyses.
- Town of Narragansett (Narragansett) – \$30,000 to support the continued implementation of the Town's redevelopment plan. Funding will support the Town in hiring a consultant to make and review revisions and integrate them into the Town's Zoning Ordinance.

The Wastewater Treatment Resilience Fund, administered by the Department of Environmental Management, provides funding to municipalities and/or quasi-state entities to protect government-owned wastewater collection systems and treatment facilities against the effects of climate change and flooding. Projects harden, relocate, repair or replace, and/or provide redundancy infrastructure. Approximately \$5,000,000 of funding has been provide, with recent projects including:

- Barrington: Barrington/Warren Joint Portable Generators - Barrington
- Cranston: Installation of Watertight Doors at the Pontiac, Worthington, and Youlden Pump Stations
- East Greenwich: Construction of Wastewater Treatment Resilience Fund resiliency measures and emergency generator replacement
- East Providence: Silver Street Pump Station Floodproofing
- Narragansett: Scarborough Wastewater Treatment Resilience Fund Outfall Groin Restoration
- Newport: Long Wharf Pump Station Flood Protection
- New Shoreham: Ocean Avenue Pump Station

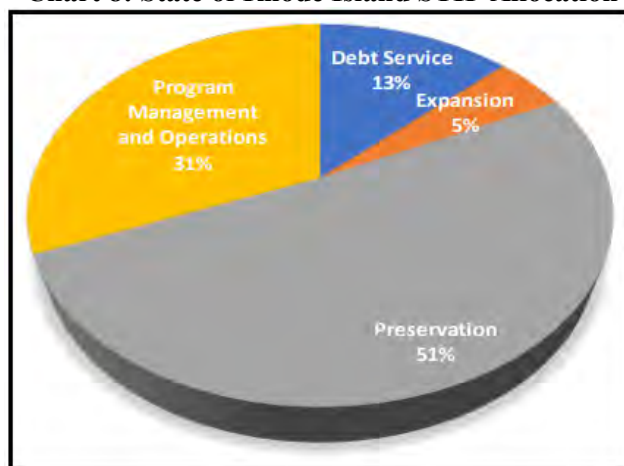
- Quonset Development Corporation: Quonset Wastewater Treatment Resilience Fund Resiliency Action Plan, Phase R1
- Smithfield: Smithfield Wastewater Treatment Resilience Fund – Grit Removal System
- South Kingstown: Middlebridge Pump Station Resiliency/Hardening
- Warren: Barrington/Warren Joint Portable Generators - Warren
- Warren: Wood Street Pump Station Resiliency Improvements
- Westerly: New Canal Street Pump Station Flood Retaining Wall
- Woonsocket: Woonsocket Wastewater Treatment Resilience Fund Building Hardening

The redevelopment of roads and bridges plays a crucial role in hazard mitigation by incorporating resilient design, enhancing safety measures, and improving the overall ability of transportation infrastructure to withstand and recover from natural and human-made disasters. Redevelopment projects can enhance drainage systems to mitigate the impact of heavy rainfall and reduce the risk of flooding. A properly designed drainage infrastructure helps prevent road and bridge damage caused by water accumulation. Vulnerability to hazards like flooding, sea level rise, and storm surge can be mitigated by upgrading bridge piers and foundations. Strengthening these critical components enhances the overall stability and durability of the infrastructure.

RIDOT, the Rhode Island Public Transit Authority, and the Rhode Island Division of Statewide Planning oversee all transportation planning and investment within the state. As part of this planning, the State Transportation Improvement Program (STIP) is a list of transportation projects the State of Rhode Island intends to implement using U.S. Department of Transportation funds. The STIP presents a 10-year program of over \$6,000,000,000 in funding for transportation projects from federal fiscal year 2022 to 2031. Projects completed under STIP are expected to decrease the vulnerability of state infrastructure as they will be constructed to newer building code standards, which plays a crucial role in enhancing hazard resilience by promoting the construction of structures that can withstand and mitigate the impacts of hazards.

The majority of the STIP (53%) is dedicated to keeping and maintaining the state’s existing infrastructure in a state of good repair, as indicated by the following chart:

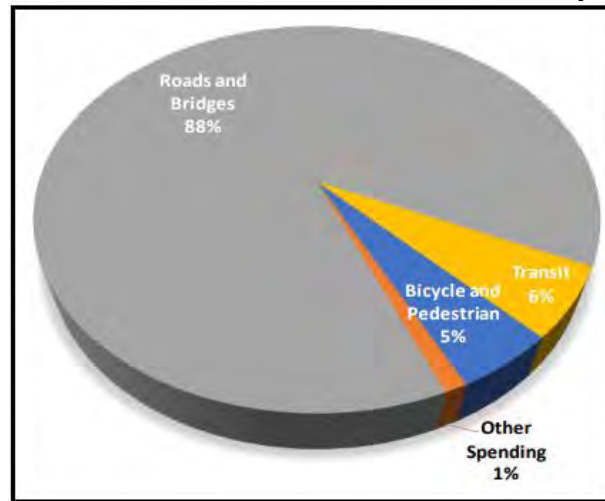
Chart 8: State of Rhode Island STIP Allocation



Source: RIDOT

Highways remain the primary mode of travel in Rhode Island. Highway projects receive the greatest share of STIP resources, with the focus on state of good repair and system preservation, as indicated by the following chart:

Chart 9: State of Rhode Island STIP Allocation by Mode



Source: RIDOT

A full list of funded and identified STIP projects may be found in Section 3 of the Division of Statewide Planning's State of Rhode Island Transportation Improvement Program FFY 2022-2031 report.

In addition, numerous large infrastructure projects are underway in the state. The United States Department of Transportation define regionally significant projects as "transportation projects that are on a facility which serves regional transportation needs (such as access to and from the area outside of the region, major activity centers in the region, major planned developments such as new retail malls, sports complexes, etc., or transportation terminals as well as most terminals themselves) and would normally be included in the modeling of a metropolitan area's transportation network, including at a minimum all principal arterial highways and all fixed guideway transit facilities that offer an alternative to regional highway travel."

Recent significant projects within the State of Rhode Island include:

- **Route 6/10 Redesign/Reconstruction:** In 2018, RIDOT began work on the project which involves nine bridge replacements. The project also creates 1.4 miles of new bike path and opens more than four acres of developable land in Providence. The project is scheduled to be completed in 2023.
- **Interstate 95 Northbound Viaduct** In 2019, RIDOT was awarded a federal Infrastructure for Rebuilding America grant to transform the I-95 Northbound section of I-95 in Providence. The project will reconstruct the Viaduct's six bridges, rehabilitate five bridges, and construct three new bridge structures, along with the reconfiguration of a series of ramps to separate conflicting lanes of traffic. The project is scheduled to be completed in 2023
- **Route 146 Reconstruction:** This project has several major components, one of the largest of which is the construction of an overpass at the intersection of Route 146 at Sayles Hill Road. Project components also include bridge rehabilitation and bridge replacement. The project is scheduled to be completed in 2023.
- **Washington Bridge Rehabilitation and Redevelopment:** This project will address the structural deficiencies of the westbound portion of the Washington Bridge, which carries I-195 over the Seekonk River between East Providence and Providence. The project also includes a new off-ramp to support economic development opportunities and address chronic congestion issues on the Interstate. The completion date of this project is yet to be determined.
- **Route 37 Improvements:** Improvements to Route 37 will be made in two separate projects that will rehabilitate or replace 22 bridges and make safety and congestion improvements in this critical east-west freeway in central Rhode Island, linking the cities of Cranston and Warwick, major retail, office and residential areas, Interstate highways and Rhode Island T.F. Green International Airport. Phase 2 of this project is scheduled to be completed in 2026.
- **Amtrak Rail Platform at Warwick Station (T.F. Green Airport):** RIDOT has started a preliminary design and environmental study to assess what would be necessary to build the infrastructure necessary to accommodate

Amtrak intercity service at the Rhode Island T.F. Green International Airport in Warwick. The preliminary design study is projected to be completed in 2023.

- Pawtucket/Central Falls Train Station: The Pawtucket/Central Falls Train Station is an infill commuter rail train station located along the Northeast Corridor. The station was completed and opened in January of 2023.

Additionally, the following projects are currently being funded or are under construction for the State of Rhode Island:

- RIEMA has applied for a \$17,000,000 grant (under COVID-19 disaster declaration) to raise the docks in both Narragansett and Wickford to an operational height of 5.06 above Mean High Water. This Federally funded project will be constructed to provide the ability to further raise the docks as sea level continues to rise and will provide protection to these important critical assets to continue to offer economic stability and growth.
- As of this plan, a new Public Health Laboratory is currently under construction in Providence. The building will also be home to a new Center for Biological Sciences that will function as a dedicated Genomics Sequencing Core Laboratory. The laboratory will enable the Rhode Island State Health Laboratory to continue providing high-quality scientific testing through genomic sequencing of infectious disease.

State of Rhode Island assets, to include state-owned and operated buildings, infrastructure, community lifelines, and critical facilities, as defined by the FEMA State Mitigation Planning Policy Guide, have seen no major changes in the past five years, with only modest repairs and upgrades being conducted, other than the projects noted above. As such, the overall risk to state assets has remained static since the completion of the 2019 SHMP. Additionally, apart from the above-mentioned projects, no large-scale facility projects were noted in any agency budgets or reports.

A review of data from jurisdictional Local Hazard Mitigation Plans, including data on potentially vulnerable infrastructure, and a review of infrastructure data in this SHMP indicates the following State of Rhode Island jurisdiction's infrastructure is most vulnerable to the hazards identified in this plan:

Table 16: State of Rhode Island Jurisdictions with Most Vulnerable Infrastructure

County	Jurisdiction
Bristol County	Bristol
Kent County	East Greenwich
Newport County	Newport
Providence County	Providence
Washington County	Charleston
Washington County	Narragansett
Washington County	South Kingstown
Washington County	Westerly

Building permit data from the Rhode Island Department of Housing was researched to gain an understanding of residential growth within the State. Available data did not provide an indication as to which permits were for renovations or additions and which were for new projects. However, any project is expected to adhere to all State of Rhode Island Building Codes (currently the International Building Code, 2018 Edition). The following table indicates the number of building permits issues in 2022 (the latest full year data available), by city:

Table 17: Rhode Island 2022 Building Permits by City

Jurisdiction	Single Family	Two or More Family	Accessory Dwelling Unit	Total
Barrington	15	0	2	17
Bristol	23	0	0	23
Burrillville	69	0	0	69
Central Falls	0	6	0	6
Charlestown	40	0	0	40
Coventry	55	16	0	71

Table 17: Rhode Island 2022 Building Permits by City

Jurisdiction	Single Family	Two or More Family	Accessory Dwelling Unit	Total
Cranston	46	4	0	50
Cumberland	52	0	0	52
East Greenwich	32	48	0	80
East Providence	19	24	0	43
Exeter	10	40	0	50
Foster	9	0	0	9
Glocester	27	2	0	29
Hopkinton	16	0	0	16
Jamestown	15	5	0	20
Johnston	39	0	0	39
Lincoln	53	42	0	95
Little Compton	13	0	0	13
Middletown	10	0	0	10
Narragansett	12	2	0	14
New Shoreham	9	0	0	9
Newport	11	0	0	11
North Kingstown	65	12	0	77
North Providence	6	0	0	6
North Smithfield	15	0	0	15
Pawtucket	14	6	0	20
Portsmouth	35	0	0	35
Providence	6	122	0	128
Richmond	22	0	0	22
Scituate	26	0	0	26
Smithfield	28	0	0	28
South Kingstown	80	17	7	104
Tiverton	38	0	0	38
Warren	7	15	0	22
Warwick	40	40	0	80
West Greenwich	21	0	0	21
West Warwick	6	29	0	35
Westerly	27	8	2	37
Woonsocket	35	0	0	35
State Total	1,046	438	11	1,495

Source: Rhode Island Department of Housing 2022 Integrated Housing Report

For simplification, those permits were combined into a county level summary. The following table indicates the number of building permits issues in 2022 (the latest full year data available), by county:

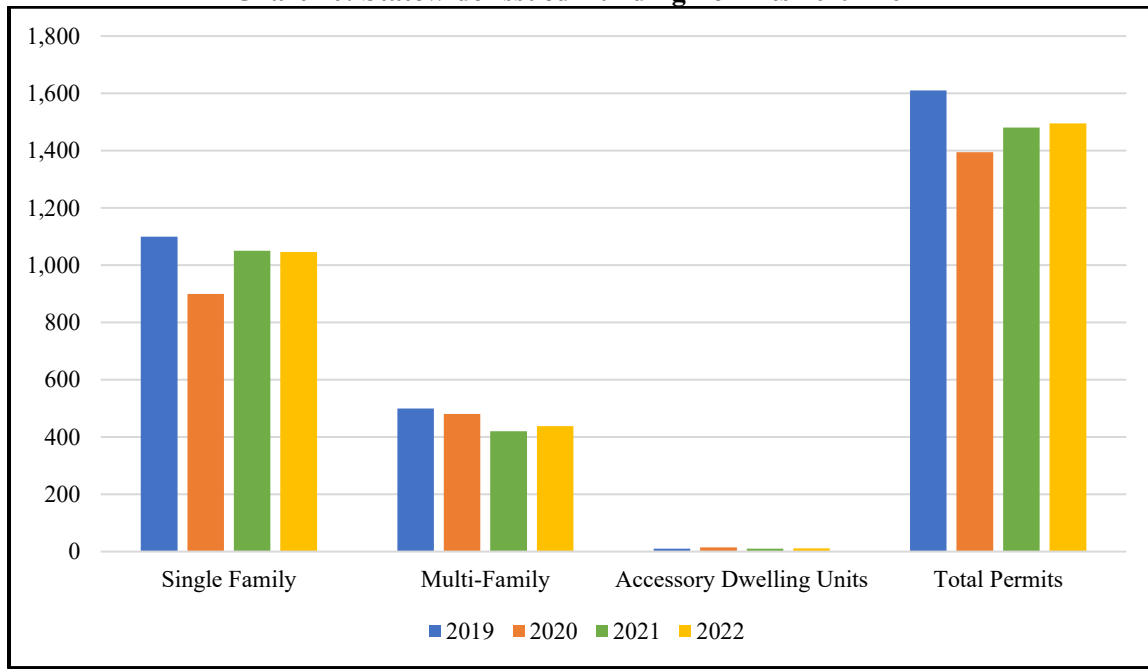
Table 18: Rhode Island 2022 Building Permits by County

Jurisdiction	Single Family	Two or More Family	Accessory Dwelling Unit	Total
Bristol County	107	0	2	109
Kent County	193	26	0	219
Newport County	113	114	0	227
Providence County	351	189	0	540
Washington County	282	109	9	400
State Total	1,046	438	11	1,495

Source: Rhode Island Department of Housing 2022 Integrated Housing Report

For comparison, the following graph from the Rhode Island Department of Housing indicates trends in issued building permits for the years 2019 to 2022:

Chart 10: Statewide Issued Building Permits 2019 -2022



Source: Rhode Island Department of Housing 2022 Integrated Housing Report

Detailed information concerning land use trends may be found in the Rhode Island State Land Use Policies and Plan, Land Use 2025, State Guide Plan, Element 121, by the Rhode Island Department of Administration Division of Planning Statewide Planning Program. This document, the most recent available, indicates the following trends for land development and use in Rhode Island:

- Rhode Island’s use of land in the past few decades has set a trajectory that cannot be sustained.
- Current, continued development is characterized by the diffuse, low density land use pattern described by planners as sprawl.
- Spawl development continues in spite of major changes to the State’s planning enabling legislation that calls for detailed local comprehensive plans and land management regulations.

On ground observation suggests that Rhode Island’s use of land in the past decade continues to follow land use patterns described by planners as sprawl. On average, about 30% of the land identified as undeveloped in 1995 has been built upon over the last ten years. In some of the State’s more rapidly developing communities, this recent building activity has consumed as much as 75% of the developable land that was identified as vacant in 1995. Based upon tracking of recent major capital investments by the Economic Development Corporation, it is anticipated that these development trends will continue.

3.8 Agricultural Data

Agriculture forms a very important part of both the economic and social fabric of Rhode Island. USDA National Agricultural Statistics Service data from 2007, 2012, and 2017 (the latest available data) was used to develop agricultural information for the State, as detailed in the following table:

Table 19: Rhode Island Agricultural Data

Jurisdiction	Year	Number of Farms	Farm Acreage	Market Value of Products Sold
Bristol County	2007	51	1,665	\$2,831,000
	2012	42	-	\$2,669,000
	2017	40	1,331	\$1,005,000
Kent County	2007	143	9,673	\$3,764,000
	2012	126	-	\$4,353,000

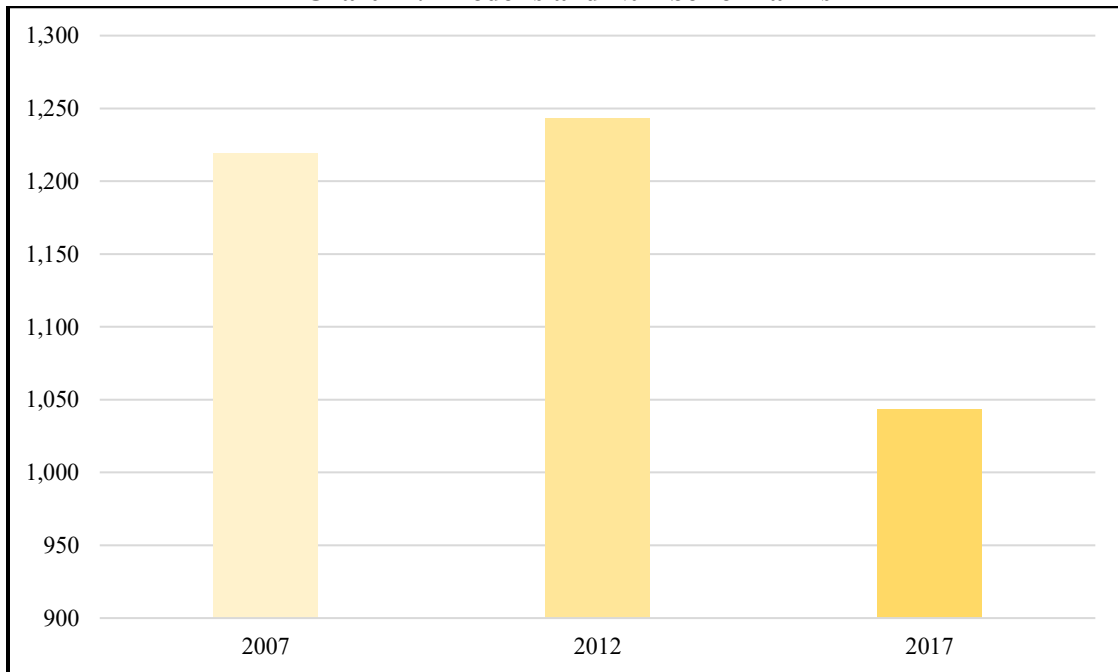
Table 19: Rhode Island Agricultural Data

Jurisdiction	Year	Number of Farms	Farm Acreage	Market Value of Products Sold
	2017	111	9,626	\$3,092,000
Newport County	2007	187	10,147	\$18,299,000
	2012	214	11,559	\$14,630,000
	2017	196	9,713	\$19,280,000
Providence County	2007	469	21,683	\$15,072,000
	2012	425	-	\$14,079,000
	2017	377	16,328	\$12,432,000
Washington County	2007	369	24,651	\$25,941,000
	2012	436	27,305	\$23,921,000
	2017	319	19,866	\$22,190,000
State Rhode Island	2007	1,219	67,819	\$65,908,000
	2012	1,243	69,589	\$59,652,000
	2017	1,043	56,864	\$57,998,000

Source: USDA National Agricultural Statistics Service

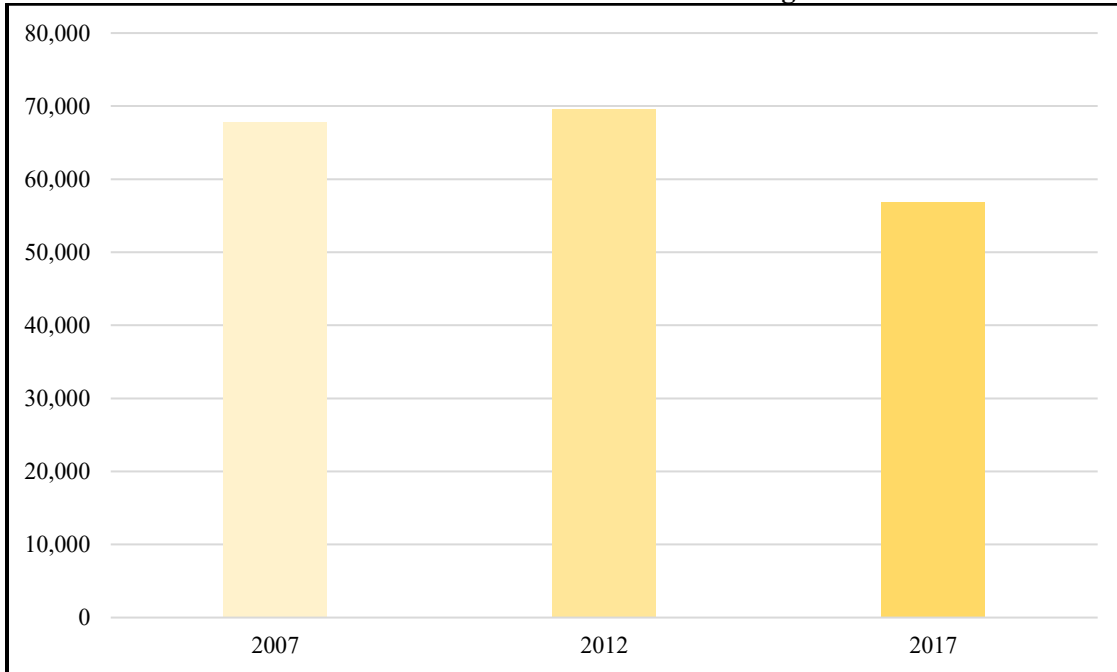
-: Data unavailable

As indicated by the data, Rhode Island has seen a decrease in the number farms, farm acreage, and the market value of products sold from 2007 to 2017. The following charts detail information from the preceding table:

Chart 11: Rhode Island Number of Farms

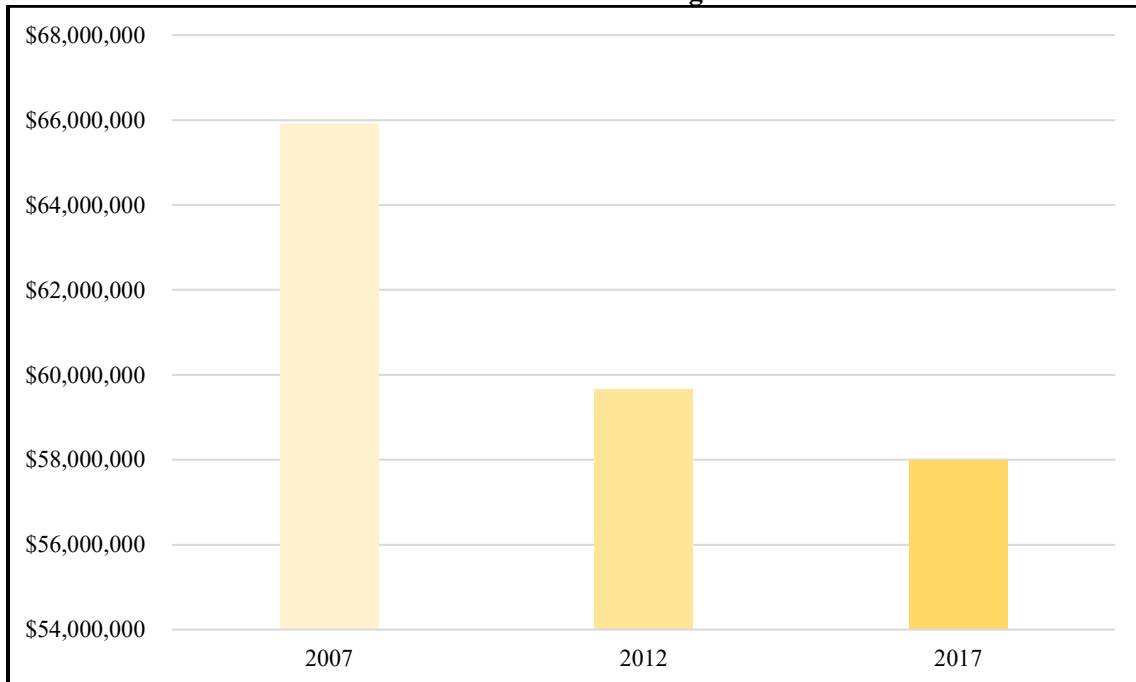
Source: USDA National Agricultural Statistics Service

Chart 12: Rhode Island Farm Acreage



Source: USDA National Agricultural Statistics Service

Chart 13: Rhode Island Market Value of Agricultural Products Sold



Source: USDA National Agricultural Statistics Service

3.9 Statewide Economic Activity Patterns

According to the 2022 Overview of the Rhode Island Economy report, completed by the Executive Office of Commerce, the economy rebounded with strength in 2021 after the COVID-related downturn in 2020. The state's economy grew in 2021 with the 20th highest Gross Domestic Product growth rate in the country. At the same time, the unemployment rate continued to decline in 2021, falling from a pandemic peak of 18.4% in April 2020 to 4.5% by December 2021. Per capita personal income and weekly earnings in Rhode Island continued to grow, and the state's housing market experienced a strong year, with an increase in total units authorized by permits as well as median single-family home prices. However, housing construction remains below historical levels, and housing affordability remains a challenge.

International trade also rebounded in 2021, with increases in Rhode Island's exports and imports as the global economy reopened. Finally, the state's population grew by 4.2% between 2010 and 2020.

3.10 Potential Impacts of Climate Change

There is a scientific consensus that climate change is occurring, and recent climate modeling results indicate that extreme weather events may become more common. Rising average temperatures produce a more variable climate system which may result in an increase in the frequency and severity of some extreme weather events including longer and hotter heat waves (and by correlation, an increased risk of wildfires), higher wind speeds, greater rainfall intensity, and increased tornado activity. Where applicable, and with proper scientific evidence, potential climate change factors will be addressed in subsequent sections for relevant identified hazards.

Climate change management and initiatives in Rhode Island are directed by the Executive Climate Change Coordinating Council (EC4). Members of the EC4 include CRMC, Department of Administration, RIDEM, RIDOH, RIDOT, Division of Planning, RIEMA, Office of Health and Human Services, RIIB, Office of Energy Resources, Rhode Island Public Transit Authority, Division of Public Utilities and Carriers, and Rhode Island Commerce.

As part of this, the Rhode Island EC4 Advisory Board was established pursuant to the Resilient Rhode Island Act of 2014. The EC4 Advisory Board has thirteen full members, appointed by the Governor, House and Senate. Members include municipal representatives as well as those with expertise in economic policy, workforce development, protection of natural/cultural resources, energy planning, education, public health, and housing.

According to the Rhode Island Executive Climate Change Coordinating Council (EC4) 2022 Climate Update report, the impacts of climate change upon Rhode Island's built and natural environments are wide-ranging, discernible, and documented, and, in many cases growing in severity. Rhode Island will experience warmer air and water temperatures, more extreme weather events such as droughts, intense precipitation, severe storms, and flooding, increasing rates of sea level rise, shorter winters and longer summers, and less snowfall and ice coverage. Climate change has the potential to pose significant risks for Rhode Island's water, wastewater, surface transportation, and energy infrastructures and utilities, our natural environment, and our health, welfare, and economic well-being.

Data from the NOAA National Centers for Environmental Information Rhode Island 2022 State Climate Summary indicates the following concerning the climate change in the state:

- Temperatures have risen approximately 4° Fahrenheit since the beginning of the 20th century.
- An increase in the intensity of future heat waves is projected.
- A decrease in the number of cold waves is projected.
- Annual precipitation has increased since 1895.
- Extreme precipitation events have increased since 1950.
- Continued increase in both the frequency and intensity of precipitation are predicted.
- Sea level has risen more than nine inches in Newport since 1930, faster than the global average.
- Sea levels are projected to rise in the state from one to four feet by 2100.
- Sea level rise will likely increase both coastal flooding and erosion, especially during tropical and extratropical storm events.

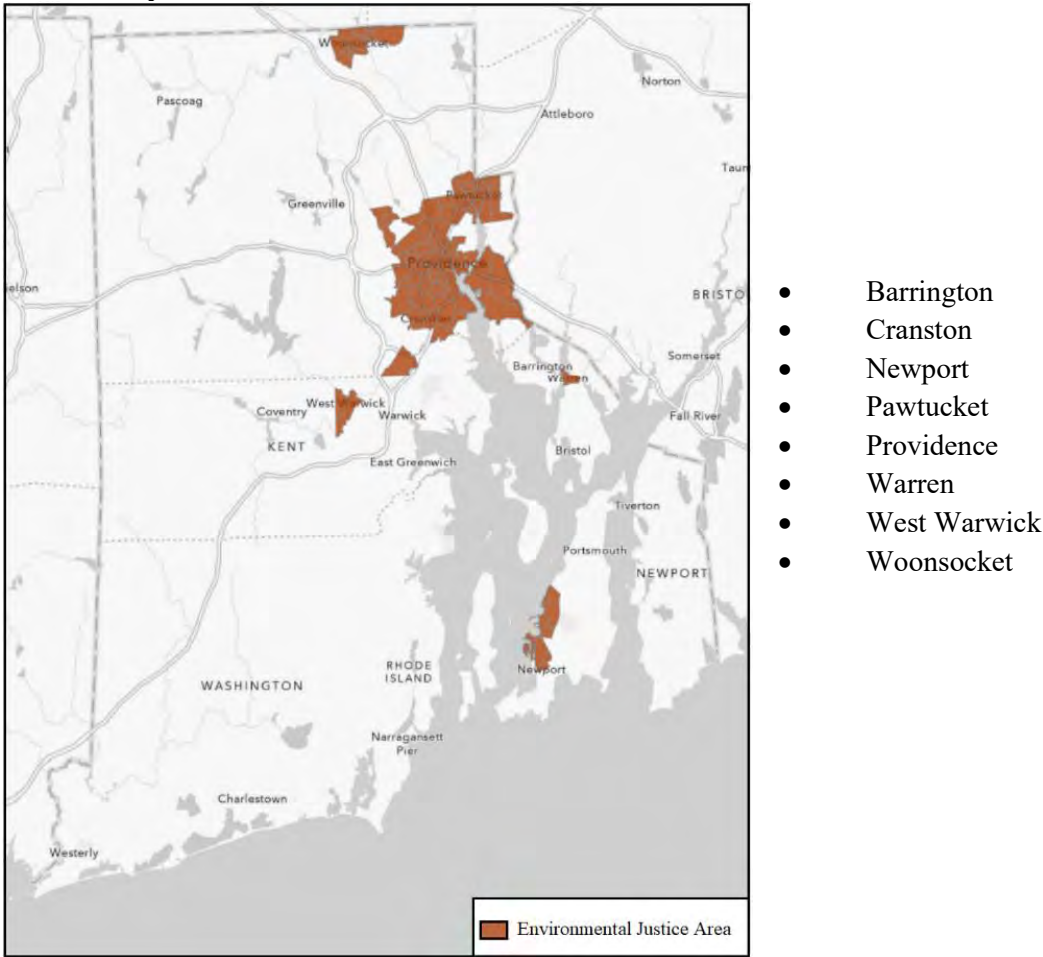
RIDEM has developed an environmental justice policy to help guide all programs. This environmental justice policy has been created to represent the commitment to the inclusion of equity and justice within all programs. RIDEM intends to use data, resources, and information to provide ample opportunity for members in environmental justice focus areas to influence policy that will advance environmental justice for all by implementing and enforcing environmental laws and regulations, preventing pollution, addressing climate change and its effects, and working to clean up legacy pollution that is harming human health and the environment. A key element of supporting and serving communities is building capacity and supporting efforts for all communities to have meaningful involvement, providing people with a sense of collective ownership through recognition of community autonomy, and effectively participate in conversations and decisions about environmental issues of interest. These issues are wide ranging in nature and should be thought of to include access to the state's abundance of natural resources, not just protection from environmental harm. This will assist in problem solving and improve the natural and economic conditions across all of Rhode Island's neighborhoods.

As part of this effort, the RIDEM has delineated areas within Rhode Island as Environmental Justic Focus Areas. To be considered as one of these areas one of the following criteria must be met:

- Annual median household income is not more than 65% of the statewide annual median household income
- Minority population is equal to or greater than 40% of the population
- 25% or more of the households lack English language proficiency
- Minorities comprise more that 25% of the population and the annual median household income of the municipality in the proposed area does not exceed 150% of the statewide annual median household income

The following map, from RIDEM, indicates areas of the State classified as Environmental Justice Areas:

Map 2: RIDEM Environmental Justice Areas



Source: DEM

3.11 Statewide Actions to Combat Climate Change

As a coastal state, Rhode Island is particularly vulnerable to the impacts of climate change. The need to take concrete and measurable action to both reduce greenhouse gas emissions and to prepare for climate change impacts is clear and has been prioritized by the highest levels of government. The State of Rhode Island is committed to developing practical solutions to climate change through reductions in greenhouse gas emissions, making preparations for the effects of current and future changes in the state’s climate. The following charts indicate a full ten-year timeline of climate action taken by the State of Rhode Island:

Chart 14: Timeline of Climate Action In Rhode Island



Chart 14: Timeline of Climate Action In Rhode Island



The following detail some of the Executive actions listed in the chart above:

- On May 9th, 2023, the Governor signed Executive Order 23-06 directing any state agency with 15 or more employees to appoint a Lead by Example coordinator and to collaborate with the Lead by Example program in the Office of Energy Resources to lower state energy consumption. The order also sets targets for state agency fossil fuel emissions reductions, site energy use intensity reductions, use of zero-emission vehicles, and installation of electric vehicle charging stations.

- On May 23, 2023, the Governor signed Executive Order 23-07 directing the Department of Environmental Management to recruit and hire an individual to serve as the Rhode Island Chief Resilience Officer.
- On January 17, 2020, the Governor signed Executive Order 20-012 directing the Rhode Island Office of Energy Resources to conduct economic and energy market analysis and develop viable policy and programmatic pathways to meet one hundred percent (100%) of the state's electricity demand with renewable energy resources by 2030.

On April 14, 2021, the 2021 Act on Climate was signed into Rhode Island law. The Act sets enforceable climate emissions reduction mandates and updates the 2014 Resilient Rhode Island Act. As part of the act, the following emission mandates were established:

- 45% below 1990 levels by 2030
- 80% below 1990 levels by 2040
- Net-zero emissions by 2050

The EC4 is currently developing a plan to reduce climate emissions with the 2025 Climate Action Strategy. This strategy is the next step in the implementation of the Act on Climate which requires the state to incrementally reduce climate emissions to net-zero by 2050 and consists of three required reports, the Priority Climate Action Plan (2024), The Comprehensive Climate Action Plan (2025 Strategy) (2025), and the Status Report (2027). These plans and reports will be utilized and incorporated into the next iteration of the SHMP.

Section 4 – Hazard Identification and Risk Assessment

4.1 Introduction

The goal of this hazard mitigation is to reduce the future impacts of hazards, including deaths and injuries, property damage, and disruption to local and state economies, and to further reduce the amount of public and private funds spent to assist recovery. To complete this goal, hazard mitigation decision-making in this plan has been based on a robust risk assessment, completed to identify natural, human caused, and technological hazards that represent a risk to the State of Rhode Island. The following provide a definition of the risk assessment terms used during this assessment:

- **Hazard:** An act or phenomenon that has the potential to produce harm or other undesirable consequences to a person or thing.
- **Exposure:** The people, property, systems, or functions that could be lost to a hazard. Generally, exposure includes what lies in the area the hazard could affect.
- **Vulnerability:** Vulnerability is susceptibility to physical injury, harm, damage, or economic loss. It depends on an asset’s construction, contents, and economic value of its functions.
- **Risk:** A function of hazard, vulnerability, and exposure. It refers to the likelihood of an event resulting in an adverse condition that causes injury or damage.

In order to accomplish this assessment, all relevant natural, human caused and technological, potential vulnerabilities, and exposures were identified. As potential hazards, vulnerabilities, and exposure are identified the State of Rhode Island can continue to develop a strategy to identify and prioritize mitigation action to defend against these potential risks.

4.2 Identified Potential Hazards

The first step in developing a hazard assessment is to identify the hazards that have a reasonable risk of occurring. Proper identification allows for appropriate and well-planned action in order to mitigate the extent and cascading impacts of an incident. Furthermore, while not all disaster contingencies can be planned for, applying an all-hazards approach to the mitigation process does yield greater awareness and better preparedness for unforeseen hazard incidents overall.

The SIHMPC met to discuss previously identified hazards and deliberate on any changes or additions to the State hazard profile. A thorough and comprehensive revision of data for each hazard was completed as part of this plan update. Additionally, this plan has worked, as per FEMA recommendations, to merge similar hazards together with the aim of both simplifying the usage of the plan and reducing duplication of effort.

The SIHMPC confirmed 11 natural hazards that may impact the State Rhode Island, as listed below:

Table 20: State of Rhode Island Identified Natural Hazards

Hazard	Included in 2019 SHMP	Notes
Dam Failure	Yes	-
Drought	Yes	-
Earthquake	Yes	-
Extreme Temperatures	Yes	Combined extreme cold and extreme heat
Flood	Yes	-
Sea Level Rise	Yes	-
Severe Thunderstorms	Yes	Combined hail, lightning, and high and thunderstorm winds
Severe Winter Weather	Yes	-
Tornado	Yes	-
Tropical/Extratropical Storms	Yes	-
Wildfire	Yes	Renamed with greater focus on wildfires

The SIHMC confirmed eight human caused and technological hazards that may impact the State of Rhode Island, as listed below:

Table 21: State of Rhode Island Identified Human Caused and Technological Hazards

Hazard	Included in 2019 SHMP	Notes
Biological Incident	Yes	-
Civil Disorder	Yes	-
Cybersecurity Incident	Yes	-
Hazardous Materials Incident	Yes	Renamed from chemical incident
Infrastructure Failure	Yes	-
Radiological Incident	Yes	-
Terrorism	Yes	-
Transmissible Disease	Yes	Renamed from infectious disease

As in the 2019 SHMP, coastal erosion is no longer profiled as an independent hazard and is discussed in various hazard profiles as an additional impact.

4.3 Hazard Planning Significance

For the purposes of this plan, hazard planning significance refers to the relevance of the identified hazard to the State of Rhode Island when calculating risk and vulnerability. In order to help quantify the planning significance for a hazard, data was reviewed on three levels, federal (National Risk Index data), state (researched plan data relevant to occurrence and vulnerability on a state level), and local (local hazard mitigation plans). This allowed for a comparison between data sets for each hazard type and allowed for a summation at the state level. It is recognized that inconsistencies in methodologies and data make it difficult to make a direct comparison across all data levels. However, as possible, collected data was translated into a unified model that accounted for any variability in data and methodologies.

The result of this assessment provides a larger scale snapshot of how the State of Rhode Island views risk and allowed for integration of hazard data in to the SHMP. The levels of planning significance were identified, as follows:

- Low: Hazards that have a low rate of historical occurrence and/or a low amount of documented damage.
- Moderate: Hazards that have a sporadic rate of historical occurrence and/or cause a lower number of documented damages. Additionally, moderate hazards have the potential to become more frequent or impactful due to climate change.
- High: Hazards that have a high rate of historical occurrence and/or cause a high amount of documented damage.

The following table presents a summation of the planning significance for all natural hazards:

Table 22: State of Rhode Island Natural Hazard Planning Significance Index

Hazard	Planning Significant
Dam Failure	Moderate
Drought	Moderate
Earthquake	Low
Extreme Temperatures	Moderate
Flood	High
Sea Level Rise	Moderate
Severe Thunderstorms	Moderate
Severe Winter Weather	High
Tornado	Low
Tropical/Extratropical Storms	High
Wildfire	Moderate

The large majority of local hazard mitigation plans reviewed did not analyze technological or human caused hazards. As such, a summation of a review of local level plan ratings could not be completed. Additionally, the FEMA NRI

database does not include human caused or technological hazards. As such, a review of all applicable current data concerning recurrence intervals, potential increases in intensity, duration, or frequency, potential impacts, and identified vulnerabilities was completed to determine the planning significance of human caused and technological hazards for the State of Rhode Island. The following table presents these ratings:

Table 23: State of Rhode Island Human Caused and Technological Hazard Planning Significance Index

Hazard	Planning Significant
Biological Incident	Low
Civil Disorder	Low
Cybersecurity Incident	High
Hazardous Materials Incident	Moderate
Infrastructure Failure	Moderate
Radiological Incident	Low
Terrorism	Low
Transmissible Disease	Moderate

Additionally, and in order to help codify potential vulnerability, state level community lifelines and assets are mapped against known hazards in Appendix D.

As in the 2019 SHMP, the coastal erosion hazard is considered under multiple causal hazards such as flood and sea level rise. Based on discussion with the SIHMC, a lack of identified risk or history, and/or geographic improbability, the FEMA identified hazards avalanche, landslide, and volcanic activity were not included in the scope of this plan.

4.4 Local Jurisdiction Hazard Rating and Vulnerability Methodology

In an effort to understand local vulnerability data and determine which jurisdictions are most threatened by the identified hazards and most vulnerable to damage and loss from hazard events with regards to populations, structures, and infrastructure, each approved local Hazard Mitigation Plan (LHMP) was researched to provide an overview of local hazard ratings. It is important to note that the LHMP hazard ratings take into account potential losses. In researching this data, it is recognized that inconsistencies in methodologies and data make it difficult to make a direct comparison across all plans. As such, the State of Rhode Island elected to use multiple tools to help understand local vulnerability and determine any potential inconsistencies in current understanding.

For each jurisdiction with an approved LHMP, data was sourced concerning the rating of each identified hazard in the plan. Their hazard rating was an overall assessment of jurisdictional risk, and is comprised of indices detailing vulnerability, probability of occurrence, vulnerable populations and critical facilities, and the potential magnitude and losses from an event. As there is no standardized methodology for the rating of hazard planning significance, a unified model was created to allow for comparison and comprehension. The model translated both numerical and verbal ratings into the following hazard planning significance categories:

- High: Hazards that were determined to be of the greatest concern to the jurisdiction, as expressed in the plan
- Moderate: Hazards that were determined to be of the moderate concern to the jurisdiction, as expressed in the plan
- Low: Hazards that were determined to be of the low concern to the jurisdiction, as expressed in the plan

Additionally, the State of Rhode Island has determined that utilizing FEMA’s National Risk Index (NRI) provides a method of understating high and local level jurisdictional vulnerability. FEMA’s NRI dataset and online tool was used to help determine local community risk for identified natural hazards in this HMP.

The risk equation behind the Risk Index includes three components, Expected Annual Loss (EAL), social vulnerability (previously discussed), and community resilience (previously discussed). The dataset supporting EAL provides estimates measured in 2022 U.S. dollars. The datasets supporting the social vulnerability and community resilience components have been standardized using a minimum-maximum normalization approach prior to being incorporated into the NRI risk calculation.

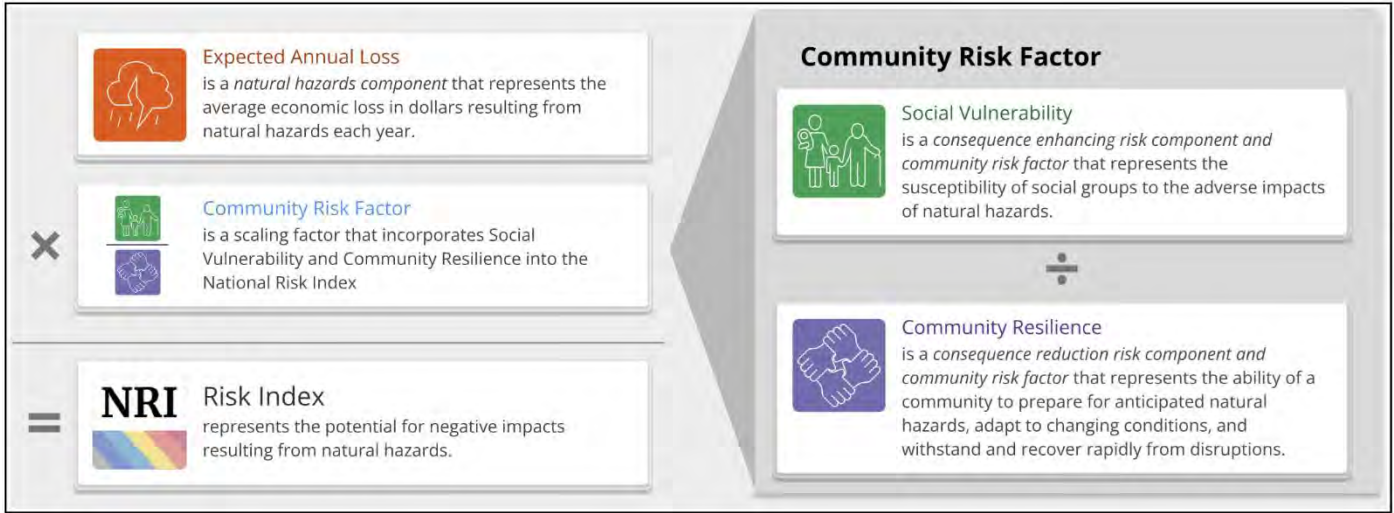
As part of the NRI, EAL represents the average economic loss in dollars resulting from a hazard each year. It quantifies loss for relevant consequence types, buildings, people, and agriculture. An EAL score and rating represent a community's relative level of expected losses each year when compared to all other communities at the same level. EAL is calculated using an equation that includes exposure, annualized frequency, and historic loss ratio risk factors. Exposure is a factor that measures the building value, population, and agriculture value potentially exposed to a natural hazard occurrence. Annualized frequency is a factor that measures the expected frequency or probability of a hazard occurrence per year. Historic loss ratio is a factor that measures the percentage of the exposed consequence type value (building, population, or agriculture) expected to be lost due to an occurrence. EAL represents the average economic loss in dollars resulting from natural hazards each year and is proportional to a community's risk.

To calculate Risk Index values, the NRI generates a Community Risk Adjustment to scale EAL values up or down, depending on their community risk factors, increasing with social vulnerability and decreases with community resilience. For a jurisdiction, a higher social vulnerability results in a higher Risk Index value while higher community resilience results in a lower Risk Index value.

Using these three components, Risk Index values are calculated for each jurisdiction (county and Census tract). The calculated Risk Index values form an absolute basis for measuring Risk within the NRI, and they are used to generate Risk Index percentiles and ratings across communities.

The risk equation behind the NRI is as follows:

Figure 1: FEMA NRI



For both the Risk Index and EAL there is a qualitative rating that describes the nature of a community's score in comparison to all other communities at the same level, ranging from "Very Low" to "Very High." Because all ratings are relative, there are no specific numeric values that determine the rating.

Additionally, jurisdiction level community lifelines and critical facilities are mapped against known hazards in Appendix D.

4.5 Declared Federal Disasters

The Robert T. Stafford Disaster Relief and Emergency Assistance Act (42 U.S.C. §§ 5121-5206) provides for the Federal support of State and local governments and their citizens when impacted by an overwhelming disaster. The Robert T. Stafford Disaster Relief and Emergency Assistance Act, as amended, establishes the process for requesting a Presidential Disaster Declaration and defines the type of assistance available.

If it is apparent that a Presidential Disaster Declaration may be necessary to assist in the recovery of an impacted area, Rhode Island and FEMA Regional I will conduct a Preliminary Damage Assessment (PDA). This assessment is used to determine:

- The extent of the event.
- The impact of the event on individuals and public facilities.
- The types of federal assistance that may be needed.

Once the PDA is complete, and if a determination is made that the damages exceed available State of Rhode Island resources, the Governor may submit through FEMA Region I a declaration request to the President.

A major disaster declaration provides a wide range of federal assistance programs for individuals and public infrastructure, including funds for both emergency and permanent work. Not all programs, however, are activated for every disaster. The determination of which programs are authorized is based on the types of assistance specified in the Governor's request and the needs identified during the initial and subsequent PDAs. FEMA disaster assistance programs may include:

- Individual Assistance
- Public Assistance
- Hazard Mitigation

To recognize and encourage mitigation, FEMA considers the extent to which mitigation measures contributed to the reduction of disaster damages. This could be especially significant in those disasters where, because of mitigation, the estimated public assistance damages fell below the per capita indicator.

Historical events of significant magnitude or impact can result in a Presidential Disaster Declaration. The SIHMC reviewed the historical federal disaster declarations to assist in hazard identification. The following table details Disaster Declarations for the State of Rhode Island.

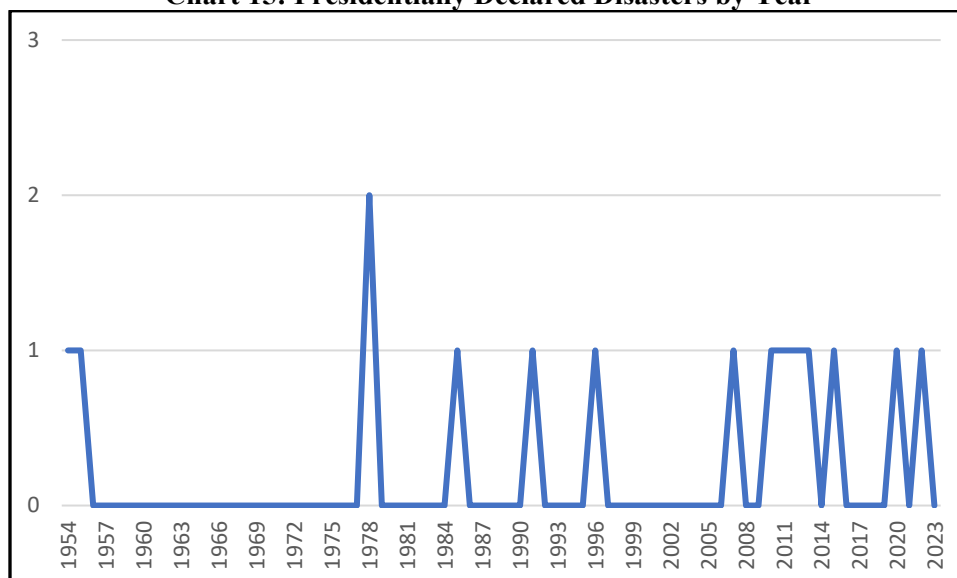
Table 24: State of Rhode Island Presidentially Declared Disasters

Designation	Declaration Date	Incident Type	Individual Assistance	Public Assistance
DR-4653-RI	05/12/2022	Severe Winter Storm and Snowstorm	-	\$8,901,805
DR-4505-RI	03/30/2020	COVID-19 Pandemic	\$11,381,189	\$763,887,593
DR-4212-RI	4/3/2015	Severe Winter Storm and Snowstorm	\$8,897,067	\$1,327,144
DR-4107-RI	3/22/2013	Severe Winter Storm and Snowstorm	\$7,929,498	\$1,070,561
DR-4089-RI	11/3/2012	Hurricane Sandy	\$421,341	\$8,629,001
DR-4027-RI	9/3/2011	Tropical Storm Irene	\$8,341,293	\$1,608,921
DR-1894-RI	3/29/2010	Severe Storms and Flooding	\$37,157,115	\$24,263,849
DR-1704-RI	5/25/2007	Severe Storms and Island/Coastal Flooding	\$605,080	\$529,557
DR-1091-RI	01/24/1996	Blizzard	-	-
DR-913-RI	08/26/1991	Hurricane Bob	-	-
DR-748-RI	10/15/1985	Hurricane Gloria	-	-
DR-3058-RI	02/07/1978	Blizzards and Snowstorms	-	-
DR-548-RI	02/16/1978	Snow and Ice	-	-
DR-39-RI	08/20/1955	Hurricane and Flood	-	-
DR-23-RI	09/02/1954	Hurricane	-	-

Source: FEMA

Note: -: Data unavailable

The following chart represents Presidentially Declared Disasters in the State of Rhode Island by year, starting in 1954:

Chart 15: Presidentially Declared Disasters by Year

Source: FEMA

The President can declare an emergency for any occasion or instance when the President determines federal assistance is needed. Emergency Declarations supplement State and local or Indian tribal government efforts in providing emergency services, such as the protection of lives, property, public health, and safety, or to lessen or avert the threat of a catastrophe. The total amount of assistance provided for in a single emergency may not exceed \$5,000,000. The following types of assistance are available under an Emergency Declaration:

- Public Assistance, Categories A (debris removal) and B (emergency protective measures)
- Individual Assistance, the Individuals and Households Program

The SIHMC reviewed the historical federal disaster declarations to assist in hazard identification. The following table details Emergency Declarations for the State of Rhode Island.

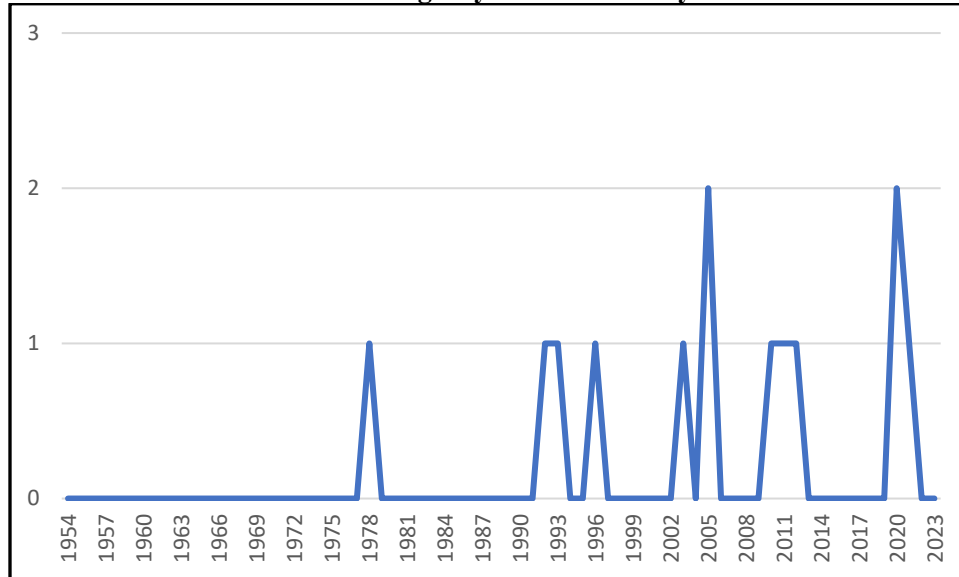
Table 25: State of Rhode Island Emergency Declarations

Designation	Declaration Date	Incident Type	Public Assistance
EM-3563-RI	08/21/2021	Hurricane Henri	-
EM-3440-RI	03/13/2020	Rhode Island COVID-19	-
EM-3515-RI	03/13/2020	Narragansett Indian Tribe COVID-19	-
EM-3355-RI	08/29/2012	Hurricane Sandy	-
EM-3334-RI	8/27/2011	Hurricane Irene	-
EM-3311-RI	3/30/2010	Severe Storms and Flooding	
EM-3255-RI	9/19/2005	Hurricane Katrina Evacuation	\$1,110,010
EM-3203-RI	2/17/2005	Record Snow	\$6,273,609
EM-3182-RI	3/27/2003	Snowstorm	\$2,002,984
EM-3123-RI	11/19/1996	Major Water Main Break	\$1,539,100
EM-3102-RI	03/16/1993	Blizzards, High Winds, and Record Snowfall	\$167,897
EM-3094-RI	09/16/1992	Water Contamination	\$1,994,343
EM-3058-RI	02/07/1978	Blizzards and Snowstorms	-

Source: FEMA

Note: -: Data unavailable

The following chart represents Emergency Declarations in the State of Rhode Island by year, starting in 1954:

Chart 16: Emergency Declarations by Year

Source: FEMA

The Governor, or the Governor's Authorized Representative, may submit a request for a fire management assistance declaration as required. FEMA will approve declarations for fire management assistance when it is determined that a fire or fire complex on public or private forest land or grassland threatens such destruction as would constitute a major disaster.

The SIHMC reviewed the historical fire management declarations to assist in hazard identification. The following table details fire management declarations for the State of Rhode Island.

Table 26: State of Rhode Island Fire Management Declarations

Designation	Declaration Date	Incident Type
FM-5464-RI	04/14/2023	Brushfire

Source: FEMA

A state of emergency can be declared by executive order or proclamation of the governor if he or she finds a disaster has occurred, or the threat thereof, is imminent. The state of disaster emergency shall continue until the governor finds that the threat or danger has passed, or the disaster has been dealt with to the extent that emergency conditions no longer exist and terminates the state of disaster emergency by executive order or proclamation. No state of disaster emergency may continue for longer than 30 days unless renewed by the governor.

Table 27: State of Rhode Governor's Executive Order Declarations of Disaster Emergency

Executive Order	Declaration Date	Incident Type
23-08	06/13/2023	Failure of Cliff Walk in Newport
22-10	01/29/2022	Winter Storm Consisting of High Winds, Heavy Snow, and Blizzard Conditions
21-109	10/12/2021	New COVID-19 variants (extended)
21-88	08/20/2021	Hurricane Henri
21-86	08/19/2021	New COVID-19 Variants (extended)
20-02	03/09/2020	COVID-19 (extended)
19-01	01/21/2019	Disruption of Low-Pressure Gas Line Service
15-02	01/26/2015	Winter Storm Consisting of High Winds, Heavy Snow, and Blizzard Conditions

Source: FEMA

4.6 Assessment Data

NOAA's National Centers for Environmental Information (NCEI) Storm Events Database was used as the primary source of information for previous occurrences of storm events. It is worth noting that damage estimates indicated by the NCEI are often artificially low. This underreporting is a result of the way the events are reported to the NCEI, often by the local and/or National Weather Service (NWS) office. When reporting an event oftentimes the NWS office does not have access to the actual damage assessment resulting from that event. As such, the report often details a very low amount or zero-dollar amount for damages. Most of the events from NCEI are not associated with a federal emergency or disaster. If the event occurred at the same time as an event that was later determined to be a federal emergency or disaster, it is included with the NCEI data even if it occurred in a county not included in the federal declaration.

Environmental Systems Research Institute ArcGIS v10 was used to determine which critical and state owned or operated facilities were located within the boundaries of identified hazards (when applicable, and if data was available). Data was provided by the following agencies:

- Rhode Island Geographical Information System (RIGIS)
- RIEMA
- FEMA's National Flood Hazard Layer
- University of Wisconsin–Madison Spatial Analysis for Conservation and Sustainability Silvis Lab
- NOAA

Data was also obtained and utilized using Hazus-MH, Version 2.2 SP1, a program administered by the FEMA used to model losses. Modelling for hazards uses Hazus analysis to estimate losses and projected impacts from historical and annualized hazard events. Hazus default data was used in the analysis, including the 2020 Census and other State and Federal government facility databases. A level I analysis was run in Hazus for flood, hurricane, and earthquake, meaning the default population, building stock, and critical infrastructure data within the program was used to calculate losses and damages. Multiple hazard scenarios were run to estimate losses for the identified hazards. For the earthquake and hurricane hazards, historic event scenarios and probabilistic scenarios were run. Flood losses were analyzed using the 100 and 500 years return scenarios as well as a probabilistic scenario.

4.7 State Assets and Critical Facilities

Data from RIEMA and RIGIS was used to locate state assets, defined as state-owned or operated buildings, infrastructure, community lifelines, and critical facilities throughout the state. It is important to note that the data provided included both state-owned or operated critical facilities and non-state owned or operated critical facilities. Based on this data, a complete mapping profile was created. The following table details the number of critical facilities by county in the State of Rhode Island:

Table 28: State of Rhode Critical Facilities

Jurisdiction	Hospitals	Fire Stations	Law Enforcement	Schools	Power Generation
Bristol County	0	12	3	34	1
Kent County	1	23	9	116	14
Newport County	2	12	9	65	6
Providence County	15	77	35	375	44
Washington County	2	36	14	87	18
State Totals	18	124	56	590	65

Source: RIEMA and RIGIS and Local Hazard Mitigation Plans

Data was also provided concerning all state-owned facilities by RIEMA and RIGIS. The following table details the complete inventory of assets, by city and by facility construction type, along with the most recent valuation for both the facility envelope and assets housed with the facility for the State of Rhode Island:

Table 29: State of Rhode Island Facilities and Assets by City

Jurisdiction	Brick	Concrete	Frame (Wood)	Manufactured	Masonry and Block	Metal	Not Defined	Stone	Valuation
Barrington	1	0	5	0	0	0	0	0	\$10,243,057
Bristol	0	0	16	1	1	0	2	0	\$132,214,526

Table 29: State of Rhode Island Facilities and Assets by City

Jurisdiction	Brick	Concrete	Frame (Wood)	Manufactured	Masonry and Block	Metal	Not Defined	Stone	Valuation
Burrillville	17	0	14	4	3	1	1	0	\$117,245,554
Central Falls	0	0	1	0	0	0	0	0	\$573,157
Charlestown	0	0	39	0	7	1	0	0	\$15,874,479
Coventry	0	0	17	0	1	0	1	0	\$23,173,062
Cranston	48	4	17	2	25	13	9	12	\$1,171,484,112
Cumberland	0	0	4	0	1	0	0	0	\$4,260,336
East Greenwich	0	0	5	0	1	1	0	0	\$11,440,248
East Providence	0	0	9	0	2	0	2	0	\$12,527,196
Exeter	1	0	22	1	6	1	3	0	\$24,912,017
Foster	0	0	2	0	2	2	0	0	\$2,842,796
Glocester	0	2	18	0	5	1	1	0	\$11,149,317
Hopkinton	0	0	6	0	2	0	1	0	\$7,115,927
Jamestown	0	4	4	0	5	0	0	0	\$8,973,197
Johnston	0	0	20	0	3	0	1	0	\$16,569,006
Lincoln	3	0	21	0	7	3	0	0	\$26,189,656
Little Compton	0	0	1	0	1	0	0	0	\$1,649,481
Middletown	1	0	9	0	1	1	0	0	\$14,184,428
Narragansett	2	9	145	0	13	0	1	0	\$71,207,712
New Shoreham	0	0	0	0	0	0	0	0	-
Newport	7	0	16	0	3	0	1	1	\$72,614,245
North Kingstown	2	1	16	0	12	0	2	0	\$94,726,030
North Providence	0	1	3	0	1	0	0	0	\$4,701,484
North Smithfield	1	0	6	0	0	0	0	0	\$7,590,441
Pawtucket	3	0	4	0	1	0	0	0	\$9,552,112
Portsmouth	3	0	16	1	1	0	0	0	\$23,105,003
Providence	38	3	14	0	16	1	3	3	\$990,545,842
Richmond	1	0	37	0	1	0	2	0	\$17,547,103
Scituate	4	0	7	0	7	0	1	0	\$117,595,341
Smithfield	0	0	12	0	0	0	0	0	\$9,940,534
South Kingstown	8	0	45	0	2	0	0	0	\$37,167,835
Tiverton	0	0		0	1	0	0	0	\$6,827,483
Warren	1	0	4	0	1	0	0	0	\$7,958,227
Warwick	2	0	33	0	31	1	0	0	\$153,277,534
Westerly	1	0	13	0	6	0	1	0	\$26,691,944
West Greenwich	0	0	1	0	1	0	0	0	\$443,826
West Warwick	1	0	1	0	2	0	0	0	\$5,799,647
Woonsocket	3	0	6	0	5	0	0	0	\$23,826,172
State Total	148	24	609	9	177	26	32	16	\$3,293,740,067

Source: RIEMA

Note: Valuations include structure and assets

For simplification, the following table details the complete inventory of assets, by county and by facility construction type, along with the most recent valuation for both the facility envelope and assets housed with the facility for the State of Rhode Island:

Table 30: State of Rhode Island Facilities and Assets by County

Jurisdiction	Brick	Concrete	Frame (Wood)	Manufactured	Masonry and Block	Metal	Not Defined	Stone	Valuation
Bristol County	2	0	25	1	2	0	2	0	\$150,415,810
Kent County	3	0	69	0	40	2	2	0	\$215,026,614
Newport County	11	4	46	1	12	1	1	1	\$127,353,837
Providence County	117	10	158	6	78	21	18	15	\$2,526,593,056
Washington County	15	10	311	1	45	2	9	0	\$274,350,750
State Total	148	24	609	9	177	26	32	16	\$3,293,740,067

Source: RIEMA

Note: Valuations include structure and assets

A comparison of state owned or operated facilities was conducted to determine if there have been any major changes to the inventory since the last SHMP was completed. Other than an increase in valuation, the number has remained steady with no major additions or subtractions.

The age of a structure could have a significant impact on its resilience to a hazard as an older structure may be less resilient due to the following factors:

- **Building Codes and Standards:** Older buildings may not have been constructed to meet modern building codes, which often incorporate lessons learned from previous disasters. Newer buildings are typically designed to be more resilient to known hazards.
- **Construction Materials and Techniques:** Older buildings may have been constructed using materials and techniques that are less resilient to hazards. Modern construction materials and methods, such as reinforced concrete and steel frames, are often more resilient to hazards.

The following table indicates the average year of construction for state-owned or operated facilities with a known completion date, by city, for the State of Rhode Island:

Table 31: Average Year of Construction for State of Rhode Island Owned or Operated Facilities by City

Jurisdiction	Average Year of Construction
Barrington	1957
Bristol	1973
Burrillville	1955
Central Falls	1928
Charlestown	1979
Coventry	1982
Cranston	1954
Cumberland	1988
East Greenwich	1976
East Providence	1959
Exeter	1967
Foster	1961
Glocester	1958
Hopkinton	1979
Jamestown	1959
Johnston	1966
Lincoln	1979
Little Compton	1934
Middletown	1969
Narragansett	1958
New Shoreham	-

Table 31: Average Year of Construction for State of Rhode Island Owned or Operated Facilities by City

Jurisdiction	Average Year of Construction
Newport	1904
North Kingstown	1973
North Providence	1962
North Smithfield	1983
Pawtucket	1961
Portsmouth	1957
Providence	1932
Richmond	1953
Scituate	1928
Smithfield	1983
South Kingstown	1983
Tiverton	1929
Warren	1569
Warwick	2004
Westerly	1972
West Greenwich	1972
West Warwick	1984
Woonsocket	1964
State Buildings Average Year of Construction	1969

Source: RIEMA

Note: Average age based on buildings with known construction date

For simplification, the following table indicates the average year of construction for state-owned or operated facilities with a known completion date, by county, for the State of Rhode Island:

Table 32: Average Year of Construction for State of Rhode Island Owned or Operated Facilities by County

Jurisdiction	Average Year of Construction
Bristol County	1966
Kent County	1983
Newport County	1978
Providence County	1961
Washington County	1964
State Buildings Average Year of Construction	1969

Source: RIEMA

Note: Average age based on buildings with known construction date

All critical facility maps, including mapping related to state assets, critical facilities, and community lifelines in identified hazards, are presented in Appendix D. Where appropriate, state owned or operated facilities are addressed per identified hazard.

4.8 Hazard Profiles

Each identified hazard is profiled in the subsequent sections, with the level of detail varying based on available information. Sources of information are cited in the detailed hazard profiles below.

With each update of this plan, new information will be incorporated to provide for better evaluation and prioritization of the hazards.

The following hazards are presented in alphabetical order, and not by planning significance, for ease of reference. Please note that natural hazards are presented in order first, followed by human caused and technological hazards.

4.9 Dam Failure

4.9.1 Hazard Description

A dam is a barrier across flowing water that obstructs, directs, or slows down the flow, often creating a reservoir, lake, or impoundment. Most dams have a section called a spillway or weir, over or through, which water flows, either intermittently or continuously. Dams commonly come in two types, embankment (the most common) and concrete (gravity, buttress, and arch), as well as sizes. They also serve a number of purposes and provide essential benefits, including drinking water, irrigation, hydropower, flood control, and recreation.

Large or small, dams have a powerful presence that is frequently overlooked until a failure occurs. Dams fail in two ways, a controlled spillway release done to prevent full failure, or the partial or complete collapse of the dam itself. In each instance, an overwhelming amount of water, and potentially debris, is released. Dam failures are rare, but when they do occur, they can cause loss of life and immense damage to property, critical infrastructure, and the environment.



Possible reasons for dam failure include but are not limited to:

- Overtopping caused by excess rainfall
- Sub-standard construction materials/techniques
- Spillway design error
- Geological instability caused by changes to water levels during filling or poor surveying
- Sliding of a mountain into the reservoir
- Poor maintenance, especially of outlet pipes
- Human, computer, or design error
- Internal erosion, especially in earthen dams
- Earthquakes
- Terrorism

There are three classifications of dam failure, hydraulic, seepage, and structural. The following is an explanation of each these failure classifications:

- **Hydraulic:** This failure is a result of an uncontrolled flow of water over and around the dam structure as well as the erosive action on the dam and its foundation. The uncontrolled flow causing the failure is often classified as wave action, toe erosion, or gullyng. Earthen dams are particularly susceptible to hydraulic failure because earthen materials erode more quickly than other materials, such as concrete and steel. This type of failure constitutes approximately 40% of all dam failures.
- **Seepage:** Seepage is the velocity of an amount of water controlled to prevent failure. This occurs when the seepage occurs through the structure to its foundation, where it begins to erode within. This type of failure accounts for approximately 4% of all dam failures.
- **Structural:** A failure that involves the rupture of the dam or the foundation by water movement, earthquake, or sabotage. When weak materials construct dams (large, earthen dams) are the primary cause of this failure. Structural failure occurs with approximately 30% of dam failures.

4.9.2 Location and Extent

The Rhode Island Department of Environmental Management (RIDEM) is responsible for causing dams to be inspected to determine their condition, reviewing and approving plans for construction or substantial alteration of a dam, and ordering repairs or other action to address unsafe conditions. To meet that mandate, RIDEM established the Dam Safety Program, which is housed in the Bureau of Environmental Protection within the Office of Compliance and Inspection.

There are 670 inventoried dams in Rhode Island, which are ranked by a Dam Hazard Classification system. The classifications are based on the downstream impacts expected to occur if the dam failed. In addition, dams are also rated by the Dam Safety Program during a visual inspection as to their condition, Good, Fair, Poor, Unsafe, or May Be Unsafe. Dam Hazard Classifications are detailed in the following table:

Table 33: Dam Hazard Potential Classification

Hazard Potential	Definition	Inspection Timeline	Number of Dams in Category	Number of Unsafe Dams in Category
High	Failure or mis-operation will result in a probable loss of human life.	Two Years	95	32
Significant	Failure or mis-operation results in no probable loss of human life but can cause major economic loss, disruption of lifeline facilities or impact other concerns detrimental to the public's health, safety, or welfare.	Five Years	81	35
Low	Failure or mis-operation results in no probable loss of human life and low economic losses.	Downstream conditions reassessed to determine if reclassification of dam is necessary	494	Not Inspected

Source: Rhode Island Department of Environmental Management

The following table represents a breakdown of identified dams within the state by county and jurisdiction.

Table 34: State of Rhode Island Dams

County	Jurisdiction	High	Significant	Low	Total
Bristol	Barrington	0	0	2	2
	Bristol	0	1	2	3
	Warren	0	2	2	4
Kent	Coventry	6	6	49	61
	East Greenwich	0	1	13	14
	East Greenwich/North Kingstown	0	0	1	1
	Warwick	2	1	15	18
	West Greenwich	0	0	27	27
	West Warwick	4	1	8	13
Newport	Jamestown	0	2	4	6
	Little Compton	1	2	8	11
	Middletown	3	0	3	6
	Newport/Middletown	1	0	0	1
	Portsmouth	3	1	4	8
	Tiverton	1	2	1	4
Providence	Burrillville	5	6	30	41
	Central Falls/Cumberland	0	0	1	1
	Central Falls/Pawtucket	0	0	1	1
	Cranston	5	1	17	23
	Cranston/Warwick	0	0	1	1
	Cumberland	3	3	11	17
	Cumberland/Lincoln	0	0	4	4
	East Providence	2	0	8	10

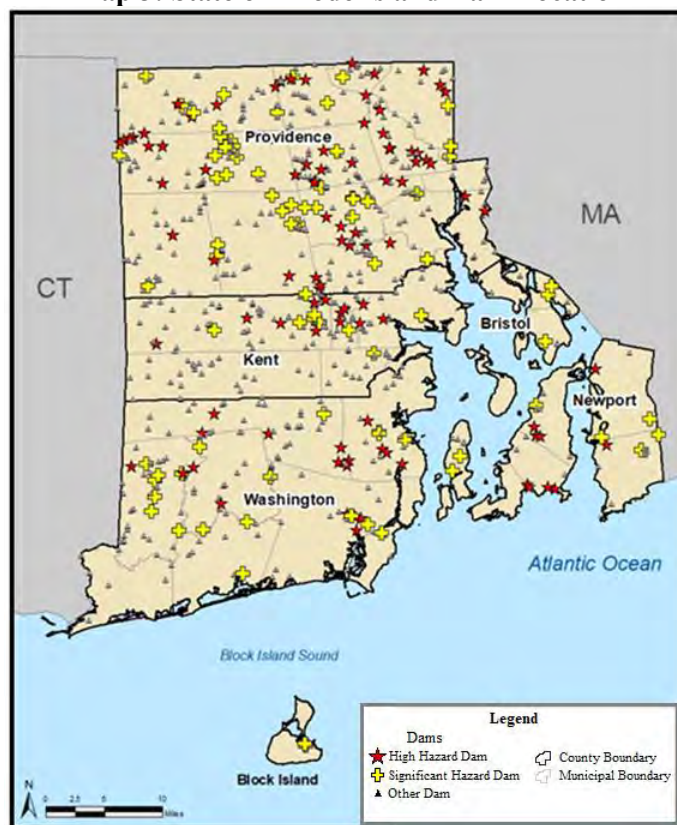
Table 34: State of Rhode Island Dams

County	Jurisdiction	High	Significant	Low	Total
	Foster	2	1	23	26
	Foster/Scituate	0	0	1	1
	Glocester	8	10	32	50
	Glocester/Smithfield	1	0	0	1
	Johnston	5	5	10	20
	Johnston/North Providence	0	0	2	2
	Johnston/Providence	0	0	1	1
	Lincoln	9	0	10	19
	Lincoln/North Smithfield	1	0	0	1
	North Providence	2	0	4	6
	North Providence/Johnston	0	0	1	1
	North Smithfield	3	2	15	20
	North Smithfield/Smithfield	1	0	0	1
	Pawtucket	0	0	5	5
	Providence	0	2	7	9
	Scituate	1	8	18	27
	Smithfield	6	2	22	30
	Woonsocket	2	1	10	13
Washington	Charlestown	0	1	6	7
	Charlestown/Richmond	0	1	2	3
	Charlestown/Hopkinton	0	0	1	1
	Exeter	5	1	30	36
	Hopkinton	2	6	26	34
	Hopkinton/Richmond	1	2	2	5
	Hopkinton/Westerly	0	0	1	1
	Narragansett	0	1	4	5
	New Shoreham	0	2	0	2
	North Kingstown	6	3	13	22
	Richmond	1	1	10	12
	South Kingstown	4	3	17	24
	Westerly	0	0	4	4
	Westerly/CT	0	0	1	1

Source: RIEMA

The following map details the locations of identified State of Rhode Island dams.

Map 3: State of Rhode Island Dam Location



Source: RIEMA

According to the Federal Guidelines for Dam Safety, two factors influence the severity of a dam failure, the amount of water impounded by the dam and the density, type, and value of development and infrastructure located downstream. Most dam failures are described by their potential for area inundation. The magnitude and severity of dam failures may be assessed in terms of damage, including loss of life and injury, damage to infrastructure, loss of services or utilities, debris, and economic damage.

4.9.3 Previous Occurrences

Data from the National Performance of Dams Program at Stanford University indicates that Rhode Island has had 109 dam incidents listed in the reporting database. The reported causes of these incidents have varied and include lack of maintenance (bush or tree growth), erosion, inflow flood, piping failure, inoperable gates, and settlement. Many of these reported incidents were minor, resulting in no reported deaths, injuries, or damage to property. However there have been four events that have resulted in the uncontrolled release of the reservoir, and one that resulted in the loss of three lives, as detailed below:

Table 35: State of Rhode Island Dam Incidents

Dam Name	Incident Type	Uncontrolled Release of Reservoir	Year	Deaths
Spring Lake	Piping	Yes	1889	3
Randall's Pond	Inflow Flood - Hydrologic Event	Yes	1901	None
Peace Dale Pond	Inflow Flood - Hydrologic Event	Yes	1998	None
Sweet's Mill	Lack of Maintenance and Embankment Erosion	Yes	2002	None

Source: Stanford University National Performance of Dams Program

Additionally, RIDEM records indicate that in 2010 there were five dam failures due to excessive rain, two significant hazard dams (Blue Pond and Geneva Ponds), and three low hazard dams. No injuries or fatalities were reported.

4.9.4 Probability of Future Incidents

Despite the infrequent historical occurrences of dam failure resulting in an uncontrolled release of the reservoir, there remains a significant concern due to the large number of significant and high hazard dams throughout the state. The probability of dam failure events is not easily measured, but may align with:

- The probability of future flood events
- Preventative measure taken by dam owners and operators, maintenance, and repair
- Frequent condition inspections
- Proper operating procedures

RIDEM's Dam Safety Program conducts routine monitoring and inspection of dams within the state on the previously identified schedule, with priority placed on those dams which pose the greatest potential threat. However, to fully determine the probability of a future event, a full engineering inspection would need to be completed on each dam, something beyond the scope of this plan.

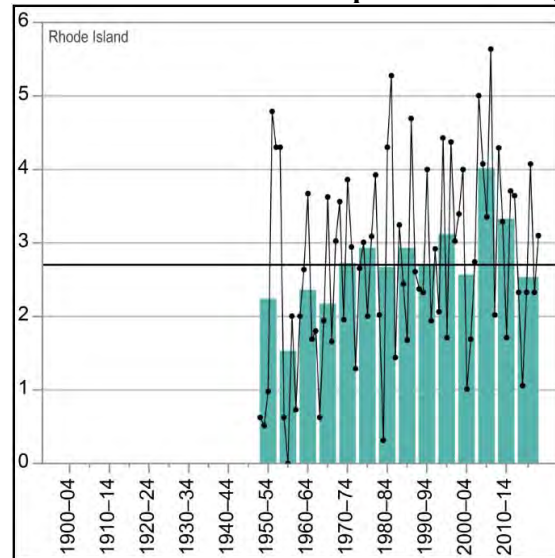
Dams undergoing repair and/or reconstruction are required to be designed to pass at least the 1%-annual-chance rainfall event with one foot of freeboard. The most critical and hazardous dams are required to meet a spillway design standard much higher than passing the runoff from a 1%-annual-chance rainfall event. Although not all the dams have been shown to withstand the 1%-annual-chance rainfall event, most of the dams meet this standard due to original design requirements or recent spillway upgrades.

4.9.5 Projected Changes in Hazard Location, Intensity, Frequency, and Duration

The 2018 National Climate Assessment report indicates that much of the water infrastructure in the northeast portion of the United States, including dams, is nearing the end of its planned life expectancy. As indicated in the report: "Aging and deteriorating dams and levees also represent an increasing hazard when exposed to extreme or, in some cases, even moderate rainfall. Several recent heavy rainfall events have led to dam, levee, or critical infrastructure failures, including the Oroville emergency spillway in California in 2017, Missouri River levees in 2017, 50 dams in South Carolina in October 2015 and 25 more dams in the state in October 2016, and New Orleans levees in 2005 and 2015. The national exposure to this risk has not yet been fully assessed."

A potential outcome of changing climate in Rhode Island is an increase in extreme precipitation events which may lead to more severe floods and a greater risk of dam failure. Additional projected greater periods of drought conditions and high heat may result in ground cracking, a reduction of soil strength, erosion, and subsidence in earthen dams.

At present there is no comprehensive assessment of the climate-related vulnerability and risks to existing dams. Additionally, there are no common design standards concerning the repair or modification of existing dams nor for the designed and construction of new dams operated in the face of changing climate risk. The NOAA NCEI State Climate Summary 2022 for Rhode Island suggests that the number of extreme precipitation events are projected to increase. These extreme events will likely place increased stress on dams within the State.

Chart 17: Rhode Island Number of Extreme Precipitation Events (Greater Than 2 Inches)

Source: NOAA State Climate Summary 2022 for Rhode Island

4.9.6 Vulnerability and Impact

RIDEM makes available dam safety inspection reports for review. These reports provide a general description of downstream conditions and potential failure impacts. However, a GIS analysis has not been completed to quantify the impacts of a high or significant dam failure on state assets, populations, infrastructure, or housing. As a result, dam vulnerability analysis was not possible for this SHMP. As part of an ongoing project to address this deficiency, RIDEM and RIEMA are working to compile and disseminate relevant information concerning potential vulnerabilities to assist both state and local planners with fully understanding risk. It is anticipated that this data will be available for the next iteration of the SHMP.

The following table presents identified conditions below identified high hazard dams, including number of identified bridges (and roads), occupancy characteristics, and potentially impacted downstream dams from RIDEM dam safety inspections. These conditions have not been mapped using GIS to determine vulnerable population or facility values, and are only observational:

Table 36: State of Rhode Island High Hazard Dams Observed Downstream Conditions

County	Jurisdiction	Name	Dam Number	Bridges (Roads)	Downstream Dams	Downstream Usage
Kent	Coventry	Pearce Pond	371	1	1	High density residential and industrial
	Coventry	Arnold Pond	561	2	3	Low density residential
	Coventry	Black Rock Reservoir	159	1	1	Residential
	Coventry	Coventry Reservoir	176	2	3	Industrial, commercial, and residential
	Coventry	Flat River Reservoir	167	0		Low to high density residential, campgrounds
	Coventry	Tiogue Lake	177	0	2	Commercial and residential
	Warwick	Camp Warwick Pond	462	2	2	Light residential
	Warwick	Grist Mill Apartments	764	2		Medium density residential
	West Warwick	Riverpoint Pond Upper	147	2	2	Light commercial, residential, and industrial
	West Warwick	Centerville Pond	149	None	1	Large scale industrial
	West Warwick	Natick Pond	145	1	2	Light development

Table 36: State of Rhode Island High Hazard Dams Observed Downstream Conditions

County	Jurisdiction	Name	Dam Number	Bridges (Roads)	Downstream Dams	Downstream Usage
Newport	Little Compton	Harold E Watson Reservoir	485	1	0	Residential
	Middletown	Easton Pond North	584	1	1	Easton beach
	Middletown	Easton Pond South	585	1	0	Easton beach
	Middletown	Gardiner Pond	583	1	0	Trailer Park and beach parking
	Middletown	Nelson Pond	582	1	0	Beach parking for Middletown Second Beach
	Portsmouth	Lawton Valley Reservoir	325	4	0	Undeveloped woodland
	Portsmouth	Sisson Pond	580	5	1	Residential, agricultural, and commercial
	Portsmouth	St. Mary's Pond	581	2	2	Residential, agricultural, and commercial
	Tiverton	Creamer	742	3	0	Medium density residential
Providence	Burrillville	Harrisville Pond	8	3	0	Commercial and residential
	Burrillville	Ocean State Power	766	1	1	Commercial properties
	Burrillville	Pascoag Reservoir Upper	16	3	5	Commercial and residential
	Burrillville	Wilbur Pond	572	1	1	Primarily undeveloped
	Burrillville	Wilson Reservoir	3	5	3	Low density residential
	Cranston	Cranston Printworks Pond	172	5	2	Residential, commercial, and industrial
	Cranston	Curran Lower Reservoir	198	3	2	Moderate residential, light industrial
	Cranston	Curran Upper Reservoir	166	1	0	Light residential
	Cranston	Stone Pond	320	1	0	Primarily undeveloped
	Cumberland	Diamond Hill Reservoir	77	2	8	Residential
	Cumberland	Micoe Lake	74	2	1	Residential and commercial
	Cumberland	Pawtucket Reservoir	78	2	2	Residential and commercial
	East Providence	James V Turner Reservoir	407	3	0	Residential, commercial, and sports fields
	Foster	Westconnaug Reservoir	163	4	0	Light residential
	Gloicester	Bowdish Lower Dam	727	4	2	Light residential
	Gloicester	Bowdish Reservoir Upper	566	2	3	Light residential
	Gloicester	Clarksville Pond	556	3	1	Residential
	Gloicester	Keech Pond	22	2	2	Very light residential

Table 36: State of Rhode Island High Hazard Dams Observed Downstream Conditions

County	Jurisdiction	Name	Dam Number	Bridges (Roads)	Downstream Dams	Downstream Usage
	Glocester	Lake Washington	401	2	4	Residential
	Glocester	Ponaganset Reservoir	165	2	0	Primarily undeveloped
	Glocester	Waterman Lake	111	2	3	Residential and commercial
	Johnson	Almy Reservoir	169	3	5	Residential and commercial
	Johnson	Hughesdale Pond Upper	313	1	1	Medium density residential and commercial
	Johnson	Oak Swamp Reservoir	168	3	1	Light residential
	Johnson	Simmons Lower Reservoir	171	2	1	Medium density residential and commercial
	Johnson	Simmons Upper Reservoir	170	2	2	Medium density residential and commercial
	Lincoln	Bleachery Pond	104	2	1	Industrial
	Lincoln	Bridlewood Pond	408	2	0	Very light residential
	Lincoln	Butterfly Pond	97	2	3	Low density residential
	Lincoln	Olney Pond	102	1	2	Residential
	North Providence	Louisquisset Flood Control	760	3	0	Light residential
	North Providence	Wenscott Reservoir	84	4	2	Residential and recreational
	North Smithfield	Forestdale Pond	48	2	0	Commercial and residential
	North Smithfield	Slatersville Reservoir Upper	43	2	3	Commercial, residential, two schools
	North Smithfield	Slatersville Upper Intermediate	46	2	2	Commercial, residential, two schools
	North Smithfield	Woonsocket Reservoir #1	70	2	0	Very light density residential
	North Smithfield	Woonsocket Reservoir #3	68	2	0	Light residential and commercial
	Scituate	Gainer Memorial	161	2	0	Residential
	Smithfield	Georgiaville Pond	126	3	10	Residential, commercial, industrial, and sports fields
	Smithfield	Hopkins Pond	116	0	0	Residential
	Smithfield	Slack Reservoir	115	1	0	Moderate density commercial and residential
	Smithfield	Sprague Upper Reservoir	120	2	0	Residential
	Smithfield	Stillwater Reservoir	108	2	0	Residential
	Woonsocket	Harris Pond	73	1	9	Commercial, residential, sports fields
	Woonsocket	Holley Lane Pond	621	1	2	Commercial, residential, schools
Washington	Exeter	Boone Lake	219	3	2	Primarily undeveloped
	Exeter	Metcalf Wildlife Marsh	527	4	8	Low to moderate density residential and agriculture

Table 36: State of Rhode Island High Hazard Dams Observed Downstream Conditions

County	Jurisdiction	Name	Dam Number	Bridges (Roads)	Downstream Dams	Downstream Usage
	Exeter	Slocum Reservoir	239	1	1	Residential
	Exeter	Yorker Mill Pond	240	2		Low density residential, agricultural, and industrial
	Hopkinton	Langworthy Pond	285	3	1	Commercial and residential
	Hopkinton	Wyoming Upper	216	2	1	Primarily undeveloped
	Hopkinton	Yawgood Pond	226	2	6	Low to medium density residential
	North Kingstown	Carr Pond	513	1	0	Light residential
	North Kingstown	Rodman Mill	615	1	2	Commercial
	North Kingstown	Shady Lea Mill	708	1	1	Commercial and industrial
	North Kingstown	Silver Spring Lake	444	1	2	Residential
	North Kingstown	Slocum Road Upper	710	1	1	Light residential and agricultural
	North Kingstown	Slocum Woods	693	1	0	Light residential and agricultural
	Richmond	White's Pond	261	1	1	Light residential
	South Kingstown	Asa Pond	549	1	2	Residential and industrial
	South Kingstown	Hefler Pond Farm	525	1	0	Commercial, residential, and industrial
	South Kingstown	Peace Dale Pond	426	1	0	Commercial, residential, and industrial
	South Kingstown	Wakefield Pond	425	1	0	Commercial and residential

Source: RIDEM

In the five years since the previous SHMP, seven dams with unsafe conditions were properly addressed and are now considered compliant. As of the latest available data, the following dams in unsafe condition need to be addressed:

Table 37: State of Rhode Island Unsafe Significant and High Hazard Dams

County	Jurisdiction	Dam Number	Dam Name	Hazard Class
Bristol	Warren	480	Warren Upper	Significant
Kent	Coventry	176	Coventry Reservoir	High
	Coventry	371	Pearce Pond	High
	Coventry	498	Hopkins Farm	Significant
	Coventry	645	Centre of New England #1	Significant
	West Warwick	455	Boucher Farm	Significant
Newport	Jamestown	575	Jamestown Lower	Significant
	Little Compton	746	Adamsville	Significant
	Middletown	582	Nelson	High
	Middletown	583	Gardiner	High
	Middletown	584	Easton North	High
	Middletown	585	Easton South	High
Providence	Burrillville	003	Wilson	High
	Burrillville	027	Sucker	Significant
	Burrillville	565	Ross	Significant
	Burrillville	572	Wilbur	High
	Cranston	198	Curran Lower	High
	Cranston	320	Stone	High
	Foster	163	Westconnaug	High

Table 37: State of Rhode Island Unsafe Significant and High Hazard Dams

County	Jurisdiction	Dam Number	Dam Name	Hazard Class
	Foster	349	Spear	High
	Glocester	018	Burlingame	High
	Glocester	354	Coomer	Significant
	Glocester	401	Lake Washington	High
	Glocester	727	Bowdish Lower	High
	Glocester	111	Waterman	High
	Johnston	127	Belknap	Significant
	Johnston	169	Almy	High
	Johnston	170	Simmons Upper	High
	Johnston	171	Simmons Lower	High
	Johnston	313	Hughesdale Upper	High
	Johnston	346	Kimball	Significant
	North Providence	084	Wenscott	High
	Scituate	345	Jordan	Significant
	Scituate	361	Pine Swamp No. 1	Significant
	Smithfield	108	Stillwater	High
	Smithfield	109	Stillwater	Significant
	Smithfield	121	Sprague Lower	High
Washington	Exeter	221	Browning	High
	Exeter	238	Edward's	Significant
	Hopkinton	216	Wyoming Upper	High
	New Shoreham	424	Rod and Gun Club	Significant
	North Kingstown	444	Silver Spring	High
	Richmond	273	Wood River Junction	Significant

Source: RIDEM

In addition to the above-identified unsafe dams, and of further concern, RIDEM has identified the following unsafe dams with unknown owners.

Table 38: State of Rhode Island Unsafe Significant and High Hazard Dams with Unknown Owners

County	Jurisdiction	Dam Number	Dam Name	Hazard Class
Kent	Coventry	152	Mill	Significant
	Coventry	177	Tiogoe	High
	Coventry	185	Black Rock	High
	Warwick	669	Dave's Marketplace	Significant
	Warwick	764	Grist Mill Apartments	High
	West Warwick	145	Natick	High
	West Warwick	149	Centerville	High
Providence	Burrillville	015	Union Mill	Significant
	Burrillville	016	Pascoag West Dike	High
	Burrillville	039	Spring	Significant
	Burrillville	051	Nichols	Significant
	Foster	526	Gorham	Significant
	Glocester	021	Cherry Valley	Significant
	Glocester	029	Mower	Significant
	Glocester	032	Snakeskin	Significant
	Hopkinton	227	Ashville	Significant
	Johnston	323	Cesarville	Significant
	Johnston	504	Dexter Farm	Significant
	Lincoln	649	Bridelwood Upper End	High
	North Providence	760	Louisquisset	Significant

Table 38: State of Rhode Island Unsafe Significant and High Hazard Dams with Unknown Owners

County	Jurisdiction	Dam Number	Dam Name	Hazard Class
	North Smithfield	048	Forestdale	High
	Scituate	160	Hope	Significant
	Smithfield	123	Hawkins	Significant
Washington	Charlestown	758	Cross Mills	Significant
	Hopkinton	227	Ashville	Significant
	North Kingstown	550	Hamilton	Significant
	North Kingstown	704	Secret	Significant
	South Kingstown	579	Rocky Brook	Significant

Source: RIDEM

Rhode Island General Laws Section 46-19-9 requires the municipality where either a High Hazard or Significant Hazard dam is located to complete an Emergency Action Plan and submit it to RIEMA for approval. RIDEM works with RIEMA in reviewing and these plans. Data indicates that only 68 dams have approved plans (including seven of 36 unsafe High Hazard dams). Recognizing this deficiency, RIDEM has made resources available to local jurisdictions, including access to staff and online guides. For jurisdictions with High Hazards dams without Emergency Action Plans, RIEMA has actively encouraged the inclusion of mitigation actions that address the problem in local Hazard Mitigation Plans.

A dam failure can be triggered by various cascading impacts, potentially involving a combination of natural and human-induced factors. The specific causes depend on the type of dam, its construction, maintenance, and the surrounding environment, and may include:

- **Hydraulic Loading and Over-Topping:** Excessive rainfall, rapid snowmelt, or a combination of both can lead to increased water inflow into the reservoir. If the rate of inflow exceeds the dam's discharge capacity, the water level may rise, leading to over-topping. Over-topping can cause erosion of the dam crest, ultimately resulting in failure.
- **Slope Instability and Landslides:** Landslides in the dam's vicinity can impact the stability of the dam structure. The sliding of soil or rock materials into the reservoir can displace water, causing waves that may over-top the dam or lead to structural damage.
- **Seismic Events:** Earthquakes can induce ground shaking, which may compromise the integrity of a dam. Seismic forces can lead to foundation movement, slope instability, or structural failure, resulting in dam failure.
- **Foundation Failure:** Weaknesses or faults in the dam's foundation can result in structural failure. Prolonged seepage, erosion, or subsurface geological issues can compromise the foundation's ability to support the dam's weight and the reservoir's water pressure.
- **Piping and Internal Erosion:** Piping occurs when water flows through internal pathways within the dam, gradually eroding and weakening the structure. If left unaddressed, piping can compromise the dam's integrity and lead to failure.
- **Dam Aging and Deterioration:** Over time, dams may experience aging and deterioration due to various factors such as freeze-thaw cycles, chemical reactions, and exposure to environmental conditions. This aging process can weaken materials and reduce the dam's structural stability.
- **Inadequate Maintenance:** Poor maintenance practices, including neglecting repairs and inspections, can contribute to the degradation of a dam's components. Cracks, leaks, and other defects left unaddressed can worsen over time, increasing the risk of failure.
- **Ice Dam Formation:** In cold climates, ice dams can form at the base of a dam or within the reservoir. The accumulation of ice can obstruct the flow of water, leading to over-topping and structural stress.
- **Human-Induced Factors:** Human activities, such as excavation, construction, or changes in land use near the dam, can influence its stability. Unauthorized modifications or alterations to the dam structure can compromise its integrity.

Additionally, a dam failure can cause a number of rapidly cascading impacts both down and up-stream of the failure. The following sections discuss these potential impacts:

People and Communities

A dam failure event can have devastating and wide-ranging impacts on both people and communities. The severity of these impacts depends on the volume of water released and the location of the dam in relation to communities, and may include:

- **Loss of Life:** The sudden release of a large volume of water can result in flooding downstream, leading to drowning and casualties. The loss of life can be particularly high if a dam failure occurs in highly populated areas or when people are unable to evacuate in time.
- **Long Term Displacement:** People living downstream may be forced to evacuate their homes leading to displacement and requiring long-term shelter assistance.
- **Economic Consequences:** Both property damage and the disruption of transportation and utilities could affect local economies.
- **Psychological Trauma:** Survivors of dam failure events may experience psychological trauma, including post-traumatic stress disorder, anxiety, and depression. Witnessing the loss of lives and property can have long-lasting emotional effects on individuals and communities.

In general, when considering a dam failure event concern exists with an increasing population. Available data indicates that the State of Rhode Island will see a slightly increasing population until 2035 and then will begin to see a gradual decrease.

Facilities

The impact of a dam failure event on facilities, including critical facilities, can be wide-ranging. The extent of the impact depends on the proximity of the facilities to the dam and the volume of water released, and may include:

- **Structural Damage:** Facilities located downstream could sustain severe structural damage, structural failures, and damaging foundations. This can render facilities inoperable or unsafe for use.
- **Equipment Damage:** Sensitive equipment can be severely damaged or destroyed by floodwaters and debris. This can include electrical systems, machinery, data centers, and communication equipment.
- **Disruption of Operations:** Flooding can disrupt the normal operations of critical facilities, including hospitals, emergency response centers, power plants, and water treatment plants. This disruption can have cascading effects on public services and infrastructure.
- **Long-Term Recovery:** The recovery process could be lengthy and resource intensive. It may involve rebuilding damaged infrastructure, restoring functionality, and implementing measures to prevent future vulnerabilities.

Environment

The environmental impact of dam failures depends on the circumstances of the failure. After a dam failure occurs, the resulting flooding and moving debris can affect wildlife and natural habitats. The spread of pollution and hazardous materials can have negative impacts on the environment. Ecosystems and natural habitats may be destroyed, causing the migration or death of local wildlife. Depending on the timing and location of the failure, it can result in rapid changes in water temperature downstream. This can be harmful to temperature-sensitive aquatic species and ecosystems. Dam failures can disrupt natural ecological processes, such as nutrient cycling, sediment transport, and flow regimes. These disruptions can have cascading effects on ecosystems.

State Operations and Assets

Mapping of state assets within dam failure inundation zones has yet to be completed. However, any state owned or operated building within an identified inundation zone will be immediately impacted potentially causing a cessation of all operations. The extent of the impact depends on multiple factors, and may include:

- **Emergency Response and Management:** State agencies, such as RIEMA and the National Guard, may be called upon to respond to a dam failure event.

- **Infrastructure Damage and Maintenance:** RIDOT may need to assess and repair damage to roads, bridges, and other critical infrastructure affected by floodwaters and debris resulting from the dam failure. This can strain resources and disrupt transportation networks.
- **Environmental Oversight and Regulation:** RIDEM is responsible for assessing the environmental impact of a dam failure, monitoring water quality, and coordinating cleanup efforts. They may also be involved in addressing long-term environmental consequences.
- **Water Resource Management:** Multiple Rhode Island agencies may need to manage and allocate water resources differently in the aftermath of a dam failure, especially if the failure affects water supplies, water quality, or flood control systems.
- **Public Health and Safety:** RIDOH would provide support for public health needs during and after a dam failure, including managing emergency shelters and addressing potential health risks from waterborne diseases.
- **Financial Impact:** A dam failure event can strain state budgets due to the costs associated with emergency response, infrastructure repair, environmental cleanup, and long-term recovery efforts.

Consequence Analysis

This consequence analysis lists the potential impacts of this hazard on various elements of community and state infrastructure. The impact of this hazard is evaluated in terms of disruption of operations, recovery challenges, and overall wellbeing to all Rhode Island residents and first responder personnel. The consequence analysis supplements the hazard profile by analyzing specific impacts.

Table 39: Dam Failure Consequence Analysis

Subject	Potential Impacts
Impact on the Public	Heavy flooding can cause power loss, property damage, injury, and death, and the displacement of populations. Standing water can also pose a public health risk due to the reproduction of disease vectors such as mosquitos.
Impact on Responders	Heavy flooding may cause inaccessibility of roadways for first responders as well as damage of materials and resources. First responders will also have to facilitate evacuation measures to move people from the flooded area.
Continuity of Operations	Rhode Island agencies maintain continuity plans which can be enacted as necessary based on the situation. Flooding caused by dam failure may create power outages, debris damage, and road closures.
Delivery of Services	Delivery of services may be disrupted due to flood-damaged bridges and roadways. Transit systems may face closures due to public safety concerns. The ability to deliver food, drinking water, and services will be heavily disrupted. Flooding may also interrupt communications and transportation due to power failure and accessibility challenges.
Property, Facilities, and Infrastructure	Flooding from dam failures impact roads and bridges, businesses, hospitals, and other critical entities. Water and sewer systems may also be damaged. Homes and businesses may be completely destroyed if situated close to the failure point.
Impact on Environment	Flooding and moving debris can affect natural areas and wildlife, spreading pollution and hazardous materials. Ecosystems and natural habitats may be completely destroyed, causing migration or death of wildlife.
Economic Conditions	There is a fiscal impact on the state government after a dam failure due to disruption of travel and commerce routes and employee's ability to travel to work. Recourses at all levels are utilized impacting the ability to access resources long-term.
Public Confidence in Governance	Direct, immediate, and effective actions must be taken in order to maintain public confidence. Response activities must include all levels of government.

4.9.7 Jurisdictional Vulnerability and Planning Significance

Data from FEMA's NRI was unavailable concerning dam failure. However, the following table details the hazard ratings from Local Hazard Mitigation Plan (LHMP) analysis:

Table 40: State of Rhode Jurisdictional Dam Failure Planning Significance

County	Jurisdiction	Dam Failure LHMP Rating
Bristol	Barrington	Low
	Bristol	-
	Warren	No Rating
Kent	Coventry	-
	East Greenwich	Low
	Warwick	Low
	West Greenwich	-
	West Warwick	Low
Newport	Jamestown	-
	Little Compton	Moderate
	Middletown	Low
	Newport	Low
	Portsmouth	Low
	Tiverton	-
Providence	Burrillville	Moderate
	Central Falls	Low
	Cranston	Low
	Cumberland	-
	East Providence	Moderate
	Foster	Low
	Glocester	Moderate
	Johnson	Moderate
	Lincoln	Low
	North Providence	Low
	North Smithfield	Not Rated
	Pawtucket	Low
	Providence	High
	Scituate	-
	Smithfield	-
	Woonsocket	Low
Washington	Charlestown	-
	Exeter	-
	Hopkinton	-
	Narragansett	Not Rated
	New Shoreham	-
	North Kingstown	Moderate
	Richmond	Moderate
	South Kingstown	Not Rated
	Westerly	Moderate

Source: LHMPs

Note: - indicates no current plan available for review

The following jurisdictions, identified as having high hazard dams listed as unsafe and as unsafe high hazard dams with no known owner, may be at higher risk to a dam failure event:

Table 41: State of Rhode Island Jurisdictions with Unsafe High Hazard Dams

County	Jurisdiction	Unsafe High Hazard Dams	Unsafe High Hazard Dams with Unknown Owner
Kent County	Coventry	2	2
Newport County	Middletown	4	4

Table 41: State of Rhode Island Jurisdictions with Unsafe High Hazard Dams

County	Jurisdiction	Unsafe High Hazard Dams	Unsafe High Hazard Dams with Unknown Owner
Providence County	Burrillville	2	2
	Cranston	2	1
	Foster	2	
	Glocester	4	4
	Johnston	4	4
	Lincoln	0	1
	North Providence	1	1
	North Smithfield	0	1
	Smithfield	2	0
Washington County	Exeter	1	0
	Hopkinton	1	0
	North Kingstown	1	0

Source: RIDEM

4.9.8 Hazard Planning Significance

The SIHMC determined dam failure to have a moderate planning significance for Rhode Island, greater than local level assessment due to the indication that in-state dams are nearing the end of their planned life expectancy and the possible impact of climate change on extreme precipitation events.

4.10 Drought

4.10.1 Hazard Description

Drought is defined as an abnormally dry period lasting months or years when an area has a deficiency of water and precipitation in its surface and/or underground water supply. It is, however, a normal, seasonal, and recurrent feature of climate that occurs in virtually all climate zones—typically in late spring through early fall. The duration of drought varies widely. There are cases when drought develops relatively quickly and lasts a very short period of time, exacerbated by extreme heat and/or wind, and there are other cases when drought spans multiple years, or even decades. The hydrological imbalance can be grouped into the following non-exclusive categories:



- Agricultural: When the amount of moisture in the soil no longer meets the needs of previously grown crops
- Hydrological: When surface and subsurface water levels are significantly below their normal levels
- Meteorological: When there is a significant departure from the normal levels of precipitation
- Socio-Economic: When the water deficiency begins to significantly affect the population

When below average, little or no rain falls, soil can dry out, and plants can die. If unusually dry weather persists and water supply problems develop, the period is defined as a drought. Human activity such as over-farming, excessive irrigation, deforestation, and poor erosion controls can exacerbate a drought’s effects. It can take weeks or months before the effects of below average precipitation on bodies of water are observed. Depending upon the region, droughts can happen more quickly, and be noticed sooner, or have their effects naturally mitigated. The more humid and wet an area is, the faster the effects will be realized. A naturally dry region, which typically relies more on subsurface water will take more time to actualize its effects.

Periods of drought can have significant environmental, agricultural, health, economic, and social consequences. The effects vary depending upon vulnerability and regional characteristics. Droughts can also reduce water quality through a decreased ability for natural rivers and streams to dilute pollutants and increase contamination. The most common effects are diminished crop yield, increased erosion, dust storms, ecosystem damage, reduced electricity production due to reduced flow through hydroelectric dams, shortage of water for industrial production, and increased risk of wildland fires.

4.10.2 Location and Extent

The Rhode Island Drought Steering Committee assigns drought levels based on six indices including the Palmer Drought Index, the Crop Moisture Index, precipitation data, groundwater levels, stream flow, and surface water reservoir levels.

The PDI, devised in 1965, was the first drought indicator to assess moisture status comprehensively. It uses temperature and precipitation data to circulate water supply and demand; incorporates soil moisture; and is considered most effective for non-irrigated cropland. It primarily reflects long-term drought and has been used extensively to initiate drought relief.

Table 42: Palmer Drought Severity Index

Category	Range (Per Year)
Extremely Wet	4.0 or more
Very Wet	3.0 to 3.99
Moderately Wet	2.0 to 2.99
Slightly Wet	1.0 to 1.99
Incipient Wet Spell	0.5 to 0.99
Near Normal	0.49 to -0.49
Incipient Dry Spell	-0.5 to -0.99

Table 42: Palmer Drought Severity Index

Category	Range (Per Year)
Mild Drought	-1.0 to -1.99
Moderate Drought	-2.0 to -2.99
Severe Drought	-3.0 to -3.99
Extreme Drought	-4.0 or less

Source: U.S. Drought Monitor

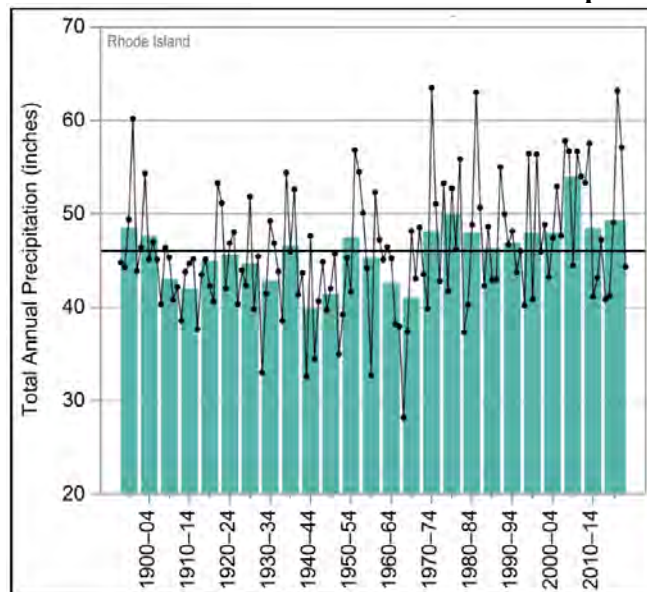
The Crop Moisture Index gives the short-term or status of purely agricultural drought or moisture surplus and can change rapidly from week to week. The Crop Moisture Index indicates general conditions and not local variations caused by isolated rain. Input to the calculations include the weekly precipitation total and average temperature, division constants and previous history of the indices. The Crop Moisture Index can be used to measure the status of dryness or wetness affecting warm season crops and field activities.

Table 43: Crop Moisture Index

Category	Crop Moisture Index
Excessively Wet	3.0+
Wet	2.0 to 3.0
Abnormally Moist	1.0 to 1.9
Slightly Dry/Favorably Moist	-0.9 to 0.9
Abnormally Dry	-1.0 to -1.9
Excessively Dry	-2.0 to -2.9
Severely Dry	-3.0 or less

Source: U.S. Drought Monitor

Precipitation data is collected by the NWS at eight data points and two U.S. Climate Reference Network sites. Additional rainfall data is also collected by the NWS through citizen weather rainfall sites. A drought phase determination is based on conditions relative to normal in 2-, 3-, 6-, and 12-month intervals. The following chart indicates annual precipitation averages for Rhode Island from 1895 to 2020:

Chart 18: Rhode Island Observed Annual Precipitation

Source: NOAA State Climate Summary 2022 for Rhode Island

Data from the RIDEM indicates that annual precipitation averages 42 to 46 inches over most of the State, with a tendency for decreasing amounts from west to east. While there are no pronounced wet and dry months as in other climates, the months of May, June and July are relatively dry. The average total precipitation for each of these months is 2.5 to three inches. October and February have an average of slightly more than three inches over most of the State.

The remaining months each yield from 3.5 to four inches. Measurable precipitation falls on an average of one day in three.

Groundwater level data is monitored by the United State Geological Survey (USGS) from 27 observation wells. A drought phase determination is based on the number of months groundwater levels are below normal. Local water suppliers also monitor public wells to make seasonal water availability comparisons.

Stream flow conditions are monitored by the USGS from 16 near-real-time stations with 30 or more years of record and seven near real-time stations with 10 years or more of record. A drought phase determination is based on the number of months that stream flow levels are below normal compared to historical trend data.

Surface water reservoir levels data is typically reported by water suppliers, relative to normal conditions. A drought phase determination considers historic monthly averages of small, medium, and large index reservoirs.

Based on these criteria, Normal, Advisory, and Watch drought levels are issued statewide, and Warning and Emergency drought levels are issued on a regional basis. The following table details State of Rhode Island drought classifications:

Table 44: State of Rhode Island Drought Classifications

Drought Level	PDI	CMI	Precipitation	Ground Water	Stream Flow	Surface Water Reservoirs
Normal	-1.0 to -1.99	0.0 to -1.0 Slightly dry	Slightly dry	1 month below normal	2 consecutive months below normal	Reservoir levels at or near normal for the time of year
Advisory	-2.0 to -2.99	-1.0 to -1.9 Abnormally dry	2 months cumulative below 65% of normal	At least 2 out of 3 months below normal	3 consecutive months below normal	Small index reservoir below normal
Watch	-3.0 to -3.99	-2.0 to -2.9 Excessively dry	1 of the following criteria met: 3-month cum. <65% or 6 months cum. <70% or 12 months cum. <70%	4-5 consecutive months below normal	At least 4 out of 5 consecutive months below normal	Medium index reservoir normal
Warning	-4.0 and below	>-2.9 Severely dry	2 of 3 of the above criteria met: 3 months cum. <65% and 6 months cum. <65% and 12 months cum. <65% or 3 months cum. <65% and 12 months <65%	6-7 consecutive months below normal	At least 6 out of 7 consecutive months below normal	Large index reservoir below normal
Emergency	-4.0 and below	>-2.9 Severely dry	Same criteria as Warning and previous month was a warning.	7 months below normal observation wells recording monthly record lows	>7 months below normal	Continuation of previous month's conditions

Source: RIEMA

An additional indicator of historic drought periods is provided by the U.S. Drought Monitor, which lists weekly drought conditions for the State of Rhode Island. The following table details the U.S. Drought Monitor categories.

Table 45: U.S. Drought Monitor Categories

Rating	Described Condition
None	No drought conditions
D0	Abnormally Dry
D1	Moderate Drought
D2	Severe Drought
D3	Extreme Drought

Table 45: U.S. Drought Monitor Categories

Rating	Described Condition
D4	Exceptional Drought

Source: U.S. Drought Monitor

Current drought conditions in Rhode Island are tracked by the U.S. Drought Monitor, who provide graphic and text summaries of current and projected drought conditions which are updated on a weekly basis through the Drought Monitor website.

4.10.3 Previous Occurrences

One of the best indicators of historic drought periods is provided by the U.S. Drought Monitor, which lists weekly drought conditions for the Rhode Island. Historical data was gathered from the U.S. Drought Monitor weekly reports for the period between 2000 and 2023 (totaling 1,233 weeks' worth of data). This data was compiled and aggregated to provide a weekly estimate of the percentage of each Rhode Island County in each Drought Monitor category for the period.

Table 46: Weekly Percentage Rhode Island Counties in U.S. Drought Monitor Category, 2000 -2023

County	None	D0-D4	D1-D4	D2-D4	D3-D4	D4
Bristol	74.0%	26.0%	11.4%	3.3%	0.6%	0.0%
Kent	74.7%	25.2%	11.7%	3.1%	0.8%	0.0%
Newport	75.3%	24.7%	11.1%	3.3%	1.0%	0.0%
Providence	73.6%	26.4%	12.4%	3.4%	0.5%	0.0%
Washington	73.4%	26.6%	12.2%	3.9%	0.9%	0.0%
State Average	74.2%	25.8%	11.8%	3.4%	0.8%	0.0%

Source: U.S. Drought Monitor

The Secretary of Agriculture is authorized to designate counties as disaster areas to make emergency loans available to producers suffering losses in those counties and in counties that are contiguous to a designated county. USDA Secretarial disaster designations must be requested of the Secretary of Agriculture by a governor or the governor's authorized representative, and there is an expedited process for drought. The following table represents the total number of Secretarial Disaster Declarations, by County, for the State of Rhode Island:

Table 47: Secretarial Drought Disaster Declarations, 2012 -2023

Year	Bristol County	Kent County	Newport County	Providence County	Washington County	State Total
2023	0	0	0	0	0	0
2022	1	1	1	1	1	5
2021	0	0	0	0	0	0
2020	1	1	1	1	1	5
2019	0	0	0	0	0	0
2018	0	0	0	0	0	0
2017	0	0	0	0	0	0
2016	1	1	1	1	1	5
2015	0	0	0	0	0	0
2014	0	0	0	0	0	0
2013	0	0	0	0	0	0
2012	0	0	0	0	0	0

Source: USDA Farm Service Agency

In addition, the National Centers for Environmental Information identified seven drought events in Rhode Island from 1950 to 2023.

4.10.4 Probability of Future Events

Historically, drought has affected Rhode Island on a reoccurring basis. In reviewing historical data from the U.S. Drought Monitor weekly reports for each Rhode Island County from 2000 through 2023 (1,233 weeks) a weekly average can be created indicating the percentage time in each Drought Monitor category for the State. This average can be used to extrapolate the potential likelihood of future drought conditions.

Table 48: Estimated Weekly Probability of Rhode Island Being in U.S. Drought Monitor Category

None	D0-D4	D1-D4	D2-D4	D3-D4	D4
74.2%	25.8%	11.8%	3.4%	0.8%	0.0%

Data: U.S. Drought Monitor

Rhode Island can experience rapid droughts, with a sudden onset of intense dry periods following a period of normal precipitation. While these conditions may last only a few months, they can result in agricultural losses, water supplies shortages, and low stream and river volume.

While predicting drought provides many challenges, NOAA’s National Integrated Drought Information System provides the Northeast Drought Early Warning System to improve drought early warning capacity. The system is a network of regional and national partners that share information and coordinate actions to help communities in the region cope with drought. Developing and implementing the system allows Rhode Island to quickly respond to emerging drought conditions Through developing regional systems, the National Integrated Drought Information System is building the foundation for a nationwide system to improve drought forecasting.

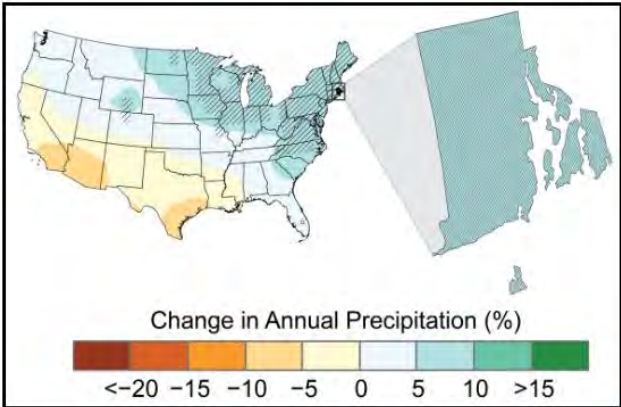
State Guide Plan Element #724, Rhode Island Drought Management Plan, was developed to provide government with a framework for coordinated response in times of drought. An interdisciplinary Steering Committee meets regularly to assess current conditions and make determinations regarding phases of drought. Rhode Island works closely with the NWS and the US Geological Survey to coordinate drought phases and indices with neighboring states.

4.10.5 Projected Changes in Location, Intensity, Frequency, and Duration

According to the National Institutes of Health National Center for Biotechnology Information publication Global Drought Trends and Future Projections “Drought is one of the most difficult natural hazards to quantify and is divided into categories (meteorological, agricultural, ecological and hydrological), which makes assessing recent changes and future scenarios extremely difficult.” However, using long term data estimates of future drought conditions can be determined through a combination of climate modeling, historical data analysis, and scientific assessments. This modelling takes into account factors such as temperature, precipitation, soil moisture, and other relevant variables.

Current modelling from the NOAA State Climate Summary 2022 for Rhode Island suggests that annual average precipitation, as well as extreme precipitation events, are projected to increase for Rhode Island. Although increased precipitation is projected, naturally occurring droughts are projected to be more intense because higher temperatures will increase evaporation rates. The following map indicates the expected annual increase in precipitation for Rhode Island:

Map 4: Rhode Island Change in Annual Precipitation



Source: NOAA State Climate Summary 2022 for Rhode Island

Additionally, higher temperatures associated with climate change can increase the rate of evaporation from soil, water bodies, and vegetation. This can contribute to soil moisture depletion and more rapid drying of surface water sources during dry periods.

The Public Utilities Commission regulates for-profit water supply utilities as well as municipal water utility boards that serve areas outside the boundaries of their respective municipality. Specifically, regulation is found in Title 815 – Division of Public Utilities and Carriers, Chapter 40 – Water Utilities. The following are water utility providers for the State of Rhode Island:

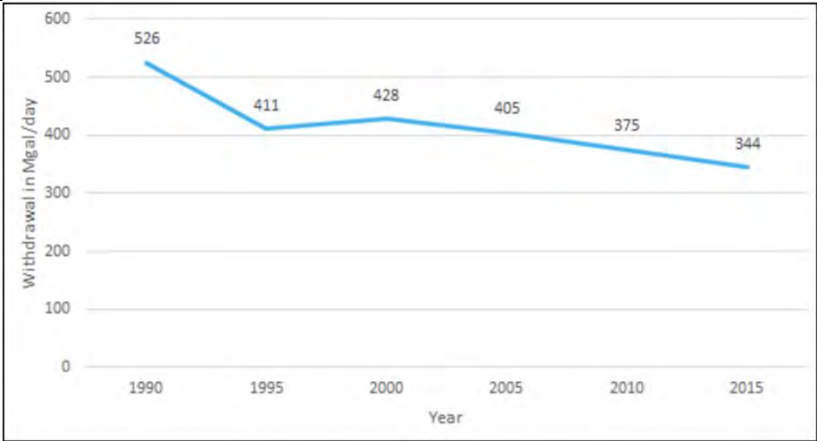
- Kent County Water Authority
- City of Newport Water Department
- Pawtucket Water Supply Board
- Providence Water Supply Board
- Veolia Water Rhode Island, Inc.
- Woonsocket Water Department

Additionally, RIDOH provides testing and oversight of all water systems and private wells within the State of Rhode Island. RIDOH reports the following concerning public water systems in Rhode Island:

- There are 481 public water systems across the state
- Surface water reservoirs supply approximately 85 % of public water
- The Scituate Reservoir is the largest source of Rhode Island’s surface water supply, providing approximately 60% of Rhode Islanders with drinking water.
- Approximately 150,000 Rhode Islanders have groundwater supplied by a private well
- 95% of the public water systems serve fewer than 3,300 people.

Data from the USGS indicates that in 2015 (the latest available data), approximately 344,000,000 gallons per day of water were withdrawn from groundwater and surface-water sources in Rhode Island. Total water withdrawals declined steadily between 1990 and 2015 by approximately 35 %. The following graph shows this decline in usage:

Graph X: State of Rhode Island Withdraws of Ground and Surface Water



Source: USGS

**4.10.6 Vulnerability and Impact
People and Communities**

Droughts are rarely a direct cause of death, though the associated heat, dust, and stress can all contribute to increased mortality. However, drought can severely challenge a public water supplier through depletion of the raw water supply and greatly increased customer water demand. Even if the raw water supply remains adequate, problems due to limited treatment capacity or limited distribution system capacity may be encountered. Water supply planning is the key to minimizing the effects of drought on the population. Public water suppliers should continue to work to identify

vulnerabilities and develop infrastructure, conservation plans, and partnerships to reduce the likelihood of running out of water during a drought.

Of particular concern, socially vulnerable and at-risk populations that may have difficulty with medical issues, poverty, extremes in age, and communications due to language barriers are often at increased susceptibility to the impacts of hazards. These populations suffer a disproportionate impact due to:

- Inadequate infrastructure, especially after an event
- Constrained financial resources
- Limited access to healthcare
- Limited access to information

Facilities and Critical Infrastructure

In general, critical facilities and infrastructure are not directly vulnerable to losses as a result of drought. However, there is a potential that operations could be impacted by power failures caused by either increased utility demand or damaged power delivery infrastructure. In addition, drinking water infrastructure may be specifically vulnerable to the impacts of drought. The U.S. Geological Survey has estimated that 27,000,000 gallons per day of groundwater are used in the state for drinking water and other uses, with groundwater supplies approximately 26% of the state's population. The Department of Environmental Management provides oversight to all 686 public wells in Rhode Island. Any decrease in groundwater supplies would stress this infrastructure and may cause shortages or rationing. The Scituate Reservoir provides water to a large percent of the State population living in the Providence metropolitan region and its surrounding suburbs. Drought conditions would also stress this asset and may result in shortages and rationing.

Environment

Drought conditions can cause significant environmental and agricultural impacts. In addition to obvious losses in yields in both crop and livestock production, drought is associated with increases in insect infestations, plant disease, and wind erosion. As of 2017, and as indicated by the USDA National Agricultural Statistics Service Census of Agriculture, Rhode Island has:

- 1,043 farms
- 56,864 acres in agriculture
- \$57,998,000 in agricultural products sold

Droughts also bring increased problems with insects and disease to forests and reduce growth. The incidence of wildfires increases substantially during extended droughts, which in turn places both human and wildlife populations at higher levels of risk. Although environmental losses are difficult to quantify, increasing public awareness and concern for environmental quality has forced public officials to focus greater attention and resources on these effects. Environmental losses are the result of damage to plant and animal species, wildlife habitat, and air and water quality, wildfires, degradation of landscape quality, loss of biodiversity, and soil erosion. Some of the effects are short-term and conditions quickly return to normal following the end of the drought. Other environmental effects linger for some time or may even become permanent. Wildlife habitat, for example, may be degraded through the loss of wetlands, lakes, and vegetation. However, many species will eventually recover from it if it is a temporary aberration. However, the degradation of landscape quality, with increased soil erosion, may lead to a more permanent loss of biological productivity of the landscape.

State Operations and Assets

State operations and assets will likely experience minimal impacts or dollar losses from drought conditions, unless there are substantial power, communications, or water outages. However, reduced water availability would likely have an immediate impact on firefighting efforts in urban and suburban areas as fire suppression equipment requires a minimum level of water pressure to activate. There has been no documented damage (dollar losses) to state owned buildings, infrastructure, or critical facilities at the time of this SHMP update. Expected future drought events are expected to have no impact (dollar losses) on state facilities or assets.

Consequence Analysis

This consequence analysis summarizes the potential impacts of a hazard on various elements of community and state infrastructure. The impact of each hazard is evaluated in terms of disruption of operations, recovery challenges, and overall wellbeing to all Rhode Island residents and first responder personnel. The consequence analysis supplements the hazard profile by analyzing specific impacts.

Table 49: Drought Consequence Analysis

Subject	Potential Impacts
Impact on the Public	If the drought coincides with warmer months, vulnerable populations may face an increased risk of dehydration, death, heat-related illness, heat stroke. Lower quantities of water may also increase the likelihood of contamination due to higher concentrations of bacteria. During droughts, dry soils and wildfires increase the number of airborne particles, such as pollen and smoke, which can worsen chronic respiratory illnesses.
Impact on Responders	Reduced water availability would likely complicate firefighting efforts in urban and suburban areas where wildfire-fighting tactics such as chemical retardants and controlled burns are less suitable. Some fire suppression equipment requires a minimum level of water pressure to activate. If the drought coincides with warm months, first responders may face increased risk of heat-related injuries or death.
Continuity of Operations	Rhode Island agencies maintain continuity plans which can be enacted as necessary based on the situation. While the expectation is minimal, this threat may impact an agency's ability to implement their continuity plan based on the hazard's potential to impact power, communications, or water outages. Critical life-saving activities and fire suppression will be directly impacted by these outages.
Delivery of Services	Waterways can recede, impacting the size of ships that can navigate shallower waters. Droughts may also impact the delivery of goods and services if there are shortages of raw materials.
Property, Facilities, and Infrastructure	Drought conditions may threaten levels or quality of municipal public water supplies or impact small communities and/or private potable water wells.
Impact on Environment	The potential of drought-related impacts could have significant impacts on supplies of animal feed, livestock, meat and dairy products, and processed grain products, and on crop production. Drought conditions may also increase the potential for fires. Drought is also associated with insect infestations, plant disease, wind erosion of soil, and decrease in levels of water produced by natural aquifers.
Economic Conditions	The economic impacts from a drought could be significant. Droughts have the potential to drain state, and local resources, which will have a significant fiscal impact on the local government.
Public Confidence in Governance	Droughts can adversely affect the public, first responders, infrastructure, agriculture, economy, and overall state operations. Direct, effective, and timely response by all levels of government is required for public confidence in the state's governance, especially in recognizing and mitigating economic impacts of the drought.

4.10.7 Jurisdictional Vulnerability and Planning Significance

Utilizing data from FEMA's NRI and from LHMPs, the following tables detail the hazard ratings for jurisdictions, by census tract, for drought. Data concerning both risk index and EAL is reported by census tract. As many State of Rhode Island jurisdictions encompass multiple tracts, an average of tract ratings was completed to determine the overall jurisdictional rating. The following table details the hazard rating by Local Hazard Mitigation Plan (LHMP) analysis, FEMA NRI, and EAL by jurisdiction for the State of Rhode Island:

Table 50: State of Rhode Jurisdictional Drought Planning Significance

County	Jurisdiction	Drought LHMP Rating	Drought NRI	Drought EAL
Bristol	Barrington	Low	Very Low	Very Low
	Bristol	-	No rating	No rating

Table 50: State of Rhode Jurisdictional Drought Planning Significance

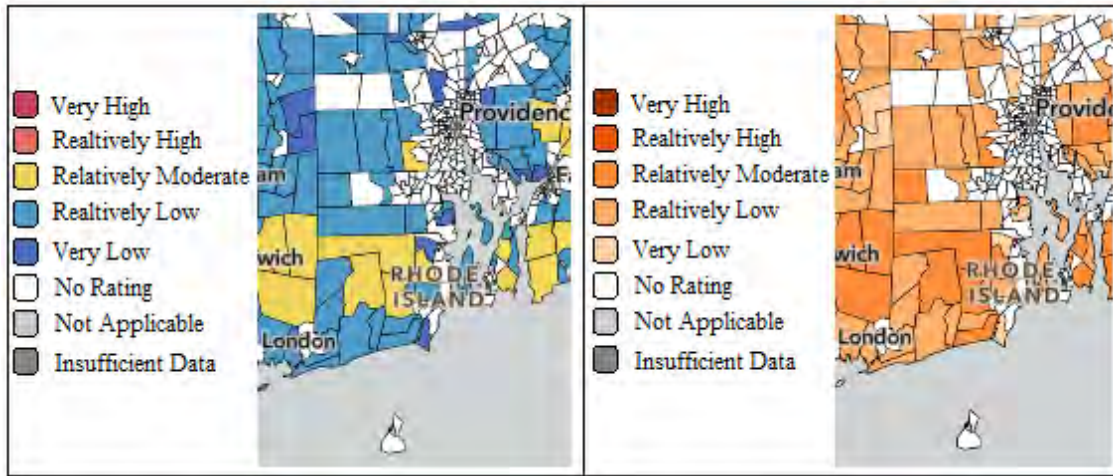
County	Jurisdiction	Drought LHMP Rating	Drought NRI	Drought EAL
	Warren	Low	Very Low	Very Low
Kent	Coventry	-	Very Low	Very Low
	East Greenwich	Moderate	Very Low	Very Low
	Warwick	Low	No Rating	No Rating
	West Greenwich	-	Relatively Low	Relatively Low
	West Warwick	Low	No Rating	No Rating
	Jamestown	-	Relatively Low	Relatively Low
Newport	Little Compton	Not rated	Relatively Moderate	Relatively Moderate
	Middletown	Low	Very Low	Very Low
	Newport	Moderate	No rating	No rating
	Portsmouth	Moderate	Relatively Low	Relatively Low
	Tiverton	-	Very Low	Very Low
	Burrillville	Moderate	Very Low	Very Low
Providence	Central Falls	Low	No Rating	No Rating
	Cranston	Low	No Rating	No Rating
	Cumberland	-	No Rating	No Rating
	East Providence	Moderate	No Rating	No Rating
	Foster	Moderate	Relatively Low	Relatively Low
	Glocester	Not rated	No Rating	No Rating
	Johnson	Moderate	No Rating	Very Low
	Lincoln	Low	No Rating	No Rating
	North Providence	Low	No Rating	No Rating
	North Smithfield	Not rated	Relatively Low	Relatively Low
	Pawtucket	Low	No Rating	No Rating
	Providence	Low	No Rating	No Rating
	Scituate	-	Relatively Low	Relatively Low
	Smithfield	-	Relatively Low	Very Low
	Woonsocket	Low	No Rating	No Rating
Washington	Charlestown	-	Relatively Low	Relatively Low
	Exeter	-	Relatively Moderate	Relatively Moderate
	Hopkinton	-	Relatively Low	Relatively Low
	Narragansett	Low	No Rating	No Rating
	New Shoreham	-	No Rating	No Rating
	North Kingstown	Moderate	Very Low	Very Low
	Richmond	Moderate	Relatively Moderate	Relatively Moderate
	South Kingstown	Not Rated	Relatively Low	Relatively Low
	Westerly	Low	Very Low	Very Low

Source: LHMPs and FEMA NRI

Note: - indicates no current plan available for review

The following maps indicate the FEMA NRI and EAL for drought by census tract for the state of Rhode Island. The

Map 5: FEMA NRI and EAL for Drought by Census Tract for Rhode Island



Source: FEMA NRI

As the greatest impact of drought conditions is likely to be felt by the agricultural community of Rhode Island, the following table is presented to understand and determine the most vulnerable communities. As agricultural data, as provided by the 2017 USDA National Agricultural Statistics Service Census of Agriculture, is only available on a county level a jurisdiction specific analysis was not possible.

Table 51: Potential Monetary Impacts of Drought Conditions on Agricultural Sales

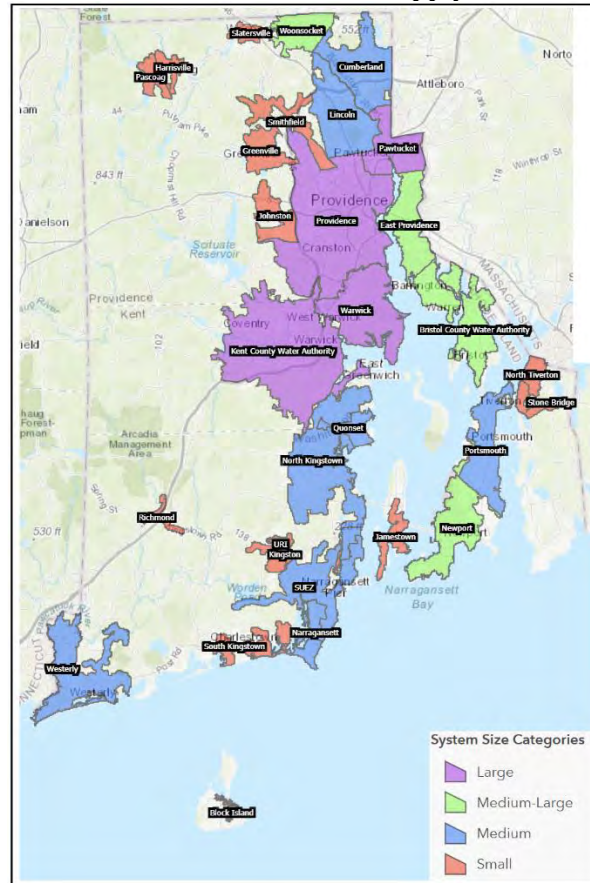
Jurisdiction	Market Value	25% Losses	50% Losses	75% Losses
Bristol County	\$1,005,000	\$251,250.00	\$502,500	\$753,750.00
Kent County	\$3,092,000	\$773,000.00	\$1,546,000	\$2,319,000.00
Newport County	\$19,280,000	\$4,820,000.00	\$9,640,000	\$14,460,000.00
Providence County	\$12,432,000	\$3,108,000.00	\$6,216,000	\$9,324,000.00
Washington County	\$22,190,000	\$5,547,500.00	\$11,095,000	\$16,642,500.00
State Rhode Island	\$16,477,000.00	\$32,954,000.0	\$49,431,000	\$16,477,000.00

Source: USDA National Agricultural Statistics Service

:- Data unavailable

Additionally, communities served by private wells rather than water supply districts may be at higher risk to drought conditions. The following map, from the Rhode Island Water Resources Board, shows water utility coverage for the state:

Map 6: State of Rhode Island Water Supply District Coverage



Source: Rhode Island Water Resources Board

Based on lack of water supply district coverage, the following communities may be at greater risk to drought conditions:

- Charlestown, Washington County
- Foster, Providence County
- Glocester, Providence County
- Hopkinton, Washington County
- Richmond, Washington County
- Scituate, Providence County
- West Greenwich, Kent

4.10.8 State of Rhode Island Hazard Planning Significance

Based on a review of all available data, the SIHMC determined drought to have a moderate planning significance for Rhode Island, differing from local assessments due to the likely future impacts of climate change on precipitation conditions.

4.11 Earthquake

4.11.1 Hazard Description

An earthquake is the result of a sudden release of energy in the Earth's crust that creates seismic waves that are typically caused by the rupturing of geological faults. A fault is a fracture or zone of fractures between two blocks of rock. Faults allow the blocks to move relative to each other, which, when rapidly occurring, causes an earthquake. When stresses in the crust exceed the strength of the surrounding rock, a rupture or break may occur in a fault plane. The point of origin of an earthquake is known as the hypocenter, which may be deep beneath the surface. The point at the surface directly above the hypocenter is known as the epicentre. Seismic waves radiate out from the hypocenter causing the ground to shake. These waves can travel long distance, but in general are strongest near the epicenter.



Earthquakes tend to occur along faults, which can be divided into three categories:

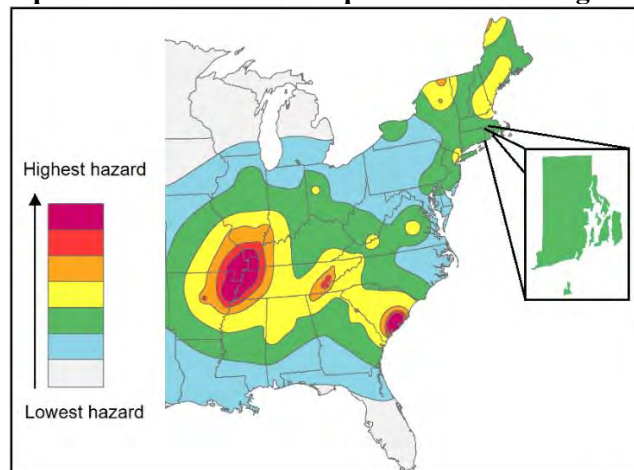
- Normal Fault: Resulting from pulling or tension with the overlying block moving down the dip of the fault plane
- Thrust (Reverse) Fault: Resulting from squeezing or compression, with the overlying block moving up the dip of the fault plane
- Strike-Slip (Lateral) Fault: Resulting from either type of stress, with the blocks moving horizontally past one another

4.11.2 Location and Extent

Rhode Island is located on the North Atlantic tectonic plate, an area of very low seismic activity. Additionally, the underlying geology of the State is largely composed of unsorted rock of varying size that is considered geologically stable and not prone to seismic amplification. However, areas surrounding Narragansett Bay have been identified as having Outwash Deposits, which can be prone to seismic amplification, increasing the intensity of the seismic event.

The following map, from the USGS, indicates the earthquakes hazard designation for the State of Rhode Island:

Map 7: Rhode Island Earthquake Hazard Designation



Source: USGS

Two scales are used when referring to earthquake activity. Estimating the total force of an earthquake is the Richter scale, and the observed damage from an earthquake is the Modified Mercalli Intensity Scale. Additionally, both Peak Ground Acceleration (%g) and Velocity (cm/s) can be used to measure and quantify force and movement. Peak Ground Acceleration is a measure of the maximum acceleration experienced by a point on the Earth's surface during an

earthquake. It quantifies the intensity of ground shaking at a specific location and is a crucial parameter for assessing seismic hazard. Peak Ground Acceleration is typically measured in units of gravity (g), where one g is approximately equal to the acceleration due to Earth's gravity (about 9.81 meters per second squared or 32.2 feet per second squared). So, if the PGA at a location is 0.2 g, it means the ground acceleration during the earthquake was 20% of the acceleration due to gravity. Peak Ground Acceleration can vary significantly from one location to another during the same earthquake event. Factors that influence Peak Ground Acceleration include the earthquake's magnitude, depth, distance from the epicenter, and local geological conditions.

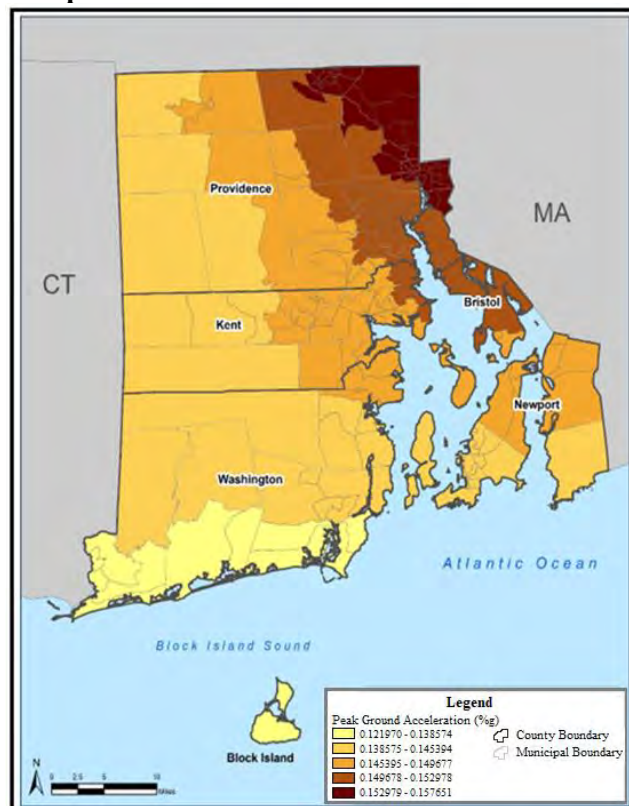
The following table equates the above referenced earthquake scales.

Table 52: Earthquake Magnitude Scale Comparison

Mercalli Scale Intensity	Verbal Description	Richter Scale Magnitude	Acceleration (%g)	Velocity (cm/s)	Witness Observations
I	Instrumental	1 to 2	0.17%	<0.1	None
II	Feeble	2 to 3	1.40%	1.1	Noticed only by sensitive people
III	Slight	3 to 4	1.40%	1.1	Resembles vibrations caused by heavy traffic
IV	Moderate	4	3.90%	3.4	Felt by people walking, rocking of free-standing objects
V	Rather Strong	4 to 5	9.20%	8.1	Sleepers awakened; bells ring
VI	Strong	5 to 6	18.00%	16	Trees sway, some damage from falling objects
VII	Very Strong	6	34.00%	31	General alarm, cracking of walls
VIII	Destructive	6 to 7	65.00%	60	Chimneys fall and some damage to building
IX	Ruinous	7	124.00%	116	Ground crack, houses begin to collapse, pipes break
X	Disastrous	7 to 8	>124.0%	>116	Ground badly cracked, many buildings destroyed. Some landslides
XI	Very Disastrous	8	>124.0%	>116	Few buildings remain standing, bridges destroyed.
XII	Catastrophic	8 or greater	>124.0%	>116	Total destruction, objects thrown in air, shaking and distortion of ground

The following map indicates probabilistic Peak Ground Acceleration for Rhode Island:

Map 8: Rhode Island Peak Ground Acceleration



Source: RIEMA

4.11.3 Previous Occurrences

The USGS and the Boston College Weston Observatory maintain earthquake records for the State of Rhode Island. Data indicates that while the State has had numerous earthquakes, the largest on record occurred in June 1951 in Kingstown and was measured at a 4.6 on the Richer Scale. For both the USGS and Weston Observatory, reported earthquakes before 1951 had no recorded measured intensity. The following table details notable earthquake events, in descending order of magnitude, for the State of Rhode Island:

Table 53: State of Rhode Island Earthquakes

Year	Month	Richter Scale Magnitude	Location
1951	June	4.6	Kingstown
1967	February	3	Narragansett Bay
1976	March	3.5	5 km SE of Portsmouth
1996	March	3.1	2 km NE of Bristol
1978	September	2.8	Rhode Island Sound
1976	March	2.7	New Bedford, MA
1981	April	2.7	Portsmouth
1965	December	2.6	Warwick
1974	October	2.5	West Warwick
2015	July	2.3	5.0km S of Providence
1983	February	2.2	NW of Newport
2015	February	2	7km SE of Newport
1978	October	1.9	South of Dartmouth, MA
2002	March	1.9	1 km WNW of East Greenwich
2020	January	1.9	6 km WSW of Exeter
1976	March	1.8	Portsmouth
1982	November	1.8	West of Providence
1983	February	1.8	SW of New Bedford, MA

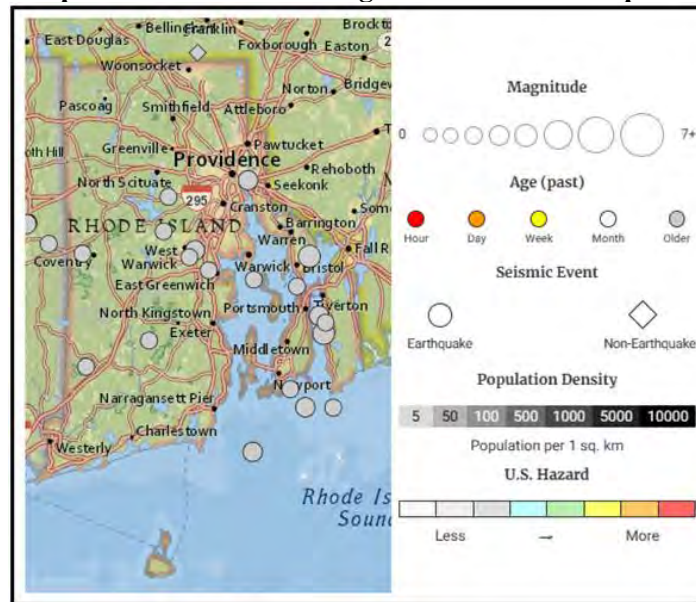
Table 53: State of Rhode Island Earthquakes

Year	Month	Richter Scale Magnitude	Location
2003	October	1.8	5 km NW of West Warwick
2013	April	1.8	2km WSW of Hope Valley
2018	June	1.8	Near Greene
2019	December	1.8	3 km SE of Newport
2014	March	1.71	3 km NNE of Woonsocket
1978	April	1.7	Block Island Sound
2017	January	1.6	1km SE of West Warwick
2022	June	1.6	4 km S of Bristol
2014	September	1.5	2.0km N of Melville
2018	June	1.3	12km ESE of Newport
2018	July	1.3	5km W of Coventry
2015	February	1.1	7.0km NE of Narragansett Pier
2012	December	1	Near Exeter
2016	February	1	3.0km NNW of Providence

Source: USGS and Weston Observatory

The following map from the USGS illustrates the location and intensity of the largest recorded earthquakes:

Map 9: Rhode Island's Largest Recorded Earthquakes



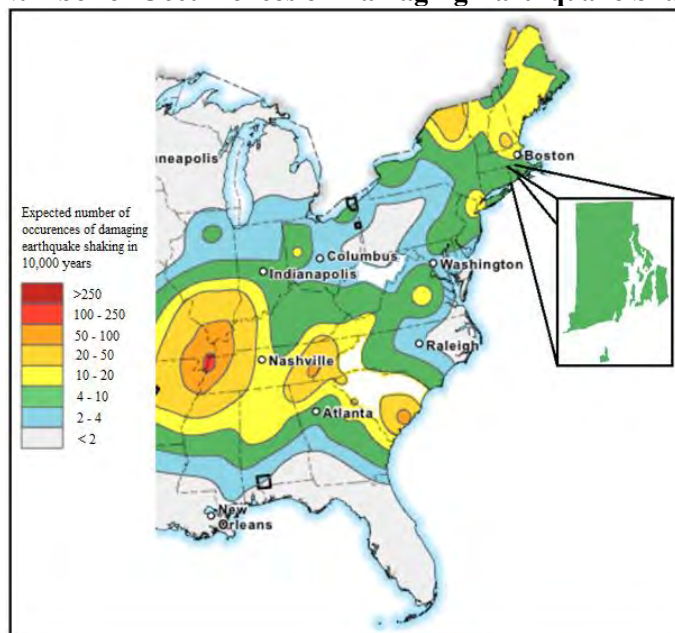
Source: USGS

While earthquake events do occur in Rhode Island, they tend to occur at a much lower intensity than elsewhere in the region. Additionally, earthquake events felt in Rhode Island are likely the result of an earthquake that occurs in the surrounding region.

4.11.4 Probability of Future Events

Predicting the occurrence of earthquakes is tremendously challenging due to the large number of factors involved. However, mapping from the USGS can help detail future earthquake probability. The following map, from the USGS, illustrates potential earthquake hazard for Rhode Island.

Map 10: Expected Number of Occurrences of Damaging Earthquake Shaking in 10,000 Years



Source: USGS

4.11.5 Projected Changes in Location, Intensity, Frequency, and Duration

Due to the very long-term nature of geological process, including earthquakes, the seismic hazard for the State of Rhode Island is not to change during the life of this plan.

4.11.6 Vulnerability and Impact

People and Communities

Although earthquakes occur infrequently in Rhode Island, a large magnitude quake could cause significant impacts. In a larger magnitude earthquake, ground movement can lead to building and infrastructure collapse. Additionally, concurrent hazards caused by earthquakes may include fire, hazardous material release, tsunamis (if in an offshore environment), and dam failure. These concurrent events will likely result in deaths and injuries.

Of particular concern, socially vulnerable and at-risk populations that may have difficulty with medical issues, poverty, extremes in age, and communications due to language barriers are often at increased susceptibility to the impacts of hazards. These populations suffer a disproportionate impact due to:

- Inadequate infrastructure, especially after an event
- Constrained financial resources
- Limited access to information

Facilities and Critical Infrastructure

Structures in the State of Rhode Island may be particularly vulnerable to the effect of a moderate to large earthquake as seismic design criteria are not required for either new building construction or old building renovation. Of particular concern to Rhode Island are unreinforced masonry buildings. An unreinforced masonry building is constructed of brick or masonry with no steel reinforcing bars. Because these buildings were not built using modern building codes, they are much more likely to experience damage or collapse during an earthquake. Surveys conducted by the Northeast States Emergency Consortium estimate that Rhode Island has approximately 108 unreinforced masonry school buildings alone. Additionally, certain areas, buildings, and infrastructure are at increased risk due to underlying soil types and methods of construction.

Critical facilities are at potential risk to earthquakes, especially those that store or transport handle hazardous materials. Unauthorized releases from these facilities may cause health impacts, environmental damage, or force road or area closures. Infrastructure is also at high risk for earthquakes, as extensive ground movement can either compress or rupture the infrastructure (pipelines, underground utilities) or cause a collapse (above ground utilities, bridges). Roads and other

transportation infrastructure damaged during an earthquake would initially be unusable until inspected for structural integrity.

Recent earthquakes worldwide depict a pattern of steadily increasing damage and losses that are due to significant growth in earthquake-prone urban areas and vulnerability of older building stock, including buildings constructed within the past 20 years. In 2017 FEMA released the Hazus Estimated Annualized Earthquake Losses for the United States that reported recent earthquakes show a pattern of steadily increasing damages and losses due to:

- Significant growth in earthquake-prone urban areas
- Vulnerability of the older building stock

Data in the report details the Annualized Earthquake Loss (AEL) measures, the annualized earthquake losses (for property) in any single year, and the Annualized Earthquake Loss Ratio (AELR), which addresses seismic risk in relation to the value of the buildings in the study area. In order to calculate the AELR the AEL were divided by building replacement values. The report notes that Rhode Island, in relation to other states, is the:

- 42nd state for expected annualized earthquake losses at \$1,944,000
- 33rd for the Annualized Earthquake Loss Ratio of 14.5 (\$/million \$)

The Hazus Estimated Annualized Earthquake Losses for the United States report also notes that Rhode Island can expect:

- 23,000 tons of debris from an estimated 250-year event (39th in nation.)
- 134,000 tons of debris from an estimated 1,000-year event (39th in nation)
- 34 people requiring short term shelter from an estimated 250-year event (37th in nation)
- 284 people requiring short term shelter from an estimated 1,000-year event (37th in nation)
- No minor, life threatening or fatal injuries from any modeled event

Further data was obtained from a probabilistic earthquake scenario completed to estimate the annualized property losses expected from an earthquake in Rhode Island. Annualized losses for each county are estimates as follows:

- Bristol County: \$155,000
- Kent County: \$531,000
- Newport County: \$277,000
- Providence: \$2,195,000
- Washington: \$3,578,000

State Operations and Assets

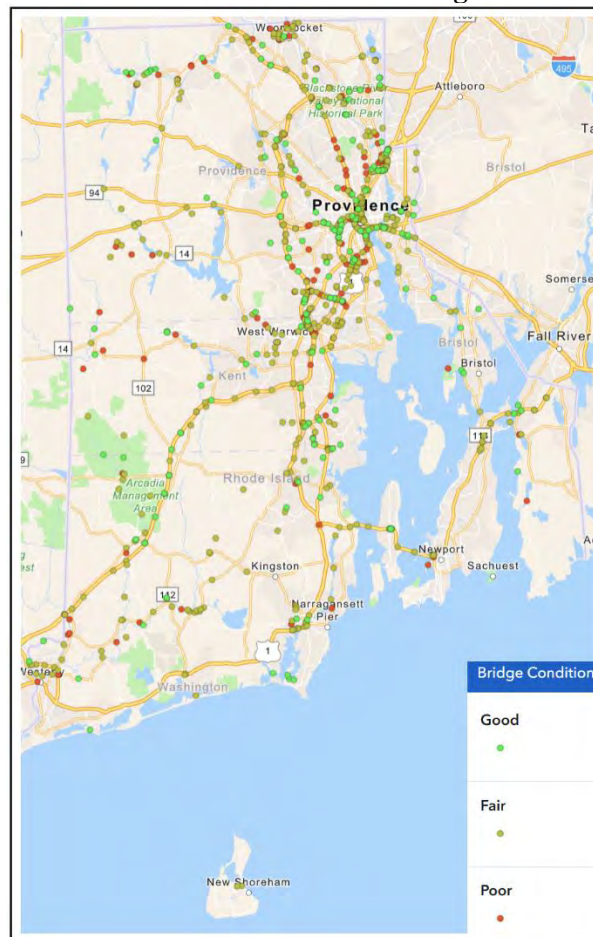
The risk to state operations and facilities from earthquakes can vary significantly depending on the level of seismic activity. Potential risks to state operations and buildings from earthquakes include:

- **Structural Damage:** Earthquakes can cause significant structural damage. The extent of damage depends on the earthquake's magnitude, depth, and proximity to populated areas.
- **Injury and Loss of Life:** Earthquakes can result in injuries and loss of life among state employees, particularly if buildings are not constructed to withstand seismic forces or if there are insufficient emergency evacuation plans and procedures in place.
- **Disruption of Government Operations:** Earthquakes can disrupt the normal functioning of state operations. Damaged buildings may need to be evacuated or temporarily closed for repairs, which can affect the delivery of services.
- **Communication Disruptions:** Seismic activity can damage communication infrastructure, including phone lines, cell networks, and data networks. This can hinder the ability to communicate internally and with the public during and after an earthquake.

- **Power Outages:** Earthquakes can lead to power outages by damaging electrical infrastructure such as substations and power lines. State buildings may lose power, affecting critical operations and services.
- **Loss of Records and Data:** Earthquakes can result in the loss of important records and data. This can have legal and operational implications.
- **Emergency Response:** State agencies may need to activate emergency response plans, deploy first responders, and coordinate relief efforts in the aftermath of a significant earthquake.
- **Budgetary Impact:** The costs associated with repairing and retrofitting state buildings and infrastructure after an earthquake can be substantial.

The structural stability of bridges and overpasses remains a concern during strong earthquake events. Bridges inspected and found to be in poor condition are considered at a higher risk of damage or failure during strong ground movement. The following map, from RIDOT, indicates all state-maintained bridges within the State of Rhode Island and their latest condition assessment:

Map 11: State of Rhode Island Maintained Bridge Condition Assessment



Source: RIDOT

The United State Department of Transportation Federal Highway Administration 2020 Bridge Replacement Unit Costs report indicates that the replacement cost of a bridge in Rhode Island to be \$373 per square foot. Using this cost, the estimated 2020 total costs for the replacement of all bridges found to be in poor condition is \$497,017,198. The estimated 2020 total costs for the rehabilitation of all bridges found to be in poor condition is \$ 337,971,695.

All state assets are vulnerable to damage from earthquakes, The following table details the complete inventory of assets, by county and by facility construction type, along with the most recent valuation for both the facility envelope and assets housed with the facility for the State of Rhode Island:

Table 54: State of Rhode Island Facilities and Assets by County

Jurisdiction	State Owned or Operated Facilities	Valuation
Bristol County	32	\$150,415,810
Kent County	116	\$215,026,614
Newport County	77	\$127,353,837
Providence County	423	\$2,526,593,056
Washington County	393	\$274,350,750
State Total	1,041	\$3,293,740,067

Source: RIEMA

Note: Valuations include structure and assets

Of particular concern to the state are unreinforced masonry buildings as they may be particularly vulnerable to the effect of a moderate to large earthquake. An unreinforced masonry building is constructed of block, brick, masonry, or stone with no steel reinforcing bars. Because these buildings were not built using modern building codes, they are much more likely to experience damage or collapse during an earthquake. Data was provided concerning all state-owned facilities classified as masonry buildings, including block, brick, joisted masonry, masonry, and stone buildings, by RIEMA and RIGIS. It is important to note that the inventory did not identify if the building was classified as unreinforced. The following table details the complete inventory of all masonry buildings, by city, along with the most recent valuation for both the facility envelope and assets housed with the facility for the State of Rhode Island:

Table 55: State of Rhode Island Masonry Facilities by Jurisdiction

Jurisdiction	Number of Masonry Buildings	Valuation
Barrington	1	\$3,779,049
Bristol	1	\$1,043,574
Burrillville	20	\$105,089,917
Central Falls	0	\$0
Charlestown	7	\$3,011,496
Coventry	1	\$3,066,667
Cranston	85	\$998,190,750
Cumberland	1	\$2,156,093
East Greenwich	1	\$2,533,491
East Providence	2	\$5,896,751
Exeter	7	\$7,523,589
Foster	2	\$2,956,069
Glocester	5	\$4,153,295
Hopkinton	2	\$3,284,845
Jamestown	5	\$6,519,441
Johnston	3	\$2,682,821
Lincoln	10	\$5,036,180
Little Compton	1	\$633,129
Middletown	2	\$4,908,818
Narragansett	15	\$28,514,833
New Shoreham	0	\$0
Newport	11	\$57,082,954
North Kingstown	14	\$89,729,123
North Providence	1	\$3,270,502
North Smithfield	1	\$2,102,383
Pawtucket	4	\$6,796,876
Portsmouth	4	\$4,649,444
Providence	57	\$814,326,533
Richmond	2	\$3,253,098
Scituate	11	\$110,064,936
Smithfield	1	\$3,029,717

Table 55: State of Rhode Island Masonry Facilities by Jurisdiction

Jurisdiction	Number of Masonry Buildings	Valuation
South Kingstown	10	\$21,177,010
Tiverton	1	\$119,995
Warren	1	\$119,995
Warwick	33	\$111,590,571
Westerly	7	\$18,936,993
West Greenwich	1	\$285,490
West Warwick	3	\$5,292,759
Woonsocket	8	\$20,036,430
State Total	341	\$2,442,809,187

Source: RIEMA

Note: Valuations include structure and assets

Utilizing information provided by the State of Rhode Island, and removing identifying addresses, the following facilities have been determined to be most vulnerable to potential earthquakes due to the age of construction and/or the total valuation of the building. for the State of Rhode Island:

Table 56: State of Rhode Island Masonry Facilities Most Vulnerable to Earthquakes

County	Jurisdiction	Valuation	Year Constructed
Providence County	Providence	\$10,459,508	1763
Newport County	Newport	\$6,875,212	1824
Providence County	Providence	\$11,051,412	1874
Providence County	Cranston	\$17,406,342	1878
Providence County	Cranston	\$17,406,342	1878
Providence County	Cranston	\$13,544,843	1890
Providence County	Cranston	\$10,224,779	1890
Providence County	Cranston	\$9,758,371	1890
Providence County	Cranston	\$8,161,594	1890
Providence	Providence	\$205,125,220	1900

Source: RIEMA

Note: Valuations include structure and assets

Consequence Analysis

This consequence analysis lists the potential impacts of this hazard on various elements of community and state infrastructure. The impact of this hazard is evaluated in terms of disruption of operations, recovery challenges, and overall wellbeing to all Rhode Island residents and first responder personnel. The consequence analysis supplements the hazard profile by analyzing specific impacts.

Table 57: Earthquake Consequence Analysis

Subject	Potential Impacts
Impact on the Public	Earthquakes may cause injury or death to people from vehicle accidents, falling objects, or structural failure. There may be a large number of people seeking treatment for traumatic injuries. Ground shaking may result in broken service lines or pipelines, triggering the release of hazardous materials or waste materials.
Impact on Responders	The extent of the damage to infrastructure such as roads and bridges and communications can greatly impact first responders' ability to access or transport victims. Equipment, facilities, or other assets may be damaged and restrict first responders' capacity to respond to calls for assistance.
Continuity of Operations	Rhode Island agencies maintain continuity plans which can be enacted as necessary based on the situation. Earthquakes could potentially impact critical infrastructure resulting in power outages, access to roadways or public transportation, damage to facilities or infrastructure, including alternate locations.

Table 57: Earthquake Consequence Analysis

Subject	Potential Impacts
Delivery of Services	Delivery of services may be impacted by dangerous transportation conditions, causing food, water, and resource systems to be delayed or halted. Waterway infrastructure may be damaged or malfunction, stopping barge and ship traffic.
Property, Facilities, and Infrastructure	Rhode Island has many unreinforced masonry structures that are inherently vulnerable to seismic forces. All critical facilities and transportation corridors and pipelines can be impacted. Ground shaking can lead to the collapse of buildings and bridges, and disrupt gas, lifelines, electric, and phone service.
Impact on Environment	Earthquakes have the potential to trigger secondary hazards such as fire, flash flooding, hazardous materials release, slope failure, dam failures, and tsunamis, all potentially devastating to the environment. These secondary hazards can completely wipe out habitats and environments, cause significant injury to animals or livestock, or contaminate certain components of the environment.
Economic Conditions	Earthquakes pose a fiscal impact on the local and state governments, even if some of those costs can be recouped through federal grant reimbursements. Local, county, and state resources may be drained by response and recovery efforts. Additionally, a severe earthquake would affect the ability of businesses to maintain operations. If the private sector is not able to re-establish operations this would also impact the state economy.
Public Confidence in Governance	Governmental response, on all levels – state and local – requires direct actions that must be immediate and effective to maintain public confidence. If the state takes a long time to begin recovery operations, or for the public to see recover operations, this will have a negative impact on the public’s confidence in the state’s governance.

4.10.7 Jurisdictional Vulnerability and Planning Significance

Utilizing data from FEMA’s NRI and from LHMPs, the following tables detail the hazard ratings for jurisdictions, by census tract, for earthquake. Data concerning both risk index and EAL is reported by census tract. As many State of Rhode Island jurisdictions encompass multiple tracts, an average of tract ratings was completed to determine the overall jurisdictional rating. The following table details the hazard rating by LHMP analysis, FEMA NRI, and EAL by jurisdiction for the State of Rhode Island:

Table 58: State of Rhode Jurisdictional Earthquake Planning Significance

County	Jurisdiction	Earthquake LHMP Rating	Earthquake NRI	Earthquake EAL
Bristol	Barrington	Low	Very Low	Very Low
	Bristol	-	Very Low	Very Low
	Warren	Not rated	Very Low	Very Low
Kent	Coventry	-	Very Low	Very Low
	East Greenwich	Low	Very Low	Relatively Low
	Warwick	Low	Very Low	Very Low
	West Greenwich	-	Very Low	Very Low
	West Warwick	Low	Very Low	Very Low
Newport	Jamestown	-	Very Low	Very Low
	Little Compton	Not Rated	Very Low	Very Low
	Middletown	Low	Very Low	Very Low
	Newport	Low	Very Low	Very Low
	Portsmouth	Low	Very Low	Very Low
	Tiverton	-	Very Low	Very Low
Providence	Burrillville	Moderate	Moderate	Very Low
	Central Falls	Low	Low	Very Low
	Cranston	Low	Low	Very Low
	Cumberland	-	-	Very Low
	East Providence	Moderate	Moderate	Very Low
	Foster	Low	Moderate	Very Low

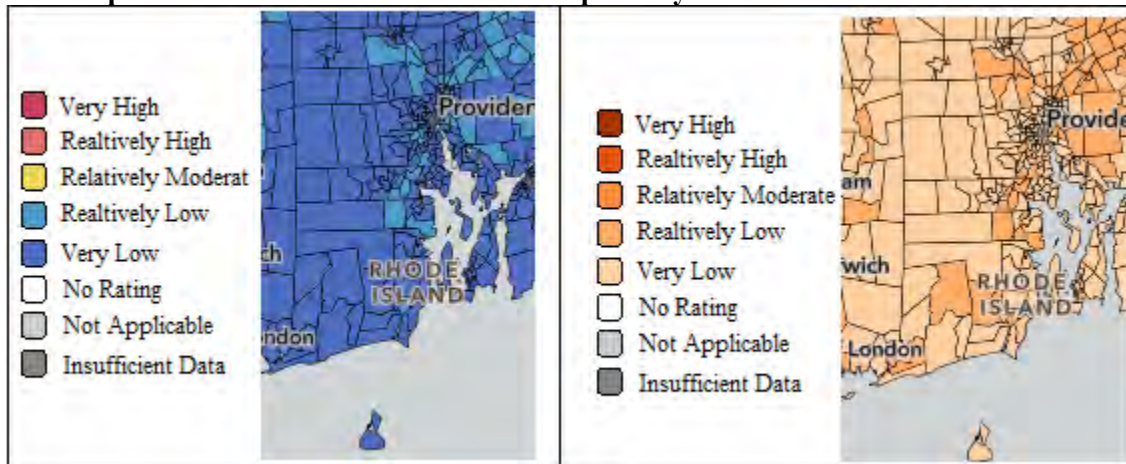
Table 58: State of Rhode Jurisdictional Earthquake Planning Significance

County	Jurisdiction	Earthquake LHMP Rating	Earthquake NRI	Earthquake EAL
	Glocester	Not Rated	Not Rated	Very Low
	Johnson	Moderate	Moderate	Very Low
	Lincoln	Low	Low	Relatively Low
	North Providence	Low	Low	Very Low
	North Smithfield	Moderate	Not Rated	Very Low
	Pawtucket	Low	Low	Very Low
	Providence	Low	Low	Very Low
	Scituate	-	-	Very Low
	Smithfield	-	-	Very Low
	Woonsocket	Low	Low	Very Low
Washington	Charlestown	-	Very Low	Very Low
	Exeter	-	Very Low	Very Low
	Hopkinton	-	Very Low	Very Low
	Narragansett	Not rated	Very Low	Very Low
	New Shoreham	-	Very Low	Very Low
	North Kingstown	Moderate	Very Low	Very Low
	Richmond	Low	Very Low	Relatively Low
	South Kingstown	Low	Very Low	Very Low
	Westerly	Moderate	Very Low	Very Low

Source: LHMPs and FEMA NRI

Note: - indicates no current plan available for review

The following maps indicate the FEMA NRI and EAL for earthquake by census tract for the state of Rhode Island. The

Map 12: FEMA NRI and EAL for Earthquake by Census Tract for Rhode Island

Source: FEMA NRI

While all of Rhode Island will be impacted by a large earthquake, a disproportionate impact may be felt by jurisdictions with older housing units as they may have been constructed using materials and techniques that are less resilient to hazards. Modern construction materials and methods, such as reinforced concrete and steel frames, are often more resilient. The following table Rhode Island Department of Housing 2022 Integrated Housing Report from the Rhode Island Department of Housing 2022 Integrated Housing Report indicates jurisdictions with greater than 40% of their housing units constructed before 1939:

Table 59: Jurisdictions at Greater Risk to Earthquakes Due to Age of Housing

County	Jurisdiction	1939 and later
Providence	Providence	57.70%
Newport	Newport	55.20%

Table 59: Jurisdictions at Greater Risk to Earthquakes Due to Age of Housing

County	Jurisdiction	1939 and later
Providence	Central Falls	50.40%
Providence	Woonsocket	43.50%
Providence	Pawtucket	43.30%
Bristol	Warren	41.40%

Source: Rhode Island Department of Housing 2022 Integrated Housing Report

Based on FEMA NRI social vulnerability ratings, and on U.S. Census Bureau data concerning percentage of the population below the 100% poverty who may difficulty seismically retrofitting for their homes for earthquakes and may be unable to evacuate if homes are damage or destroyed, the following communities may be at greater risk to earthquakes:

Table 60: Jurisdictions at Greater Risk to Earthquakes Due to Poverty and Social Vulnerability

County	Jurisdiction	Population Below 100% of the Poverty Level	Social Vulnerability Rating
Providence County	Central Falls	29.9%	Very High
Providence County	Providence	23.0%	Relatively High
Providence County	Woonsocket	21.0%	Relatively Moderate
Washington County	Narragansett	19.3%	Relatively High
Newport County	Newport	14.4%	Relatively Low
Kent County	West Warwick	12.1%	Relatively Low
Providence County	Pawtucket	11.6%	Relatively High

Source: FEMA NRI and U.S. Census Bureau

4.11.8 State of Rhode Island Hazard Planning Significance

Based on a review of all available data, the SIHMC determined earthquake to have a low planning significance for Rhode Island.

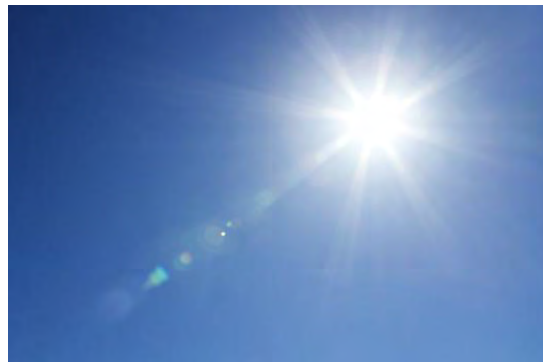
4.12 Extreme Temperatures

4.12.1 Hazard Description

Extreme temperature events occur when climate conditions produce temperatures well outside of the predicted norm. These extremes can have severe impacts on human health and mortality, natural ecosystems, agriculture, and other economic sectors.

The CDC identifies the following six groups as being especially vulnerable to extreme temperatures:

- Older Adults (aged 65)
- Infants and Children
- Individuals with Chronic Conditions
- Low-income Individuals
- Athletes
- Outdoor workers



4.12.2 Location and Extent

In general, most of Rhode Island has a humid continental climate, with warm summers and cold winters. The state's southern coastal portions are the broad transition zone into subtropical climates, with hot summers and cool winters with a mix of rain and snow. The mean annual temperature ranges from 48° Fahrenheit (° F) to 51° F in Narragansett Bay and around Providence.

The average daily minimum temperature in January and February is 20° F for much of the State, and around 25° F in immediate coastal areas. The number of days with temperatures at or below 0° F averages one or less per year in Narragansett Bay and coastal areas and around five per year in much of the State. The lowest recorded temperature in Rhode Island was -28° F in 1942.

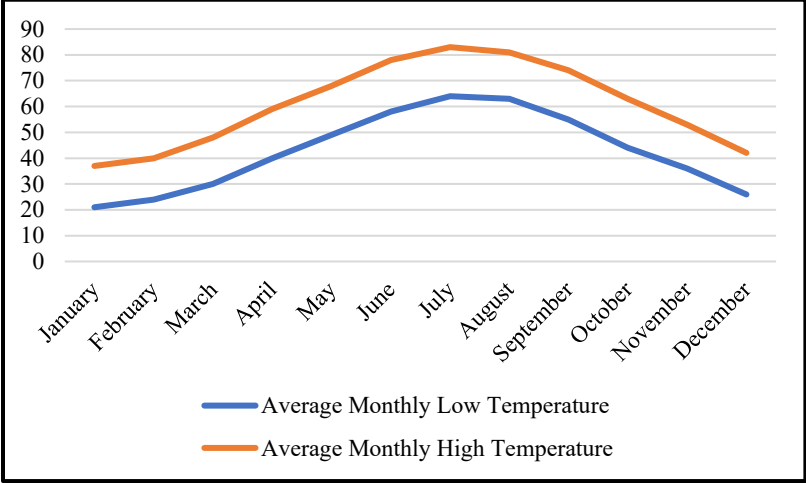
Metropolitan areas and portions of Northern Rhode Island can expect eight to 10 days of temperatures of 90° F or higher. Coastal areas can expect fewer hot days, with an average of one 90° F. Temperatures of 100° F or higher have been recorded in the northern interior occasionally. The highest recorded temperature in Rhode Island was 104° F in 1975.

All of Rhode Island is at risk to extreme temperatures, defined as:

- Extreme heat is defined as temperatures that hover 10 degrees or more above the average high temperature for the region and last for several weeks. Ambient air temperature is one component of heat conditions, with relative humidity being the other. Humid or muggy conditions, which add to the discomfort of high temperatures, occur when an area of high atmospheric pressure traps moisture laden air near the ground.
- Although no specific definition exists for extreme cold, an extreme cold event can generally be defined as temperatures at or below freezing for an extended period of time. Extreme cold events are usually part of winter storm events but can occur during anytime of the year and can have devastating effects on agricultural production.

The following chart details monthly temperature averages for Rhode Island:

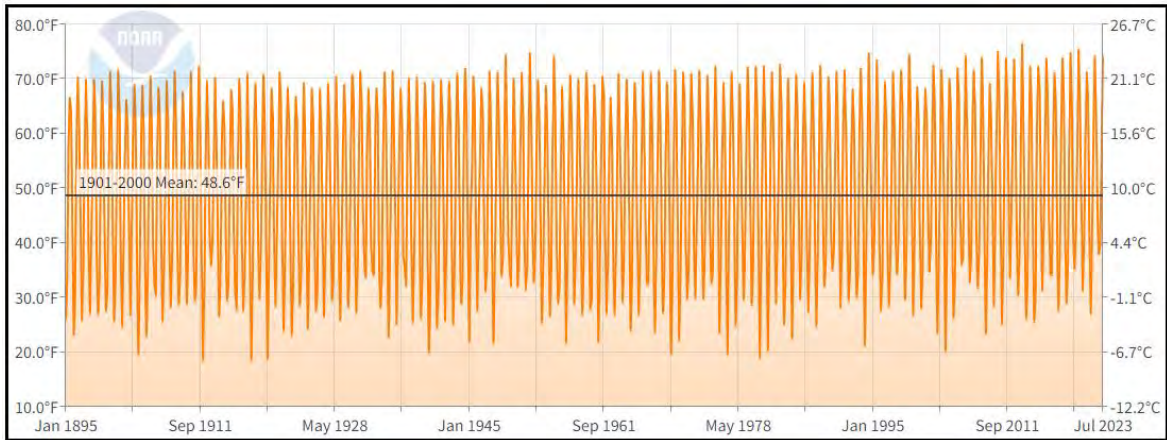
Chart 19: Rhode Island Average Monthly Temperatures (Degrees Fahrenheit)



Source: NOAA

The following chart details the average temperature for Rhode Island from 1895 to 2023.

Chart 20: Rhode Island Average Temperature, 1895 - 2023

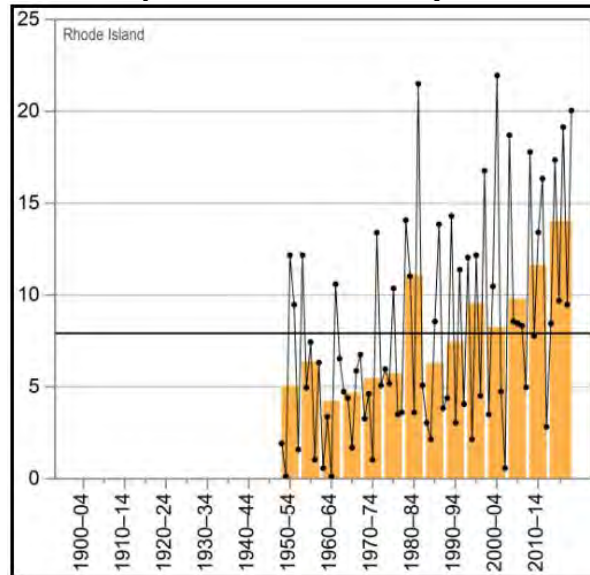


Source: NOAA

4.12.3 Previous Occurrences

The following chart details the annual number of hot days (maximum temperature of 90°F or higher) for Rhode Island from 1950 to 2020. Data indicates that the number of hot days has been above average since the 1990s, with the highest number hot days occurring from 2015–2020 (an average of 14 hot days per year).

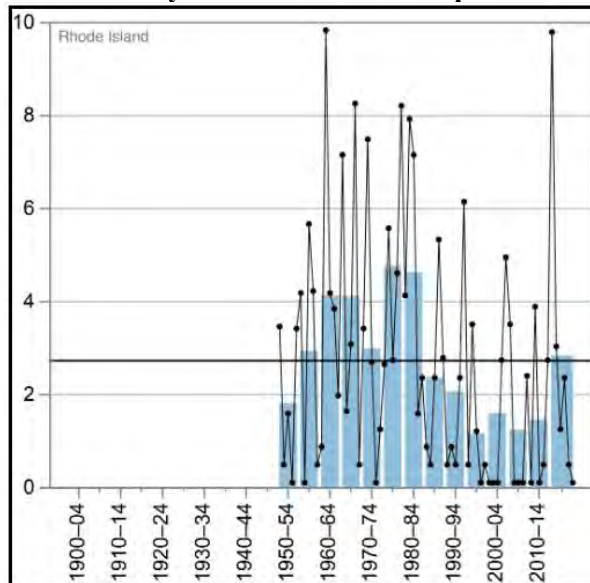
Chart 21: Number of Days with Maximum Temperature of 90° F or Higher



Source: NOAA State Climate Summary 2022 for Rhode Island

The following chart details the annual number of very cold days (minimum temperature of 0°F or lower) for Rhode Island from 1950 to 2020. The average number of very cold days was highest in the late 1970s and early 1980s. Beginning in the 1980s, the annual number of very cold days was below average. Recently, from 2015 to 2020, the number of very cold days has been near average.

Chart 22: Number of Days with Minimum Temperature of 0° F or Less



Source: NOAA State Climate Summary 2022 for Rhode Island

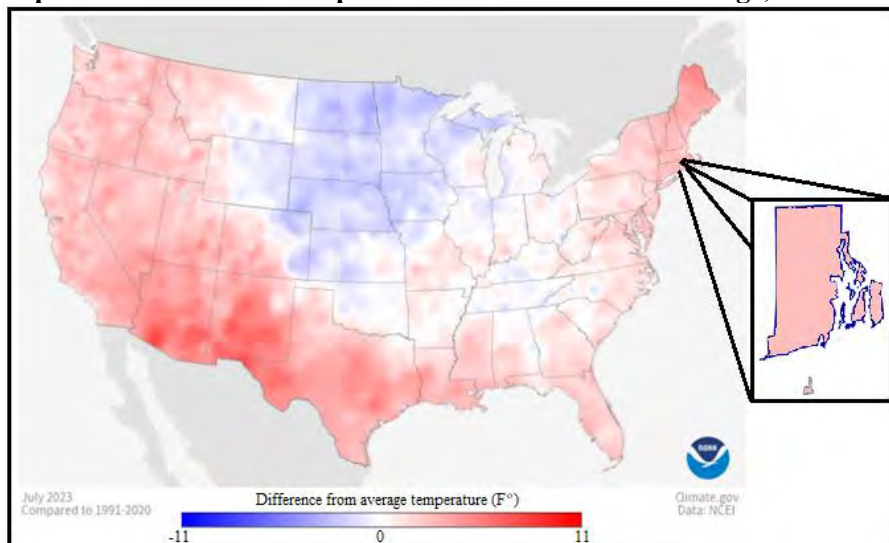
Additionally, data from the National Centers for Environmental Information from 1950 through 2023 indicates the following recorded extreme temperature events:

- Excessive Heat Events: Six
- Extreme Cold Events: Two

4.12.4 Probability of Future Events

Predicting the probability of extreme temperature occurrences is tremendously challenging due to the large number of factors involved. Available data suggests that both the average high temperatures and the record high temperature will likely increase over the coming years as indicated by the following map:

Map 13: Rhode Island Temperature Difference from Average, 1990 - 2020



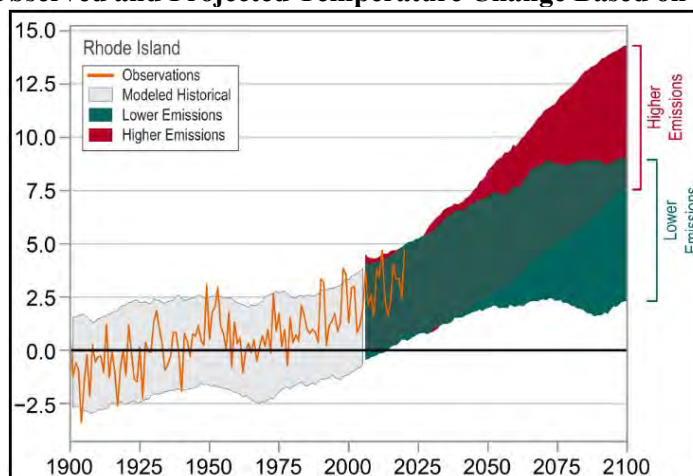
Source: NOAA

Temperatures in Rhode Island have risen by 4° F since the early 1900s, with the number of hot days above the long-term average since the 1990s. Additionally, the greatest number of warm nights has been recorded over the 2015–2020 period. Very cold days have been mostly below average since the 1980s.

4.12.5 Projected Changes in Location, Intensity, Frequency, and Duration

When discussing extreme temperatures, climate change should be considered as it may markedly change future events. Recent climate modeling results indicate that extreme temperature events may become more common for Rhode Island, especially heat. The following chart indicates the projected temperature change for Rhode Island utilizing two global climate models. One model utilizes information in which greenhouse gas emissions continue to increase (higher emissions), with the other model utilizing information in which greenhouse gas emissions increase at a slower rate (lower emissions). Temperatures in Rhode Island, detailed by the orange line, have risen 4° F since the beginning of the early 1900s. Based on both the higher emission and lower emission models, continued warming is projected throughout this century.

Chart 23: Rhode Island Observed and Projected Temperature Change Based on Greenhouse Gas Emissions



Source: NOAA State Climate Summary 2022 for Rhode Island

Rising average temperatures produce a more variable climate system which may result in an increase in the frequency and severity of some extreme temperature events. Additionally, rising temperatures can harm air quality and amplify existing threats to health. Warmer weather can increase the production of ground-level ozone, a pollutant that causes lung and heart problems. Heat stress is expected to increase with hotter summer temperatures and more humidity. Children, the elderly, the sick, and those living below the poverty line may be especially vulnerable.

As previously noted, State of Rhode Island facilities have seen no major changes in the past five years, with only modest repairs and upgrades being conducted and no major rehabilitation or construction projects completed. As such, the risk to state assets has remained static since the completion of the 2019 SHMP. Additionally, with the exception of the two previously identified rehabilitation and construction projects, no large-scale facility projects were noted in any agency budgets or reports.

4.12.6 Vulnerability and Impact

People and Communities

A primary concern with this hazard is human health safety issues, as extreme temperatures can be a direct cause of death. Specific at-risk groups include outdoor workers, farmers, young children, and senior citizens. Compounding these concerns is the potential loss of electric power due to increased strain on power generation and distribution due to increased air conditioning or heating needs.

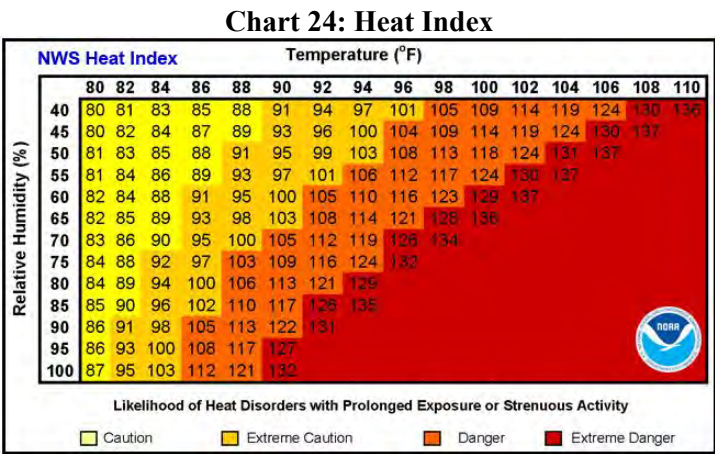
Extreme temperature impacts on humans can be measured for both heat and cold. The following table discusses potential impacts on human health related to excessive heat.

Table 61: Extreme Heat Impacts on Human Health

Heat Index Temperature	Potential Impact on Human Health
80-90° F	Fatigue possible with prolonged exposure and/or physical activity
90-105° F	Sunstroke, heat cramps, and heat exhaustion possible
105-130° F	Heatstroke/sunstroke highly likely with continued exposure

Source: National Weather Service Heat Index Program

Exposure to direct sun can increase Heat Index values by as much as 15°F. The zone above 105°F corresponds to a Heat Index that may cause increasingly severe heat disorders with continued exposure and/or physical activity. The following graph, from the NWS, indicates Heat Index values.



Source: NWS

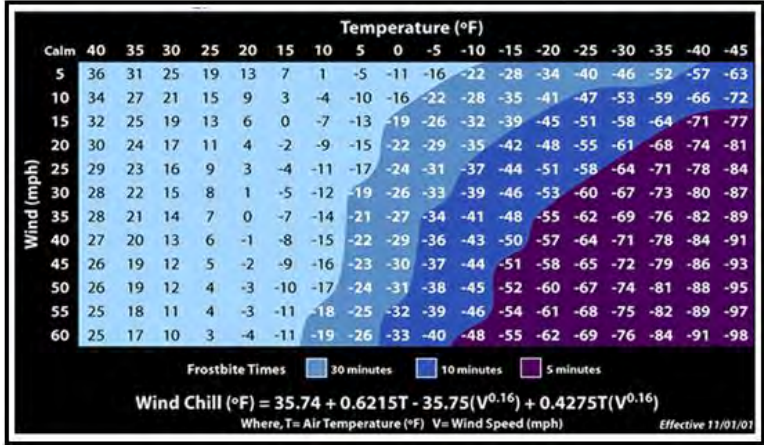
Extreme cold temperatures can result in a variety of concerns, including:

- Frostbite: The freezing of skin and the body tissue just beneath it
- Hypothermia: Dangerously low body temperature (and the most common winter weather killer)

When extremely cold temperatures are accompanied by strong winds the result can be potentially lethal wind chills. Wind chill is the temperature your body feels when the air temperature is combined with the wind speed and is based

on the rate of heat loss from exposed skin caused by the effects of wind and cold. As the speed of the wind increases, it can carry heat away from your body much more quickly, causing skin temperature to drop. The wind chill chart shows the difference between the actual air temperature and the perceived temperature due to wind, and amount of time until frostbite occurs.

Chart 25: Wind Chill Chart



Source: NOAA

Of particular concern, socially vulnerable and at-risk populations that may have difficulty with medical issues, poverty, extremes in age, and communications due to language barriers are often at increased susceptibility to the impacts of hazards. These populations suffer a disproportionate impact due to:

- Inadequate access to heating or cooling
- Constrained financial resources
- Limited access to healthcare
- Limited access to information

Facilities and Critical Facilities

Extreme temperatures, whether they are excessively high or low, can have differing impacts on homes and businesses. These effects depend on both the severity and duration of the event. Increased temperatures can lead to heat stress inside buildings, making spaces uncomfortable and potentially posing health risks, especially for vulnerable populations. Prolonged exposure to high temperatures can damage building materials, such as warping or cracking of wood, deterioration of roofing materials, and fading of paint. Additionally, hot and dry conditions can elevate the risk of wildfires, posing a direct threat to homes and businesses.

Extremely low temperatures can cause water pipes to freeze, leading to burst pipes and water damage when the pipes thaw. The accumulation of ice on roofs, known as ice dams, can lead to water damage as melting snow cannot properly drain off the roof. Extreme cold can also impact the structural integrity of a house, especially if it is not designed to withstand freezing temperatures.

Extreme temperatures can have significant impacts on critical facilities, with the specific effects varying depending on whether the extreme temperatures are high or low. During extremely high temperatures, the overheating of electronic equipment, such as servers and data storage systems, is a concern. A consistent strain on power infrastructure due to cooling demands can strain the power grid, leading to potential power outages. High temperatures can stress physical infrastructure, such as roads and bridges, potentially leading to structural damage.

Extremely low temperatures can cause water in pipes and equipment to freeze, leading to equipment failure and potential damage. Cold temperatures can also reduce the performance of batteries, affecting backup power systems and potentially leading to power outages. Roads, runways, and other transportation infrastructure may be affected by ice and snow, impacting the delivery of critical supplies and personnel.

Environment

Extreme heat can cause significant damage to the local environment by dehydrating vegetation and wildlife, which may result in cascading effects to the surrounding environment, such as drought, wildfires, mudslides, or landslides. Extreme temperatures may severely decrease the yield of Rhode Island’s agricultural sector. The yield of cash crops may be reduced, livestock may be adversely impacted by extreme heat, or grazing losses may be incurred by farmers or ranchers; potentially resulting in decreased food security. In the event of significant agricultural losses caused by extreme heat or drought, some assistance may be available to impacted farms or ranches.

State Operations and Assets

Extreme temperatures can pose various risks to state operations and facilities, and may include:

- **Health and Safety Risks:** High temperatures, especially during heatwaves, can pose significant health risks to state employees. Heat-related illnesses such as heat exhaustion and heatstroke can occur, potentially leading to hospitalizations or fatalities. Cold temperatures can also lead to cold-related illnesses and injuries, such as frostbite and hypothermia.
- **Power Grid Strain:** Extreme temperatures, whether hot or cold, can lead to increased demand for electricity. This can strain the power grid, potentially causing power outages, which can disrupt state operations, including the functioning of critical infrastructure such as hospitals, emergency services, and data centers.
- **Infrastructure Stress:** State buildings and infrastructure can suffer damage due to extreme temperatures. For example, prolonged exposure to high temperatures can lead to structural damage, such as cracking and warping. Extreme cold can freeze and damage pipes, leading to water leaks and flooding when temperatures rise.
- **Transportation Disruptions:** Extreme heat can cause pavement to soften and buckle, leading to road closures and transportation disruptions. Extreme cold can result in icy road conditions and reduce visibility, making travel hazardous.
- **Emergency Response:** State agencies may need to respond to extreme weather events, such as providing emergency shelter during heatwaves or responding to weather-related accidents and emergencies. These responses can strain resources and personnel.
- **Budgetary Impact:** The costs associated with responding to and mitigating the effects of extreme temperatures can strain state budgets. This includes expenses related to emergency response, infrastructure repairs, and healthcare.

Of concern to the state are wood framed buildings as they may be particularly vulnerable to negative impacts of temperature extremes. Wood has the tendency to expand and contract with changes in temperature. In hot temperatures, wood absorbs moisture from the air, causing it to expand. This expansion and contraction can lead to warping, cracking, and distortion of wooden components in the building. Wood has a relatively high coefficient of thermal expansion, meaning it expands and contracts significantly with temperature changes. This thermal movement can affect the structural integrity of a building, especially if not accounted for in the design and construction.

The following table details the complete inventory of all wood framed buildings, by city, along with the most recent valuation for both the facility envelope and assets housed with the facility for the State of Rhode Island:

Table 62: State of Rhode Island Wood Framed Facilities by City

Jurisdiction	Number of Wood Framed Buildings	Valuation
Barrington	5	\$6,464,007
Bristol	16	\$126,216,260
Burrillville	14	\$10,008,460
Central Falls	1	\$573,157
Charlestown	39	\$10,905,022
Coventry	17	\$20,106,395
Cranston	17	\$21,623,231
Cumberland	4	\$2,104,243
East Greenwich	5	\$5,526,595

Table 62: State of Rhode Island Wood Framed Facilities by City

Jurisdiction	Number of Wood Framed Buildings	Valuation
East Providence	9	\$6,630,445
Exeter	20	\$9,685,127
Foster	2	\$1,564,162
Glocester	18	\$6,798,862
Hopkinton	6	\$3,831,082
Jamestown	4	\$2,453,756
Johnston	20	\$13,886,184
Lincoln	21	\$19,087,783
Little Compton	1	\$1,016,351
Middletown	9	\$7,739,128
Narragansett	144	\$42,692,879
New Shoreham	0	\$0
Newport	16	\$13,502,570
North Kingstown	14	\$5,033,308
North Providence	3	\$1,430,982
North Smithfield	6	\$5,488,058
Pawtucket	4	\$2,755,235
Portsmouth	16	\$9,526,961
Providence	14	\$79,641,720
Richmond	37	\$14,167,169
Scituate	7	\$7,530,405
Smithfield	6	\$6,910,817
South Kingstown	43	\$15,990,825
Tiverton	8	\$6,707,488
Warren	4	\$2,224,020
Warwick	33	\$21,581,255
Westerly	13	\$7,754,951
West Greenwich	1	\$158,337
West Warwick	1	\$506,889
Woonsocket	6	\$3,789,742
State Total	604	\$523,613,859

Source: RIEMA

Note: Valuations include structure and assets

The age of a state owned or operated structure could have a significant impact on its resilience to extreme temperatures as an older structure may be less resilient due to the following factors:

- **Building Codes and Standards:** Older buildings may not have been constructed to meet modern building codes, which often incorporate lessons learned from previous disasters. Newer buildings are typically designed to be more resilient to known hazards.
- **Construction Materials and Techniques:** Older buildings may have been constructed using materials and techniques that are less resilient to hazards. Modern construction materials and methods, such as reinforced concrete and steel frames, are often more resilient to hazards.

Utilizing information provided by the State of Rhode Island, and removing identifying addresses, the following wood framed facilities have been determined to be most vulnerable to potential extreme temperature events due to the age of construction and/or the total valuation of the building, for the State of Rhode Island:

Table 63: State of Rhode Island Wood Framed Facilities Most Vulnerable to Extreme Temperatures

County	Jurisdiction	Valuation	Year Constructed
Providence County	Providence	\$447,093.75	1707

Table 63: State of Rhode Island Wood Framed Facilities Most Vulnerable to Extreme Temperatures

County	Jurisdiction	Valuation	Year Constructed
Washington County	Richmond	\$422,882.64	1790
Providence County	Scituate	\$2,362,303.84	1799
Providence County	Providence	\$2,030,638.26	1850
Providence County	Scituate	\$241,356.48	1850
Providence County	Woonsocket	\$856,206.69	1875
Providence County	East Providence	\$1,070,644.45	1890
Providence County	Johnston	\$859,381.12	1890
Newport County	Tiverton	\$526,023.09	1890
Newport County	Tiverton	\$2,053,952.92	1890
Newport County	Portsmouth	\$825,347.53	1895

Source: RIEMA

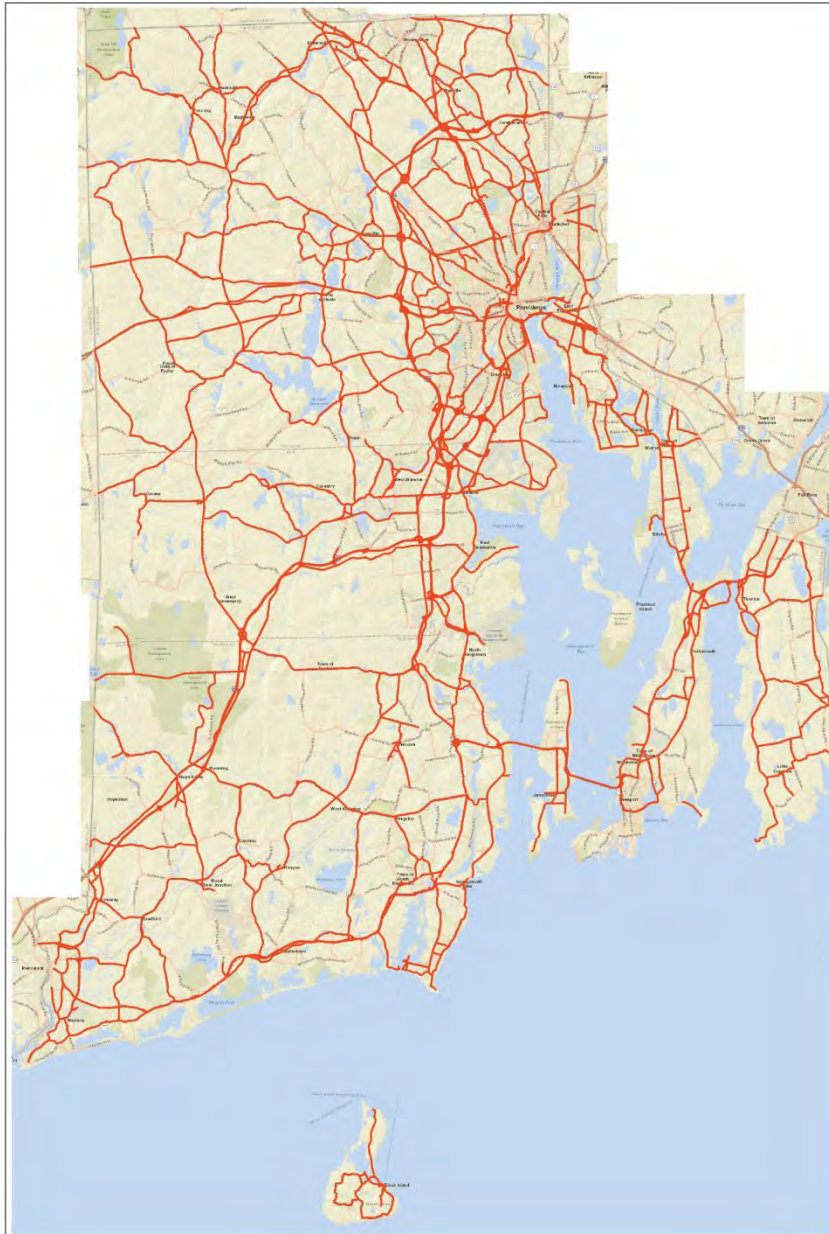
Note: Valuations include structure and assets

Of particular concern during extremely cold temperatures is continued access to roads for travel. Approximately 80% of all the miles of roads in Rhode Island (approximately 5,000 miles) are maintained by cities and towns. RIDOT maintains the balance of roads, with approximately 1,100 miles of state roads under its jurisdiction. RIDOT has a fleet of 178 plow trucks, and access to 150 vendors who help bring the total available number of plow trucks up to 500. During a typical winter, RIDOT from its 20 statewide stockpile locations, consumes approximately:

- 125,000 tons of salt
- 10,000 tons of sand
- 14,000 gallons of brine

The following map, from RIDOT, details state-maintained roads:

Map 14: RIDOT Maintained Roads



Source: RIDOT

Consequence Analysis

This consequence analysis lists the potential impacts of this hazard on various elements of community and state infrastructure. The impact of this hazard is evaluated in terms of disruption of operations, recovery challenges, and overall wellbeing to all Rhode Island residents and first responder personnel. The consequence analysis supplements the hazard profile by analyzing specific impacts.

Table 64: Extreme Temperature Consequence Analysis

Subject	Potential Impacts
Impact on the Public	Extreme temperatures can have severe consequences for health, particularly for the elderly and young. Loss of electricity may impact heating or air conditioning leading to poorly tolerated indoor temperatures. Physical effects of extreme temperatures can cause major health problems and may lead to injury or death.
Impact on Responders	Without proper mitigation efforts, responders may be susceptible to temperature related illness. Extreme temperatures may also damage instruments or equipment

Table 64: Extreme Temperature Consequence Analysis

Subject	Potential Impacts
	necessary for response activities. First responders may face dangerous road conditions leading to accidents and prolonged response times.
Continuity of Operations	Rhode Island agencies maintain continuity plans which can be enacted as necessary based on the situation. This hazard may impact an agency's ability to implement continuity operations based due to power outages. If the activation of alternate facilities was required, continuity of operations may be difficult due to lack of computer/network access during power outages.
Delivery of Services	Extreme temperatures can impact efficient delivery or inability of goods or services due to potential health impacts on workers. Equipment and vehicles may be damaged, and the delivery of services may be delayed due to poor travel conditions
Property, Facilities, and Infrastructure	Facility integrity is at risk with regards to power cables and stations being overused and limiting operations. This could lead to limits on facility heating or cooling.
Impact on Environment	Extreme temperatures can cause significant damage to the local environment and result in habitat loss, invasive species, and changes in migration. Extreme temperatures may severely decrease the yield of cash crops. Livestock are adversely affected by extreme temperatures and may suffer medical problems or death.
Economic Conditions	Extreme temperatures drain state and local resources. Under some conditions, some of the costs can be recouped through federal grant reimbursements.
Public Confidence in Governance	Governmental response, on all levels, requires direct actions that must be immediate and effective to maintain public confidence.

4.12.7 Jurisdictional Vulnerability and Planning Significance

Utilizing data from FEMA's NRI and from LHMPs, the following tables detail the hazard ratings for jurisdictions, by census tract, for extreme temperatures. Data concerning both risk index and EAL is reported by census tract. As many State of Rhode Island jurisdictions encompass multiple tracts, an average of tract ratings was completed to determine the overall jurisdictional rating. The following table details the hazard rating by LHMP analysis, FEMA NRI, and EAL by jurisdiction for the State of Rhode Island:

Table 65: State of Rhode Jurisdictional Extreme Temperature Planning Significance

County	Jurisdiction	Extreme Temperature LHMP Rating	Heatwave NRI	Heatwave EAL	Cold Wave NRI	Cold Wave EAL
Bristol	Barrington	High	Relatively Low	Relatively Low	Relatively High	Relatively Low
	Bristol	-	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Warren	No Rating	Relatively Low	Relatively Low	Relatively Low	Relatively Low
Kent	Coventry	-	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	East Greenwich	Moderate	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Warwick	Moderate	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	West Greenwich	-	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	West Warwick	Moderate	Relatively Low	Relatively Low	Relatively Low	Relatively Low
Newport	Jamestown	-	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Little Compton	Moderate	Relatively Low	Relatively Low	Very Low	Relatively Low
	Middletown	Not Rated	Very Low	Very Low	Very Low	Very Low
	Newport	Moderate	Very Low	Very Low	Very Low	Very Low
	Portsmouth	Moderate	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Tiverton	-	Very Low	Relatively Low	Very Low	Relatively Low
Providence	Burrillville	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Central Falls	Moderate	Relatively Low	Relatively Low	Relatively High	Relatively Moderate
	Cranston	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate

Table 65: State of Rhode Jurisdictional Extreme Temperature Planning Significance

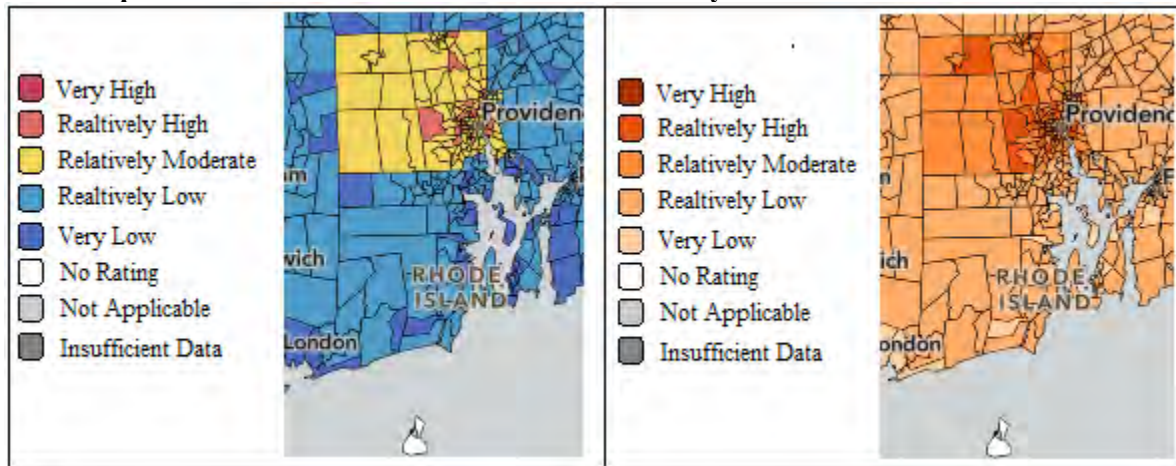
County	Jurisdiction	Extreme Temperature LHP Rating	Heatwave NRI	Heatwave EAL	Cold Wave NRI	Cold Wave EAL
	Cumberland	-	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	East Providence	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Foster	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Glocester	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Johnson	Moderate	Relatively Low	Relatively Moderate	Relatively High	Relatively Moderate
	Lincoln	Low	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	North Providence	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	North Smithfield	Not Rated	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Pawtucket	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Providence	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Scituate	-	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Smithfield	-	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Woonsocket	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
Washington	Charlestown	-	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Exeter	-	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Hopkinton	-	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Narragansett	Moderate	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	New Shoreham	-	No Rating	No Rating	No Rating	No Rating
	North Kingstown	Moderate	Very Low	Relatively Low	Very Low	Relatively Low
	Richmond	Not Rated	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	South Kingstown	Moderate	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Westerly	Moderate	Relatively Low	Relatively Low	Relatively Low	Relatively Low

Source: LHMPs and FEMA NRI

Note: - indicates no current plan available for review

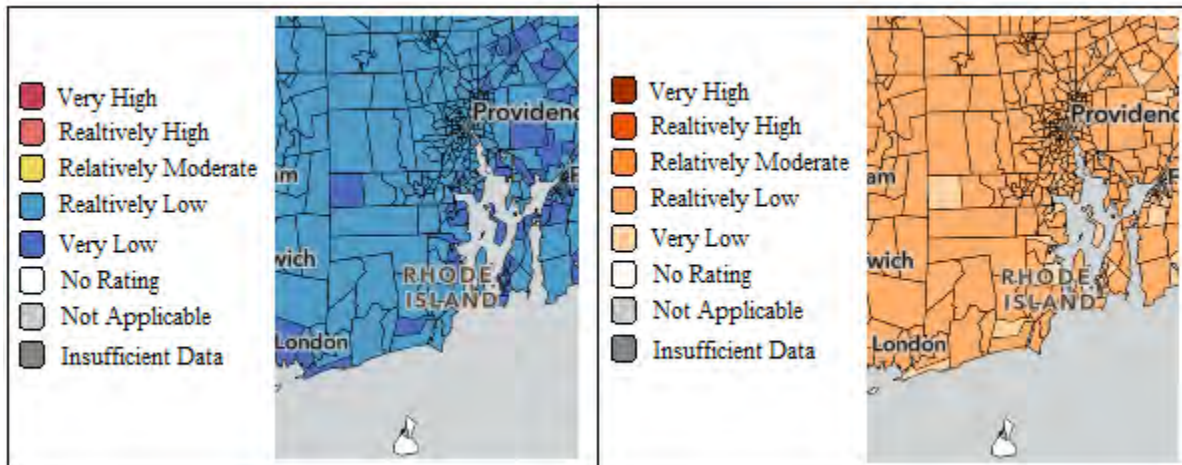
The following maps indicate the FEMA NRI and EAL for extreme heat and extreme cold by census tract for the state of Rhode Island.

Map 15: FEMA NRI and EAL for Extreme Heat by Census Tract for Rhode Island



Source: FEMA NRI

Map 16: FEMA NRI and EAL for Extreme Cold by Census Tract for Rhode Island



Source: FEMA NRI

Based on FEMA NRI social vulnerability ratings, and on U.S. Census Bureau data concerning percentage of the population below the 100% poverty who may difficulty heating and cooling the residences, the following communities may be at greater risk to extreme temperature conditions:

Table 66: Jurisdictions at Greater Risk to Extreme Temperatures Due to Poverty and Social Vulnerability

County	Jurisdiction	Population Below 100% of the Poverty Level	Social Vulnerability Rating
Providence County	Central Falls	29.9%	Very High
Providence County	Providence	23.0%	Relatively High
Providence County	Woonsocket	21.0%	Relatively Moderate
Washington County	Narragansett	19.3%	Relatively High
Newport County	Newport	14.4%	Relatively Low
Kent County	West Warwick	12.1%	Relatively Low
Providence County	Pawtucket	11.6%	Relatively High

Source: FEMA NRI and U.S. Census Bureau

A household is considered cost burdened if it spends 30% or more of its income on housing costs, with cost burdens being an indication of rents increasing faster than incomes. Cost burdened citizens may have great difficulty paying heating and electric bills. The following table, using data from the Housing and Urban Development's Comprehensive

Housing Affordability Strategy Data for Rhode Island, indicates jurisdictions with the most cost burdened renters who may be at greater risk to extreme temperatures due to economic constraints.

Table 67: Jurisdictions at Greater Risk to Severe Winter Storms due to Housing Cost Burden

County	Jurisdiction	Cost Burdened Single Families
Kent County	Exeter	73%
Kent County	Narragansett	65%
Bristol County	Barrington	64%
Providence County	Cumberland	54%
Newport County	Portsmouth	54%
Providence County	Smithfield	53%
Providence County	Central Falls	52%
Bristol County	Warren	52%
Kent County	Warwick	50%

Source: Housing and Urban Development's Comprehensive Housing Affordability Strategy Data

4.12.8 State of Rhode Island Hazard Planning Significance

Based on a review of all available data, the SIHMC determined extreme temperatures to have a moderate planning significance for Rhode Island.

4.13 Flood

4.13.1 Hazard Description

Floodin, as defined by the NFIP is a general and temporary condition of partial or complete inundation of two or more acres of normally dry land area or of two or more properties from overflow of inland or tidal waters. Flooding is often caused by heavy rainfall, snowmelt, storm surges, or the failure of natural or artificial barriers. Flooding can lead to the inundation of homes, roads, farmland, and other areas, causing damage to property, disruption of daily life, and potential threats to human safety and the environment.

A floodplain is a flat or gently sloping area adjacent to a river, stream, or other water body. These areas act as a buffer during periods of heavy rainfall or snowmelt, absorbing excess water and preventing it from rushing downstream too quickly. In its common usage, a floodplain refers to areas inundated by the 100-year flood, the flood that has a 1% chance of being equaled or exceeded in any given year, and the 500-year flood, the flood that has a 0.2% chance of being equaled or exceeded in any given year. The 1% annual chance flood is the national minimum standard to which communities regulate their floodplains through the NFIP.



4.13.2 Location and Extent

A variety of factors affect the severity of flooding within the State. These include topography, weather characteristics, development, and geology. Intense flooding will create havoc in any jurisdiction affected.

Coastal Flooding

Coastal flooding is generally a result of storm surge or wind-driven waves, often caused by hurricanes, extra-tropical storms, and Nor'easters. Storm surges may overrun barrier islands, force sea water up rivers and inlets, and block the downstream flow of upland water. Coastal flooding may cutoff escape routes on barrier islands and low coastal areas and could inundate low lying agriculture both saltwater and freshwater.

Flash Flooding

Flash flooding occurs during heavy or extended periods of rain, generally when the ground is unable to rapidly absorb the water. Most flash flooding in Rhode Island is caused by hurricanes or extra-tropical storms, Nor'easters, or stationary thunderstorms. Heavy sustained rain can create rapid flooding very quickly, and flooding can occur miles away from where the rain fell. Factors that can contribute to the severity of flash flooding include rainfall intensity, duration, drainage condition, and ground conditions (paved or unpaved). Flash floods are particularly dangerous to people and property, as six inches of moving water can knock a person down and two feet can lift a vehicle. As there is often little warning of a flash flood event, they are the cause of most flood fatalities.

Riverine Flooding

Riverine flooding refers to the overflow of water from a river or a stream onto adjacent land areas. This type of flooding occurs when the water level in a river or stream rises significantly and exceeds its banks, inundating the surrounding areas. The severity of riverine flooding can be influenced by the amount and intensity of rainfall in the watershed, the size, shape, and slope of the river or stream channel, and the presence of dams on the river system.

Urban Flooding

FEMA defines urban flooding as "the inundation of property in a built environment, particularly in more densely populated areas, caused by rain falling on increased amounts of impervious surfaces and overwhelming the capacity of drainage systems." In Rhode Island, urban flooding has consistently increased due to a number of factors, including the filling for development of natural wetlands and waterways, the reduction of permeable surfaces, and the aging and insufficient capacity of stormwater systems.

To establish floodplains, FEMA adopted the Base Flood Elevation (BFE), which is the computed elevation that floodwater is anticipated to rise during a flood that has a 1% chance of occurring in any given year. The BFE establishes the regulatory requirement for the elevation or floodproofing of structures, and the relationship between the BFE and a given structure's elevation determines the flood insurance premium through the NFIP.

FEMA, through the Risk Mapping, Assessment, and Planning (Risk MAP) program, works with partners to assess and map these flood risks producing Flood Insurance Rate Maps (FIRMs). As an additional benefit, the FIRMs serve as the basis for NFIP regulations and flood insurance purchase requirements.

SFHAs are defined as the area that will be inundated by the flood event having a 1% chance of being equaled or exceeded in any given year. The 1% annual chance flood is also referred to as the base flood. The FIRM depicts the SFHA, including the 1%-annual-chance flood. These areas are labeled on the map as zone, as explained in the following table:

The following table details FEMA's FIRM flood zone classifications.

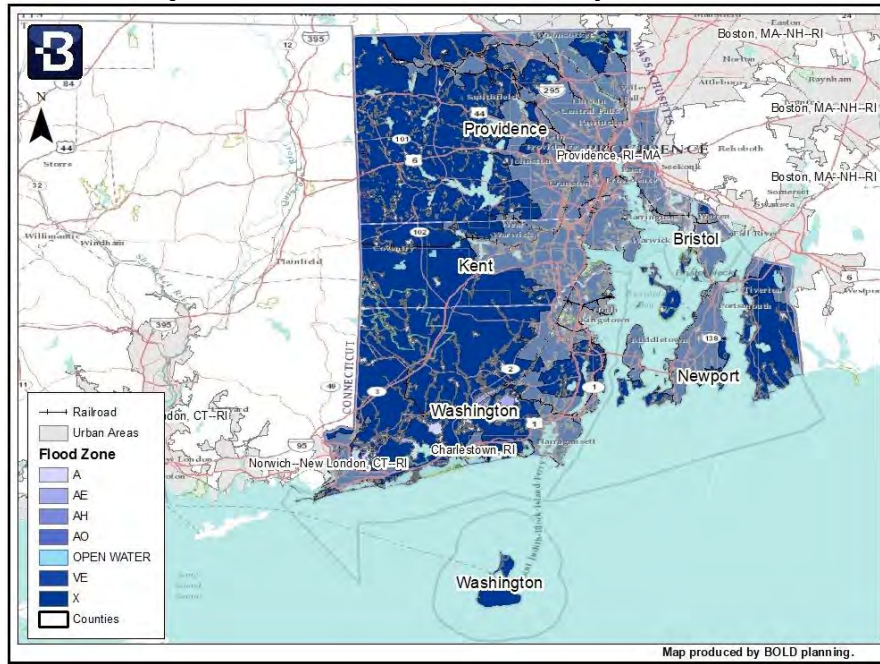
Table 68: Flood Zone Classifications

Zone	Description
A	The 1%-annual-chance or base floodplain. There are six (6) types of A Zones.
AE	The base floodplain where base flood elevations are provided.
AH	Shallow flooding base floodplain. BFEs are provided.
AO	The base floodplain with sheet flow, ponding, or shallow flooding. Base flood depths (feet above ground) are provided.
AR	The base floodplain that results from the decertification of a previously accredited flood protection system that is in the process of being restored to provide a 1%-annual-chance or greater level of flood protection.
A99	Area to be protected from base flood by levees or Federal Flood Protection Systems under construction. BFEs are not determined.
B or Shaded X	Areas between the limits of the base flood and the 0.2% annual-chance (or 500-year) flood.
C or Unshaded X	Areas of minimal flood hazard, which are the areas outside the SFHA and higher than the elevation of the 0.2% annual-chance flood
V	The coastal area subject to a velocity hazard (wave action) where BFEs are not determined on the FIRM.
VE	The coastal area subject to a velocity hazard (wave action) where BFEs are provided on the FIRM.

Source: FEMA

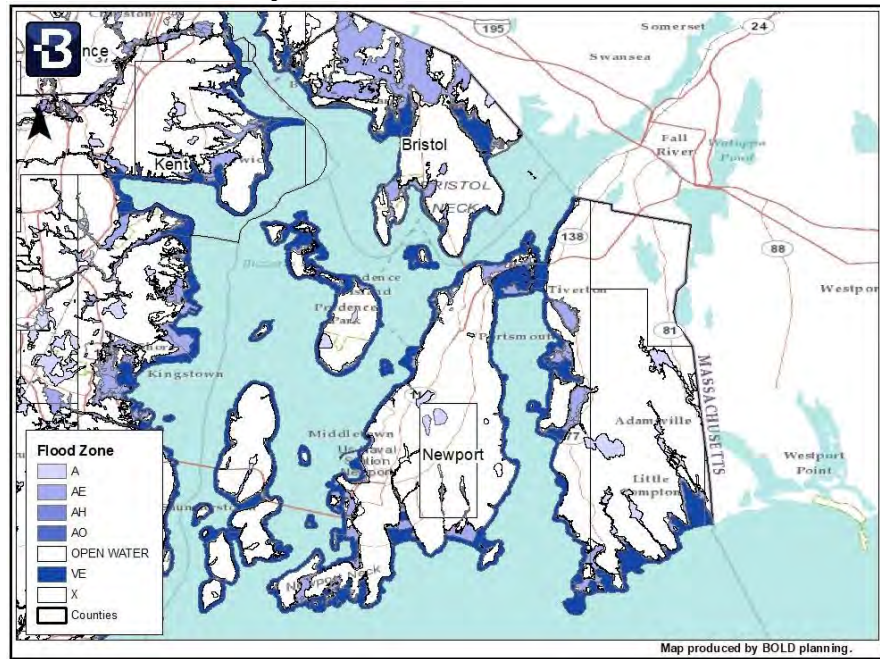
The following maps use FEMA FIRM data to depict the location of identified flood zones within Rhode Island.

Map 17: State of Rhode Island County Flood Zones



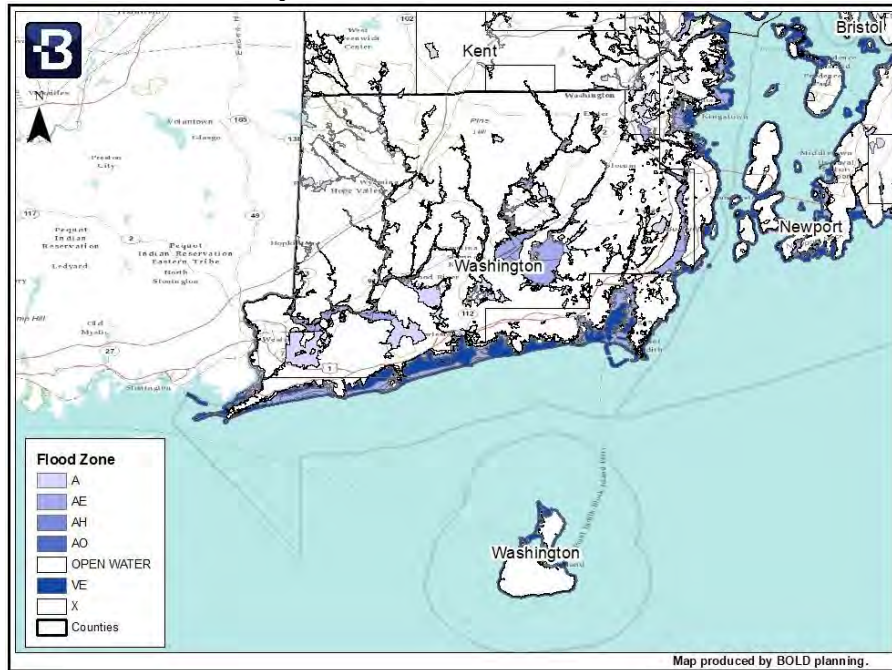
Source: FEMA

Map 18: Rhode Island Flood Zones



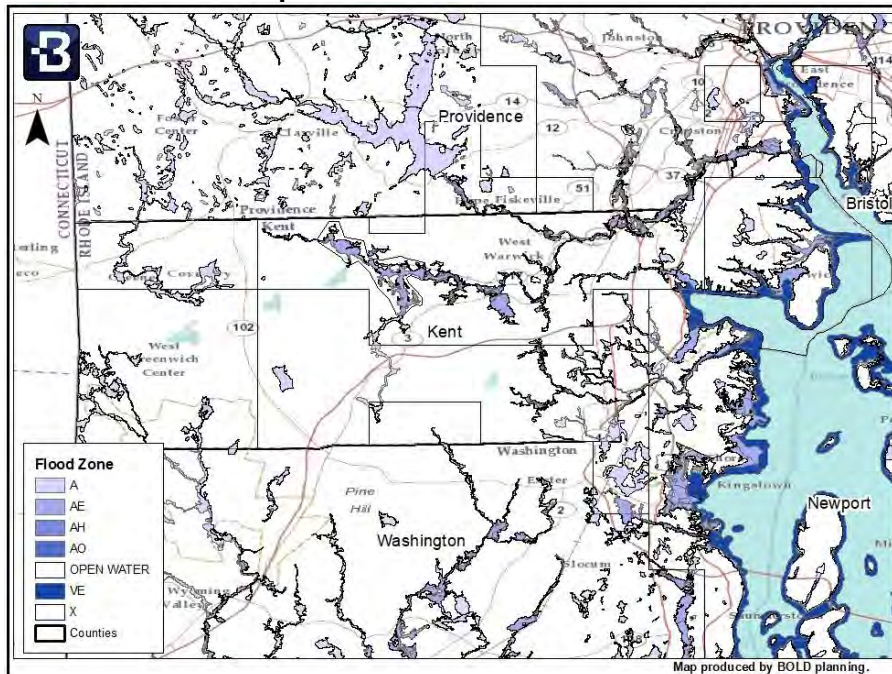
Source: FEMA

Map 19: Rhode Island Flood Zones



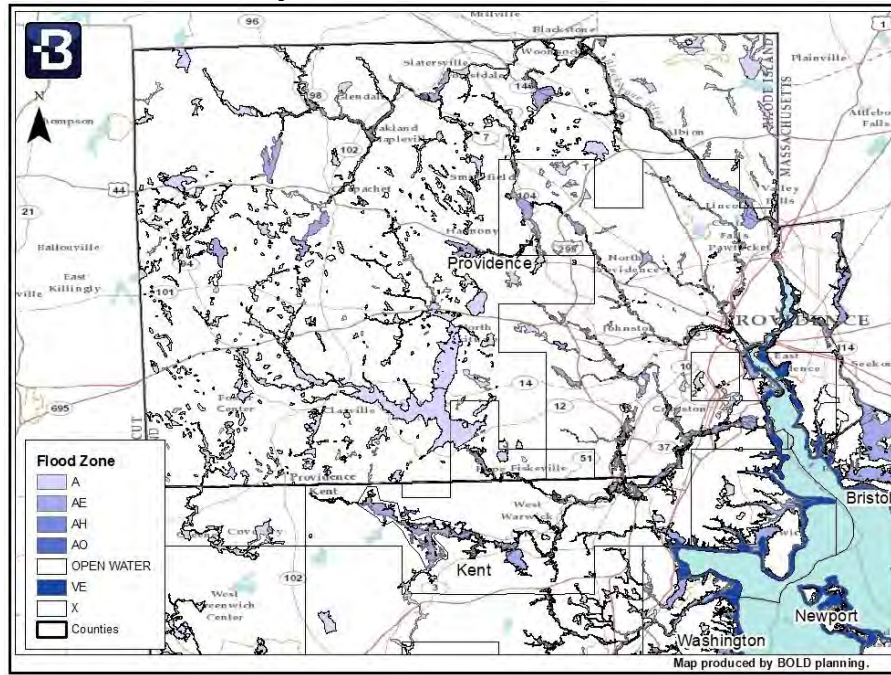
Source: FEMA

Map 20: Rhode Island Flood Zones



Source: FEMA

Map 21: Rhode Island Flood Zones



Source: FEMA

The following table provides a square mile estimate of the area within each type of flood hazard area based on an analysis of available mapping.

Table 69: State of Rhode Areas Within Identified Flood Zones (Square Miles)

County	Zone A	Zone AE	Zone AO	Zone AH	Zone VE	Zone X	Percentage Total
Bristol	0.08	7.6	0.002	0	3.9	3.2	61.3%
Kent	6.5	8.3	0.007	0	4.1	3.2	13.1%
Newport	1.8	5.8	0	0	23.5	4.3	34.6%
Providence	24.9	13.3	0	0.008	2.5	6.0	11.4%
Washington	30.5	20.7	0	0	21.2	10.0	25.0%

Rhode Island currently does not have mapping illustrating frequent areas of flash or urban flooding. However, specific areas prone to coastal and riverine flooding have been identified. These areas include:

- Beavertail Road at Mackerel Cove, Matunuck Beach Road, and Arnolds Neck Road: Areas identified as being prone to coastal flooding and as having vulnerable access routes.
- Nausauket, Oakland Beach, Buttonwoods, and Conimicut, the Port of Providence area, the Newport Harbor area, Narragansett Bay, the Bristol Harbor, and New Shoreham: Areas vulnerable to coastal flooding.
- Pawtuxet River: The Pawtuxet River runs through the communities of Coventry, Warwick, West Warwick, and Cranston. Frequent flooding occurs due to the river's shallow depths from sediment buildup, increased development, lack of pervious surface, and other localized drainage issues. The Pawtuxet River experienced the most significant flooding in recorded history during the March 2010 floods. Flood levels hit historic levels, with a record of 20.79 feet observed at the Cranston USGS river gauge on March 31, 2010. Sewer plants in West Warwick and Warwick sustained \$23,000,000 in damages collectively, business and homes were flooded, and I-95 was shut down in addition to many other local roads in the state.
- Pawcatuck River: The Pawcatuck River and tributaries run west to east through the southern reaches of the state, including portions of Westerly, Richmond, and North Kingstown. There is limited development in these areas; however, during significant flood events, flooding damages do occur. The Pawcatuck River experienced the most significant flooding in recorded history during the March 2010 floods. Flood levels hit historic levels,

with a record of 15.38 feet observed at the Westerly and 11.16 feet observed at the Wood River Junction USGS river gages on March 31, 2010, as well as 13.72 feet observed at the Wood River at Hope Valley USGS river gauge on March 30, 2010.

- **Woonasquatucket River:** The Woonasquatucket River runs from Smithfield to Providence. The areas surrounding this river are densely populated and developed with a combination of mixed-use lots, including residential and commercial properties.
- **Blackstone River:** The Blackstone River forms in Worcester, Massachusetts and then follows a southeast course through Woonsocket, Cumberland, Lincoln, Central Falls, and Pawtucket, where the river reaches Pawtucket Falls before becoming tidal and turning into the Seekonk River. During late March through early April 2010, pre-existing elevated river levels and saturated soil conditions combined with heavy rainfall caused significant flooding along the river in northern Rhode Island. The flooding of 2010 caused flooding on many roads, as well as in business and residential areas in the Cumberland, Lincoln, and Pawtucket areas.
- **Natick Area:** Natick is an area located within the Town of West Warwick. This area is relatively flat and has a combination of new and historic mixed-use structures, including commercial, industrial, and residential, as well as the West Warwick Regional Wastewater Treatment Facility. Natick was hit incredibly hard during the March 2010 floods. NFIP staff assisted with building inspections, technical support, and site visits after the flood receded. Many of the structures, including local infrastructure, such as the old Natick Bridge and the wastewater treatment plant, were significantly damaged when the 2010 floods inundated much of Natick.
- **East Providence:** There are several low-lying areas in East Providence that routinely get flooded, including Riverside and Marsh Street off Waterman Avenue. Marsh Street and the surrounding area are adjacent to freshwater wetlands, which fill up and create a bathtub effect during significant rain events.

4.13.3 Previous Occurrences

Historical events of significant magnitude or impact can result in a Presidential Disaster Declaration. Rhode Island has experienced three Presidential Disaster Declarations related to flooding, reflected in the following table.

Table 70: State of Rhode Island Presidentially Declared Disasters

Designation	Declaration Date	Incident Type	Individual Assistance	Public Assistance
DR-1894-RI	3/29/2010	Severe Storms and Flooding	\$37,157,115	\$24,263,849
DR-1704-RI	5/25/2007	Severe Storms and Island/Coastal Flooding	\$605,080	\$529,557
DR-39-RI	08/20/1955	Hurricane and Flood	-	-

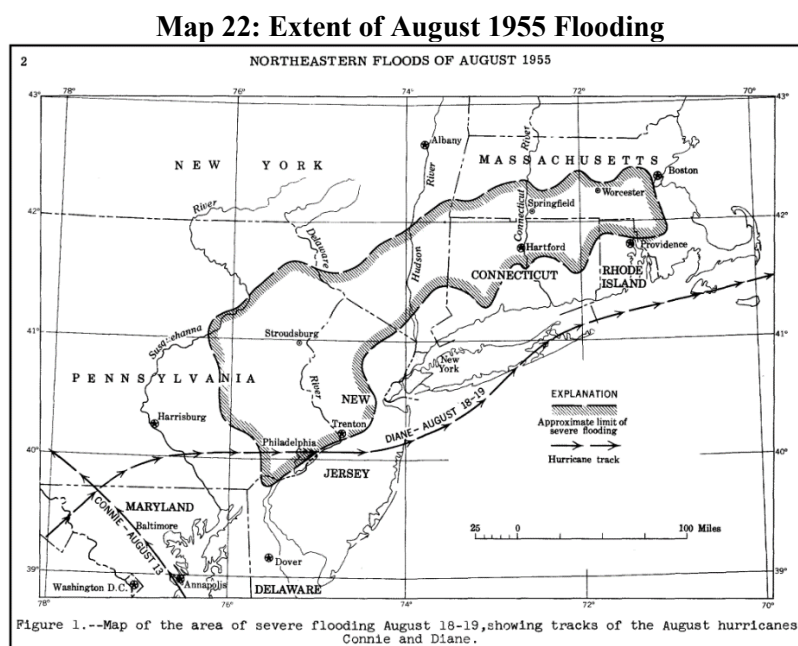
Source: FEMA

Note: -: Data unavailable

The following provide a brief narrative of these events:

- **DR-1894-RI, March 29, 2010:** The largest recent flood event in Rhode Island occurred in March and early April of 2010, and was caused by a series of moderate to heavy rainfall events over a five-week period from late February through March. The length and successive nature of the rainfall saturated soils and kept rivers and streams at or near flood stage. Then, on March 29 through 31, six to ten inches of rain fell over Rhode Island which resulted in record levels of river flooding across the State. The Pawtuxet River in Cranston broke its record crest by an elevation of 6 feet resulting in the West Warwick Wastewater Treatment Plant and the Warwick Mall being inundated by floodwaters. In Cranston, I-95 and the airport connector were shut down due to flooding. The Pawcatuck River in Westerly reached a new record level and did not recede below flood stage until April 12. In Westerly, a mile of train track in Westerly was inundated, resulting in a suspension of Amtrak services. In addition, Chapman Pond was flooded so badly that it effectively shut down Route 91 and Pound Road, causing a prolonged blocked access to an entire neighborhood. Blue Pond Dam failed in the headwaters of the Wood River in Hopkinton, with damage to infrastructure in the area. Nearly 26,000 residents of Rhode Island applied for assistance, with \$79,000,000 approved for individuals and business owners.

- **DR-1704-RI, March 9, 2007:** A combination of heavy rains, frozen ground, and high tide caused extensive flooding in the Silver Creek area of Bristol. The flooding resulted in major roads closures, flooding of area businesses and a local cemetery.
- **DR-39-RI, August 20, 1955:** In August 1955, record flooding along the main rivers in the Blackstone River Basin in Rhode Island and Massachusetts, as well as the Thames River Basin (predominantly in Connecticut), occurred, resulting from torrential rainfall accompanying Hurricane Diane. Damages from the 1955 flood were estimated at approximately \$28 million for Rhode Island, with the Woonsocket area hardest hit in the state. The following map, from the Department of the Interior Geological Survey Circular Number 377, details the extent of the August 1955 flooding event.



Source: Department of Interior

In addition to the Presidentially Declared Disasters, the following table presents NCEI identified flood events in Rhode Island from 1950 to 2023:

Table 71: State of Rhode Island NCEI Flood Events, 1950 - 2023

Jurisdiction	Coastal Flood Events	Flash Flood Events	Flood Events	Injuries and Deaths
Bristol County	7	5	13	0
Kent County	5	16	24	0
Newport County	8	7	8	0
Providence County	3	31	55	0
Washington County	9	6	18	0
State Total	32	65	118	0

Source: National Centers for Environmental Information

Total damage from these reported events was estimated at \$107,923,000. It is worth noting that damage estimates indicated by the NCEI are often artificially low. This underreporting is a result of the way the events are reported to the NCEI, often by the local and/or NWS office. When reporting an event oftentimes the NWS office does not have access to the actual damage assessment resulting from that event. As such, the report often details a very low amount or zero-dollar amount for damages.

4.13.4 Probability of Future Incidents

Based on historical occurrences, Rhode Island will continue to experience flood events on an annual basis. The definition of each flood zone's classification is used for the purpose of calculating the yearly probability of a riverine

flood. Jurisdictions with property in a 100-year floodplain can expect a 1% annual chance of flooding within the designated areas. Jurisdictions with property in a 500-year floodplain can expect a 0.2% annual chance of flooding within the designated areas.

Table 72: Flood Recurrence Probability

Recurrence Interval	Annual Chance of Occurrence	Recurrence Interval Chance of Occurrence	30-Year Chance of Occurrence (Typical Residential Mortgage Term)
10-Year	10%	65%	95.8%
50-Year	2%	64%	45.5%
100-Year	1%	63.4%	26%
500-Year	0.2%	63.2%	5.8%

Source: National Centers for Environmental Information

FEMA FIRMs can be consulted to provide assistance in determining flooding probability specific mapped locations within Rhode Island.

4.13.5 Projected Changes in Location, Intensity, Frequency, and Duration

The location, intensity, frequency, and duration of flooding are influenced by a combination of natural and human-induced factors.

Continued urbanization, deforestation, and changes in land use can alter natural drainage patterns. The conversion of natural landscapes to impervious surfaces, such as roads and buildings, reduces the ability of the land to absorb water, leading to increased runoff and the potential for urban flooding. Alterations to river channels, including channelization and dam construction, can influence the flow of water. Modifications may lead to changes in river behavior, affecting the potential for both upstream and downstream flooding. Poorly planned infrastructure, inadequate stormwater management, and the lack of effective drainage systems in urban areas can contribute to localized flooding. The increase in impervious surfaces reduces natural infiltration, leading to more runoff during rainfall events.

Current available research indicates that sea levels are rising, which can have a significant impact on coastal flooding, amplifying the frequency, extent, and severity of events. Rising sea levels elevate the baseline water levels along coastlines. As sea levels increase, even under calm weather conditions, the water level is higher which makes coastal areas more susceptible to flooding during storm events or high tide cycles. Additionally, storm surges, which are temporary increases in sea level caused by intense storms, are exacerbated by rising sea levels. When a storm makes landfall, the elevated baseline sea level combines with the surge generated by the storm, resulting in more extensive and severe coastal flooding. Higher sea levels accelerate coastal erosion as waves and tides reach farther inland. This can lead to the loss of beaches, coastal cliffs, and protective natural barriers and human infrastructure further exacerbating the vulnerability of coastal communities to flooding.

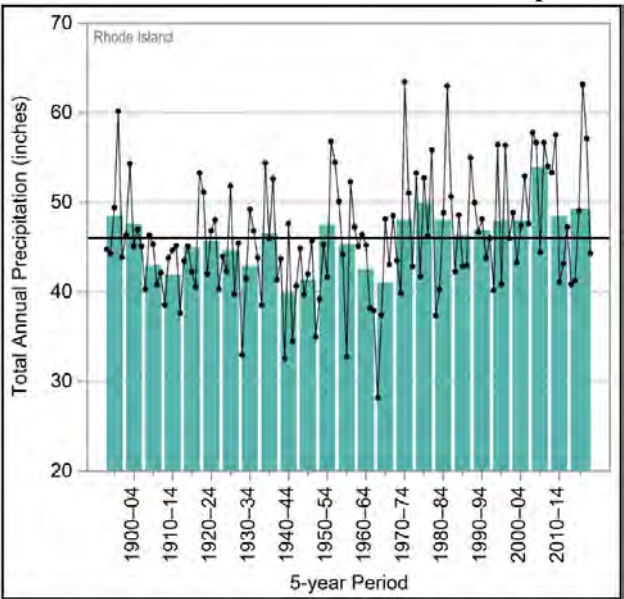
Data from the Rhode Island Coastal Resources Management Council’s Shoreline Change Special Area Management Plan indicates that:

- Since 1930, sea levels in Rhode Island have increased by an average of 1” per decade over the past half century
- Sea levels in the northeast have been increasing three to four times faster than the global average rate
- Rhode Island could see a potential for one foot of sea level rise by 2050
- Sea level is projected to increase by three to five feet above 1990 levels in Rhode Island by 2100

As per the State of Rhode Island Climate Change portal, the impacts of climate change upon Rhode Island’s built and natural environments are wide-ranging, discernible and documented, and, in many cases growing in severity. Related to both riverine, flash, and urban flooding, Rhode Island will experience more extreme and intense precipitation events. Rhode Island’s precipitation rates are climbing an inch almost every 10 years, and 2018 was the third wettest year on record. In 2018, Rhode Island saw a record number of days with over an inch of rain. Under a higher emissions pathway, it is expected that Rhode Island will see a continued increase in frequency and intensity of extreme precipitation events.

The NOAA State Climate Summary 2022 for Rhode Island indicates that the total annual precipitation for Rhode Island has generally been above average in recent decades, as indicated by the following chart:

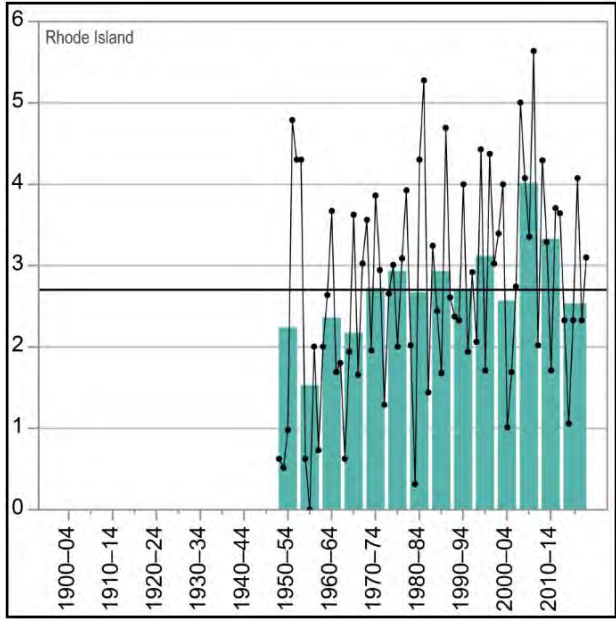
Chart 26: Rhode Island Total Annual Precipitation



Source: NOAA State Climate Summary 2022 for Rhode Island

Additionally, the NOAA NCEI State Climate Summary 2022 for Rhode Island suggests that the number of extreme precipitation events are projected to increase for Rhode Island. These extreme events will likely increase the incidence of flooding within the State.

Chart 27: Rhode Island Number of Extreme Precipitation Events (Greater Than 2 Inches)



Source: NOAA State Climate Summary 2022 for Rhode Island

As previously noted, State of Rhode Island facilities have seen no major changes in the past five years, with only modest repairs and upgrades being conducted and no major rehabilitation or construction projects completed. As such, the risk to state assets has remained static since the completion of the 2019 SHMP. Additionally, with the exception of the two previously identified rehabilitation and construction projects, no large-scale facility projects were noted in any agency budgets or reports.

4.13.6 Vulnerability and Impact

The results of the HAZUS analysis were utilized to estimate potential losses for flooding. The intent of this analysis was to enable the State of Rhode Island to estimate where flood losses could occur and the degree of severity using a consistent methodology. The HAZUS model helps quantify risk along known flood-hazard corridors as well as lesser streams and rivers that have a drainage area of ten square miles or more. Hazus-MH, Version 2.2 SP1, was used to perform the analysis for Rhode Island using essential facility data available through Hazus databases and Homeland Infrastructure Foundation-Level Data. For this hazard, the risk assessment data and maps involved were from an analysis of 1% annual chance flood event. The reported losses are based upon essential facility and census data as part of Hazus. Hazus determines the displaced population based on the inundation area, not necessarily impacted buildings. As a result, there may be a population vulnerable to displacement even if the structure is not vulnerable to damage. Individuals and households will be displaced from their homes even when the home has suffered little or no damage either because they were evacuated or there was no physical access to the property because of flooded roadways.

Flood sheltering needs are based on the displaced population, not the damage level of the structure. Hazus determines the number of individuals likely to use government-provided short-term shelters through determining the number of displaced households as a result of the flooding. To determine how many of those households and the corresponding number of individuals will seek shelter in government-provided shelters, the number is modified by factors accounting for income and age. Displaced people using shelters will most likely be individuals with lower incomes and those who do not have family or friends within the immediate area. Since the income and age factors are taken into account, the proportion of displaced population and those seeking shelter will vary from county to county.

Additionally, Hazus considers flood depth when modeling damage (based on FEMA’s depth-damage functions). Generated reports capture damage by occupancy class (in terms of square footage impacted) by damage percent classes. Occupancy classes include agriculture, commercial, education, government, industrial, religion, and residential. Damage percent classes are grouped by 10% increments up to 50%. Buildings that sustain more than 50% damage are considered to be substantially damaged.

The following table provides the HAZUS results for vulnerable populations and the population estimated to seek short term shelter as well as the numbers of damaged and substantially damaged buildings for Rhode Island.

Table 73: State of Rhode Island Hazus Flood Scenario Displaced Population Building Damages

County	Displaced Population	Person Seeking Shelter	Damaged Buildings	Destroyed Buildings
Bristol	6,814	644	365	21
Kent	6,305	678	269	14
Newport	208	3,689	125	7
Providence	9,247	1,770	214	10
Washington	350	5,851	668	91

Source: FEMA Hazus

Especially critical is timely evacuation orders, and adherence to those orders. If evacuation is not heeded, or flood waters rise quickly enough, citizens could drown or become trapped for extended periods of time with no access to services or medical care. Of special concern are long term care and medical facilities where it can take longer to evacuate, or evacuation may be impossible. Additionally, lower income citizens may not have the means to relocate, whether it be lack of transportation or lack of resources to afford temporary shelter. Expected impacts of flooding on citizens may include:

- Loss of Life: Flooding is one of the leading causes of weather-related fatalities worldwide. Fast-rising floodwaters can lead to drowning and other water-related accidents, resulting in the tragic loss of lives.
- Injuries: Floods can cause injuries due to waterborne diseases, contaminated floodwaters, debris, and accidents during evacuation or rescue operations.
- Displacement: Many people may be forced to evacuate their homes during floods and will require emergency shelter or temporary housing. Prolonged displacement can be emotionally and economically challenging.

- **Health Risks:** Floodwaters often contain pollutants, sewage, and hazardous materials. Exposure to contaminated water can lead to waterborne diseases, infections, and other health risks.
- **Mental Health Effects:** Survivors of floods may experience a range of emotional and psychological challenges, including post-traumatic stress disorder, anxiety, depression, and grief.
- **Food and Water Shortages:** Floods can contaminate water supplies and disrupt the distribution of food. This can lead to shortages of clean drinking water and essential food items.
- **Impact on Vulnerable Populations:** Vulnerable populations, including the elderly, children, people with disabilities, and those living in poverty, are often disproportionately affected by floods due to limited resources and mobility challenges.
- **Long-Term Consequences:** Some flood impacts, such as mold growth, structural damage, and land degradation, can have long-term consequences that persist even after the floodwaters recede.

The Hazus analysis also provides an estimate of the repair costs for impacted buildings as well as the associated loss of building contents and business inventory. Building damage can also cause additional losses to a community by restricting a building's ability to function properly. Income loss data accounts for losses such as business interruption and rental income losses as well as the resources associated with damage repair and job and housing losses. These losses are calculated by Hazus using a methodology based on the building damage estimates.

The damaged building counts generated by Hazus are susceptible to rounding errors and are likely the weakest output of the model due to the use of census blocks for analysis. Generated reports include this disclaimer: "Unlike the earthquake and hurricane models, the flood model performs its analysis at the census block level. This means that the analysis starts with a small number of buildings within each census block and applies a series of distributions necessary for analyzing the potential damage. The application of these distributions and the small number of buildings make the flood model more sensitive to rounding errors that introduces uncertainty into the building count results." Additionally, losses are not calculated for individual buildings, but instead are based on the performances of entire classes of buildings obtained from the general building stock data. In the flood model, the number of grid cells (pixels) at each flood depth value is divided by the total number of grid cells in the census block. The result is used to weight the flood depths applied to each specific occupancy type in the general building stock. First floor heights are then applied to determine the damage depths to analyze damages and losses.

The following table provides the Hazus results for building damages and business interruption loss due to these damages.

Table 74: Rhode Island Hazus Flood Scenario Structural Damage and Income Loss

County	Structural Damage	Contents Damage	Inventory Loss	Total Direct Loss	Total Business Interruption Loss	Total Loss
Bristol	\$178,100,000	\$253,790,000	\$16,470,000	\$448,360,000	\$506,050,000	\$954,410,000
Kent	\$168,480,000	\$319,370,000	\$47,620,000	\$535,470,000	\$574,710,000	\$1,110,180,000
Newport	\$95,790,000	\$155,930,000	\$14,070,000	\$265,790,000	\$462,910,000	\$728,700,000
Providence	\$321,480,00	\$733,330,000	\$92,380,000	\$1,147,180,000	\$1,283,660,000	\$2,430,840,000
Washington	\$313,510,000	\$440,950,000	\$34,180,000	\$788,640,000	\$744,700,000	\$1,533,340,000

Source: FEMA Hazus

Facilities

Floods can have significant and often costly impacts on facilities and critical infrastructure. These impacts can disrupt essential services, damage infrastructure, and pose safety risks. The extent of the impact depends on factors such as the severity of the flood, the preparedness of the infrastructure, and the effectiveness of flood management measures. Here are some of the common impacts of floods on facilities and critical infrastructure:

- **Structural Damage:** Floodwaters can cause extensive damage to buildings, including critical infrastructure such as power plants, water treatment facilities, hospitals, and transportation hubs. The force of moving water can weaken foundations, erode structural elements, and compromise the integrity of buildings.

- **Electrical and Mechanical Systems:** Floodwaters can damage electrical systems, including transformers, switchgear, and electrical panels, leading to power outages and the disruption of critical services. Mechanical systems, such as heating, ventilation, and air conditioning, may also be affected.
- **Water and Wastewater Infrastructure:** Floods can overwhelm water supply and wastewater treatment systems. Contamination of drinking water sources can lead to water shortages and health risks, while damage to wastewater treatment plants can result in the discharge of untreated sewage into water bodies.
- **Transportation Networks:** Floods can damage roads, bridges, railways, and airports, making transportation difficult or impossible.
- **Communication Infrastructure:** Floods can disrupt telecommunications and internet services, hindering communication among emergency responders and the public. Loss of communication can impede coordination and response efforts.
- **Healthcare Facilities:** Damage to healthcare infrastructure can limit the capacity to provide medical care during a crisis.
- **Energy Infrastructure:** Floods can damage power generation facilities, including hydroelectric dams and power plants, leading to power outages and potential safety hazards.

The Hazus model indicated that the following number of critical facilities are estimated to be damaged or suffer loss of use from the flood scenario.

Table 75: Rhode Island Hazus Flood Scenario Number of Critical Facilities Damaged or Impacted

County	Emergency Operations Centers	Fire Stations	Hospitals	Police Stations	Schools
Bristol	0	1	0	1	1
Kent	0	0	0	0	0
Newport	0	1	0	0	1
Providence	0	0	0	2	3
Washington	0	3	0	0	0

Source: FEMA HAZUS

Environment

Environmental impacts from flooding can be far reaching. Of particular concern is flood related runoff, potentially carrying sewage, pesticides, or hazardous chemicals, which can cause long lasting environmental harm. Expected negative outcomes could include changes in habitat, a decrease of available food, and an increase in the spread of vector-associated disease due to standing water.

State Operations and Assets

Floods can pose significant risks to state operations, as they can result in a wide range of immediate and long-term consequences including:

- **Emergency Response and Management:** RIEMA and the National Guard may be mobilized to respond to floods. They would coordinate rescue operations, evacuations, and disaster response efforts to mitigate immediate risks to human life and property.
- **Infrastructure Damage and Maintenance:** RIDOT may need to assess and repair damage to roads, bridges, and other critical infrastructure affected by floodwaters and debris. This can strain resources and disrupt transportation networks.
- **Environmental Oversight and Regulation:** RIDEM responsible for assessing the environmental impact of floods, monitoring water quality, and coordinating cleanup efforts. They may also be involved in addressing long-term environmental consequences.
- **Water Resource Management:** State water resource agencies may need to manage and allocate water resources differently in the aftermath of floods, especially if the flood affects water supplies, water quality, or flood control systems.

- Public Health and Safety: RIDOH may coordinate support for public health needs during and after a flood, including potential health risks from contaminants or waterborne diseases.
- Food and Water Distribution: State agencies may be involved in the distribution of emergency food and clean drinking water to affected communities.
- Long-Term Recovery: State agencies play a critical role in long-term recovery efforts, including securing federal disaster assistance, providing financial support to affected communities, and helping with the rebuilding and restoration of infrastructure.
- Public Information and Communication: State agencies are responsible for disseminating timely and accurate information to the public during a flood event.

Table 76: Critical and State Facilities in Flood Plains

Facility Type	Flood Zone					
	VE	A	AE	AH	AO	Total
Fire Stations	0	0	4	0	0	4
Law Enforcement	0	0	0	0	0	0
Power Plants	0	2	1	0	0	3
Natural Gas Pipelines	3	14	8	0	0	25
Airports	4	0	6	0	0	10
Bus Stops	15	10	208	0	5	238
Wastewater Facilities	6	0	9	0	0	15
Hazardous Material/Tier II Facilities	11	6	51	0	0	68
Flood Control Dams	2	125	168	0	0	295
FDIC Insured Banks	1	0	11	0	0	12
Schools	0	0	8	0	0	8
State Facilities	0	3	12	0	0	15
Hospitals	0	0	0	0	0	0
Nursing Facility	0	0	2	0	0	2
Total	42	162	489	0	5	698

Source: RIEMA and RIGIS

State owned assets in identified flood plains were matched, when possible, to the master state facility list for valuation purpose. Some data discrepancies were discovered during this process causing a valuation not to be determined for facilities in Barrington and Lincoln. The following table present the valuation of the state assets identified in flood zones:

Table 77: Valuation of State of Rhode Island Facilities in Flood Plains

Jurisdiction	County	Valuation
Cranston	Providence	\$21,218
Narragansett	Washington	\$14,366,393
Newport	Newport	\$1,868,147
Pawtucket	Providence	\$2,937,066
Providence	Providence	\$21,218
South Kingstown	Washington	\$86,720
Warwick	Kent	\$4,828,709
Westerly	Kent	\$1,083,187

Source: RIEMA and RIGIS

Using GIS, flood zones were overlaid on the critical facility and state assest location data. Maps are included in Appendix D.

Roads within the State are also susceptible to the negative impacts from flooding, which can range from immediate and localized damage to long-term effects on infrastructure integrity. The severity of these impacts depends on factors such as the intensity and duration of the flooding, the design and construction of the roads, and the effectiveness of drainage

systems. Notable impacts can include surface erosion, where water can erode the road surface leading to potholes, rutting, and an uneven road surface, and undermining and subsidence, which can cause the failure of foundation layers and potential structural failure. The following table details the mileage of State Maintained roads in the State of Rhode Island in flood identified flood areas:

Table 78: State of Rhode State Maintained Roads in Flood Zones

Jurisdiction	Miles of Roadway in Each Zone				
	VE	AO	AH	AE	A
Bristol County	0.0	4.9	0.0	169.5	125.3
Kent County	12.0	1.1	0.0	10.9	0.0
Newport County	79.4	0.0	0.0	43.7	4.0
Providence County	0.0	0.0	0.0	122.6	56.2
Washington County	29.5	0.0	0.0	15.7	12.0
State Total	120.9	6.0	0.0	362.4	197.5

Source: RIEMA

Bridges crossing rivers can pose significant concerns during flooding events due to the increased risk of structural failure. Floodwaters can exert powerful hydraulic forces on bridge structures, with the flow of water, debris, and floating objects impacting the bridge's substructure and foundation. Scouring, the removal of soil or sediment around bridge foundations can increase during a flood event increasing the risk of failure. Floodwater can also cause the deformation and misalignment of bridge components. As water levels rise and fall, the structural elements may undergo stress and strain, potentially leading to long-term damage and misalignment.

Of particular concern are bridges identified by RIDOT as structurally deficient. A structurally deficient bridge is a bridge that has one or more components showing signs of deterioration or damage, affecting its overall structural integrity. A structurally deficient bridge is not necessarily in immediate danger of collapse, but rather contains elements that require attention and may need repair, rehabilitation, or replacement. The following table details structurally deficient bridges crossing over rivers in the State of Rhode Island:

Table 79: Structurally Deficient Bridges Crossing Rivers

Jurisdiction	Location	Year Constructed	Average Daily Crossings	Status
Providence	Interstate 95 over Seekonk River	1969	80,500	Under repair
Providence	Route 37 over Pawtuxet River	1963	36,090	Under repair

Source: Federal Highway Administration National Bridge Inventory

Of additional concern when considering coastal flooding is coastal erosion, a dynamic process that can be influenced by a combination of factors. Waves generated by wind are a primary force driving coastal erosion. The energy of waves can impact shorelines, wearing away coastal features, and transporting sediment. The intensity and direction of wave action are influenced by wind patterns and storm events. Tidal currents, associated with the rise and fall of tides, can also contribute to coastal erosion. The movement of water during tidal cycles can transport sediment, especially in areas with strong tidal flows.

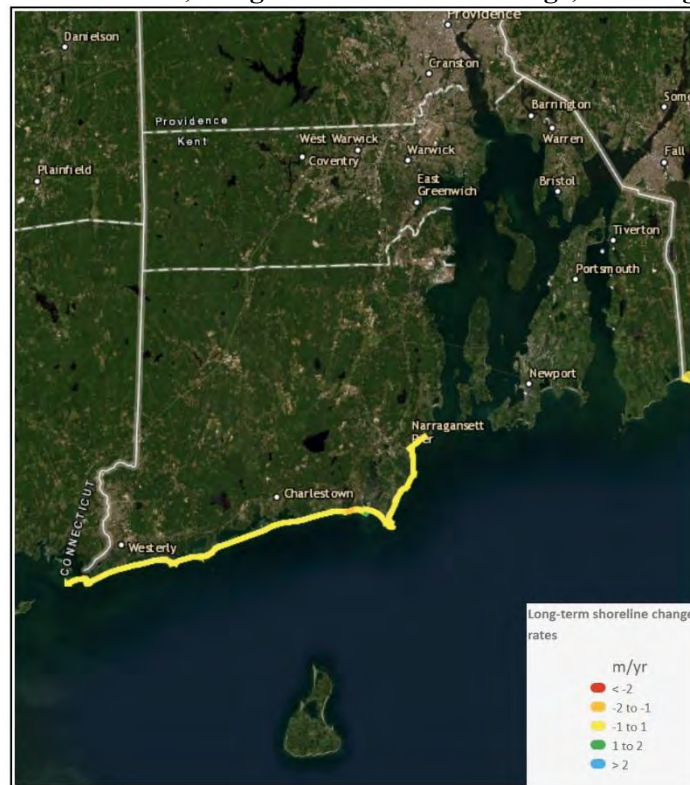
Storm surges, which are temporary increases in sea level during storms, can lead to intensified wave action and coastal erosion. Storm surges are often associated with tropical cyclones, hurricanes, or severe weather events.

Rising sea levels, linked to global climate change, can exacerbate coastal erosion. Higher sea levels increase the reach of waves and storm surges, leading to more frequent and severe erosion along vulnerable coastlines.

Anthropogenic factors, including coastal development, can alter natural coastal processes and exacerbate erosion. The construction of coastal structures such as seawalls, groins, and jetties can influence erosion patterns. While these structures may provide protection to specific areas, they can also lead to unintended consequences, such as increased erosion in adjacent areas.

Extensive mapping concerning coastal erosion is available with the Rhode Island Coastal Resources Management Council. The following map provides an example of these maps, detailing coastal erosion for Washington County:

Map 23: Coastal Erosion, Long Term Shoreline Change, Washington County



Source: Rhode Island Coastal Resources Management Council

When detailing state assets at risk to coastal flooding and coastal erosion, consideration must be given to ports. The Coastal Resources Division is responsible for the development, management, and maintenance of state owned and/or operated ports, including:

- Port of Galilee (Narragansett, Washington County)
- State Pier #9 (Newport, Newport County))
- State Pier #4 (Jerusalem, Washington County)
- State Pier #5 (Narragansett. Washington County)
- Chase Marina (Tiverton, Newport County)

These properties comprise 41 piers and 24 lease holdings with an approximate combined value of over \$10,000,000. The Port of Galilee is one of the largest commercial fishing ports on the East Coast and is the home port to over 250 commercial fishing vessels. The following map details shoreline changes due to coastal erosion in areas surrounding the port:

Map 24: Port of Galilee Coastal Erosion



Source: Rhode Island Coastal Resources Management Council

Consequence Analysis

This consequence analysis lists the potential impacts of this hazard on various elements of community and state infrastructure. The impact of this hazard is evaluated in terms of disruption of operations, recovery challenges, and overall wellbeing to all Rhode Island residents and first responder personnel. The consequence analysis supplements the hazard profile by analyzing specific impacts.

Table 80: Flood Consequence Analysis

Subject	Potential Impacts
Impact on the Public	Significant flooding events can lead to the damage and loss of homes, property, and businesses. Flash flooding and excessive rainfall may lead to dangerous conditions on roadways. Closures of medical facilities is a major public health concern if flooding damages those facilities. Water sources may become contaminated, and water or sewer systems may be disrupted. Vector-associated disease may increase.
Impact on Responders	Fire, police, and emergency responders may be called on to evacuate people from impacted areas, as well as close roads, attend to the injured, and direct traffic away from the flooded area and roads. First responders may face challenges with transportation and access to a location. Flash floods and mudslides due to heavy rainfall can also injure first responders, as well as delay response operations.
Continuity of Operations	Rhode Island agencies maintain continuity plans which can be enacted as necessary based on the situation. Floods which create power outages, debris damage, and road closures are not uncommon in Rhode Island. This threat may impact an agency's ability to maintain continuity of operations based on the incidents impact on power, communications and the potential to damage equipment and records within primary and alternate facilities.

Table 80: Flood Consequence Analysis

Subject	Potential Impacts
Delivery of Services	Flooding can cause road and bridge closures, as well as disrupt transit services, impacting the ability to deliver goods and services. Exposure to flood waters may also damage or destroy physical goods such as food, clothing, and hygiene products.
Property, Facilities, and Infrastructure	Flooding can cause significant property destruction. Floods can disrupt normal daily activities due to the potential impact on schools, hospitals, and other public infrastructure. Transportation infrastructure can be damaged which could impact the freedom of movement or provision of utilities. Water sources can become contaminated. Water and sewer systems may be disrupted. Solid-waste collection and disposal may also be impacted, causing dangerous public health risks.
Impact on Environment	Rising waters from flooding impact the environment by spreading pollution, inundating water and wastewater treatment plants, carrying debris, and disrupting wildlife and reserve areas. Standing water following a flooding event can facilitate the spread of vector-associated diseases.
Economic Conditions	Significant and repeated flooding can lower property value throughout the state, which can have a deleterious effect on the tax base. Furthermore, flooding drains response resources, which can be costly during a large flooding event for disaster reimbursement.
Public Confidence in Governance	Ineffective flooding response can decrease the public's confidence in the state's ability to respond and govern. Multi-level government response requires direct actions that must be immediate and effective to maintain public confidence. Efficiency in response and recovery operations is critical in keeping public confidence high.

4.13.7 Jurisdictional Vulnerability and Planning Significance

Utilizing data from FEMA's NRI and from LHMPs, the following tables detail the hazard ratings for jurisdictions, by census tract, for flooding. Data concerning both risk index and EAL is reported by census tract. As many State of Rhode Island jurisdictions encompass multiple tracts, an average of tract ratings was completed to determine the overall jurisdictional rating. The following table details the hazard rating by LHMP analysis, FEMA NRI, and EAL by jurisdiction for the State of Rhode Island:

Table 81: State of Rhode Jurisdictional Flood Planning Significance

County	Jurisdiction	Flood LHMP Rating	Coastal Flood NRI	Coastal Flood EAL	Riverine Flood NRI	Riverine Flood EAL
Bristol	Barrington	High	Relatively Moderate	Relatively Moderate	Relatively Moderate	Relatively Moderate
	Bristol	-	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Warren	High	Relatively Moderate	Relatively Moderate	Relatively Moderate	Relatively Moderate
Kent	Coventry	-	No Rating	No Rating	Relatively Low	Relatively Moderate
	East Greenwich	High	Very Low	Very Low	Relatively Moderate	Relatively Moderate
	Warwick	Moderate	Very Low	Very Low	Relatively Low	Relatively Moderate
	West Greenwich	-	No Rating	No Rating	Relatively Low	Relatively Low
	West Warwick	Moderate	No Rating	No Rating	Relatively Moderate	Relatively Moderate
Newport	Jamestown	-	Relatively Moderate	Relatively Moderate	Relatively Low	Relatively Low
	Little Compton	Moderate	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Middletown	High	Very Low	Very Low	Very Low	Very Low
	Newport	High	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Portsmouth	High	Relatively Low	Relatively Low	Relatively Low	Relatively Low

Table 81: State of Rhode Jurisdictional Flood Planning Significance

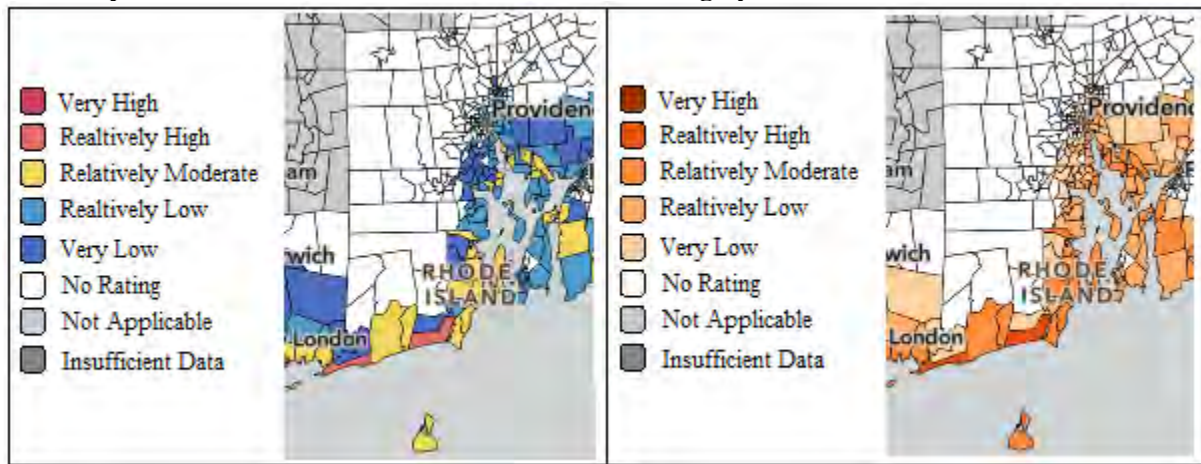
County	Jurisdiction	Flood LHMP Rating	Coastal Flood NRI	Coastal Flood EAL	Riverine Flood NRI	Riverine Flood EAL
	Tiverton	-	Relatively Low	Relatively Low	Relatively Low	Relatively Low
Providence	Burrillville	High	No Rating	No Rating	Relatively Moderate	Relatively Moderate
	Central Falls	High	Very Low	Very Low	Relatively Moderate	Relatively Low
	Cranston	High	Very Low	Very Low	Relatively Low	Relatively Low
	Cumberland	-	No Rating	No Rating	Relatively Moderate	Relatively Moderate
	East Providence	Moderate	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Foster	Moderate	No Rating	No Rating	Relatively Moderate	Relatively Moderate
	Glocester	Moderate	No Rating	No Rating	Relatively Moderate	Relatively Moderate
	Johnson	High	No Rating	No Rating	Relatively Moderate	Relatively Moderate
	Lincoln	High	No Rating	No Rating	Relatively Moderate	Relatively Low
	North Providence	High	No Rating	No Rating	Relatively Low	Relatively Low
	North Smithfield	Moderate	No Rating	No Rating	Relatively Moderate	Relatively Moderate
	Pawtucket	High	No Rating	No Rating	Very Low	Very Low
	Providence	High	Very Low	Very Low	Very Low	Very Low
	Scituate	-	No Rating	No Rating	Relatively Low	Relatively Moderate
	Smithfield	-	No Rating	No Rating	Relatively Moderate	Relatively Moderate
	Woonsocket	Low	No Rating	No Rating	Relatively Low	Relatively Low
Washington	Charlestown	-	Relatively Moderate	Relatively Moderate	Relatively Moderate	Relatively High
	Exeter	-	No Rating	No Rating	Relatively Low	Relatively Low
	Hopkinton	-	No Rating	No Rating	Relatively Low	Relatively Low
	Narragansett	Moderate	Relatively Moderate	Relatively Moderate	Relatively Moderate	Relatively Moderate
	New Shoreham	-	Relatively Moderate	Relatively Moderate	Relatively Moderate	Relatively Moderate
	North Kingstown	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Richmond	High	No Rating	No Rating	Relatively Low	Relatively Low
	South Kingstown	High	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Westerly	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate

Source: LHMPs and FEMA NRI

Note: - indicates no current plan available for review

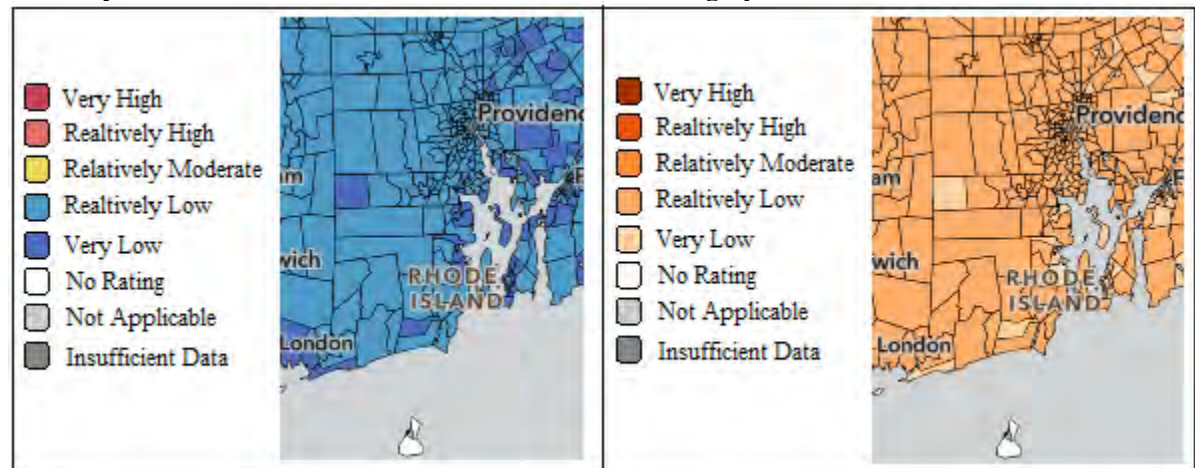
The following maps indicate the FEMA NRI and EAL for coastal flood and riverine flood by census tract for the state of Rhode Island.

Map 25: FEMA NRI and EAL for Coastal Flooding by Census Tract for Rhode Island



Source: FEMA NRI

Map 26: FEMA NRI and EAL for Riverine Flooding by Census Tract for Rhode Island



Source: FEMA NRI

Based on FEMA NRI social vulnerability ratings, and on U.S. Census Bureau data concerning percentage of the population below the 100% poverty who may relocating during and after a flood event, the following communities may be at greater risk to flooding:

Table 82: Jurisdictions at Greater Risk to Flood Events Due to Poverty and Social Vulnerability

County	Jurisdiction	Population Below 100% of the Poverty Level	Social Vulnerability Rating
Providence County	Central Falls	29.9%	Very High
Providence County	Providence	23.0%	Relatively High
Providence County	Woonsocket	21.0%	Relatively Moderate
Washington County	Narragansett	19.3%	Relatively High
Newport County	Newport	14.4%	Relatively Low
Kent County	West Warwick	12.1%	Relatively Low
Providence County	Pawtucket	11.6%	Relatively High

Source: FEMA NRI and U.S. Census Bureau

Additionally, coastal jurisdictions at risk from sea level rise will be more vulnerable to coastal flooding. The following map, from the Socioeconomics of Sea Level Rise, Technical Paper 168, from the Rhode Island Department of Administration Division of Planning details the locations of the 21 Rhode Island coastal jurisdictions vulnerable to sea level rise: These jurisdictions are:

Map 27: Rhode Island Coastal Jurisdictions



- Barrington
- New Shoreham (Block Island)
- Bristol
- Charlestown
- Cranston
- East Greenwich
- East Providence
- Jamestown
- Little Compton
- Middletown
- Narragansett
- Newport
- North Kingstown
- Pawtucket
- Portsmouth
- Providence
- South Kingstown
- Tiverton
- Warren
- Warwick
- Westerly

Source: Sea Level Rise, Technical Paper 168

4.13.8 National Flood Insurance Program and Community Rating System Communities

The NFIP is a federal program, managed by FEMA, which exists to provide flood insurance for property owners in participating communities, to improve floodplain management practices, and to develop maps of flood hazard areas. The following table presents NFIP participating communities.

Table 83: State of Rhode Island NFIP Communities

ID	Community Name	County	Initial FHMB Identified	Initial FIRM Identified	Current Effective Map Date
445392#	Barrington	Bristol	5/15/1970	4/16/1971	7/7/2014
445393#	Bristol (Town)	Bristol	-	12/2/1972	7/7/2014
445408#	Warren	Bristol	-	7/28/1972	7/7/2014
440004E	Coventry	Kent	6/28/1974	9/1/1978	7/19/2023
445397E	East Greenwich	Kent	2/8/1973	2/9/1973	10/2/2015
445409E	Warwick	Kent	-	4/6/1973	10/2/2015
440037E	West Greenwich	Kent	1/24/1975	1/3/1986	07/19/23(M)
440007E	West Warwick	Kent	4/13/1973	2/1/1978	10/2/2015
445399#	Jamestown	Newport	-	4/20/1972	9/4/2013
440035C	Little Compton	Newport	7/19/1974	8/17/1981	7/6/2021
445401#	Middletown	Newport	-	4/9/1971	9/4/2013
445403#	Newport (City)	Newport	-	6/17/1970	9/4/2013
445405#	Portsmouth	Newport	-	8/24/1973	9/4/2013
440012C	Tiverton	Newport	5/24/1974	5/2/1977	7/6/2021
440013E	Burrillville	Providence	9/13/1974	7/2/1979	7/19/2023
445394E	Central Falls	Providence	-	5/27/1971	10/2/2015
445396E	Cranston	Providence	-	8/28/1971	10/2/2015
440016E	Cumberland	Providence	1/3/1975	12/16/1980	10/2/2015

Table 83: State of Rhode Island NFIP Communities

ID	Community Name	County	Initial FHMB Identified	Initial FIRM Identified	Current Effective Map Date
445398E	East Providence	Providence	6/5/1970	5/18/1973	10/2/2015
440033E	Foster	Providence	9/13/1974	12/4/1985	07/19/23(M)
440034E	Glocester	Providence	10/18/1974	8/15/1979	7/19/2023
440018E	Johnston	Providence	11/22/1977	9/1/1978	10/2/2015
445400E	Lincoln	Providence		11/30/1973	10/2/2015
440020E	North Providence	Providence	4/13/1973	12/15/1977	10/2/2015
440021E	North Smithfield	Providence	6/14/1974	8/1/1978	10/2/2015
440022E	Pawtucket	Providence	-	7/16/1971	10/2/2015
445406E	Providence (City)	Providence	12/15/1970	12/15/1970	10/2/2015
440024E	Scituate	Providence	9/6/1974	1/2/1981	10/2/2015
440025E	Smithfield	Providence	8/9/1974	3/1/1977	10/2/2015
445411#	Woonsocket	Providence	-	10/13/1971	3/2/2009
445395D	Charlestown	Washington	-	7/13/1972	4/3/2020
440032D	Exeter	Washington	3/14/1975	3/1/1982	7/19/2023
440028D	Hopkinton	Washington	5/31/1974	3/16/1981	4/3/2020
445414D	Narragansett Indian Tribe	Washington	-	7/13/1972	4/3/2020
445402D	Narragansett	Washington	-	12/7/1971	4/3/2020
440036#	New Shoreham	Washington	1/3/1975	4/3/1985	10/16/2013
445404D	North Kingstown	Washington	-	7/14/1972	4/3/2020
440031D	Richmond	Washington	5/31/1974	11/5/1980	4/3/2020
445407D	South Kingstown	Washington	-	7/13/1972	4/3/2020
445410D	Westerly	Washington	-	7/28/1972	4/3/2020

Notes: (M): No elevation determined, all Zone A, C, and X

FHMB: Flood hazard boundary map

:- No data

The Community Rating System (CRS) is a voluntary incentive program that recognizes and encourages community floodplain management practices that exceed the minimum requirements of the NFIP. In CRS communities, flood insurance premium rates are discounted to reflect the reduced flood risk resulting from the community's efforts that address the three goals of the program:

- Reduce and avoid flood damage to insurable property
- Strengthen and support the insurance aspects of the NFIP
- Foster comprehensive floodplain management•

As of this plan, there are a total of 11 communities in Rhode Island participating in the CRS:

Table 84: State of Rhode Island CRS Communities

Community Name	County	CRS Entry Date	Current Effective Date	Current Class	Percentage Discount SFHA	Percentage Discount Non-SFHA
Bristol	Bristol	5/1/2013	5/1/2018	7	15%	5%
Warwick	Kent	10/1/2016	10/1/2016	9	5%	5%
Middletown	Newport	10/1/1991	5/1/2000	7	15%	5%
Newport (City)	Newport	10/1/2017	10/1/2017	7	15%	5%
East Providence	Providence	5/1/2014	5/1/2014	8	10%	5%
Pawtucket	Providence	10/1/2014	10/1/2014	8	10%	5%
Charlestown	Washington	5/1/2015	4/1/2022	6	20%	10%
Narragansett	Washington	10/1/1992	5/1/2017	7	15%	5%
New Shoreham	Washington	5/1/2019	5/1/2019	8	10%	5%

Table 84: State of Rhode Island CRS Communities

Community Name	County	CRS Entry Date	Current Effective Date	Current Class	Percentage Discount SFHA	Percentage Discount Non-SFHA
North Kingstown	Washington County	10/1/1993	10/1/1993	9	5%	5%
Westerly	Washington County	5/1/2013	10/1/2018	7	15%	5%

Source: FEMA

4.13.9 FEMA Flood Policy and Loss Data

Rhode Island flood policy information was sourced from FEMA’s Flood Insurance Data and Analytics. The number of flood insurance policies in effect may not include all structures at risk of flooding, and it is likely that some properties are under-insured. The flood insurance purchase requirement is for flood insurance in the amount of federally backed mortgages, not the entire value of the structure. Additionally, contents coverage is not required.

The following table shows the details of NFIP policy statistics for the State of Rhode Island.

Table 85: State of Rhode Island NFIP Coverage

County	Jurisdiction	Policies in Force	Total Coverage
Bristol	Barrington	755	\$234,439,000
	Bristol	271	\$71,833,000
	Warren	262	\$57,834,000
Kent	Coventry	107	\$33,305,000
	East Greenwich	76	\$23,228,000
	Warwick	740	\$200,031,000
	West Greenwich	5	\$1,610,000
	West Warwick	64	\$16,723,000
Newport	Jamestown	177	\$53,697,000
	Little Compton	96	\$30,401,000
	Middletown	66	\$23,514,000
	Newport	752	\$193,027,000
	Portsmouth	339	\$90,368,000
	Tiverton	93	\$25,576,000
Providence	Burrillville	31	\$9,302,000
	Central Falls	50	\$6,813,000
	Cranston	275	\$79,634,000
	Cumberland	43	\$14,409,000
	East Providence	160	\$50,703,000
	Foster	4	\$1,240,000
	Glocester	17	\$4,353,000
	Johnston	40	\$13,657,000
	Lincoln	215	\$55,429,000
	North Providence	47	\$15,485,000
	North Smithfield	19	\$6,604,000
	Pawtucket	82	\$21,600,000
	Providence	515	\$143,221,000
	Scituate	21	\$5,429,000
	Smithfield	122	\$28,243,000
	Woonsocket	155	\$40,422,000
Washington	Charlestown	466	\$138,402,000
	Exeter	9	\$2,344,000
	Hopkinton	14	\$3,383,000

Table 85: State of Rhode Island NFIP Coverage

County	Jurisdiction	Policies in Force	Total Coverage
	Narragansett Indian Tribe	1	\$215,000
	Narragansett	957	\$280,661,000
	New Shoreham	51	\$17,998,000
	North Kingstown	508	\$154,746,000
	Richmond	18	\$7,738,000
	South Kingstown	524	\$153,634,000
	Westerly	593	\$172,511,000

Source: FEMA Flood Insurance Data and Analytics

The following table details NFIP claim statistics for the State of Rhode Island

Table 86: State of Rhode Island NFIP Claim Statistics

Category	Data
Claims Closed with Payment	4,847
Open Claims	25
Claims Closed without Payment	1,864
Net Building Payments	\$98,997,142
Net Contents Payments	\$25,019,479
Net Increased Cost of Compliance Payments	\$657,233
Net Total Payments	\$124,673,854
Average Net Payment (Open and Closed with Payment)	\$642,942
Number of Records	6,736

Source: RIEMA

4.13.10 Repetitive Loss Structures

A high priority for the State of Rhode Island is the mitigation of, and/or the reduction of losses to, Repetitive Loss (RL) and Severe Repetitive Loss (SRL) structures. The NFIP defines a RL property as:

- Any insurable building for which two or more claims of more than \$1,000 were paid by the NFIP within any rolling 10-year period, since 1978. At least two of the claims must be more than 10 days apart.

The definition of severe repetitive loss as applied to this program was established in section 1361A of the National Flood Insurance Act, as amended, 42 U.S.C. 4102a. An SRL property is defined as a residential property that is covered under an NFIP flood insurance policy and:

- That has at least four NFIP claim payments (including building and contents) over \$5,000 each, and the cumulative amount of such claims payments exceeds \$20,000; or
- For which at least two separate claims payments (building payments only) have been made with the cumulative amount of the building portion of such claims exceeding the market value of the building.

For both of the above, at least two of the referenced claims must have occurred within any ten-year period and must be greater than ten days apart.

The following table details information concerning RL and SRL identified properties in Rhode Island:

Table 87: Rhode Island RL and SRL Properties

Jurisdiction	Number of RL Properties	Number of SRL Properties
Bristol County	36	3
Kent County	95	8
Newport County	36	6
Providence County	193	22
Washington County	127	23

Table 87: Rhode Island RL and SRL Properties

Jurisdiction	Number of RL Properties	Number of SRL Properties
State Total	487	62

Source: RIEMA

The following tables detail information on RL and SRL claims and payments for Rhode Island.

Table 88: Rhode Island RL Claims

Jurisdiction	RL Claims	RL Building Payments	RL Contents Payments
Bristol County	100	\$1,484,358.54	\$142,223.43
Kent County	245	\$6,127,140.30	\$3,139,477.38
Newport County	101	\$4,279,706.48	\$277,598.34
Providence County	569	\$17,412,365.27	\$10,063,356.38
Washington County	406	\$16,328,612.58	\$3,080,190.45
State Total	1,421	\$45,632,183.17	\$16,702,845.98

Source: RIEMA

Table 89: Rhode Island SRL Claims

Jurisdiction	SRL Claims	SRL Building Payments	SRL Contents Payments
Bristol County	23	\$809,107.53	\$44,144.03
Kent County	43	\$736,518.53	\$737,803.70
Newport County	23	\$2,310,583.37	\$56,167.72
Providence County	144	\$7,378,996.01	\$5,830,643.89
Washington County	131	\$8,887,357.43	\$1,961,774.21
State Total	364	\$20,122,562.87	\$8,630,533.55

Source: RIEMA

4.13.11 Hazard Planning Significance

Based on a review of all available data, the SIHMC determined flood to have a high planning significance for Rhode Island.

4.14 Sea Level Rise

4.14.1 Hazard Description

Sea level rise is an increase in the ocean's surface height relative to the land in a specific location. The primary causes of sea level rise are the melting of polar ice caps and thermal expansion of ocean water. Thermal expansion occurs when water gets warmer, causing the volume to increase.

Melting water from ice sheets and glaciers, caused by rising air temperatures, is the main contributor to sea level rise. This melting process is accelerated by warmer ocean water eroding ice shelves from below. Additionally, ocean measurements indicate that the temperature of the ocean's surface has risen approximately 1.5°F since the late 1800s. This rising temperatures is a result of increased absorption of excess heat trapping greenhouse gases from the atmosphere. Another factor in rising sea levels is coastal land subsistence, often caused by natural geologic processes. It is estimated that Rhode Island has been sinking at a rate of four inches per 100 years.



The State of Rhode Island has completed several sea level rise vulnerability assessments and reports concerning sea level rise. These assessments and reports include:

- Technical Paper 164: Vulnerability of Transportation Assets to Sea Level Rise (2014): Utilizing data from the National Oceanic and Atmospheric Administration, combined with analysis techniques developed at University of Rhode Island, this report examined where the state's transportation infrastructure might be exposed to sea level rise, and devised a technique for assessing the vulnerability of those assets. Using a GIS-based methodology, the study analyzes the transportation assets potentially exposed under one, three, and five feet of sea level rise.
- Technical Paper 167: Vulnerability of Municipal Transportation Assets to Sea Level Rise and Storm Surge (2016): This project adapted the methods of Technical Paper 164 to incorporate new data, a municipal level outlook, and an easier to understand presentation. The project utilized the STORMTOOLS data set created by the Coastal Resources Management Council, which offered the opportunity to bring information about storm surge and an additional sea level rise scenario into the analysis. A broad summary of the results of this report are presented in factsheets tailored to each municipality threatened by sea level rise and storm surge.
- Socioeconomics of Sea Level Rise project: This project uses the best available data in an effort to identify the social, economic and demographic makeup of the populations located within the 1-, 3-, 5- and 7-foot sea level rise inundation zones in the state's 21 coastal communities.
- The Shoreline Change (Beach) Special Area Management Plan: The Rhode Island Coastal Resources Management Council, in coordination with the University of Rhode Island, has prepared a plan to help find solutions these problems, which are threatening communities in many areas along the state's coast. The Beach Special Area Management Plan aims to improve the coastal resiliency of cities and towns throughout Rhode Island to the threats of erosion and flooding caused by storm events or sea level rise. Using up-to-date research and expertise from a wide range of public and private sectors, the Beach Special Area Management Plan will help inform state and local policy and planning initiatives. Part of this partnership includes the maintenance of the Beach Special Area Management Plan web site by Rhode Island Sea Grant.

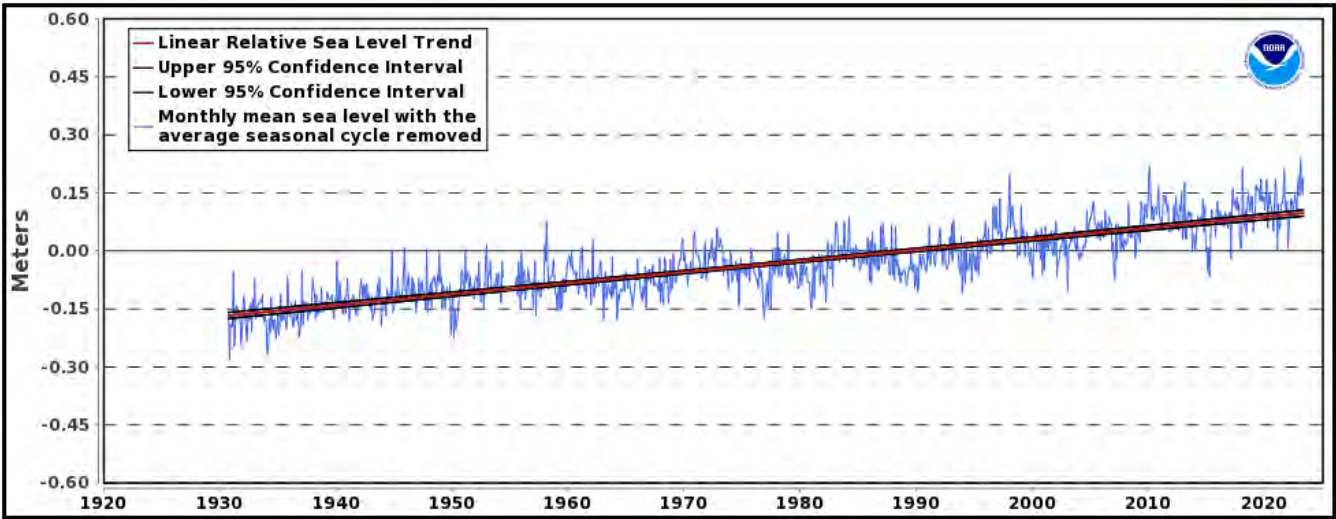
4.14.2 – Location and Extent

Data from the Shoreline Change (Beach) Special Area Management Plan indicates that:

- Since 1930, sea levels in Rhode Island have increased by an average of 1" per decade over the past half century.
- Sea levels in the northeast have been increasing three to four times faster than the global average rate.
- Rhode Island could see a potential for one foot of sea level rise by 2050.
- Sea level is projected to increase by three to five feet above 1990 levels in Rhode Island by 2100.

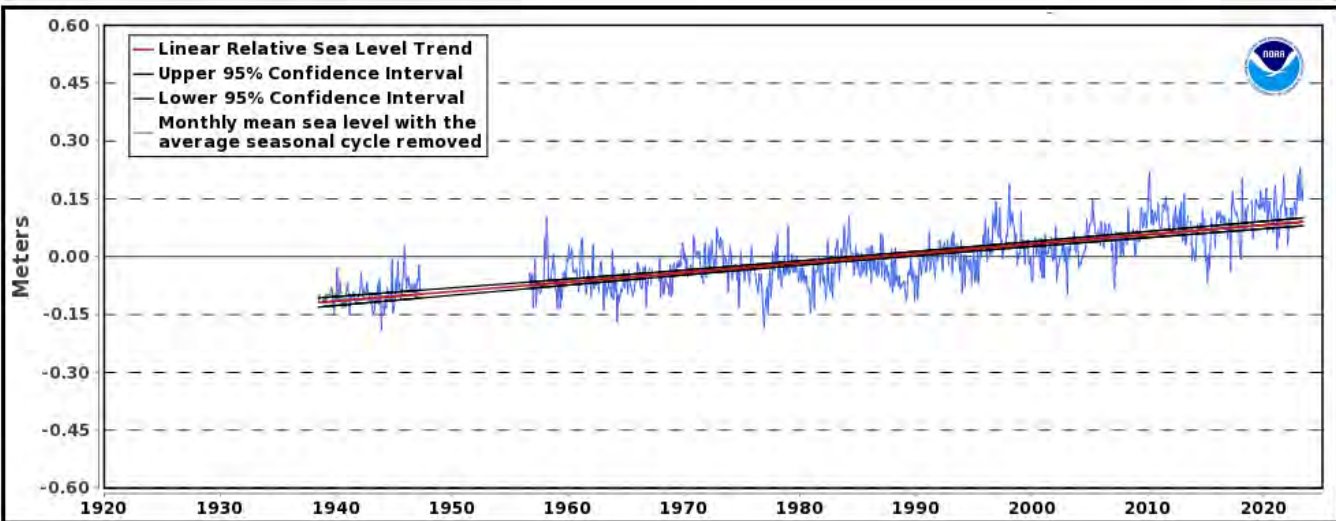
NOAA tracks sea level data at two locations in Rhode Island, the Newport and Providence tidal gauges: Data from these tidal gauges indicates that they have seen a steady increase in sea level over the past 90 years, as indicated in the following charts.

Chart 28: Newport Rhode Island Sea Level Trend



Source: NOAA

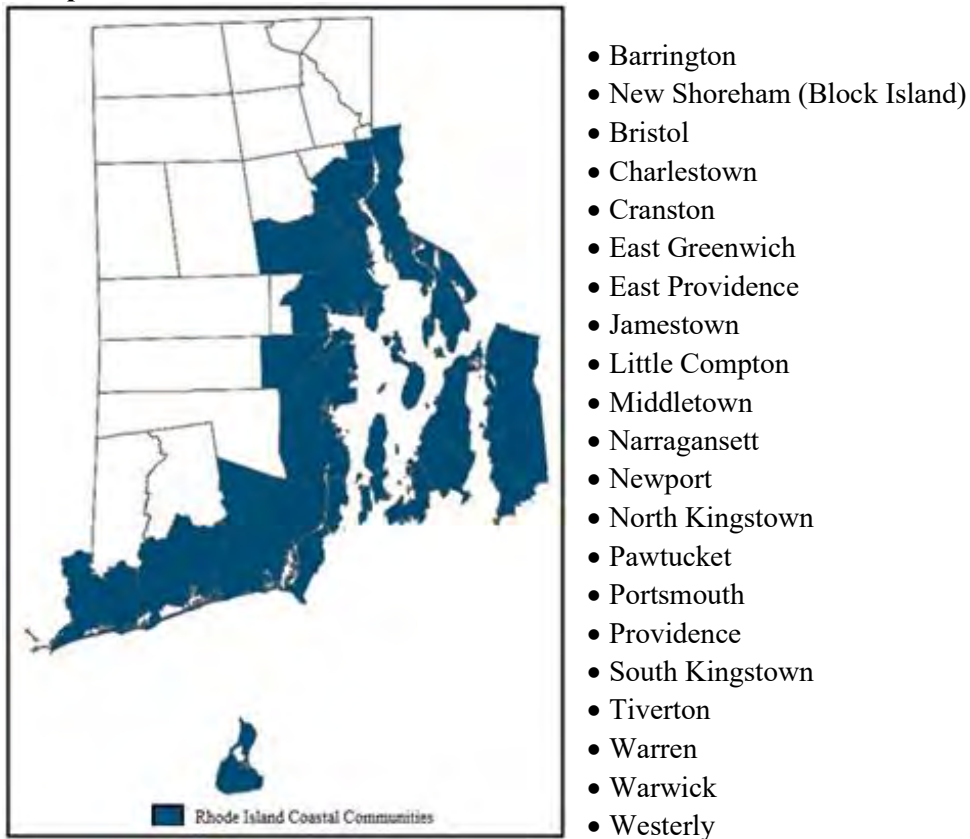
Chart 29: Providence Rhode Island Sea Level Trend



Source: NOAA

The following map, from the Socioeconomics of Sea Level Rise, Technical Paper 168, from the Rhode Island Department of Administration Division of Planning details the locations of the 21 Rhode Island coastal jurisdictions vulnerable to sea level rise: These jurisdictions are:

Map 28: Rhode Island Coastal Jurisdictions



Source: Sea Level Rise, Technical Paper 168

4.14.3 Previous Occurrences

Sea level rise is an on-going concern for the State of Rhode Island. Historically:

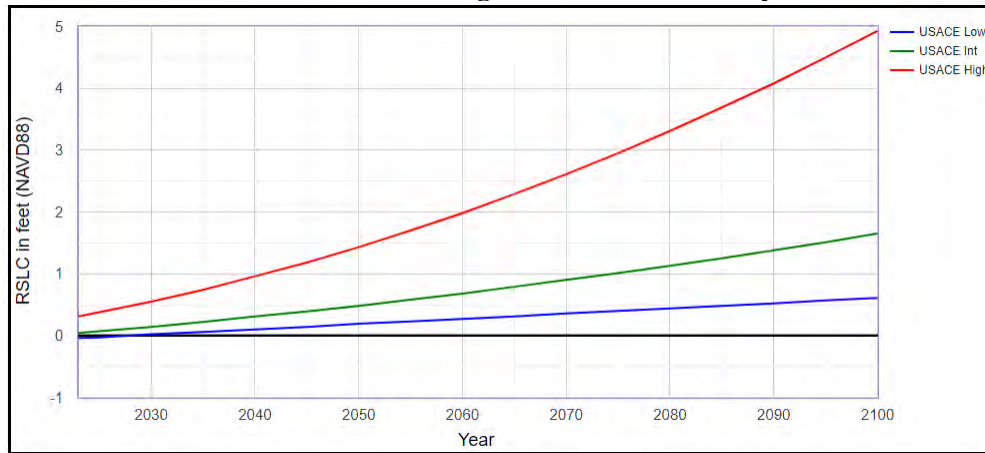
- Sea levels have risen more than 10 inches in Rhode Island since 1930
- Since 1930 the sea level in Rhode Island has increased by an average of one inch per decade over the past half century
- The water in Narragansett Bay has increased 3° F in the last 100 years, increasing thermal expansion

The Rhode Island Coastal Resources Management Council. provides a detailed mapping program indicating shoreline changes for Coastal Communities in Rhode Island.

4.14.4 Probability of Future Events

The USACE Sea-Level Change Curve Calculator was used to determine potential changes for Newport and Providence Rhode Island Tide Gauge Stations. The calculator projects three categories of change, Low, Intermediate, and High to illustrate potential sea level rise scenarios. The following charts and tables indicate potential sea level rise scenarios:

Chart 30: Estimated Sea Level Change Predictions for Newport Rhode Island



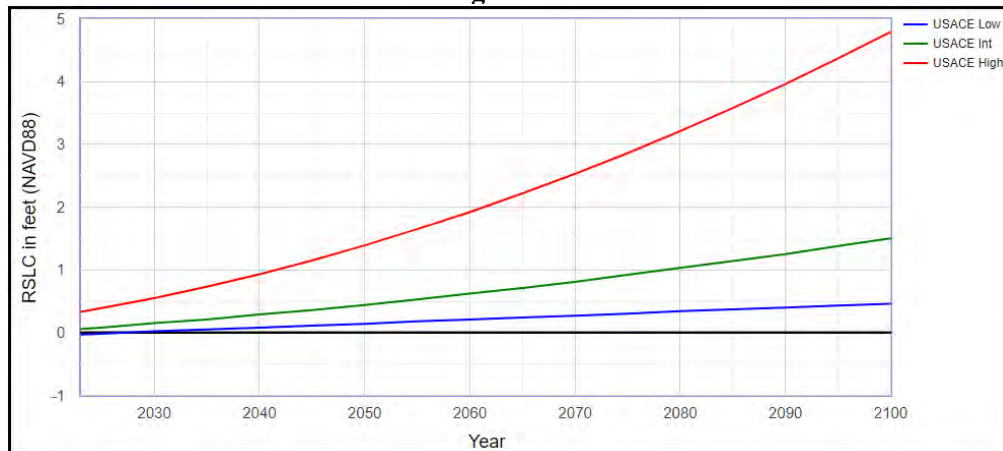
Source: USACE Sea-Level Change Curve Calculator

Table 90: Estimated Sea Level Change Predictions for Newport Rhode Island

Year	USACE Low	USACE Medium	USACE High
2023	-0.04	0.04	0.31
2025	-0.03	0.07	0.38
2030	0.02	0.14	0.55
2035	0.06	0.22	0.74
2040	0.10	0.31	0.96
2045	0.14	0.39	1.18
2050	0.19	0.48	1.43
2055	0.23	0.58	1.70
2060	0.27	0.68	1.98
2065	0.31	0.79	2.29
2070	0.36	0.90	2.61
2075	0.40	1.01	2.95
2080	0.44	1.13	3.31
2085	0.48	1.25	3.69
2090	0.52	1.38	4.08
2095	0.57	1.51	4.50
2100	0.61	1.65	4.93

Source: USACE Sea-Level Change Curve Calculator

Chart 31: Estimated Sea Level Change Predictions for Providence Rhode Island



Source: USACE Sea-Level Change Curve Calculator

Table 91: Estimated Sea Level Change Predictions for Providence Rhode Island

Year	USACE Low	USACE Medium	USACE High
2023	-0.03	0.06	0.33
2025	-0.02	0.08	0.39
2030	0.02	0.15	0.55
2035	0.05	0.21	0.73
2040	0.08	0.29	0.93
2045	0.11	0.36	1.15
2050	0.14	0.44	1.39
2055	0.18	0.53	1.65
2060	0.21	0.62	1.92
2065	0.24	0.71	2.22
2070	0.27	0.81	2.53
2075	0.30	0.92	2.86
2080	0.34	1.03	3.21
2085	0.37	1.14	3.58
2090	0.40	1.25	3.96
2095	0.43	1.38	4.37
2100	0.46	1.50	4.79

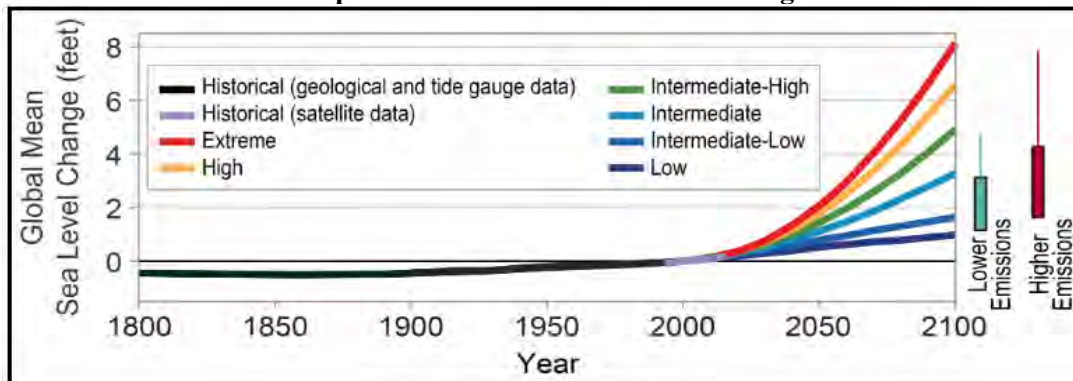
Source: USACE Sea-Level Change Curve Calculator

4.14.5 Projected Changes in Location, Intensity, Frequency, and Duration

Climate change has had, and will continue to have, significant impacts on sea level rise in Rhode Island. Rhode Island is particularly vulnerable due to its extensive coastline and low-lying coastal regions. Unless abated, Rhode Island can expect the following impacts from climate change:

Rhode Island has experienced an acceleration in sea level rise in recent decades, which is consistent with global trends. This acceleration is primarily driven by the thermal expansion of seawater as it warms and the melting of polar ice caps and glaciers. These rising sea levels exacerbate coastal erosion, which is a significant issue in Rhode Island. Coastal communities, homes, and infrastructure are increasingly vulnerable to erosion and storm damage as the shoreline recedes. Higher sea levels elevate the baseline for storm surges and high tides. This results in more frequent and severe coastal flooding events, which can disrupt daily life, damage property, and impact infrastructure.

Ongoing efforts to reduce greenhouse gas emissions on a global scale remain critical to slow the rate of sea level rise for both Rhode Island and the world. The following graph presents sea level rise projections from the U.S. Interagency Sea Level Rise Task Force relative to historical geological, tide gauge, and satellite altimeter as related to lower and higher greenhouse gas emissions.

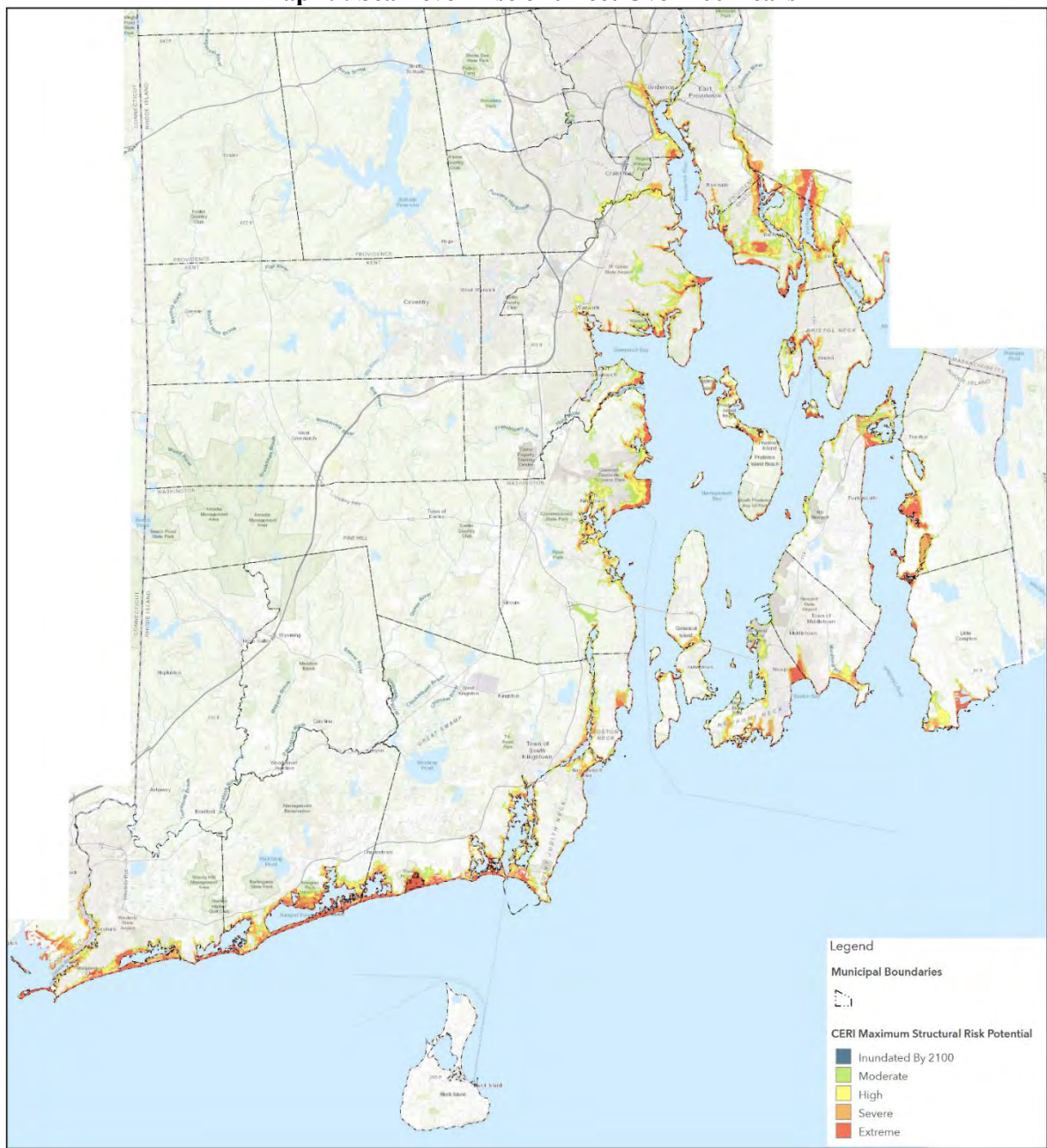
Graph X: Global Mean Sea Level Change

Source: U.S. Interagency Sea Level Rise Task Force

STORMTOOLS is a method to map storm inundation, with and without sea level rise, for varying return period storms that covers all of Rhode Island's coastal waters. Predictions are provided that show water extent and depth at any given

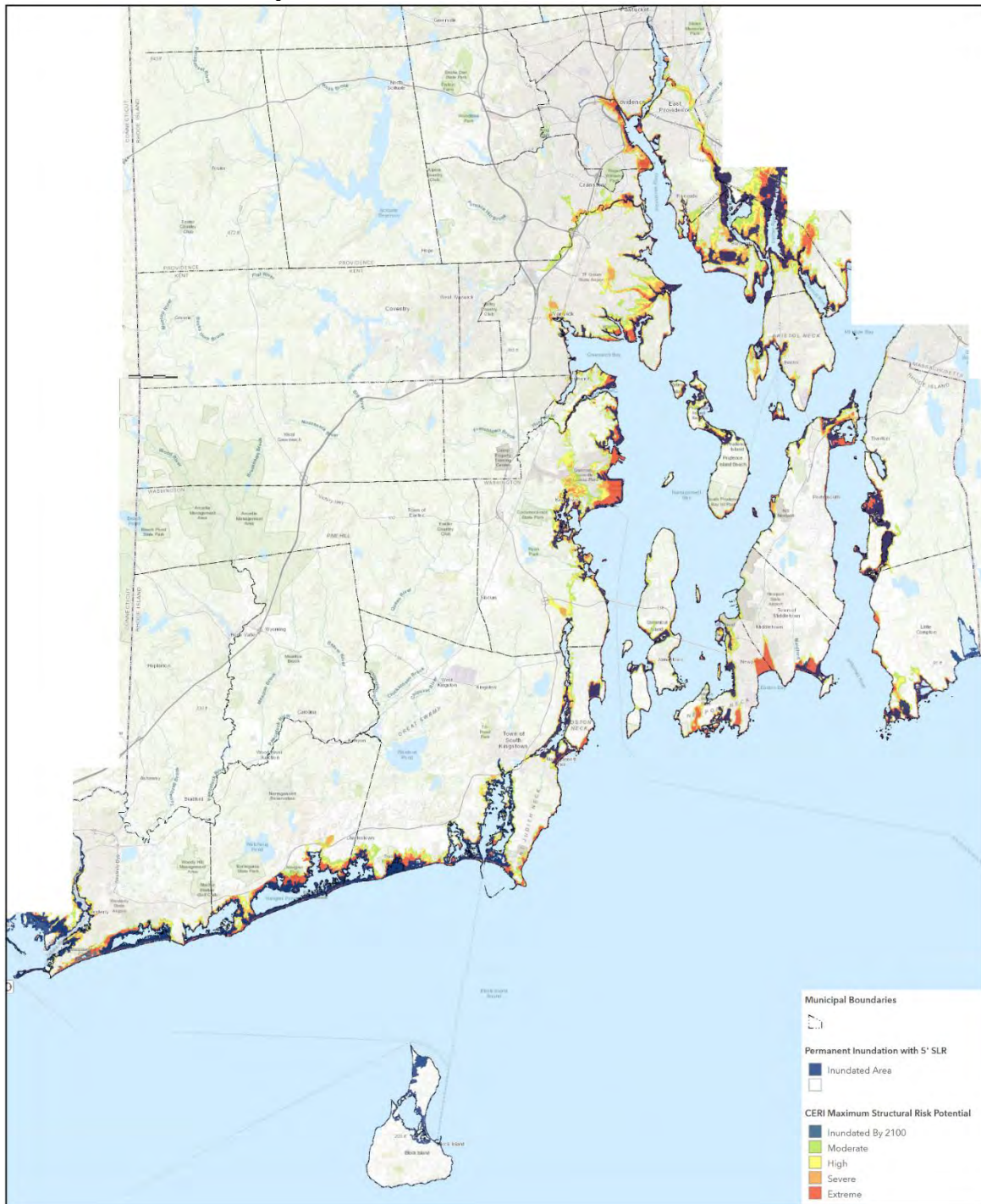
point for nuisance floods (1-, 3-, 5-, and 10-year recurrence intervals) and 25-, 50-, 100-, and 500-year storm scenarios at a 95% confidence interval. Sea level rise of 1, 2, 3, 5, and 7 feet on their own as well as combined with each storm scenario are also modeled. The following maps, using STORMTOOLS data, detail areas vulnerable to zero feet rise over 100 years and five feet rise over 100 years sea level rise:

Map 29: Sea Level Rise of 0 Feet Over 100 Years



Source: STORMTOOLS

Map 30: Sea Level Rise of 5 Feet Over 100 Years



Source: STORMTOOLS

As previously noted, State of Rhode Island assets have seen no major changes in the past five years, with only modest repairs and upgrades being conducted and no major rehabilitation or construction projects completed. As such, the risk to state assets has remained static since the completion of the 2019 SHMP. Additionally, with the exception of the two previously identified rehabilitation and construction projects, no large-scale facility projects were noted in any agency budgets or reports.

4.14.6 Vulnerability and Impact

People and Communities

Sea level rise can have a range of significant impacts on people and communities, especially those living in coastal areas. These impacts can vary depending on the rate and extent of sea level rise and the effectiveness of adaptation and mitigation measures, and may include:

- **Displacement and Migration:** As sea levels rise, low-lying coastal communities may become uninhabitable due to increased flooding and erosion. This can lead to the displacement of populations, resulting in climate-induced migration.
- **Loss of Coastal Property:** Sea level rise can lead to the gradual loss of coastal properties and infrastructure, including homes, businesses, and critical facilities. This can result in substantial economic losses for individuals and communities.
- **Increased Flooding:** Higher sea levels exacerbate coastal flooding during storms and high tides. This can lead to more frequent and severe coastal inundation events, damaging homes, infrastructure, and agriculture.
- **Saltwater Intrusion:** Rising seas can cause saltwater to intrude into freshwater sources, contaminating drinking water supplies and agricultural land. This can threaten the availability of safe drinking water and harm food production.
- **Economic Impact:** Coastal regions often host vital economic activities, such as tourism, fisheries, and ports. Sea level rise can disrupt these industries, leading to job losses and economic downturns.
- **Cultural Heritage Loss:** Coastal communities often have rich cultural and historical heritage tied to their coastal locations. Sea level rise can threaten these cultural assets.
- **Insurance and Property Values:** As the risks associated with sea level rise become more evident, property values in vulnerable coastal areas may decline, and the availability and affordability of insurance can become a challenge.
- **Social Equity:** Vulnerable and marginalized populations, including low-income communities and minority groups, often bear a disproportionate burden from the impacts of sea level rise due to limited resources and social inequalities.

The Socioeconomics of Sea Level Rise, Technical Paper 168, from the Rhode Island Department of Administration Division of Planning details the following concerning populations potentially impacted by sea level rise:

Table 92: Three Foot Sea Level Rise Scenario Population Impact Assessment

Jurisdiction	21 Coastal Jurisdictions Potentially Impacted	All of Rhode Island
Age (Largest Representation at 45-64)	34%	28%
Individuals with a Disability	7%	10%
Limited English Proficiency Households	7%	9%
Unemployment Rate	7%	9%
Below the Poverty Level	9%	14%
Occupied Housing Units	58%	89%

Source: Socioeconomics of Sea Level Rise, Technical Paper 168

Facilities and Critical Infrastructure

The following table, again using data from the Socioeconomics of Sea Level Rise, Technical Paper 168 details the number of housing units per potentially impacted communities for multiple sea level rise scenarios.

Table 93: Estimated Vulnerability of Housing Units to Sea Level Rise

Jurisdiction	1 Foot Rise	3 Foot Rise	5 Foot Rise	7 Foot Rise
Barrington	0	19	85	263
New Shoreham (Block Island)	0	3	10	19
Bristol	0	10	59	119
Charlestown	0	39	185	396
Cranston	0	1	8	22

Table 93: Estimated Vulnerability of Housing Units to Sea Level Rise

Jurisdiction	1 Foot Rise	3 Foot Rise	5 Foot Rise	7 Foot Rise
East Greenwich	0	0	0	1
East Providence	0	0	0	5
Jamestown	0	0	8	28
Little Compton	0	0	5	12
Middletown	0	0	0	3
Narragansett	0	56	153	421
Newport	0	14	225	438
North Kingstown	1	7	84	235
Pawtucket	0	0	0	0
Portsmouth	0	13	103	216
Providence	0	0	0	2
South Kingstown	7	62	229	429
Tiverton	2	25	61	89
Warren	0	8	181	368
Warwick	0	15	179	497
Westerly	0	61	316	556

Source: Socioeconomics of Sea Level Rise, Technical Paper 168

The following data from Technical Paper 164: Vulnerability of Transportation Assets to Sea Level Rise by the Rhode Island Department of Administration Division of Planning details potentially exposed transportation infrastructure. The vulnerability assessment was an exercise in determining the relative vulnerability of transportation assets under state jurisdiction.

Table 94: Estimated Vulnerability of Transportation Systems to Sea Level Rise

Impacted Infrastructure	1 Foot Rise	3 Foot Rise	5 Foot Rise
All Roads	2.3 miles	28 miles	84 miles
All Rail	10 feet	145 feet	3,685 feet
Rhode Island Public Transit Authority Routes	55 feet	2.0 miles	11.0 miles
Intermodal Terminals	2	6	7
Park and Rides	0	0	3
Ports and Harbors	17.4 acres	56.5 acres	220 acres
All Bike Infrastructure	54 feet	4,342 feet	3.1 miles

Source: Technical Paper 164: Vulnerability of Transportation Assets to Sea Level Rise

Environmental

Sea level rise can cause a variety of environmental impacts, particularly to the near-shore region due to severe instances of flooding, storm surge, and coastal erosion. The following impacts could occur:

- **Coastal Habitat Loss:** Coastal habitats like salt marshes, mangroves, and coral reefs are vulnerable to sea level rise. These habitats provide essential breeding and feeding grounds for various species, and their loss can disrupt ecosystems and threaten biodiversity.
- **Saline Intrusion:** Rising seas can push saltwater into freshwater sources like rivers and aquifers, contaminating drinking water supplies and affecting agriculture by making soil less fertile.
- **Impact on Agriculture:** Coastal agriculture is at risk due to sea level rise, as saline intrusion can reduce soil productivity.
- **Impact on Fisheries:** Rising seas can alter the distribution and abundance of marine species, affecting fisheries. Changes in water temperature and chemistry can also impact fish and shellfish populations.
- **Increased Coastal Pollution:** Flooding from sea level rise can lead to the release of pollutants stored in coastal areas, including industrial chemicals and sewage, which can harm marine ecosystems.

State Operations and Assets

State assets and operations, including infrastructure and public services, can be significantly impacted by sea level rise. The specific damage and consequences will depend on the location of the facilities, the rate of sea level rise, and the level of preparedness and adaptation measures in place. Impacts to state assets and operations may include:

- **Flooding:** Severe flooding in low-lying coastal areas would impact state assets located in these areas. The facilities may experience damage to the structure, electrical systems, and equipment due to saltwater inundation.
- **Coastal Erosion:** Coastal erosion can threaten state-owned properties near shorelines. Buildings, roads, and other infrastructure may be at risk of collapse or damage as the land they are built on is eroded away.
- **Saltwater Intrusion:** Rising sea levels can increase the risk of saltwater intrusion into freshwater aquifers and water supply sources, potentially compromising the quality and availability of drinking water for state assets.
- **Transportation Infrastructure:** Coastal roads, bridges, and public transportation systems can be impacted by sea level rise. Flooding and damage to transportation infrastructure can disrupt the movement of people and goods.
- **Emergency Services:** Sea level rise can affect the operations of state emergency services. Flooded roads and infrastructure can impede the response to natural disasters and emergencies.
- **Port Facilities:** State owned or operated port facilities may be impacted, disrupting trade and commerce. Sea level rise can impact the functionality and capacity of ports, affecting the state's economy.
- **Coastal Infrastructure:** State infrastructure such as seawalls and barrier projects may require significant upgrades or extensions to protect against sea level rise and storm surges.

The following table details information concerning the number of critical and state-owned or operated facilities in potential sea level rise areas:

Table 95: Critical and State Assets in Sea Level Rise Areas

Facility Type	Sea Level Rise Scenarios					Total
	MHHW	MHHW+1ft	MHHW+3ft	MHHW+5ft	MHHW+7ft	
Fire Stations	0	0	0	1	0	1
Law Enforcement	0	0	0	0	1	1
Power Plants	0	0	0	0	0	0
Natural Gas Pipelines	4	5	5	5	5	24
Electric Transmission Facilities	16	17	18	23	30	104
Airports	0	0	1	2	6	9
Bus Stops	0	1	9	85	109	204
Wastewater Facilities	12	0	0	0	1	13
Hazardous Material/Tier II Facilities	0	0	3	11	15	29
Flood Control Dams	3	1	3	2	2	11
FDIC Insured Banks	0	0	0	0	14	14
Schools	0	0	1	1	1	3
State Facilities	0	0	1	3	3	7
Hospitals	0	0	0	0	0	0
Nursing Facility	0	0	0	0	0	0
Communications Network Facilities	0	0	0	0	0	0
Total	35	24	41	133	187	420

Source: RIEMA and RIGIS

MHHW: Mean Higher High Water

State owned facilities in identified sea level rise areas were matched, when possible, to the master state facility list for valuation purpose. Some data discrepancies were discovered during this process causing a valuation not to be determined for all facilities in Barrington and Providence. The following table present the valuation of the state facilities identified in sea level rise areas:

Table 96: Valuation of State of Rhode Island Facilities in Sea Level Rise Areas

Jurisdiction	County	Valuation
Newport	Newport	\$1,868,147
Providence	Providence	\$11,051,413
South Kingstown	Washington	\$86,720

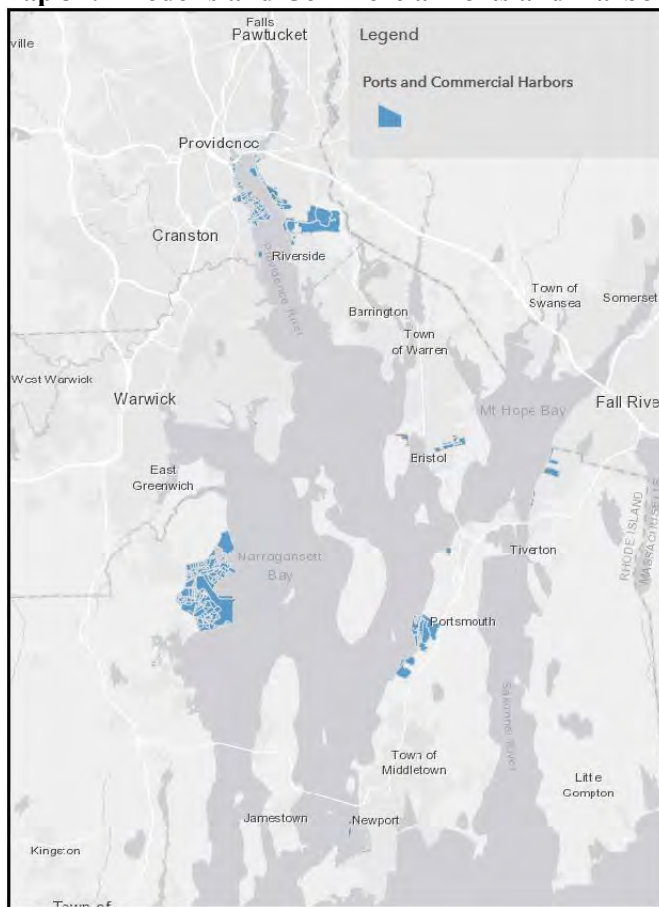
Source: RIEMA and RIGIS

When detailing state assets at risk to sea level rise (along with coastal flooding and coastal erosion), consideration must be given to ports. The Coastal Resources Division is responsible for the development, management, and maintenance of state owned and/or operated ports, including:

- Port of Galilee (Narragansett, Washington County)
- State Pier #9 (Newport, Newport County))
- State Pier #4 (Jerusalem, Washington County)
- State Pier #5 (Narragansett, Washington County)
- Chase Marina (Tiverton, Newport County)

These properties comprise 41 piers and 24 lease holdings with an approximate combined value of over \$10,000,000. The Port of Galilee is one of the largest commercial fishing ports on the East Coast and is the home port to over 250 commercial fishing vessels.

The following map details the location of ports and commercial harbors within the State of Rhode Island:

Map 31: Rhode Island Commercial Ports and Harbors

Source: RIGIS

In 2018, the state's ports moved a combined total of 9,000,000 tons of freight. The Port of Providence moves the largest volume of freight, accounting for nearly 8,500,000 tons. The Port of Providence ranks 60th in the nation by tonnage

moved, handling a variety of products and the port receiving fuel products and refined petroleum from facilities in New Jersey, Pennsylvania, Canada, and Europe. The Port of Davisville is one of the top ten auto ports in North America. The Port of Davisville supports nearly 1,700 jobs, \$98,900,000 in household earnings, and \$374,300,000 in business output annually. The Port also generates \$45,500,000 in total state tax revenue as well as \$10,500,000 in local taxes. Operated and managed by RIDEM, the Port of Galilee is home to a commercial fishing fleet with 40 piers providing dockage for 250 fishing vessels. Newport's State Pier #9 is the only state-owned commercial fishing facility and supports many commercial fishing vessels. Ports are an important source of revenue for the State of Rhode Island. The Economic Impact of Rhode Island's Fisheries and Seafood Sector from the University of Rhode Island indicates that approximately 428 fisheries firms provide 3,147 jobs and generate \$538,000,000 in gross sales.

Using GIS, sea level rise areas were overlaid on the critical facility and state facility location data. Maps are included in Appendix D.

Consequence Analysis

This consequence analysis lists the potential impacts of this hazard on various elements of community and state infrastructure. The impact of this hazard is evaluated in terms of disruption of operations, recovery challenges, and overall wellbeing to all Rhode Island residents and first responder personnel. The consequence analysis supplements the hazard profile by analyzing specific impacts.

Table 97: Sea Level Rise Consequence Analysis

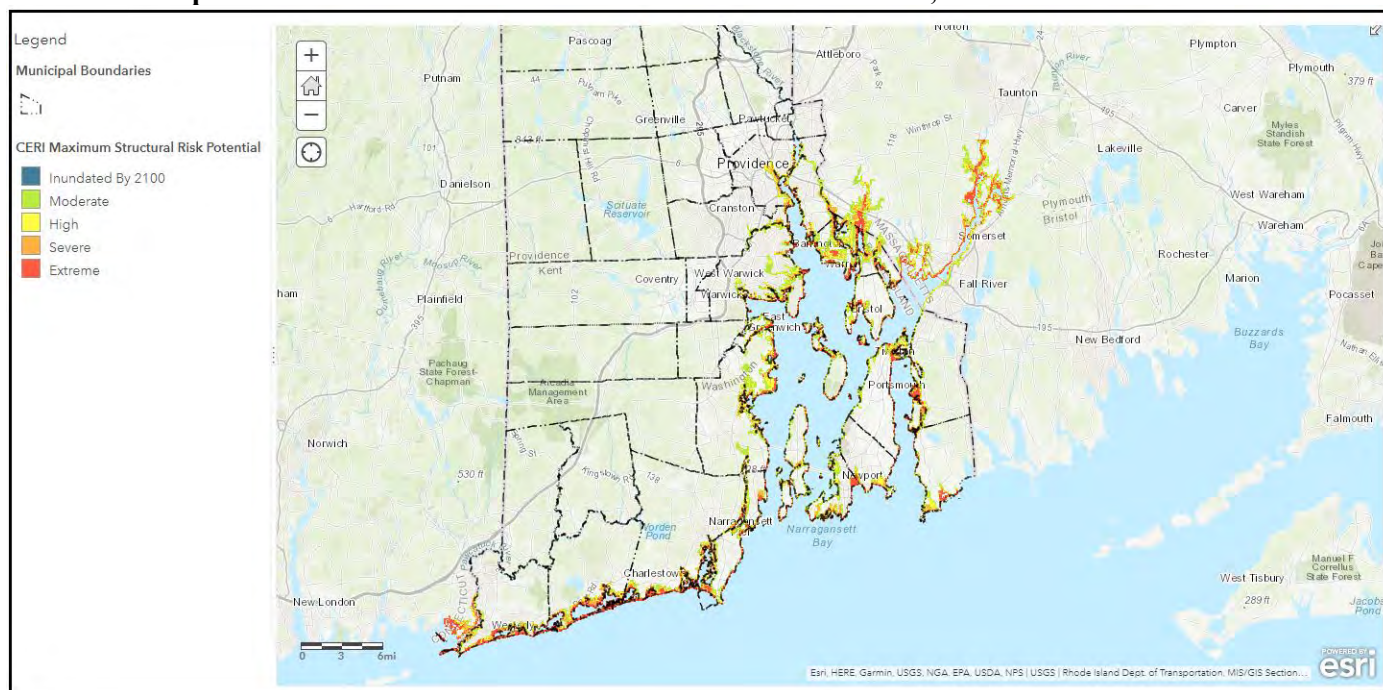
Subject	Potential Impacts
Impact on the Public	Vulnerability is restricted to the coastal communities and shoreline. Closures of hospitals and clinics is a major public health concern if overland flooding due to sea level rise causes the buildings to be uninhabitable. Water sources may also become contaminated, and water or sewer systems may be completely disrupted.
Impact on Responders	Roadways could become inundated, possibly resulting in essential facilities becoming isolated or emergency services rerouted in response to calls. Flooding may result in significant delays in responding to emergency calls.
Continuity of Operations	Rhode Island agencies maintain continuity plans which can be enacted as necessary based on the situation. This threat may impact an agency's ability to maintain continuity of operations based on the hazard's potential to cause flooding and outages of critical infrastructure. Flooding may result in limited access to primary and alternate facilities.
Delivery of Services	Delivery of services may be impacted by secondary effects, such as flooding or storm surge, causing deliveries to be delayed or halted. Waterway infrastructure may be impacted by rising waters, changing the capacity of ports to accommodate barges or ships. Flooding may also damage or destroy goods if exposed for longer periods of time.
Property, Facilities, and Infrastructure	As the sea level rises and inundation zones increase, the number of residential units at potential risk increases. Unprotected critical infrastructure may be susceptible to damage over time. If critical facilities are damaged, state services may be disrupted or more expensive given increased costs of maintaining or repairing damaged facilities.
Impact on Environment	Flooding caused by sea level rise may cause contamination of food or water resources, and water or sewer systems may be completely disrupted. Storm surge may also cause flooding (and subsequent contamination of resources).
Economic Conditions	A fiscal impact on the local and state governments may be felt. Additionally, the temporary disruption of services caused by secondary flooding may drive up the costs of response and recovery. Local, county, and state resources may not be sufficient to cover the costs of implementing mitigation projects.
Public Confidence in Governance	The public's confidence in the state's governance is affected by the immediate provision of effective response actions. Efficiency in response and recovery operations is critical in keeping public confidence high. Earning the public's trust will also depend on the state's ability to demonstrate that mitigation projects are effective.

4.14.7 Jurisdictional Vulnerability and Planning Significance

Data from FEMA's NRI concerning sea level rise was not available for review. Additionally, a review of LHMPs indicates that planners elected to cover the subject under other identified hazards, largely flooding. To facilitate an assessment of jurisdictional risk to sea level rise, STORMTOOLS, an interactive mapping tool developed by the Rhode Island Coastal Resource Management Council and the University of Rhode Island was used. The system uses modelling tools to predict storm surge and wave, combined with shoreline change maps (erosion), and damage functions to construct a Coastal Environmental Risk Index. Access to the state emergency database (E-911) provides information on structure characteristics. Coastal Environmental Risk Index has been designed as a GIS based tool, and hence is fully compatible with current flooding maps, including those from FEMA. The basic framework and associated GIS methods can be readily applied to any coastal area.

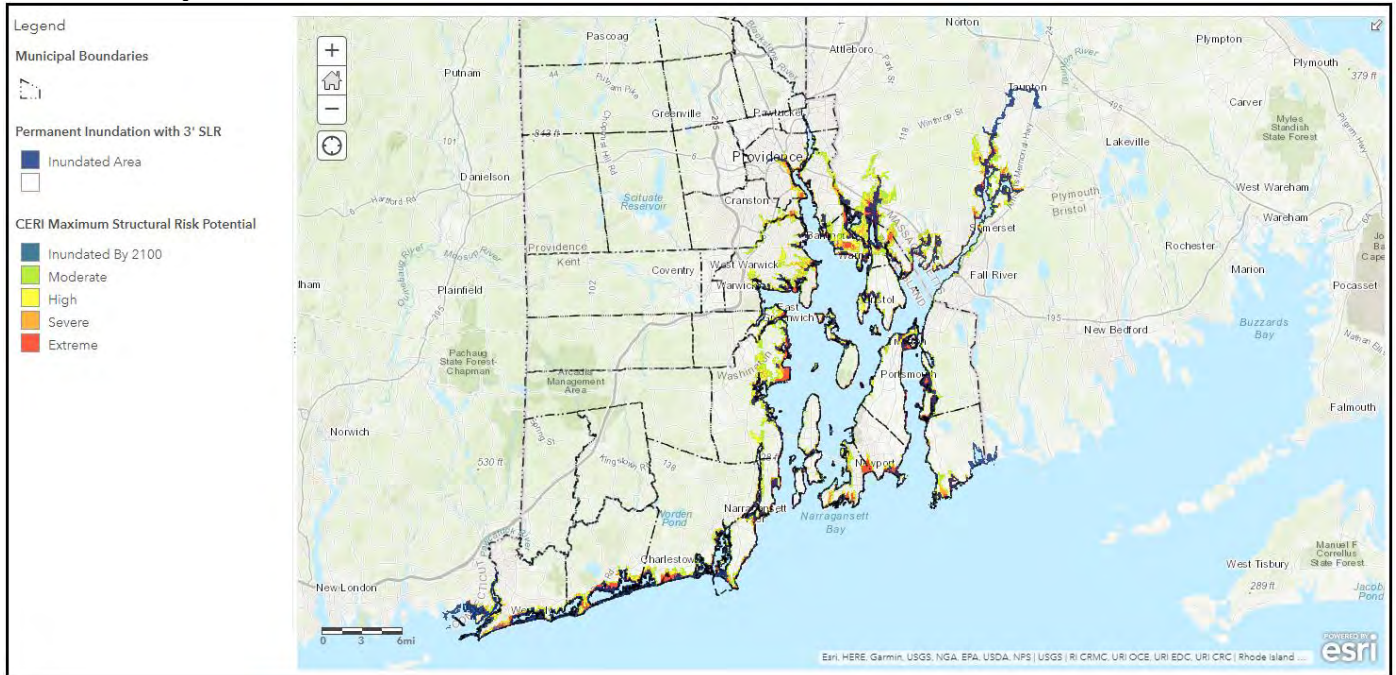
The following STORMTOOLS maps detail the Coastal Environmental Risk Index maximum structural risk potential for three sea level rise scenarios, a sea level rise of zero feet, a sea level rise of three feet, and a sea level size of five feet in order to provide best and worst-case vulnerability scenarios:

Map 32: STORMTOOLS Maximum Structural Risk Potential, Zero Feet Sea Level Rise



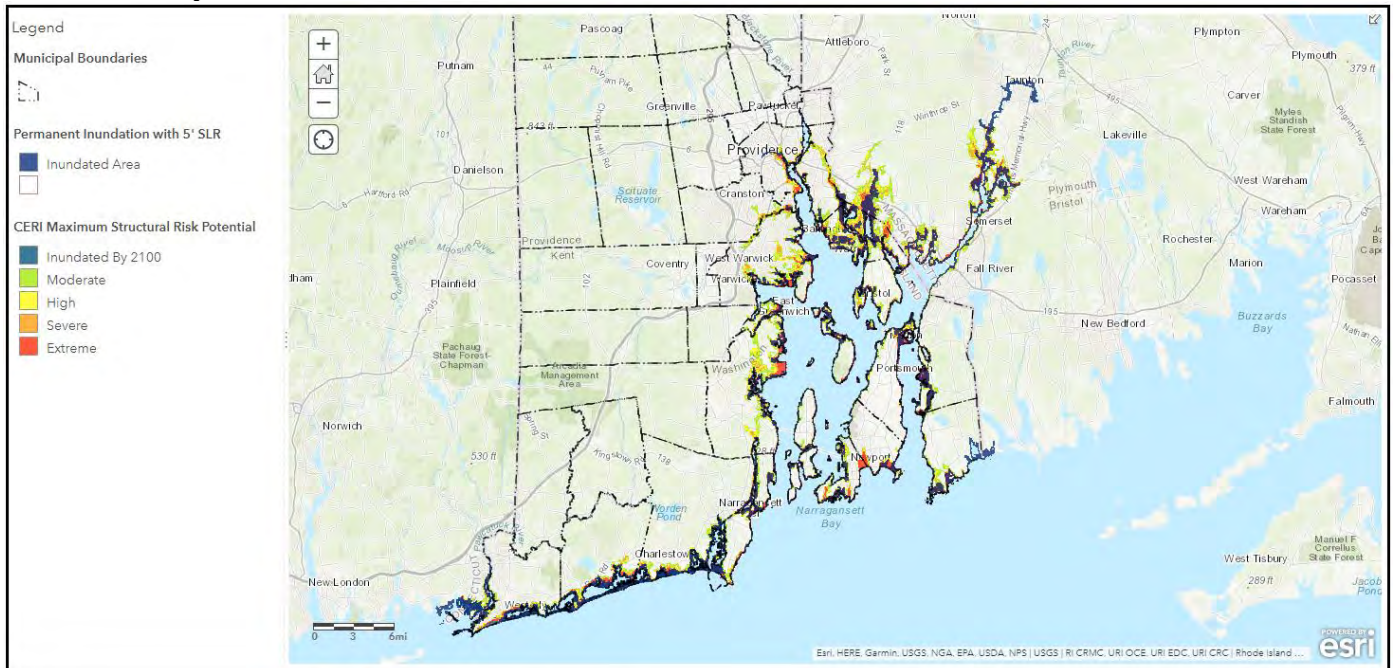
Source: STORMTOOLS

Map 33: STORMTOOLS Maximum Structural Risk Potential, Three Feet Sea Level Rise



Source: STORMTOOLS

Map 34: STORMTOOLS Maximum Structural Risk Potential, Five Feet Sea Level Rise



Source: STORMTOOLS

Sea level rise vulnerability is restricted to the coastal communities of Rhode Island. As indicated on the above maps, and in the data above indicating the 21 Rhode Island Communities vulnerable to seal level rise. Washington County has potential vulnerability and losses in North Kingstown, Narragansett, South Kingstown, Charlestown, and Westerly. All municipalities within Newport County are vulnerable to inundation and losses; these include Jamestown, Newport, Middletown, Portsmouth, Tiverton, and Little Compton. Barrington, Warren, and Bristol within Bristol County are also vulnerable, with the entire extent of Barrington and Warren at eminent risk due to the elevation of the community in relation to the potential inundation.

A household is considered cost burdened if it spends 30% or more of its income on housing costs, with cost burdens being an indication of rents increasing faster than incomes. Cost burdened citizens may have great difficulty moving residences in the event of sea level rise due to economic constraints. The following table, using data from the Housing and Urban Development’s Comprehensive Housing Affordability Strategy Data for Rhode Island, indicates jurisdictions located in potential sea level rise areas with the most cost burdened renters who may be at greater risk to due to economic constraints.

Table 98: Jurisdictions at Greater Risk to Severe Winter Storms due to Housing Cost Burden

County	Jurisdiction	Cost Burdened Single Families
Washington County	Narragansett	65%
Bristol County	Barrington	64%
Newport County	Portsmouth	54%
Bristol County	Warren	52%
Kent County	Warwick	50%
Providence County	East Providence	49%
Newport County	Jamestown	47%
Washington County	New Shoreham	47%
Providence County	Providence	47%
Washington County	Westerly	47%

Source: Housing and Urban Development’s Comprehensive Housing Affordability Strategy Data

4.14.8 Hazard Planning Significance

Based on a review of all available data, the SIHMC determined sea level to have a moderate planning significance for Rhode Island.

4.15 Severe Thunderstorm

4.15.1 Hazard Description

Severe thunderstorms comprise the hazardous and damaging weather effects often found in violent storm fronts. They can occur together or separate, they are common and usually not hazardous, but on occasion they can pose a threat to life and property.

This plan defines severe thunderstorms as a combination of the following severe weather effects as defined by NOAA and the NWS:



- **Hail:** Precipitation in the form of irregular pellets or balls of ice more than 5 mm in diameter, falling from a cumulonimbus cloud.
- **Lightning:** A visible electrical discharge produced by a thunderstorm. The discharge may occur within or between clouds, between the cloud and air, between a cloud and the ground or between the ground and a cloud.
- **Thunderstorm Winds:** The same classification as high or strong winds but accompanies a thunderstorm. It is also referred to as a straight-line wind to differentiate from rotating or tornado associated wind. Additionally, these winds can rapidly create dust storms that severely impact visibility.

Severe thunderstorms have been so consistent throughout modern history that much of the vulnerability is mitigated. However, this section is not concerned with everyday wind, lightning in the sky, or mild precipitation. This section is concerned with common storm elements when they behave such that they pose a threat to property and life.

4.15.2 – Location and Extent

Severe thunderstorms can rapidly descend on an area, but in many cases are predictable. Most weather forecasts focus on more than just temperature but on quickly changing conditions that may lead to the onset of severe storms. All of Rhode Island is susceptible to severe thunderstorms.

The NWS classifies thunderstorms, often the generator of hail, lightning, and high winds, using the following categories.

- **Marginal:** Isolated severe thunderstorms, limited in duration and/or coverage and/or intensity
- **Slight:** Scattered severe storms possible, short-lived and/or not widespread, isolated intense storms possible
- **Enhanced:** Numerous severe storms possible, more persistent and/or widespread, a few intense
- **Moderate:** Widespread severe storms likely, long-lived, and intense
- **High:** Widespread severe storms expected, long-lived and particularly intense

In the United States, hail causes billions of dollars in damage to property each year. Vehicles, roofs of buildings and homes, and landscaping are most commonly damaged by hail. Hail has been known to cause injury and the occasional fatality to humans, often associated with traffic accidents.

Based on information provided by the National Weather Service concerning size, the following table describes potential damage impacts of the various sizes of hail.

Table 99: Hail Size Comparison and Damage Descriptions

Diameter (inches)	Size Description	Potential Damage Impacts
1/4	Pea Size	No damage
1/2	Mothball, peanut, USB Plug	Slight damage to vegetation
3/4	Penny Size	Increased damage to crops and vegetation
7/8	Nickel Size	Severe damage to crops and vegetation, damage begins to glass and plastic
1	Quarter Size	Increased glass damage, damage begins to bodies of vehicles

Table 99: Hail Size Comparison and Damage Descriptions

Diameter (inches)	Size Description	Potential Damage Impacts
1 1/4	Half Dollar Size	Large scale glass damage, begin roof damage, risk of injury to exposed persons
1 1/2	Ping Pong Ball Size	Large scale glass damage, begin roof damage, increased risk of injury to exposed persons
1 3/4	Golf Ball Size	Severe roof damage, risk of serious injuries to exposed persons
2	Lime or medium sized Hen Egg	Potential structural damage, risk of very severe injuries to exposed persons
2 1/2	Tennis Ball Size	Extensive structural damage, risk of very severe injuries or death to exposed persons

Source: National Weather Service

A recent report by the Insurance Information Institute says lightning strikes caused \$1,300,000,000 in damage across the United States in 2021. There is currently no scale to indicate the severity of a lightning strike, but data from NOAA indicates that there approximately 25,000,000 cloud-to-ground lightning strikes per year in the United States.

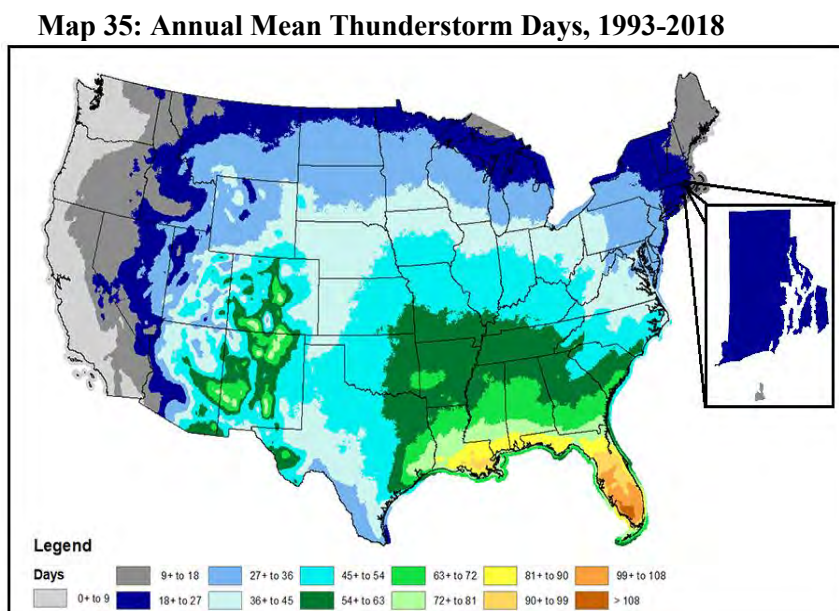
To measure wind speed and its correlating potential for damage, experts use the Beaufort scale as shown below.

Table 100: Beaufort Scale

Beaufort Number	Wind Speed (mph)	Effects on Land	Effects on Water
0	Under 1	Calm, smoke rises vertically	Sea surface smooth and mirror-like
1	1-3	Smoke drift indicates wind direction, vanes do not move	Scaly ripples, no foam crests
2	4-7	Wind felt on face, leaves rustle, vanes begin to move	Small wavelets, crests glassy, no breaking
3	8-12	Leaves, small twigs in constant motion. Light flags extended.	Large wavelets, crests begin to break, scattered whitecaps
4	13-18	Dust, leaves, and loose paper raised up; small branches move	Small waves (1-4 feet) becoming longer, numerous whitecaps
5	19-24	Small trees begin to sway	Moderate waves (4-8 feet) taking longer form, many whitecaps, some spray
6	25-31	Large branches of trees in motion, whistling heard in wires	Larger waves (8-13 feet), whitecaps common, more spray
7	32-38	While trees in motion, resistance felt in walking against the wind	Sea heaps up, waves 13-19 feet, white foam streaks off breakers
8	39-46	Twigs and small branches broken off trees	Moderately high (18-25 feet) waves of greater length, edges of crests begin to break into spindrift, foam blown in streaks
9	47-54	Slight structural damage occurs, slate blown from roofs	High waves (23-32 feet), sea begins to roll, dense streaks of foam, spray reduces visibility
10	55-63	Seldom experienced on land, trees broken, structural damage occurs	Very high waves (29-41 feet) with overhanging crests, sea white with densely blown foam, heavy rolling, lowered visibility
11	64-72	Very rarely experienced on land, usually with widespread damage	Exceptionally high (37-52 feet) waves, foam patches cover sea, visibility more reduced
12	73 or higher	Violence and destruction	Air filled with foam, waves over 45 feet, sea completely white with driving spray, visibility greatly reduced

Source: NOAA

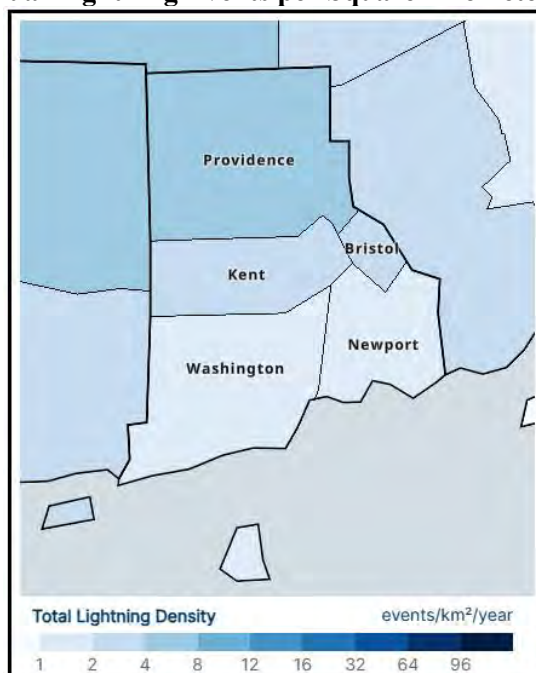
The widespread and frequent nature of thunderstorms makes hail, lightning, and high wind a relatively common occurrence for Rhode Island. The following map, from NOAA, indicates annual mean thunderstorm days from 1993 to 2018:



Source: NOAA

The following map, from Vaisala, indicates the average annual light events per square kilometer per year for Rhode Island:

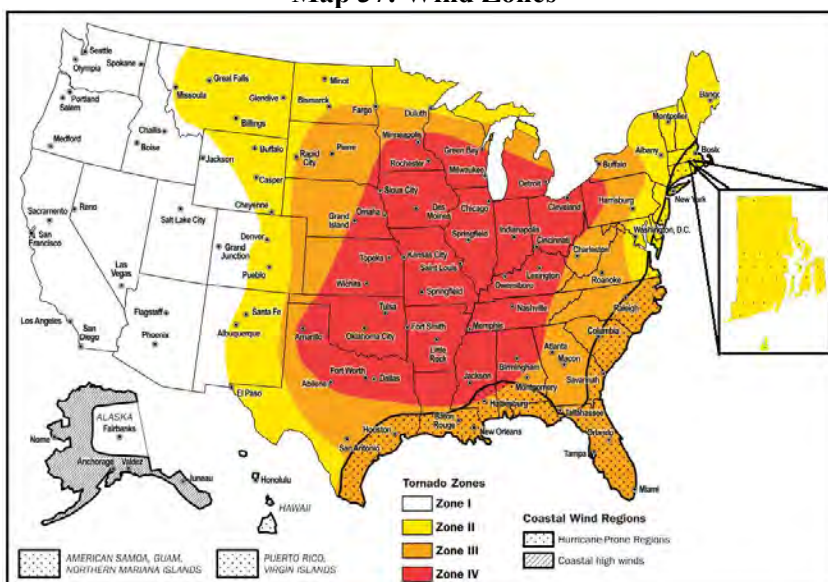
Map 36: Average Annual Lightning Events per Square Kilometer per Year, 2016 - 2022



Source: Vaisala

The following map from FEMA indicate the highest possible expected wind speeds for Rhode Island:

Map 37: Wind Zones



Source: FEMA

4.15.3 Previous Occurrences

Historical events of significant magnitude or impact can result in a Presidential Disaster Declaration. Rhode Island has experienced two Presidential Disaster Declarations related to severe storms, reflected in the following table:

Table 101: State of Rhode Island Presidentially Declared Disasters

Designation	Declaration Date	Incident Type	Individual Assistance	Public Assistance
DR-1894-RI	3/29/2010	Severe Storms and Flooding	\$37,157,115	\$24,263,849
DR-1704-RI	5/25/2007	Severe Storms and Island/Coastal Flooding	\$605,080	\$529,557

Source: FEMA

In addition to the Presidentially Declared Disasters, the following table presents NCEI identified severe thunderstorm events and the resulting damage totals in Rhode Island from 1950 to 2023:

Table 102: NCEI Rhode Island Thunderstorm Events

County	Event Type	Number of Days with Events	Property Damage	Deaths and Injuries
Bristol County	Hail	4	\$0	0
	Lightning	5	\$82,000	1
	Winds	89	\$366,600	0
Kent County	Hail	24	\$0	0
	Lightning	8	\$464,000	9
	Winds	156	\$1,252,000	0
Newport County	Hail	9	\$0	0
	Lightning	7	\$73,000	0
	Winds	114	\$645,900	1
Providence County	Hail	41	\$20,000	0
	Lightning	23	\$665,000	9
	Winds	209	\$2,741,000	7
Washington County	Hail	17	\$0	0
	Lightning	7	\$49,000	2
	Winds	169	\$1,143,000	1

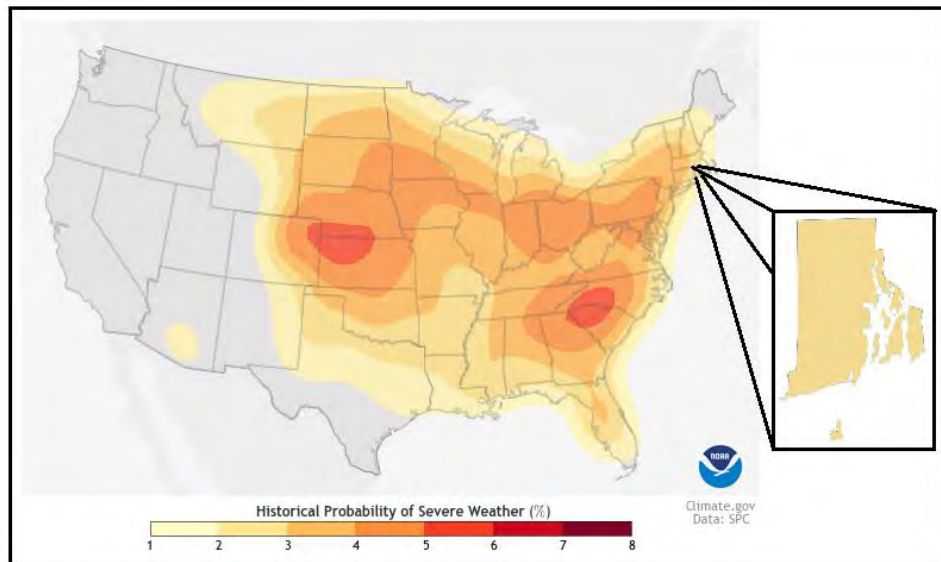
Source: National Centers for Environmental Information

It is worth noting that damage estimates indicated by the NCEI are often artificially low. This underreporting is a result of the way the events are reported, often by the local and/or NWS office. When reporting an event oftentimes the NWS office does not have access to the actual damage assessment resulting from that event. As such, the report often details a very low amount or zero-dollar amount for damages. Additionally, deaths and injuries may be underreported as they may be a result of a concurrent event, such as a person driving unsafely during heavy rain and passing away.

4.15.4 Probability of Future Events

Predicting the probability of severe thunderstorm occurrences is tremendously challenging due to the large number of factors involved and the random nature of formation. Data from NOAA indicates that Rhode Island can expect between 18 to 27 thunderstorms per year. Additionally, the following map from NOAA provides a probability snapshot for the probability of a severe weather event on a summer day.

Map 38: Historic Probability of a Severe Weather Summer Event In Rhode Island



Source: NOAA

4.15.5 Projected Changes in Location, Intensity, Frequency, and Duration

Climate change can have several impacts on severe thunderstorms, although the precise details can vary depending on regional climate patterns and other factors. In general, it is believed that climate change can alter the timing and seasonality of severe thunderstorms. In some cases, this may mean more thunderstorms occurring earlier or later in the year.

Climate change can lead to increased temperatures and moisture levels in the atmosphere, which can provide favorable conditions for the development of severe thunderstorms. This can result in a higher frequency of severe thunderstorm events and an increase in their intensity. As a result of increased temperatures, warmer air can hold more moisture, leading to increased rainfall during severe thunderstorms. This can elevate the risk of flash flooding, particularly in areas prone to heavy precipitation. Changes in atmospheric circulation patterns associated with climate change can lead to stronger winds within thunderstorms. This can result in more powerful wind gusts, increasing the risk of wind damage and downed trees and power lines.

Climate change can influence the conditions necessary for hail formation within thunderstorms. Warmer temperatures at the surface and greater instability in the atmosphere can contribute to larger and more damaging hailstones. Additionally, changes in atmospheric conditions can affect the frequency and distribution of lightning strikes associated with thunderstorms. More lightning can increase the risk of wildfires in dry regions.

It is important to note that while there is evidence linking climate change to changes in weather patterns that can influence severe thunderstorms, predicting specific thunderstorm events remains challenging. Climate models provide

valuable insights into long-term trends, but individual thunderstorm events are influenced by a complex interplay of factors.

As previously noted, State of Rhode Island facilities have seen no major changes in the past five years, with only modest repairs and upgrades being conducted and no major rehabilitation or construction projects completed. As such, the risk to state has remained static since the completion of the 2019 SHMP. Additionally, with the exception of the two previously identified rehabilitation and construction projects, no large-scale facility projects were noted in any agency budgets or reports.

4.15.6 Vulnerability and Impact

People and Communities

Severe thunderstorms can have a wide range of effects on people, often posing significant risks to life, property, and general well-being. In the absence of proper shelter, hail, lightning, and high winds can cause serious injury. In general, if potentially exposed persons take shelter in a solid, well-constructed structure protection from these severe thunderstorm components would be provided. However, old or poorly constructed facilities may be more prone to damage, potentially increasing the impact on economically disadvantaged populations. Some of the potential effects of severe thunderstorms on people may include:

- **Death and Injury:** Thunderstorms produce lightning and strong winds driving debris. Both of these elements can cause injuries or fatalities.
- **Power Outages:** Lightning strikes, strong winds, and falling trees can lead to power outages, disrupting daily life, and potentially affecting essential services, such as medical equipment and refrigeration.
- **Mental Health Impact:** Severe thunderstorms can be frightening and stressful, leading to anxiety and post-traumatic stress disorder in some individuals. The emotional toll of property damage and loss can also be significant.
- **Displacement:** People may need to evacuate their homes or be temporarily displaced due to storm damage, requiring emergency shelter and support.
- **Economic Costs:** Severe thunderstorms result in economic costs, including repair and recovery expenses, insurance claims, and potential loss of income due to property damage or work disruptions.
- **Public Safety Response:** Severe thunderstorms can strain public safety resources, including emergency services, law enforcement, and medical facilities.

Facilities and Critical Infrastructure

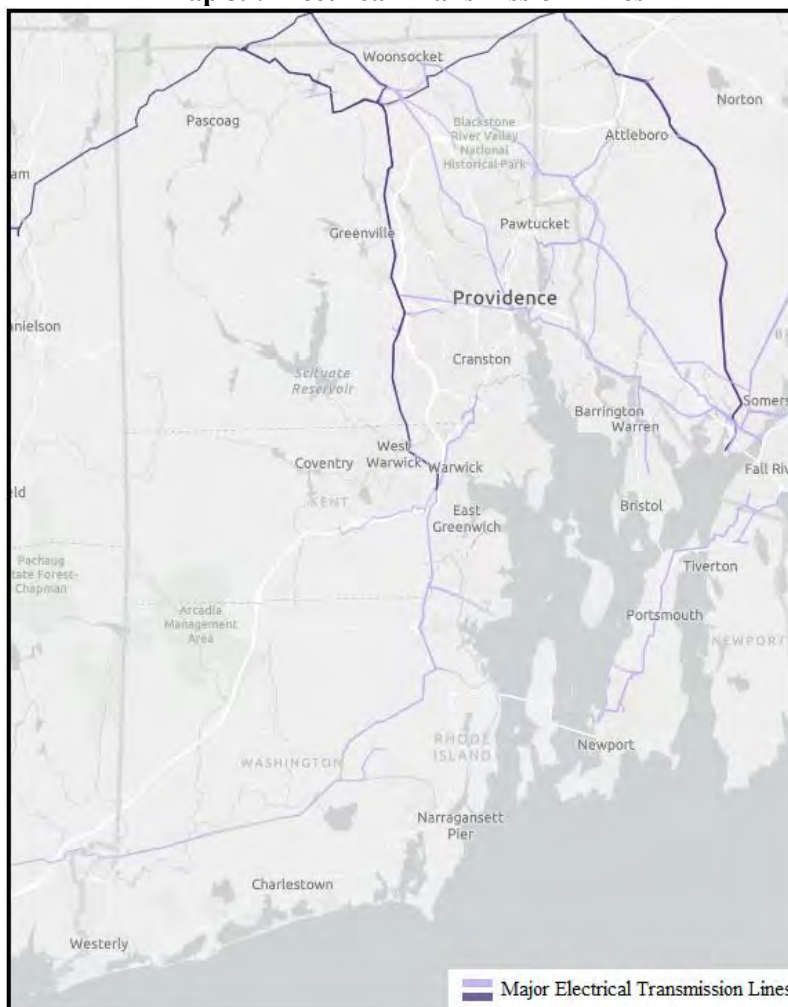
All facilities within Rhode Island can be impacted by severe thunderstorms, including critical facilities. However, the construction, and date of construction, of the facility will have a significant impact on the vulnerability. In general, older structures would be considered to be at higher risk to negative impacts due to the application of less stringent building codes. Some of the potential impacts include:

- **Electrical Infrastructure Damage:** Severe thunderstorms can damage electrical infrastructure, including power lines, transformers, and substations. This can result in widespread power outages, affecting homes, businesses, hospitals, and other critical facilities.
- **Communication Disruptions:** Thunderstorms can disrupt telecommunications infrastructure, including cell towers, data centers, and communication networks. This can impact emergency communication and coordination efforts.
- **Transportation Disruptions:** Heavy rain, strong winds, and flooding can damage roads, bridges, and transportation networks. This can lead to transportation disruptions, accidents, and delays, affecting the movement of goods and people.
- **Airport Closures:** Thunderstorms can force the closure of airports due to safety concerns, affecting air travel and cargo shipments.
- **Water and Wastewater Systems:** Severe storms can overwhelm water treatment plants and wastewater facilities, leading to contamination and water supply disruptions. Flooding can also damage water infrastructure.

- **Critical Facilities:** Hospitals, emergency response centers, and other critical facilities may be affected by power outages, flooding, and damage to infrastructure. This can impact the ability to provide essential services during and after the storm.
- **Energy Generation:** Thunderstorms can disrupt energy generation facilities, such as wind farms and solar installations, and damage conventional power plants. This can affect the availability of electricity.
- **Safety Risks:** Damage to infrastructure can pose safety risks to workers and the public. Fallen power lines, damaged buildings, and debris can be hazardous.

Electrical utilities within the State of Rhode Island are privately managed and maintained. GIS analysis indicates that there are approximately 419 miles of major electrical transmission lines within the State. The following map details electrical transmission lines within the state, all of which are vulnerable to the effects of severe thunderstorms:

Map 39: Electrical Transmission Lines



Source: U.S. Energy Information Administration

Environment

Severe thunderstorms can pose various risks to the environment. These risks can have both short-term and long-term impacts on natural ecosystems. Severe thunderstorms can produce heavy rainfall over a short period of time, leading to flash floods and riverine flooding. This can result in soil erosion, damage to aquatic habitats, and the displacement of aquatic organisms. Large hailstones can damage crops, vegetation, and natural habitats. Hail can strip leaves from trees and plants, reducing their ability to photosynthesize and grow. It can also damage wildlife habitats. Severe thunderstorms often produce strong straight-line winds. These winds can uproot trees, damage forests, and disrupt animal habitats. They can also scatter debris and cause structural damage to buildings, which can lead to further environmental issues if hazardous materials are released. Lightning is a common occurrence in thunderstorms and can

spark wildfires. These wildfires can have significant ecological impacts, including habitat destruction, loss of wildlife, and changes in the local ecosystem.

State Operations and Assets

Severe thunderstorms can pose various risks to state operations and facilities, with all facilities being vulnerable. These risks can have significant economic and operational consequences, and can include:

- **Structural Damage:** High winds, hail, and tornadoes associated with severe thunderstorms can cause significant damage to state buildings and infrastructure. This can result in costly repairs and disruptions.
- **Power Outages:** Severe thunderstorms can lead to power outages by damaging electrical infrastructure such as power lines and substations. State buildings may lose power, affecting critical operations and services.
- **Flooding:** Heavy rainfall during severe thunderstorms can lead to flooding, which can damage state buildings and disrupt operations. Flood damage may require extensive repairs and cleanup.
- **Communication Disruptions:** Lightning strikes can damage communication equipment, including telephone lines and computer systems. This can hinder communication between state agencies and the public.
- **Transportation Disruptions:** Severe thunderstorms can make roads impassable due to flooding or fallen trees. This can impact the ability of employees to commute to work and can disrupt the delivery of goods and services.
- **Emergency Response:** Severe thunderstorms may require the activation of emergency response plans. This can strain resources and personnel, especially if the storms lead to widespread damage or evacuations.
- **Loss of Records and Data:** Flooding or equipment damage can result in the loss of important records and data. This can have legal and operational implications.
- **Budgetary Impact:** The costs associated with repairing and restoring state buildings and infrastructure after severe thunderstorms can strain budgets.

Depending on the severity of the thunderstorm, all state owned or operated facilities are vulnerable to potential impacts. The following table indicates the number and valuation of the building envelope and building contents for all state owned or operated facilities:

Table 103: State of Rhode Island Facilities and Assets by County

Jurisdiction	State Owned or Operated Facilities	Valuation
Bristol County	32	\$150,415,810
Kent County	116	\$215,026,614
Newport County	77	\$127,353,837
Providence County	423	\$2,526,593,056
Washington County	393	\$274,350,750
State Total	1,041	\$3,293,740,067

Source: RIEMA

Note: Valuations include structure and assets

The age of a state owned or operated structure could have a significant impact on its resilience to a severe thunderstorm, especially high winds, as an older structure may be less resilient due to the following factors:

- **Building Codes and Standards:** Older buildings may not have been constructed to meet modern building codes, which often incorporate lessons learned from previous disasters. Newer buildings are typically designed to be more resilient to known hazards.
- **Construction Materials and Techniques:** Older buildings may have been constructed using materials and techniques that are less resilient to hazards. Modern construction materials and methods, such as reinforced concrete and steel frames, are often more resilient to hazards.

The following table indicates the average year of construction for state-owned or operated facilities with a known completion date, by county, for the State of Rhode Island:

Table 104: Average Year of Construction for State of Rhode Island Owned or Operated Facilities by County

Jurisdiction	Average Year of Construction
Bristol County	1966
Kent County	1983
Newport County	1978
Providence County	1961
Washington County	1964
State Buildings Average Year of Construction	1969

Source: RIEMA

Note: Average age based on buildings with known construction date

Of particular concern, a subset of state owned or operated facilities consists of manufactured buildings. Data from the NOAA National Severe Storms Laboratory reports that people working in manufactured buildings are especially at risk for injury and death as even anchored manufactured buildings can be seriously damaged when winds gust over 80 miles per hour. The following table indicates the number and valuation of the building envelope and building contents for all state owned or operated manufactured buildings:

Table 105: State of Rhode Island Manufactured Facilities and Assets by City and County

Jurisdiction	County	Number of Manufactured Buildings	Valuation
Bristol	Bristol County	1	\$174,599
Burrillville	Providence County	4	\$431,464
Cranston	Providence County	2	\$859,871
Exeter	Washington County	1	\$749,043
Portsmouth	Newport County	1	\$8,928,598
State Total	-	6	\$11,143,575

Source: RIEMA

Note: Valuations include structure and assets

Consequence Analysis

This consequence analysis lists the potential impacts of this hazard on various elements of community and state infrastructure. The impact of this hazard is evaluated in terms of disruption of operations, recovery challenges, and overall wellbeing to all Rhode Island residents and first responder personnel. The consequence analysis supplements the hazard profile by analyzing specific impacts.

Table 106: Severe Thunderstorm Consequence Analysis

Subject	Potential Impacts
Impact on the Public	Thunderstorms can cause extensive property damage, loss of utility service, and injury to the public. Those most at-risk are low-income and homeless individuals without proper shelter.
Impact on Responders	First responders may be unable to access roadways due to flooding, trees, or debris. Exposure to lightning, flooding, and high winds may cause injuries to first responders. Vehicles and resources may be damaged, leading to impaired response activities. In addition, road conditions may become hazardous as a result of the by-products
Continuity of Operations	Rhode Island agencies maintain continuity plans which can be enacted as necessary based on the situation. Severe thunderstorms may impact an agency's ability to maintain continuity of operations due to power outages, flooding, and wind damage. If the activation of alternate facilities was required, travel may be difficult as well as computer/network access due to long-term power outages caused by thunderstorms.
Delivery of Services	Delivery of services may be impaired by flooding, obstruction, and damage to roadways and resources. The ability to deliver goods and services will be impacted locally, regionally, or statewide depending on the magnitude of the event. Goods, equipment, and vehicles may become damaged during transport.
Property, Facilities, and Infrastructure	Power lines and power generators are most at risk from thunderstorms and impacts could result in isolated power outages or full-scale blackouts. Building and vehicle

Table 106: Severe Thunderstorm Consequence Analysis

Subject	Potential Impacts
	damage can occur from hail and other debris created by thunderstorms. Properties and critical facilities also may face foundational and physical damage due to flooding, lightning strike, or excessive winds, delaying response and recovery operations.
Impact on Environment	Waste and debris from damage treatment infrastructure or hazardous materials facilities could contaminate sources of water and food. Debris and by-products of thunderstorms can impact and contaminate wildlife and natural areas. Lightning strikes may also ignite fires, leading to destruction of agricultural crops, critical ecosystems, and natural habitats.
Economic Conditions	Flooding, high winds, lightning, and hail can stress state and local resources. Even if some of the costs can be recouped through federal reimbursements (federal disaster declaration), there is a fiscal impact on the local government.
Public Confidence in Governance	Ineffective thunderstorm response can decrease the public's confidence in the state's ability to respond and govern. Governmental response across local, state, regional, and federal levels require direct actions that must be immediate and effective to maintain public confidence.

4.15.7 Jurisdictional Vulnerability and Planning Significance

Utilizing data from FEMA's NRI and from LHMPs, the following tables detail the hazard ratings for jurisdictions, by census tract, for severe thunderstorms. Data concerning both risk index and EAL is reported by census tract. As many State of Rhode Island jurisdictions encompass multiple tracts, an average of tract ratings was completed to determine the overall jurisdictional rating. The following table details the hazard rating by LHMP analysis, FEMA NRI, and EAL by jurisdiction for the State of Rhode Island:

Table 107: State of Rhode Jurisdictional Severe Thunderstorm Planning Significance

County	Jurisdiction	Severe T-Storm LHMP Rating	Hail NRI	Hail EAL	Lightning NRI	Lightning EAL	Strong Wind NRI	Strong Wind EAL
Bristol	Barrington	Moderate	Very Low	Very Low	Relatively Moderate	Relatively Moderate	Relatively Low	Relatively Low
	Bristol	-	Very Low	Very Low	Relatively Moderate	Relatively Moderate	Relatively Low	Relatively Low
	Warren	Moderate	Very Low	Very Low	Relatively Moderate	Relatively Moderate	Relatively Low	Relatively Low
Kent	Coventry	-	Very Low	Very Low	Relatively Moderate	Relatively Moderate	Relatively Low	Relatively Low
	East Greenwich	Moderate	Very Low	Very Low	Relatively Moderate	Relatively Moderate	Relatively Low	Relatively Low
	Warwick	Moderate	Very Low	Very Low	Relatively Moderate	Relatively Moderate	Relatively Low	Relatively Low
	West Greenwich	-	Very Low	Very Low	Relatively Moderate	Relatively High	Relatively Low	Relatively Low
	West Warwick	Moderate	Very Low	Very Low	Relatively Moderate	Relatively Moderate	Relatively Low	Relatively Low
Newport	Jamestown	-	Very Low	Very Low	Relatively Low	Relatively Moderate	Relatively Low	Relatively Low
	Little Compton	Moderate	Very Low	Very Low	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Middletown	High	Very Low	Very Low	Relatively Low	Relatively Low	Very Low	Very Low
	Newport	Moderate	Very Low	Very Low	Relatively Low	Relatively Low	Very Low	Relatively Low
	Portsmouth	Moderate	Very Low	Very Low	Relatively Moderate	Relatively Moderate	Relatively Low	Relatively Low

Table 107: State of Rhode Jurisdictional Severe Thunderstorm Planning Significance

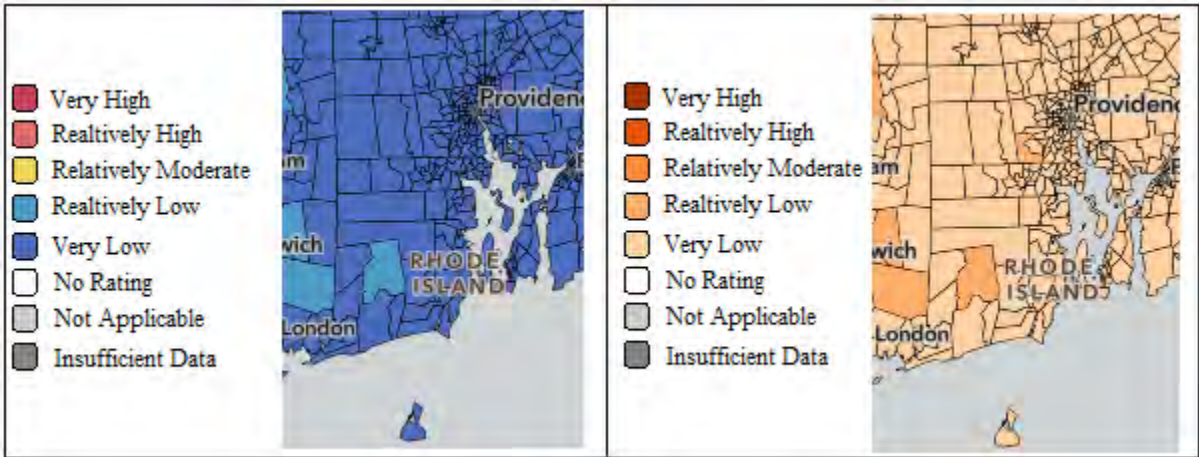
County	Jurisdiction	Severe T-Storm LHP Rating	Hail NRI	Hail EAL	Lightning NRI	Lightning EAL	Strong Wind NRI	Strong Wind EAL
	Tiverton	-	Very Low	Very Low	Relatively Low	Relatively Moderate	Relatively Low	Relatively Low
Providence	Burrillville	Moderate	Very Low	Very Low	Relatively Low	Relatively Low	Very Low	Very Low
	Central Falls	Moderate	Very Low	Very Low	Relatively Low	Relatively Low	Very Low	Very Low
	Cranston	Moderate	Very Low	Very Low	Relatively Low	Relatively Low	Very Low	Very Low
	Cumberland	-	Very Low	Very Low	Relatively Low	Relatively Low	Very Low	Very Low
	East Providence	Moderate	Very Low	Very Low	Relatively Low	Relatively Low	Very Low	Very Low
	Foster	Moderate	Very Low	Very Low	Relatively Low	Relatively Low	Very Low	Very Low
	Glocester	Moderate	Very Low	Very Low	Relatively Low	Relatively Low	Very Low	Very Low
	Johnson	Moderate	Very Low	Very Low	Relatively Moderate	Relatively Moderate	Relatively Low	Relatively Low
	Lincoln	Moderate	Very Low	Very Low	Relatively Low	Relatively Low	Very Low	Very Low
	North Providence	Moderate	Very Low	Very Low	Relatively Low	Relatively Low	Very Low	Very Low
	North Smithfield	Moderate	Very Low	Very Low	Relatively Low	Relatively Low	Very Low	Very Low
	Pawtucket	Moderate	Very Low	Very Low	Relatively Low	Relatively Low	Very Low	Very Low
	Providence	High	Very Low	Very Low	Relatively Low	Relatively Low	Very Low	Very Low
	Scituate	-	Very Low	Very Low	Relatively Low	Relatively Low	Very Low	Very Low
	Smithfield	-	Very Low	Very Low	Relatively Low	Relatively Low	Very Low	Very Low
	Woonsocket	Moderate	Very Low	Very Low	Relatively Low	Relatively Low	Very Low	Very Low
Washington	Charlestown	-	Very Low	Very Low	Relatively Moderate	Relatively Moderate	Relatively Low	Relatively Low
	Exeter	-	Very Low	Very Low	Relatively Moderate	Relatively Moderate	Relatively Low	Relatively Low
	Hopkinton	-	Very Low	Very Low	Relatively Moderate	Relatively Moderate	Very Low	Relatively Low
	Narragansett	Low	Very Low	Very Low	Relatively Low	Relatively Moderate	Very Low	Very Low
	New Shoreham	-	Very Low	Very Low	Relatively Low	Relatively Low	Very Low	Very Low
	North Kingstown	Moderate	Very Low	Very Low	Relatively Low	Relatively Moderate	Very Low	Relatively Low
	Richmond	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively High	Relatively Low	Relatively Low
	South Kingstown	Low	Very Low	Very Low	Relatively Moderate	Relatively Moderate	Relatively Low	Relatively Low
	Westerly	Moderate	Very Low	Very Low	Relatively Moderate	Relatively Moderate	Relatively Low	Relatively Low

Source: LHPs and FEMA NRI

Note: - indicates no current plan available for review

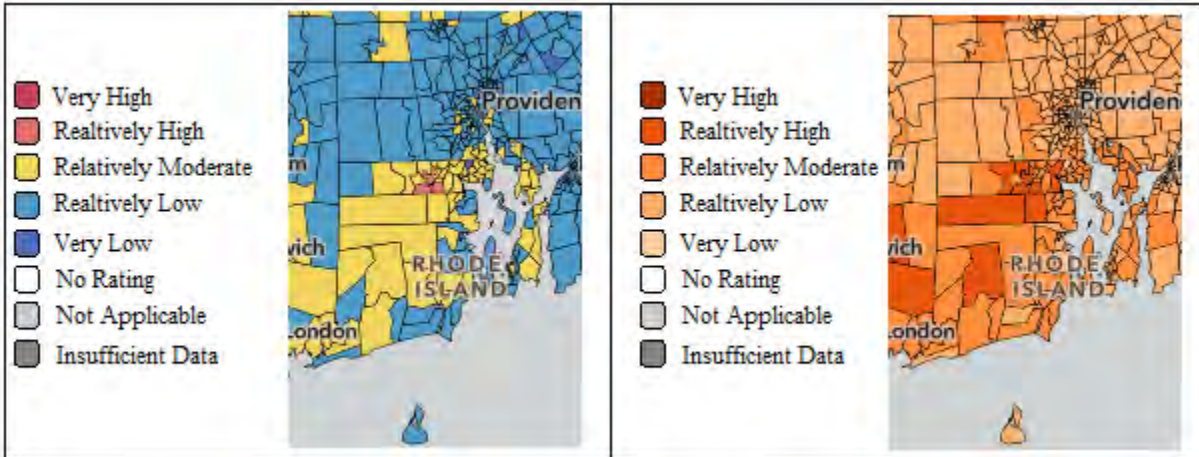
The following maps indicate the FEMA NRI and EAL for hail, lightning, and strong winds, the components of severe thunderstorms, by census tract for the state of Rhode Island.

Map 40: FEMA NRI and EAL for Hail by Census Tract for Rhode Island



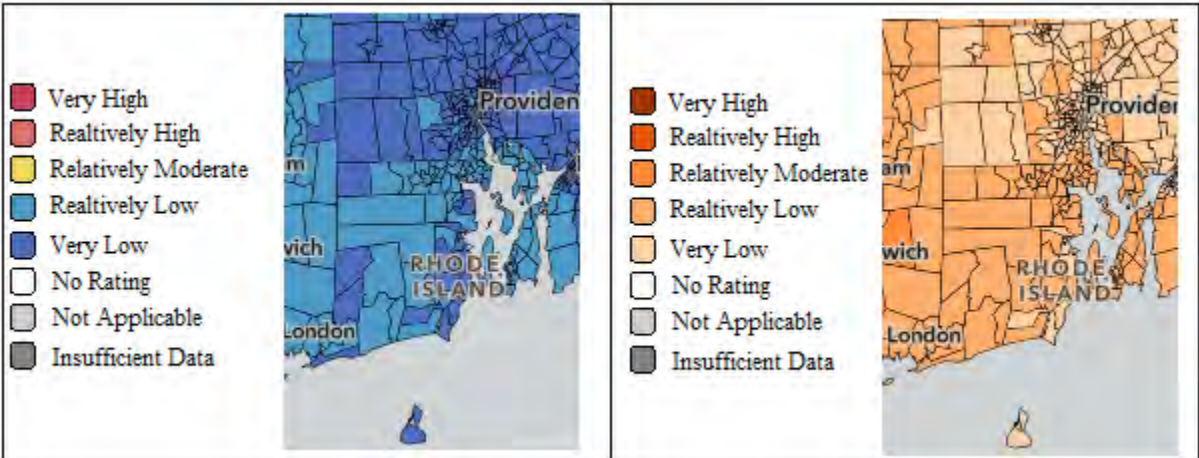
Source: FEMA NRI

Map 41: FEMA NRI and EAL for Lightning by Census Tract for Rhode Island



Source: FEMA NRI

Map 42: FEMA NRI and EAL for Strong Wind by Census Tract for Rhode Island



Source: FEMA NRI

Of particular concern when considering housing data is manufactured buildings (often referred to as mobile homes). Data from the NOAA National Severe Storms Laboratory reports that people living in mobile homes are especially at

risk for injury and death as even anchored mobile homes can be seriously damaged when winds gust over 80 miles per hour. Additionally, study data from Michigan State University reported that the two biggest factors related to wind event fatalities were housing quality (measured by mobile homes as a proportion of housing units) and income level. When a strong wind strikes, a county with double the number of mobile homes as a proportion of all homes will experience 62% more fatalities than a county with fewer mobile homes, according to the study data. The following indicates the percentage of mobile homes for each Rhode Island County:

Table 108: Rhode Island 2020 Census Mobile Home Data

Jurisdiction	Number of Mobile Homes	Percentage Of Housing Stock as Mobile Homes
Bristol County	0	0%
Kent County	902	1.2%
Newport County	701	1.6%
Providence County	1,059	0.4%
Washington County	793	1.2%
State of Rhode Island	3,455	0.7%

Source: United States Census Bureau

A disproportionate impact from severe thunderstorms may be felt by jurisdictions with older housing units as they may have been constructed using materials and techniques that are less resilient to hazards. Modern construction materials and methods, such as reinforced concrete and steel frames, are often more resilient. The following table Rhode Island Department of Housing 2022 Integrated Housing Report from the Rhode Island Department of Housing 2022 Integrated Housing Report indicates jurisdictions with greater than 40% of their housing units constructed before 1939:

Table 109: Jurisdictions at Greater Risk to Severe Thunderstorms due to Age of Housing

County	Jurisdiction	1939 and later
Providence	Providence	57.70%
Newport	Newport	55.20%
Providence	Central Falls	50.40%
Providence	Woonsocket	43.50%
Providence	Pawtucket	43.30%
Bristol	Warren	41.40%

Source: Rhode Island Department of Housing 2022 Integrated Housing Report

Based on FEMA NRI social vulnerability ratings, and on U.S. Census Bureau data concerning percentage of the population below the 100% poverty who may have difficulty conducting repairs after a severe thunderstorm event due to inadequate insurance or lack of expendable income, the following communities may be at greater risk to severe thunderstorm events:

Table 110: Jurisdictions at Greater Risk to Severe Thunderstorms Due to Poverty and Social Vulnerability

County	Jurisdiction	Population Below 100% of the Poverty Level	Social Vulnerability Rating
Providence County	Central Falls	29.9%	Very High
Providence County	Providence	23.0%	Relatively High
Providence County	Woonsocket	21.0%	Relatively Moderate
Washington County	Narragansett	19.3%	Relatively High
Newport County	Newport	14.4%	Relatively Low
Kent County	West Warwick	12.1%	Relatively Low
Providence County	Pawtucket	11.6%	Relatively High

Source: FEMA NRI and U.S. Census Bureau

4.15.8 Hazard Planning Significance

Based on a review of all available data, the SIHMC determined severe thunderstorms to have a moderate planning significance for Rhode Island.

4.16 Severe Winter Weather

4.16.1 Hazard Description

A winter storm encompasses multiple effects caused by winter weather. Included are strong winds, ice storms, heavy or prolonged snow, sleet, and extreme temperatures. Winter storms can be increasingly hazardous in areas and regions that only see winter storms intermittently.

This plan defines winter storms as a combination of the following winter weather effects as defined by NOAA and the NWS.



- **Ice Storm:** An ice storm is used to describe occasions when damaging accumulations of ice are expected during freezing rain situations. Significant accumulations of ice pull down trees and utility lines resulting in loss of power and communication. These accumulations of ice make walking and driving extremely dangerous. Significant ice accumulations are usually accumulations of 1/4" or greater.
- **Heavy Snow:** This generally means snowfall accumulating to 4" or more in depth in 12 hours or less; or snowfall accumulating to 6" or more in depth in 24 hours or less.
- **Winter Storm:** Hazardous winter weather in the form of heavy snow, freezing rain, or heavy sleet. It may also include extremely low temperatures and increased wind.
- **Cold Wave/Extreme Cold:** As described by NWS, a cold wave is a rapid fall in temperature within a 24-hour period requiring substantially increased protection to agriculture, industry, commerce, and social activities. As evidenced by past incidents across the U.S., extreme cold can cause impact to human life and property.

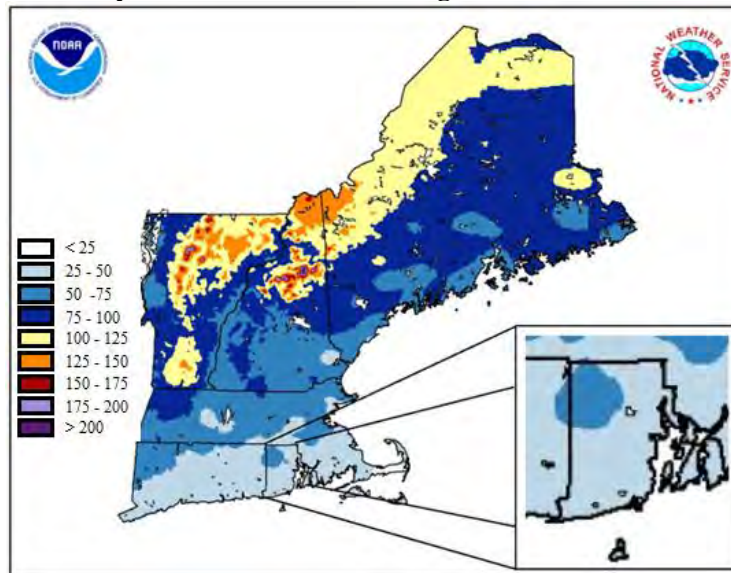
4.16.2 – Location and Extent

Winter storms occur regularly throughout Rhode Island. These events occur on a large geographic scale, often affecting multiple counties, regions, and states. Winter storms typically form with warning and are often anticipated. Like other large storm fronts, the severity of a storm is not as easily predicted and when it is, the window of notification is up to a few hours to under an hour. Although meteorologists estimate the amount of snowfall a winter storm will drop, it is not known exactly how much snow will fall, whether or not it will form an ice storm, or how powerful the winds will be until the storm is already affecting a community.

The average annual snowfall in Rhode Island increases from about 20 inches on Block Island and along the southeast shores of Narragansett Bay to 40 to 55 inches in the western third of the State. Areas near the western and northern shores of the Bay, including greater Providence, have an average range from 25 to 30 inches of snow per year. During mild winters these snowfall totals can be significantly less.

The following map from NOAA and the NWS indicates that the average annual snowfall for Rhode Island is between 25 – 75 inches per year:

Map 43: Rhode Island Average Annual Snowfall



Source: NOAA

The Northeast Snowfall Impact Scale is a scale used to assess and rank the impact of snowfall events in the northeastern United States. It was developed by NOAA to provide a standardized way of measuring the societal and economic impacts of snowstorms. The scale takes into account factors such as snowfall amount, population density, and the area affected by the storm to determine its impact. The scale has five categories, each with its own associated impacts:

Table 111: Northeast Snowfall Impact Scale

Category	Description	Impacts
1	Notable	Light to moderate snowfall. Limited impacts on transportation and daily life. Typically localized to small areas.
2	Significant	Moderate to heavy snowfall. Widespread impacts on transportation, including delays and disruptions. Some school and business closures. Widespread power outages are rare.
3	Major	Heavy snowfall, often exceeding one foot or more. Significant transportation disruptions, including major highway closures. Widespread school and business closures. Power outages may occur, especially in areas with wet, heavy snow.
4	Crippling	Extreme snowfall, often exceeding two feet or more. Severe and prolonged transportation disruptions, including highway closures. Widespread school and business closures for an extended period. Widespread and prolonged power outages, especially in areas with ice accumulation.
5	Extreme	Exceptional snowfall, often exceeding three feet or more. Complete paralysis of transportation systems, including major highways and airports. Extended school and business closures. Widespread and prolonged power outages with significant damage to the electrical infrastructure.

Source: NOAA

The scale provides information for emergency management, public safety agencies, and the public to understand the potential impacts of a snowstorm and to prepare accordingly. It helps to quantify and communicate the severity of winter weather events, especially where snowfall can have a major impact on daily life and the economy.

Ice storms are characterized by the accumulation of freezing rain or freezing drizzle, which coats surfaces with a layer of ice. These storms can have significant impacts on transportation, infrastructure, and the environment. Ice storms occur when there's a layer of warm air above a layer of cold air near the surface. Precipitation falls as rain in the warm

layer and then freezes upon contact with surfaces at or below freezing temperatures in the cold layer. The most common type of precipitation during an ice storm is freezing rain. This is rain that falls as a liquid but freezes upon contact with cold surfaces, forming a layer of ice.

The Sperry–Piltz Ice Accumulation Index is an ice accumulation and ice damage prediction index that, when combined with NWS data, predicts the projected footprint, total ice accumulation, and resulting potential damage from approaching ice storms.

Figure 2: Sperry–Piltz Ice Accumulation Index

ICE DAMAGE INDEX	DAMAGE AND IMPACT DESCRIPTIONS
0	Minimal risk of damage to exposed utility systems; no alerts or advisories needed for crews, few outages.
1	Some isolated or localized utility interruptions are possible, typically lasting only a few hours. Roads and bridges may become slick and hazardous.
2	Scattered utility interruptions expected, typically lasting 12 to 24 hours. Roads and travel conditions may be extremely hazardous due to ice accumulation.
3	Numerous utility interruptions with some damage to main feeder lines and equipment expected. Tree limb damage is excessive. Outages lasting 1 – 5 days.
4	Prolonged & widespread utility interruptions with extensive damage to main distribution feeder lines & some high voltage transmission lines/structures. Outages lasting 5 – 10 days.
5	Catastrophic damage to entire exposed utility systems, including both distribution and transmission networks. Outages could last several weeks in some areas. Shelters needed.

Source: Sperry–Piltz Ice Accumulation Index

4.16.3 Previous Occurrence

Historical events of significant magnitude or impact can result in a Presidential Disaster Declaration. Rhode Island has experienced six Presidential Disaster Declarations related to severe winter storms, reflected in the following table.

Table 112: State of Rhode Island Presidentially Declared Disasters

Designation	Declaration Date	Incident Type	Individual Assistance	Public Assistance
DR-4653-RI	05/12/2022	Severe Winter Storm and Snowstorm	-	\$8,901,805
DR-4212-RI	4/3/2015	Severe Winter Storm and Snowstorm	\$8,897,067	\$1,327,144
DR-4107-RI	3/22/2013	Severe Winter Storm and Snowstorm	\$7,929,498	\$1,070,561
DR-1091-RI	01/24/1996	Blizzard	-	-
DR-3058-RI	02/07/1978	Blizzards and Snowstorms	-	-
DR-548-RI	02/16/1978	Snow and Ice	-	-

Source: FEMA

Note: -: Data unavailable

The following provide a brief narrative of these events:

- **DR-4653, May 12, 2022 (Winter Storm Kenan):** Blizzard conditions were reached at several locations throughout Rhode Island. On January 29, 2022. Snowfall amounts of one to two feet were recorded along with wind gusts reaching up to hurricane force in some areas. FEMA provided more than \$2,600,000 to reimburse the Rhode Island Department of Transportation for some of the costs of plowing and treating roads after January 2022.
- **DR-4212, April 3, 2015 (Winter Storm Juno):** This Severe Winter Storm and Snowstorm occurred during the period of January 26, 2015, to January 28, 2015. The Governor requested that federal assistance funds be

available to the state and eligible local governments and private nonprofit organizations on a cost-sharing basis for emergency work and the repair or replacement of damaged facilities in Bristol, Kent, Newport, Providence, and Washington counties. In addition, the disaster declaration authorized snow assistance for a period of 48 hours for Bristol, Kent, Newport, Providence, and Washington counties.

- **DR-4107, March 22, 2013:** Reports indicated that this storm stretched from New Jersey to Maine and into Canada. More than two feet of snow fell in Rhode Island overnight. PPL Energy estimated more than 180,000 customers lost power.

In addition to the Presidentially Declared Disasters, the following table presents NCEI identified ice storm and winter storm events and the resulting damage totals in Rhode Island from 1950 to 2023:

Table 113: NCEI Rhode Island Winter Storm Events

Jurisdiction	Event Type	Number of Days with Events	Property Damage	Deaths and Injuries
Bristol County	Blizzard	2	\$0	0
	Ice Storm	0	\$0	0
	Winter Storm	15	\$35,000	0
Kent County	Blizzard	3	\$10,000	0
	Ice Storm	0	\$0	0
	Winter Storm	22	\$220,000	0
Newport County	Blizzard	5	\$52,000	0
	Ice Storm	0	\$0	0
	Winter Storm	16	\$35,000	0
Providence County	Blizzard	2	\$0	0
	Ice Storm	0	\$0	0
	Winter Storm	29	\$145,000	0
Washington County	Blizzard	4	\$0	0
	Ice Storm	0	\$0	0
	Winter Storm	21	\$61,900	0

Source: National Centers for Environmental Information

It is worth noting that damage estimates indicated by the NCEI are often artificially low. This underreporting is a result of the way the events are reported to the NCEI, often by the local and/or NWS office. When reporting an event oftentimes the NWS office does not have access to the actual damage assessment resulting from that event. As such, the report often details a very low amount or zero-dollar amount for damages. Additionally, deaths and injuries may be underreported as they may be a result of a concurrent event, such as a person driving unsafely during heavy rain and passing away.

Additionally, Rhode Island has been impacted by numerous nor'easters bring severe winter conditions including the following:

- **The Presidents' Day Storm of 2003:** In February 2003, a powerful nor'easter, often referred to as the Presidents' Day Storm, impacted the northeastern United States, including Rhode Island. This storm brought heavy snowfall, with some areas in Rhode Island receiving over 20 inches of snow. Strong winds contributed to drifting snow and power outages. The storm disrupted travel and daily life for several days.
- **March 2018 Nor'easter:** A powerful nor'easter struck the eastern United States, affecting Rhode Island. This storm brought heavy snowfall, coastal flooding, and strong winds to the region. Some parts of Rhode Island received more than a foot of snow, and coastal areas experienced inundation and erosion.
- **April 2007 Nor'easter:** A late-season nor'easter brought heavy snowfall to Rhode Island and other parts of the northeast. The storm was notable for its unusual timing, as it occurred in the spring. Some areas received more than a foot of snow, causing travel disruptions and power outages.

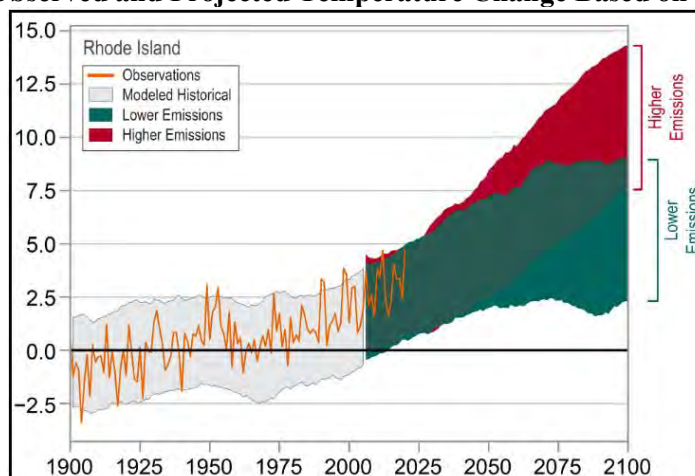
4.16.4 Probability of Future Events

Predicting the probability of winter storm occurrences is tremendously challenging due to the large number of factors involved and the random nature of formation. Data from NOAA and the NWS indicate that Rhode Island can expect an average annual snowfall of between 25 – 75 inches per year. Additionally, data from the NCEI indicates that Rhode Island can expect at least two winter storm events per year.

4.16.5 Projected Changes in Location, Intensity, Frequency, and Duration

Climate change can lead to greater variability in precipitation patterns. In Rhode Island, this may result in more erratic and intense winter storms with periods of heavy snowfall followed by rain or freezing rain. These mixed precipitation events can make winter storms more challenging to predict and can lead to a greater risk of ice accumulation. Additionally, Rhode Island may experience milder winters as average temperatures rise due to climate change. While this could lead to a decrease in the frequency of traditional snowstorms, it may also increase the likelihood of winter storms that produce mixed precipitation, including freezing rain and sleet. Warmer temperatures can lead to a higher snowfall threshold, meaning that storms that would have produced snow in the past may now bring more rain or a mix of precipitation types. This can affect the accumulation of snow in the state. The following chart indicates the projected temperature change for Rhode Island utilizing two global climate models. One model utilizes information in which greenhouse gas emissions continue to increase (higher emissions), with the other model utilizing information in which greenhouse gas emissions increase at a slower rate (lower emissions). Temperatures in Rhode Island, detailed by the orange line, have risen 4° F since the beginning of the early 1900s. Based on both the higher emission and lower emission models, continued warming is projected throughout this century.

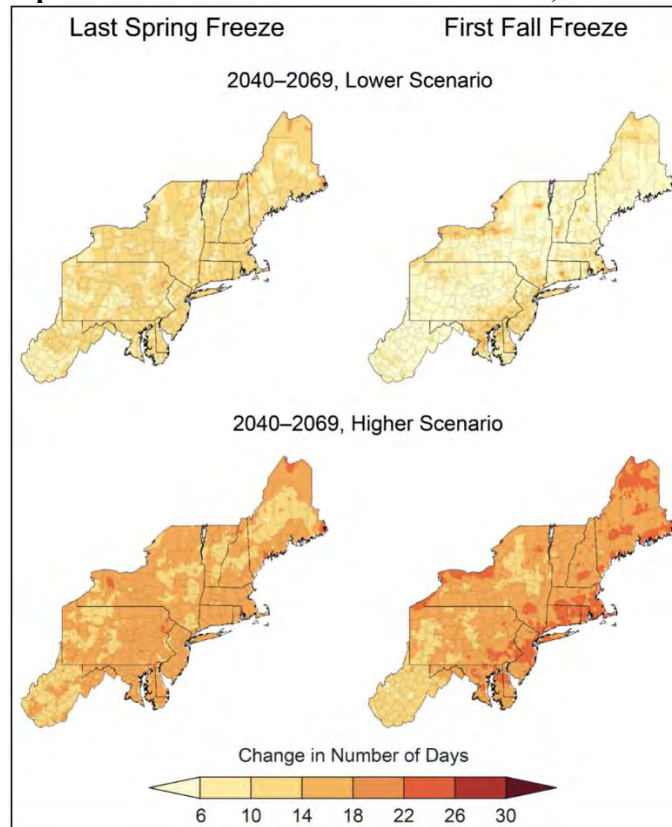
Chart 32: Rhode Island Observed and Projected Temperature Change Based on Greenhouse Gas Emissions



Source: NOAA State Climate Summary 2022 for Rhode Island

The following map from the U.S. Global Change Research Program Fourth National Climate Assessment show projected shifts in the date of the last spring freeze and the date of the first fall freeze under the lower scenario and the higher scenario emission pathways.

Map 44: Shift in First and Last Freeze Dates, 2040 - 2069



Source: U.S. Global Change Research Program Fourth National Climate Assessment

Research indicates that by the middle of the century freeze-free periods may lengthen by two weeks under the lower emissions scenario and by up to three weeks under the higher emissions scenario.

As previously noted, State of Rhode Island facilities have seen no major changes in the past five years, with only modest repairs and upgrades being conducted and no major rehabilitation or construction projects completed. As such, the risk to state assets has remained static since the completion of the 2019 SHMP. Additionally, with the exception of the two previously identified rehabilitation and construction projects, no large-scale facility projects were noted in any agency budgets or reports.

4.16.6 Vulnerability and Impact

People and Communities

Extremely cold temperatures are a threat to anyone exposed to them. Extreme cold can cause frostbite and hypothermia. Bitterly cold temperatures can also burst water and create an excessive demand on providers to deliver energy for household heating. There are also fire dangers associated with home heating. Heavy snow and/or ice can paralyze communities. Roads can become hazardous which may cause accidents, disrupted flow of supplies, and challenges in the delivery of emergency and medical services. Additional impacts on people and the community may include:

- **Injuries and Fatalities:** Slippery sidewalks, roads, and driveways can lead to slip and fall accidents, vehicle crashes, and pedestrian injuries. Exposure to extreme cold temperatures can cause frostbite, hypothermia, and cold-related illnesses, which can be life-threatening.
- **Power Outages:** Heavy snow, ice, and freezing rain can bring down power lines and disrupt electricity supply. Power outages can lead to heating and lighting challenges, particularly in extreme cold conditions.
- **Transportation Disruptions:** Winter storms can make roads and highways treacherous, leading to travel delays, accidents, and stranded motorists. Public transportation services may be disrupted, affecting commuters and essential travel.

- **Stranded or Isolated Communities:** Severe winter weather can leave communities isolated and cut off from emergency services and supplies. Residents may need to shelter in place or rely on local resources until conditions improve.
- **Health Risks:** Exposure to extreme cold can lead to a range of health risks, including frostbite, hypothermia, and cold-related illnesses. Individuals with pre-existing health conditions may face exacerbated risks.
- **Increased Heating Costs:** Cold weather can result in higher heating costs, which can be a financial burden for many households. Low-income individuals and families may struggle to afford adequate heating.
- **Disruption of Essential Services:** Severe winter weather can disrupt essential services such as healthcare, emergency response, and utilities. Hospitals may face increased patient volumes due to weather-related injuries and illnesses.

Of particular concern, socially vulnerable and at-risk populations that may have difficulty with medical issues, poverty, extremes in age, and communications due to language barriers are often at increased susceptibility to the impacts of hazards. These populations suffer a disproportionate impact due to:

- Constrained financial resources
- Limited access to healthcare
- Limited access to information

The following table details identified socially vulnerable and at-risk populations for the State of Rhode Island:

Table 114: Rhode Island 2020 Vulnerable Populations

Jurisdiction	Under Five	Over 65	Language Barrier	Poverty	Disability
Bristol County	1,976	8,372	6,552	3,403	4,753
Kent County	8,148	26,437	16,525	11,925	24,694
Newport County	3,641	13,819	7,794	8,307	9,105
Providence County	36,846	84,768	205,490	90,522	87,498
Washington County	4,760	18,797	7,790	11,815	13,832
State of Rhode Island	55,370	152,183	240,326	125,101	139,927

Source: US Census Bureau

Facilities

Severe winter storms can have significant and wide-ranging impacts on critical facilities and infrastructure. The extent of the impact depends on the severity of the storm, the preparedness of the facilities and infrastructure, and the effectiveness of response and recovery efforts, and may include:

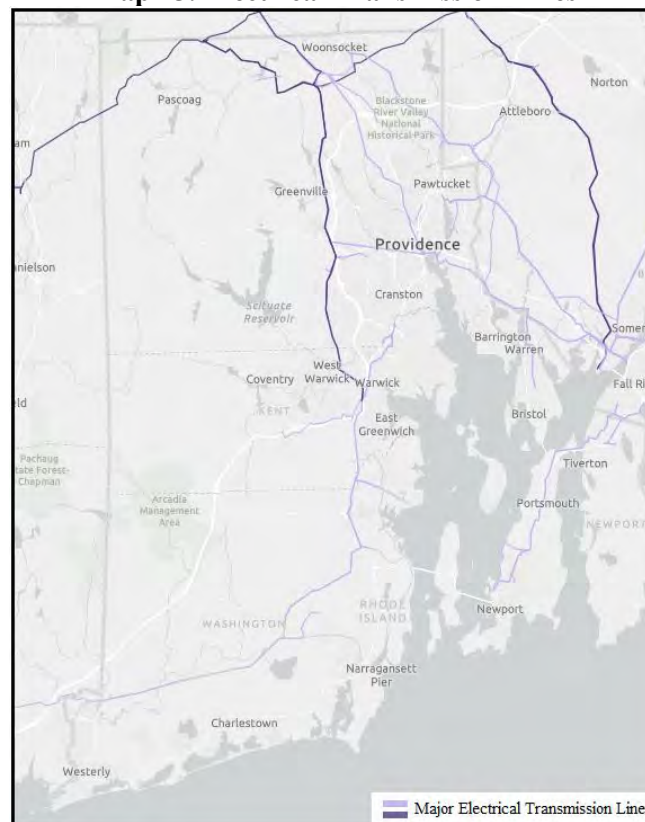
- **Power Outages:** Severe winter storms can cause power outages by bringing down power lines, causing ice accumulation on electrical infrastructure, or overloading the electrical grid due to increased demand for heating. Critical facilities such as hospitals, emergency response centers, and data centers may rely on backup generators to maintain essential operations during outages.
- **Communication Disruptions:** Ice and freezing rain can damage communication infrastructure, including cell towers, telephone lines, and data centers, leading to disruptions in phone and internet services. This can hinder emergency communication and coordination, affecting critical response efforts.
- **Transportation Disruptions:** Snow and ice accumulation on roads, runways, and railways can disrupt transportation networks, leading to travel delays, accidents, and closures. Critical facilities may face challenges in receiving essential supplies and personnel during and after the storm.
- **Healthcare System Strain:** Hospitals and healthcare facilities may experience increased demand for medical services due to storm-related injuries and illnesses, including those related to slips and falls, traffic accidents, and cold exposure.
- **Water Supply Interruptions:** Freezing temperatures can cause water pipes to burst, leading to water supply interruptions and damage to water infrastructure. Critical facilities such as hospitals and emergency response centers rely on a continuous supply of clean water for various purposes, including patient care and firefighting.

- **Wastewater Systems:** Cold temperatures can affect wastewater treatment plants, leading to potential operational disruptions and contamination risks.
- **Fuel Supply Disruptions:** Snow and ice can disrupt fuel supply chains, leading to shortages of gasoline, diesel, and heating oil. Critical facilities may rely on fuel for backup power generators and heating systems.
- **Property Damage:** Severe winter storms can result in property damage, including roof collapses due to heavy snow accumulation, ice damming, and frozen pipes.

Winter storms can have significant impacts on utility lines, causing disruptions to essential services such as electricity, heating, and communication. The severity of these impacts depends on the intensity of the storm and the operating electrical utility's preparedness. Freezing rain and heavy snow can accumulate on power lines and utility equipment, causing them to sag or break, resulting in power outages. Additionally, accumulations of ice or heavy snow on trees can cause branches to break and fall onto power lines, leading to outages.

Electrical utilities within the State of Rhode Island are privately managed and maintained. Electrical utilities within the State of Rhode Island are privately managed and maintained. GIS analysis indicates that there are approximately 419 miles of major electrical transmission lines within the State. The following map details electrical transmission lines within the state, all of which are vulnerable to the effect of winter storms:

Map 45: Electrical Transmission Lines



Source: U.S. Energy Information Administration

Environment

Winter storms can have various impacts on the environment, particularly in regions prone to cold and snowy winters. These impacts can affect ecosystems, wildlife, and natural resources and can include habitat disruption, reduction of food sources, changes in migration patterns, and damage to foliage (especially if a spring storm). Additionally, the use of salt and de-icing chemicals on roads and sidewalks can have negative environmental impacts. These chemicals can find their way into nearby water bodies, leading to water pollution and harm to aquatic ecosystems. Snowmelt can also introduce pollutants from roadways and urban areas into rivers and streams, leading to reduced water quality. Elevated sediment levels and changes in water temperature can also affect aquatic life.

State Operations and Assets

Severe winter weather can pose risks to state operations and assets (with all state assets being vulnerable), and can disrupt functions, damage critical infrastructure, and strain resources. Some of the risks to state operations and facilities include:

- **Transportation Disruptions:** Snow and ice accumulation on roads and highways can hinder transportation, making it difficult for state agencies and personnel to travel and respond to emergencies. RIDOT must allocate resources to plow and salt roads, clear snow and ice, and repair potholes caused by freezing and thawing. These efforts are costly and resource intensive.
- **School Closures:** Winter storms often lead to school closures, which can affect state-run education programs and services. State agencies may need to coordinate with school districts to ensure the safety of students.
- **Emergency Response and Public Safety:** Winter storms can result in increased demands for emergency services, including responses to traffic accidents, medical emergencies, and stranded motorists. State and local agencies must allocate additional resources to address these needs.
- **Economic Impact:** Winter storms can result in economic losses due to reduced economic activity, transportation disruptions, property damage, and increased spending on emergency response and recovery efforts.
- **Emergency Shelter Operations:** RIEMA may need to coordinate emergency shelters during winter storms to provide shelter and resources to vulnerable populations, including those experiencing homelessness.
- **Resource Allocation:** State agencies must allocate resources, including personnel, equipment, and stockpiled supplies, to support emergency response efforts and maintain public services.
- **Communication Challenges:** Winter storms can disrupt communication networks, hindering the ability of state agencies to communicate internally and with the public. This can impact emergency notifications and coordination efforts.
- **Budgetary Impact:** The costs associated with snow removal, road maintenance, emergency response efforts, and infrastructure repair can strain state budgets.
- **Governance and Administrative Challenges:** State government offices and facilities may experience closures or reduced staffing during severe winter weather, affecting administrative functions, regulatory processes, and public services.

Depending on the severity and duration of a winter storm, all state owned or operated facilities are vulnerable to potential impacts. The following table indicates the number and valuation of the building envelope and building contents for all state owned or operated facilities:

Table 115: State of Rhode Island Facilities and Assets by County

Jurisdiction	State Owned or Operated Facilities	Valuation
Bristol County	32	\$150,415,810
Kent County	116	\$215,026,614
Newport County	77	\$127,353,837
Providence County	423	\$2,526,593,056
Washington County	393	\$274,350,750
State Total	1,041	\$3,293,740,067

Source: RIEMA

Note: Valuations include structure and assets

The age of a state owned or operated structure could have a significant impact on its resilience to a winter storm, especially if a heavy snow lingered on roofs, as an older structure may be less resilient due to the following factors:

- **Building Codes and Standards:** Older buildings may not have been constructed to meet modern building codes, which often incorporate lessons learned from previous disasters. Newer buildings are typically designed to be more resilient to known hazards.

- Construction Materials and Techniques: Older buildings may have been constructed using materials and techniques that are less resilient to hazards. Modern construction materials and methods, such as reinforced concrete and steel frames, are often more resilient.

The following table indicates the average year of construction for state-owned or operated facilities with a known completion date, by county, for the State of Rhode Island:

Table 116: Average Year of Construction for State of Rhode Island Owned or Operated Facilities by County

Jurisdiction	Average Year of Construction
Bristol County	1966
Kent County	1983
Newport County	1978
Providence County	1961
Washington County	1964
State Buildings Average Year of Construction	1969

Source: RIEMA

Note: Average age based on buildings with known construction date

Utilizing information provided by the State of Rhode Island, and removing identifying addresses, the following facilities have been determined to be most vulnerable to potential severe winter weather due to the age of construction and/or the total valuation of the building. for the State of Rhode Island:

Table 117: Rhode Island Buildings Vulnerable to Severe Winter Weather due to Age

County	Jurisdiction	Construction Type	Valuation	Year Built
Providence	Providence	Wood Frame	\$447,094	1707
Newport	Newport	Brick	\$6,107,811	1739
Providence	Providence	Brick	\$95,176	1763
Providence	Providence	Brick	\$34,290	1763
Providence	Providence	Brick	\$10,459,508	1763
Washington	Richmond	Wood Frame	\$422,883	1790
Providence	Scituate	Wood Frame	\$2,362,304	1799
Newport	Newport	Masonry Non-Combustible	\$6,875,212	1824
Providence	Providence	Stone	\$2,583,581	1839
Providence	Scituate	Brick	\$1,623,657	1850
Providence	Providence	Wood Frame	\$2,030,638	1850
Providence	Scituate	Wood Frame	\$241,356	1850
Providence	Lincoln	Frame	\$379,034	1860
Kent	Warwick	Frame	\$77,606	1860
Kent	Warwick	Frame	\$65,746	1860
Kent	Warwick	Frame	\$81,639	1860
Kent	Warwick	Frame	\$564,246	1860
Kent	Warwick	Frame	\$382,313	1860
Newport	Newport	Frame	\$2,282,592	1873
Newport	Newport	Frame	\$2,736,474	1873
Providence	Providence	Masonry Non-Combustible	\$11,051,413	1874
Providence	Woonsocket	Wood Frame	\$856,207	1875
Providence	Cranston	Stone	\$17,406,343	1878
Providence	Woonsocket	Block	\$1,360,264	1882
Newport	Tiverton	Block	\$119,995	1890
Providence	Providence	Brick	\$67,256	1890
Providence	Cranston	Stone	\$2,384,845	1890
Providence	Cranston	Stone	\$1,217,563	1890
Providence	Cranston	Stone	\$8,161,595	1890
Providence	Cranston	Stone	\$9,758,371	1890

Table 117: Rhode Island Buildings Vulnerable to Severe Winter Weather due to Age

County	Jurisdiction	Construction Type	Valuation	Year Built
Providence	Cranston	Stone	\$10,224,779	1890
Providence	Cranston	Stone	\$13,544,843	1890
Providence	East Providence	Wood Frame	\$1,070,644	1890
Providence	Johnston	Wood Frame	\$859,381	1890
Newport	Tiverton	Wood Frame	\$526,023	1890
Newport	Tiverton	Wood Frame	\$2,053,953	1890
Newport	Portsmouth	Wood Frame	\$825,348	1895
Providence	Providence	Wood Frame	\$1,305,996	1895

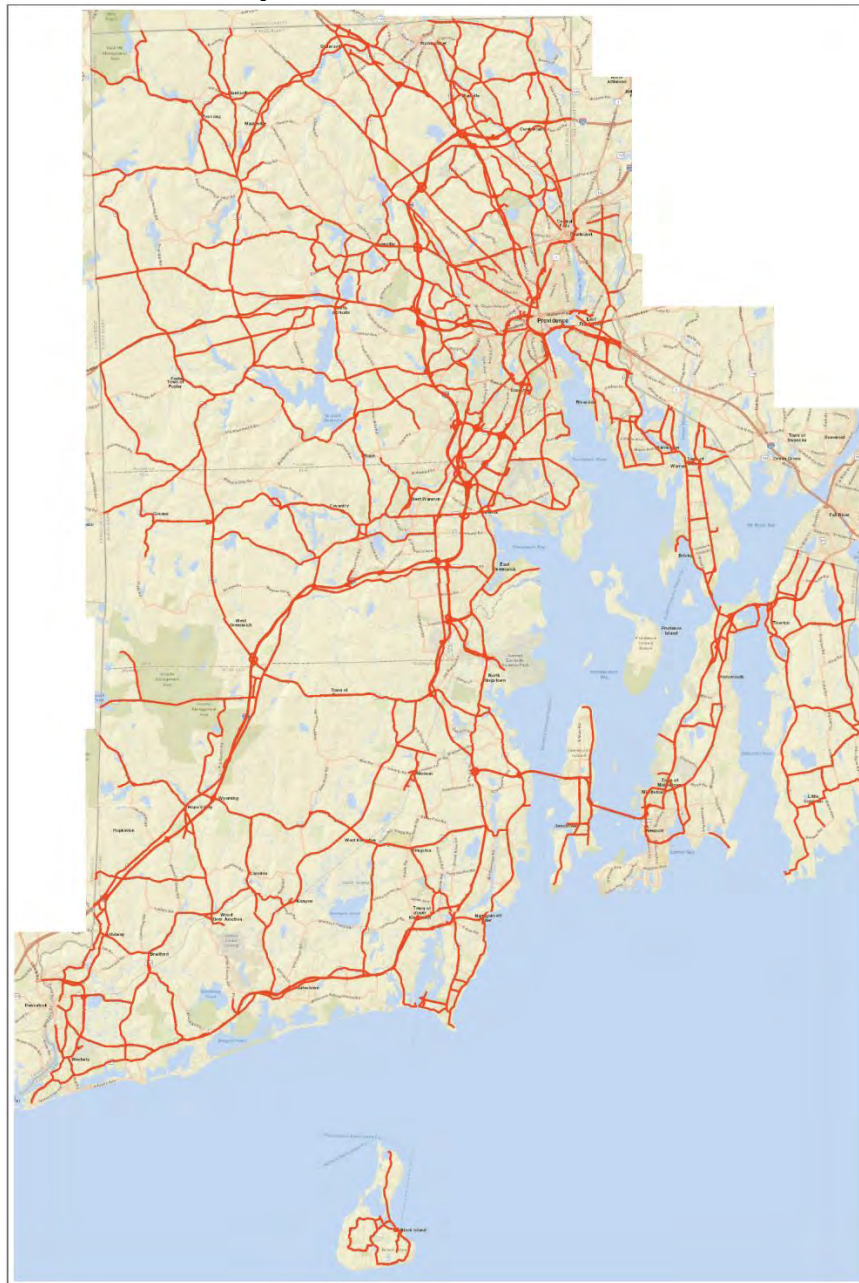
Source: US Census Bureau

Of particular concern during winter weather is continued access to roads for travel. Approximately 80% of all the miles of roads in Rhode Island (approximately 5,000 miles) are maintained by cities and towns. RIDOT maintains the balance of roads, with approximately 1,100 miles of state roads under its jurisdiction. RIDOT has a fleet of 178 plow trucks, and access to 150 vendors who help bring the total available number of plow trucks up to 500. Additionally, RIDOT budgets approximately \$20,000,000 per year on winter maintenance operations. During a typical winter, RIDOT from its 20 statewide stockpile locations, consumes approximately:

- 125,000 tons of salt
- 10,000 tons of sand
- 14,000 gallons of brine

The following map, from RIDOT, details state-maintained roads:

Map 46: RIDOT Maintained Roads



Source: RIDOT

Consequence Analysis

This consequence analysis lists the potential impacts of this hazard on various elements of community and state infrastructure. The impact of this hazard is evaluated in terms of disruption of operations, recovery challenges, and overall wellbeing to all Rhode Island residents and first responder personnel. The consequence analysis supplements the hazard profile by analyzing specific impacts.

Table 118: Severe Winter Weather Consequence Analysis

Subject	Potential Impacts
Impact on the Public	Freezing temperatures coupled with heavy snow accumulation can cause dangerous travel conditions, leading to accidents and road closures. Downed power lines can lead to a loss of electricity and heat, with the young and the elderly especially vulnerable. Extremely cold temperatures may lead to hypothermia and death.
Impact on Responders	Dangerous road conditions create a transportation challenge for first responders. First responders will need to control their own exposure to the elements for prolonged

Table 118: Severe Winter Weather Consequence Analysis

Subject	Potential Impacts
	periods of time and will need to continuously seek heat and shelter to stay warm. Equipment may also be damaged or destroyed due to cold temperatures, heavy wind, ice, and heavy snow fall, which may lead to a decrease in response capabilities.
Continuity of Operations	Rhode Island agencies maintain continuity plans which can be enacted as necessary based on the situation. Severe winter weather may impact an agency's ability to maintain continuity of operations due to power outages and transportation difficulties. If the activation of alternate facilities was required, travel may be difficult. Additionally, computer/network and other communication access may be impacted due to power outages caused by severe winter weather.
Delivery of Services	The ability to deliver services can be impacted locally, regionally, or statewide depending on the severity of the severe winter weather event. Dangerous road conditions may lead to roadway and bridge closures, as well as transit service disruptions. Businesses and places of commerce may completely shut down, which leads to the disruption of goods and services.
Property, Facilities, and Infrastructure	Transportation, governmental operations, and communications may be heavily disrupted. Roads and bridges may be heavily impacted by severe winter weather, and may be completely obstructed by downed trees, powerlines, and snow accumulation. Snow and ice can impact access to homes and critical facilities such as hospitals, schools, and supermarkets. Power loss can lead to disruption of critical infrastructure and technology.
Impact on Environment	Heavy snow and ice accumulation can weigh down and damage vegetation, tree limbs, and power lines. Flooding may also occur after the rapid melting of a heavy snowfall, causing bodies of water to flood, damaging the surrounding areas. Exposure to extreme winter weather may result in animal death. Chemicals used to treat roadways may contaminate natural environments and water reservoirs if used in large quantities.
Economic Conditions	Severe winter weather poses a fiscal impact on the local and state governments, even if some of those costs can be recouped through federal grant reimbursements. Local, county, and state resources may be drained by a severe winter weather event.
Public Confidence in Governance	The public's confidence in the state's governance is affected by immediate local and state response through direct and effective actions. Efficiency in response and recovery operations is critical in keeping public confidence high.

4.16.7 Jurisdictional Vulnerability and Planning Significance

Utilizing data from FEMA's NRI and from LHMPs, the following tables detail the hazard ratings for jurisdictions, by census tract, for severe winter weather. Data concerning both risk index and EAL is reported by census tract. As many State of Rhode Island jurisdictions encompass multiple tracts, an average of tract ratings was completed to determine the overall jurisdictional rating. The following table details the hazard rating by LHMP analysis, FEMA NRI, and EAL by jurisdiction for the State of Rhode Island:

Table 119: State of Rhode Jurisdictional Severe Winter Weather Planning Significance

County	Jurisdiction	Severe Winter Weather LHMP Rating	Ice Storm NRI	Ice Storm EAL	Winter Weather NRI	Winter Weather EAL
Bristol	Barrington	High	Relatively Low	Relatively Low	Relatively High	Relatively Low
	Bristol	-	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Warren	No Rating	Relatively Low	Relatively Low	Relatively Low	Relatively Low
Kent	Coventry	-	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	East Greenwich	Moderate	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Warwick	Moderate	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	West Greenwich	-	Relatively Low	Relatively Low	Relatively Low	Relatively Low

Table 119: State of Rhode Jurisdictional Severe Winter Weather Planning Significance

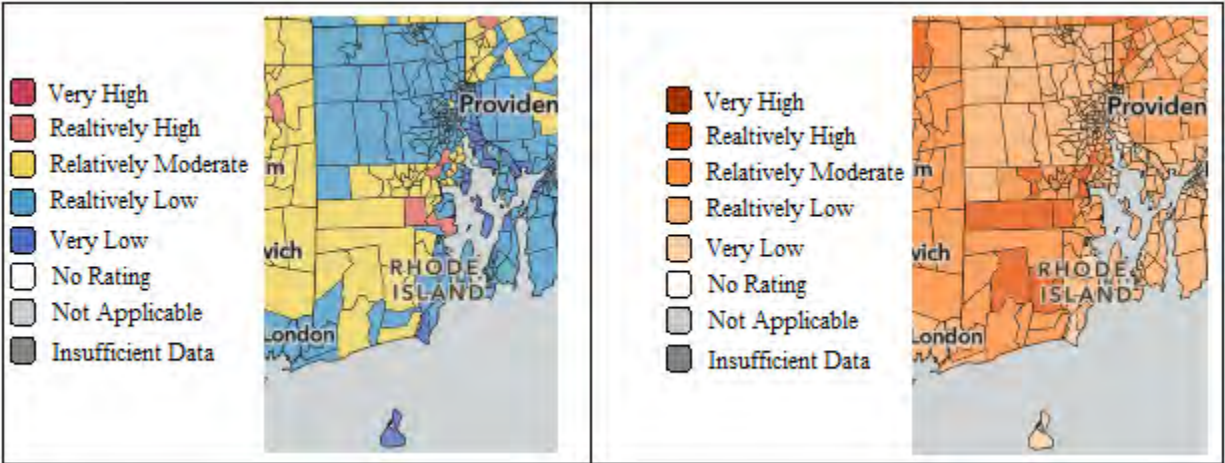
County	Jurisdiction	Severe Winter Weather LHP Rating	Ice Storm NRI	Ice Storm EAL	Winter Weather NRI	Winter Weather EAL
	West Warwick	Moderate	Relatively Low	Relatively Low	Relatively Low	Relatively Low
Newport	Jamestown	-	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Little Compton	Moderate	Relatively Low	Relatively Low	Very Low	Relatively Low
	Middletown	Not Rated	Very Low	Very Low	Very Low	Very Low
	Newport	Moderate	Very Low	Very Low	Very Low	Very Low
	Portsmouth	Moderate	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Tiverton	-	Very Low	Relatively Low	Very Low	Relatively Low
Providence	Burrillville	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Central Falls	Moderate	Relatively Low	Relatively Low	Relatively High	Relatively Moderate
	Cranston	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Cumberland	-	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	East Providence	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Foster	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Glocester	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Johnson	Moderate	Relatively Low	Relatively Moderate	Relatively High	Relatively Moderate
	Lincoln	Low	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	North Providence	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	North Smithfield	Not Rated	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Pawtucket	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Providence	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Scituate	-	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Smithfield	-	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
	Woonsocket	Moderate	Relatively Low	Relatively Low	Relatively Moderate	Relatively Moderate
Washington	Charlestown	-	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Exeter	-	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Hopkinton	-	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Narragansett	Moderate	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	New Shoreham	-	No Rating	No Rating	No Rating	No Rating
	North Kingstown	Moderate	Very Low	Relatively Low	Very Low	Relatively Low
	Richmond	Not Rated	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	South Kingstown	Moderate	Relatively Low	Relatively Low	Relatively Low	Relatively Low
	Westerly	Moderate	Relatively Low	Relatively Low	Relatively Low	Relatively Low

Source: LHMPs and FEMA NRI

Note: - indicates no current plan available for review

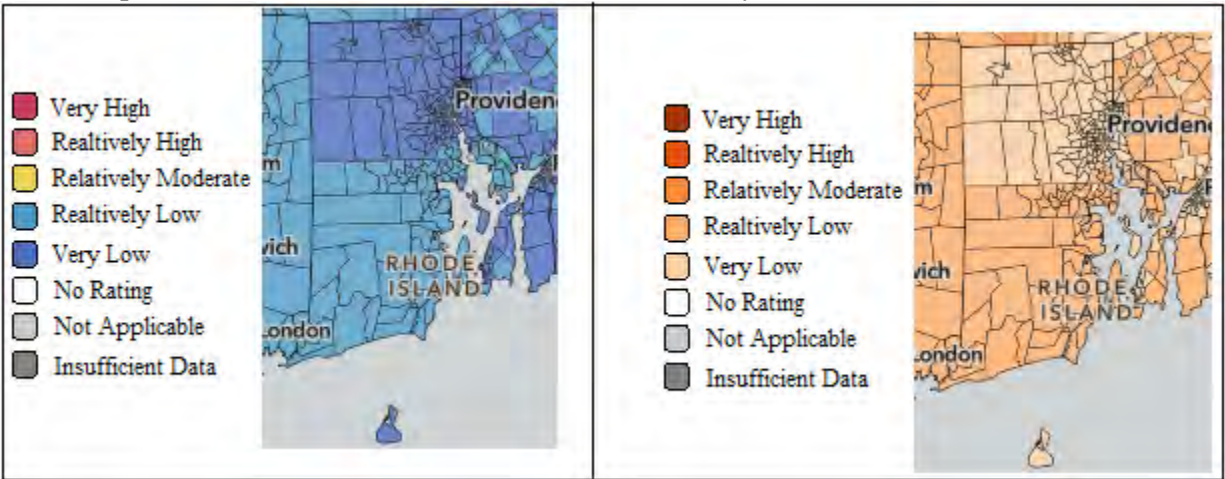
FEMA NRI was utilized to determine the risk and expected annual loss due to severe winter storms (including ice storms and winter weather) by census tract for the state of Rhode Island. The following maps detail these parameters:

Map 47: FEMA NRI and EAL for Ice Storm by Census Tract for Rhode Island



Source: FEMA NRI

Map 48: FEMA NRI and EAL for Winter Weather by Census Tract for Rhode Island



Source: FEMA NRI

Based on FEMA NRI social vulnerability ratings, and on U.S. Census Bureau data concerning percentage of the population below the 100% poverty who may difficulty heating their residences, the following communities may be at greater risk to severe winter storms conditions:

Table 120: Jurisdictions at Greater Risk to Winter Weather due to Poverty and Social Vulnerability

County	Jurisdiction	Population Below 100% of the Poverty Level	Social Vulnerability Rating
Providence County	Central Falls	29.9%	Very High
Providence County	Providence	23.0%	Relatively High
Providence County	Woonsocket	21.0%	Relatively Moderate
Washington County	Narragansett	19.3%	Relatively High
Newport County	Newport	14.4%	Relatively Low
Kent County	West Warwick	12.1%	Relatively Low
Providence County	Pawtucket	11.6%	Relatively High

Source: FEMA NRI and U.S. Census Bureau

A disproportionate impact from severe winter storms may be felt by jurisdictions with older housing units as they may have been constructed using materials and techniques that are less resilient to ice and snow loading on roofs. Modern construction materials and methods, such as reinforced concrete and steel frames, are often more resilient. The following table Rhode Island Department of Housing 2022 Integrated Housing Report from the Rhode Island Department of Housing 2022 Integrated Housing Report indicates jurisdictions with greater than 40% of their housing units constructed before 1939:

Table 121: Jurisdictions at Greater Risk to Severe Winter Storms due to Age of Housing

County	Jurisdiction	1939 and later
Providence County	Providence	57.70%
Newport County	Newport	55.20%
Providence County	Central Falls	50.40%
Providence County	Woonsocket	43.50%
Providence County	Pawtucket	43.30%
Bristol County	Warren	41.40%

Source: Rhode Island Department of Housing 2022 Integrated Housing Report

A household is considered cost burdened if it spends 30% or more of its income on housing costs, with cost burdens being an indication of rents increasing faster than incomes. Cost burdened citizens may have great difficulty paying heating and electric bills. The following table, using data from the Housing and Urban Development's Comprehensive Housing Affordability Strategy Data for Rhode Island, indicates jurisdictions located in WUI areas with the most cost burdened renters who may be at greater risk to wildfires due to economic constraints.

Table 122: Jurisdictions at Greater Risk to Severe Winter Storms due to Housing Cost Burden

County	Jurisdiction	Cost Burdened Single Families
Kent County	Exeter	73%
Kent County	Narragansett	65%
Bristol County	Barrington	64%
Providence County	Cumberland	54%
Newport County	Portsmouth	54%
Providence County	Smithfield	53%
Providence County	Central Falls	52%
Bristol County	Warren	52%
Kent County	Warwick	50%
Providence County	Woonsocket	50%

Source: Housing and Urban Development's Comprehensive Housing Affordability Strategy Data

4.16.8 Hazard Planning Significance

Based on a review of available data, the SIHMC determined severe winter weather to have a high planning significance for Rhode Island.

4.17 Tornado

4.17.1 Hazard Description

A tornado is a violent, dangerous, rotating column of air that is in contact with both the surface of the earth and a cumulonimbus cloud or, in rare cases, the base of a cumulus cloud. Tornadoes come in many shapes and sizes but are typically in the form of a visible condensation funnel, whose narrow end touches the earth and is often encircled by a cloud of debris and dust.



Tornadoes can cause several kinds of damage to buildings. Tornadoes have been known to lift and move objects weighing more than three tons, toss homes more than 300 feet from their foundations, and siphon millions of tons of water. However, less spectacular damage is much more common. Houses and other obstructions in the path of the wind cause the wind to change direction. This change in wind direction increases pressure on parts of the building. The combination of increased pressures and fluctuating wind speeds creates stress on the building that frequently causes connections between building components, roofing, siding, and windows to fail. Tornadoes can also generate a tremendous amount of flying debris. If wind speeds are high enough, airborne debris can be thrown at buildings with enough force to penetrate windows, roofs, and walls.

4.17.2 – Location and Extent

Tornadoes can strike anywhere in Rhode Island. However, the areas at greatest risk for touchdowns run from northwestern to northeastern of the state. A tornado may arrive with a squall line or cold front and touch down quickly. Smaller tornadoes can strike without warning. Other times tornado watches and sirens will alert communities of high potential tornado producing weather or an already formed tornado and its likely path.

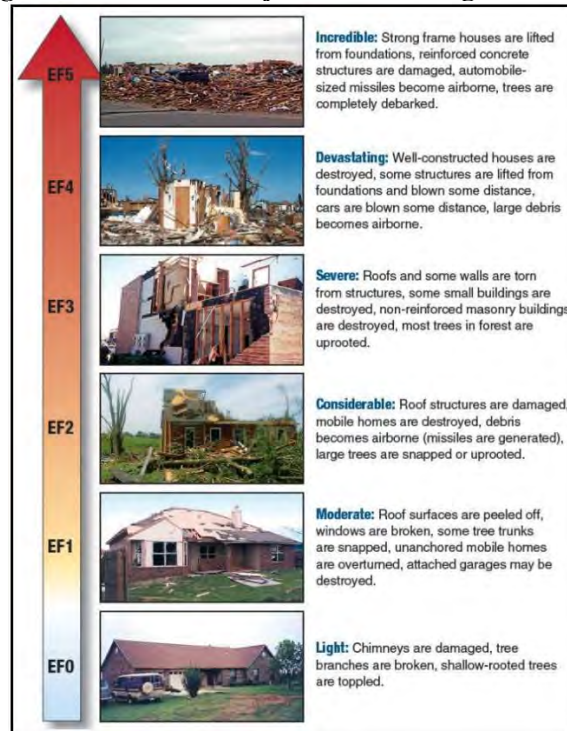
Since 2007, the United States uses the Enhanced Fujita (EF) Scale to categorize tornadoes. The scale correlates wind speed values per F level and provides a rubric for estimating damage.

Table 123: Enhanced Fujita Scale

Scale	Wind Speed (mph)	Relative Frequency	Potential Damage
EF0	65-85	53.5%	Light. Removes surface off some roofs, some damage to gutters or siding, branches broken off trees, shallow-rooted trees pushed over. Confirmed tornadoes with no reported damage (i.e., those that remain in open fields) are always rated EF0.
EF1	86-110	31.6%	Moderate. Roofs severely stripped; mobile homes overturned or badly damaged; loss of exterior doors; windows and other glass broken.
EF2	111-135	10.7%	Considerable. Roofs torn off well-constructed houses; foundations of frame homes shifted; mobile homes complete destroyed; large trees snapped or uprooted; light object missiles generated; cars lifted off ground.
EF3	136-165	3.4%	Severe. Entire stores of well-constructed houses destroyed; severe damage to large buildings such as shopping malls; trains overturned; trees debarked; heavy cars lifted off the ground and thrown; structures with weak foundations blown away some distance.
EF4	166-200	0.7%	Devastating. Well-constructed houses and whole frame houses completely leveled; cars thrown, and small missiles generated.
EF5	>200	<0.1%	Explosive. Strong frame houses leveled off foundations and swept away; automobile-sized missiles fly through the air in excess of 300 ft.; steel reinforced concrete structure badly damaged; high rise buildings have significant structural deformation; incredible phenomena will occur.

Source: NOAA Storm Prediction Center

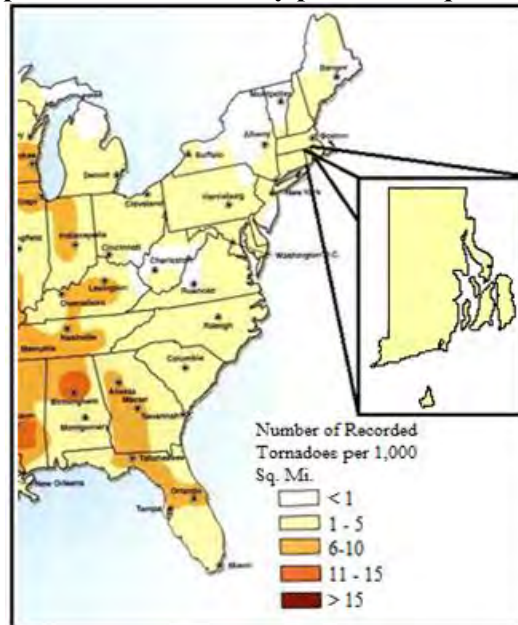
Figure 3: Enhanced Fujita Scale Damage Estimates



Source: FEMA

The following map, from FEMA, indicates that Rhode Island can expect on average one to five tornadoes per 1,000 square miles, a moderate category.

Map 49: Tornado Activity per 1,000 Square Miles



Source: FEMA

4.17.3 Previous Occurrences

The following table presents NCEI identified tornado events and the resulting damage totals in Rhode Island from 1950 to 2023.

Table 124: Rhode Island Tornado Events

County	Number of Events	Property Damage	Deaths	Injuries	Strongest Tornado
Bristol County	2	\$45,000	0	0	EF1
Kent County	2	\$250,000	0	0	F1
Newport County	1	\$5,000	0	0	EF0
Providence County	7	\$4,260,000	0	23	F2
Washington County	2	\$150,000	0	0	EF1

Source: National Centers for Environmental Information

It is worth noting that damage estimates indicated by the NCEI are often artificially low. This underreporting is a result of the way the events are reported to the NCEI, often by the local and/or NWS office. When reporting an event oftentimes the NWS office does not have access to the actual damage assessment resulting from that event. As such, the report often details a very low amount or zero-dollar amount for damages. Additionally, deaths and injuries may be underreported as they may be a result of a concurrent event, such as a person driving unsafely during heavy rain and passing away.

The largest tornado to strike Rhode Island occurred on August 7, 1986, in Providence County. The tornado touched down near Cranston and moved in a north-easterly direction through south Providence. Up to 6,000 customers lost power, 20 injuries from flying debris were reported, and property damages were estimated at \$2,500,000.

NOAA has been tracking tornadoes in Rhode Island for decades. This following map, which contains data from 1950 to 2023, pinpoints where tornadoes have touched down and traces its path.:

Map 50: Rhode Island Tornado Paths

Source: NOAA

4.17.4 Probability of Future Events

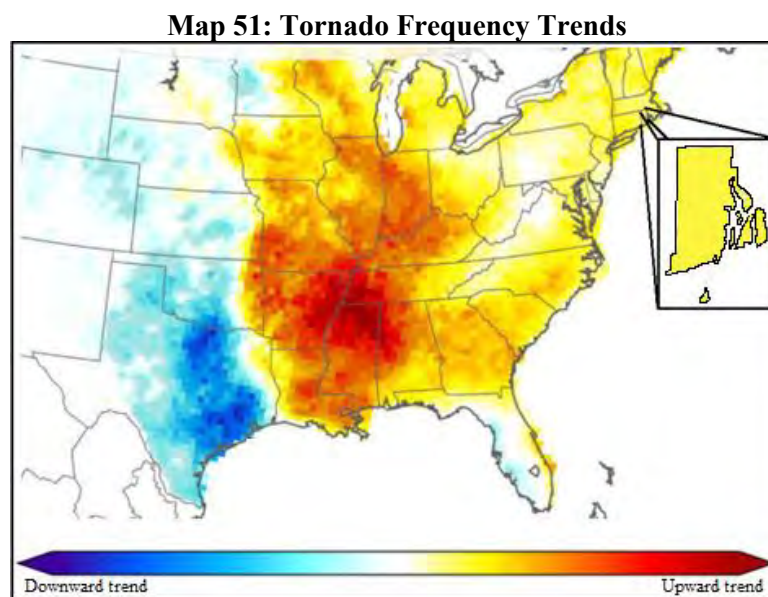
Predicting the probability of tornado occurrences is tremendously challenging due to the large number of factors involved and the random nature of formation. Data from the NCEI indicates that Rhode Island can expect infrequent tornado events based on the 14 events recorded from 1950 to 2023 (equating to less than one event per year). Available

historical tornado data suggests that Rhode Island can expect future tornadoes to range from EF0 to EF2 on the Enhanced Fujita Scale.

4.17.5 Projected Changes in Location, Intensity, Frequency, and Duration

The relationship between climate change and tornadoes is complex, and while there is ongoing research in this area, it is not fully understood. Tornadoes are small-scale, short-lived weather phenomena that can be influenced by a variety of atmospheric factors, including temperature, humidity, wind patterns, and atmospheric instability. Climate change can influence some of these factors, which may, in turn, affect tornado activity. Tornadoes typically form when warm, moist air near the surface clashes with cooler, drier air aloft, creating atmospheric instability. Climate change can alter temperature and humidity patterns, potentially affecting the conditions necessary for tornado formation. Additionally, climate change can lead to more extreme and variable weather patterns. While this may not necessarily increase the overall number of tornadoes, it could lead to more unpredictable and severe tornado events when they do occur. Some research suggests that climate change could lead to longer tornado seasons, with tornadoes occurring outside of their typical timeframes.

Research conducted by the National Severe Storms Lab looked at Significant Tornado Parameters to help determine future tornado probability. Significant Tornado Parameters are a measurement of the major parameters of tornado conditions, including wind speed and direction, wind at differing altitudes, unstable air patterns, and humidity. The following map, generated by Northern Illinois University and compiled from Significant Tornado Parameter data, indicates that Rhode Island may see an increasing number of tornadoes:



Source: Northern Illinois University

It's important to emphasize that while there may be some links between climate change and tornado activity, these links are not fully understood, and it is difficult to attribute specific tornado events to climate change. Tornadoes are influenced by a complex interplay of factors, and any changes in tornado patterns may vary by region.

As previously noted, State of Rhode Island facilities have seen no major changes in the past five years, with only modest repairs and upgrades being conducted and no major rehabilitation or construction projects completed. As such, the risk to state assets has remained static since the completion of the 2019 SHMP. Additionally, with the exception of the two previously identified rehabilitation and construction projects, no large-scale facility projects were noted in any agency budgets or reports.

4.17.6 Vulnerability and Impact

People and Communities

Tornadoes can have significant and often devastating impacts on people and communities. These impacts can vary depending on the tornado's intensity, size, path, and may include:

- **Injuries and Fatalities:** Tornadoes can cause a wide range of injuries, from minor cuts and bruises to severe trauma. Flying debris, structural damage, and the force of the wind can lead to injuries or fatalities among those directly affected by the tornado. Prompt medical care is essential to treat injuries effectively and save lives.
- **Mental Health Effects:** Tornadoes can be extremely traumatic events, causing psychological distress and emotional trauma for survivors. Individuals may experience post-traumatic stress disorder, anxiety, depression, and grief. Mental health support and counseling services are often needed to help survivors cope with these emotional challenges.
- **Displacement:** People may need to evacuate their homes or be temporarily displaced due to tornado damage, requiring emergency shelter and support.

After a tornado, health risks may arise due to contaminated water, debris, and unsafe conditions. Inadequate sanitation and exposure to harsh weather can exacerbate health issues. Children, the elderly, and individuals with disabilities or limited mobility may face additional challenges in evacuating to safety and accessing needed resources.

Facilities and Critical Infrastructure

Tornadoes can have significant and wide-ranging impacts on facilities, critical facilities, and critical infrastructure. These risks can have significant economic and operational consequences, and can include:

- **Critical Facility Damage:** Hospitals and healthcare facilities may be damaged or rendered inoperable during tornadoes, affecting the ability to provide medical care during a disaster. Fire stations, police stations, and emergency operation centers may be damaged or destroyed, impacting the ability of first responders to coordinate disaster response efforts. Damage to emergency shelters and housing facilities can disrupt services which are critical for providing temporary shelter to displaced individuals and families.
- **Infrastructure Damage:** High winds and blowing debris can cause considerable damage to infrastructure. This can result in costly repairs and disruptions to daily lives for an extended period.
- **Power Outages:** Tornadoes can cause power outages by bringing down power lines and damaging electrical infrastructure. Critical facilities such as hospitals, emergency response centers, and data centers may rely on backup generators to maintain essential operations during outages.
- **Communication Disruptions:** Tornadoes can damage communication infrastructure, including cell towers, telephone lines, and data centers, leading to disruptions in phone and internet services. This can hinder emergency communication and coordination, affecting critical response efforts.
- **Transportation Disruptions:** Debris and fallen trees on roads, runways, and railways can disrupt transportation networks, leading to travel delays, accidents, and closures. Critical facilities may face challenges in receiving essential supplies and personnel during and after the storm.
- **Healthcare System Strain:** Hospitals and healthcare facilities may experience increased demand for medical services due to storm-related injuries and illnesses.
- **Water and Wastewater System Interruptions:** Tornadoes can damage water treatment plants, pumping stations, and water distribution systems. This can lead to a loss of clean drinking water and sanitation services, posing health risks to affected communities. Damage to wastewater treatment facilities and sewer systems can result in the release of untreated sewage, creating environmental hazards and public health concerns.
- **Fuel Supply Disruptions:** Tornadoes disrupt fuel supply chains, leading to shortages of gasoline, diesel, and heating oil. Critical facilities may rely on fuel for backup power generators and heating systems.
- **Property Damage:** Tornadoes can result in property damage, up to and including complete structural collapse.

Environment

Tornadoes can have significant impacts on the environment. These impacts are often destructive and can affect ecosystems, wildlife, natural resources, and even the local climate. Tornadoes can disrupt natural habitats by uprooting or damaging trees, destroying vegetation, and altering landscapes. This can affect the habitat suitability for wildlife and plant species. Tornadoes can harm or displace wildlife, resulting in injury or death. Nesting birds, burrowing mammals, and other species can be particularly vulnerable. As tornadoes can transport plant seeds, insects, and other organisms over long distances, in the aftermath it is possible for invasive species to take root in new areas, especially those impacted by wildfires caused by downed utility lines.

State Operations and Assets

Tornadoes can have significant and wide-ranging impacts on state operations and assets (with all state assets vulnerable). When tornadoes strike, they can disrupt government functions, damage critical infrastructure, and strain resources. Some of the key impacts of tornadoes on state operations and facilities may include:

- **Emergency Response and Public Safety:** Tornadoes can lead to a surge in emergency calls for services related to accidents, injuries, and damaged structures. State agencies involved in emergency response, such as RIEMA and the Rhode Island State Police, must mobilize additional resources to handle these demands.
- **Emergency Operations Centers:** Tornadoes often require the activation of state Emergency Operations Centers to coordinate emergency response efforts. These centers serve as hubs for communication, resource allocation, and decision-making during disasters.
- **Emergency Shelters and Services:** Tornadoes may require the establishment of emergency shelters and services for displaced residents. State agencies must coordinate the setup and operation of these facilities.
- **Education Disruption:** Tornadoes can lead to school closures, affecting state-run education programs and services. State agencies may need to coordinate with school districts to ensure the safety of students.
- **Budgetary Impact:** The costs associated with emergency response efforts, disaster recovery, and infrastructure repair can strain state budgets.
- **Resource Allocation:** State governments must allocate resources, including personnel, equipment, and stockpiled supplies, to support emergency response and recovery efforts.
- **Communication Challenges:** Tornadoes can disrupt communication networks, hindering the ability of state agencies to communicate internally and with the public. This can impact emergency notifications and coordination efforts.
- **Administrative and Governance Challenges:** State government offices and facilities may experience closures or reduced staffing during tornadoes, affecting administrative functions, regulatory processes, and public services.
- **Economic Impact:** The destruction of infrastructure and businesses can have significant economic consequences for the state and local communities, including job losses and reduced economic activity.
- **Public Services:** Tornadoes can disrupt the delivery of public services, including transportation, utilities, and social services, affecting the well-being of residents.

Depending on the intensity of the tornado, all state owned or operated facilities are vulnerable to potential impacts. The following table indicates the number and valuation of the building envelope and building contents for all state owned or operated facilities:

Table 125: State of Rhode Island Facilities and Assets by County

Jurisdiction	State Owned or Operated Facilities	Valuation
Bristol County	32	\$150,415,810
Kent County	116	\$215,026,614
Newport County	77	\$127,353,837
Providence County	423	\$2,526,593,056
Washington County	393	\$274,350,750
State Total	1,041	\$3,293,740,067

Source: RIEMA

Note: Valuations include structure and assets

The age of a state owned or operated structure could have a significant impact on its resilience to a tornado as an older structure may be less resilient due to the following factors:

- **Building Codes and Standards:** Older buildings may not have been constructed to meet modern building codes, which often incorporate lessons learned from previous disasters. Newer buildings are typically designed to be more resilient to known hazards.

- **Construction Materials and Techniques:** Older buildings may have been constructed using materials and techniques that are less resilient to hazards. Modern construction materials and methods, such as reinforced concrete and steel frames, are often more resilient to hazards.

The following table indicates the average year of construction for state-owned or operated facilities with a known completion date, by county, for the State of Rhode Island:

Table 126: Average Year of Construction for State of Rhode Island Owned or Operated Facilities by County

Jurisdiction	Average Year of Construction
Bristol County	1966
Kent County	1983
Newport County	1978
Providence County	1961
Washington County	1964
State Buildings Average Year of Construction	1969

Source: RIEMA

Note: Average age based on buildings with known construction date

Of particular concern, a subset of state owned or operated facilities consists of manufactured buildings. Data from the NOAA National Severe Storms Laboratory reports that people working in manufactured buildings are especially at risk for injury and death as even anchored manufactured buildings can be seriously damaged when winds gust over 80 miles per hour. The following table indicates the number and valuation of the building envelope and building contents for all state owned or operated manufactured buildings:

Table 127: State of Rhode Island Manufactured Facilities and Assets by City and County

Jurisdiction	County	Number of Manufactured Buildings	Valuation
Bristol	Bristol County	1	\$174,599
Burrillville	Providence County	4	\$431,464
Cranston	Providence County	2	\$859,871
Exeter	Washington County	1	\$749,043
Portsmouth	Newport County	1	\$8,928,598
State Total	-	6	\$11,143,575

Source: RIEMA

Note: Valuations include structure and assets

Consequence Analysis

This consequence analysis lists the potential impacts of this hazard on various elements of community and state infrastructure. The impact of this hazard is evaluated in terms of disruption of operations, recovery challenges, and overall wellbeing to all Rhode Island residents and first responder personnel. The consequence analysis supplements the hazard profile by analyzing specific impacts.

Table 128: Tornado Consequence Analysis

Subject	Potential Impacts
Impact on the Public	High wind speeds can cause automobiles to become airborne, destroy homes, and turn debris into projectiles, which may cause injury or death. An increased demand for medical treatment for traumatic injuries caused by the tornado would be anticipated. Significant portions of the population may be displaced by the destruction and those individuals may not have access to personal documents or medical records.
Impact on Responders	First responders may be injured as the tornado passes, resulting in employee absenteeism that impacts the overall capacity to respond to the event. The deposit of debris on major roadways, the location of the event, and/or damage to equipment or facilities may increase the response times. Exposed wires or hazardous materials may cause injury to first responders during search and rescue operations.
Continuity of Operations	Rhode Island agencies maintain continuity plans which can be enacted as necessary based on the situation. Tornadoes may impact an agency's ability to maintain

Table 128: Tornado Consequence Analysis

Subject	Potential Impacts
	continuity of operations due to impacts on power or communications infrastructure, or equipment at primary or alternate facilities. If the activation of alternate facilities was required, travel may be difficult due to reduced transportation options, power outages, or damage to facilities.
Delivery of Services	Delivery of services may be impacted by dangerous conditions or disruption to transportation systems, causing food, water, and resource systems to be delayed or halted. Waterway infrastructure may be damaged or malfunction, stopping barge and ship traffic. Goods may be damaged, destroyed, or carried off by high winds.
Property, Facilities, and Infrastructure	Damages from lower intensity tornadoes can range from chimney damage to uprooted shallow trees. A significant tornado (EF-2) would cause damage to roofs on frame houses, complete destruction of mobile homes and large trees and utility lines snapping. A devastating tornado (EF-4) would result in well-constructed houses being leveled, weak foundations blown away, and cars thrown away. Communications or power infrastructure may be damaged or destroyed.
Impact on Environment	Tornadoes may cause significant damage to the environment by exposing hazardous materials, causing contamination of water or food sources, or uprooting vegetation. Animals may be injured by flying debris or being lifted by the tornado. Agricultural crops may be lost due to contamination or being uprooted.
Economic Conditions	Tornadoes pose a fiscal impact on the local and state governments, even if some of those costs can be recouped through federal grant reimbursements. Fiscal resources may be drained by the occurrence of a tornado.
Public Confidence in Governance	The public's confidence in the state's governance is affected by immediate local and state response through direct and effective actions. Efficiency in response and recovery operations is critical in keeping public confidence high.

4.17.7 Jurisdictional Vulnerability and Planning Significance

Utilizing data from FEMA's NRI and from LHMPs, the following tables detail the hazard ratings for jurisdictions, by census tract, for severe winter weather. Data concerning both risk index and EAL is reported by census tract. As many State of Rhode Island jurisdictions encompass multiple tracts, an average of tract ratings was completed to determine the overall jurisdictional rating. The following table details the hazard rating by LHMP analysis, FEMA NRI, and EAL by jurisdiction for the State of Rhode Island:

Table 129: State of Rhode Jurisdictional Tornado Event Planning Significance

County	Jurisdiction	Tornado LHMP Rating	Tornado NRI	Tornado EAL
Bristol	Barrington	Moderate	Very Low	Very Low
	Bristol	-	Very Low	Very Low
	Warren	Moderate	Very Low	Very Low
Kent	Coventry	-	Relatively Low	Relatively Low
	East Greenwich	Low	Relatively Low	Relatively Low
	Warwick	Low	Very Low	Relatively Low
	West Greenwich	-	Relatively Low	Relatively Low
	West Warwick	Low	Relatively Low	Relatively Low
Newport	Jamestown	-	Very Low	Relatively Low
	Little Compton	Moderate	Very Low	Very Low
	Middletown	Not Rated	Very Low	Very Low
	Newport	Low	Very Low	Very Low
	Portsmouth	Low	Very Low	Very Low
	Tiverton	-	Very Low	Very Low
Providence	Burrillville	Low	Relatively Low	Relatively Low
	Central Falls	Low	Relatively Low	Relatively Low
	Cranston	Low	Relatively Low	Relatively Low

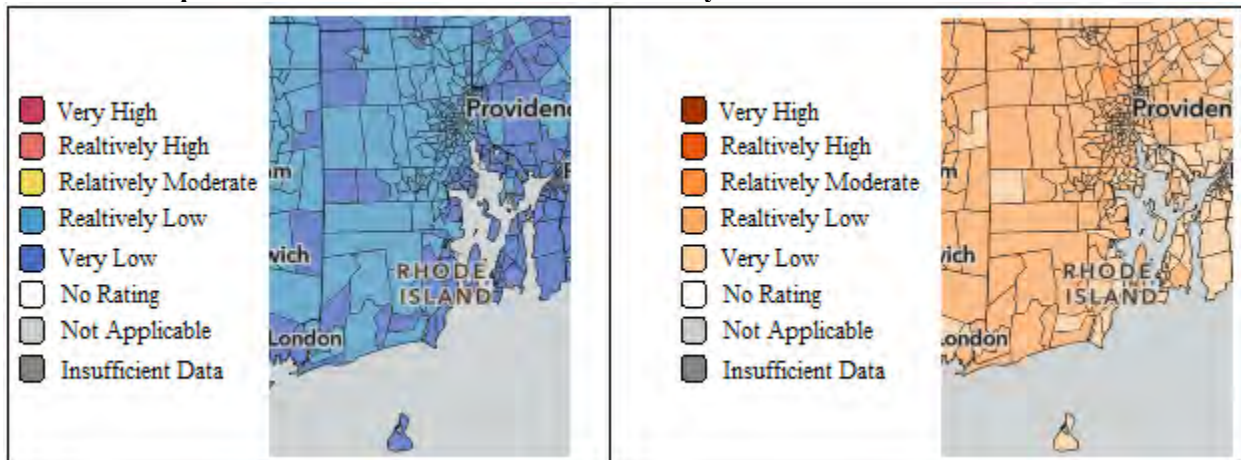
Table 129: State of Rhode Jurisdictional Tornado Event Planning Significance

County	Jurisdiction	Tornado LHMP Rating	Tornado NRI	Tornado EAL
	Cumberland	-	Relatively Low	Relatively Low
	East Providence	Moderate	Very Low	Relatively Low
	Foster	Low	Relatively Low	Relatively Low
	Glocester	Not Rated	Relatively Low	Relatively Low
	Johnson	Moderate	Relatively Low	Relatively Low
	Lincoln	Moderate	Relatively Low	Relatively Low
	North Providence	Low	Relatively Low	Relatively Low
	North Smithfield	Not Rated	Relatively Low	Relatively Low
	Pawtucket	Low	Relatively Low	Relatively Low
	Providence	Low	Relatively Low	Relatively Low
	Scituate	-	Relatively Low	Relatively Low
	Smithfield	-	Relatively Low	Relatively Low
	Woonsocket	Low	Relatively Low	Very Low
Washington	Charlestown	-	Relatively Low	Relatively Low
	Exeter	-	Relatively Low	Relatively Low
	Hopkinton	-	Relatively Low	Relatively Low
	Narragansett	Not Rated	Very Low	Very Low
	New Shoreham	-	Very Low	Very Low
	North Kingstown	Moderate	Very Low	Relatively Low
	Richmond	Low	Relatively Low	Relatively Low
	South Kingstown	Low	Relatively Low	Relatively Low
	Westerly	Moderate	Relatively Low	Relatively Low

Source: LHMPs and FEMA NRI

Note: - indicates no current plan available for review

FEMA NRI was utilized to determine the risk and expected annual loss due to tornadoes by census tract for the state of Rhode Island. The following maps detail these parameters:

Map 52: FEMA NRI and EAL for Tornado by Census Tract for Rhode Island

Source: FEMA NRI

Of particular concern when considering housing data is manufactured buildings (often referred to as mobile homes). Data from the NOAA National Severe Storms Laboratory reports that people living in mobile homes are especially at risk for injury and death as even anchored mobile homes can be seriously damaged when winds gust over 80 miles per hour. Additionally, study data from Michigan State University reported that the two biggest factors related to wind event fatalities were housing quality (measured by mobile homes as a proportion of housing units) and income level. When a strong wind strikes, a county with double the number of mobile homes as a proportion of all homes will

experience 62% more fatalities than a county with fewer mobile homes, according to the study data. The following indicates the percentage of mobile homes for each Rhode Island County:

Table 130: Rhode Island 2020 Census Mobile Home Data

Jurisdiction	Number of Mobile Homes	Percentage Of Housing Stock as Mobile Homes
Bristol County	0	0%
Kent County	902	1.2%
Newport County	701	1.6%
Providence County	1,059	0.4%
Washington County	793	1.2%
State of Rhode Island	3,455	0.7%

Source: United States Census Bureau

Based on FEMA NRI social vulnerability ratings, and on U.S. Census Bureau data concerning percentage of the population below the 100% poverty who may have difficulties in either relocating or repairing their residence, the following communities may be at greater risk to tornado events:

Table 131: Jurisdictions at Greater Risk to Tornadoes Due to Poverty and Social Vulnerability

County	Jurisdiction	Population Below 100% of the Poverty Level	Social Vulnerability Rating
Providence County	Central Falls	29.9%	Very High
Providence County	Providence	23.0%	Relatively High
Providence County	Woonsocket	21.0%	Relatively Moderate
Washington County	Narragansett	19.3%	Relatively High
Newport County	Newport	14.4%	Relatively Low
Kent County	West Warwick	12.1%	Relatively Low
Providence County	Pawtucket	11.6%	Relatively High

Source: FEMA NRI and U.S. Census Bureau

A disproportionate impact from tornadoes may be felt by jurisdictions with older housing units as they may have been constructed using materials and techniques that are less resilient to strong winds. Modern construction materials and methods, such as reinforced concrete and steel frames, are often more resilient. The following table Rhode Island Department of Housing 2022 Integrated Housing Report from the Rhode Island Department of Housing 2022 Integrated Housing Report indicates jurisdictions with greater than 40% of their housing units constructed before 1939:

Table 132: Jurisdictions at Greater Risk to Tornadoes due to Age of Housing

County	Jurisdiction	1939 and later
Providence	Providence	57.7%
Newport	Newport	55.2%
Providence	Central Falls	50.4%
Providence	Woonsocket	43.5%
Providence	Pawtucket	43.3%
Bristol	Warren	41.4%

Source: Rhode Island Department of Housing 2022 Integrated Housing Report

4.17.8 Hazard Planning Significance

Based on a review of available data, the SIHMC determined tornado to have a low planning significance for Rhode Island.

4.18 Tropical and Extratropical Storms

4.18.1 Hazard Description

Tropical and extratropical storms are two distinct types of weather systems characterized by their formation, structure, and associated meteorological conditions.

Tropical storms originate over warm ocean waters, typically in tropical and subtropical regions. They form when warm, moist air rises, creating an area of low pressure at the surface. Warm ocean waters provide the heat and moisture necessary for their development. Tropical storms have a well-defined center called the

"eye" surrounded by bands of thunderstorms. They have warm core temperatures, meaning that the strongest winds and heaviest rainfall are concentrated near the center. Their primary energy source is the heat released when moist air rises and condenses into clouds. Tropical storms are often driven by easterly trade winds and move generally west and north. However, their paths can be influenced by various atmospheric factors.



Tropical storms are classified as:

- **Tropical Depression:** An organized system of clouds and thunderstorms with a defined circulation and maximum sustained winds of 38 mph or less
- **Tropical Storm:** An organized system of strong thunderstorms with a defined circulation and maximum sustained winds of 39 mph to 73 mph
- **Hurricane:** An intense tropical weather system with a well-defined circulation, producing maximum sustained winds of 74 mph or greater

Extratropical storms, such as Nor'easters, form in regions outside the tropics. They result from the interaction of contrasting air masses, often where cold, polar air meets warm, moist air from lower latitudes. Unlike tropical storms, extratropical storms lack a well-defined eye and typically have a frontal boundary associated with them. Their primary energy source is the temperature contrast between warm and cold air masses. The movement of extratropical storms can be influenced by the polar jet stream, but they generally move from west to east.

Associated with both tropical and extratropical storms, a storm surge is a sudden and significant rise in sea level. It occurs when strong onshore winds and low atmospheric pressure cause ocean water to pile up and move inland. Storm surges can be extremely dangerous and damaging, as they can inundate coastal areas, leading to flooding, erosion, and significant property damage. The following are key factors that contribute to storm surges:

- **Pressure:** Low atmospheric pressure at the center of the storm works to push water toward the coastline.
- **Wind Setup:** Strong onshore winds generate a push effect on the ocean's surface, causing water to pile up along the coast.
- **Shallow Coastal Slope:** The shallow slope characteristics of Rhode Island tend to cause more significant surges because there is less space for the water to disperse.
- **Tide Interaction:** The timing of a storm's arrival relative to the natural tidal cycle can greatly affect the severity of a storm surge.
- **Bathymetry:** The shape of the coastline, underwater topography, and the presence of bays, estuaries, and other geographical features can influence how the surge propagates and amplifies.

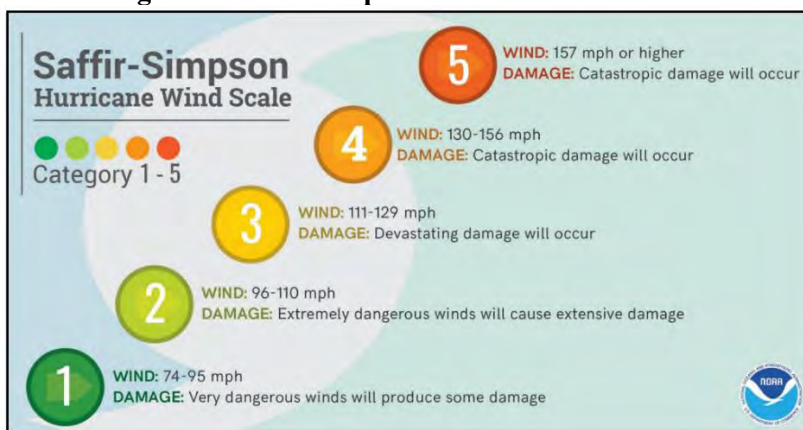
4.18.2 – Location and Extent

Rhode Island is susceptible to both tropical and extratropical storms, although the extent and impacts of these storms can vary. Rhode Island is vulnerable to tropical storms and hurricanes, particularly during the Atlantic hurricane season, which runs from June 1st to November 30th. While Rhode Island does not experience hurricanes as frequently as some southern coastal states, it is not immune to the effects of these powerful storms. Coastal regions of Rhode Island, including cities like Newport and Narragansett, are most at risk from hurricanes and tropical storms. The coastline is

susceptible to storm surges, high winds, heavy rainfall, and coastal erosion. Inland areas can also experience the impacts of hurricanes and tropical storms, including flooding from heavy rainfall and strong winds that can topple trees and power lines.

The Saffir-Simpson Hurricane Wind Scale is a classification system used to categorize hurricanes and tropical cyclones based on their maximum sustained wind speeds. The Saffir-Simpson Scale consists of five categories, each associated with a range of wind speeds and expected levels of damage, as follows:

Figure 4: Saffir-Simpson Hurricane Wind Scale



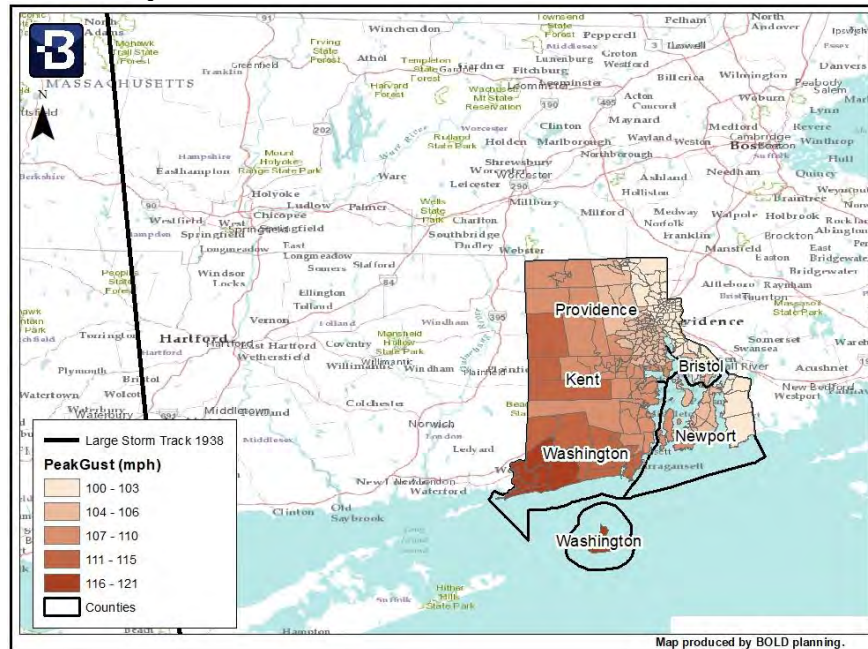
Source: Saffir-Simpson

The following provide a general overview of potential damages associated with each Saffir-Simpson hurricane category

- **Category 1 (74-95 mph)**
 - Minimal damage to well-constructed buildings
 - Damage to unanchored mobile homes, signs, and trees
 - Some power outages, primarily due to downed tree limbs
- **Category 2 (96-110 mph)**
 - Moderate damage to roofs, windows, and doors
 - Considerable damage to mobile homes and poorly constructed buildings
 - Extensive tree damage, power outages, and potential for water supply disruption
- **Category 3 (111-129 mph)**
 - Extensive damage to roofs, windows, and doors of well-built homes
 - Well-constructed buildings may incur structural damage
 - Many trees will be uprooted or snapped
 - Widespread power outages and water supply disruptions
- **Category 4 (130-156 mph)**
 - Catastrophic damage to roofs, walls, and windows of well-constructed buildings
 - Severe damage to homes, including total roof failure and wall collapse
 - Most trees and power poles will be downed
 - Prolonged power outages, water supply disruption, and extensive structural damage
- **Category 5 (157 mph or higher)**
 - Catastrophic damage to well-built homes and buildings
 - Complete roof failure, walls may collapse
 - Widespread destruction of trees, power poles, and other infrastructure
 - Extended power outages, water supply disruption, and severe structural damage

The following map shows wind intensity through the state for a modelled 1938 hurricane (often referred to as the great 1938 hurricane) storm tracking to the West of Rhode Island:

Map 53: Peak Wind Gust of the 1938 Great Hurricane



Source: NOAA and NWS

In addition to the above noted impacts, hurricanes can also bring other hazards that contribute to overall destruction including storm surge: heavy rainfall resulting in flooding, and tornadoes.

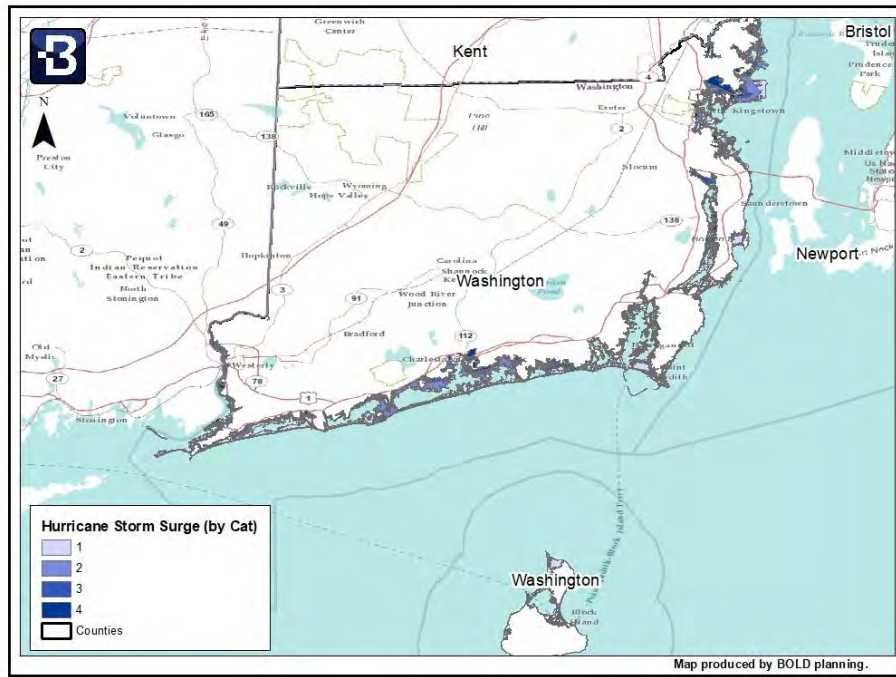
Rhode Island is also susceptible to extratropical storms, including Nor'easters. These storms are more common during the fall, winter, and early spring. Extratropical storms can affect both coastal and inland areas of Rhode Island. Coastal regions can experience coastal flooding and erosion, while inland areas may face heavy snowfall, rain, and strong winds. Nor'easters, a type of extratropical storm, can bring significant snowfall to the state, particularly in the western and northern parts of Rhode Island. This can lead to transportation disruptions and power outages. High winds associated with extratropical storms can lead to tree damage, power outages, and hazardous driving conditions.

Currently there is no standardized scale for measuring the intensity of Nor'easters. However, meteorological agencies can use the following criteria and measures to assess the strength and potential impact:

- **Central Pressure:** A lower central pressure generally indicates a stronger storm.
- **Wind Speed:** Higher wind speeds can lead to more significant coastal erosion, infrastructure damage, and structural damage.
- **Precipitation:** The amount and type of precipitation (rain, snow, sleet) can also be used to assess the storms' potential impact.
- **Storm Surge:** The extent and height of the storm surge are important for assessing coastal flooding risks.
- **Duration:** Prolonged periods of strong winds and precipitation can lead to more damage and disruptions.

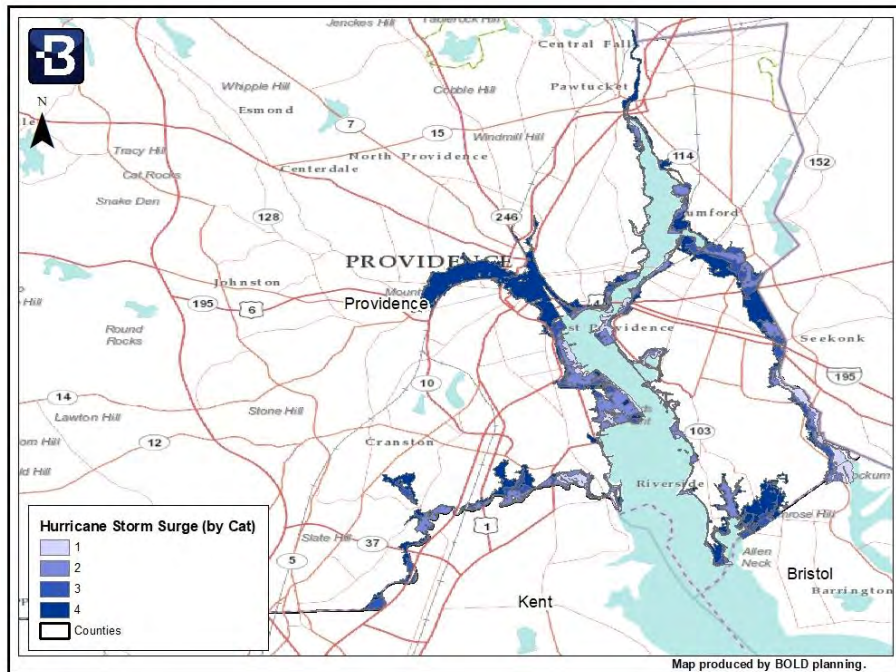
Tropical and extratropical storm surge heights and inundation levels can be estimated using the Sea, Lake, and Overland Surges from Hurricanes. (SLOSH) model, developed by NOAA. The SLOSH model takes into account a variety of factors, including the hurricane's track, intensity, size, forward speed, and atmospheric pressure, as well as the local coastal geography, bathymetry, and tidal conditions. By simulating these variables, the model can provide forecasts of storm surge heights at various locations along the coast in the form of inundation maps. The following maps indicate potential inundation areas from tropical storms systems:

Map 54: Rhode Island Storm Surge Inundation Areas



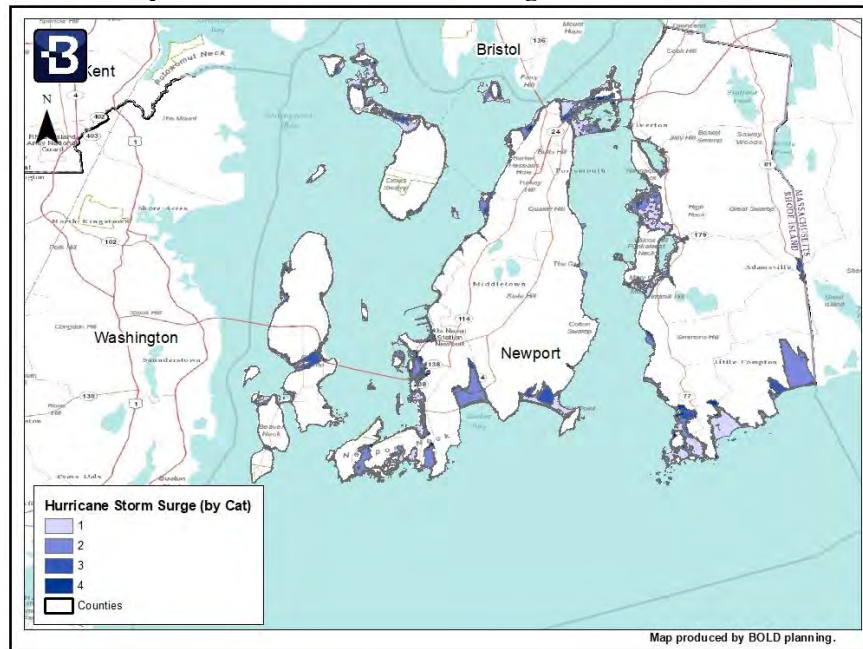
Source: SLOSH

Map 55: Rhode Island Storm Surge Inundation Areas



Source: SLOSH

Map 56: Rhode Island Storm Surge Inundation Areas



Source: SLOSH

4.18.3 Previous Occurrences

Historical events of significant magnitude or impact can result in a Presidential Disaster Declaration. Rhode Island has experienced six Presidential Disaster Declarations related to extratropical and tropical storms, reflected in the following table.

Table 133: State of Rhode Island Presidentially Declared Disasters

Designation	Declaration Date	Incident Type	Individual Assistance	Public Assistance
DR-4089-RI	11/3/2012	Hurricane Sandy	\$421,341	\$8,629,001
DR-4027-RI	9/3/2011	Tropical Storm Irene	\$8,341,293	\$1,608,921
DR-913-RI	08/26/1991	Hurricane Bob	-	-
DR-748-RI	10/15/1985	Hurricane Gloria	-	-
DR-39-RI	08/20/1955	Hurricane and Flood	-	-
DR-23-RI	09/02/1954	Hurricane	-	-

Source: FEMA

Note: -: Data unavailable

The following table details information concerning the four most recent hurricane occurrences in Rhode Island:

- **DR-4809-RI:** On October 29, 2019, Hurricane Sandy swept through the region causing significant damage. Beaches along Westerly were devastated and almost unrecognizable and more than 122,000 people lost power. It is estimated that more than \$39,400,000 was provided to Rhode Island from four federal disaster relief programs, the majority of which is from the NFIP at \$31,100,000.
- **DR-4207-RI:** On August 27-29, 2011: Hurricane Irene knocked down trees and power lines, leaving up to half of Rhode Island residents without power. Gusts of wind up to 71 mph were reported, and storm surge in Narragansett Bay caused some coastal damage. The storm surge experienced along the coast was generally in the two-to-four-foot range with a high of 4.78 feet at Fox Point in Providence. The highest sustained wind speed was 62 mph at the Physical Oceanographic Real Time System station at Conimicut Light in Narragansett Bay.
- **DR-913-RI:** Hurricane Bob struck Rhode Island on August 18, 1991. Damage throughout Southern New England were reported at \$1,500,000,000 and over 60% of residents across Southeastern New England lost power. Storm surge height was recorded at six to 10 feet in Narragansett Bay. Additionally, two unconfirmed

- **DR-748-RI:** Hurricane Gloria struck Rhode Island and the Northeast on September 27, 1985. Property damage estimated at \$1,800,000 and 8,596 electric customers lost power. Before the landfall of the hurricane, an estimated 23,700 people evacuated.

- **The Blizzard of 1978:** The Blizzard of 1978 is one of the most infamous nor'easters in Rhode Island's history. It struck the region in late January 1978 and brought record-breaking snowfall, high winds, and coastal flooding. Snowfall totals exceeded 20 inches in many areas, with some locations reporting more than 30 inches. The combination of heavy snow and strong winds resulted in massive snowdrifts and made transportation nearly impossible. Coastal areas experienced significant flooding, and the storm caused extensive damage to homes and infrastructure.
- **The Presidents' Day Storm of 2003:** In February 2003, a powerful nor'easter, often referred to as the Presidents' Day Storm, impacted the northeastern United States, including Rhode Island. This storm brought heavy snowfall, with some areas in Rhode Island receiving over 20 inches of snow. Strong winds contributed to drifting snow and power outages. The storm disrupted travel and daily life for several days.
- **March 2018 Nor'easter:** A powerful nor'easter struck the eastern United States, affecting Rhode Island. This storm brought heavy snowfall, coastal flooding, and strong winds to the region. Some parts of Rhode Island received more than a foot of snow, and coastal areas experienced inundation and erosion.
- **April 2007 Nor'easter:** A late-season nor'easter brought heavy snowfall to Rhode Island and other parts of the northeast. The storm was notable for its unusual timing, as it occurred in the spring. Some areas received more than a foot of snow, causing travel disruptions and power outages.

The following map, from NOAA, show the tracks of all recorded tropical and extratropical storms that have impacted Rhode Island:

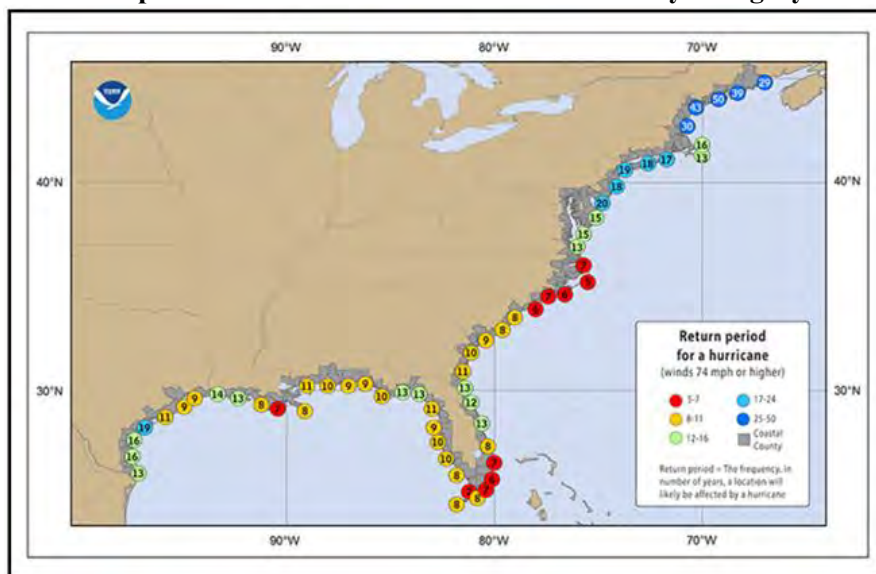
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4.18.4 Probability of Future Events

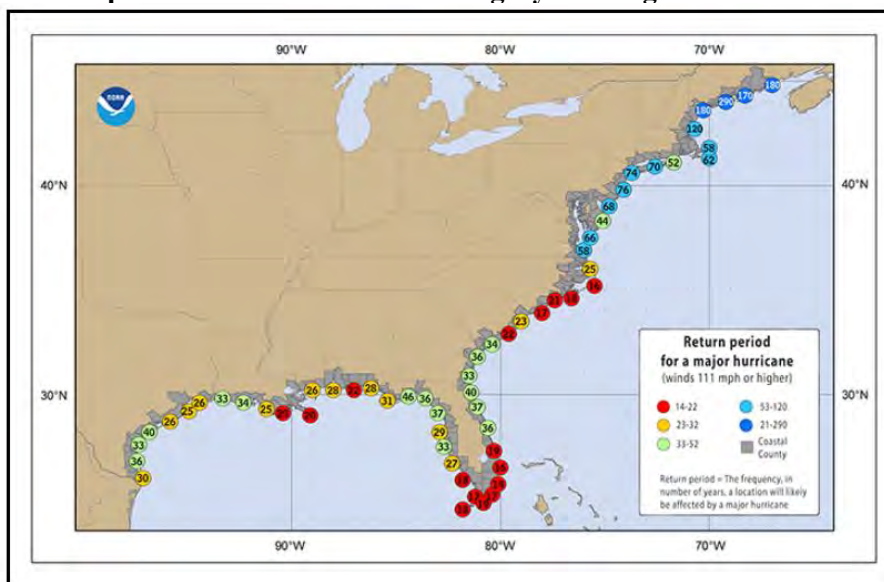
Predicting the probability of tropical or extratropical storm occurrences is tremendously challenging due to the large number of factors involved and the random nature of formation. However, the following maps depict the return period for a hurricane of any category on the Saffir-Simpson Hurricane Wind Scale (Rhode Island 17 years), and the return period for a major hurricane of Category 3 and higher (Rhode Island 52 years).

Map 58: Return Period for a Hurricane of Any Category



Source: NOAA

Map 59: Return Period for a Category 3 or Higher Hurricane



Source: NOAA

4.18.5 Projected Changes in Location, Intensity, Frequency, and Duration

Climate change can have several significant impacts on both tropical and extratropical storms in Rhode Island. While the precise effects can vary it is expected that climate change will result in increased storm intensity. For tropical storms and hurricanes, this increased intensity is caused by warmer sea surface temperatures which provide more energy to the systems. This can result in higher wind speeds and more rainfall, increasing the risk of damage and flooding. For extratropical storms, particularly Nor'easters, the increase in intensity is caused by changes in atmospheric conditions, including temperature gradients, which can affect the strength and track of these storms.

Both tropical and extratropical storms are expected to produce heavier rainfall in a warmer climate. This can lead to more significant inland flooding and exacerbate the risk of river and urban flooding. Compounding flooding, sea-level rise, driven by climate change, can enhance the impacts of storm surges associated with tropical and extratropical storms. Higher sea levels mean that storm surges can penetrate further inland, causing more extensive coastal flooding and coastal erosion.

Vulnerable populations, including low-income communities and those with limited access to resources, may be disproportionately affected by the impacts of climate change on storms. These communities often have fewer resources to prepare for and recover from storm-related disasters.

It's important to note that while there is a scientific consensus that climate change is contributing to these impacts, the exact magnitude of these changes can vary based on future greenhouse gas emissions, regional climate patterns, and other factors. To address these challenges, Rhode Island, like other states, is working on climate adaptation and mitigation strategies to enhance its resilience to the effects of climate change, including those related to tropical and extratropical storms.

As previously noted, State of Rhode Island facilities have seen no major changes in the past five years, with only modest repairs and upgrades being conducted and no major rehabilitation or construction projects completed. As such, the risk to state assets has remained static since the completion of the 2019 SHMP. Additionally, with the exception of the two previously identified rehabilitation and construction projects, no large-scale facility projects were noted in any agency budgets or reports.

4.18.6 Vulnerability and Impact

People and Communities

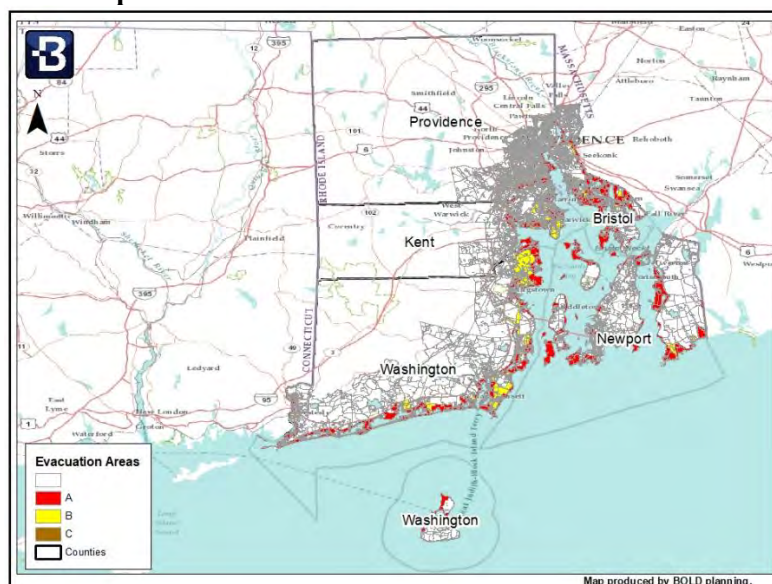
Tropical and extratropical storms, including hurricanes, typhoons, and nor'easters, can have a range of significant impacts on people and communities. The specific impacts can vary depending on the type and intensity of the storm, its location, and the preparedness of impacted areas, and may include:

- **Injuries and Fatalities:** Strong winds, storm surge, and heavy rainfall can lead to injuries and fatalities.
- **Storm Surge:** Coastal areas are vulnerable to storm surge, which can result in heavy flooding.
- **Flooding:** Heavy rainfall or snowfall can lead to flooding, including flash floods and river flooding, causing property damage, and posing risks to residents.
- **Property Damage:** Extensive damage to homes, businesses, and infrastructure, resulting in financial losses and displacement is possible.
- **Power Outages:** High winds, falling trees, and debris can lead to widespread power outages, affecting homes, hospitals, and critical infrastructure.
- **Communication Disruptions:** These storms can damage or disrupt communication infrastructure, including cell towers and telephone lines, making it challenging for residents to contact emergency services and loved ones.
- **Health Risks:** Flooding and contaminated water can pose health risks, including waterborne diseases. Exposure to mold and allergens in water-damaged homes can also lead to health problems.
- **Transportation Disruptions:** These storms can damage roads, bridges, and transportation networks, making it difficult for people to travel to safety or access essential services.
- **Food and Water Shortages:** Storm-related disruptions to supply chains can lead to shortages of food, water, and other essential goods, creating challenges for residents and emergency response efforts.
- **Emergency Response Challenges:** These storms can overwhelm emergency response resources, leading to challenges in providing timely assistance to affected communities.

Evacuations may be ordered for tropical and extratropical storms. This can lead to temporary displacement as individuals and families seek safety in emergency shelters or with friends and family. The following maps depict Rhode Island hurricane evacuation areas A, B, and C, which are part of the state's emergency preparedness and evacuation planning. These evacuation zones are designated based on the potential impact of storm surge and coastal flooding in the event of a hurricane or other coastal storm.

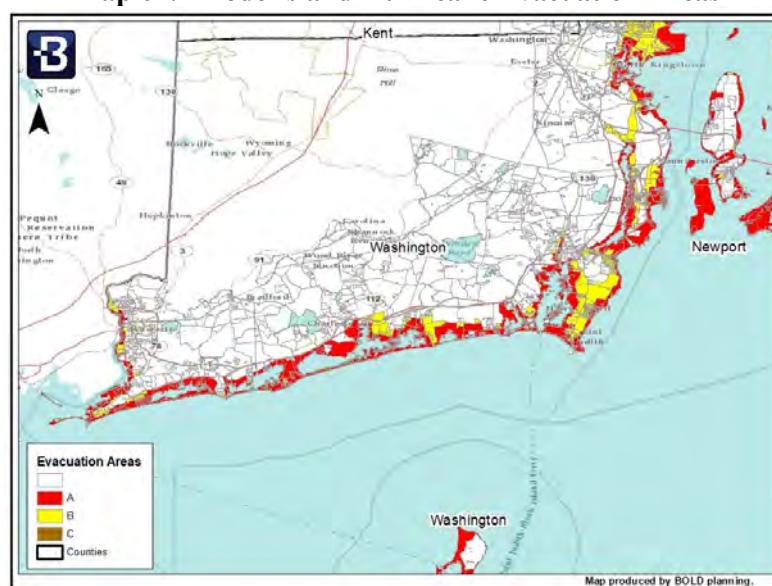
- **Evacuation Zone A:** This zone is typically the area most at risk of storm surge and coastal flooding during a hurricane or tropical storm. It includes low-lying coastal areas and communities that are highly vulnerable to coastal inundation. Residents in Zone A are usually among the first to receive evacuation orders when a hurricane threatens the state. Evacuation shelters and routes are pre-identified for residents in Zone A.
- **Evacuation Zone B:** This zone represents areas that are less vulnerable to coastal flooding but may still be at risk in certain hurricane scenarios. It often includes a broader coastal region beyond Zone A and may encompass more inland areas. Residents in Zone B may be advised to evacuate in advance of a hurricane, especially if the storm's track and intensity warrant it.
- **Evacuation Zone C:** This zone is typically considered a lower-risk area for storm surge and coastal flooding during hurricanes. It includes inland areas and neighborhoods farther from the coast. Residents in Zone C may be advised to take precautionary measures, such as securing outdoor items, but evacuation orders are less common for this zone.

Map 60: Rhode Island Hurricane Evacuation Areas



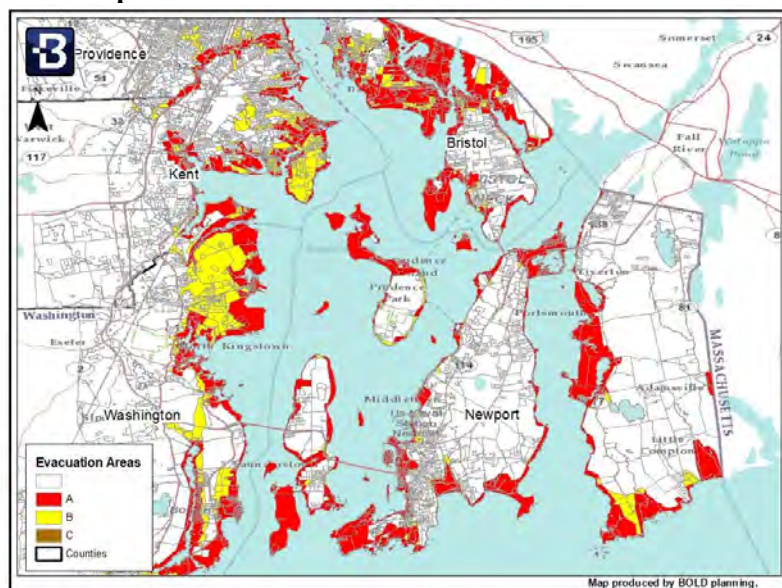
Source: RIEMA

Map 61: Rhode Island Hurricane Evacuation Areas



Source: RIEMA

Map 62: Rhode Island Hurricane Evacuation Areas



Source: RIEMA

Of particular concern, socially vulnerable and at-risk populations that may have difficulty with medical issues, poverty, extremes in age, and communications due to language barriers are often at increased susceptibility to the impacts of hazards. These populations suffer a disproportionate impact due to:

- Constrained financial resources, causing evacuation difficulties
- Susceptibility to impacts of infrastructure failure hampering transportation and evacuation
- Limited access to healthcare
- Limited access to information due to language difficulties

The following table details identified socially vulnerable and at-risk populations for the State of Rhode Island:

Table 134: Rhode Island 2020 Vulnerable Populations

Jurisdiction	Language Barrier	Poverty	Disability
Bristol County	6,552	3,403	4,753
Kent County	16,525	11,925	24,694
Newport County	7,794	8,307	9,105
Providence County	205,490	90,522	87,498
Washington County	7,790	11,815	13,832
State of Rhode Island	240,326	125,101	139,927

Source: US Census Bureau

Facilities and Critical Infrastructure

These storms can have significant and wide-ranging impacts on critical infrastructure and facilities, which are essential for maintaining public safety, providing essential services, and supporting the economy. The specific impacts can vary depending on the type and intensity of the storm, its location, and the preparedness of the location, but may include:

- **Power Outages:** High winds, flying debris, and flooding can lead to widespread power outages, disrupting critical facilities such as hospitals, emergency response centers, water treatment plants, and data centers. Backup power systems may be necessary to maintain essential operations.
- **Communication Disruptions:** Damage to communication infrastructure, including cell towers and telephone lines, can disrupt emergency communication and coordination, affecting the ability of critical facilities to respond effectively.

- **Transportation Disruptions:** These storms can damage transportation infrastructure, including roads, bridges, airports, and ports, impacting the movement of goods and personnel to and from critical facilities.
- **Water Supply Interruptions:** Flooding and damage to water treatment plants and pipelines can lead to water supply interruptions, affecting hospitals, firefighting capabilities, and other critical facilities that rely on a continuous supply of clean water.
- **Wastewater Systems:** Flooding and stormwater runoff can overwhelm wastewater treatment plants, potentially causing contamination and posing health risks to communities.
- **Healthcare Facilities:** Hospitals and healthcare facilities may experience increased patient volumes due to storm-related injuries and illnesses, necessitating additional resources and capacity.
- **Emergency Shelters:** The demand for emergency shelters may increase, requiring additional facilities and resources to accommodate displaced residents.
- **Transportation Safety:** Damaged transportation infrastructure can pose safety risks to workers and travelers, including emergency responders traveling to critical facilities.
- **Fuel and Energy Supply:** Disruptions to fuel supply chains and energy infrastructure can affect the availability of gasoline, diesel, heating oil, and electricity, impacting critical facilities and emergency response efforts.
- **Food and Medical Supplies:** Supply chain disruptions can lead to shortages of food, medical supplies, and pharmaceuticals, affecting critical facilities like hospitals and emergency response centers.

The Fox Point Hurricane Protection Barrier in Providence provides virtually complete protection against tidal flooding from hurricanes and other coastal storms to about 280 acres of downtown Providence. The protected area includes the commercial and industrial center, transportation facilities, public utilities, and many homes. The barrier is a 700-foot-long concrete structure, 25 feet high, that extends westerly across the Providence River. During a flood situation, the barrier's pumping station discharges floodwaters into the bay.

Electrical transmission lines are vulnerable to the high winds generated by hurricanes. Maps indicating the location of major electrical transmission lines are available in the severe storm thunderstorm section of this plan.

Environment

Tropical and extratropical storms, including hurricanes, tropical storms, and nor'easters pose several risks to the environment due to their strong winds, heavy rainfall, and storm surges. These risks can have both short-term and long-term impacts on ecosystems and natural resources. Some of the potential impacts to the environment include coastal erosion, habitat disruption (especially in critical habitats such as wetlands, marshes, and barrier islands), and water quality issues due to runoff, carrying pollutants from urban areas and agriculture. These storms can disrupt marine ecosystems by stirring up sediment and reducing water quality, which can be harmful to fish and other aquatic species, or by temporarily increasing salinity in freshwater ecosystems, affecting aquatic species that are sensitive to salinity fluctuations. Compounding this damage, storms can facilitate the spread of invasive species by transporting them to new areas via floodwater and ocean currents. This can have long-term ecological impacts by altering native ecosystems.

State Operations and Assets

These storms can pose significant risks to state operations and assets (all state assets are vulnerable). These risks can disrupt government functions, damage critical infrastructure, and strain resources. Some of the key risks to state operations and facilities include:

- **Emergency Response and Public Safety:** Storms can lead to a surge in emergency calls for services related to accidents, medical emergencies, and stranded motorists.
- **Emergency Operations:** State agencies involved in emergency response, such as law enforcement, fire departments, and emergency medical services, may need to mobilize additional resources to handle storm-related incidents.
- **Infrastructure Damage:** Strong winds, heavy rainfall, and flooding can lead to structural damage to state buildings and facilities, including offices, shelters, and emergency operation centers.
- **Transportation Infrastructure:** Roads, bridges, and airports may require repair and maintenance due to the impact of storm-related hazards.

- **Power Outages:** Storms can cause power outages due to downed power lines, transformer failures, and flooding. State assets, including emergency operations centers and healthcare facilities, may rely on backup power sources.
- **Water and Gas Supply:** Utility services such as water and natural gas may be disrupted, affecting state buildings and services.
- **Emergency Response Costs:** The costs associated with emergency response efforts, including search and rescue, debris removal, and disaster recovery, can strain state budgets.
- **Economic Impact:** Storms can result in economic losses due to reduced economic activity, property damage, and increased spending on emergency response and recovery efforts.
- **Interruption of Communication Networks:** Storms, especially those with strong winds, can disrupt communication networks, hindering the ability of state agencies to communicate internally and with the public.
- **Data Loss:** Power outages and equipment failures can result in data loss, affecting state operations and data integrity.
- **Service Interruptions:** Storm-related disruptions can lead to interruptions in state provided services, impacting citizens' access to critical services and information.
- **Delayed Operations:** Closure or reduced staffing of state offices can delay administrative functions, regulatory processes, and public services.

The following table details information concerning the number of critical and state-owned or operated facilities in potential storm surge areas:

Table 135: Critical and State Assets in Hurricane Storm Surge Areas

Facility Type	Storm Surge by Hurricane Category Level				
	1	2	3	4	Total
Fire Stations	1	2	4	7	14
Law Enforcement	0	0	1	6	7
Power Plants	0	1	0	3	4
Natural Gas Pipelines	4	5	4	5	18
Electric Transmission Facilities	18	25	27	44	114
Airports	3	6	3	4	16
Bus Stops	64	100	101	466	731
Wastewater Facilities	0	2	1	0	3
Hazardous Material/Tier II Facilities	14	37	24	62	137
Flood Control Dams	4	8	11	14	37
FDIC Insured Banks	1	5	8	21	35
Schools	3	5	5	19	32
State Facilities	3	2	4	34	43
Hospitals	0	0	0	1	1
Nursing Facility	1	1	2	1	5
Communications Network Facilities	0	0	0	0	0
Total	116	199	195	687	1,197

Source: RIEMA and RIGIS

State owned facilities in identified storm surge areas were matched, when possible, to the master state facility list for valuation purpose. Some data discrepancies were discovered during this process causing a valuation estimate for some facilities based on the best available information. The following table present the valuation of the state facilities identified in storm surge areas:

Table 136: Valuation of State of Rhode Island Facilities in Storm Surge Areas

Jurisdiction	Valuation
Bristol	\$1,030,000
Kent	\$3,798,709

Table 136: Valuation of State of Rhode Island Facilities in Storm Surge Areas

Jurisdiction	Valuation
Newport	\$1,868,147
Providence	\$142,973,544
Washington	\$19,366,393
State Total	\$169,036,793

Source: RIEMA and RIGIS

Note: Valuations include structure and assets

Consequence Analysis

This consequence analysis lists the potential impacts of this hazard on various elements of community and state infrastructure. The impact of this hazard is evaluated in terms of disruption of operations, recovery challenges, and overall wellbeing to all Rhode Island residents and first responder personnel. The consequence analysis supplements the hazard profile by analyzing specific impacts.

Table 137: Tropical and Extratropical Storm Consequence Analysis

Subject	Potential Impacts
Impact on the Public	Immediate damage to homes, businesses, infrastructure, government facilities, and roadways would cause major disruptions in response operations. High winds, flooding, and flying debris may cause injury or death. Hazardous waste and material may be introduced into flooding waters, contaminating water supplies as well as standing water. Disease vectors such as mosquitos may cause an increase in transmissible disease.
Impact on Responders	Coordinating an evacuation in advance of a significant tropical and extratropical storm event requires enhanced response coordination and causes a substantial strain on resources. First responders may face flooding, high winds, and storm surge, which may lead to injury or death. Critical roadways and response facilities may flood, lose power, or become damaged or destroyed. Response equipment and vehicles may be damaged further complicating response and recovery operations.
Continuity of Operations	Rhode Island agencies maintain continuity plans which can be enacted as necessary based on the situation. Tropical and extratropical storms may impact an agency's ability to maintain continuity of operations based on storm produced flooding and high winds which may impact a government building or agency directly through damage or destruction. Continuity of operations could be impacted due to this damage and the agency would need to relocate to their alternate facility. Damage or destruction to primary and alternate facilities could also affect operations.
Delivery of Services	Tropical and extratropical storms can cause road and bridge closures and transit disruptions. Many businesses may shut down to prepare safely for the storm. Impacting the delivery of goods and services. Goods and delivery vehicles may become damaged, destroyed, or inoperable due to event impacts.
Property, Facilities, and Infrastructure	Tropical and extratropical storms can cause widespread destruction to property, facilities, and infrastructure. Large coastal storms can also cause power outages and disrupt transportation and communications infrastructure, such as roads, bridges, telecommunications towers, and internet connectivity.
Impact on Environment	Hurricanes and tropical storms can cause beach erosion and sea level rise, both of which negatively impact the environment in the long and short term. Strong winds may be generated that completely defoliate forest canopies and cause structural changes in wooded ecosystems. Changes in habitat and food availability due to flooding, winds, and storm surge can heavily impact the health of animals, as well as cause death.
Economic Conditions	A major tropical and extratropical storm could cause damage to beaches, historical sites, and other areas disrupting state tourist activity, a major economic driver. In addition, the costs associated with response and recovery, although reimbursable during a federal declaration, are significant and can have cascading impacts on the state economy at large.

Table 137: Tropical and Extratropical Storm Consequence Analysis

Subject	Potential Impacts
Public Confidence in Governance	Ineffective response both before and after a tropical and extratropical storm can decrease the public's confidence in the state's ability to respond and govern. Governmental response across local, state, regional, and federal levels require direct actions that must be immediate and effective to maintain public confidence.

Using GIS, storm surge areas were overlaid on the critical facility and state facility location data. Maps are included in Appendix D.

4.18.7 Jurisdictional Vulnerability and Planning Significance

Utilizing data from FEMA's NRI and from LHMPs, the following tables detail the hazard ratings for jurisdictions, by census tract, for severe winter weather. Data concerning both risk index and EAL is reported by census tract. As many State of Rhode Island jurisdictions encompass multiple tracts, an average of tract ratings was completed to determine the overall jurisdictional rating. The following table details the hazard rating by LHMP analysis, FEMA NRI, and EAL by jurisdiction for the State of Rhode Island:

Table 138: State of Rhode Jurisdictional Tropical/Extratropical Storm Event Planning Significance

County	Jurisdiction	Tropical and Extratropical LHMP Rating	Tropical and Extratropical NRI	Tropical and Extratropical EAL
Bristol	Barrington	High	Relatively Moderate	Relatively Moderate
	Bristol	-	Relatively Moderate	Relatively Moderate
	Warren	High	Relatively Moderate	Relatively Moderate
Kent	Coventry	-	Relatively Moderate	Relatively Moderate
	East Greenwich	High	Relatively Moderate	Relatively Moderate
	Warwick	High	Relatively Moderate	Relatively Moderate
	West Greenwich	-	Relatively Moderate	Relatively Moderate
	West Warwick	High	Relatively Moderate	Relatively Moderate
Newport	Jamestown	-	Relatively Moderate	Relatively Moderate
	Little Compton	Moderate	Relatively Moderate	Relatively Moderate
	Middletown	High	Relatively Moderate	Relatively Moderate
	Newport	High	Relatively Moderate	Relatively Moderate
	Portsmouth	High	Relatively Moderate	Relatively Moderate
	Tiverton	-	Relatively Low	Relatively Moderate
Providence	Burrillville	High	Relatively Moderate	Relatively Moderate
	Central Falls	High	Relatively Moderate	Relatively Low
	Cranston	High	Relatively Moderate	Relatively Moderate
	Cumberland	-	Relatively Moderate	Relatively Moderate
	East Providence	Moderate	Relatively Moderate	Relatively Moderate
	Foster	High	Relatively Moderate	Relatively Moderate
	Glocester	Moderate	Relatively Moderate	Relatively Moderate
	Johnson	Moderate	Relatively High	Relatively High
	Lincoln	High	Relatively Moderate	Relatively Moderate
	North Providence	Moderate	Relatively Moderate	Relatively Low
	North Smithfield	Moderate	Relatively Moderate	Relatively Moderate
	Pawtucket	High	Relatively Low	Relatively Low
	Providence	High	Relatively Moderate	Relatively Low
	Scituate	-	Relatively Moderate	Relatively Moderate
	Smithfield	-	Relatively Moderate	Relatively Moderate
	Woonsocket	Moderate	Relatively Low	Relatively Low
Washington	Charlestown	-	Relatively Moderate	Relatively Moderate
	Exeter	-	Relatively Moderate	Relatively Moderate
	Hopkinton	-	Relatively Moderate	Relatively Moderate
	Narragansett	Moderate	Relatively Moderate	Relatively Moderate
	New Shoreham	-	Relatively Moderate	Relatively Moderate

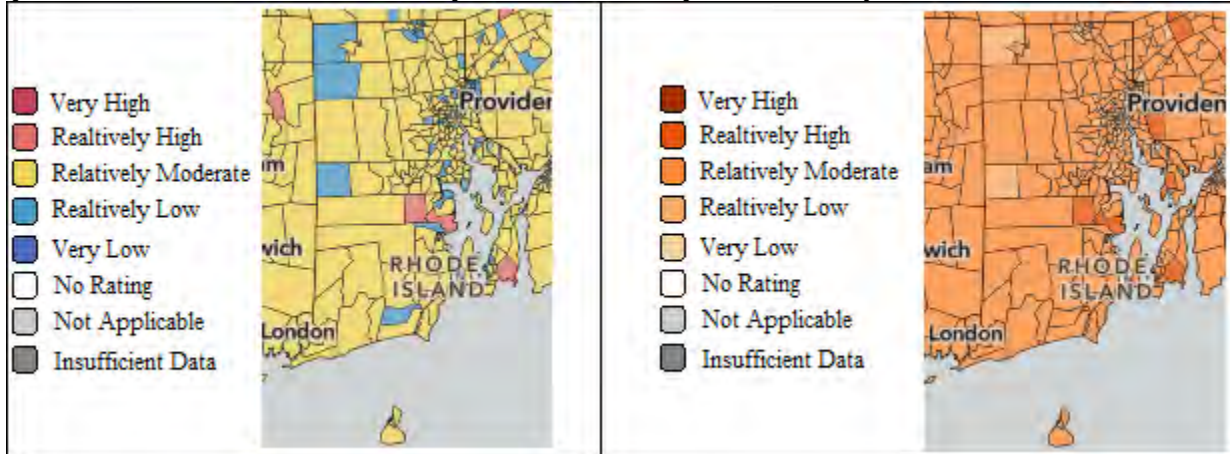
Table 138: State of Rhode Jurisdictional Tropical/Extratropical Storm Event Planning Significance

County	Jurisdiction	Tropical and Extratropical LHMP Rating	Tropical and Extratropical NRI	Tropical and Extratropical EAL
	North Kingstown	Moderate	Relatively Moderate	Relatively Moderate
	Richmond	High	Relatively Moderate	Relatively Moderate
	South Kingstown	High	Relatively Moderate	Relatively Moderate
	Westerly	Moderate	Relatively Moderate	Relatively Moderate

Source: LHMPs and FEMA NRI

Note: - indicates no current plan available for review

FEMA NRI was utilized to determine the risk and expected annual loss due to tropical and extratropical storms by census tract for the state of Rhode Island. The following maps detail these parameters:

Map 63: FEMA NRI and EAL for Tropical and Extratropical Storms by Census Tract for Rhode Island

Source: FEMA NRI

Using GIS, storm surge areas were overlaid on the critical facility and state facility location data. Maps are included in Appendix D.

Of particular concern when considering housing data is manufactured buildings (often referred to as mobile homes). Data from the NOAA National Severe Storms Laboratory reports that people living in mobile homes are especially at risk for injury and death as even anchored mobile homes can be seriously damaged when winds gust over 80 miles per hour. Additionally, study data from Michigan State University reported that the two biggest factors related to wind event fatalities were housing quality (measured by mobile homes as a proportion of housing units) and income level. When a strong wind strikes, a county with double the number of mobile homes as a proportion of all homes will experience 62% more fatalities than a county with fewer mobile homes, according to the study data. The following indicates the percentage of mobile homes for each Rhode Island County:

Table 139: Rhode Island 2020 Census Mobile Home Data

Jurisdiction	Number of Mobile Homes	Percentage Of Housing Stock as Mobile Homes
Bristol County	0	0%
Kent County	902	1.2%
Newport County	701	1.6%
Providence County	1,059	0.4%
Washington County	793	1.2%
State of Rhode Island	3,455	0.7%

Source: United States Census Bureau

Based on FEMA NRI social vulnerability ratings, and on U.S. Census Bureau data concerning percentage of the population below the 100% poverty who may have difficulties in evacuating before an event and/ or repairing their residence, the following communities may be at greater risk to tropical and extratropical storm events:

Table 140: Jurisdictions at Greater Risk to Tropical and Extratropical Storms due to Socioeconomic Conditions

County	Jurisdiction	Population Below 100% of the Poverty Level	Social Vulnerability Rating
Providence County	Central Falls	29.9%	Very High
Providence County	Providence	23.0%	Relatively High
Providence County	Woonsocket	21.0%	Relatively Moderate
Washington County	Narragansett	19.3%	Relatively High
Newport County	Newport	14.4%	Relatively Low
Kent County	West Warwick	12.1%	Relatively Low
Providence County	Pawtucket	11.6%	Relatively High

Source: FEMA NRI and U.S. Census Bureau

A disproportionate impact from tropical and extratropical storms may be felt by jurisdictions with older housing units as they may have been constructed using materials and techniques that are less resilient to strong winds and flooding. Modern construction materials and methods, such as reinforced concrete and steel frames, are often more resilient. The following table Rhode Island Department of Housing 2022 Integrated Housing Report from the Rhode Island Department of Housing 2022 Integrated Housing Report indicates jurisdictions with greater than 40% of their housing units constructed before 1939:

Table 141: Jurisdictions at Greater Risk to Tropical and Extratropical due to Age of Housing

County	Jurisdiction	1939 and later
Providence	Providence	57.70%
Newport	Newport	55.20%
Providence	Central Falls	50.40%
Providence	Woonsocket	43.50%
Providence	Pawtucket	43.30%
Bristol	Warren	41.40%

Source: Rhode Island Department of Housing 2022 Integrated Housing Report

A household is considered cost burdened if it spends 30% or more of its income on housing costs, with cost burdens being an indication of rents increasing faster than incomes. Cost burdened citizens may have great difficulty evacuating in the event of a tropical or extratropical storm due to economic constraints. The following table, using data from the Housing and Urban Development's Comprehensive Housing Affordability Strategy Data for Rhode Island, indicates jurisdictions located in coastal communities with the most cost burdened renters who may be at greater risk to due to economic constraints.

Table 142: Jurisdictions at Greater Risk to Tropical and Extratropical Storms due to Housing Cost Burden

County	Jurisdiction	Cost Burdened Single Families
Washington County	Narragansett	65%
Bristol County	Barrington	64%
Newport County	Portsmouth	54%
Bristol County	Warren	52%
Kent County	Warwick	50%
Providence County	East Providence	49%
Newport County	Jamestown	47%
Washington County	New Shoreham	47%
Providence County	Providence	47%
Washington County	Westerly	47%

Source: Housing and Urban Development's Comprehensive Housing Affordability Strategy Data

4.18.8 Hazard Planning Significance

Based on a review of available data, the SIHMC determined tropical and extratropical storms to have a high planning significance for Rhode Island.

4.19 Wildfire

4.19.1 Hazard Description

The NWS defines a wildfire as any free burning uncontrollable wildland fire not prescribed for the area which consumes the natural fuels and spreads in response to its environment. They can occur naturally, by human accident, and on rare occasions by human action. Population de-concentration in the U.S. has resulted in rapid development in the outlying fringe of metropolitan areas and in rural areas with attractive recreational and aesthetic amenities, especially forests. This expansion has increased the likelihood that wildfires will threaten life and property.



According to the National Park Service there three classifications of wildfires:

- **Surface Fire:** Burning which may spread rapidly and ignite leaf litter, fallen branches and other fuels located at ground level.
- **Ground Fire:** Burning of organic matter in the soil beneath the surface.
- **Crown Fire:** Burning through the top layer (canopy) of trees. Crown fires, which can be very intense and difficult to contain, require strong winds, steep slopes, and large amounts of fuel to burn.

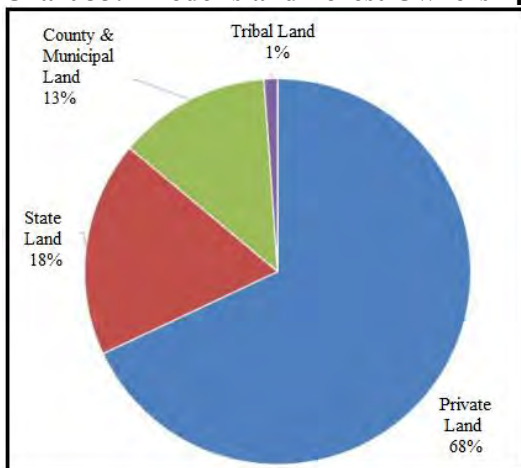
Wildfires are strongly influenced by multiple factors, including:

- **Weather:** Factors such as relative humidity, wind speed, ambient temperature and precipitation all influence the formation and growth of wildfires.
- **Topography:** Natural features, such as canyons or ridges, can increase the spread rate of a fire by funneling or drawing heated air and fire.
- **Fuel Type, Distribution and Moisture:** Available fuels, the spacing and density of available fuels, and fuel moisture content can determine spread rates and intensity of wildfires.
- **Drought Conditions:** Drought tends to increase both the likelihood and severity of wildfires.

4.19.2 – Location and Extent

Rhode Island's forests are owned and managed by a combination of federal agencies and programs, state agencies and programs, national and local land trusts and other conservation organizations, and private landowners. The 2020 State and Private Forestry Fact Sheet for Rhode Island estimates that approximately 68% of forest land is privately owned and managed by an estimated 38,000 landowners, including conservation organizations and nonprofits. The following chart details forest ownership within the State:

Chart 33: Rhode Island Forest Ownership



Source: 2020 State and Private Forestry Fact Sheet

RIDEM permanently protects 73,324 acres of forest land, owning 47,384 acres of forest land in fee, and holding additional interests on 25,940 acres through conservation easements, deeds to development rights, and recreation easements.

The wildland/urban interface (WUI) is the area where human improvements such as homes, ranches and farms come in contact with the wildlands. The WUI creates an environment in which fire can move readily between structure and vegetation fuels, often resulting in massive fires, or conflagrations, which may lead to widespread evacuations. The expansion of the WUI in recent decades has significant implications for wildfire management and its impact. There are two types of WUI, intermixed and interface. Intermix WUI are areas where housing and vegetation intermingle, and interface WUI are areas with housing in the vicinity of dense, contiguous wildland vegetation.

While Rhode Island is the 2nd-most densely populated state per capita, the western part of the state is still largely forested. Although parcel size is typically small, and tracts of forest land are surrounded by non-forest uses, the continued encroachment into the forested areas also expands the WUI.

Based on available mapping, the following table presents the amount of each Rhode Island County in WUI classified areas.

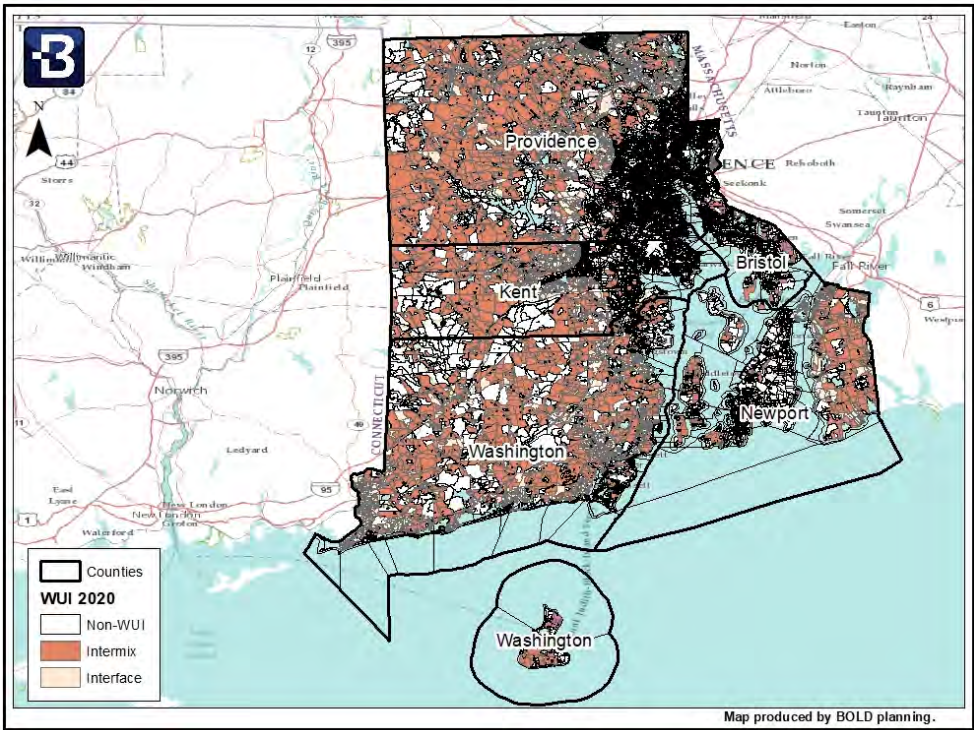
Table 143: Rhode Island WUI Area

Jurisdiction	Intermix (Square Miles)	Interface (square miles)	Total (square miles)
Bristol County	2.85	0	24.72
Kent County	78.85	8.49	173.96
Newport County	41.02	8.27	107.49
Providence County	213.48	34.57	429.95
Washington County	185.77	41.13	341.86
State Total	521.97	92.46	1077.98

Source: RIEMA

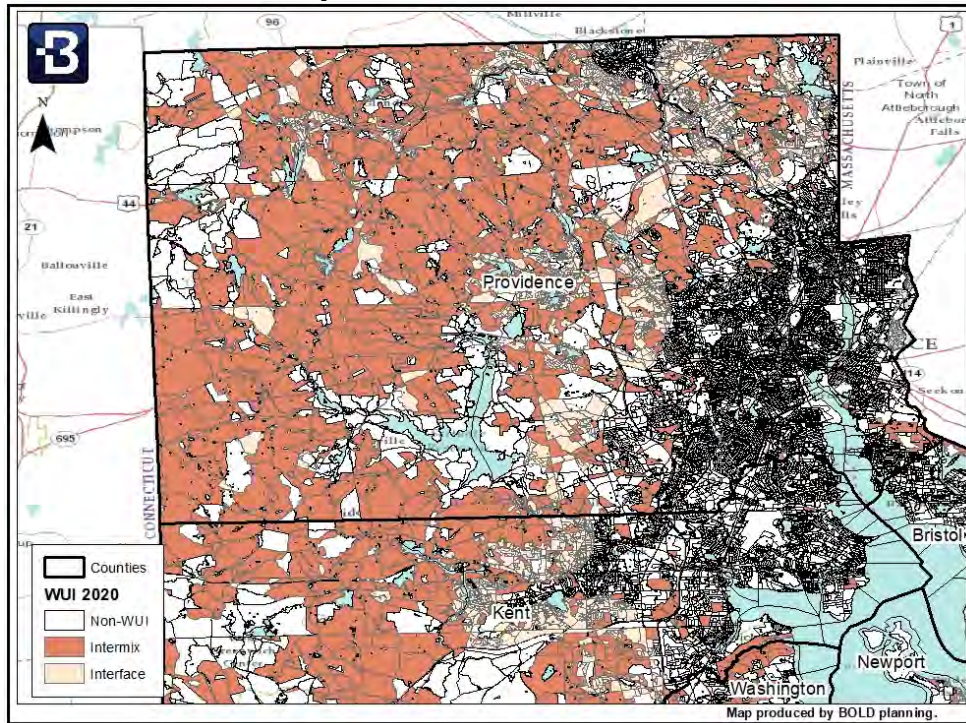
The following map illustrates WUI areas throughout the State:

Map 64: Rhode Island WUI Areas



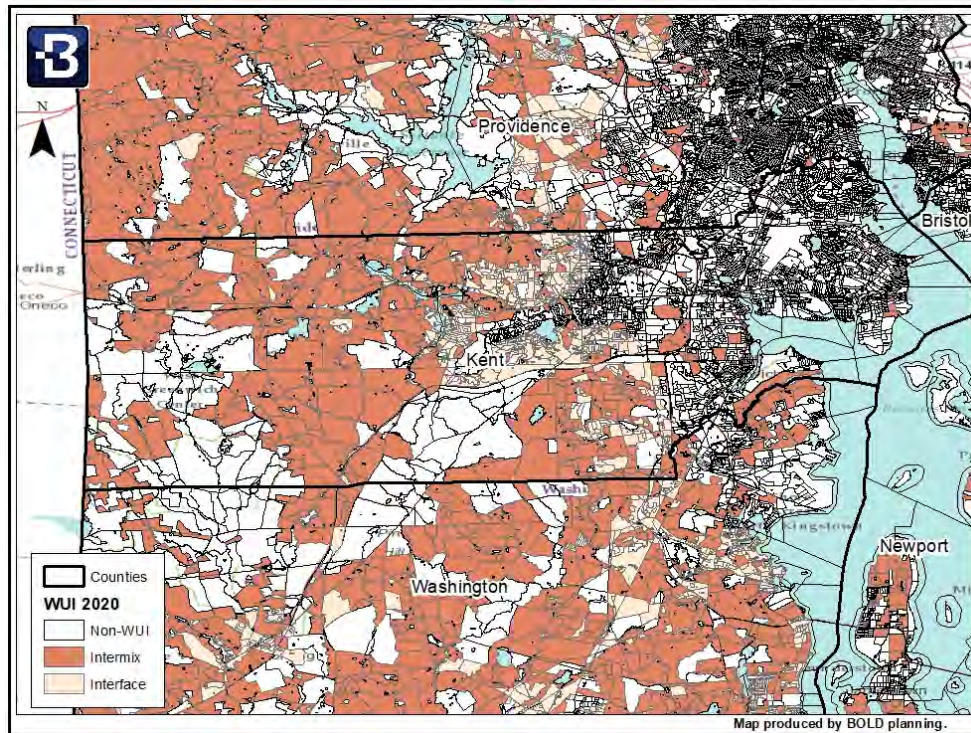
Source: RIEMA and RIGIS

Map 65: Rhode Island WUI Areas



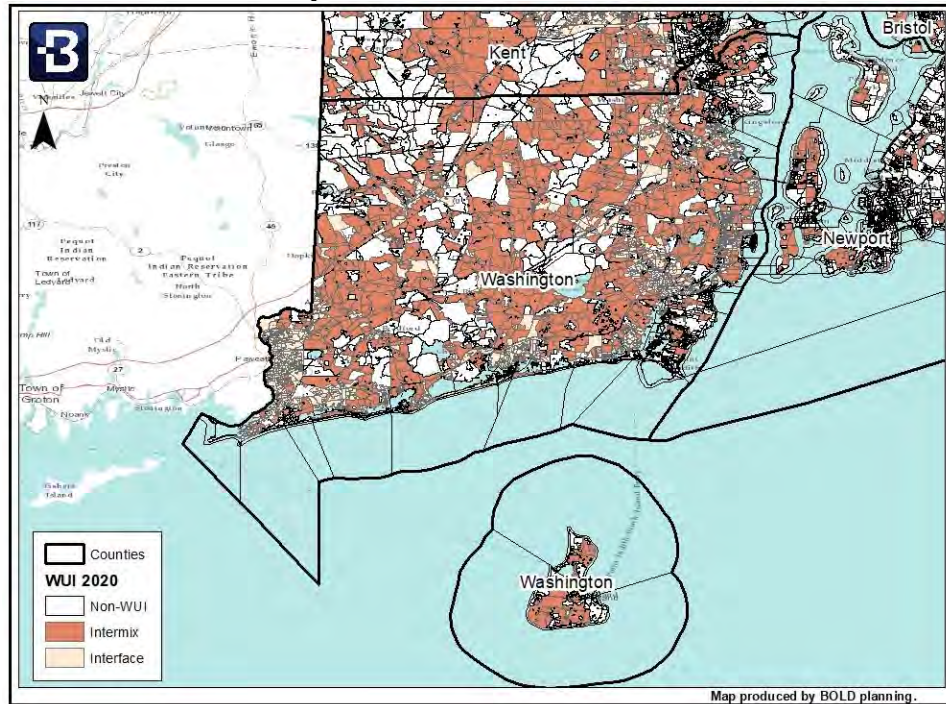
Source: RIEMA and RIGIS

Map 66: Rhode Island WUI Areas



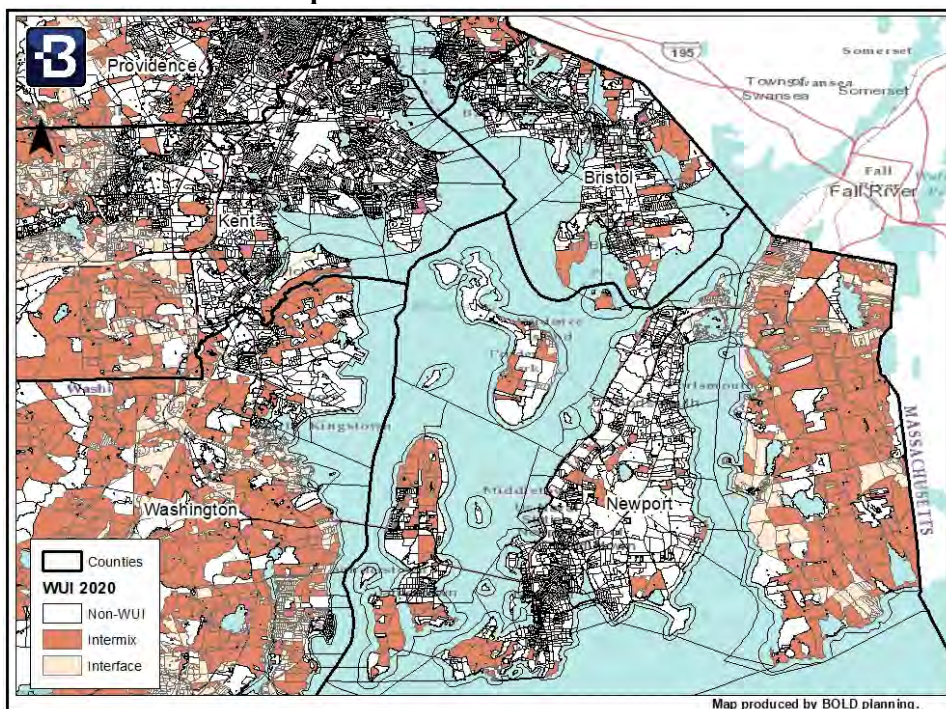
Source: RIEMA and RIGIS

Map 67: Rhode Island WUI Areas



Source: RIEMA and RIGIS

Map 68: Rhode Island WUI Areas

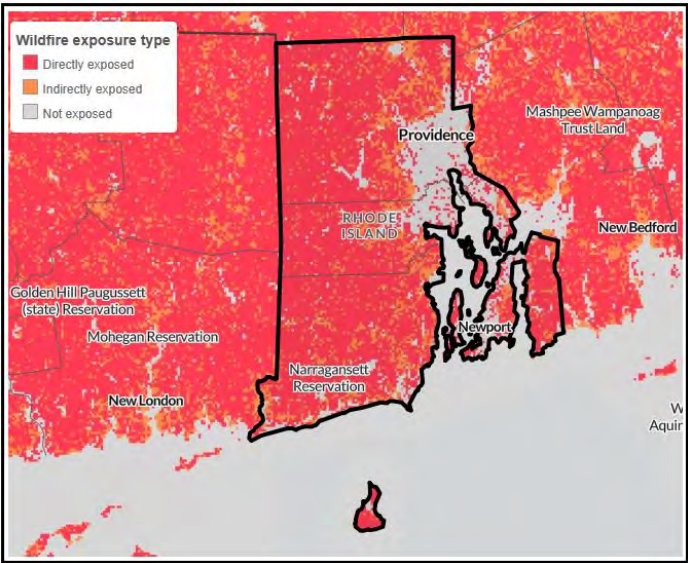


Source: RIEMA and RIGIS

Exposure is the intersection of wildfire likelihood and intensity with communities. Communities can be directly exposed to wildfire from adjacent wildland vegetation, or indirectly exposed to wildfire from embers and home-to-home ignition. Communities that are not exposed are not likely to be subjected to wildfire from either direct or indirect sources. Wildfire exposure is calculated based on wildfire likelihood and proximity to large areas of flammable wildland vegetation. Any community that is located where there is a chance wildfire could occur (in other words, where wildfire likelihood is greater than zero) is exposed to wildfire. Directly exposed homes are located in an area considered to be covered by flammable wildland vegetation. Indirectly exposed homes are located within one mile of a large area

considered to be covered by flammable wildland vegetation. Non-exposed homes are located more than one mile from a large area considered to be covered by flammable wildland vegetation. The following map, from NOAA’s Wildfire Risk to Communities, indicates the wildfire exposure for Rhode Island:

Map 69: Rhode Island Wildfire Exposure Type



Source: NOAA’s Wildfire Risk to Communities

The duration of a wildfire depends on the weather conditions, how dry it is, the availability of fuel to spread, and the ability of responders to contain and extinguish the fire. Historically, some wildfires have lasted only hours, while other fires have continued to spread and grow for an entire season. They spread quickly and often begin unnoticed until they have grown large enough to signal by dense smoke. If fuel is available, and high wind speeds hit, a wildfire can spread over a large area in a very short amount of time. These factors make the difference between small upstart fires easily controlled by local fire services to fires destroying thousands of acres requiring multiple state and federal assets for containment and suppression.

The National Fire Danger Rating System allows fire managers to estimate today's or tomorrow's fire danger for a given area. It combines the effects of existing and expected states of selected fire danger factors into one or more qualitative or numeric indices that reflect an area's fire protection needs. It links an organization's readiness level (or pre-planned fire suppression actions) to the potential fire problems of the day. The following is a brief explanation of the different fire danger levels based on criteria established by the National Fire Danger Rating System.

Table 144: National Fire Danger Rating System

Rating	Description
Low	Fuels do not ignite easily from small embers, but a more intense heat source, such as lightning, may start fires in duff or dry rotten wood. Fires in open, dry grasslands may burn easily a few hours after a rain, but most wood fires will spread slowly, creeping or smoldering. Control of fires is generally easy.
Moderate	Fires can start from most accidental causes, but the number of fire starts is usually pretty low. If a fire does start in an open, dry grassland, it will burn and spread quickly on windy days. Most wood fires will spread slowly to moderately. Average fire intensity will be moderate except in heavy concentrations of fuel, which may burn hot. Fires are still not likely to become serious and are often easy to control.
High	Fires can start easily from most causes and small fuels (such as grasses and needles) will ignite readily. Unattended campfires and brush fires are likely to escape. Fires will spread easily, with some areas of high intensity burning on slopes or concentrated fuels. Fires can become serious and difficult to control unless they are put out while they are still small.
Very High	Fires will start easily from most causes. The fires will spread rapidly and have a quick increase in intensity, right after ignition. Small fires can quickly become large fires and

Table 144: National Fire Danger Rating System

Rating	Description
	exhibit extreme fire intensity, such as long-distance spotting and fire whirls. These fires can be difficult to control and will often become much larger and longer-lasting fires.
Extreme	Fires of all types start quickly and burn intensely. All fires are potentially serious and can spread very quickly with intense burning. Small fires become big fires much faster than at the "very high" level. Spot fires are probable, with long-distance spotting likely. These fires are very difficult to fight and may become very dangerous and often last for several days.

Source: Wildfire Fire Assessment System

The severity of wildfire depends on several quickly changing environmental factors. It is impossible to strategically estimate the severity of a wildfire as these factors, including drought conditions and wind speed, have such a great influence on the wildfire conditions. The Characteristic Fire Intensity Scale within the Southern Wildfire Risk Assessment Summary Report specially identifies areas where significant fuel hazards and associated dangerous fire behavior potential exist.

The following table details the range of wildfire intensity:

Table 145: Characteristic Fire Intensity Scale

Class	Description
Class 1 - Very Low	Very small, discontinuous flames, usually less than 1 foot in length; very low rate of spread; no spotting. Fires are typically easy to suppress by firefighters with basic training and non-specialized equipment.
Class 2 - Low	Small flames, usually less than two feet long; small amount of very short-range spotting possible. Fires are easy to suppress by trained firefighters with protective equipment and specialized tools.
Class 3 - Moderate	Flames up to 8 feet in length; short-range spotting is possible. Trained firefighters will find these fires difficult to suppress without support from aircraft or engines, but dozer and plows are generally effective. Increasing potential for harm or damage to life and property.
Class 4 - High	Large Flames, up to 30 feet in length; short-range spotting common; medium range spotting possible. Direct attack by trained firefighters, engines, and dozers are generally ineffective, indirect attack may be effective. Significant potential for harm or damage to life and property
Class 5 - Very High	Very large flames up to 150 feet in length; profuse short-range spotting, frequent long-range spotting; strong fire-induced winds. Indirect attack marginally effective at the head of the fire. Great potential for harm or damage to life and property.

Source: Southern Wildfire Risk Assessment Summary Report

Urban fires can occur in any building, facility, or infrastructure across the state. Small urban fires occur routinely and are resolved quickly by first responders.

4.19.3 Previous Occurrences

FEMA can approve declarations for fire management assistance when the Administrator determines that a fire or fire complex on public or private forest land or grassland threatens such destruction as would constitute a major disaster. The following table details the one fire management declaration for the State of Rhode Island.

Table 146: State of Rhode Island Fire Management Declarations

Designation	Declaration Date	Incident Type
FM-5464-RI	04/14/2023	Brushfire

Source: FEMA

The following provides information on the declaration:

- **FM-5464-RI:** A brush fire scorched between 575-600 acres in the Queen River Nature Preserve. The fire forced the temporary evacuation of nearby residences, and more than 200 firefighters from departments as far as

Central Connecticut worked for hours to get the fire under control. The Rhode Island National Guard's Black Hawk team was also called in to air drop water onto the flames. The brush fire is the largest the state has seen since 1942, but no deaths, injuries, or property damage were recorded.

Other large wildfires of note include:

- **Coventry Fire:** On April 30, 1942, the largest wildfire in Rhode Island occurred. The fire burned over 24,000 acres and 180 people were injured.
- **Wood River Fire:** On May 2, 1951, this fire burned 7,400 acres and caused one fatality.

Wildfires are a frequent occurrence in Rhode Island but are generally small and quickly contained. The worst year for wildfires was 1930, when 37,400 acres burned. Recent fire occurrences have burned a much smaller acreage due to quicker response times, better spotting practices, and stronger forest management policies. The following table details recent Rhode Island wildfire information:

Table 147: Rhode Island Wildfires 2000- 2023

Year	Number of Events	Acres Burned
2000	110	211
2001	202	241
2002	128	213
2003	83	90
2004	66	95
2005	90	100
2006	113	96
2007	105	38
2008	143	96
2009	54	60
2010	50	31
2011	38	29
2012	55	42
2013	46	40
2014	51	54
2015	95	133
2016	83	57
2017	35	31
2018	32	14
2019	43	31
2020	115	86
2021	95	188
2022	88	51
2023	74	582

Source: Rhode Island Department of Environmental Management

The following table details Rhode Island wildfire information by county for the period 2019 through 2023 (year to date data):

Table 148: Rhode Island Wildfires 2019 – 2023*

Jurisdiction	Number of Events	Acres Burned	Structures Burned	Injuries	Fatalities
Bristol County	8	5.7	0	0	0
Kent County	99	547.3	3	5	0
Newport County	10	61.8	4	1	0
Providence County	172	232.9	7	1	0

Table 148: Rhode Island Wildfires 2019 – 2023*

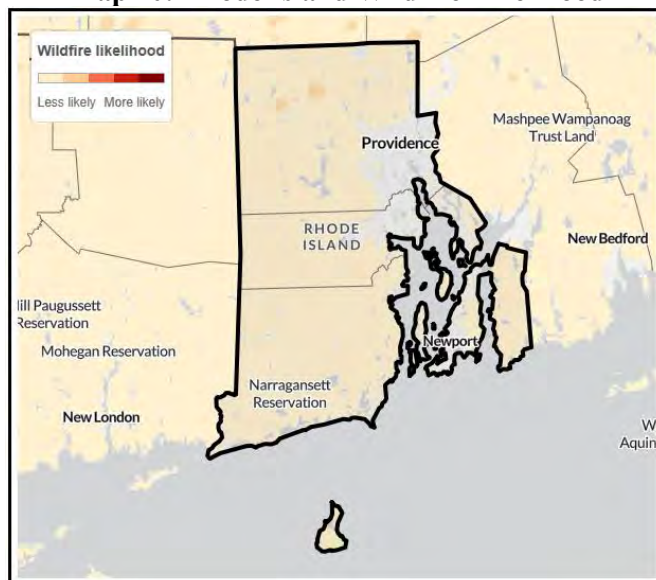
Jurisdiction	Number of Events	Acres Burned	Structures Burned	Injuries	Fatalities
Washington County	118	81.4	13	6	0
State Total	407	929.1	27	13	0

Source: Rhode Island Department of Environmental Management

Note: *: Year to date data

4.19.4 Probability of Future Events

Predicting the probability of wildfire occurrences is tremendously challenging due to the large number of factors involved and the random nature of formation. NOAA's Wildfire Risk to Communities mapping, which uses the best available science to identify risk, was used to help determine the probability of future wildfires within the State. As indicated on the following map, the likelihood of a wildfire within the State of Rhode Island is considered less likely:

Map 70: Rhode Island Wildfire Likelihood

Source: NOAA's Wildfire Risk to Communities

4.19.5 Projected Changes in Location, Intensity, Frequency, and Duration

Climate change can result in a significant increase in the likelihood and severity of wildfires. The occurrence of more frequent and longer lasting droughts due to climate change can increase the availability of fuels for wildfires through the drying of vegetation. Additionally, both the increased occurrence and continued decline of native species due to lack of precipitation can cause the proliferation of invasive species which can provide quick-burning fuels that contribute to the start and spread of fire.

Climate change may impact the frequency and magnitude of wildfires in the following ways:

- **Increased Frequency:** Warmer temperatures and prolonged periods of drought associated with climate change create conditions that favor more frequent wildfires. Extended fire seasons are becoming the new norm in many regions.
- **Greater Intensity:** Higher temperatures and drier conditions can lead to more intense wildfires. These fires burn hotter and spread more rapidly, making them more challenging to control and extinguish.
- **Longer Fire Seasons:** Climate change is extending the length of fire seasons, leading to earlier starts and later endings. This puts additional stress on firefighting resources and increases the risk of wildfires overlapping with other disasters.
- **Altered Precipitation Patterns:** Changes in precipitation patterns, including more intense rainfall events followed by extended dry periods, can promote the growth of vegetation, which can then become fuel for wildfires during subsequent dry periods.

- **Drought Conditions:** Prolonged droughts associated with climate change reduce soil moisture levels and the availability of water sources. Dry conditions increase the susceptibility of vegetation to ignition.
- **Vegetation Changes:** Climate change can alter the distribution and composition of vegetation, such as the expansion of drought-tolerant species. This can change fuel availability and make ecosystems more fire prone.
- **Insect Infestations:** Warmer temperatures can lead to increased insect infestations in forests. Infested and dead trees provide additional fuel for wildfires.
- **Wildfire Behavior:** Climate change can lead to changes in wildfire behavior, including more extreme fire events and increased spotting (the spread of embers ahead of the main fire).

The Department of Environmental Management's Division of Agriculture and Forest Environment shares responsibility with the Division of Fish & Wildlife for over 40,000 acres of state-owned forest land. To help reduce future occurrences of wildfires, management practices include forest thinning to prevent crowding of trees that can make them more susceptible to diseases and pests, as trees killed by disease and pest may potentially be more prone to ignition, and prescribed fire to reduce the growth of invasive plants and other competing vegetation. The Rhode Island Department of Environmental Management has announced it will ramp up its use of prescribed burns on state-owned lands in the future.

In an effort to minimize future wildfires, four areas under Department of Environmental Management oversight have been prioritized to see near future controlled burns:

- Dutch Island in Jamestown
- Nicholas Farm Management Area in Coventry
- Pratt Farm, part of the Arcadia Management Area in Exeter
- Prudence Island in Portsmouth

A State of Rhode Island Commission has been convened to "formulate and put into effect a forest fire plan and integrate the forest fire plan with the regional plan recommended by the compact commission. The plan shall include a reasonable effort to establish and maintain any adequate forest fire fighting services that are consistent with the requirements of the state and to make available those services to other signatory states consonant with maintaining reasonably safe protection to this state at all times."

As previously noted, State of Rhode Island facilities have seen no major changes in the past five years, with only modest repairs and upgrades being conducted and no major rehabilitation or construction projects completed. As such, the risk to state assets has remained static since the completion of the 2019 SHMP. Additionally, with the exception of the two previously identified rehabilitation and construction projects, no large-scale facility projects were noted in any agency budgets or reports.

4.19.6 Vulnerability and Impact

People and Communities

Wildfires can have significant and often devastating impacts on people and communities. These impacts can vary depending on the wildfire's intensity, size, path, and the preparedness of the affected area, and may include.

- **Injuries and Fatalities:** Wildfires can lead to injuries and fatalities among residents, firefighters, and emergency responders due to burns, smoke inhalation, and accidents during firefighting efforts.
- **Evacuations and Displacement:** Wildfire damage can force people to leave their homes, leading to temporary or even long-term displacement. Some may require emergency shelter and assistance from relief organizations.
- **Property Loss:** Wildfires can cause extensive property damage to homes, businesses, and vehicles.
- **Health Risks:** Smoke from wildfires can contain harmful pollutants, including fine particulate matter and toxic gases, which can lead to respiratory problems and exacerbate pre-existing health conditions. Vulnerable populations, such as children and the elderly, are at higher risk.
- **Mental Health Impact:** The trauma and stress associated with experiencing a wildfire, evacuations, property loss, and the challenges of recovery can have a significant impact on mental health, including anxiety, depression, and post-traumatic stress disorder.

- **Emergency Response Challenges:** Wildfires can strain emergency response resources, including firefighting personnel, equipment, and medical facilities. First responders may be faced with many emergency calls.
- **Economic Costs:** Wildfires result in economic costs, including property damage and insurance claims.

Of particular concern, socially vulnerable and at-risk populations that may have difficulty with medical issues, poverty, extremes in age, and communications due to language barriers are often at increased susceptibility to the impacts of hazards. These populations suffer a disproportionate impact due to:

- Constrained financial resources, causing evacuation difficulties.
- Susceptibility to impacts of infrastructure failure hampering transportation and evacuation.
- Susceptibility to health effect of smoke.
- Limited access to healthcare.
- Limited access to information due to language difficulties.

The following table details identified socially vulnerable and at-risk populations for the State of Rhode Island:

Table 149: Rhode Island 2020 Vulnerable Populations

Jurisdiction	Under Five	Over 65	Language Barrier	Poverty	Disability
Bristol County	1,976	8,372	6,552	3,403	4,753
Kent County	8,148	26,437	16,525	11,925	24,694
Newport County	3,641	13,819	7,794	8,307	9,105
Providence County	36,846	84,768	205,490	90,522	87,498
Washington County	4,760	18,797	7,790	11,815	13,832
State of Rhode Island	55,370	152,183	240,326	125,101	139,927

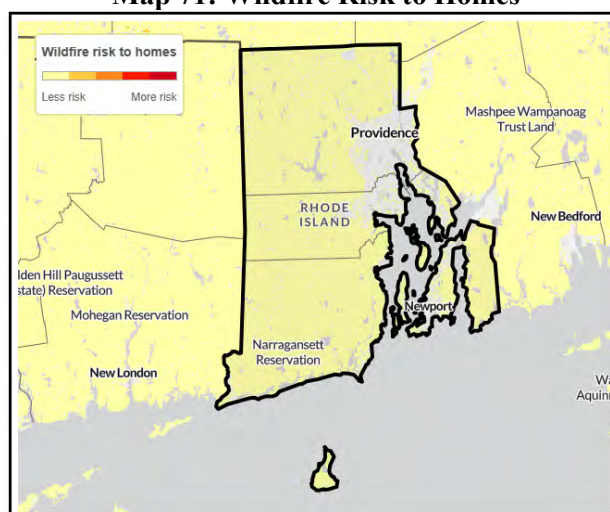
Source: US Census Bureau

Facilities and Critical Facilities

Wildfires can devastate communities and homes, causing various types of property damage including burning structures, charring of exterior surfaces, and damage to roofs, walls, and windows. The heat generated by wildfires can weaken or melt building materials. In extreme cases, wildfires can destroy homes, reducing them to ashes and rubble. Homes that may not have been directly impacted by the fire may also be affected. Wildfires can damage utility infrastructure, including power lines and gas pipelines, leading to utility interruptions that affect homes and residents. They can damage or contaminate water supply infrastructure, affecting access to clean water for drinking, firefighting, and sanitation.

The following map, from NOAA's Wildfire Risk to Communities, indicates the wildfire risk to homes in Rhode Island:

Map 71: Wildfire Risk to Homes



Source: NOAA's Wildfire Risk to Communities

Wildfires can have wide-ranging impacts on critical infrastructure. They can damage electrical transmission and distribution lines, transformers, and power substations. This can lead to widespread power outages, affecting homes, businesses, hospitals, and emergency response capabilities. Damage cell towers, telephone lines, and other communication infrastructure can hinder emergency response efforts, as well as the ability of individuals to call for help or communicate with loved ones. Wildfires can block roads with debris, making them impassable and hindering emergency response and evacuation efforts.

Hospitals and healthcare facilities may be damaged or rendered inoperable during wildfires, affecting the ability to provide medical care during a disaster. Fire stations, police stations, and emergency operation centers may be damaged or destroyed, impacting the ability of first responders to coordinate disaster response efforts. Damage to emergency shelters and housing facilities can disrupt services which are critical for providing temporary shelter to displaced individuals and families.

Environment

Wildfires can have varied impacts on the environment. These impacts are often destructive and can affect ecosystems, wildlife, natural resources, and even the local climate. They can destroy natural habitats, including forests, grasslands, wetlands, and shrublands. This can have devastating effects on wildlife species that depend on these ecosystems for shelter, food, and breeding. Wildfires can harm or displace wildlife, resulting in injury or death. They can force wildlife to flee their habitats, leading to displacement and potential conflicts with human populations. Animals may struggle to find suitable new habitats. Post-fire landscapes are often vulnerable to colonization by invasive plant species, which can outcompete native vegetation and disrupt ecosystem functions.

State Operations and Assets

Wildfires can have significant impacts on state operations, which may include:

- **Emergency Response and Public Safety:** Wildfires can lead to a surge in emergency calls for services related to accidents, injuries, and damaged structures. State agencies involved in emergency response, such as RIEMA the State of Rhode Island Police, fire departments, and emergency medical services, must mobilize additional resources to handle these demands.
- **Emergency Operations Centers:** Wildfire often require the activation of state Emergency Operations Centers to coordinate emergency response efforts. These centers serve as hubs for communication, resource allocation, and decision-making during disasters.
- **Infrastructure Damage:** Wildfires can cause extensive damage to critical infrastructure, including roads, bridges, schools, state buildings, and utility facilities. This damage can disrupt state operations and hinder transportation and communication.
- **Emergency Shelters and Services:** Wildfires may require the establishment of emergency shelters and services for displaced residents. State agencies (RIEMA) must coordinate the setup and operation of these facilities.
- **Budgetary Impact:** The costs associated with emergency response efforts, disaster recovery, and infrastructure repair can strain state budgets.
- **Resource Allocation:** State governments must allocate resources, including personnel, equipment, and stockpiled supplies, to support emergency response and recovery efforts.
- **Communication Challenges:** Wildfires can disrupt communication networks, hindering the ability of state agencies to communicate internally and with the public. This can impact emergency notifications and coordination efforts.
- **Administrative and Governance Challenges:** State government offices and facilities may experience closures or reduced staffing during wildfires, affecting administrative functions, regulatory processes, and public services.
- **Economic Impact:** The destruction of infrastructure and businesses can have significant economic consequences for the state and local communities, including job losses and reduced economic activity.
- **Public Services:** Wildfires can disrupt the delivery of public services, including transportation, utilities, and social services, affecting the well-being of residents.

The following table details information concerning the number of critical and state-owned or operated facilities in WUI areas:

Table 150: Critical and State Assets in WUI Areas

Facility Type	WUI Area		
	Intermix	Interface	Total
Fire Stations	26	43	69
Law Enforcement	8	15	23
Power Plants	23	11	34
Natural Gas Pipelines	33	16	49
Electric Transmission Facilities	101	64	165
Airports	3	4	7
Bus Stops	252	447	699
Wastewater Facilities	1	0	1
Hazardous Material/Tier II Facilities	73	87	160
Flood Control Dams	187	55	242
FDIC Insured Banks	7	45	52
Schools	65	110	175
State Facilities	18	25	43
Hospitals	0	3	3
Nursing Facility	9	16	25
Communications Network Facilities	9	3	12
Total	815	944	1,759

Source: RIEMA and RIGIS

State owned facilities in identified WUI areas were matched, when possible, to the master state facility list for valuation purpose. Some data discrepancies were discovered during this process causing a valuation estimate for some facilities based on the best available information. The following table present the valuation of the state facilities identified in WUI intermix areas:

Table 151: Valuation of State of Rhode Island Facilities in WUI Areas

Jurisdiction	Valuation
Bristol County	\$831,534
Kent County	\$13,932,178
Newport County	\$0
Providence County	\$3446060
Washington County	\$59,325,126
State Total	\$77,534,898

Source: RIEMA and RIGIS

Of concern to the state are wood framed buildings as they may be particularly vulnerable to negative impacts of wildfires. The combustible nature of wood makes it vulnerable to ignition during a wildfire, and once ignited, the fire can spread more easily within and through the structure.

The following table details the complete inventory of all wood framed buildings, by county, along with the most recent valuation for both the facility envelope and assets housed with the facility for the State of Rhode Island:

Table 152: State of Rhode Island Wood Framed Facilities by Jurisdiction

Jurisdiction	Number of Wood Framed Buildings	Valuation
Bristol County	25	\$134,904,288
Kent County	70	\$55,634,421
Newport County	54	\$40,946,254
Providence County	152	\$189,823,484

Table 152: State of Rhode Island Wood Framed Facilities by Jurisdiction

Jurisdiction	Number of Wood Framed Buildings	Valuation
Washington County	303	\$102,305,412
State Total	604	\$523,613,859

Source: RIEMA

Note: Valuations include structure and assets

The age of a state owned or operated structure could have a significant impact on its resilience to wildfires as an older structure may be less resilient due to the following factors:

- **Building Codes and Standards:** Older buildings may not have been constructed to meet modern building codes, which often incorporate lessons learned from previous disasters. Additionally, older buildings may not utilize fire suppression equipment and materials. Newer buildings are typically designed to be more resilient to known hazards.
- **Construction Materials and Techniques:** Older buildings may have been constructed using materials and techniques that are less resilient to hazards. Modern construction materials and methods, such as fire-resistant cladding and steel frames, are often more resilient to hazards.

Utilizing information provided by the State of Rhode Island, and removing identifying addresses, the following wood framed facilities have been determined to be most vulnerable to potential wildfire events due to the age of construction and/or the total valuation of the building, for the State of Rhode Island:

Table 153: State of Rhode Island Wood Framed Facilities Most Vulnerable to Wildfires

County	Jurisdiction	Valuation	Year Constructed
Providence	Scituate	\$2,362,303.84	1799
Providence	Scituate	\$241,356.48	1850
Providence	Scituate	\$972,337.13	1935
Providence	Burrillville	\$747,983.95	1936
Providence	Burrillville	\$1,006,697.99	1936
Providence	Burrillville	\$1,263,331.82	1936
Providence	Burrillville	\$984,752.44	1940
Providence	Scituate	\$159,592.00	1948
Providence	Burrillville	\$632,423.94	1965
Providence	Glocester	\$624,352.63	1966

Source: RIEMA

Note: Valuations include structure and assets

Using GIS, WUI areas were overlaid on the critical facility and state facility location data. Maps are included in Appendix D.

Consequence Analysis

This consequence analysis lists the potential impacts of this hazard on various elements of community and state infrastructure. The impact of this hazard is evaluated in terms of disruption of operations, recovery challenges, and overall wellbeing to all Rhode Island residents and first responder personnel. The consequence analysis supplements the hazard profile by analyzing specific impacts.

Table 154: Wildfire Consequence Analysis

Subject	Potential Impacts
Impact on the Public	People located in the immediate area of the fire face the risk injury or death if not evacuated in time. Once evacuated, they may face lengthy period of relocation. Fires can release toxic components which can cause adverse health effects including respiratory and cardiovascular system impacts. Psychological and psychiatric concerns may arise due to exposure to the traumatic event. Young children and the elderly are especially vulnerable to health issues stemming from fire and smoke exposure.

Table 154: Wildfire Consequence Analysis

Subject	Potential Impacts
Impact on Responders	Fire, police, and emergency responders may be called to evacuate people from the fire area, close roads, create fire breaks, attend to the injured, and direct traffic. Firefighters are at a higher risk of smoke inhalation, burns, and health problems due to working in close proximity to fires and the subsequent smoke.
Continuity of Operations	Rhode Island agencies maintain continuity plans which can be enacted as necessary based on the situation. Wildfires may impact an agency's ability to maintain continuity of operations due to impacts on critical infrastructure. If the activation of alternate facilities was required, ensuring that facilities are not in the path of a fire would be critical to continuing operations.
Delivery of Services	Fires can cause disruption of services, including the ability to deliver goods and services. Impacts on operations could lead to a reduction or cessation of services. Goods and facilities may be damaged or destroyed by fire, smoke, or extremely high temperatures.
Property, Facilities, and Infrastructure	Fire can damage or completely destroy property and critical facilities, as well as lead to interruption of the power supply system. A fire of significant strength can cause major damage to buildings or farmland. Large fires may also interrupt transportation systems such as train and bus lines, creating a challenge for public transit and evacuation.
Impact on Environment	Fires can cause significant impact to the environment by spreading pollution, damaging agricultural crops, and disturbing the wildlife and natural areas. Water and soil pollution caused by fire can cause longer term threats to ecosystem health. Fire damage may also affect soil formation, nutrient cycling, and carbon sequestration and storage.
Economic Conditions	Fires can cause a fiscal impact on the local government, even if costs can be recouped by federal grants. Rhode Island's agriculture, tourism and aquaculture are a major component of the local, county and state economy, and major fires could cause significant impact to those sectors. Costs may be associated with loss of income, damage to property, firefighting can be significant.
Public Confidence in Governance	Governmental response, on all levels, state and local, would require direct action that must be immediate and effective to maintain public confidence.

4.19.7 Jurisdictional Vulnerability and Planning Significance

Utilizing data from FEMA's NRI and from LHMPs, the following tables detail the hazard ratings for jurisdictions, by census tract, for severe winter weather. Data concerning both risk index and EAL is reported by census tract. As many State of Rhode Island jurisdictions encompass multiple tracts, an average of tract ratings was completed to determine the overall jurisdictional rating. The following table details the hazard rating by LHMP analysis, FEMA NRI, and EAL by jurisdiction for the State of Rhode Island:

Table 155: State of Rhode Jurisdictional Wildfire Event Planning Significance

County	Jurisdiction	Wildfire LHMP Rating	Wildfire NRI	Wildfire EAL
Bristol	Barrington	Low	Relatively Low	Relatively Low
	Bristol	-	Very Low	Very Low
	Warren	Medium	Relatively Low	Relatively Low
Kent	Coventry	-	Relatively Low	Relatively Low
	East Greenwich	High	Relatively Low	Relatively Low
	Warwick	Low	Very Low	Very Low
	West Greenwich	-	Relatively Low	Relatively Low
	West Warwick	Low	Relatively Low	Very Low
Newport	Jamestown	-	Relatively Low	Relatively Low
	Little Compton	Not Rated	Relatively Low	Relatively Low
	Middletown	Low	Very Low	Very Low
	Newport	Low	Very Low	Very Low
	Portsmouth	Low	Relatively Low	Relatively Low

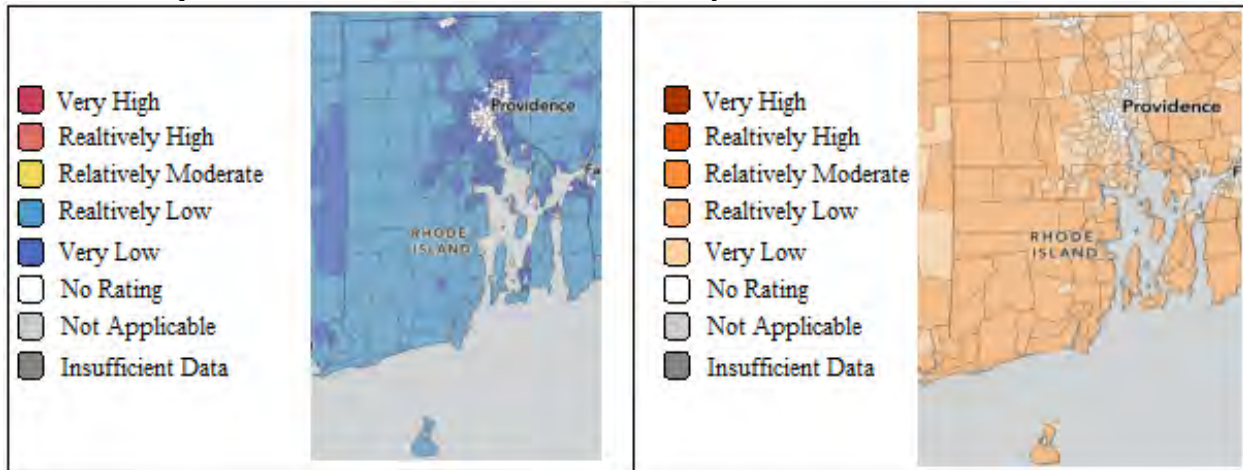
Table 155: State of Rhode Jurisdictional Wildfire Event Planning Significance

County	Jurisdiction	Wildfire LHMP Rating	Wildfire NRI	Wildfire EAL
	Tiverton	-	Relatively Low	Relatively Low
Providence	Burrillville	Low	Relatively Low	Relatively Low
	Central Falls	Low	Very Low	Very Low
	Cranston	Low	Very Low	Very Low
	Cumberland	-	Very Low	Very Low
	East Providence	Moderate	Very Low	Very Low
	Foster	Moderate	Relatively Low	Relatively Low
	Glocester	Not Rated	Relatively Low	Relatively Low
	Johnson	Moderate	Relatively Low	Relatively Low
	Lincoln	Low	Relatively Low	Relatively Low
	North Providence	Low	Very Low	Very Low
	North Smithfield	Not Rated	Very Low	Very Low
	Pawtucket	Low	No Rating	No Rating
	Providence	Low	No Rating	No Rating
	Scituate	-	Relatively Low	Relatively Low
	Smithfield	-	Relatively Low	Relatively Low
	Woonsocket	Low	Very Low	Very Low
Washington	Charlestown	-	Relatively Low	Relatively Low
	Exeter	-	Relatively Low	Relatively Low
	Hopkinton	-	Relatively Low	Relatively Low
	Narragansett	Not Rated	Relatively Low	Relatively Low
	New Shoreham	-	Relatively Low	Relatively Low
	North Kingstown	Moderate	Relatively Low	Relatively Low
	Richmond	Moderate	Relatively Low	Relatively Low
	South Kingstown	Moderate	Relatively Low	Relatively Low
	Westerly	Moderate	Relatively Low	Relatively Low

Source: LHMPs and FEMA NRI

Note: - indicates no current plan available for review

FEMA NRI was utilized to determine the risk and expected annual loss due to wildfires by census tract for the state of Rhode Island. The following maps detail these parameters:

Map 72: FEMA NRI and EAL for Wildfires by Census Tract for Rhode Island

Source: FEMA NRI

A disproportionate impact from wildfires may be felt by jurisdictions in identified WUI areas with older housing units as they may have been constructed using materials and techniques that are less resilient to fires. Modern construction materials and methods, such as reinforced concrete and steel frames, sprinkler systems, and fire-resistant exterior

materials, are often more resilient. The following table Rhode Island Department of Housing 2022 Integrated Housing Report from the Rhode Island Department of Housing 2022 Integrated Housing Report indicates jurisdictions with greater than 40% of their housing units constructed before 1949 in identified WUI areas:

Table 156: Jurisdictions at Greater Risk to Wildfires due to Age of Housing

County	Jurisdiction	1949 and later
Kent	West Greenwich	36.00%
Providence	Burrillville	34.50%
Washington	Hopkinton	34.10%
Providence	Scituate	32.50%
Providence	Foster	26.90%
Providence	Glocester	24.30%
Washington	South Kingstown	23.10%
Providence	Coventry	19.80%
Kent	Exeter	14.50%
Washington	Richmond	13.50%
Kent	West Greenwich	36.00%
Providence	Burrillville	34.50%

Source: Rhode Island Department of Housing 2022 Integrated Housing Report

A household is considered cost burdened if it spends 30% or more of its income on housing costs, with cost burdens being an indication of rents increasing faster than incomes. Cost burdened citizens may have great difficulty evacuating and may have challenges maintaining adequate insurance coverage. The following table, using data from the Housing and Urban Development's Comprehensive Housing Affordability Strategy Data for Rhode Island, indicates jurisdictions located in WUI areas with the most cost burdened renters who may be at greater risk to wildfires due to economic constraints.

Table 157: Jurisdictions at Greater Risk to Wildfires Housing Cost Burden

County	Jurisdiction	Cost Burdened Single Families
Kent	Exeter	73%
Providence	Smithfield	53%
Kent	West Greenwich	47%
Providence	Scituate	45%
Kent	East Greenwich	44%
Washington	Hopkinton	42%

Source: Housing and Urban Development's Comprehensive Housing Affordability Strategy Data

4.19.8 Hazard Planning Significance

Based on a review of available data, the SIHMC determined wildfire to have a moderate planning significance for Rhode Island.

4.20 Biological Incident

4.20.1 Hazard Description

A biological incident is the intentional or accidental release of harmful biological agents (such as bacteria, viruses, toxins, or fungi).

An intentional release is done with the intent to cause illness, death, fear, or societal disruption. Potentially released biological agents can be naturally occurring or engineered for malicious purposes. Key factors helping to characterize an intentional biological release include:



- **Deliberate Intent:** The release of biological agents is intentional and aimed at causing harm or disruption.
- **Biological Agents:** The attackers use microorganisms, toxins, or other biological substances that have the potential to cause disease or harm.
- **Spread of Disease:** The primary goal of a biological attack is often to infect individuals with a specific disease or toxin. Depending on the agent, this can result in a wide range of illnesses, from mild to severe.
- **Fear and Panic:** In addition to causing illness and death, biological attacks can create fear, panic, and social disruption. This psychological impact is a significant part of the attacker's strategy.
- **Variety of Agents:** Biological agents that can be used in attacks include anthrax, smallpox, botulinum toxin, plague, and many others. The choice of agent may depend on factors like lethality, ease of dissemination, and availability.
- **Modes of Transmission:** Biological agents can be disseminated in various ways, including through the air (aerosolization), water, food, or direct contact.
- **Stealth:** Biological attacks can be difficult to detect and may not produce immediate symptoms, allowing for the potential for widespread infection before an outbreak is recognized.

Additionally, biological agents can be released accidentally through improper handling in a laboratory environment or through mishandling during transportation. These accidental releases would be easier to contain as they would be immediately reported, and quarantine and contact tracing procedures instituted immediately to contain any potential spread.

4.20.2 – Location and Extent

The entire State of Rhode Island is vulnerable to biological incidents. Several biological agents have been historically considered as potential candidates for use in intentional biological agent release due to their virulence, ease of dissemination, and potential for mass casualties. Some of the most likely agents to be used include:

- **Anthrax (*Bacillus anthracis*):** Anthrax spores are highly resilient and can be spread in various forms, making it a significant concern. Inhaling anthrax spores can result in a severe and often fatal respiratory infection.
- **Botulinum Toxin (*Clostridium botulinum*):** Botulinum toxin is one of the most potent toxins known to humans. It can cause paralysis and death when ingested or inhaled.
- **Plague (*Yersinia pestis*):** Plague bacteria can cause bubonic, septicemic, or pneumonic forms of the disease. Pneumonic plague, which affects the respiratory system, is particularly concerning due to its potential for person-to-person transmission.
- **Smallpox (*Variola virus*):** Smallpox was eradicated in the 1970s, but the virus still exists in laboratory settings. Its potential for rapid spread and high mortality rate makes it a significant concern.
- **Tularemia (*Francisella tularensis*):** Tularemia can be transmitted through inhalation, ingestion, or contact with contaminated materials. The inhalation form can be severe and life-threatening.
- **Viral Hemorrhagic Fevers (Ebola, Marburg):** These viruses can cause severe hemorrhagic fevers with high mortality rates. They are a concern due to their potential for rapid transmission and the difficulty of containment.
- **Ricin:** Ricin is a toxin derived from castor beans and can be lethal when ingested, inhaled, or injected. It is relatively easy to produce and disseminate.

The seriousness of an intentional biological attack release varies significantly based on several factors, including:

- **Type of Biological Agent:** Some agents are highly lethal, while others may cause less severe illnesses. The agent's virulence, ease of transmission, and availability can determine the seriousness of the attack.
- **Mode of Transmission:** The method of transmission plays a significant role in determining the attack's seriousness. Agents that can be easily aerosolized and inhaled, leading to person-to-person transmission, pose a greater threat than those requiring direct contact or ingestion.
- **Concentration and Quantity:** The amount and concentration of the biological agent released can affect the scale of the attack. Larger quantities and higher concentrations can lead to more widespread infections and a higher number of casualties.
- **Target Population:** The vulnerability of the exposed population is a critical factor. Attacks on densely populated urban areas or vulnerable groups (e.g., children, the elderly) can result in more severe consequences.
- **Preparedness and Response:** The level of preparedness and the effectiveness of the response infrastructure in place can significantly influence the seriousness of an attack. Rapid detection, isolation, treatment, and contact tracing can mitigate the impact.
- **Medical Countermeasures:** The availability of vaccines, treatments, and medical countermeasures specific to the biological agent can affect the seriousness of an attack. The ability to provide prompt medical care can save lives.
- **Vaccination Coverage:** The percentage of the population that is vaccinated or immune to the biological agent can limit the spread and severity of an outbreak.
- **Timely Detection:** Early detection and reporting of a biological attack are crucial. Delayed detection can lead to wider transmission and increased severity.
- **Public Health Infrastructure:** The strength of the public health infrastructure, including surveillance systems, laboratories, and healthcare facilities, influences the ability to respond effectively to an attack.

Laboratories that work with select agents and toxins are subject to federal regulations under the Federal Select Agent Program, as administered by the CDC and the Animal and Plant Health Inspection Service. Laboratories working with select agents and toxins must register, conduct security risk assessments, and implement strict security and access control measures. Additionally, the operation of laboratories within the state is regulated by the State of Rhode Island General Laws, Title 23 Health, and Safety Chapter 16.2 Laboratories.

4.20.3 Previous Occurrences

There has not been a significant example of a human-caused biological incident in the State of Rhode Island. However, a number of white powder incidents (which may resemble anthrax in appearance) have occurred in Rhode Island. The latest incident in January 2020 involved an employee of the Bristol-Warren Regional School District who discovered white powder on a letter brought from home. This caused the district's administration building to be shut down for about five hours.

None of the recorded incidents within the state were confirmed to contain live biological agents. But while these incidents did not cause physical harm, they did place a strain on resources and the mental health on those impacted.

There have been no reported unauthorized or accidental releases of biological agents from laboratory or transportation settings in the State of Rhode Island.

4.20.4 Probability of Future Events

Although no major biological incident has occurred within Rhode Island, the presence of major research laboratories within the State could increase the potential probability of occurrence.

To help minimize the probability of occurrence, the Rhode Island State Health Laboratories Center for Biological Sciences provides an extensive array of testing to quickly identify biological agents of public health concern, including potential agents of bioterrorism, in human and animal clinical specimens as well as in food. Serving as a reference laboratory within the CDC Laboratory Response Network they also perform testing on unknown powders and other suspicious environmental substances to rule out or confirm the presence of biothreat agents. RIDOH, per state

regulations, requires the following diseases and conditions be reported on the day of recognition or strong suspicion of disease:

- Anthrax
- Botulism
- Brucellosis
- Clostridium Perfringens (Epsilon Toxin)
- Ebola Hemorrhagic Fever (Ebola)
- Glanders
- Melioidosis
- Plague; Bubonic, Septicemic, Pneumonic (Plague)
- Q-Fever
- Ricin Poisoning
- Smallpox
- Staphylococcal Food Poisoning Enterotoxin - B Poisoning (Staph Food Poisoning)
- Tularemia (Rabbit fever)
- Viral Hemorrhagic Fever (Ebola, Lassa, Marburg)

4.20.5 Projected Changes in Location, Intensity, Frequency, and Duration

No large-scale changes in the number of entities handling biological agents are projected during the life of this plan. Increases in out of state and out of country visitors to the State of Rhode Island could increase the frequency of event occurrence if screening and testing standards are not maintained.

4.20.6 Vulnerability and Impact

People and Communities

Numerous factors influence a person's susceptibility to infection, and the potential severity of illness, from a biological agent. These vulnerabilities can be influenced by individual, societal, and environmental factors, and may include:

- Lack of Immunity: Many biological agents cause diseases for which people have little to no immunity.
- Vaccination Status: People who are not vaccinated or have not received booster shots may be more vulnerable to specific biological agents.
- Age: Infants, young children, and the elderly often have weaker immune systems, making them more susceptible to infections and complications.
- Underlying Health Conditions: Individuals with underlying health conditions, such as immunodeficiency disorders, chronic diseases, or respiratory conditions, may be more vulnerable to severe illness.
- Medication and Treatment Availability: The availability of medications, antitoxins, or treatments specific to biological agents can impact vulnerability.
- Proximity to the Source: People in close proximity to the source of the biological agent release or those directly exposed to the agent are at higher risk of infection.
- Access to Healthcare: The availability and accessibility of healthcare services, including diagnostic testing and medical treatment, can significantly impact the outcome.
- Public Awareness: People who are unaware of the risks associated with biological agents or who do not know how to protect themselves may be more vulnerable.
- Behavioral Factors: People's behavior, such as adherence to public health guidelines, can influence vulnerability.

The release of harmful biological agent can have severe and far-reaching impacts on human health, and can include:

- Illness and Death: The most immediate impact of exposure to a biological agent is the potential for widespread illness and death. Depending on the agent used, the severity of illness can range from mild to life-threatening. In some cases, diseases caused by biological agents have high mortality rates.

- **Healthcare Overload:** A large-scale release can quickly overwhelm healthcare systems, leading to shortages of medical supplies, hospital beds, and healthcare personnel. The ability to provide timely medical care may be compromised.
- **Social Disruption:** Social disruption can occur due to isolation and quarantine measures, as well as the need for social distancing. Schools, businesses, and public gatherings may be canceled or limited, affecting daily life and routines.
- **Psychological Trauma:** Survivors may experience long-lasting psychological trauma due to the fear of infection, the loss of loved ones, and the overall trauma of the event.
- **Long-Term Health Effects:** Some biological agents can cause long-term health effects in survivors, including chronic illnesses and disabilities.

Facilities and Critical Infrastructure

The direct risk or vulnerability to property and facilities from a biological agent release is limited. Impacts include restricting access to a facility due to an existing or potential communicable biological disease. Direct risk and vulnerability to actual structures is limited, however, while unlikely, biological agents could be transported through a facility's ventilation system. An incident like this would not pose a direct risk to the structure's integrity, however considerable contamination of the facility may occur requiring hazardous materials response and potential loss of access to the building for a considerable length of time.

Critical facilities and infrastructure generally will not suffer direct impacts from a biological incident. Employee absenteeism due to a large-scale biological incident could indirectly impact the ability for a critical facility to operate. Without necessary operators, critical infrastructure may be susceptible to indirect failure.

The type and scale of a biological incident will determine the severity of the effect on the environment. Diseases that cause widespread human deaths can impact the environment during the disposal of human remains and the handling of biohazardous waste. Similar impacts would be seen in the aftermath of a biological incident causing widespread deaths of animals. Whether the infected animals are buried, burned, or left in place, a large quantity and concentration of carcasses may have an impact on air, soil, and groundwater quality, if disposal is not planned and executed appropriately.

State Operations and Assets

A biological incident can have significant and wide-ranging impacts on state operations, affecting state government functions and public safety. These impacts can disrupt state operations, strain resources, and pose challenges to maintaining public order. Some of the impacts of a biological attack on state operations include:

- **Emergency Response and Healthcare:** The state would need to rapidly mobilize emergency response teams, medical personnel, and healthcare facilities to address the outbreak. The surge in demand for medical resources could strain healthcare systems, including hospitals, clinics, and emergency services.
- **Public Health Services:** RIDOH would play a critical role in disease surveillance, contact tracing, and public health messaging. A biological agent release may result in RIDOH requiring additional personnel and resources to manage the outbreak.
- **Resource Allocation:** Rhode Island may need to allocate resources for medical supplies, pharmaceuticals, personal protective equipment, and vaccine distribution. Competition for limited resources could lead to shortages and increased costs.
- **Transportation and Supply Chain Disruption:** Quarantine measures, travel restrictions, and supply chain disruptions could affect the movement of essential goods and services, including medical supplies, food, and fuel.
- **Economic Impact:** The economic consequences of a biological agent release could be severe. Business closures, reduced consumer confidence, and trade disruptions can lead to financial losses, unemployment, and economic instability.
- **Education Disruption:** School closures and disruptions to education could affect students' learning and parental work arrangements, leading to social and economic consequences.

- **Public Services:** Essential public services, such as law enforcement, fire services, and sanitation, could be impacted due to the demands of responding to the outbreak.
- **Long-Term Recovery:** Recovering from a biological attack can be a long-term process. Rebuilding public infrastructure, restoring economic stability, and addressing the long-term health needs of survivors are complex challenges.

Consequence Analysis

This consequence analysis lists the potential impacts of this hazard on various elements of community and state infrastructure. The impact of this hazard is evaluated in terms of disruption of operations, recovery challenges, and overall wellbeing to all Rhode Island residents and first responder personnel. The consequence analysis supplements the hazard profile by analyzing specific impacts.

Table 158: Biological Incident Consequence Analysis

Subject	Potential Impacts
Impact on the Public	Cities with dense populations, particularly along major travel routes, are the most vulnerable. Infants and young children in day-care centers, children in schools, the elderly in residential facilities, and hospital patients are most vulnerable to biological incidents. Protective measures, including isolation and quarantine, will need to be taken particularly for those of the most vulnerable communities.
Impact on Responders	Biological incidents will create a dangerous environment and significant challenge for first responders. First responders may need to have direct contact with individuals who are suffering from a communicable disease. First responders must control their own exposure to the biological incident and ensure correct protective equipment is utilized.
Continuity of Operations	Rhode Island agencies maintain continuity plans which can be enacted as necessary based on the situation. A biological incident may impact an agency's ability to maintain continuity of operations based on a high amount of employee absenteeism.
Delivery of Services	The ability to deliver services can be impacted locally, regionally, or statewide during a biological incident. Transit services may be directly impacted due to the potential spread of communicable diseases during a biological incident. Businesses and places of commerce may completely shut down due to biological incidents and the resulting employee absenteeism, which leads to the disruption of goods and services.
Property, Facilities, and Infrastructure	Transportation, governmental operations, and infrastructure facilities may be directly or indirectly disrupted during a significant biological incident due to the facilities exposed during the biological incident or due to absenteeism of employees after the incident.
Impact on Environment	Agriculture crops and livestock are extremely susceptible to the adverse effects of biological incidents. Biological incidents may impact the environment long-term by disturbing or killing wildlife and adversely affecting nature preserves.
Economic Conditions	Biological incidents pose a fiscal impact on the local and state governments. Local, county, and state resources may be required during a biological incident therefore reducing their availability for future events. Significant employee absenteeism during a biological incident will have a direct and significant impact on the state economy.
Public Confidence in Governance	Confidence in the state's governance is affected by immediate response through direct and effective actions. The ability for the state to managed mass care and proper medical treatment for those affected during response operations is critical.

4.20.7 Hazard Planning Significance

Based on a review of available data, the SIHMC determined biological incident to have a low planning significance for Rhode Island.

4.21 Civil Disorder

4.21.1 Hazard Description

Civil disorder refers to a situation in which there is a breakdown of public order and safety within a community or society. It typically involves a significant disturbance, often characterized by acts of violence, rioting, and public disobedience. Civil disorder can occur for various reasons, including social, political, economic, or racial tensions, and it may be triggered by specific events or grievances.



4.21.2 – Location and Extent

Historically civil disorder has been most associated with urban areas and college campuses. In general, civil disorder usually accompanies, or is started by, a gathering of people for an event. And while most events occur with no violence, violence can occur with little warning or cause. Unfortunately, large crowds can be subject to control by skillful troublemakers who are often able to incite behavior from members of the crowd that they usually would not consider. When a crowd begins to exhibit signs of disorder, it can be categorized in three categories:

- **Public disorder:** Public disorder is a basic breach of civic order. Individuals or small groups assembling tend to disrupt the normal flow of things around them.
- **Public disturbance:** Public disturbance is designed to cause turmoil on top of the disruption. Individuals and groups assembling into a crowd begin chanting, yelling, singing, and voicing individual or collective opinions.
- **Riot:** A riot is a disturbance that turns violent. Assembled crowds become a mob that violently expresses itself by destroying property, assaulting others, and creating an extremely volatile environment.

Civil disorder usually has the following characteristics:

- **Unrest:** Civil disorder involves a state of unrest and agitation among a significant portion of the population. People may engage in protest activities, demonstrations, or other forms of collective action.
- **Violence and Property Damage:** Acts of violence, vandalism, property damage, and clashes with law enforcement or counter-protesters are common features of civil disorder.
- **Disruption of Order:** Civil disorder disrupts the normal functioning of a community or society. This disruption can affect transportation, commerce, education, and other essential services.
- **Law Enforcement Response:** Law enforcement agencies are typically called upon to respond to civil disorder. They may use various tactics, including crowd control measures and, in some cases, the declaration of a state of emergency.
- **Potential for Escalation:** Civil disorder has the potential to escalate if not effectively managed. It can lead to more significant violence, injuries, and damage if left unchecked.

Civil disorder can have underlying causes, such as social inequality, injustice, discrimination, economic hardship, or political grievances. It's important to note that not all forms of protest or public demonstrations are considered civil disorder. Peaceful protests, which are protected by the principles of free speech and assembly in many democratic societies, are distinct from civil disorder. However, civil disorder typically involves a departure from peaceful and lawful means of expressing dissent or grievances, often involving illegal activities and a significant risk to public safety.

Rhode Island has many large public events a year that could provide a venue for civil disorder. These events include:

- Gaspee Days
- Newport Folk Festival
- Newport Jazz Festival
- WaterFire Providence
- Bristol Fourth of July Parade
- Charlestown Seafood Festival

- PVD Festival
- Scituate Arts Festival

4.21.3 Previous Occurrences

Rhode Island has experienced several local incidents related to civil disturbances. Demonstrations as part of the Occupy Wall Street and Black Lives Matter movements impacted citizens and local operations. In November 2014, significant protests and demonstrations led to shutdowns on I-95 outside Providence as protesters demonstrated against the death of Michael Brown in Missouri.

In June of 2020 protests involving approximately 150 people over the death of George Floyd in Providence turned violent as spread throughout downtown. Approximately 60 Providence and Rhode Island State Police vehicles were deployed, and protestors broke into a nearby mall and other stores throughout the downtown area were vandalized and looted. During the event, a Providence Police vehicle was burned. As a result, a curfew was declared in Providence for the next few nights.

4.21.4 Probability of Future Events

Civil disturbances will continue to occur in the future, however the probability of occurrence of these events is difficult to predict. Some forms of civil disturbance can be anticipated with more accuracy than others. For example, it is reasonable to assume that a riot may occur after the verdict is read in a high-profile legal case. In many instances, there is some amount of reasonable warning available to help prepare for a civil disturbance.

4.21.5 Projected Changes in Location, Intensity, Frequency, and Duration

Future political and social conditions can impact the occurrence of this hazard. However, predicting these conditions is challenging due to the large number of factors involved.

Climate change can have complex and far-reaching effects on society, including the potential to contribute to civil disorder and social unrest. While climate change itself does not directly cause civil disorder, it can exacerbate existing vulnerabilities and stressors within societies, which may increase the likelihood of civil unrest.

As previously noted, State of Rhode Island facilities have seen no major changes in the past five years, with only modest repairs and upgrades being conducted and no major rehabilitation or construction projects completed. As such, the risk to state assets has remained static since the completion of the 2019 SHMP. Additionally, with the exception of the two previously identified rehabilitation and construction projects, no large-scale facility projects were noted in any agency budgets or reports.

4.21.6 Vulnerability and Impact

People and Communities

Civil disorder events can have significant impacts on individuals and communities. These impacts can vary depending on the scale, duration, and intensity of the event, but may include:

- **Fatalities:** In some cases, civil disorder events can result in fatalities, including deaths of protesters, law enforcement officers, or innocent bystanders. This loss of life is a tragic consequence of extreme civil disorder.
- **Physical Injury:** Individuals involved in civil disorder events, whether as participants, bystanders, or law enforcement personnel, can suffer physical injuries. These injuries may range from minor cuts and bruises to more severe injuries like gunshot wounds or head injuries.
- **Emotional Trauma:** Witnessing or experiencing violence and chaos during civil disorder can lead to psychological trauma, including post-traumatic stress disorder, anxiety, and depression.

Additionally, community impacts may include:

- **Vandalism:** Civil disorder often involves property damage, including vandalism, looting, and arson. Businesses, homes, and public infrastructure may be damaged or destroyed.

- **Economic Impact:** The destruction of businesses and property can have economic consequences, including job losses and reduced property values.
- **Curfews and Restrictions:** Local authorities may impose curfews and restrictions on movement, affecting people's ability to carry out daily activities.
- **School Closures:** Schools and educational institutions may close temporarily to ensure the safety of students and staff.
- **Reduced Emergency Services:** Emergency response services may be stretched thin during civil disorder events, potentially affecting the response to other emergencies.
- **Social Divisions:** Civil disorder events can deepen social and political divisions within communities, straining relationships among neighbors and community members.
- **Trust Erosion:** Trust in institutions, including law enforcement, may erode as a result of civil disorder, affecting community-police relations.

Vulnerable populations, such as low-income individuals and minority communities, may be disproportionately affected by civil disorder events. Vulnerable individuals may have limited access to essential services during and after civil disorder events.

Facilities and Critical Infrastructure

Civil disorder events can have significant impacts on facilities, including public and private buildings, infrastructure, and critical facilities. The extent of these impacts can vary depending on the scale and intensity of the civil disorder, as well as the effectiveness of security measures in place, and may include:

- **Property Damage:** Civil disorder often involves acts of vandalism, such as graffiti, broken windows, and defacement of buildings.
- **Looting:** Businesses may be targeted for looting, resulting in theft of merchandise and property damage.
- **Arson:** In some cases, facilities may be set on fire, leading to extensive damage or total destruction.
- **Temporary Closures:** Businesses and commercial facilities may need to close temporarily during civil disorder events to ensure the safety of employees and customers.
- **Loss of Revenue:** Extended closures or damage to inventory can result in financial losses for businesses.
- **Damage to Public Infrastructure:** Civil disorder can lead to damage or destruction of public infrastructure such as roads, bridges, and transportation facilities.

Additionally specific critical facilities may be especially prone to the impacts of a civil disorder event. These facilities would include:

- **State assets,** including courthouses, government buildings, and law enforcement locations, may be targeted for vandalism or attack.
- **Schools and universities** may experience damage to buildings, equipment, and facilities.
- **Hospitals and Healthcare Facilities:** Healthcare facilities may need to implement security measures to protect patients, staff, and property.
- **Public Safety Facilities:** Damage to police stations or fire departments can affect emergency response capabilities.

Repairing and restoring facilities can be expensive and time-consuming, placing a financial burden on property owners and municipalities. Property owners may need to file insurance claims to cover the costs of repairs and replacements. Additionally, facilities may need to invest in enhanced security measures, such as hiring security personnel or installing surveillance systems.

Environment

The debris and other byproducts of civil disturbances can pollute the environment, damage sewage and wastewater infrastructure, and potentially destroy green spaces.

State Operations and Assets

Civil disorder events can have direct impacts on state operations, particularly on agencies responsible for maintaining public safety. The extent of these impacts depends on the scale, duration, and intensity of the civil disorder, as well as the effectiveness of emergency management and response efforts, and may include:

- **Emergency Response:** State agencies, including the Rhode Island State Police, RIEMA, and the Rhode Island National Guard, may need to allocate resources and personnel to respond to the event.
- **Public Safety:** Ensuring public safety and managing potentially violent or dangerous situations can become a top priority.
- **Infrastructure Damage:** Civil disorder events may result in damage to public infrastructure, including roads, bridges, and public transportation facilities.
- **Transportation Disruptions:** Disruptions in transportation networks can impact the movement of people and resources.
- **Legal Proceedings:** Investigations into civil disorder events may lead to legal proceedings and prosecutions of individuals involved in criminal activities.
- **Policy Review:** State agencies may need to review and adapt policies, regulations, and laws related to civil unrest and emergency response.
- **Resource Diversion:** State agencies may need to divert resources and personnel from other operations to manage the event and maintain public safety.
- **Erosion of Community-Police Relations:** Civil disorder events can strain community-police relations, requiring efforts to rebuild trust and improve community engagement.

Consequence Analysis

This consequence analysis lists the potential impacts of this hazard on various elements of community and state infrastructure. The impact of this hazard is evaluated in terms of disruption of operations, recovery challenges, and overall wellbeing to all Rhode Island residents and first responder personnel. The consequence analysis supplements the hazard profile by analyzing specific impacts.

Table 159: Civil Disorder Consequence Analysis

Subject	Potential Impacts
Impact on the Public	Civil unrest events are more likely to occur when large groups of people are congregated together at localized events creating a situation where local medical resources and response capabilities may be overwhelmed. Civil unrest will have varying effects on the population including injury, death, and significant psychological impacts. These impacts may be immediate or long-term depending on the incident.
Impact on Responders	Civil unrest can create a dangerous environment for first responders. First responders may have to manage the evacuation of people from the area, as well as direct traffic, close down roads, and take care of the injured. First responders may also become the direct target of the incident themselves. Equipment may also be damaged or destroyed, which may lead to a decrease in response capabilities.
Continuity of Operations	Rhode Island agencies maintain continuity plans which can be enacted as necessary based on the situation. A civil disorder event may impact an agency’s ability to maintain continuity of operations based on the disruptions potential to cause a significant injury to staff or impede travel to an alternate facility if relocation was required.
Delivery of Services	The ability to deliver services can be impacted locally, regionally, or statewide depending on the size of the event due to roadway and bridge closures. Businesses and places of commerce may completely shut down.
Property, Facilities, and Infrastructure	Transportation, governmental operations, and infrastructure facilities may be disrupted during a significant civil unrest. Roads and bridges may be impacted due to roadblocks or closures. Civil unrest can impact access to homes and critical facilities such as

Table 159: Civil Disorder Consequence Analysis

Subject	Potential Impacts
	hospitals, schools, and supermarkets. If power loss occurs during an incident, it may lead to disruption of critical infrastructure and technology.
Impact on Environment	Civil unrest can impact the environment in the form of smoke and destruction of vegetation. Debris and byproducts of the disturbance have the potential to cause pollution the natural environment.
Economic Conditions	Local, county, and state resources may be required during a civil unrest incident therefore reducing their availability for future events. Private businesses may not be able to maintain operations during if the area of unrest was near or involved their business, which would impact the economy.
Public Confidence in Governance	If government employees or facilities are targeted directly, it will have a significant impact on the state's ability to govern. The public's confidence in the state's governance is affected by immediate response through direct and effective actions. Efficiency in response and recovery operations is critical in keeping public confidence high.

4.21.7 Hazard Planning Significance

Based on a review of available data, the SIHMC determined civil disorder to have a low planning significance for Rhode Island.

4.22 Cybersecurity Event

4.22.1 Hazard Description

Cybersecurity attack refers to a deliberate and malicious attempt to compromise the security of computer systems, networks, devices, or data. The primary objectives of cyberattacks can vary widely and may include gaining unauthorized access, stealing sensitive information, disrupting operations, or extorting payment. Cybersecurity threat actors can be classified as:

- **Hacktivists:** Loosely organized groups known for conducting distributed denial-of-service attacks and defacing websites to promote political or social causes.
- **Ransomware Operators:** Criminal groups use ransomware to encrypt victims' data and demand ransoms for decryption keys.
- **Malware Developers:** Individuals or groups create and distribute malicious software (malware) for profit.
- **Organized Crime:** Criminal organizations may engage in various forms of cybercrime, such as identity theft, credit card fraud, and hacking for profit.
- **Advanced Persistent Threat Groups:** Nation-state-sponsored groups are among the most sophisticated threat actors. They conduct long-term, highly targeted cyber espionage campaigns.



4.22.2 – Location and Extent

The entire state is vulnerable to cybersecurity incidents. As most day-to-day activities rely on the internet in one aspect or another, any person or infrastructure is susceptible to cybersecurity threats. Cyber-attacks can take various forms, each with its own tactics and techniques, and include:

- **Malware Attacks:** Malicious software, such as viruses, worms, Trojans, ransomware, and spyware, is used to infect and compromise a computer or network. Malware can cause damage, steal information, or provide unauthorized access.
- **Phishing Attacks:** Phishing attacks involve tricking individuals into revealing sensitive information, such as passwords or financial details, by posing as a legitimate entity. Phishing emails, websites, and messages are common tools for attackers.
- **Denial-of-Service Attack:** An attack that overwhelms a target system or network with traffic, rendering it inaccessible.
- **Distributed Denial-of-Service Attack:** An attack that involves multiple compromised devices (a botnet) flooding a target with traffic, making it impossible to function effectively.
- **Man-in-the-Middle Attacks:** In these attacks, an attacker intercepts and possibly alters communications between two parties without their knowledge. This can lead to data interception, eavesdropping, or impersonation.
- **SQL Injection Attacks:** Attackers inject malicious SQL code into input fields of a web application to manipulate a database, potentially gaining unauthorized access or extracting data.
- **Zero-Day Vulnerabilities:** Attackers leverage security vulnerabilities in software or hardware that are not yet known to the vendor or public. These vulnerabilities are known as "zero-days."
- **Brute Force:** Attackers attempt to gain access to an account or system by trying all possible password combinations until the correct one is found.
- **Dictionary Attacks:** Attackers use precompiled lists of common passwords to guess login credentials.
- **Social Engineering Attacks:** This involves manipulating individuals into divulging confidential information or performing actions that compromise security. It often relies on psychological manipulation.
- **Ransomware Attacks:** Attackers encrypt a victim's data and demand a ransom in exchange for the decryption key. Payment does not guarantee data recovery, and it encourages further attacks.
- **Insider Attacks:** Malicious or negligent actions by individuals within an organization can pose significant cybersecurity risks, as they may have access to sensitive information and systems.

- Supply Chain Attacks: Attackers target suppliers, vendors, or partners to compromise the security of products or services, which can affect downstream organizations and consumers.
- Internet of Things Attacks: Devices connected to the internet, such as smart appliances and sensors, can be targeted to gain unauthorized access or control.

4.22.3 Previous Occurrences

Rhode Island has experienced numerous cyber-attacks over the past few years, including:

- 2023: CharterCARE Hospital System (ransomware)
- 2023: Roger Williams Medical Center and Our Lady of Fatima (ransomware)
- 2023: North Kingstown (ransomware)
- 2022: Providence (hacking, data breach)
- 2022: Narragansett Bay Commission, Providence County (ransomware)
- 2022: Rhode Island Public Transit Authority (hacking, data breach)
- 2020: Care New England, Providence County (malware)
- 2020: City of Exeter (ransomware)
- 2019: Town of East Greenwich (ransomware)
- 2019: Town of Greenwich and East Greenwich Police Department (ransomware)
- 2019: Pawtucket Fire Department (ransomware)
- 2018: Department of Children, Youth, and Families, Department of Human Services, and Department of Behavioral Healthcare, Developmental Disabilities and Hospitals (phishing)
- 2018: State Treasurer's Office (phishing)
- 2018: Thundermist Health Center, Providence County (ransomware)
- 2016: Rhode Island State Police (ransomware)
- 2016: City of Providence (hack)
- 2016: University Gastroenterology (ransomware)

It is important to note that this is likely only a subset of actual attacks, as many go unreported.

4.22.4 Probability of Future Events

The continued evolution of cyber criminals and nation sponsored groups indicates that the probability of future events is significant. Although the State has not experienced a large-scale cybersecurity incident, large-scale attacks occur worldwide on a regular basis. The number of attacks is projected to increase, especially on critical infrastructure. Additionally, due to the widespread use of computers, email, and the internet, and the reliance on technology to support daily functions, the risks of cybersecurity incidents will continue to grow.

4.22.5 Projected Changes in Location, Intensity, Frequency, and Duration

Predicting the exact future changes in the frequency and intensity of cyber-attacks is challenging due to the rapidly evolving nature of threats, the expanding diversity of attack vectors, and the dynamic landscape of technology. Cyber criminals are likely to continue evolving their tactics, techniques, and procedures to become more sophisticated. This includes the use of advanced malware, ransomware, and targeted attacks that exploit vulnerabilities in both technology and human behavior.

Future geopolitical landscape conditions can influence the location and targeting of cyber-attacks. Nation-state actors may shift their focus based on political tensions, economic interests, or strategic objectives. Critical infrastructure, government entities, and corporations may be primary targets.

As previously noted, State of Rhode Island facilities have seen no major changes in the past five years, with only modest repairs and upgrades being conducted and no major rehabilitation or construction projects completed. As such, the risk to state assets has remained static since the completion of the 2019 SHMP. Additionally, with the exception of the two previously identified rehabilitation and construction projects, no large-scale facility projects were noted in any agency budgets or reports.

4.22.6 Vulnerability and Impact

People and Communities

Cybersecurity attacks can have a range of potential impacts on individuals, both direct and indirect, often affecting their finances and privacy. Some of the potential impacts of a cybersecurity attack may include:

- **Theft of Funds:** Attackers may steal money from victims' bank accounts or cryptocurrency wallets.
- **Credit Card Fraud:** Stolen credit card information can be used for unauthorized purchases.
- **Identity Theft:** Attackers may steal personal data, such as Social Security numbers, addresses, and dates of birth, to commit identity theft.
- **Opening Fraudulent Accounts:** Cybercriminals can use stolen information to open credit cards, loans, or other financial accounts in the victim's name.
- **Data Exposure:** Personal or sensitive information may be exposed, leading to loss of privacy and potential embarrassment or harm.
- **Blackmail or Extortion:** Attackers may use compromising information to blackmail or extort victims.

The impact of a cybersecurity attack on people can be far-reaching, affecting various aspects of their lives. Timely detection, response, and preventive measures, such as strong passwords, cybersecurity awareness, and software updates, are essential to mitigate these risks.

Facilities and Critical Facilities

Cybersecurity attacks can have wide-ranging impacts on facilities, including critical infrastructure, industrial facilities, state government buildings, and data centers. The extent of these impacts depends on the type and sophistication of the attack, the facility's level of cybersecurity preparedness, and the criticality of the systems and operations involved. Potential impacts may include:

- **Disruption of Operations:** Cyberattacks can lead to the disruption of facility operations, causing downtime that can be costly and disruptive.
- **Production Delays:** Manufacturing and industrial facilities may experience delays in production processes, affecting supply chains and delivery schedules.
- **Revenue Loss:** Downtime and operational disruptions can result in financial losses due to lost sales, contracts, or customer trust.
- **Remediation Costs:** Facilities must invest in cybersecurity measures and incident response efforts, incurring additional costs.
- **Data Breach:** Facilities that store sensitive data, such as customer information or proprietary research, may suffer data breaches, leading to data loss or theft.
- **Regulatory Penalties:** Compliance violations and regulatory fines may be imposed for failing to protect sensitive data.
- **Physical Safety Risks:** Attacks on critical infrastructure facilities, such as power plants or water treatment plants, can pose physical safety risks to the public and the environment.
- **Industrial Accidents:** Industrial control systems attacks can lead to accidents or malfunctions with serious safety implications.
- **Loss of Control:** Cyberattacks targeting operational technology systems can lead to a loss of control over critical processes, affecting safety and efficiency.

Rhode Island has four data centers, run by two organizations, which may be at a higher susceptibility to cyber-attacks. All data centers are located in Providence.

Environment

Attacks on facilities with environmental controls can lead to environmental damage, such as chemical spills or pollution which can affect the surrounding ecosystem and wildlife.

State Operations and Assets

Cyberattacks on state operations can have wide-ranging impacts on state agencies and the services provided to citizens. The effects of these attacks can vary depending on factors like the type of attack, the target's level of cybersecurity readiness, and the criticality of the systems involved, and may include:

- Disruption of Government Services: Cyberattacks can disrupt state services, leading to delays in processing applications, issuing licenses, or providing essential public services.
- Website Downtime: State websites may become inaccessible, hindering access to information and online services.
- Financial Costs: States may incur significant expenses related to incident response, system recovery, and cybersecurity improvements.
- Loss of Revenue: Disrupted services can lead to revenue losses, impacting state budgets and financial stability.
- Confidential Data Exposure: Cyberattacks can result in the exposure of sensitive citizen and employee data, including Social Security numbers, health records, and financial information.
- Regulatory Penalties: Non-compliance with data protection regulations can lead to penalties and legal consequences.
- Election Integrity: Attacks on election systems can compromise the integrity of elections, eroding trust in the democratic process.
- Emergency Response: Cyberattacks on public safety and emergency response systems can hinder responses to disasters or crises.
- Reputation Damage: Publicized cyberattacks can damage citizens' trust in state agencies and institutions.
- Legal and Regulatory Consequences: States may face legal liability for cybersecurity incidents, leading to lawsuits, fines, and settlements.

Consequence Analysis

This consequence analysis lists the potential impacts of this hazard on various elements of community and state infrastructure. The impact of this hazard is evaluated in terms of disruption of operations, recovery challenges, and overall wellbeing to all Rhode Island residents and first responder personnel. The consequence analysis supplements the hazard profile by analyzing specific impacts.

Table 160: Cybersecurity Incident Consequence Analysis

Subject	Potential Impacts
Impact on the Public	The public is heavily reliant on technology for daily life. Any disruption to this technology caused by a cyber incident could impair the ability to conduct activities such as communications and mobile banking. Although mostly indirect, public health impacts may include loss of access of important medical information and services, personal information, and unwanted sharing of that information. Disruption in attaining medical help or resources may delay receiving proper medical attention or care.
Impact on Responders	If a cybersecurity incident were to directly impact the communications infrastructure relied upon by first responders, it would create severe disruptions in the ability to provide response services. If a cybersecurity event were to affect the 911 operations, response capabilities would be impacted significantly increasing critical response times.
Continuity of Operations	Rhode Island agencies maintain continuity plans which can be enacted as necessary based on the situation. A cybersecurity event may impact an agency's ability to maintain continuity of operations based on the hazard's potential to impact power or communications infrastructure. Specifically, agencies that rely on electronic backup of critical files are vulnerable to cyber incidents. A cyber incident that disrupts access to technology at both the primary and alternative facilities would be catastrophic.
Delivery of Services	The delivery of goods and services is heavily reliant on technology for the facilitation of transactions. A cyber incident could significantly disrupt the delivery of goods and services for businesses that rely on technology for the delivery of their materials.

Table 160: Cybersecurity Incident Consequence Analysis

Subject	Potential Impacts
Property, Facilities, and Infrastructure	Property and facilities may become unusable because of a cyber incident, particularly if their infrastructure is reliant on technology for sustainability. A significant majority of critical infrastructure systems are tied to technology through virtual operations and supervisory control and data acquisition systems. A cyber incident could disable many systems which control critical infrastructure, as well as traffic control, dispatch, utility, and response systems.
Impact on Environment	Targeted cyber incidents can impact water or wastewater treatment facilities. The disruption of the systems tied to this infrastructure could cause water pollution or contamination. In addition, a cyber incident could impact the environment if a release of a hazardous material was triggered as a cascading effect of the incident.
Economic Conditions	A significant cyber incident could have ramifications on the state economy. Society is heavily reliant on electronic-based commerce through mobile banking, automated teller machines, and electronic trading. Any disruption to daily activities by a cyber incident could effectively halt the ability to conduct transactions electronically.
Public Confidence in Governance	In the case of a cyber incident in which significant amounts of data is stolen, the government's inability to protect confidential personal data would impact confidence in the state. Such an incident would also subsequently cause pause regarding the security of using electronic systems for government services.

4.22.7 Hazard Planning Significance

Based on a review of available data, the SIHMC determined cybersecurity incident to have a high planning significance for Rhode Island.

4.23 Hazardous Material Incident

4.23.1 Hazard Description

Hazardous materials are any substances that pose a risk to health, life, or property when released or improperly handled. Generally, the term refers to materials with hazardous chemical or physical properties, though sometimes biological agents can fall under this category. The basic types of hazardous materials may be categorized according to more than six different systems; but the categories of U.S. Emergency Planning and Community Right-to-Know Act (42 U.S.C. 11002) provide a general guide to hazardous materials:



- **Extremely Hazardous Substances:** Materials that have acutely toxic chemical or physical properties and may cause irreversible damage or death to people or harm the environment if released or used outside their intended use.
- **Hazardous Substances:** Materials posing a threat to human health and/or the environment, or any substance designated by the EPA to be reported if a designated quantity of the substance is spilled into waterways, aquifers, or water supplies or is otherwise released into the environment.

4.23.2 – Location and Extent

The entire State of Rhode Island is vulnerable to hazardous materials incidents. Hazardous materials incidents are generally classified as:

- **Fixed Facility Incidents:** Commercial Facilities and Superfund Sites
- **Transportation Incidents:** Highway, Railway, Pipeline, Air, and Water

Tier II facilities, also known as Tier II Reporting facilities, refer to certain types of industrial or commercial establishments that are required to report information about the hazardous chemicals they store or use. This reporting is mandated under the Emergency Planning and Community Right-to-Know Act under Section 312. Key factors in Tier II reporting include:

- **Hazardous Chemicals:** Tier II facilities are those that store or use hazardous chemicals in quantities that meet or exceed specific thresholds established by Emergency Planning and Community Right-to-Know Act. Hazardous chemicals can include substances such as flammable liquids, toxic gases, and corrosive materials.
- **Reporting Thresholds:** Facilities must report if they have a quantity of any hazardous chemical at the facility that equals or exceeds established thresholds. These thresholds can vary depending on the specific chemical and are typically set in terms of pounds (or a lower threshold for Extremely Hazardous Substances).
- **Reporting Frequency:** Tier II reports must be submitted annually to the State Emergency Response Commission, the Local Emergency Planning Committee, and local fire department.
- **Information Required:** Tier II reports must include detailed information about the hazardous chemicals stored or used at the facility, including the chemical name, location on the site, quantities, and specific health and physical hazards.
- **Community Right-to-Know:** In addition to assisting emergency responders, Tier II reporting also serves the "Community Right-to-Know" aspect of Emergency Planning and Community Right-to-Know Act, allowing the public to access information about hazardous chemicals in their communities. This information is typically made available through public databases.
- **Enforcement:** Non-compliance with Tier II reporting requirements can result in penalties and fines. Facilities are responsible for ensuring accurate and timely reporting.

Transportation-related hazardous materials incidents can encompass a wide range of scenarios involving the transportation of hazardous materials, including chemicals, flammable substances, radioactive materials, and other

dangerous goods. These incidents can occur during the movement of these materials by road, rail, air, or sea. These transportation-related hazardous materials incidents can include:

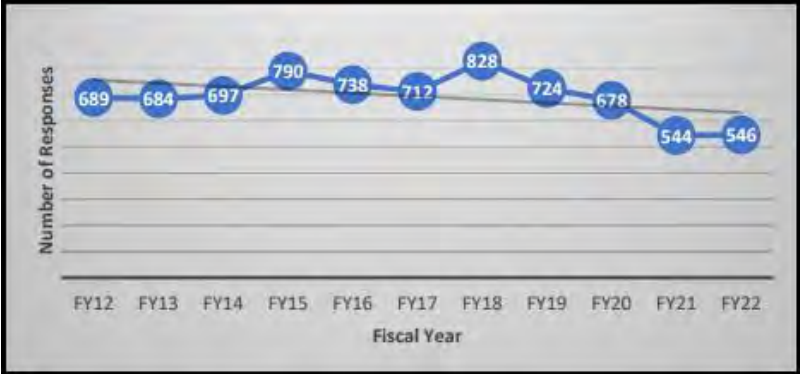
- **Chemical Spills on Highways:** Accidents involving trucks carrying hazardous chemicals can result in spills on highways. This can lead to the release of toxic, flammable, or corrosive substances, posing risks to people, the environment, and emergency responders.
- **Train Derailments:** Train derailments can result in the release of hazardous materials from tanker cars. These incidents can occur on both freight and passenger rail lines and may involve chemicals, fuels, or other hazardous substances.
- **Aircraft Hazmat Incidents:** Cargo planes and commercial aircraft can carry hazardous materials as cargo. Incidents may involve leaks, fires, or other issues related to hazardous materials on board.
- **Marine Spills:** Incidents involving the transport of hazardous materials by sea can lead to marine spills. These spills may involve oil, chemicals, or other substances, and can have significant environmental and economic consequences.
- **Pipeline Leaks:** Pipelines transport hazardous liquids and gases over long distances. Leaks or ruptures in pipelines can result in the release of hazardous materials into the environment.
- **Radiological Transport Incidents:** The transport of radioactive materials, including medical isotopes and nuclear fuel, carries the risk of accidents that can result in the release of radioactive substances. These incidents can have serious health and environmental consequences.
- **Chemical Fires in Transit:** Fires in transit vehicles carrying hazardous chemicals can be particularly challenging to control. The fire may cause chemical reactions, leading to toxic smoke or explosions.
- **Cargo Container Incidents:** Shipping containers transported by truck or rail can contain hazardous materials. Incidents involving these containers may include leaks, fires, or chemical reactions.
- **Intermodal Transport Incidents:** When goods are transferred between different modes of transportation (e.g., ship to truck), there is the potential for mishandling or spills during these transfers.

Counties with multiple chemical facilities, such as Providence, experience a greater risk of a chemical incident than other locations. However, almost every community in Rhode Island has at least one fixed facility that stores, produces, or utilizes hazardous material. Hundreds of thousands of hazardous materials shipments move through Rhode Island annually. These shipments can occur at any time, day or night, and by means of road, rail, air, and water, and often through areas with urbanized, high volume traffic routes.

4.23.3 Previous Occurrences

The RIDEM Office of Emergency Response operates as an all-hazard response program, incorporating the oil spill prevention and response functions carried out in Rhode Island. In 2022, there were 546 emergency response cases. Below is a graph showing the number of incidents across the past decade.

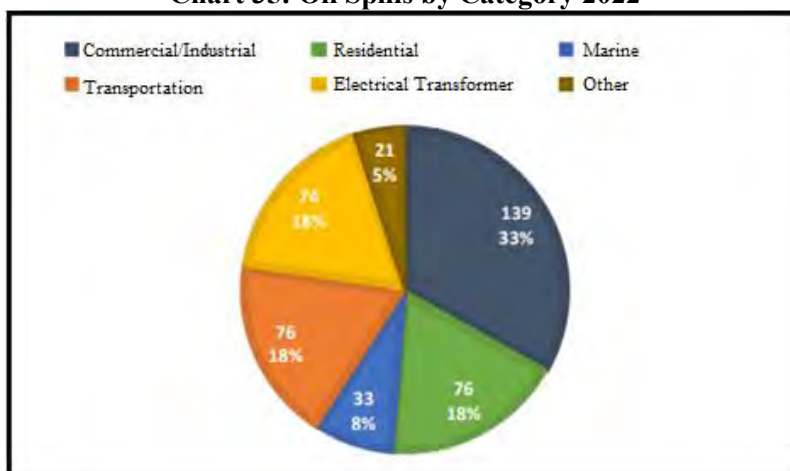
Chart 34: Office of Emergency Response Incidents 2012-2022



Source: Oil Spill Prevention, Administration and Response Reports 2022

Circumstances causing oil spills and their consequential environmental impacts vary. The primary categories of oil spills and their relative percentages for 2022 are charted below:

Chart 35: Oil Spills by Category 2022



Source: Oil Spill Prevention, Administration and Response Reports 2022

The greatest percentage of oil spills (33%) during 2022 were caused by commercial/industrial incidents.

Additionally, the RIDEM reported the following recent hazardous materials incidents in Rhode Island:

- Oil Tanker Spill, Providence:** On October 3, 2018, a tanker truck rolled over on the I-95 north bound ramp off Allens Avenue in Providence. The truck was carrying 11,500 gallons of gasoline, and all but approximately 400 gallons were released into the surrounding area! Contaminated areas included a stream connected to the Providence River, grass bordering the east side of the street and the storm drain system discharging directly into the Providence River. Remediation efforts lasted several days, and extremely high levels of gasoline vapors were present in the storm drain system following the spill. These hazardous levels reached higher than 99% of the lower explosive limit, meaning the air was capable of combustion. Over 6,000 gallons of firefighting foam, water and gasoline were pumped out of the underground line beneath Allens Avenue, and a total of 17,900 gallons was collected throughout the entire cleanup process.
- Fire, Warwick:** On January 4, 2019, a fire started in a parking garage at the T.F. Green Airport in Warwick. Multiple cars on the third floor erupted in flames, and gasoline combusted as well. Firefighting foam and water runoff travelled into storm drains located near the garage and required removal. Five gallons of concentrated firefighting foam was used to completely extinguish the fire. A total of 650 gallons of firefighting foam, water and gasoline were collected throughout the entire cleanup process.
- Unknown Substances Incident, Pawtucket:** On March 20, 2019, a RIDEM emergency responder travelled to a scene located on Pine Street in Pawtucket. Upon arrival, the responder discovered a large, unoccupied warehouse littered with unknown substances and ample debris. Nine chemical drums filled with unknown liquids were found near storm drains, displaying evidence of contamination to the surrounding ground. It was certainly feasible that the oily substances seeped into the storm drain, potentially posing an environmental threat. In addition to those drums, the main courtyard contained many large piles of assorted solid waste, multiple additional drums, and five pallet mounted totes which appeared to release unknown materials due to vandalism. The unknown substances were reported to be old beer and liquor, but the consistency of the substances did not align with those reports.

The United States Department of Transportation Pipeline and Hazardous Materials Safety Administration is a federal agency responsible for regulating the safe and secure transportation of hazardous materials by all modes of transportation, including pipelines, trucks, trains, ships, and aircraft. Their primary mission is to protect people and the environment from the risks associated with the transportation of hazardous materials. They play a crucial role in safeguarding public safety, protecting the environment, and ensuring the integrity of the nation's hazmat transportation

infrastructure. Its work encompasses a wide range of hazardous materials, including chemicals, radioactive materials, explosives, and more. The agency collaborates with industry stakeholders, state and local governments, and other federal agencies to achieve its safety and security objectives.

For the five-year period from 2018 to 2023, United States Department of Transportation Pipeline and Hazardous Materials Safety Administration has reported 275 hazardous materials incidents in Rhode Island. Of these incidents, four were considered as a serious bulk release, and one required a serious evacuation. No fatalities or injuries were reported, and no radiological incidents were reported.

4.23.4 Probability of Future Events

Data from the Rhode Island Department of Environmental Management and from United States Department of Transportation Pipeline and Hazardous Materials Safety Administration indicates that the probability of a hazardous material incident during any given year is 100%. However, data indicates that the large majority of these incident will be small in scale and cause no evacuation, injuries, or deaths, Rhode Island's first line of defense in protecting public health, safety, and welfare in an environmental emergency is the RIDEM. Their emergency responders are prepared to handle incidents of great variety including everything from a spill of a few gallons to an entire tanker full of petroleum; from a single abandoned drum to biological and chemical weapons. Highly trained first responders are on-call 24 hours a day, seven days a week.

4.23.5 Projected Changes in Location, Intensity, Frequency, and Duration

Projecting specific changes in the location, intensity, and frequency of hazardous materials events involves numerous variables, including future industrial activities, changes in transportation systems, and more stringent regulatory measures. The location of hazardous materials events is often influenced by urbanization and industrialization. The vulnerability of communities to hazardous materials incidents may change based on demographic shifts, land-use changes, and socioeconomic factors. Population density and proximity to industrial sites influence the potential impact of such incidents.

The continued transportation of hazardous materials by road, rail, air, and sea poses inherent risks. Changes in transportation patterns, such as increased volumes or altered routes, can impact the potential for accidents and spills. However, the adoption of new technological solutions, such as sensor technologies, remote monitoring, and safety measures, can contribute to the mitigation of hazardous materials risks.

Changes in climate patterns, such as extreme weather events, floods, or wildfires, can influence the frequency and intensity of hazardous materials incidents. Events like hurricanes, floods, or wildfires may impact facilities handling hazardous materials.

As previously noted, State of Rhode Island facilities have seen no major changes in the past five years, with only modest repairs and upgrades being conducted and no major rehabilitation or construction projects completed. As such, the risk to state assets has remained static since the completion of the 2019 SHMP. Additionally, with the exception of the two previously identified rehabilitation and construction projects, no large-scale facility projects were noted in any agency budgets or reports.

4.23.6 Vulnerability and Impact

People and Communities

A hazardous materials release can have serious and immediate impacts on human health and safety, as well as long-term effects depending on the nature of the hazardous materials involved, the release's magnitude, and the proximity of individuals to the incident. Acute health effects from a hazardous materials release can include:

- **Chemical Exposure:** Depending on the type of hazardous material, exposure can lead to symptoms such as respiratory distress, skin burns, eye irritation, nausea, vomiting, and headaches.
- **Toxicity:** Exposure to highly toxic substances can cause severe poisoning, organ damage, and even death.
- **Asphyxiation:** Some hazardous materials, like certain gases, can displace oxygen and lead to asphyxiation when inhaled in high concentrations.

- **Physical Injuries:** Explosive releases or fires involving hazardous materials can cause physical injuries such as burns, cuts, and blunt force trauma.
- **Psychological Trauma:** Witnessing or being affected by a hazardous materials incident can lead to psychological trauma, including post-traumatic stress disorder and anxiety.

Long-term health effects from a hazardous materials release can include:

- **Chronic Illnesses:** Exposure to hazardous materials may lead to chronic health conditions, including cancer, respiratory diseases, neurological disorders, and reproductive problems.
- **Delayed Effects:** Some hazardous substances have delayed health effects, with symptoms appearing days, months, or even years after exposure.

Additionally, a hazardous material release can result in impacted populations requiring:

- **Evacuation:** To protect public safety, authorities may order evacuations of affected areas, displacing residents from their homes.
- **Temporary Shelter:** Evacuated individuals may require temporary shelter, food, and medical care.

While all citizens are vulnerable to the effects of hazardous materials, of particular concern is socially vulnerable and at-risk populations. These populations may have difficulty with medical issues, poverty, extremes in age, and communications due to language barriers. These populations suffer a disproportionate impact due to:

- Constrained financial resources potentially hampering evacuation
- Limited access to healthcare
- Limited access to information

The following table details identified socially vulnerable and at-risk populations for the State of Rhode Island:

Table 161: Rhode Island 2020 Vulnerable Populations

Jurisdiction	Under Five	Over 65	Language Barrier	Poverty	Disability
Bristol County	1,976	8,372	6,552	3,403	4,753
Kent County	8,148	26,437	16,525	11,925	24,694
Newport County	3,641	13,819	7,794	8,307	9,105
Providence County	36,846	84,768	205,490	90,522	87,498
Washington County	4,760	18,797	7,790	11,815	13,832
State of Rhode Island	55,370	152,183	240,326	125,101	139,927

Source: US Census Bureau

Facilities and Critical Infrastructure

The direct risk or vulnerability to property and facilities from a hazardous materials incident is generally limited. Impacts include restricting access to a facility or potential damage to the facility from corrosive agents. Direct risk and vulnerability to actual structures is limited due to the characteristics of a hazardous materials incident.

Critical facilities and infrastructure may suffer secondary impacts from a hazardous materials incident. Access may be restricted due to closures, causing employee absenteeism which could indirectly impact the ability of a critical facility to operate. Without necessary operators, critical infrastructure may be susceptible to indirect failure.

Environment

A hazardous materials release can have significant and lasting impacts on the environment, depending on the type and quantity of hazardous materials involved, the location of the release, and the effectiveness of response and cleanup efforts. Environmental impacts can range from immediate and localized effects to long-term ecological damage and may include:

- **Soil Contamination:** Hazardous materials can seep into the soil, contaminating it with toxic substances. This can affect soil quality and fertility.
- **Agricultural Damage:** Contaminated soil may harm crops, leading to reduced agricultural yields or the need to abandon affected fields.
- **Surface Water Contamination:** Hazardous materials can enter rivers, lakes, and streams, leading to water pollution. This can harm aquatic ecosystems, fish, and wildlife.
- **Groundwater Contamination:** Contaminants can infiltrate underground aquifers, potentially affecting drinking water supplies and requiring costly remediation efforts.
- **Habitat Destruction:** Contamination can harm natural habitats, disrupt ecosystems, and threaten the survival of plant and animal species.
- **Bioaccumulation:** Toxic substances can accumulate in the food chain, leading to health issues for wildlife and potentially impacting humans who consume contaminated organisms.

Some hazardous materials are persistent and can remain in the environment for extended periods, causing ongoing harm. Environmental recovery from hazardous materials releases can be slow and challenging, requiring extensive remediation efforts.

State Operations and Assets

A hazardous materials incident can have wide-ranging impacts on state operations. These impacts can disrupt state operations, strain resources, and pose challenges to maintaining public order. Some of the impacts of a hazardous materials release on state operations include:

- **Emergency Response and Healthcare:** State of Rhode Island agencies must rapidly mobilize emergency response teams, medical personnel, and healthcare facilities to address a release. The surge in demand for medical resources can strain healthcare systems, including hospitals, clinics, and emergency services.
- **Resource Allocation:** State agencies, including RIEMA, may need to allocate resources to respond to the incident. This includes personnel, equipment, and facilities.
- **Transportation and Supply Chain Disruption:** Transportation infrastructure closures can affect the movement of essential goods and services, including medical supplies, food, and fuel.
- **Economic Impact:** The economic consequences of a hazardous materials incident can be severe. Business closures, reduced consumer confidence, and trade disruptions can lead to financial losses, unemployment, and economic instability.
- **Public Services:** Essential public services, such as law enforcement, fire services, and sanitation, may be stretched thin due to the demands of responding to an incident.
- **Agency Coordination:** Coordination and communication among various state agencies and with federal authorities will be tested during a hazardous materials incident. RIEMA will activate emergency response plans and incident command structures.

Hazardous materials response is conducted by Rhode Island's Hazardous Materials and Mass Decontamination Team, which is made up of six Hazardous Material Teams (Cranston, East Providence, Hope Valley, Providence, Warwick, and Woonsocket), and four Decontamination Teams (Hopkins Hill, Kingston, North Kingstown, and North Providence) overseen by RIEMA. The teams are used to assess and manage the consequences of a hazardous materials

Consequence Analysis

This consequence analysis lists the potential impacts of this hazard on various elements of community and state infrastructure. The impact of this hazard is evaluated in terms of disruption of operations, recovery challenges, and overall wellbeing to all Rhode Island residents and first responder personnel. The consequence analysis supplements the hazard profile by analyzing specific impacts.

Table 162: Hazardous Materials Incident Consequence Analysis

Subject	Potential Impacts
Impact on the Public	Cities within Rhode Island with dense populations, particularly along major travel routes, are the most vulnerable. Particularly vulnerable groups include infants and young children in day-care centers, children in schools, the elderly in residential facilities, and hospital patients. Varying materials will have different effects on the population. Protective measures will need to be taken particularly for those of the most vulnerable communities.
Impact on Responders	Hazardous material incidents can create a dangerous environment and significant challenge for first responders. First responders may have to manage the evacuation of people from the area impacted by an incident, as well as direct traffic, close roads, operate shelters, and take care of the injured and sick. First responders must control their own exposure to the incident and ensure the correct equipment is utilized. Equipment may also be damaged or destroyed due to the impact of the incident, which may lead to a decrease in response capabilities.
Continuity of Operations	Rhode Island agencies maintain continuity plans which can be enacted as necessary based on the situation. A hazardous materials incident may impact an agency's ability to maintain continuity of operations based on the incidents potential to cause workforce absenteeism, contamination, or destruction of public facilities.
Delivery of Services	The ability to deliver services can be impacted locally, regionally, or statewide depending on the characteristics of the incident. To reduce the public's potential exposure to dangerous materials, roadway and bridge closures may be required, as well as transit service disruptions. Businesses and places of commerce may completely shut down due to chemical incidents, which leads to the disruption of goods and services.
Property, Facilities, and Infrastructure	Transportation, governmental operations, and infrastructure facilities may be disrupted during a significant incident. Roads and bridges can be completely obstructed and require cleanup. Incidents can impact access to homes and critical entities such as hospitals, schools, and supermarkets, as well as other critical facilities. Safe access to homes, vehicles, structures, and resources may adversely affect response activities. Power loss can lead to disruption of critical infrastructure and technology.
Impact on Environment	Crops and livestock are extremely susceptible to the adverse effects of hazardous materials incidents. Hazardous materials incidents may impact the environment long-term by disturbing or killing wildlife and adversely affecting nature preserves.
Economic Conditions	Hazardous materials incidents pose a fiscal impact on the local and state governments. Local, county, and state resources may be required during a large incident therefore reducing their availability for future events. Additionally, private businesses may not be able to maintain operations during or after an incident if they are impacted, which would impact the economy.
Public Confidence in Governance	The public's confidence in the state's governance is affected by immediate local and state response through direct and effective actions. Efficiency in response and recovery operations is critical in keeping public confidence high.

4.23.7 Hazard Planning Significance

Based on a review of available data, the SIHMC determined hazardous materials incident to have a moderate planning significance for Rhode Island.

4.24 Infrastructure Failure

4.24.1 Hazard Description

Infrastructure failure refers to the malfunction, breakdown, or collapse of critical infrastructure systems or components that are essential for the functioning of the State. These failures can disrupt essential services, impact public safety, and lead to economic losses. There are many potential causes of infrastructure failure, including:



- **Aging Infrastructure:** Many infrastructure systems, such as bridges, roads, and water pipelines, have exceeded their designed lifespan. Over time, the materials degrade, and the risk of failure increases.
- **Earthquakes:** Seismic events can damage or destroy buildings, bridges, dams, and utility systems.
- **Floods:** Flooding can damage electrical systems, disrupt transportation, and contaminate water supplies.
- **Hurricanes and Cyclones:** High winds, storm surges, and heavy rainfall can damage infrastructure, especially in coastal regions.
- **Extreme Heat:** Prolonged periods of extreme heat can cause roads to buckle, power lines to sag, and strain electrical grids.
- **Freezing Temperatures:** Cold weather can lead to frozen water pipes, which can burst and disrupt water supply.
- **Design Flaws and Poor Maintenance:** Inadequate design, construction, or maintenance practices can result in structural weaknesses or deteriorating infrastructure.
- **Corrosion and Erosion:** Infrastructure components, particularly those involving metals, can deteriorate due to corrosion over time. Erosion of natural landscapes, such as coastal erosion, can damage infrastructure like seawalls and levees.
- **Material Failures:** Inadequate materials or the use of substandard materials during construction can lead to premature infrastructure failure.
- **Overloading and Overuse:** Bridges, roads, and other structures can fail if they are subjected to loads beyond their designed capacity. Water and wastewater systems can fail if they are overwhelmed by excessive demand.
- **Cyberattacks:** Critical infrastructure systems, such as power grids, water treatment plants, and transportation systems, can be vulnerable to cyberattacks, which can disrupt operations and compromise safety.
- **Terrorism and Sabotage:** Deliberate acts of terrorism or sabotage can target critical infrastructure, leading to failures and disruptions.
- **Environmental Changes:** Long-term environmental changes, such as sea-level rise due to climate change, can threaten coastal infrastructure.

Infrastructure failures can have significant consequences, including economic losses, public safety risks, and disruptions to daily life. Preventing such failures and ensuring the resilience of critical infrastructure often require proactive measures such as regular maintenance, improvements in design and construction practices, disaster preparedness, and investments in modernization and upgrades.

4.24.2 – Location and Extent

Details concerning Rhode Island’s infrastructure were sourced from the 2020 Report Card for Rhode Island’s Infrastructure from the Rhode Island Section of the American Society of Civil Engineers (ASCE). The report provides information on infrastructure components and provides a letter grade to indicate condition. Grades are issued based on the following scale:

Table 163: ASCE Infrastructure Grade System

Grade	Description
A: Exceptional	The infrastructure in the system or network is generally in excellent condition, typically new or recently rehabilitated, and meets capacity needs for the future. A few elements show signs of general deterioration that

Table 163: ASCE Infrastructure Grade System

Grade	Description
	require attention. Facilities meet modern standards for functionality and are resilient to withstand most disasters and severe weather events.
B: Adequate for Now	The infrastructure in the system or network is in good to excellent condition; some elements show signs of general deterioration that require attention. A few elements exhibit significant deficiencies. Safe and reliable with minimal capacity issues and minimal risk.
C: Mediocre, Requires Attention	The infrastructure in the system or network is in fair to good condition; it shows general signs of deterioration and requires attention. Some elements exhibit significant deficiencies in conditions and functionality, with increasing vulnerability to risk.
D: Poor, At Risk	The infrastructure is in poor to fair condition and mostly below standard, with many elements approaching the end of their service life. A large portion of the system exhibits significant deterioration. Condition and capacity are of significant concern with strong risk of failure.
F: Failing/Critical, Unfit for Purpose	The infrastructure in the system is in unacceptable condition with widespread advanced signs of deterioration. Many of the components of the system exhibit signs of imminent failure.

Source: ASCE

The following table indicates the grades by the State of Rhode Island received for infrastructure components:

Table 164: ASCE Rhode Island Infrastructure Grades

Infrastructure Component	Grade
Roads	D
Bridges	D-
Drinking Water	C+
Wastewater	C
Energy	C+
Rail	B-
Ports	C
Overall Grade	C-

Source: RIEMA

Rhode Island has a network of more than 6,000 miles of public roads and highways owned by local, state, and federal entities. About 23 % consists of rural roads while 77 % are urban. The following table details the conditions of these roads:

Table 165: Condition of Rhode Island of Transportation Owned and Maintained Highway Systems

Highway Systems	Good	Fair	Poor
Interstates	55.05%	44.95%	0%
Non-Interstates	18.01%	62.08%	19.91%

Source: ASCE

Rhode Island is home to the highest percentage of structurally deficient bridges in the country, with 17.5% reported as structurally deficient. Of the state's ten most traveled structurally deficient bridges, five are on Interstate 95. A structurally deficient bridge is a bridge that has one or more components showing signs of deterioration or damage, affecting its overall structural integrity. A structurally deficient bridge is not necessarily in immediate danger of collapse, but rather contains elements that require attention and may need repair, rehabilitation, or replacement. The following table details structurally deficient bridges in the State of Rhode Island:

Table 166: Structurally Deficient Bridges Crossing Rivers

County	Location	Year Constructed	Average Daily Crossings
Kent	Route 37 EB over Amtrak	1963	47,277
Kent	Route 37 EB & WB over US 1 Post Road	1965	46,000
Kent	Route 113 Main Avenue over Amtrak	2006	34,118
Providence	I-95 NB & SB over Amtrak	1964	134,500
Providence	I-95 NB & SB over Wellington Avenue	1964	134,500
Providence	I-95 NB & SB over US 1 Elmwood Avenue	1965	134,500
Providence	I-195 WB over Seekonk River	1969	80,500
Providence	Route 146 over Route 15 Mineral Spring Avenue	1957	69,109
Providence	Route 146 Ed Dowl Highway over Branch Avenue	1957	65,800
Providence	US 6 EB & WB over US 6A Hartford Avenue Ramp	1969	53,619
Providence	Route 37 EB & WB over Route 2 New London Avenue	1965	47,871
Providence	Route 146 Ed Dowl Hwy over Cobble Hill Road	1956	47,393
Providence	Route 146 Ed Dowl Hwy over Route 246 Charles Street	1956	47,208
Providence	Route 37 EB over Pawtuxet River	1963	36,090
Providence	Route 10 Huntington Exp over I-95 NB & SB, Amtrak	1965	34,764
Providence	Ramp over Amtrak, Woon River, Ramp Road	1988	31,200
Providence	Dean St over US 6 Harris Avenue & Amtrak	1972	31,112
Providence	Route 37 EB & WB over Wash Sec Bike Path	1968	30,529
Providence	Route 37 EB & WB over Cranston Street	1968	30,528
Providence	US 44 Smith St over I-95 NB & SB	1967	30,299
Providence	I-295 NB over Route 146 Eddie Dowling Highway	1969	27,992
Providence	Route 37 EB & WB over I-295 NB	1968	27,900
Providence	Route 37 EB & WB over I-295 SB	1969	27,900
Providence	Route 10 Huntington Exp over US 1 Elmwood Avenue	1965	26,767
Washington	US 1 Tower Hill Road over Route 138	1969	46,800

Source: Federal Highway Administration National Bridge Inventory

While RIDOT manages the majority of the state's 782 bridges, 191 are managed by towns, cities, other state agencies, private companies, and the Rhode Island Turnpike and Bridge Authority. The following table indicates bridge ownership within the State and the number of bridges in each condition category:

Table 167: Condition of Rhode Island of Bridges

Bridge Owner	Total Bridges	Condition		
		Good	Fair	Poor
Rhode Island Department of Transportation	588	89	369	130
Others	191	49	98	44

Source: ASCE

RIDOH reports the following concerning public water systems in Rhode Island:

- There are 481 public water systems across the state
- Surface water reservoirs supply approximately 85 % of public water
- The Scituate Reservoir is the largest source of Rhode Island's surface water supply, providing approximately 60% of Rhode Islanders with drinking water.
- Approximately 150,000 Rhode Islanders have groundwater supplied by a private wells
- 95% of the public water systems serve fewer than 3,300 people.

There are 19 major wastewater treatment systems in Rhode Island, composed of treatment systems and thousands of miles of underground piping and approximately 240 pumping stations. Approximately 73% of Rhode Island's citizens

are served by these 19 municipally owned treatment facilities. Research conducted for the Rhode Island Water Quality 2035 report indicates that:

- The average treatment plant is 69 years old
- 16 out of 19 facilities have received major upgrades since 2000
- the age of up to 2,600 miles of pipe in the collection system was unknown, but portions were “decades old”

Major sources of energy for Rhode Island include natural gas, fuel oil, and renewable energy. Data indicates that the commercial/industrial sector consumes approximately 36.8% of all generated energy, while the residential sector consumes 30.7% and the transportation sector consumes 32.5%.

In Rhode Island, the energy sector is regulated by the Rhode Island Public Utilities Commission. The Public Utilities Commission is an independent state agency responsible for regulating various utility services, including electricity, natural gas, water, and telecommunications. The Public Utilities Commission establishes rates, rules, and regulations for utility companies operating within Rhode Island. Additionally, energy regulation can involve federal entities as certain aspects of the energy sector may fall under the jurisdiction of federal agencies like the Federal Energy Regulatory Commission.

Rhode Island is home to approximately two gigawatts of electric generating capacity, with six power plants over 50 megawatts. Virtually all major generating facilities in the state use natural gas as the primary fuel, accounting for approximately 98% of in-state generation capacity. Many small, distributed renewable energy systems also exist in the state, with approximately 2,150 distributed renewable energy systems representing approximately 100 megawatts of wind, solar, and hydropower capacity.

The following table presents major electric generating stations in Rhode Island:

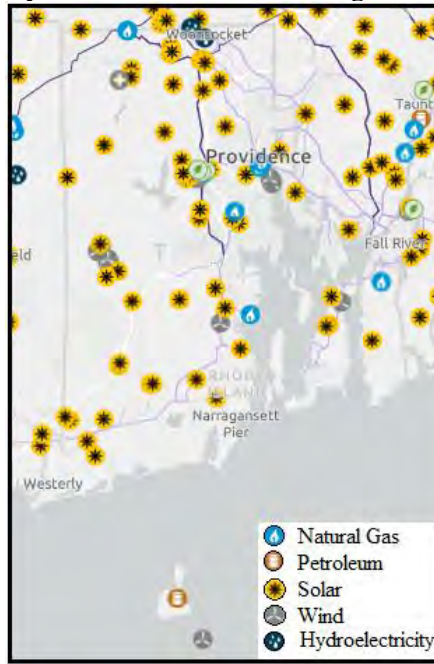
Table 168: Rhode Island Electric Generating Stations

Generating Plant	County	Generating Capacity (Megawatts)	Plant Type
Central Power Plant	Providence County	11	Natural Gas Fired Combustion Turbine
Manchester Street	Providence County	515	Natural Gas Fired Combined Cycle
Ocean State Power	Providence County	254.2	Natural Gas Fired Combined Cycle
Ocean State Power II	Providence County	254.2	Natural Gas Fired Combined Cycle
Pawtucket Power Associates	Providence County	68.8	Natural Gas Fired Combined Cycle
Rhode Island Hospital	Providence County	10.4	Natural Gas Steam Turbine
Rhode Island State Energy Center	Providence County	596	Natural Gas Fired Combined Cycle
Tiverton Power Plant	Newport County	272.5	Natural Gas Fired Combined Cycle
Toray Plastic America's CHP Plant	Washington County	20	Natural Gas Fired Combustion Turbine

Source: U.S. Energy Information Administration

The following map details the location of these stations, along with smaller generating stations and green energy generating stations:

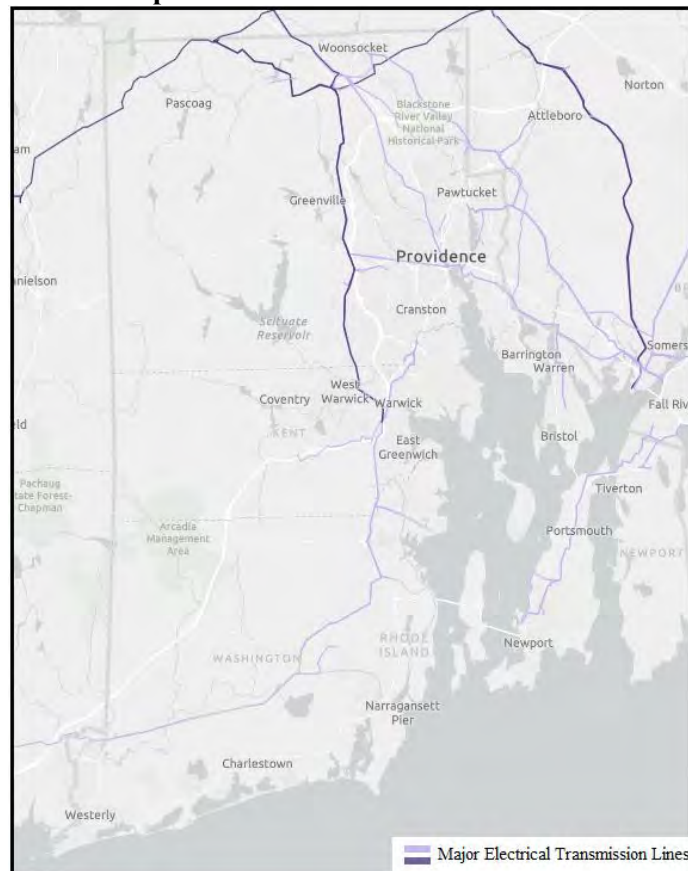
Map 73: Electrical Generating Stations



Source: U.S. Energy Information Administration

Electrical utilities within the State of Rhode Island are privately managed and maintained. The following map details electrical transmission lines within the state, all of which are vulnerable to the effect of winter storms:

Map 74: Electrical Transmission Lines

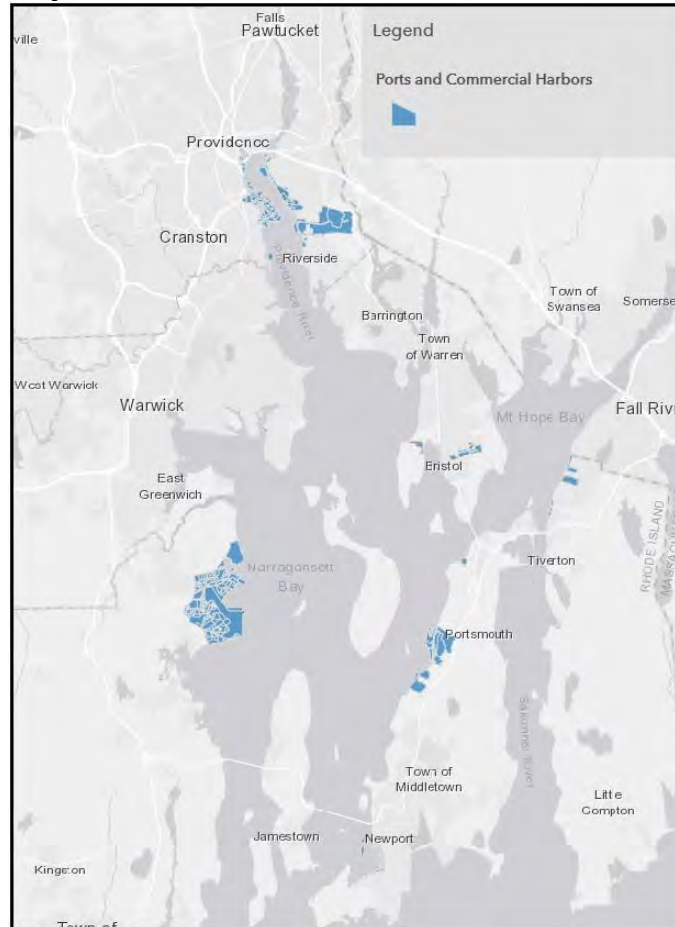


Source: U.S. Energy Information Administration

The following major Rhode Island ports serve the state and surrounding region:

- Newport Harbor
- Port of Galilee
- Port of Providence
- Port of Davisville
- Tiverton Marine Terminal

Map 75: Rhode Island Commercial Ports and Harbors



Source: RIGIS

In 2018, the state's ports moved a combined total of 9,000,000 tons of freight. The Port of Providence moves the largest volume of freight, accounting for nearly 8,500,000 tons. The Port of Providence ranks 60th in the nation by tonnage moved, handling a variety of products and the port receiving fuel products and refined petroleum from facilities in New Jersey, Pennsylvania, Canada, and Europe. The Port of Davisville is one of the top ten auto ports in North America. The Port of Davisville supports nearly 1,700 jobs, \$98,900,000 in household earnings, and \$374,3000,000 in business output annually. The Port also generates \$45,500,000 in total state tax revenue as well as \$10,500,000 in local taxes. Operated and managed by RIDEM, the Port of Galilee is home to a commercial fishing fleet with 40 piers providing dockage for 250 fishing vessels. Newport's State Pier #9 is the only state-owned commercial fishing facility and supports many commercial fishing vessels. Ports are an important source of revenue for the State of Rhode Island. The Economic Impact of Rhode Island's Fisheries and Seafood Sector from the University of Rhode Island indicates that approximately 428 fisheries firms provide 3,147 jobs and generate \$538,000,000 in gross sales.

4.24.3 Previous Occurrences

Small scale infrastructure failures occur as a secondary impact from a natural disaster, such as a temporary power outage due to a thunderstorm or a communications outage from downed lines following a hurricane. Rhode Island experiences these minor disruptions routinely and manages them through coordination across agencies and with the private sector.

Specifically, when utility and/or infrastructure failure does occur, utility providers generally respond quickly to restore service. However, depending on the cause of the utility disruption, events of prolonged outages do occur.

4.24.4 Probability of Future Events

The probability of a utility failure can vary depending on a range of factors, including the type of utility, the condition of the infrastructure, weather conditions, and maintenance practices. Utility providers typically have systems and protocols in place to minimize the risk of utility failures, and they work to respond quickly to any outages or disruptions. The probability of a utility failure may also vary seasonally or during extreme weather events.

4.24.5 Projected Changes in Location, Intensity, Frequency, and Duration

Climate change can influence the frequency, intensity, and patterns of extreme weather events. An increase in these events can cause a commensurate increase in infrastructure failures. It is expected that climate change will impact infrastructure in the following ways:

- **Increased Frequency of Extreme Weather Events:** Climate change is associated with an increased frequency and intensity of extreme weather events, such as hurricanes, heatwaves, heavy rainfall, and wildfires. These events can damage utility infrastructure, leading to outages.
- **Heatwaves and Electrical Grids:** Rising temperatures can lead to more frequent and prolonged heatwaves. High temperatures can strain electrical grids, leading to increased demand for electricity for cooling and potentially causing power outages.
- **Increased Storm Intensity and Utility Damage:** Hurricanes and tropical storms may become more intense due to warming oceans. Stronger storms can damage power lines, transformers, and other electrical infrastructure, resulting in widespread electricity outages.
- **Sea-Level Rise and Coastal Infrastructure:** Sea-level rise, a consequence of climate change, can threaten coastal infrastructure, including power plants, wastewater treatment facilities, and transportation systems. It can lead to saltwater intrusion, erosion, and damage to critical infrastructure.
- **Flooding and Water Utilities:** More frequent and severe flooding events can impact water supply and wastewater treatment facilities, causing contamination and disruptions in water services.
- **Wildfires and Power Lines:** Climate change can contribute to more extensive and intense wildfires. In regions prone to wildfires, power lines and electrical equipment are at risk of igniting fires, leading to power outages and infrastructure damage.
- **Extreme Weather and Gas Pipelines:** Extreme weather events, including extreme cold or heat, can impact natural gas pipelines. Cold temperatures can freeze pipelines, while heatwaves can affect gas compressors and transmission systems.
- **Changing Precipitation Patterns:** Altered precipitation patterns, such as more intense rainfall or prolonged droughts, can affect the availability and quality of water resources, impacting water utilities and hydropower generation.
- **Increased Storm Surge and Coastal Erosion:** Rising sea levels can exacerbate storm surges during hurricanes and coastal storms, leading to flooding and damage to utility infrastructure located along coastlines.

As previously noted, State of Rhode Island facilities have seen no major changes in the past five years, with only modest repairs and upgrades being conducted and no major rehabilitation or construction projects completed. As such, the risk to state assets has remained static since the completion of the 2019 SHMP. Additionally, with the exception of the two previously identified rehabilitation and construction projects, no large-scale facility projects were noted in any agency budgets or reports.

4.24.6 Vulnerability and Impact

Infrastructure failure can have significant and immediate impacts on people. The specific impacts can vary depending on the type of utility that fails (electricity, water, gas) and the duration of the outage, and may include:

- **Disruption of Daily Life:** Utility failures can disrupt daily routines, including cooking, bathing, heating, or cooling homes, and using electronic devices. Lack of electricity can also disrupt businesses, schools, and healthcare facilities.
- **Safety Concerns:** Utility failures, particularly in electrical and gas systems, can pose safety risks such as fires, electrical hazards, and gas leaks. Lack of electricity can result in the loss of lighting, increasing the risk of accidents and falls.
- **Health Implications:** Medical equipment that relies on electricity can become non-functional, posing risks to individuals with medical conditions. Lack of access to clean water can impact hygiene and health. Utility failures in healthcare facilities can impact the ability to provide medical care and support for patients. Prolonged utility failures, especially during extreme weather events, can lead to stress, anxiety, and discomfort. Vulnerable populations, such as the elderly, children, and those with special needs, may be particularly affected.

Utility failures can have significant impacts on critical infrastructure and facilities. The specific impacts can vary depending on the type of utility affected, the duration of the outage, and the criticality of the infrastructure, and may include:

- **Disruption of Operations:** Utility failures can disrupt the normal operations of critical facilities, including hospitals, emergency response centers, data centers, and transportation hubs.
- **Compromised Safety and Security:** Loss of electricity can impact security systems, including surveillance cameras and alarm systems. Critical facilities may rely on backup power sources to maintain safety and security.
- **Loss of Communication:** Utility failures can disrupt communication systems, affecting the ability of critical facilities to coordinate responses and communicate with staff and the public.
- **Healthcare Impacts:** Hospitals and healthcare facilities may experience disruptions in patient care due to power outages, affecting the health and safety of patients. Medical equipment may require backup power to continue functioning.
- **Water and Sanitation Services:** Water utility failures can disrupt water supply to critical facilities, impacting sanitation services, firefighting capabilities, and patient care. Wastewater treatment plants may be affected, posing environmental and health risks.
- **Transportation Disruptions:** Transportation infrastructure, including airports, train stations, and traffic management systems, may be impacted by utility failures, leading to travel disruptions.
- **Safety Hazards:** Gas utility failures can result in gas leaks, posing fire and explosion hazards to critical infrastructure and nearby areas. Electrical failures may lead to equipment malfunctions, increasing the risk of accidents and safety incidents.

Environment

In general, a utility failure would have little effect on the environment. However, specific circumstances of the failure, such as a chemical leak, a downed power line in a fire prone area, or loss of wastewater containment could pose a concern. The impacts from those type of events can range from relatively minor and localized effects to more significant and widespread environmental consequences, and may include:

- **Wildfires:** Electrical utility failures, such as downed power lines or equipment malfunctions, can trigger wildfires. Wildfires can have devastating effects on natural landscapes and ecosystems.
- **Water Pollution:** Water utility failures, such as sewage system overflows or treatment plant malfunctions, can lead to the release of untreated wastewater into rivers, lakes, or oceans. This can result in water pollution, harm aquatic ecosystems, and affect drinking water quality downstream.
- **Chemical Spills:** Utility failures, particularly in industrial settings, can result in chemical spills and releases. These spills can harm the environment, contaminate soil and water, and endanger wildlife.

State Operations and Assets

Infrastructure failure can have significant impacts on state operations, affecting the ability to provide essential services, respond to emergencies, and maintain critical infrastructure. The specific impacts can vary depending on the type of utility affected and the duration of the outage, and may include:

- **Disruption of Emergency Services:** Failures can disrupt the operations of emergency response agencies, including police, fire departments, and medical services. This can impede their ability to respond to accidents, fires, and medical emergencies.
- **Communication Challenges:** Failures, particularly in telecommunications and internet infrastructure, can hinder communication between state agencies, first responders, and the public. This can impact coordination during emergencies.
- **Data Loss and Information Technology Disruptions:** Electrical outages and information technology infrastructure failures can result in data loss and disrupt state operations that rely on digital records and systems.
- **Transportation Disruptions:** Transportation infrastructure, such as traffic management systems and public transit, may be impacted by utility failures, leading to travel disruptions and challenges in managing traffic flow.
- **Public Health Services:** Healthcare facilities and public health agencies may experience disruptions in patient care, vaccination programs, and disease surveillance during utility failures.
- **Safety Risks:** Failures can pose safety risks to state government employees and the public, particularly when they result in electrical hazards, gas leaks, or water contamination.
- **Economic Consequences:** The economic impact of infrastructure failures can extend to state operations, affecting state budgets and resources available for public programs and services.
- **Disaster Response and Recovery:** Failures may occur during natural disasters, adding complexity to state response and recovery efforts. Coordination among agencies becomes crucial.

Consequence Analysis

This consequence analysis lists the potential impacts of this hazard on various elements of community and state infrastructure. The impact of this hazard is evaluated in terms of disruption of operations, recovery challenges, and overall wellbeing to all Rhode Island residents and first responder personnel. The consequence analysis supplements the hazard profile by analyzing specific impacts.

Table 169: Infrastructure Failure Consequence Analysis

Subject	Potential Impacts
Impact on the Public	Critical infrastructure failures can lead to heavy flooding, power loss, property damage, injury, and even death. Roadways may be obstructed or inaccessible to the public, challenging transport and resource acquirement activities. A failure of critical infrastructure would have a direct impact on public health. Power outages, transit failures, access to clean water would create severe and immediate public health impacts.
Impact on Responders	Infrastructure failure would have a direct and immediate impact on the first responder’s ability to respond effectively. Critical infrastructure failure may cause inaccessibility of roadways. Communications system failure would impact the responders’ ability to communicate their status or response capability.
Continuity of Operations	Rhode Island agencies maintain continuity plans which can be enacted as necessary based on the situation. An infrastructure failure may impact an agency’s ability to maintain operations, including access to facility by transportation systems, and the availability of utilities, communications, energy, and water and wastewater systems.
Delivery of Services	Delivery of services will be disrupted due to critical infrastructure failure. Transit systems may face closures due to public safety concerns. The ability to deliver food, drinking-water, and services will be impacted due to problems with accessibility and transport abilities. Communications, transportation, and governmental services operations would be impacted due to power failure and accessibility challenges.

Table 169: Infrastructure Failure Consequence Analysis

Subject	Potential Impacts
Property, Facilities, and Infrastructure	Roads and bridges may be impacted, water and sewer systems may be damaged, leading to the issue of sanitation and waste collection. Property of homes and businesses may be completely destroyed if situated close to the failure point.
Impact on Environment	The impacts on the environment of critical infrastructure would vary based on the event. Failure of wastewater plants would result in spreading pollution and hazardous materials throughout the environment including large bodies of water. Ecosystems and natural habitats may be destroyed, causing migration or death of wildlife.
Economic Conditions	Critical infrastructure failure would have a direct and considerable fiscal impact on the state government, however through federal disaster may be offset. Additionally, infrastructure failure in every sector has the potential to impact the ability of businesses to operate. If the private sector was not able to maintain operability, there would be continued revenue loss until operability was restored.
Public Confidence in Governance	Critical infrastructure failure would have a direct and immediate impact on the state's ability to provide governance, maintain order, and ensure the continuity of public services. Given a prolonged failure, the public would become increasingly distrustful of the government's abilities. Direct, immediate, and effective actions must be taken in order to maintain public confidence.

4.24.7 Hazard Planning Significance

Based on a review of available data, the SIHMC determined infrastructure failure to have a moderate planning significance for Rhode Island.

4.25 Radiological Incident

4.25.1 Hazard Description

For purposes of this plan, a radiological incident is considered an accident involving a release of radioactive materials from a nuclear reactor. Radiological accidents could cause injury or death, contaminate property and valuable environmental resources, as well as disrupt the functioning of communities and their economies. Since 1980, each utility that owns a commercial nuclear power plant in the United States has been required to have both an onsite and offsite emergency response plan as a condition of obtaining and maintaining a license to operate that plant. Emergency response plans are approved by the U.S. Nuclear Regulatory Commission.

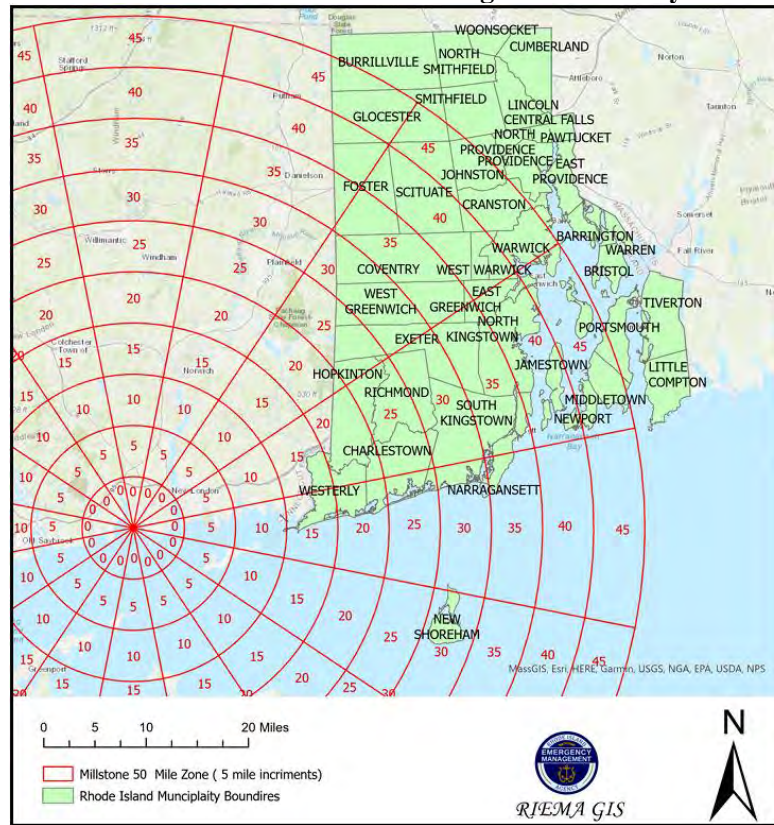


4.25.2 – Location and Extent

Although Rhode Island does not have any commercial nuclear power plants within its borders, it does fall within the 50-mile Ingestion Pathway Planning Zone for Millstone Power Station in Waterford, Connecticut. Millstone Power Station is comprised of three units, one in permanent shutdown and two operational. Millstone Unit 1 was a boiling water reactor that operated from 1970 to 1995. Millstone Unit 1 is currently in SAFSTOR status, meaning it is permanently shut down and the fuel is stored in the spent fuel pool. Millstone Unit 2 is a currently operating 884 megawatts pressurized water reactor that began operation in 1975. Millstone Unit 3 is a currently operating 1,227 megawatts pressurized water reactor that began operation in 1986. The following list and map detail the cities and towns are within the 50-mile radius:

- Burrillville
- Charlestown
- Coventry
- Cranston
- East Greenwich
- Exeter
- Foster
- Gloucester
- Hopkinton
- Jamestown
- Johnston
- Middletown
- Narragansett
- Newport
- New Shoreham
- North Kingstown
- Portsmouth
- Providence
- Richmond
- Scituate
- Smithfield
- South Kingstown
- Warwick
- Westerly
- W. Greenwich
- West Warwick

Map 76: Millstone Power Station 50-Mile Ingestion Pathway Planning Area



Source: RIEMA

Rhode Island does have one research nuclear reactor within the state, located on the University of Rhode Island's Bay campus. The reactor was built in 1960 and went critical in 1964. It is a two Mega-Watt, light water cooled, pool type reactor.

The State of Rhode Island utilizes the Nuclear Regulatory Commission emergency classification system indicating the risk of radiological incident to the public:

Table 170: U.S. Nuclear Regulatory Commission Emergency Classification System

Notification	Description
Notification of Unusual Event	Under this category, events are in process or have occurred that indicate potential degradation in the level of safety of the plant. No release of radioactive material requiring offsite response or monitoring is expected unless further degradation of the safety systems occur.
Alert	If an alert is declared, events are in process or have occurred that involve an actual or potential substantial degradation in the level of safety of the plant. Any releases of radioactive material from the plant are expected to be limited to a small fraction of the Environmental Protection Agency Protective Action Guides exposure levels.
Site Area Emergency	A site area emergency involves events in process or that have occurred that result in actual or likely major failures of plant functions needed for protection of the public. Any releases of radioactive material not expected to exceed the Environmental Protection Agency Protective Action Guides exposure levels except near the site boundary.
General Emergency	Events are in process or have occurred that involve actual or imminent substantial core degradation or melting with potential loss of containment integrity. Releases can reasonably be expected to exceed the Environmental Protection Agency Protective Action Guides exposure levels offsite for more than the immediate site area.

Source: U.S. Nuclear Regulatory Commission

4.25.3 Previous Occurrences

No impactful nuclear incidents have been reported at the Millstone Power Station. Rhode Island has had one nuclear related death. The death occurred on July 24, 1964, at the United Nuclear Corporation Fuels Recovery Plant in Wood River Junction (the plant stripped remaining enriched material from spent material through a chemical process). Through a series of mistakes and a misplaced label, Robert Peabody accidentally set off a small-scale nuclear reaction when a uranium mixture that had been in its safe geometry container was poured into a vat and physically concentrated. A nuclear reaction began instantly. Peabody was bombarded with an enormous number of neutrons as well as gamma rays. He died later from the exposure.

4.25.4 Probability of Future Events

The Millstone Power Station Units 2 and 3 operate using a pressurized water reactor, the type used by approximately two-thirds of the U.S. nuclear power plants. These reactors have several safety systems installed to protect the public from a possible accident. One notable safety feature is that if a pipe breaks and allows cooling water to escape from the metal vessel that contains the reactor core, there are several emergency pumps that will automatically start refilling lost water from an outdoor storage tank. However, the Nuclear Regulatory Commission has identified a flaw in this safety system, where debris can clog the backup pumps, preventing adequate cooling and presenting a significant hazard.

If the safety systems were to fail within the Millstone Power Station, this could precipitate a high-impact event causing immediate and long-term damage to the environment and surrounding area. However, historically the Millstone Power Station has only had minor malfunctions of the system.

The Radiation Safety Office of the Rhode Island Nuclear Science Center provides for the safe operation of the University of Rhode Island's research reactor and provides a Radiation Safety Manual to prepare prospective radiation workers.

4.25.5 Projected Changes in Location, Intensity, Frequency, and Duration

There are no expected changes to the location or frequency of occurrence during the life of this plan as no nuclear power plants have been commissioned in or near the State of Rhode Island. However, as the Millstone Power Station continues to near the end of its life expectancy, and unless the continued diligent inspection and maintenance program is continued, the potential of a failure event could increase.

Climate change is associated with an increase in extreme weather events such as hurricanes, storms, and floods. These events can potentially damage or disrupt facilities that handle radioactive materials, leading to the release of radioactive substances.

As previously noted, State of Rhode Island facilities have seen no major changes in the past five years, with only modest repairs and upgrades being conducted and no major rehabilitation or construction projects completed. As such, the risk to state assets has remained static since the completion of the 2019 SHMP. Additionally, with the exception of the two previously identified rehabilitation and construction projects, no large-scale facility projects were noted in any agency budgets or reports.

4.25.6 Vulnerability and Impact

People and Communities

A radiological release can have serious immediate and long-term impacts on human health, depending on the release's magnitude and the proximity of individuals to the incident. Acute and long-term health effects from a hazardous materials release can include:

- **Acute Radiation Sickness:** High levels of radiation exposure can lead to Acute Radiation Sickness, which can cause symptoms such as nausea, vomiting, diarrhea, fever, and fatigue. The severity of Acute Radiation Sickness depends on the radiation dose received, with higher doses resulting in more severe symptoms. However, RIEMA notes that a release from Millstone Power Station containing enough high activity material to cause acute radiation sickness is considered a near zero probability.
- **Cancer:** Prolonged or repeated exposure to ionizing radiation increases the risk of cancer, including leukemia, thyroid cancer, and solid tumors.

- **Long-Term Health Issues:** Radiation exposure can lead to chronic health problems, such as cataracts, cardiovascular disease, and thyroid disorders.
- **Chronic Illnesses:** Exposure to hazardous materials may lead to chronic health conditions, including cancer, respiratory diseases, neurological disorders, and reproductive problems.
- **Delayed Effects:** Some hazardous substances have delayed health effects, with symptoms appearing days, months, or even years after exposure.
- **Genetic Effects:** Radiation exposure can increase the risk of genetic mutations in offspring, potentially leading to hereditary health issues.

Radiological incidents can cause significant psychological stress, anxiety, and fear among affected individuals and communities. Compounding the stress, large populations may need to be evacuated from affected areas to minimize exposure and may require temporary shelter and support services.

While all citizens are vulnerable to the effects of radiation, of particular concern is socially vulnerable and at-risk populations. These populations may have difficulty with medical issues, poverty, extremes in age, and communications due to language barriers. These populations suffer a disproportionate impact due to:

- Constrained financial resources potentially hampering evacuation
- Limited access to healthcare
- Limited access to information

The following table details identified socially vulnerable and at-risk populations for the State of Rhode Island:

Table 171: Rhode Island 2020 Vulnerable Populations

Jurisdiction	Under Five	Over 65	Language Barrier	Poverty	Disability
Bristol County	1,976	8,372	6,552	3,403	4,753
Kent County	8,148	26,437	16,525	11,925	24,694
Newport County	3,641	13,819	7,794	8,307	9,105
Providence County	36,846	84,768	205,490	90,522	87,498
Washington County	4,760	18,797	7,790	11,815	13,832
State of Rhode Island	55,370	152,183	240,326	125,101	139,927

Source: US Census Bureau

Facilities and Critical Infrastructure

Radiological incidents can have significant impacts on facilities and critical infrastructure, affecting their operations, safety, and long-term functionality. These impacts can vary depending on the type and scale of the incident, the level of preparedness, and the effectiveness of mitigation efforts, but may include:

- **Radiological Contamination:** Facilities located in or near the affected area may become contaminated with radioactive materials, posing risks to employees, equipment, and operations. Facilities may require extensive decontamination efforts to ensure safe operations, which can be time-consuming and costly.
- **Operational Disruption:** Radiological incidents can lead to operational disruptions, shutdowns, and production delays.
- **Restricted Access:** Access restrictions may be imposed on facilities located in contaminated areas, affecting maintenance and repairs.
- **Supply Chain Disruption:** Facilities reliant on suppliers or vendors in affected areas may experience disruptions in the supply chain.
- **Logistics Challenges:** Transportation and logistics may be affected, hindering the delivery of essential materials and equipment.

Environment

Radiological incidents can have long-lasting impacts on the environment, affecting ecosystems, wildlife, and natural resources. The extent of these impacts depends on the type and scale of the incident, the level of radiation released, and the effectiveness of containment and mitigation efforts, and may include:

- **Soil Contamination:** Radioactive materials can contaminate soil, making it unsuitable for agriculture and affecting plant and microbial life.
- **Water Contamination:** Surface and groundwater sources can become contaminated, posing risks to aquatic ecosystems and drinking water supplies.
- **Loss of Biodiversity:** Radiological incidents can harm local flora and fauna, leading to a decline in biodiversity and disrupting food chains.
- **Habitat Destruction:** Contaminated areas may become uninhabitable for wildlife, leading to the displacement or loss of species.
- **Persistent Contamination:** Some radioactive isotopes can remain in the environment for extended periods, causing long-term environmental damage.
- **Chronic Exposure:** Ecosystems and wildlife may experience chronic radiation exposure, leading to genetic mutations and reduced reproductive success.
- **Contaminated Fisheries Waters:** Radiation can affect fish and aquatic organisms, potentially entering the food chain and impacting fisheries and seafood safety.
- **Crop Contamination:** Crops grown in contaminated soil can become radioactive, posing risks to the food supply chain.
- **Livestock Impact:** Livestock that graze on contaminated vegetation may ingest radioactive materials, affecting meat and dairy products.

Radiological materials may be persistent and can remain in the environment for extended periods, causing ongoing harm. Environmental recovery from radiological releases can be slow and challenging, requiring extensive remediation efforts.

State Operations and Assets

A radiological incident can have wide-ranging impacts on state operations, affecting state functions, public safety, and overall stability. These impacts can disrupt state operations, strain resources, and pose challenges to maintaining public order, and may include:

- **Emergency Response and Healthcare:** RIEMA and RIDOH would rapidly mobilize emergency response teams, medical personnel, and healthcare facilities to address the release. The surge in demand for medical resources can strain healthcare systems, including hospitals, clinics, and emergency services.
- **Resource Allocation:** State agencies, RIEMA, RIDOH, and a host of agencies may need to allocate resources to respond to the incident. This includes personnel, equipment, and facilities.
- **Economic Impact:** The economic consequences of a radiological incident can be severe. Business closures, reduced consumer confidence, and trade disruptions can lead to financial losses, unemployment, and economic instability.
- **Public Services:** Essential public services, such as law enforcement, fire services, and sanitation, may be stretched thin due to the demands of responding to an incident.

Of concern to the state are buildings within the 50-Mile Ingestion Pathway Planning Area for the Millstone Power Station. A release of radiological materials of sufficient quantity could impact facilities due to concern about potential radiological ingestion. The following table details the complete inventory of all buildings within the 50-Mile Ingestion Pathway Planning Area, by city, along with the most recent valuation for both the facility envelope and assets housed with the facility for the State of Rhode Island:

Table 172: State of Rhode Island Facilities and Assets by City Within 50 Mile Ingestion Pathway Planning Area

Jurisdiction	Stone	Valuation
Burrillville	40	\$117,245,554
Charlestown	47	\$15,874,479
Coventry	19	\$23,173,062
Cranston	130	\$1,171,484,112
Cumberland	5	\$4,260,336
East Greenwich	7	\$11,440,248
Exeter	34	\$24,912,017
Foster	6	\$2,842,796
Glocester	27	\$11,149,317
Hopkinton	9	\$7,115,927
Jamestown	13	\$8,973,197
Johnston	24	\$16,569,006
Middletown	12	\$14,184,428
Narragansett	170	\$71,207,712
New Shoreham	0	-
Newport	28	\$72,614,245
North Kingstown	33	\$94,726,030
Pawtucket	8	\$9,552,112
Portsmouth	21	\$23,105,003
Richmond	41	\$17,547,103
Scituate	19	\$117,595,341
Smithfield	12	\$9,940,534
South Kingstown	55	\$37,167,835
Warwick	67	\$153,277,534
Westerly	21	\$26,691,944
West Greenwich	2	\$443,826
West Warwick	4	\$5,799,647
State Total	854	\$2,068,893,345

Source: RIEMA

Note: Valuations include structure and assets

Consequence Analysis

This consequence analysis lists the potential impacts of this hazard on various elements of community and state infrastructure. The impact of this hazard is evaluated in terms of disruption of operations, recovery challenges, and overall wellbeing to all Rhode Island residents and first responder personnel. The consequence analysis supplements the hazard profile by analyzing specific impacts.

Table 173: Radiological Incident Consequence Analysis

Subject	Potential Impacts
Impact on the Public	Citizens of Rhode Island that are within the 50-mile Ingestion Pathway for Millstone Power Station are at highest risk. Radiological incidents will have varying effects on the population based on the size, location, and current environmental conditions during the incident (wind direction, temperature). Protective measures will need to be taken, particularly for those in the most vulnerable communities. The main concern for Rhode Island is preventing the ingestion of contaminated food/water products which could result in long term health effects.
Impact on Responders	Radiological incidents will create a dangerous environment and significant challenge for first responders who will have to manage the evacuation of people, direct traffic, close down roads, operate shelters, and take care of the injured and sick. First responders must control their own exposure to the radiological incident and ensure the correct protective equipment is utilized.

Table 173: Radiological Incident Consequence Analysis

Subject	Potential Impacts
Continuity of Operations	Rhode Island agencies maintain continuity plans which can be enacted as necessary based on the situation. A radiological event may impact an agency's ability to maintain continuity of operations due to high amounts of employee absenteeism and potential restricted access to primary and alternate facilities due to radiological contamination.
Delivery of Services	To reduce the public's potential exposure to radiation, roadway and bridge closures may be required depending on the plume models, as well as transit service disruptions. Businesses and places of commerce may completely shut down due to radiological incidents, which leads to the disruption of goods and services.
Property, Facilities, and Infrastructure	Transportation and infrastructure facilities may be disrupted. Roads and bridges can be closed due to radiological releases and the required security perimeter around an event. Radiological incidents can impact access to homes and critical facilities such as hospitals, schools, and supermarkets. Energy services may be temporarily disrupted due to a plant failure, causing secondary impacts to other critical facilities.
Impact on Environment	Agriculture crops and livestock are extremely susceptible to the adverse effects of radiological incidents. Radiological incidents may impact the environment long-term by disturbing or killing wildlife and contaminating soils.
Economic Conditions	Local, county, and state resources may be required during a radiological incident therefore reducing their availability for future events. Private businesses may not be able to maintain operations during or after an incident if they are impacted, which would impact the economy.
Public Confidence in Governance	The public's confidence in the state's governance is affected by immediate response through direct and effective actions to secure the radiological site and ensure the population is adequately informed of the risks posed by the incident. Efficiency in response and recovery operations is critical in keeping public confidence high.

4.25.7 Hazard Planning Significance

Based on a review of available data, the SIHMC determined radiological incident to have a low planning significance for Rhode Island.

4.26 Terrorism

4.26.1 Hazard Description

The United States does not have a standardized definition of terrorism that is agreed upon by all agencies. The Federal Bureau of Investigation generally defines terrorism as:

"The unlawful use of force and violence against persons or property to intimidate or coerce a government, the civilian population, or any segment thereof, in furtherance of political or social objectives."

Terrorism is characterized by the use of violence, intimidation, or the threat of violence to instill fear, achieve political, religious, ideological, or social objectives, and disrupt the normal functioning of a society. It often involves acts of violence deliberately targeting civilians. Key elements and characteristics of terrorism include:



- **Political or Ideological Motivation:** Terrorism is often driven by political, religious, ideological, or social goals. Perpetrators seek to advance a particular agenda or bring about change in accordance with their beliefs.
- **Use of Violence:** Terrorism involves the use of violence, which can range from bombings, shootings, and kidnappings to cyberattacks and biological threats. The intent is to cause harm and instill fear.
- **Targeting Civilians:** Terrorist acts typically target civilians or non-combatants, rather than military or government personnel. This is done to maximize the psychological impact and create a sense of vulnerability within society.
- **Psychological Impact:** The primary objective of terrorism is to create fear and anxiety within the population. The fear generated by terrorist acts can have profound psychological and societal effects.
- **Non-State Actors:** Terrorism is often associated with non-state actors, such as terrorist organizations, extremist groups, or individuals acting independently. However, some state entities have also been accused of engaging in acts that meet the criteria of terrorism.
- **Symbolism:** Terrorist acts are often symbolic in nature, targeting specific locations, landmarks, or institutions that hold significance to the perpetrators or their cause.

Terrorism can take various forms, and the nature of terrorist threats has evolved over time. Common forms of terrorism in the United States include:

- **Domestic Terrorism:** Domestic terrorism involves acts of violence or intimidation committed by individuals or groups within the United States. These acts are typically driven by extremist ideologies, such as far-right extremism, far-left extremism, or other radical beliefs. Recent examples of domestic terrorism include attacks on religious institutions, acts of violence against minority communities, and violent protests.
- **Far-Right Extremism:** Far-right extremism refers to ideologies and movements characterized by extreme nationalism, racism, and opposition to government authority. Some far-right extremists have engaged in acts of violence targeting minority communities, government officials, or perceived enemies.
- **Far-Left Extremism:** Far-left extremism encompasses a range of radical ideologies, including anarchist and socialist beliefs. While not as prevalent as far-right extremism, far-left extremists have been involved in protests, clashes with law enforcement, and acts of violence.
- **Religiously Motivated Terrorism:** Religious extremism can lead to acts of terrorism. In the United States, this has included attacks by individuals or groups inspired by extremist interpretations of Islam, Christianity, or other religions. Examples include the 1993 World Trade Center bombing and the 2009 Fort Hood shooting.
- **Single-Actor Terrorism:** Lone-wolf terrorism involves individuals who carry out acts of violence without direct affiliation with established terrorist organizations. These individuals are often self-radicalized and may be inspired by online propaganda. Examples include the 1995 Oklahoma City bombing and the 2013 Boston Marathon bombing.

- **Eco-Terrorism:** Eco-terrorism refers to acts of violence or sabotage carried out in the name of environmental activism. These acts target industries or organizations perceived as harmful to the environment. Examples include arson attacks on logging facilities or animal testing labs.
- **Cyberterrorism:** Cyberterrorism involves using computer technology to disrupt or damage critical infrastructure, institutions, or networks. While not as common as other forms of terrorism, cyberattacks pose significant risks. Cyberattacks by state-sponsored actors or independent hackers can target government agencies, corporations, and infrastructure.

The U.S. government, law enforcement agencies, and intelligence services actively monitor and address various forms of terrorism. Counterterrorism efforts include preventive measures, intelligence gathering, community engagement, and law enforcement actions. Public awareness, community outreach, and reporting suspicious activities also play a role in countering terrorism in the United States.

Whether mass shooting events (especially school shootings) are considered acts of terrorism can be a subject of debate and can vary depending on the specific circumstances and legal definitions in different jurisdictions. There is no standardized definition of a mass shooting. The United States Investigative Assistance for Violent Crimes Act defines a mass killing as three or more killings in a single incident while the Federal Bureau of Investigation defines a mass shooting as any incident in which at least four people were shot and killed. Mass shootings involve acts of violence carried out in public places, often by individuals who may have personal grievances, mental health issues, or other motivations not necessarily connected to a political or ideological agenda. While mass shootings are undoubtedly acts of violence that result in tragedy and loss of life, they may not always fit the traditional definition of terrorism, as the primary motivation is often not to advance a political or ideological cause. If the shooter's primary aim is to instill fear, advance a political agenda, or promote a particular ideology, it may be more likely to be classified as terrorism. However, if the shooter's motivation is primarily personal, such as a desire for revenge or mental health issues, the act may not be considered terrorism under many legal definitions.

4.26.2 – Location and Extent

The entire State of Rhode Island is vulnerable to terrorism, particularly in densely populated urban areas or crowded venues. However, it is nearly impossible to pinpoint the exact location of the next terrorist attack. Through information and intelligence sharing, public safety personnel at the local, state, and federal level help identify potential targets for terrorist activity. Although it is impossible to predict for certain where the next terrorist attack will take place, terrorists generally target large, crowded places, such as malls, parks, and other large public or social gatherings, in order to maximize damage. In addition, some acts of terror are conducted against critical infrastructure in an effort to weaken or cripple services such as transportation, communications, and electricity.

The extent of terrorism can vary significantly depending on a range of factors including the tactics, capabilities, and the effectiveness of counterterrorism efforts. Tactics employed may include bombings, firearm attacks, kidnappings, assassinations, cyberattacks, or a combination. The choice of targets, such as civilians, government institutions, religious sites, or critical infrastructure can also affect the extent of the terrorist threat. The extent of terrorism may also be influenced by public support or sympathy for extremist ideologies, as well as the recruitment and radicalization of individuals into terrorist organizations. Socio-economic factors, such as poverty, unemployment, and inequality, can contribute to the conditions conducive to terrorism.

The effectiveness of counterterrorism efforts by governments and international organizations can influence the extent of terrorism. Robust counterterrorism measures can disrupt terrorist networks and reduce the frequency and impact of attacks. Efforts to address terrorism typically involve a combination of security measures, intelligence sharing, diplomacy, counter-radicalization programs, and community engagement. Reducing the extent of terrorism often requires a multifaceted approach that addresses both the root causes and the immediate security threats associated with terrorism.

4.26.3 Previous Occurrences

Although there has not been a terrorist attack in the State of Rhode Island, this does not reduce the significance of the threat. There have been numerous examples of terrorism that have occurred in the United States, and specifically terrorist events that have occurred in the northeast. Of note:

- World Trade Center attack, New York City (1993)
- World Trade Center September 11 attacks, New York City (2001)
- Boston Marathon Bombing (2013)
- New York City Truck Attack (2017)

4.26.4 Probability of Future Events

Assessing the probability of a terrorist attack in Rhode Island involves complex analysis conducted by intelligence and law enforcement agencies such as the U.S. Department of Homeland Security, the Federal Bureau of Investigation, and the Rhode Island State Police. These agencies regularly provide threat assessments and security information to the public based on local, international, and geopolitical intelligence.

4.26.5 Projected Changes in Location, Intensity, Frequency, and Duration

Predicting the specific changes in the location, intensity, and frequency of terrorist events is highly challenging due to the complex and dynamic nature of terrorism. Terrorism is influenced by a multitude of factors, including political, social, economic, and ideological considerations. Additionally, responses by governments, international cooperation, and evolving global dynamics contribute to the uncertainty surrounding future projections.

The increasing reliance on technology provides terrorists with new tools and methods for conducting attacks. Cyberterrorism can be used to disrupt critical infrastructure or compromise information systems may become more prevalent. Additionally, the use of online platforms for radicalization and recruitment purposes is a growing concern. Changes in the online landscape, social media platforms, and encryption methods can influence the reach and effectiveness of extremist propaganda.

Climate change can indirectly influence terrorism by exacerbating certain conditions that may contribute to the emergence and persistence of terrorist threats. While climate change itself does not directly cause terrorism, it can interact with other factors to create a more conducive environment for terrorist activities. Climate change can lead to resource scarcity, such as water and arable land shortages, which may intensify poverty. This scarcity can create conditions that extremist groups exploit. Additionally, climate-induced displacement and migration can result from events like sea-level rise, extreme weather events, and droughts. Displaced populations can become vulnerable to recruitment by extremist groups, as they may lack basic necessities and economic opportunities.

As previously noted, State of Rhode Island facilities have seen no major changes in the past five years, with only modest repairs and upgrades being conducted and no major rehabilitation or construction projects completed. As such, the risk to state assets has remained static since the completion of the 2019 SHMP. Additionally, with the exception of the two previously identified rehabilitation and construction projects, no large-scale facility projects were noted in any agency budgets or reports.

4.26.6 Vulnerability and Impact People and Communities

Terrorism can have profound and far-reaching impacts on individuals and communities. These effects can be physical, psychological, social, and economic, and may include:

- **Loss of Life and Injury:** Terrorism often results in the loss of innocent lives and injuries to survivors. Victims may suffer physical trauma, disabilities, and long-term health issues.
- **Psychological Trauma:** Many survivors of terrorist attacks and witnesses may experience Post-Traumatic Stress Disorder, characterized by flashbacks, nightmares, anxiety, and emotional distress. Children and young people may be particularly vulnerable to the psychological effects of terrorism, which can impact their emotional and cognitive development.
- **Anxiety and Depression:** Terrorism can lead to increased anxiety and depression in affected individuals and communities.
- **Grief and Loss:** Those who lose loved ones in terrorist attacks may experience profound grief and loss, which can be long-lasting.

Terrorism can disrupt social structures and community cohesion, leading to feelings of insecurity and mistrust. Fear of future attacks may limit social activities and interactions, impacting the quality of life. Some terrorist attacks, such as bombings, can result in displacement and homelessness for those affected, leading to housing instability and further psychological stress. People may alter their daily routines, travel plans, or social activities due to fear of further attacks. This can impact personal freedom and quality of life.

Facilities and Critical Infrastructure

Critical infrastructure is often high-value and high-impact, making it an attractive target for terrorists looking to cause disruption, economic damage, and fear. Many critical infrastructure sectors are interconnected, so an attack on one sector can have cascading effects on others. For example, an attack on the power grid can impact telecommunications and transportation. Compounding the issue, certain critical infrastructure facilities are accessible to the public or located in urban areas, making them vulnerable to physical attacks, such as bombings or shootings. Specific impacts on critical infrastructure may include:

- **Disruption of Operations:** Attacks can disrupt the normal operations of critical facilities, including hospitals, emergency response centers, data centers, and transportation hubs.
- **Economic Disruption:** Attacks can lead to significant economic disruption, including damage to facilities, loss of productivity, and increased operational costs.
- **Public Safety:** Attacks on certain critical infrastructure, such as transportation hubs or healthcare facilities, can pose immediate risks to public safety, leading to injuries and loss of life.
- **Disruption of Services:** Infrastructure attacks can result in service disruptions, including power outages, water supply interruptions, and communication breakdowns.
- **Healthcare Impact:** Attacks on healthcare infrastructure, like hospitals, can limit access to medical care during emergencies, potentially leading to higher casualties.

State Operations and Assets

Terrorism can have significant impacts on state operations. These impacts can vary depending on the nature and scale of terrorist attacks, the level of preparedness and response, and the specific vulnerabilities, and may include:

- **Security and Law Enforcement:** An attack would lead to an increased demand on law enforcement agencies to prevent, investigate, and respond to terrorist threats and incidents. Allocation of significant resources to counterterrorism efforts would stretch resources.
- **Emergency Response:** RIEMA, in conjunction with federal agencies, would need to activate emergency response and management systems to coordinate response. A long-term activation could strain resources and personnel. Additionally, responders may be vulnerable to secondary devices or attacks.
- **Public Services:** An attack could lead to the disruption of public services, such as transportation, utilities, and public spaces, due to security concerns.
- **Economic Impact:** Negative economic consequences, including damage to businesses, loss of investor confidence, and reduced tourism and foreign investment can occur.
- **Surveillance and Privacy Concerns:** Expansion of surveillance capabilities may result in concerns about potential violations of privacy rights.
- **Impact on Government Operations:** An attack would likely cause the disruption of state functions, including closures of offices and facilities.
- **Psychological Impact on State Officials:** Psychological stress and burnout among state officials and first responders involved in counterterrorism efforts.
- **Public Opinion and Confidence:** Fluctuations in public opinion and confidence in the state's ability to provide security and protect citizens would occur.

Due to the random nature of terrorism, all state owned or operated facilities are potentially vulnerable to potential impacts. The following table indicates the number and valuation of the building envelope and building contents for all state owned or operated facilities:

Table 174: State of Rhode Island Facilities and Assets by County

Jurisdiction	State Owned or Operated Facilities	Valuation
Bristol County	32	\$150,415,810
Kent County	116	\$215,026,614
Newport County	77	\$127,353,837
Providence County	423	\$2,526,593,056
Washington County	393	\$274,350,750
State Total	1,041	\$3,293,740,067

Source: RIEMA

Note: Valuations include structure and assets

Consequence Analysis

This consequence analysis lists the potential impacts of this hazard on various elements of community and state infrastructure. The impact of this hazard is evaluated in terms of disruption of operations, recovery challenges, and overall wellbeing to all Rhode Island residents and first responder personnel. The consequence analysis supplements the hazard profile by analyzing specific impacts.

Table 175: Terrorism Consequence Analysis

Subject	Potential Impacts
Impact on the Public	Terrorist activities including bombings, kidnappings, shootings, and hijackings could cause considerable injury and death. An attack could kill and injure hundreds to thousands of people, which could overwhelm hospitals.
Impact on Responders	Attacks can create a dangerous environment for first responders, who may have to manage evacuations and treat the injured. First responders may be a direct target of terrorism themselves from a secondary attack during a response. Equipment may also be damaged or destroyed, which may lead to a decrease in response capabilities.
Continuity of Operations	Rhode Island agencies maintain continuity plans which can be enacted as necessary based on the situation. A terrorist event may impact an agency's ability to maintain due to the potential to cause significant injury to staff or impede travel to facilities.
Delivery of Services	The ability to deliver services can be impacted depending on the characteristics of the attack. Roadway and bridge closures may be required, as well as transit service disruptions. Businesses and places of commerce may completely shut down, which leads to the disruption of goods and services.
Property, Facilities, and Infrastructure	Transportation, governmental operations, and infrastructure facilities may be disrupted. Access to homes and critical facilities such as hospitals, schools, and supermarkets may be impossible. If power loss occurs following an attack, it may lead to disruption of critical infrastructure and technology.
Impact on Environment	Terrorist attacks involving bombings and arson pose considerable negative impacts to the environment in the form of smoke and destruction of vegetation. A terrorist attack utilizing chemical, nuclear, and biological weapons pose a significantly higher risk to the environment by causing pollution, damaging sewer and wastewater treatment plants; or disturbing or killing wildlife, and adversely affecting nature preserves.
Economic Conditions	Local, county, and state resources may be severely depleted during a terrorist attack response. Private businesses may not be able to maintain operations during or after an incident if they are impacted, which would impact the economy.
Public Confidence in Governance	If government employees or facilities are targeted directly by terrorism, it will have a significant impact on the ability to govern. The public's confidence in the state's governance is affected by immediate response through direct and effective actions. Efficiency in response and recovery operations is critical in keeping public confidence.

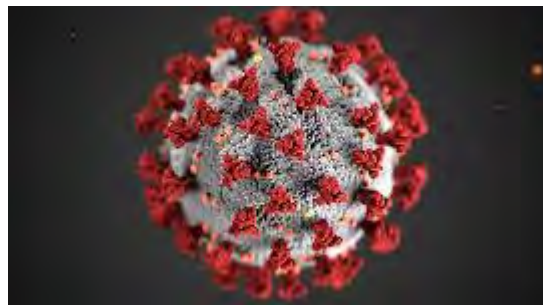
4.26.7 Hazard Planning Significance

Based on a review of available data, the SIHMC determined terrorism to have a low planning significance for Rhode Island.

4.27 Transmissible Disease

4.27.1 Hazard Description

A transmissible disease, also known as a communicable or infectious disease, is a type of illness caused by pathogens (such as bacteria, viruses, fungi, or parasites) that can be transmitted from one person or organism to another, directly or indirectly. These diseases can spread through various means, including person-to-person contact, respiratory droplets, contaminated food or water, vectors like mosquitoes, or contact with infected animals.



Transmissible diseases are characterized by their ability to pass from an infected individual to a susceptible host, leading to new cases of the disease. The transmission can occur through various routes, depending on the specific pathogen and the mode of transmission it utilizes. Examples of transmissible diseases include:

- Influenza: The flu is caused by influenza viruses and can spread through respiratory droplets when an infected person coughs or sneezes.
- West Nile virus: A mosquito-borne virus that can cause a range of illnesses in humans, from mild febrile symptoms to severe neurological disease. It is primarily transmitted to humans through the bite of infected mosquitoes.
- Malaria: Malaria is caused by Plasmodium parasites and is transmitted through the bite of infected female Anopheles mosquitoes.
- Salmonella Infection: This bacterial infection is often contracted through the consumption of contaminated food or water and can lead to gastrointestinal symptoms.
- Tuberculosis: • Tuberculosis is caused by Mycobacterium tuberculosis and can be transmitted through the inhalation of respiratory droplets from an infected person with an active disease.
- Measles: Measles is caused by the measles virus and spreads through respiratory droplets, making it highly contagious.

Of particular concern are novel transmissible diseases. This is a disease that is caused by a pathogen (such as a virus, bacterium, or other microorganism) that is newly recognized in a human population or is increasing in incidence or geographic range. These diseases are termed novel because they have not been previously identified or have not been known to affect humans in the past. Several factors can contribute to the emergence of novel transmissible diseases, including changes in human behavior, urbanization, deforestation, climate change, global travel, and the encroachment of humans into natural habitats. Defining characteristics of novel transmissible diseases: include

- New Pathogen or Strain: Novel transmissible diseases often involve a pathogen or strain of a pathogen that is new to humans. This may result from genetic mutations, cross-species transmission (zoonotic diseases), or the introduction of a pathogen to a new geographic area.
- Human Transmission: These diseases have the potential to spread from person to person, either through direct contact, respiratory droplets, contaminated surfaces, or other modes of transmission.
- Challenges in Control: Because these diseases are new and may have limited prior immunity in the population, they can pose challenges for public health authorities in terms of surveillance, diagnosis, treatment, and containment.

Novel transmissible diseases can have pandemic potential, meaning they can spread globally and affect a large portion of the world's population. Dealing with novel transmissible diseases requires a multi-pronged approach, including surveillance, early detection, containment measures, public health interventions, and research to understand the pathogen and develop effective countermeasures. It also underscores the importance of preparedness and global cooperation in responding to emerging infectious diseases.

4.27.2 – Location and Extent

Rhode Island's geographic and demographic characteristics make it vulnerable to the spread of transmissible diseases. Factors like high population density in crowded urban centers facilitating close contact can impact potential spread.

Additionally, the high number of colleges and universities in the State, bringing together students from a wide range of socioeconomic, ethnic, and cultural backgrounds, may increase rates of transmission.

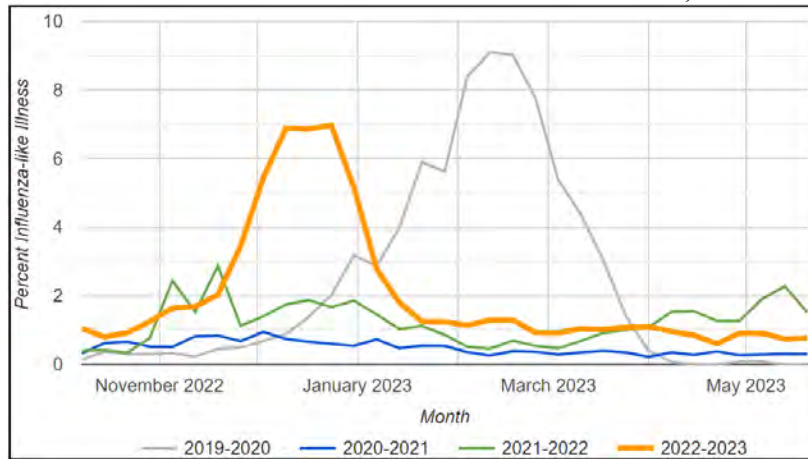
The extent of a transmissible disease can vary widely depending on several factors, including:

- **Pathogen Characteristics:** The biological properties of the infectious agent, such as its mode of transmission, incubation period, and virulence, play a significant role. Pathogens that are highly contagious and have a short incubation period are more likely to spread rapidly.
- **Human Behavior:** Human behavior and practices, such as hygiene, travel, and social interactions, can influence the extent of disease spread. For example, frequent travel and close interpersonal contact can facilitate the rapid transmission of infectious diseases.
- **Public Health Measures:** The effectiveness of public health measures, such as quarantine, isolation, contact tracing, and vaccination, can limit the extent of disease spread. Prompt and coordinated public health responses can be crucial.
- **Geographic Factors:** The geographic spread of a disease can be influenced by factors like population density, climate, and geographic barriers. Dense urban areas may experience more rapid transmission, while isolated or remote regions may be less affected.
- **Healthcare Infrastructure:** The capacity of healthcare systems to detect, treat, and isolate cases can impact the extent of an outbreak. Overwhelmed healthcare systems can lead to a larger extent of disease.
- **Pre-existing Immunity:** If a portion of the population has pre-existing immunity to the disease, either due to prior exposure or vaccination, this can limit the extent of disease transmission.
- **Global Travel:** In an era of global travel, novel infectious diseases can quickly cross international borders, affecting multiple countries and regions.
- **Vaccination:** The availability and coverage of vaccines against the disease can significantly reduce the extent of an outbreak. High vaccination rates create herd immunity, protecting even those who are not vaccinated.
- **Mutation and Variants:** Some infectious agents may undergo mutations that affect their transmissibility or virulence. New variants can lead to changes in the extent and severity of the disease.
- **Public Awareness and Compliance:** Public awareness of the disease, willingness to follow public health guidance, and compliance with preventive measures can affect disease transmission rates.
- **Timeliness of Response:** The speed with which authorities and healthcare systems respond to an outbreak can have a substantial impact. Rapid detection and containment efforts can limit the extent of spread.

The extent of a transmissible disease can range from localized outbreaks that are quickly contained to global pandemics that affect large populations across multiple countries. The management of such diseases requires a combination of robust surveillance, effective public health interventions, research, and international collaboration to minimize their impact on human health and society.

4.27.3 Previous Occurrences

One of the most common transmissible diseases within the State of Rhode Island is Influenza. Influenza, commonly known as the flu, is a contagious respiratory illness caused by influenza viruses. It can affect humans, birds, and other animals. Influenza viruses are classified into types A, B, C, and D, with types A and B being the most common in humans and responsible for seasonal flu outbreaks. RIDOH tracks flu cases throughout the state using a number of methods for gathering data. This includes gathering reports on the number of patients with influenza-like illness from a sample of sentinel physicians and collecting data from hospitals throughout the state. State Health Laboratory testing is used to determine the circulating strains of influenza. Additionally, a staff of epidemiologists work with CDC guidance to determine flu activity levels. The following chart details the incidence of flu like illness for the state from 2019 through 2023:

Chart 36: State of Rhode Island Influenza Like Illnesses, 2019 – 2023

Source: RIDOH

Additionally, the following table presents the rate of occurrence of select transmissible disease tracked by RIDOH:

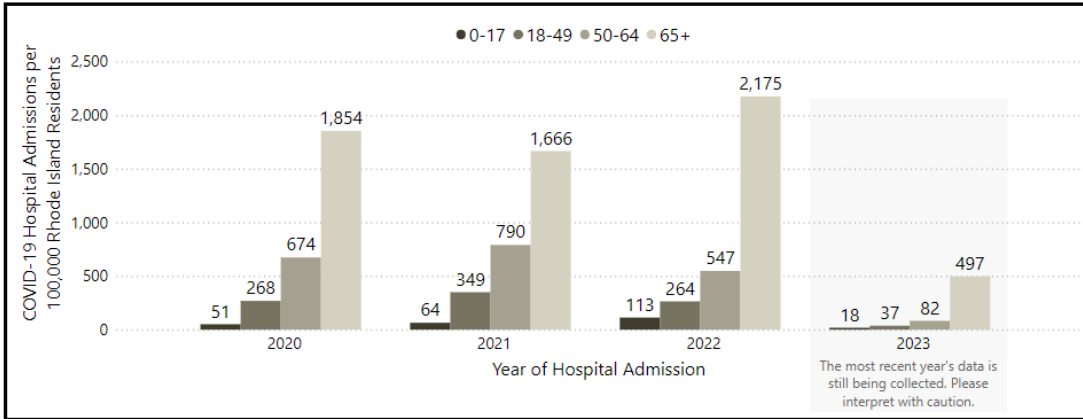
Table 176: Rhode Island Transmissible Disease Occurrence, 2017 - 2022

Disease	2017	2018	2019	2020	2021	2022	Five Year 2017-2021		Ten Year 2012-2021	
							Mean	Median	Mean	Median
Chikungunya	0	0	1	0	0	0	0	0	8	0.5
COVID-19				86,882	140,322	199,714	N/A	N/A	N/A	N/A
EEE Virus	0	0	3	0	0	0	1	0	0	0
Haemophilus Influenzae	18	25	24	12	15	6	19	18	19	18.5
Hepatitis A	6	7	6	4	1	1	5	6	5	4
Legionellosis	50	73	48	36	87	43	59	50	46	41.5
Listeriosis	7	7	2	5	6	4	5	6	5	5
Lyme Disease	1,132	1,111	971	839	980	2,222	1,007	980	871	916
Malaria	13	12	13	2	7	9	9	12	13	13
Measles	0	0	0	0	0	0	0	0	0	0
Meningococcal Disease	2	1	3	0	0	0	1	1	1	1
Mumps	29	5	9	0	2	0	9	5	5	2.5
Pertussis	84	34	44	4	0	0	33	34	67	64
Pneumococcal Disease	64	59	85	51	32	53	58	59	64	64.5
Rocky Mountain Spotted Fever	15	9	0	0	1	0	5	1	5	2
Rubella	0	0	0	0	0	0	0	0	0	0
Salmonellosis	148	175	152	115	137	141	145	148	138	138.5
Shiga-toxin positive E. coli	14	45	48	28	45	39	36	45	21	12.5
Shigellosis	44	59	85	32	49	56	54	49	65	39
Streptococcus, Group A	39	43	53	34	30	32	40	39	37	37

The most notable recent novel infectious disease to strike Rhode Island is COVID-19, also known as Coronavirus Disease 2019. COVID-19 is an infectious respiratory illness caused by a novel coronavirus known as SARS-CoV-2 (Severe Acute Respiratory Syndrome Coronavirus 2). It was first identified in December 2019 in the city of Wuhan, China, and spread globally leading to a pandemic. COVID-19 primarily spreads from person to person through respiratory droplets when an infected person coughs, sneezes, talks, or breathes. It can also spread by touching surfaces contaminated with the virus and then touching the face. Symptoms can range from mild to severe and may include fever, cough, shortness of breath, fatigue, muscle aches, loss of taste or smell, sore throat, congestion, and gastrointestinal symptoms like diarrhea. Some individuals may remain asymptomatic, meaning they carry the virus without displaying symptoms. While many people with COVID-19 experience mild to moderate symptoms and recover without hospitalization, the disease can be severe, especially among older adults and individuals with underlying health

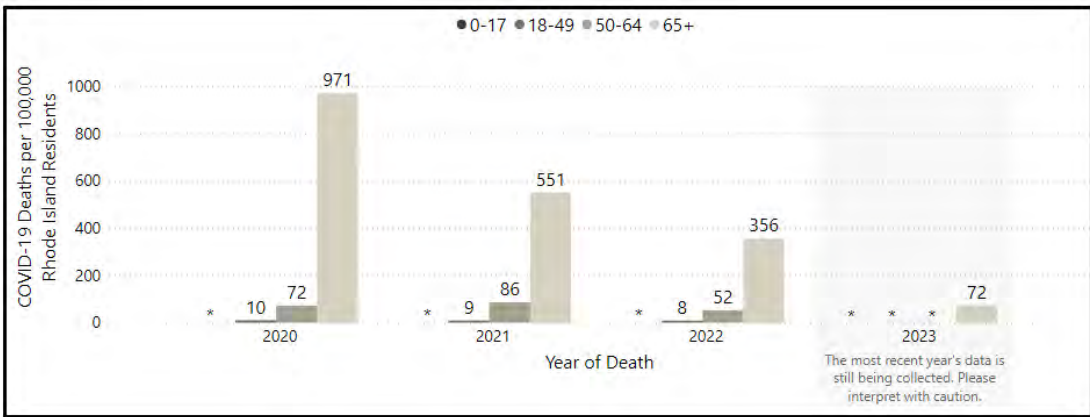
conditions. Severe cases can lead to pneumonia, acute respiratory distress syndrome, organ failure, and death. The following tables, from the Rhode Island Department of Health, detail the number of hospitalizations and deaths from COVID-19 in Rhode Island:

Chart 37: Rhode Island COVID-19 Hospitalization Rates



Source: Rhode Island Department of Health

Chart 38: Rhode Island COVID-19 Death Rates



Source: Rhode Island Department of Health

COVID-19 has had a profound impact on public health, economy, and daily life across Rhode Island. Some of the key measures taken in Rhode Island in response to the COVID-19 pandemic include:

- **Public Health Measures:** Rhode Island implemented various public health measures to slow the spread of the virus. These included stay-at-home orders, mask mandates, social distancing guidelines, and limits on gathering sizes.
- **Testing and Contact Tracing:** The state established testing sites and conducted contact tracing to identify and isolate individuals who had been exposed to the virus. Testing was widely available to the public.
- **Vaccination Efforts:** Rhode Island launched vaccination campaigns to administer COVID-19 vaccines to eligible residents. Mass vaccination sites, healthcare providers, and pharmacies played a role in the distribution of vaccines.
- **School Closures and Remote Learning:** Like many other states, Rhode Island temporarily closed schools and shifted to remote learning to minimize the risk of virus transmission among students and staff.
- **Travel and Quarantine Measures:** Rhode Island issued travel advisories and quarantine requirements for travelers coming into the state, especially from areas with high infection rates.
- **Mask Mandates and Social Distancing:** Face mask mandates and social distancing measures were enforced in indoor public spaces and in situations where social distancing was not possible.

Additionally, COVID-19 had numerous, and oftentimes severe impacts on the State, including:

- **Economic Repercussion:** Job losses, business closures, and economic strain on individuals and families were common within the State. Rhode Island, like other states, implemented economic relief measures.
- **Healthcare System Overload:** Hospitals and healthcare facilities in Rhode Island worked to increase capacity to treat COVID-19 patients. There were efforts to secure additional medical supplies and equipment.
- **Protection of Vulnerable Populations:** Efforts were made to protect vulnerable populations, including the elderly and those with underlying health conditions, who were at higher risk of severe illness from COVID-19.
- **Educational Impact:** The pandemic disrupted education, with students and teachers adapting to remote learning. Schools implemented safety measures upon reopening.

The response to COVID-19 evolved as more information became available, and measures were adjusted based on the changing circumstances of the pandemic. Rhode Island worked to balance public health concerns with the economic and social well-being of its residents. The state's response was guided by recommendations from health experts from the CDC.

4.27.4 Probability of Future Events

While it is impossible to predict with certainty when or if a transmissible disease outbreak will occur, the probability of occurrence can be estimated based on historical patterns and current global conditions. Factors to consider include:

- **Globalization:** Increased global travel and trade can facilitate the rapid spread of infectious diseases. The interconnectedness of the world means that a disease can quickly cross borders, increasing the risk of a pandemic.
- **Vaccine Coverage:** The level of vaccination coverage against preventable diseases can impact the likelihood of pandemics. Low vaccine coverage can lead to outbreaks that have pandemic potential.
- **Public Health Preparedness:** The readiness of healthcare systems, public health agencies, and governments to respond to outbreaks is crucial. Adequate preparedness can help contain outbreaks before they become pandemics.
- **Surveillance and Early Detection:** Improved surveillance systems and early detection mechanisms can help identify and contain outbreaks before they escalate to pandemics.
- **Scientific Advancements:** Advances in science and technology, such as the rapid development of vaccines and treatments, can influence our ability to respond to emerging infectious diseases.
- **Behavioral Factors:** Human behavior, including adherence to preventive measures like handwashing, mask-wearing, and vaccination, plays a role in disease transmission. Public health campaigns can influence behavior.
- **Climate Change:** Environmental changes driven by climate change can alter the geographic distribution of diseases and the behavior of vectors (like mosquitoes). This can affect disease transmission patterns and increase the risk of outbreaks.
- **Agriculture and Farming Practices:** The way animals are raised and farmed can impact the risk of zoonotic diseases, which are diseases transmitted from animals to humans. The probability of another pandemic is influenced by the frequency of spillover events (when a pathogen jumps from animals to humans). Factors like deforestation, urbanization, and increased contact with wildlife can contribute to these events.

Transmissible disease outbreaks can vary in their impact, and public health measures can mitigate their effects. Governments, international organizations, and scientists continuously monitor and assess the risk of transmissible diseases and work to improve preparedness and response capabilities.

In order to prevent the rapid spreads of transmissible diseases, RIDOH, per state regulations, requires the following diseases and conditions be reported on the day of recognition or strong suspicion of disease:

- All clusters and outbreaks (including noro-like illness, flu, and other respiratory diseases)
- Novel or emerging respiratory viruses
- Unexplained death (possibly due to unidentified infectious causes)

- Chikungunya Virus Infection (Chikungunya)
- Ciguatera (Harmful Algae Blooms)
- COVID-19 (Coronavirus Disease 2019)
- Diphtheria
- Eastern Equine Encephalitis
- Encephalitis Arboviral or parainfectious
- Hantavirus Pulmonary Syndrome
- Hepatitis A
- Measles
- Meningococcal Disease Bacterial (Meningitis, bacterial)
- Middle East Respiratory Syndrome Coronavirus
- Monkeypox Virus (MPox)
- Paralytic Shellfish Poisoning
- Poliomyelitis (Polio)
- Powassan
- Pustular Rash diseases (Smallpox, monkeypox, cowpox)
- Rabies
- Scombroid
- Staphylococcal Infection Vancomycin Intermediate
- Staphylococcal Infection Vancomycin Resistant
- Typhoid Fever Group D
- *Vibrio cholerae* (Cholera)
- Vibriosis (*Vibrio*)
- West Nile Virus
- Yellow Fever
- Zika Virus Infection (Zika)

The State of Rhode Island reports all nationally notifiable conditions to the CDC using the National Electronic Disease Surveillance System to allow for rapid and appropriate response.

RIDOH's tracking and reporting protocols are in place to lessen the potential future occurrence of any major transmissible disease.

4.27.5 Projected Changes in Location, Intensity, Frequency, and Duration

A continued increase in international travel, both to and from Rhode Island, may increase the spread of infectious disease. The movement of people across diverse geographical regions brings together individuals with different immunological profiles. This mingling creates opportunities for the emergence of novel pathogens or the introduction of diseases into populations with limited immunity.

Climate change can have several impacts on the emergence and spread of transmissible diseases. While the relationship between climate change and transmissible diseases is complex, there are several ways in which climate change can influence disease dynamics including:

- **Altered Disease Transmission Patterns in Vector-Borne Diseases:** Climate change can affect the distribution and behavior of disease vectors (mosquitoes and ticks) by influencing temperature and precipitation patterns. This can lead to the expansion of diseases like malaria, dengue fever, and Lyme disease into new geographic areas.
- **Extended Transmission Seasons:** Rising temperatures can lengthen the transmission seasons for certain diseases, allowing them to be active for a more extended period each year.
- **Changes in Pathogen Survival:** Some pathogens can survive longer in warmer and wetter conditions. This can affect the persistence of infectious agents in the environment.

- **Increased Risk of Zoonotic Diseases:** Climate change can disrupt ecosystems and alter the habitats and migration patterns of wildlife. This can lead to increased interactions between humans, domestic animals, and wildlife, potentially facilitating the transmission of zoonotic diseases (diseases that originate in animals) to humans.
- **Weakened Immune Response:** Climate-related stressors, such as extreme heat events, can weaken the immune systems of vulnerable populations, making them more susceptible to infectious diseases.

To mitigate the impacts of climate change, public health measures, adaptation strategies, and international cooperation are essential, and may include:

- Strengthening disease surveillance systems to monitor changing disease patterns.
- Implementing vector control measures in areas at risk of vector-borne diseases.
- Enhancing healthcare infrastructure resilience to climate-related disasters.
- Promoting climate-resilient agricultural practices to ensure food security.
- Supporting research on the links between climate change and infectious diseases.
- Raising awareness and educating communities about the risks and preventive measures.

4.27.6 Vulnerability and Impact

People and Communities

People can be vulnerable to transmissible diseases due to various factors that influence their susceptibility to infection and the potential severity of illness. These vulnerabilities can be influenced by individual, societal, and environmental factors, and may include:

- **Lack of Immunity:** Many transmissible diseases are ones that people have little to no immunity to.
- **Vaccination Status:** Vaccination can provide immunity against certain diseases. People who are not vaccinated or have not received booster shots may be more vulnerable.
- **Age:** Infants, young children, and the elderly often have weaker immune systems, making them more susceptible to infections and complications.
- **Underlying Health Conditions:** Individuals with underlying health conditions, such as immunodeficiency disorders, chronic diseases, or respiratory conditions, may be more vulnerable to severe illness.
- **Medication and Treatment Availability:** The availability of medications or treatments specific to the disease can impact vulnerability. Rapid access to appropriate treatments can be lifesaving.
- **Population Density:** Highly populated areas can facilitate the rapid spread of diseases, making people in densely populated regions more vulnerable.
- **Sanitation and Hygiene:** Poor sanitation and hygiene practices can increase the risk of disease transmission. Access to clean water and sanitation facilities is crucial for reducing vulnerability.
- **Access to Healthcare:** The availability and accessibility of healthcare services, including diagnostic testing and medical treatment, can significantly impact the outcome of a novel transmissible disease.
- **Public Awareness:** People who are unaware of the risks associated with a novel transmissible disease or who do not know how to protect themselves may be more vulnerable.
- **Behavioral Factors:** People's behavior, such as adherence to public health guidelines (e.g., handwashing, wearing masks), can influence vulnerability.
- **Fear and Panic:** Fear and panic can hinder effective responses, potentially increasing vulnerability.
- **Access to Information:** Timely and accurate information can empower individuals to take protective measures. Lack of information or misinformation can increase vulnerability.

The spread of a transmissible disease can have severe and far-reaching impacts on human health and society, and can include:

- **Illness and Death:** The most immediate impact is the potential for widespread illness and death. Depending on the disease, the severity of illness can range from mild to life-threatening.

- **Healthcare Overload:** A rapidly spreading disease can quickly overwhelm healthcare systems, leading to shortages of medical supplies, hospital beds, and healthcare personnel. The ability to provide timely medical care may be compromised.
- **Social Disruption:** Social disruption can occur due to isolation and quarantine measures, as well as the need for social distancing. Schools, businesses, and public gatherings may be canceled or limited, affecting daily life and routines.
- **Psychological Trauma:** Survivors of a transmissible disease may experience long-lasting psychological trauma due to the fear of infection, the loss of loved ones, and the overall trauma of the event.
- **Long-Term Health Effects:** Some diseases can cause long-term health effects in survivors, including chronic illnesses and disabilities.

While all citizens are vulnerable to the effects of transmissible disease, of particular concern is socially vulnerable and at-risk populations. These populations may have difficulty with medical issues, poverty, extremes in age, and communications due to language barriers. These populations suffer a disproportionate impact due to:

- Weakened immune systems
- Limited access to healthcare
- Limited access to information

The following table details identified socially vulnerable and at-risk populations for the State of Rhode Island:

Table 177: Rhode Island 2020 Vulnerable Populations

Jurisdiction	Under Five	Over 65	Language Barrier	Poverty	Disability
Bristol County	1,976	8,372	6,552	3,403	4,753
Kent County	8,148	26,437	16,525	11,925	24,694
Newport County	3,641	13,819	7,794	8,307	9,105
Providence County	36,846	84,768	205,490	90,522	87,498
Washington County	4,760	18,797	7,790	11,815	13,832
State of Rhode Island	55,370	152,183	240,326	125,101	139,927

Source: US Census Bureau

It is important to note that public health agencies and emergency responders work to minimize vulnerabilities by implementing preventive measures, conducting public awareness campaigns, and having response plans in place. Preparedness efforts, including vaccination programs, stockpiling of medical supplies, and coordination among healthcare providers, are critical for reducing vulnerabilities.

Critical Facilities and Infrastructure

The direct risk or vulnerability to property and critical facilities from a transmissible disease is generally limited. While unlikely, transmissible diseases could possibly be moved through a facility's ventilation system. An incident like this would not pose a direct risk to the structure's integrity; however, considerable contamination of the facility may occur, requiring decontamination and potential loss of access to the building for a considerable length of time. Critical facilities and infrastructure generally will not suffer direct impacts from a novel transmissible disease event. Employee absenteeism could indirectly impact the ability for a critical facility to operate. Without necessary operators, critical infrastructure may be susceptible to indirect failure.

Environment

Zoonotic diseases are infections that can be transmitted between animals and humans. These diseases can have significant impacts on both human and animal populations, as well as broader environmental consequences. Some diseases have caused significant declines and extinctions in affected species and can infect domesticated animals, leading to economic losses in the agricultural sector. Diseases like avian influenza and foot-and-mouth disease can result in culling of livestock to prevent disease spread. Zoonotic diseases can also influence the health and dynamics of ecosystems. Changes in wildlife populations due to disease can have cascading effects on biodiversity and ecosystem function.

State Operations and Assets

The rapid spread of a transmissible disease can have wide-ranging impacts on state operations, affecting state functions and public safety. These impacts can disrupt state operations, strain resources, and pose challenges to maintaining public order, and can include:

- Emergency Response and Healthcare: Rhode Island would need to rapidly mobilize emergency response teams, medical personnel, and healthcare facilities. The surge in demand for medical resources can strain healthcare systems, including hospitals, clinics, and emergency services.
- Public Health Services: RIDOH would play a critical role in disease surveillance, contact tracing, and public health messaging. A transmissible disease could require additional personnel and resources to manage the outbreak.
- Resource Allocation: RIDOH may need to help allocate resources for medical supplies, pharmaceuticals, personal protective equipment, and vaccine distribution. Competition for limited resources can lead to shortages and increased costs.
- Transportation and Supply Chain Disruption: Quarantine measures, travel restrictions, and supply chain disruptions can affect the movement of essential goods and services, including medical supplies, food, and fuel.
- Economic Impact: The economic consequences of a transmissible disease can be severe. Business closures, reduced consumer confidence, and trade disruptions can lead to financial losses, unemployment, and economic instability.
- Education Disruption: School closures and disruptions to education can affect students' learning and parental work arrangements, leading to social and economic consequences.
- Public Services: Essential public services, such as law enforcement, fire services, and sanitation, may be stretched thin due to the demands of responding to the outbreak.
- Social Distancing and Isolation Measures: Government directives for social distancing, isolation, and quarantine can impact daily life, social interactions, and public gatherings. The enforcement of such measures can be challenging.
- Psychological and Societal Impact: Fear and anxiety can spread rapidly during disease transmission, affecting public morale and mental health. Disinformation and rumors can compound these psychological impacts.

Consequence Analysis

This consequence analysis lists the potential impacts of this hazard on various elements of community and state infrastructure. The impact of this hazard is evaluated in terms of disruption of operations, recovery challenges, and overall wellbeing to all Rhode Island residents and first responder personnel. The consequence analysis supplements the hazard profile by analyzing specific impacts.

Table 178: Transmissible Disease Consequence Analysis

Subject	Potential Impacts
Impact on the Public	Depending on the scale of outbreak and type of disease, residents of the State of Rhode Island may be at risk of illness or death if exposed. Population density may play a role in the spread of disease, with urban areas being more likely to be impacted than rural areas. An epidemic will have significant impacts on public health. Specific impacts to residents will be dependent upon the type of disease and how it is transmitted.
Impact on Responders	Epidemics pose a unique risk to first responders because they are more likely to be exposed to a transmissible disease before it has been identified. If the novel transmissible disease infects first responders and healthcare practitioners, the provision of public safety and public health services may be significantly impacted. Rapid identification of epidemic outbreaks is critical to preventing the spread of disease and minimizing the impact to their operations.
Continuity of Operations	Rhode Island agencies maintain continuity plans which can be enacted as necessary based on the situation. A transmissible disease may impact an agency’s ability to maintain continuity of operations based on the potential to create high levels of

Table 178: Transmissible Disease Consequence Analysis

Subject	Potential Impacts
	employee absenteeism. Employee absenteeism could also hinder the ability to fulfill critical state operations as well as implementation and maintenance of the plan itself.
Delivery of Services	Epidemics may cause disruption of services in the event of employee absenteeism.
Property, Facilities, and Infrastructure	It is unlikely that an epidemic would have direct effects on critical infrastructure or other facilities or structures. However, under cases of absenteeism, it is possible that operations may be compromised, and regular maintenance or repairs would not be performed, resulting in disrepair.
Impact on Environment	In some cases, disease outbreaks are caused by infections spreading from animals to humans. Under these circumstances, infections may be spread as the result of normal care (proximity) to sick animals or consumption of byproducts of infected animals. Infected animals may die as a result of the disease; timely removal of infected animal carcasses may help to reduce the spread of the disease among animals. It is unlikely that an epidemic would have any additional direct impacts on the environment.
Economic Conditions	Depending on the scale of outbreak and type of disease, a localized infectious disease outbreak could impact the State of Rhode Island significantly. In the event residents and workers became infected from an epidemic, employee absenteeism would increase and the length of time necessary to recover could be significant.
Public Confidence in Governance	Governmental response requires direct actions that must be immediate and effective to maintain public confidence. If government functionality is reduced by absenteeism, the public's confidence in governance may be reduced. The ability RIDOH to perform critical functions will directly impact the community's perception of government. Maintenance of these operations will be critical to response and recovery operations.

4.27.7 Hazard Planning Significance

Based on a review of available data, the SIHMC determined transmissible disease to have a moderate planning significance for Rhode Island.

Section 5 – Statewide Capability Assessment

5.1 Introduction

This capability overview for the State of Rhode Island documents programs, policies, and funding mechanisms for relevant state agencies and departments. It is important to note that the State of Rhode Island has a strong and continuing history of passing and enforcing state laws, regulations, policies, and programs relating to both hazard mitigation and climate changes, continuously striving to make a safer and more resilient home for all citizens.

All listed capabilities documented in the previous SHMP were reviewed for relevance and updated to reflect the current environment, as necessary. Additionally, any programs, policies, or funding mechanisms that are no longer applicable, are outdated, or are no longer in existence have been removed. As part of this process, updated agency and department capability profiles were sent to individual agencies for review and, if necessary, further revision.

5.2 Rhode Island Policies Supporting Risk Reduction, Resiliency, and Mitigation

The state of Rhode Island has established statutory laws and regulations that address emergency management and response. These laws provide the legal framework for emergency management activities and establish the roles and responsibilities of state and local agencies during emergencies. The primary legislation governing emergency management in Rhode Island is codified under Title 30, Chapter 15 of the Rhode Island General Laws. Title 30, Chapter 15 empowers RIEMA to coordinate and oversee emergency management efforts in the state. RIEMA is responsible for developing and implementing emergency response plans, conducting preparedness activities, coordinating resources during emergencies, and providing guidance to local municipalities. Additionally, these provisions cover various aspects of emergency management, such as the declaration of emergencies, powers and authorities of the governor and local officials during emergencies, mutual aid agreements, emergency procurement procedures, and the establishment of emergency management funds.

Title 30, Chapter 15 provides a general framework for emergency management in Rhode Island, and although it does not explicitly outline hazard mitigation planning requirements, it sets the foundation for considering mitigation efforts within the broader context of emergency management planning and preparedness.

In addition, the following policies provide for hazard mitigation guidance within the state:

- Dam Environmental Protection Agency Regulation - Rhode Island Department of Environmental Management
- Public Health Emergency Preparedness Cooperative Agreement - Rhode Island Department of Health
- Hospital Preparedness Program Cooperative Agreement- Rhode Island Department of Health
- Comprehensive Planning Regulation and Land Use Act (RIGL 45-22.2) - Rhode Island Department of Administration
- State Historic Preservation Office/RIEMA Disaster Recovery Memorandum of Understanding - Rhode Island Historical Preservation and Heritage Commission
- Rhode Island Zoning Enabling Act (45-24) - Rhode Island Department of Administration
- Rhode Island Land Development and Subdivision Review Enabling Act (45-23) - Rhode Island Department of Administration
- Resilient Rhode Island Act - Rhode Island Office of the Governor
- Executive Orders 15-17, 17-06, and 17-10 - Rhode Island Office of the Governor
- Water Use and Efficiency Act - Rhode Island Water Resources Board
- State Building Code - State Building Code Commission

5.3 Regulation of Development

Comprehensive planning in the State of Rhode Island is governed by Chapter 45-22.2 of the Rhode Island General Laws, the Rhode Island Comprehensive Planning and Land Use Act (Act). The Act, which provides for a State Guide Plan and municipal comprehensive plans, is designed to:

- Maintain municipal discretion in land use decision-making.

- Establish the lead role of the State Guide Plan.
- Require local comprehensive plans to meet certain minimum standards.
- Require zoning to conform to the comprehensive plans.
- Create an incentive for municipalities by obliging the State to conform its programs and actions to municipal plans that were certified as being consistent with State goals and policies.

Some of the key features of the Act include:

- All municipalities are to prepare and adopt a single comprehensive plan which is updated every 10 years.
- Plans must include certain standard content.
- Plan forecasts, goals, and policies must be formulated looking at a minimum 20-year planning horizon.
- Public input and comment are required.
- Comprehensive plans are to serve as the foundation for municipal zoning.
- State agencies are to conform their programs and projects to municipal plans that have State approval.

In accordance with the Act, the State Planning Council has adopted certain Rules and Standards that govern State review and approval of comprehensive plans as detailed in State Planning Council Rule Four: Comprehensive Plan Reviews.

Building Codes

In Rhode Island building codes are implemented by the Rhode Island State Building Code Commission, who provided plan input through the provision and review of building code data, and local building departments to regulate the design, construction, and maintenance of buildings. The following are key aspects of building codes:

- The Rhode Island State Building Code Commission oversees the administration and enforcement of the building code.
- The state adopts and enforces the International Building Code (IBC) as the primary model code for new construction, renovation, and alteration of buildings.
- Each city and town in Rhode Island have a local building department, with local building officials responsible for enforcing the building codes within their jurisdiction.
- Rhode Island follows the accessibility requirements outlined in the IBC and the Americans with Disabilities Act.
- Rhode Island has adopted energy efficiency codes from the International Energy Conservation Code to promote sustainable building practices and reduce energy consumption.
- Building codes include provisions for fire and life safety, including requirements for fire-rated construction materials, fire alarm systems, sprinkler systems, and emergency exits.

Building codes set a reference point for the design and construction of all structures, providing minimum safe building practices to ensure occupant safety and structure resiliency. Enforced building codes are one of the most effective hazard mitigation tools available against a wide variety of hazards. The Building Code Commission is tasked with protecting public safety welfare by establishing our state's minimum standards of construction. Building codes are then enacted and enforced primarily at the local level through local building officials: Each jurisdiction appoints its own building officials or code enforcement officers responsible for reviewing building plans, issuing permits, and inspecting construction sites. The local building department is the primary authority for enforcing building codes within its jurisdiction. Building codes utilized for Rhode Island include:

- RISBC-1 Rhode Island Building Code (510-RICR-00-00-1)
Incorporates the International Building Code, 2018 Edition, by reference
- RISBC-2 Rhode Island State One- and Two-Family Dwelling Code (510-RICR-00-00-2)
Incorporates the International Residential Code, 2018 Edition, by reference
- RISBC-3 Rhode Island Plumbing Code (510-RICR-00-00-3)
Incorporates the International Plumbing Code, 2018 Edition, by reference

- RISBC-4 Rhode Island Mechanical Code (510-RICR-00-00-4)
Incorporates the International Mechanical Code, 2018 Edition, by reference
- RISBC-5 Rhode Island Electrical Code (510-RICR-00-00-5)
Incorporates the National Electrical Code, 2020 Edition, by reference
- RISBC-6 Rhode Island Property Maintenance Code (510-RICR-00-00-6)
Incorporates the International Property Maintenance Code, 2018 Edition, by reference
- RISBC-8 State of Rhode Island Energy Conservation Code (510-RICR-00-00-8)
Incorporates the International Energy Conservation Code, 2018 Edition, by reference
- RISBC-9 Enforcement and Implementation Procedure (510-RICR-00-00-9)
- RISBC-10 Code Interpretations (510-RICR-00-00-10)
- RISBC-11 Rhode Island Certification of Building Officials, Building, Electrical, Plumbing and Mechanical (510-RICR-00-00-11)
- RISBC-12 New Materials, Devices or Methods of Construction (510-RICR-00-00-12)
- RISBC-13 Standards for Existing Schools (510-RICR-00-00-13)
- Rhode Island Swimming Pool and Spa Code (510-RICR-00-00-14)
Incorporates the International Swimming Pool and Spa Code, 2018 Edition, by reference
- RISBC-17 Public Meetings Accessibility Standard (510-RICR-00-00-17)
Incorporates the Uniform Federal Accessibility Standards, as published in the Federal Register, by reference
- RISBC-18 Use of Native Lumber (510-RICR-00-00-18)
- RISBC-19 State Fuel Gas Code (510-RICR-00-00-19)
Incorporates the International Fuel Gas Code, 2018 Edition, by reference
- RISRC-1 State Rehabilitation Building and Fire Code for Existing Structures (510-RICR-00-00-20)

The Rhode Island Building Code Commission adopted the 2018 edition of International Building Code (which include hazard resistant provisions as described in the following paragraph), along with the International Residential Code, International Plumbing Code, International Mechanical Code, International Fuel Gas Code, International Energy Conservation Code, and the 2020 edition of National Electric Code as of February 1, 2022. The Rhode Island Fire Safety Code Board of Appeal and Review adopted the 2018 editions of NFPA 1 Fire Code and NFPA 101 Life Safety Code, 2019 edition of NFPA 72 Fire Alarm Code as of July 1, 2021.

Key hazard resistant building code provisions found in the 2018 International Building Code, which is utilized statewide, include:

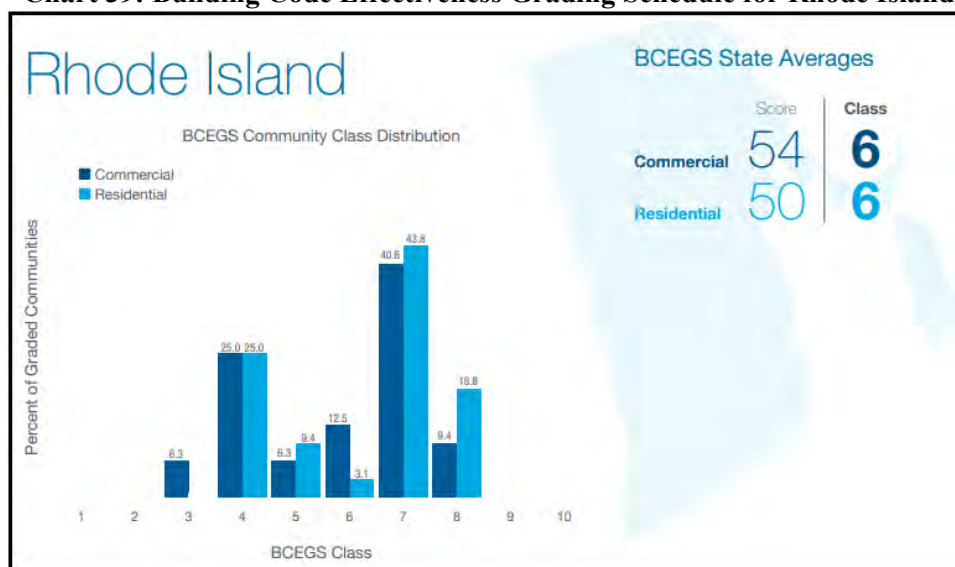
- **Structural Design Requirements:** Provides requirements for the structural design of buildings to ensure their resistance to various hazards, including earthquakes, high winds, and snow loads. These requirements are aimed at enhancing the overall structural integrity and safety of buildings.
- **Wind Design Requirements:** Provides specific provisions for wind design, considering the geographical location of the structure. Wind loads are calculated based on factors such as wind speed, exposure, and building height.
- **Seismic Design Requirements:** Incorporates seismic design provisions to address earthquake hazards. The code includes seismic design categories and requirements for the design and construction of buildings in seismic-prone regions.
- **Flood-Resistant Design Requirements:** Includes provisions related to flood-resistant design, particularly in areas prone to flooding. It may specify elevation requirements, construction materials, and other considerations to reduce the risk of flood damage. The vast majority of the regulations required by the NFIP are included within the International Building Code and the International Residential Code.
- **Fire-Resistant Construction Requirements:** Requirements for fire-resistant construction are included to mitigate the risk of fire hazards. This includes specifications for fire-resistant materials, assemblies, and building features.

- **Material and Construction Standard Requirements:** Establishes standards for building materials and construction methods to ensure the durability and safety of structures, considering various hazards.

For specific information concerning the local application and enforcement of building codes it is essential to contact the local building department for the most accurate information. It is important to note that building codes are universally and evenly enforced across the state, and no obstacles to enforcing the hazard mitigation provisions in any codes have been identified.

The Building Code Effectiveness Grading Schedule assesses the building codes in effect in a particular community and how the community enforces its building codes, with special emphasis on mitigation of losses from natural hazards. The program assigns each participating municipality a Building Code Effectiveness Grading Schedule grade of 1 (exemplary commitment to building code enforcement) to 10 (lowest rating). The following graph illustrates the combined community rating for the State of Rhode Island:

Chart 39: Building Code Effectiveness Grading Schedule for Rhode Island



Source: Building Code Effectiveness Grading Schedule

The average score for the State of Rhode Island was 54 (or a Class 6) rating for commercial, and a 50 (Class 6) for residential.

On November 9, 2009, the Green Buildings Act (RIGL §37-24) was signed into law and was further updated on June 27, 2022. The Act requires that all new construction and renovation projects over 10,000 gross square feet, constructed by a public agency, be designed and constructed to a Leadership in Energy and Environmental Design Certified or an equivalent high performance green building standard. A public agency includes all state departments, municipalities, public institutions of education, and any subdivisions of these agencies. The Rhode Island Green Buildings Advisory Committee was formed in 2010 to help implement the Green Buildings Act. The Committee, which consists of volunteers from State agencies and the public, works with the Department of Administration and the Rhode Island Building Code Commission to help ensure compliance with the law and to provide education on green building practices for Rhode Islanders.

Zoning and Land Use Planning

Zoning ordinances in Rhode Island govern land use, development, and building requirements. These ordinances work by dividing the land into different zoning districts and establishing rules and guidelines for land use, building placement, density, and setback within the zoning districts. In general, zoning ordinances establish:

- **Zoning districts:** Areas designated for specific types of land use, such as residential, commercial, industrial, agricultural, mixed-use, or special districts.

- **Land usage within a zoning district:** Specifications as to which activities, buildings, and operations are permitted in each zoning district.
- **Enforcement:** Zoning ordinances are enforced by the local building department or zoning enforcement officers.

Zoning is the traditional, and most common, tool available to local jurisdictions to control the use of land. Zoning is used to promote health, safety, and the general welfare of the community. Zoning is used to dictate the type of land use and to set minimum specifications for use such as lot size, building height and setbacks, and density of population. The following are a compilation of selected Rhode Island General Laws related to zoning and land use and planning:

- §45-22 Local Planning Board or Commission: This Act institutes the requirement that all municipalities must establish a planning board or commission. The statute also sets size limits for a board/commission as well as powers and required duties.
- §45-22.1 Joint Municipal Planning Commission: This Act authorizes two or more municipalities to establish a joint municipal planning commission.
- §45-22.2 Rhode Island Comprehensive Planning and Land Use Regulation Act: This Act establishes a system of land use planning whereby the State sets broad goals and policies through the State Guide Plan and municipalities develop comprehensive plans that reflect local desires and conditions but must be consistent with the State Guide Plan. The local comprehensive plans serve as the basis for land use regulation and establish an implementation program for achieving stated goals. The municipal zoning ordinance and map must be consistent with the comprehensive plan.
- §45-23 Rhode Island Land Development and Subdivision Review Enabling Act: This Act establishes land development and subdivision enabling authority and requires each city and town to develop land development and subdivision regulations in accordance with the community comprehensive plan, capital improvement plan, and zoning ordinance, and to ensure the consistency of all local development regulations. It also sets certain standard procedures for review and approval of land development and subdivision requests.
- §45-24 Rhode Island Zoning Enabling Act: This Act authorizes municipalities to establish zoning districts that establish regulations and standards relating to the nature and extent of uses of land and structures within the district.
- §45-24.1 Historical Area Zoning: This Act authorizes municipalities to adopt, as part of is zoning ordinance, regulations concerning the construction, alteration, repair, moving, and demolition of structures of historic or architectural value as well as historic cemeteries.
- § 45-35 Conservation Commissions: This Act grants city and town councils the authority to create a conservation commission, the purpose of which is “to promote and develop the natural resources, protect the watershed resources, and preserve natural esthetic areas within municipalities”.

State laws enable local governments to adopt and enforce zoning based upon locally developed and adopted land use plans. Under these State laws, the State sets broad goals and policies through the State Guide Plan and jurisdictions develop comprehensive plans that reflect local desires and conditions. The local comprehensive plans serve as the basis for land use regulation and establish an implementation program for achieving stated goals. The municipal zoning ordinance and map must be consistent with the comprehensive plan.

Properly applied, zoning restriction and land use planning are some of the most effective hazard mitigation tools available against a wide variety of hazards.

Rhode Island Floodplain Management Standards

Floodplain ordinances and management are one of the most effective hazard mitigation tools available against flooding. The Rhode Island Building Code Commission has adopted and incorporated the 2018 International Codes into the Rhode Island State Building Code. Many of the regulations required by the NFIP are included within the International Building Code and the International Residential Code.

While most floodplain requirements have been incorporated into the Rhode Island State Building Code, some additional provisions and regulations may be required by a community. Local floodplain ordinances, required for NFIP

participants, are often used to prevent inappropriate development in floodplains and to reduce flood hazards. In general, they allow the jurisdiction to:

- Minimize the extent of floods by preventing obstructions that inhibit water flow and increase flood height and damage.
- Prevent and minimize loss of life, injuries, and property damage in flood hazard areas.
- Promote the public health, safety, and welfare of citizens in flood hazard areas.
- Manage planned growth.
- Grant permits for use in development within special flood hazard areas that are consistent with the community ordinance and the NFIP under 44 CFR 60.3.

Rhode Island High Hazard Dam Standards

The criteria governing the administration and enforcement of Rhode Island's Dam Safety Program are contained in the General Laws of Rhode Island, Chapter 46-19. RIDEM has the responsibility to cause to be inspected dams to determine their condition, to review and approve plans for construction or substantial alteration of a dam, to order the owner to make repairs or to take other necessary action to make a dam safe.

As set forth in Rhode Island General Laws, Chapters 46-18 and 46-19, a dam owner has the responsibility for the safe operation of his/her dam and is liable for the consequences of accidents or failures of the dam. In general, a dam owner is required to use "reasonable care" in the operation and maintenance of a dam. This responsibility includes the proper operation, maintenance, repair, and rehabilitation of a dam, which are essential elements in preventing a dam failure.

Rhode Island General Laws Chapter 45-62 authorizes cities and towns to create dam management districts for the maintenance and repair of dams within their boundaries. The following two districts are currently in operation:

- Burrillville: Pascoag Upper (dam number 016)
- Exeter: Boone (dam number 219)

RIDEM does not have dam safety regulatory authority for any dam that has a Federal Energy Regulatory Commission Permit or Exemption. The following seven dams are under Federal Energy Regulatory Commission Authority:

- Central Falls / Cumberland: Valley Falls Pond (dam number 063)
- Central Falls / Pawtucket: Central Falls (dam number 064)
- North Smithfield: Blackstone (dam number 393)
- Pawtucket: Pawtucket Lower (dam number 066)
- West Warwick: Riverpoint Pond Upper (dam number 147)
- West Warwick: Arctic (dam number 148)
- Woonsocket: Woonsocket Falls (dam number 056)

Rhode Island General Laws Section 46-19-9 requires the municipality where either a High Hazard or Significant Hazard dam is located to complete an Emergency Action Plan and submit it to RIEMA for approval. The Department of Environmental Management works with RIEMA in reviewing and these Emergency Action Plans. In total, 178 dams require an Emergency Action Plan, but data in the last available reporting in 2020 indicates that only 68 dams have approved plans (including seven of 36 unsafe High Hazard dams). Recognizing this deficiency, Rhode Island RIDEM has made resources available to local jurisdictions to aid in the creation of EAPs, including access to RIDEM staff and online guides assisting in the creation of new EAPs. Additionally, for jurisdictions with High and Significant Hazard dams without EAPs, RIEMA has actively encouraged new and existing local Hazard Mitigation Plans to include mitigation actions that address the problem.

5.4 Plans and Programs Supporting Risk Reduction, Resiliency, and Hazard Mitigation

The following tables list relevant plans and programs that bolster statewide risk reduction, increase overall resiliency, and encourage mitigation measures.

Table 179: Rhode Island Plans and Programs Supporting Risk Reduction, Resiliency, and Mitigation

Agency	Plans and Programs
Rhode Island Coastal Resources Management Council	State Coastal Plan, Sea Level Affecting Marshes Model, Rhode Island Shoreline Change Special Area Management Plan, Shoreline Change Maps, Coastal Zone Buffer Program, Stormwater Design and Installation Standards Manual, Rhode Island Coastal and Estuarine Land Conservation Plan, Coastal Resources Management Program, Coastal Smart Growth Guide, Rhode Island Coastal Wetland Restoration Strategy, Coastal Landscapes, Low Impact Development Planning and Design Guidance Manual, Mapping Protocol for Eroding Bluffs
Rhode Island Department of Administration (Division of Statewide Planning)	State Plan Guide, Planning for a Climate Resilient Community Technical Assistance, Vulnerability of Transportation Assets to Sea Level Rise and Storm Surge Assessments, Socioeconomics of Sea Level Rise, 2022 Rhode Island Freight and Goods Movement Plan, Small Business Risk Reduction to Natural Hazards and Extreme Weather in Rhode Island, Moving Forward RI 2040 - the State of Rhode Island Long Range Transportation Plan and Metropolitan Transportation Plan, Transit Forward RI 2040, Protecting Our Legacy of Buildings, Places, and Culture: An Historic Preservation Plan for Rhode Island, RIGIS Strategic Plan, Public Participation and Limited English Proficiency Plan, Land Use 2025: Rhode Island's State land Use Policies and Plan
Rhode Island Department of Environmental Management	Dam Safety Program, Division of Forest Environment, Office of Water Resources (wastewater and storm water management, stream level and groundwater monitoring, watershed planning, stormwater design and installation manual, freshwater wetland programs, low impact development guidance, design guidance for considering climate change/planning and design guidance), Emergency Response Planning, (hazardous materials/oil spills, hurricane, prescribed fire burn, droughts), Implications of Climate Change for Rhode Island Wastewater Collection and Treatment Infrastructure study, Strategic Plan 2019-2022
Rhode Island Department of Health	Center for Emergency Preparedness and Response; Center for Drinking Water Quality, Center for Food Protection, Center for Health Facilities Regulation, Center for Public Health Communication, Center for Acute Infectious Disease Epidemiology, Center for HIV, Hepatitis, STD, and TB Epidemiology; Center for COVID-19 Epidemiology, Rhode Island State Health Laboratories, Office of the State Medical Examiners, Center for Preventive Services, Center for Healthy Homes and Environment, Center for Professional Licensing, Boards, and Commissions, Office of Health Regulation, Office of Public Affairs and Legislation, Center for Health Systems Policy and Planning, and the Center for Health Promotion
Rhode Island Department of Transportation	Hurricane Evacuation Routes, Strategic Highway Safety Plan and Highway safety campaigns, Stormwater management program, Infrastructure vulnerability assessments
RIEMA	NFIP, CRS, FEMA Risk MAP, Common Alerting Protocol, Rhode Island Flood Audit Task Force, Debris Management Plan Technical Assistance, Emergency Operations Plan, Infrastructure Protection Program, Protected Critical Infrastructure Information Program, State Emergency Operations Center, State Hazard Mitigation Program, Threat and Hazard Identification and Risk Assessment, Homeland Security Information Network
Rhode Island Infrastructure Bank	Resilient Rhody
Rhode Island Office of Energy Resources	Energy Assurance Plan, Energy 2035: Rhode Island State Energy Plan
Rhode Island Office of the Governor	Cybersecurity Commission
Rhode Island Office of Housing and Community Development	Disaster Housing Task Force, Housing Resources Commission, Community Development Block Grant Disaster Recovery Action plans

Table 179: Rhode Island Plans and Programs Supporting Risk Reduction, Resiliency, and Mitigation

Agency	Plans and Programs
Rhode Island Public Utilities Commission and Division of Public Utilities and Carriers	Infrastructure, Safety, and Reliability dockets for PPL Rhode Island Energy Gas and Electric, Major storm event damage summaries
Rhode Island Rivers Council	Cultural Heritage and Land Management Plan for the Blackstone River Valley, Rhode Island Rivers Policy and Classification Plan, Establishment of Riparian and Shoreline Buffers and the Taxation of Property included in Buffers report
Rhode Island Sea Grant College Program and University of Rhode Island Coastal Resources Center	Ongoing research/vulnerability assessments for coastal hazards and SLR, Technical assistance to decision makers at the state and local level, climate change adaptation planning, Rhode Island Ocean Special Area Management Plan, Rhode Island Green Infrastructure Project, Rapid PACE Mapping, Providing Resilience Education for Planning Preparedness in Rhode Island, STORMTOOLS, Block Island Shoreline Change Maps
Rhode Island State Police	Storm Preparedness Resources, Rhode Island Joint Cyber Task Force
Rhode Island Water Resources Board	Strategic Plan, Water Supply System Management Plan program, Drought Management and Steering Committee
State Building Code Commission	Building Code Standards Committee, Rebuilding After a Storm
University of Rhode Island Environmental Data Center	GIS/Mapping services, LiDAR Statewide Digital Elevation Model data, technical support

The following tables list relevant partner agencies, including federal agencies, plans and programs that bolster statewide risk reduction, increase overall resiliency, and encourage mitigation measures.

Table 180: Partner Agency Plans and Programs

Agency	Plans and Programs
PPL Rhode Island Energy	Electric Infrastructure, Safety, and Reliability Plan (includes status of flood mitigation projects), Enhanced Hazard Tree Mitigation Program, regular inspections and maintenance, storm warnings and safety/power notifications
NWS	HURREVAC/SLOSH training, Severe Weather Spotter training, StormReady, flood event summaries and studies, river and dam break modeling
USACE	Dam breach failure analysis, SLR analysis, Flood Control Projects, Habitat Restoration, Floodplain Management Services, Planning Assistance to States Wetland Identification and Restoration, Estuary Protection, inspections of infrastructure, Rhode Island Silver Jackets Team, Rhode Island Coastline Coastal Storm Risk Management Project
USDA Natural Resources Conservation Service	Emergency Watershed Protection Program, Watershed and Flood Prevention Program
USGS	Real-time monitoring of river stage, streamflow tides, and lake and pond levels, real-time monitoring of groundwater levels, real-time monitoring of water quality, coastal storm tide and storm monitoring and data collection, post flood and coastal storm data and studies, post drought data and studies. peak flow analyses and equations for estimating flood flows, streamflow analyses and equations for estimating low flows, real-time flood inundation mapping studies, real-time earthquake monitoring and studies

5.5 RIEMA Profile

RIEMA is responsible for establishing, sustaining, and coordinating the resources of Federal, State, and local governments, non-governmental organizations, and the private sector to effectively meet the challenges faced during a natural and/or man-made incident. This is accomplished by the Director through the Governor's authority under R.I. General Laws § 30-15. The organization's primary goal is to create and oversee a statewide system with an all-hazards approach to emergency management and homeland security. RIEMA is also responsible for strengthening the thirty-

two (32) core capabilities outlined by the National Preparedness Goal as established by Presidential Policy Directive 8 to reduce loss of life and property within the State of Rhode Island during natural and/or man-made incidents. The Rhode Island Emergency Management Agency and the State Emergency Operation Center (SEOC) provides a central location from which all state government at any level can provide interagency coordination and executive decision-making in support of any regional incident or local response. This is achieved through information collection and evaluation, executive decision-making, priority setting, management, coordination and tracking of resources, and interoperable communication facilitation to support overall response efforts.

As the primary State of Rhode Island agency overseeing both hazard mitigation and emergency management, RIEMA's main purpose is to protect life and property in the event of a disaster or crisis through an emergency management program of mitigation, preparedness, response, and recovery. RIEMA plays a crucial role in supporting hazard mitigation planning, with methods of support including:

- **State Hazard Mitigation Plan:** RIEMA has developed a comprehensive SHMP that serves as a framework for identifying and addressing hazards and risks across the state. This plan outlines the state's mitigation goals, strategies, and priorities, and includes input from various stakeholders and the public. This plan can be utilized by jurisdictions within the state to inform the creation of their hazard mitigation plans.
- **Technical Assistance:** RIEMA offers technical assistance to local jurisdictions, organizations, and communities to help them develop their own hazard mitigation plans. This assistance involves providing guidance on risk assessments, mitigation strategies, plan development, and implementation approaches.
- **Funding Opportunities:** RIEMA administers grant programs to provide financial assistance for hazard mitigation projects at the local level. These programs may be funded through state budgets, federal grants, or a combination of both.
- **Capacity Building and Training:** RIEMA organizes training programs, workshops, and educational initiatives to build the capacity of local jurisdictions and organizations in hazard mitigation planning and implementation. These efforts help enhance the knowledge and skills of professionals involved in mitigation activities.
- **Collaboration and Coordination:** RIEMA facilitates collaboration and coordination among different stakeholders involved in hazard mitigation, including local governments, emergency management agencies, non-profit organizations, and community groups. This collaborative approach helps ensure a comprehensive and integrated approach to mitigation planning.
- **Data and Information Sharing:** RIEMA maintains databases, maps, and other resources related to hazard risks and vulnerabilities. By making this information accessible to local jurisdictions and stakeholders, Rhode Island facilitates informed decision-making and helps ensure that mitigation efforts are based on accurate and up-to-date data.

The Planning and Mitigation Section and the State Hazard Mitigation Officer facilitate long-range planning activities that result in the development of comprehensive strategies to safeguard the public and property. This Section is responsible for the administration of the National Flood Insurance Program, statewide emergency planning, and the coordination of local and state plans, utilizing an all-hazards approach.

The Planning and Mitigation Section also focuses on current topics in emergency management, such as improving the State's resiliency to climate change and for future disasters through adaptation and mitigation programs. In general, the Section is charged with all mitigation related activities, including:

- Preventing future loss of lives and property through mitigation measures and programs
- Overseeing the development, revision, and implementation of all state hazard mitigation plans
- Assisting with the implementation and completion of identified mitigation actions and programs
- Providing direction and oversight on all hazard mitigation related funding and grants

The following highlight some of the programs that RIEMA operates and oversees:

- **State Comprehensive Emergency Management Plan:** The Comprehensive Emergency Management Plan defines the structure of emergency operations at the state level, to describe the relationships between the state and its partners at the household, local and federal levels, and to identify the various agencies and departments

with functional responsibilities. The plan addresses all hazards as identified in this SHMP and supports loss reduction by providing both emergency management and continuity of operations planning resources to the counties, other state agencies, and other organizations.

- **State Emergency Operations Center:** The State Emergency Operations Center provides a central location to provide interagency coordination and executive decision-making in support of any regional incident or local response. RIEMA's State Emergency Operations Center operates under the guidance of the National Response Framework and National Incident Management System.
- **Critical Facilities and Infrastructure Management:** This program enhances critical infrastructure protection on a state-wide basis. The program is used in supporting agency goals and objectives when deciding where to invest resources and grant allocations to enhance asset protection and resiliency. To support the program, RIEMA is an accredited Department of Homeland Security agency for the Protected Critical Infrastructure Information program, an information-protection program that enhances voluntary information sharing between infrastructure owners and operators and the government. Data is used to analyze and secure critical infrastructure and protected systems, identify vulnerabilities, develop risk assessments, and enhance recovery preparedness measures. In addition, RIEMA maintains access to the Automated Critical Asset Management System to provide a secure, online database as a management and analysis platform that allows for the collection and management of critical infrastructure asset data.
- **Rhode Island State Communications Network:** This network serves public safety agencies at the local, state, military, and federal levels. There are approximately 11,000 radios in the system, some of which are used heavily during incidents and planned events to coordinate agencies from across the state and country. There are 27 radio sites spread throughout Rhode Island, linked by microwave. All sites have a combination of diesel and propane generators.
- **National Flood Insurance Program:** Administered by RIEMA, these regulations can involve regulation of development within a designated floodway area, restrictions on activities that involve alteration of watercourses or sand dunes, and criteria for the location of mobile homes, and affords homeowners, renters, or business owners the opportunity to purchase federally backed flood insurance.
- **Community Rating System:** Administered by RIEMA, the CRS is a voluntary program that recognizes and encourages a community's efforts that exceed the NFIP minimum requirements for floodplain management.
- **Grant Program Administration:** RIEMA administers many hazard mitigation related grant programs supporting risk reduction and mitigation activities for state and local agencies:

RIEMA provides a wide variety of technical assistance to local emergency managers, businesses, and the public, including:

- GIS technical assistance
- Floodplain management assistance
- Local Hazard Mitigation Planning assistance
- Mitigation project oversight and assistance

The following charts represent the breadth of technical assistance available through RIEMA:

Chart 40: RIEMA Technical Assistance for Local Emergency Managers



Chart 41: RIEMA Technical Assistance for Businesses

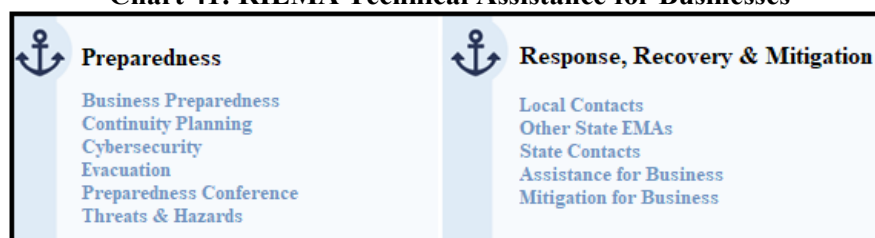


Chart 42: RIEMA Technical Assistance for Citizens



Technical assistance to communities includes planning workshops and providing GIS maps of the communities' risks and vulnerabilities. Additionally, RIEMA encourages jurisdictions to establish a hazard mitigation planning committees consisting of local subject matter experts and jurisdictional personnel to assist with identifying hazards, incorporating public input, developing, and prioritizing mitigation actions, and implementing and revising strategies.

As a stand-alone agency RIEMA is responsible for its own operational budget. The following table illustrates the budgets for 2019 through 2021 and the 2022 fiscal year budget.

Table 181: RIEMA Budget

	2019 Actuals	2020 Actuals	2021 Enacted Budget	2021 Revised Budget	2022 Recommended
Expenditures by Program					
Central Management	\$19,850,361	\$78,934,793	\$45,266,961	\$59,785,166	\$21,400,088
Total Expenditures	\$19,850,361	\$78,934,793	\$45,266,961	\$59,785,166	\$21,400,088
Expenditure by Object					
Salary and Benefits	\$3,321,401	\$3,856,700	\$3,993,821	\$4,837,789	\$4,076,676
Contract Professional Services	\$468,047	\$276,599	\$347,531	\$425,713	\$915,235
Operating Supplies and Expenses	\$4,118,669	\$65,481,832	\$36,173,940	\$43,426,589	\$4,318,934
Assistance and Grants	\$9,632,146	\$3,355,420	\$3,222,255	\$9,560,411	\$9,870,167
Capital Purchases and Equipment	\$315,684	\$3,969,828	\$1,529,414	\$1,534,664	\$2,219,076
Debt Services (Fixed Charges)	\$1,994,414	\$1,994,414	\$0	\$0	\$0
Total Expenditures	\$19,850,361	\$78,934,794	\$45,266,961	\$59,785,166	\$21,400,088
Expenditures by Source of Funds					
General Revenue	\$2,276,135	\$66,323,907	\$2,713,353	\$2,415,075	\$2,655,619
Federal Funds	\$15,703,886	\$10,927,297	\$40,506,062	\$55,321,487	\$16,472,597
Restricted Receipts	\$375,927	\$189,175	\$553,132	\$554,190	\$527,472
Transfers from Other Funds	\$1,494,414	\$1,494,414	\$1,494,414	\$1,494,414	\$1,744,400
Total Expenditures	\$19,850,361	\$78,934,794	\$45,266,961	\$59,785,166	\$21,400,088
FTE Authorization	32	32	32	32	33

Source: RIEMA

5.6 Agency Specific Profiles

Utilizing a strategy presented in the 2019 SHMP, the following table highlights the capabilities of the State as it pertains to hazard mitigation and helps identify areas of potential improvement. To evaluate the capabilities of a selected agency a weighting system was implemented to determine overall capability related to hazard mitigation. Each of the enumerated basic capabilities were reviewed and given a numerical value to demonstrate the ability of each activity to support mitigation efforts. Pre- and post-disaster capabilities were scored the highest, receiving a weighting of two. Other important but lesser valued capabilities, including funding and technical assistance resources, were weighted with a value of one and a half. Finally, the ability to regulate development in hazard prone areas and staffing resources were weighted with a value of one.

Once assessments were completed, agencies with an overall capability of nine (the highest available ranking) were given a high capability ranking. Assessment totals between seven and eight were given a medium capability ranking. A neutral capability ranking was given to agencies with scores below seven.

Table 182: Rhode Island Agency Capability Review

Agency	Pre-Disaster Capabilities	Post-Disaster Capabilities	Regulates Development in Hazard Prone Areas	Funding Capabilities	Staffing Resources	Technical Assistance	Overall Capacity Score	Capability Ranking
Rhode Island Department of Administration (Division of Statewide Planning)	2	2	1	1.5	1	1.5	9	High
RIDEM	2	2	1	1.5	1	1.5	9	High
RIDOH	2	2	1	1.5	1	1.5	9	High
RIDOT	2	2	1	1.5	1	1.5	9	High

Table 182: Rhode Island Agency Capability Review

Agency	Pre-Disaster Capabilities	Post-Disaster Capabilities	Regulates Development in Hazard Prone Areas	Funding Capabilities	Staffing Resources	Technical Assistance	Overall Capacity Score	Capability Ranking
RIIB	2	2	0	1.5	1	1.5	8	Medium
Rhode Island Water Resources Board	2	2	1	1.5	0	1.5	8	Medium
Rhode Island Coastal Resources Management Council	2	2	1	0	1	1.5	7.5	Medium
Rhode Island Office of Energy Resources	2	2	1	0	1	1.5	7.5	Medium
Rhode Island Office of Housing and Community Development	2	2	0	1.5	0	1.5	7	Medium
Rhode Island Rivers Council	2	2	0	1.5	0	1.5	7	Medium
University of Rhode Island Coastal Resources Center	2	0	1	1.5	1	1.5	7	Medium
Rhode Island Turnpike and Bridge Authority	2	1	1	2	0	1	7	Medium
Rhode Island Office of the Governor	2	2	0	1.5	1	0	6.5	Neutral
Rhode Island Public Utilities Commission and Division of Public Utilities and Carriers	2	2	0	1.5	1	0	6.5	Neutral
Rhode Island State Police	2	2	0	0	1	1.5	6.5	Neutral
University of Rhode Island Environmental Data Center	2	2	0	0	1	1.5	6.5	Neutral
Rhode Island Historical Preservation and Heritage Commission	2	2	0	0	0	1.5	5.5	Neutral
State Building Code Commission	2	2	1	0	0	0	5	Neutral

The following table writeups detail further information concerning relevant agencies with a high or medium capability ranking related to hazard mitigation efforts.

Table 183: Rhode Island Department of Administration (Division of Statewide Planning)

Capability Rating	High
Agency Description	
The Rhode Island Department of Administration (Division of Statewide Planning) provides support services to ensure the coordination and implementation of statewide programs, as well as policy direction related to administrative services. The Department of Administration's is comprised of the multiple divisions, including the Division of Statewide Planning. The Rhode Island Statewide Planning Program is tasked with preparing, adopting, and amending strategic plans for the physical, economic, and social development of the state (RIGL 42-11-10 (b) (1)), also known as the State Guide Plan. The Statewide Planning Program in Rhode Island is overseen by the State Planning Council. The State Planning Council is the designated Metropolitan Planning Organization for the state. The staff of the Division of Statewide Planning are staff of both the Statewide Planning Program as well as the Metropolitan Planning Organization. The State Guide Plan is Rhode Island's centralized and integrated long-range planning document comprising of separately published elements covering a range of topics. The State Guide Plan has four main functions, setting long-range policy, providing a means to evaluate and coordinate projects or proposals of state importance, setting standards for local comprehensive plans, and serving as a general background information source on various topics. The Rhode Island Climate Change website as well as other online resources that provide information regarding plans, programs, and policies related to resilience is also housed under Rhode Island Department of Administration. The Division of Statewide Planning also provides planning services related to Climate Change resiliency, adaptation, and mitigation.	
Available Technical Resources	
Grant application management	

Table 183: Rhode Island Department of Administration (Division of Statewide Planning)

Capability Rating	High
• Grant tracking • Review of municipal comprehensive plan • Technical assistance programs for municipalities • GIS mapping services • Planning support to other state agencies • Community engagement strategies • Guidance on legislation relating to land use, subdivision, and zoning Acts • Guidance for municipalities on how best to incorporate natural hazards and climate change into municipal comprehensive plans • Recommendations on how to assess hazard impacts and determine appropriate strategies	

Table 184: Rhode Island Department of Environmental Management

Capability Rating	High
Agency Description	
RIDEM is the chief steward of Rhode Island's natural resources. The department enforces laws that protect the environment, public health and safety, and administers a variety of programs that serve to support risk reduction and hazard mitigation and serves as the chair of the EC4. The agency is organized into two major branches, Natural Resources and Environmental Protection. The following are specific program areas within RIDEM that pertain to risk reduction and hazard mitigation: • Office of Emergency Response: Responsible for Tier II Reporting Requirement and houses the State Climatologist. • Office of Water Resources: Implements programs for the protection of surface water, groundwater and wetlands, and conducts water monitoring. • Office of Compliance and Inspection: Maintains the Dam Safety program for inspecting and evaluating the safety of dams and approves plans for future dam construction and a Hazardous Waste program that performs compliance monitoring and investigates the improper storage and/or handling of hazardous waste. • Office of Air Resources: Regulates and monitors the emission of air pollutants from stationary and mobile emission sources. • Office of Waste Management: Oversees the investigation and remediation of contaminated sites and regulates and permit facilities that accept or transport solid, medical, or hazardous waste or that store petroleum products in underground tanks. • Division of Agriculture: Works to ensure long-term agricultural viability, and oversees six primary program areas, including pesticides, animal health, plant industry, produce safety, mosquito abatement, and farmland information. • Division of Fish and Wildlife: Protects and restores habitat for resiliency and monitors fish and wildlife populations for evidence of wildlife diseases. • Division of Forest Environment: Manages all State-owned rural forestland and is responsible for tree removal and pruning, coordination of statewide forest fire protection plan, forest fire protection on State-owned lands, provision of assistance to rural volunteer fire departments, development of forest fire and wildlife management plans for private landowners, risk assessments of fires in wild land areas, and management of suppression resources and coordination with local, state, and federal entities. • Dam Safety Program, Office of Water Resources (wastewater and storm water management, stream level and groundwater monitoring, watershed planning, stormwater design and installation manual, freshwater wetland programs, low impact development guidance, design guidance for considering climate change/planning and design guidance), Emergency Response Planning, (hazardous materials/oil spills, hurricane, droughts), Division of Forest Environment (wildfire prevention, wildfire suppression)	

Table 184: Rhode Island Department of Environmental Management

Capability Rating	High
Implications of Climate Change for Rhode Island Wastewater Collection and Treatment Infrastructure study, Strategic Plan 2019-2022	
Available Technical Resources	
• Training for Hazardous Material Response Teams • Assistance related to wildfire prevention • Pre-application permitting assistance for mitigation measures • Guidance to farmers on best practices including measures needed during periods of drought and funding for the disposal of farm chemicals • Assistance with statewide wastewater resiliency improvements	

Table 185: Rhode Island Department of Health

Capability Rating	High
Agency Description	
RIDOH is the State's public health authority. The mission of RIDOH is to prevent disease and to protect and promote health and safety of the people of Rhode Island. To fulfill this mission, RIDOH operates a range of programs to address various elements of public health, including those related to public health and healthcare emergency preparedness and response, the regulation and assurance of safe drinking water and food supplies, public communications, disease surveillance and intervention, healthcare facility and professional licensure and regulation, immunizations, laboratory testing, and decedent management. RIDOH is the lead Emergency Support Function 8 agency in Rhode Island and plays a supporting role in Recovery Support Function 3. As such, it is responsible for coordinating the public health and medical aspects of disaster responses in Rhode Island and supports the efforts of the Executive Office of Health and Human Services and other partner agencies in assisting Rhode Island's recovery from major disasters. RIDOH also co-leads the Healthcare Coalition of Rhode Island, which is the State's healthcare emergency preparedness and response coalition, comprising members from hospitals, long-term care facilities, community health centers, urgent care centers, home healthcare and hospice agencies, dialysis centers, emergency medical services, and others. HCRI also includes emergency management representation among its members. The Healthcare Coalition of Rhode Island provides two main functions for its members: • A forum to facilitate information sharing among its members. Information, such as best practices, lessons learned from exercises and real-world events, details on upcoming events of interest, intelligence on new or emerging threats, etc., is routinely shared in the Healthcare Coalition of Rhode Island, both during scheduled meetings and on an ongoing basis through email and other means. • A mechanism to enhance coordination among its members and with response agencies outside the healthcare sector during emergency responses. The structure of the Healthcare Coalition of Rhode Island, with its direct connection to RIDOH and the Hospital Association of Rhode Island, provides external response agencies a single, unified means of interaction with the healthcare sector, thus ensuring an accurate and valid common operating picture for the overall response. The information sharing mechanisms within the Coalition also lend to enhanced coordination among its members during responses.	
Available Technical Resources	
• State Public Health Authority • Public health emergency preparedness and response • Healthcare emergency preparedness and response • Public information • Clinical and public health expertise • Environmental health toxicology	

Table 185: Rhode Island Department of Health

Capability Rating	High
• Disease surveillance and investigation • Disease intervention • Laboratory testing • Public drinking water regulatory oversight • Food safety regulatory oversight • Healthcare facility licensure and oversight • Decedent management, including death investigation • Emergency medical services training, licensure, and oversight • Vital records	

Table 186: Rhode Island Department of Transportation

Capability Rating	High
Agency Description	
The mission of RIDOT is to provide, maintain, and secure a safe intermodal transportation network that increases mobility opportunities for the movement of people and goods, with the goals of enabling economic development and improving quality of life. RIDOT provides several programs that are designed to protect residents, the environment, and increase economic growth. Examples include highway safety campaigns, storm water management planning, and economic recovery tools.	
Available Technical Resources	
• Engineering assistance, including bridge/structural inspections and assessments, hydraulics, geotechnical, transportation, and electrical assistance • Conducts GIS and mapping • Provides equipment and certified mechanics/technicians/operators • Guides on historic preservation during design and construction • Maintains statewide evacuation routes • Oversees and provides guidance on the usage of federal highway aid, including the location and hydraulic design of roads and bridges that encroach on floodplains	

Table 187: Rhode Island Infrastructure Bank

Capability Rating	Medium
Agency Description	
The mission of the RIIB is to actively support and finance investments in the State's infrastructure. The Infrastructure Bank does so through a variety of means, including the issuance of bonds, originating loans and making grants, and the engagement with and mobilization of sources of public and private capital. Through its activities, the Bank fosters infrastructure improvements that create jobs, promote economic development, and enhance the environment.	
Available Technical Resources	
• Home to the Chief Resilience Officer • Actively engages with Rhode Island's municipal officials about their infrastructure needs and identifying opportunities to build climate resilience. • Acts as a central hub for financing infrastructure improvements for municipalities, businesses, and homeowners including water and wastewater, road and bridge, energy efficiency and renewable energy, and brownfield remediation. • Provides the following financing programs: ○ Brownfields Revolving Loan Fund ○ Clean Water State Revolving Fund ○ C-PACE ○ CSSLP ○ Drinking Water State Revolving Fund	

Table 187: Rhode Island Infrastructure Bank

Capability Rating	Medium
- o Efficient Buildings Fund - o Facility Plan Loan Fund - o Municipal Road and Bridge Revolving Fund - o Rhode Island Water Pollution Control Revolving Fund - o Sewer Tie-In Loan Fund	

Table 188: Rhode Island Water Resources Board

Capability Rating	Medium
Agency Description	
The Rhode Island Water Resources Board regulates the proper development, protection, conservation, and use of the state's water resources and manages the withdrawals and use of the waters of the state, while providing for economic development and protection of the environment. The Rhode Island Water Resources Board implements capital projects that result from their plans and studies. These programs help strengthen water resources and supply in the state and prepare the state for emergency drought conditions, the ongoing effects of climate change, and the future water supply needs based on growth and water availability. Statewide strategic planning and risk evaluation assist in identifying risk, capacity, and mitigation and response strategies. Allocation research supports mitigation efforts and future allocation programs will consider the potential impacts of climate and hydrologic variability. They also administer the Drought Steering Committee, which assigns drought stages, engages in public communication, and coordinates with suppliers. One policy that the Rhode Island Water Resources Board oversees is the Water Use and Efficiency Act, which was passed to ensure reasonable, needed, and adequate water supplies through managing demand, reinvesting in water supply infrastructure and water supply resources, and protecting and preserving the health and ecological function of the water resources in the state.	
Available Technical Resources	
• Provides technical guidance for municipalities and water suppliers to align land management programs with water resource programs. • Assistance with early warning of drought conditions allowing for reductions in water use and preserving storage and groundwater levels. • Development regulation in hazard prone areas through the Division of Statewide Planning Program's Community Comprehensive Plan in coordination with the Division of Planning. • Completes emergency plans that are updated every five years to provide continuous assessment of hazards, risks, and responses.	

Table 189: Rhode Island Coastal Resources Management Council

Capability Rating	Medium
Description	
The CRMC plans for and manages the coastal resources of the State. Within the State Coastal Plan, there are numerous policies and programs for the protection of coastal resources, including coastal wetlands and other shoreline features (e.g., beaches, and dunes). CRMC has statutory authority to restrict the alteration of coastal wetlands for preservation purposes, and to implement regulations requiring erosion setbacks, coastal buffer zones and to prohibit construction on beaches, in dunes, and on barrier spits that are identified as undeveloped or moderately developed. The preservation of coastal wetlands, beaches and dunes from development and destruction will provide for the natural and beneficial use of coastal wetlands, beaches and dunes as related to flood retention and natural buffers from coastal storm surge and waves.	
Available Technical Resources	
• Stormwater design and installation, and coastal and estuarine land conservation support	

Table 190: Rhode Island Office of Energy Resources

Capability Rating	Medium
Agency Description	
The mission of Rhode Island Office of Energy Resources is to help Rhode Island achieve a reliable, cost-effective, and sustainable energy future. They work closely with private and public stakeholders to increase the security of the State's energy supply, reduce energy costs, and improve environmental quality. The Rhode Island Office of Energy Resources recommends and implements smart energy policies to help reduce the state's dependency on out-of-state fuels., each are discussed in further detail below.	
Available Technical Resources	
- Fuel terminal communication and response support - Manages the Energy Assurance Plan providing the State with guidance to prepare for, monitor, and mitigate energy deficiencies and disruptions - Maintains the Rhode Island State Energy Plan	

Table 191: Rhode Island Office of Housing and Community Development

Capability Rating	Medium
Agency Description	
The Rhode Island Office of Housing and Community Development coordinates housing functions and selected community development activities of State agencies. The primary purpose of the office is to promote decent, safe, and affordable housing opportunities; create a suitable living environment; and expand economic opportunities, principally for low-and moderate-income Rhode Islanders. This Office provides financial and operational support for all housing programs administered by the Housing Resources Commission. The Office works with communities to achieve their affordable housing goals. They assist the Planning staff in review of Local Comprehensive Plan Updates for consistency with State Affordable Housing Elements. They have partnered with RIEMA to create planning documents to address housing issues following a disaster.	
Available Technical Resources	
- Prepares Community Development Block Grant Action Plans after funds are allocated to the state - Guidance on federal regulations and the Community Development Block Grant program	

Table 192: Rhode Island Rivers Council

Capability Rating	Medium
Agency Description	
The Rhode Island General Assembly created the Rhode Island Rivers Council in 1991 to coordinate efforts to improve the quality of the state's rivers and their watersheds. The Rhode Island Rivers Council authorizes local watershed councils with state designation for five-year terms. This authority provides local watershed councils legal standing to represent water bodies in their watersheds before state and local government agencies. These local watershed councils are also eligible for an annual grant from the Rivers Council.	
Available Technical Resources	
- Maintains the Rhode Island Rivers Policy and Classification Plan and is charged with its implementation. - Works to strengthen the capacity and capability of local watershed councils and associations. - Participates in the Rhode Island Renewable Energy Siting Partnership regarding the hydropower element of the state energy plan. - Considers the merits of dam removal projects and hydropower projects. - Establishes riparian and shoreline buffers, including reduced taxation for buffered properties.	

Table 193: University of Rhode Island Coastal Resources Center

Capability Rating	Medium
Agency Description	
The URI CRC conducts research, training, and public outreach on priority issues of importance to coastal communities and the marine environment with the goal of providing the best available information to decision makers and the public. The program is a partnership between RISG, a NOAA funded program, and CRC, a center within the University of Rhode Island with links to a network of 30 Sea Grant university programs nationwide. The program	

Table 193: University of Rhode Island Coastal Resources Center

Capability Rating	Medium
supports research on issues such as resilient coastal development, healthy coastal ecosystems, and safe and sustainable seafood. The program has done extensive research on SLR and coastal hazards in Rhode Island. They are part of several interagency work groups and have a history of working with communities and businesses to minimize impacts of coastal and riverine flooding. The URI CRC works closely with various state agencies, including the CRMC, with a specific focus on the development of coastal policy and Special Area Management Plans.	
Available Technical Resources	
- Helps decision makers understand the science and policy implications of climate change - Works with communities to pilot climate adaptation planning - Participates in the Coastal Green and Resilient Infrastructure Project - Guidance on the STORMTOOLS and Rapid PACE Mapping tools	

Table 194: Rhode Island Turnpike and Bridge Authority

Capability Rating	Medium
Agency Description	
Rhode Island Turnpike and Bridge Authority is a quasi-public agency created by the Rhode Island General Assembly in 1954 as a corporate and body politic, with powers to construct, acquire, maintain, and operate bridge projects as defined by law. The Authority was responsible for the construction of the Claiborne Pell Bridge (formerly the Newport Bridge) which opened to traffic on June 28, 1969. The Authority has been responsible for the operation and maintenance of the Mount Hope Bridge between Bristol, Rhode Island and Portsmouth, Rhode Island and the Claiborne Pell Bridge between Newport, Rhode Island and Jamestown, Rhode Island since 1964 and 1969, respectively. On April 25, 2013, the State transferred custody, control and supervision of the land and improvements for the Jamestown Verrazzano and the Sakonnet River Bridges from the RIDOT to the Authority. Ownership and title of the bridges remains with the State. In addition to the four bridges noted above, the Authority also operates and maintains Route 138 through Jamestown and ten smaller bridges associated with this highway and the approaches to the four major bridges.	
Available Technical Resources	
- A 24/7 Traffic Management Center (with video capability for the four major bridges and roadways under care and control. - An ATMS with the ability to control Variable Message Signs. - The Maintenance department owns and operates multiple construction and inspection vehicles including an Aspen 62 under bridge inspection unit, bucket truck, Anderson Underbridge inspection platform and Truck Mounted Attenuators.	

5.7 Federal Program Integration

RIEMA works closely with FEMA Region I in all aspects of planning, response, and mitigation. To ensure understanding and cooperation, the RIEMA SHMO regularly interfaces with FEMA mitigation staff on the status of local plans, changing FEMA guidelines, and opportunities for closer working relationships. The following detail programs federal programs that are managed by RIEMA for the State of Rhode Island:

FEMA Grant Outcomes (FEMA GO)

Rhode Island is currently working with FEMA to apply the FEMA GO system to all FEMA grants. The FEMA GO system allows users to apply, track, and manage all disaster and non-disaster grants and helps improve oversight and monitoring.

FEMA Integration Team Partnership

As part of an agency-wide effort to enhance program delivery, FEMA has begun embedding staff with State, Local, and Tribal partners as part of FEMA Integration Team. The FEMA Integration Team program allows FEMA to meet the needs of each state while planning and preparing for potential catastrophic emergencies. FEMA currently has two FEMA Integration Team members in place with RIEMA.

FEMA National Risk Index

RIEMA works with FEMA to provide information and analysis for the National Risk Index. The National Risk Index is an online mapping application that identifies communities most at risk to natural hazards. The mapping service visualizes natural hazard risk metrics and includes data about expected annual losses from natural hazards, social vulnerability, and community resilience. The National Risk Index's interactive web maps are at the county and Census tract level and made available via geographic information system services for custom analyses.

National Flood Insurance Program

The NFIP is a federal program that exists to provide flood insurance for property owners in participating communities, to improve floodplain management practices, and to develop maps of flood hazard areas. RIEMA, through the NFIP Coordinator, provides state level management for the NFIP and continuously works to enroll communities into the program with the provision of technical assistance.

The Rhode Island Building Code Commission has adopted and incorporated the 2018 International Codes into the Rhode Island State Building Code. The vast majority of the regulations required by the NFIP are included within the International Building Code and the International Residential Code. While most floodplain requirements have been incorporated into the Rhode Island State Building Code, some additional provisions and regulations may be required by a community.

Communities participating in the NFIP are required to adopt, enforce, and maintain a local floodplain ordinance as a stipulation of compliance with the program. The purpose of this ordinance is to ensure public safety, minimize impact to persons and property from flooding, protect watercourses from encroachment, and maintain the capability of floodplains to retain and carry off floodwaters. The local floodplain administrator is typically the municipal official responsible for overseeing the enforcement and update of the document.

As of this plan, there are a total of 40 communities in Rhode Island participating in the NFIP:

Table 195: State of Rhode Island NFIP Communities

ID	Community Name	County	Initial FHMB Identified	Initial FIRM Identified	Current Effective Map Date
445392#	Barrington	Bristol	5/15/1970	4/16/1971	7/7/2014
445393#	Bristol (Town)	Bristol	-	12/2/1972	7/7/2014
445408#	Warren	Bristol	-	7/28/1972	7/7/2014
440004E	Coventry	Kent	6/28/1974	9/1/1978	7/19/2023
445397E	East Greenwich	Kent	2/8/1973	2/9/1973	10/2/2015
445409E	Warwick	Kent	-	4/6/1973	10/2/2015
440037E	West Greenwich	Kent	1/24/1975	1/3/1986	07/19/23(M)
440007E	West Warwick	Kent	4/13/1973	2/1/1978	10/2/2015
445399#	Jamestown	Newport	-	4/20/1972	9/4/2013
440035C	Little Compton	Newport	7/19/1974	8/17/1981	7/6/2021
445401#	Middletown	Newport	-	4/9/1971	9/4/2013
445403#	Newport (City)	Newport	-	6/17/1970	9/4/2013
445405#	Portsmouth	Newport	-	8/24/1973	9/4/2013
440012C	Tiverton	Newport	5/24/1974	5/2/1977	7/6/2021
440013E	Burrillville	Providence	9/13/1974	7/2/1979	7/19/2023
445394E	Central Falls	Providence	-	5/27/1971	10/2/2015
445396E	Cranston	Providence	-	8/28/1971	10/2/2015
440016E	Cumberland	Providence	1/3/1975	12/16/1980	10/2/2015
445398E	East Providence	Providence	6/5/1970	5/18/1973	10/2/2015
440033E	Foster	Providence	9/13/1974	12/4/1985	07/19/23(M)
440034E	Glocester	Providence	10/18/1974	8/15/1979	7/19/2023
440018E	Johnston	Providence	11/22/1977	9/1/1978	10/2/2015
445400E	Lincoln	Providence		11/30/1973	10/2/2015

Table 195: State of Rhode Island NFIP Communities

ID	Community Name	County	Initial FHMB Identified	Initial FIRM Identified	Current Effective Map Date
440020E	North Providence	Providence	4/13/1973	12/15/1977	10/2/2015
440021E	North Smithfield	Providence	6/14/1974	8/1/1978	10/2/2015
440022E	Pawtucket	Providence	-	7/16/1971	10/2/2015
445406E	Providence (City)	Providence	12/15/1970	12/15/1970	10/2/2015
440024E	Scituate	Providence	9/6/1974	1/2/1981	10/2/2015
440025E	Smithfield	Providence	8/9/1974	3/1/1977	10/2/2015
445411#	Woonsocket	Providence	-	10/13/1971	3/2/2009
445395D	Charlestown	Washington	-	7/13/1972	4/3/2020
440032D	Exeter	Washington	3/14/1975	3/1/1982	7/19/2023
440028D	Hopkinton	Washington	5/31/1974	3/16/1981	4/3/2020
445414D	Narragansett Indian Tribe	Washington	-	7/13/1972	4/3/2020
445402D	Narragansett	Washington	-	12/7/1971	4/3/2020
440036#	New Shoreham	Washington	1/3/1975	4/3/1985	10/16/2013
445404D	North Kingstown	Washington	-	7/14/1972	4/3/2020
440031D	Richmond	Washington	5/31/1974	11/5/1980	4/3/2020
445407D	South Kingstown	Washington	-	7/13/1972	4/3/2020
445410D	Westerly	Washington	-	7/28/1972	4/3/2020

Notes:

(M): No elevation determined, all Zone A, C, and X

:- No data

Community Rating System (CRS)

The CRS is a voluntary incentive program that recognizes and encourages community floodplain management practices that exceed the minimum requirements of the NFIP. In CRS communities, flood insurance premium rates are discounted to reflect the reduced flood risk resulting from the community's efforts that address the three goals of the program:

- Reduce and avoid flood damage to insurable property
- Strengthen and support the insurance aspects of the National Flood Insurance Program
- Foster comprehensive floodplain management•

As of this plan, there are a total of 11 communities in Rhode Island participating in the CRS:

Table 196: State of Rhode Island CRS Communities

Community Name	County	CRS Entry Date	Current Effective Date	Current Class	Percentage Discount SFHA	Percentage Discount Non-SFHA
Bristol	Bristol	5/1/2013	5/1/2018	7	15%	5%
Warwick	Kent	10/1/2016	10/1/2016	9	5%	5%
Middletown	Newport	10/1/1991	5/1/2000	7	15%	5%
Newport (City)	Newport	10/1/2017	10/1/2017	7	15%	5%
East Providence	Providence	5/1/2014	5/1/2014	8	10%	5%
Pawtucket	Providence	10/1/2014	10/1/2014	8	10%	5%
Charlestown	Washington	5/1/2015	4/1/2022	6	20%	10%
Narragansett	Washington	10/1/1992	5/1/2017	7	15%	5%
New Shoreham	Washington	5/1/2019	5/1/2019	8	10%	5%
North Kingstown	Washington County	10/1/1993	10/1/1993	9	5%	5%
Westerly	Washington County	5/1/2013	10/1/2018	7	15%	5%

Rhode Island Management of the NFIP

Rhode Island state agencies collaborate on the NFIP to effectively manage and mitigate flood risks within the state. In Rhode Island, various state agencies work together to implement and enforce NFIP regulations. Key aspects of their collaboration include:

- **RIDEM:** RIDEM plays a crucial role in coordinating and overseeing floodplain management efforts. This includes mapping flood hazard areas, enforcing building regulations in these zones, and promoting community resilience through outreach and education.
- **RIEMA:** RIEMA is responsible for coordinating emergency response and recovery efforts in the event of a flood. This agency works in tandem with other state departments to ensure a comprehensive and well-coordinated approach to flood management.
- **Building Code Standards Committee:** This committee, likely under the Department of Administration or a similar agency, is involved in developing and updating building codes that incorporate NFIP requirements. Ensuring that structures are built to withstand floods is a crucial component of the NFIP.
- **Coastal Resources Management Council:** Given Rhode Island's coastal nature, the Coastal Resources Management Council plays a vital role in managing development along the shoreline and implementing regulations that consider flood risks. This includes evaluating and permitting construction activities in coastal areas.
- **Department of Business Regulation:** The Department of Business Regulation may be involved in regulating insurance-related aspects of the NFIP. This includes overseeing the insurance industry's compliance with NFIP requirements and ensuring that residents have access to affordable flood insurance.
- **Local Planning and Zoning Authorities:** Municipalities play a significant role in NFIP implementation. Local planning and zoning authorities work to adopt and enforce regulations that align with NFIP requirements, such as building codes and land-use planning that mitigate flood risks.

Collaboration among these agencies is essential for a comprehensive and effective approach to NFIP implementation in Rhode Island. Regular communication, information sharing, and joint planning help the state manage flood risks, protect communities, and ensure that residents have access to affordable flood insurance.

RIEMA's Planning and Mitigation Section is responsible for administering the NFIP in Rhode Island. The program was established in 1968, with the passage of the National Flood Insurance Act, to reduce the loss of life and property associated with flooding while offering property owners an opportunity to financially protect themselves.

Participation in the NFIP is based on an agreement between the municipality and the federal government. If a municipality agrees to adopt and enforce a floodplain ordinance designed to reduce future flood risks, all citizens in the participating municipality can purchase flood insurance. Currently, all 39 municipalities and 1 Tribal Nation in Rhode Island participate in the NFIP.

In Rhode Island, as part of NFIP participation communities must:

- Use current NFIP flood maps in adopting floodplain management regulations.
- Require permits for all development in SFHAs
- Ensure that development does not increase the flood hazard on other properties.
- Meet current elevation standards. Ensuring the lowest occupied floor is elevated to or above the base flood elevation indicated on the NFIP flood map.

While most floodplain requirements have been incorporated into the Rhode Island State Building Code, some additional provisions and regulations may be required by a community. Communities participating in the NFIP are required to adopt, enforce and maintain a local floodplain ordinance as a stipulation of compliance with the program. The purpose of this ordinance is to ensure public safety, minimize impact to persons and property from flooding, protect watercourses from encroachment, and maintain the capability of floodplains to retain and carry off floodwaters. The local floodplain administrator is typically the municipal official responsible for overseeing the enforcement and update of the document.

The following figure represents both pre- and post-disaster community NFIP requirements:

Figure 5: Pre- and Post-Disaster Community NFIP requirements



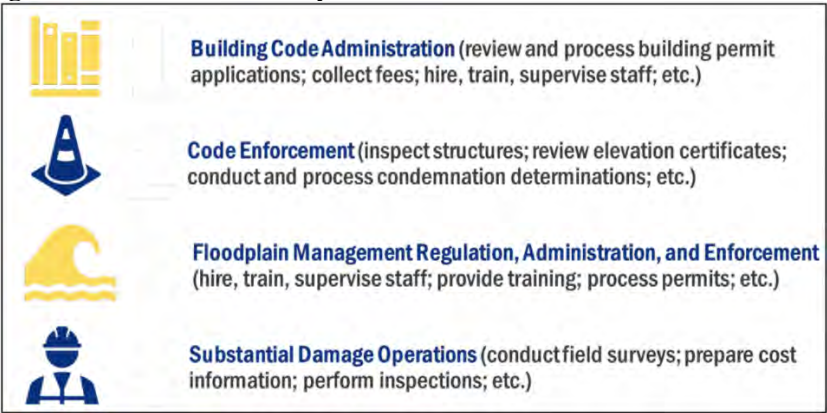
Source: FEMA

When structures located in the Special Flood Hazard Area are substantially modified (more than 50% damaged or improved) they are required to be brought into compliance with current NFIP standards and local building codes. In cases of repairs being conducted as a result of damage, jurisdictional NFIP Coordinators are responsible for substantial damage and improvement determinations. These determinations are required for compliance in the NFIP and must be completed before residents begin repairs or permits are issued.

However, the May 2020 Report to Congressional Committees on the National Flood Insurance Program by the United States Government Accountability indicates “FEMA generally does not collect or analyze the results of these assessments, limiting its ability to ensure the process operates as intended. Furthermore, FEMA has not clarified how communities can access NFIP claims data. Such data would help communities target substantial damage assessments after a flood.” This has been found to be true in the State of Rhode Island, with submitted information and data underutilized and some FEMA available data unshared and/or unadvertised. No other challenges between the State of Rhode Island and FEMA during regular coordination were identified.

Section 1206 of the Disaster Recovery Reform Act of 2018 authorizes the FEMA to provide communities with the resources to administer and enforce building code and floodplain management ordinances following a major disaster declaration through FEMA’s Public Assistance Program. To be eligible for reimbursement under the PA Program, including for the Disaster Recovery Reform Act of 2018 Section 1206, communities must be designated for Public Assistance permanent work under a major disaster declaration and be legally responsible to administer and enforce building codes or floodplain management regulations. Communities must also be in good standing with the NFIP. Available assistance includes:

Figure 6: Disaster Recovery Reform Act of 2018 Available Assistance



Source: FEMA

It is worth noting that this assistance is available for a variety of hazards occurrence types, not just flooding.

RIEMA’s Planning and Mitigation Section continuously makes information concerning flood insurance available to all citizens. However, data indicates that the number of policies and coverage has been generally declining. The following table indicates trends in NFIP coverage for participating communities. As indicated in the table, the State of Rhode Island has seen a decrease of 4,852 policies and a decrease of -\$1,094,483,800 in coverage from 2018 to 2023.

Table 197: Changes in State of Rhode Island NFIP Coverage

County	Jurisdiction	Policies in Force, 2018	Policies in Force, 2023	Change in Policies	Total Coverage, 2018	Total Coverage, 2023	Change, in Coverage
Bristol	Barrington	1,120	755	-365	\$335,639,900	\$234,439,000	-\$101,200,900
	Bristol	51	271	220	\$126,985,400	\$71,833,000	-\$55,152,400
	Warren	418	262	-156	\$85,613,300	\$57,834,000	-\$27,779,300
Kent	Coventry	190	107	-83	\$52,118,300	\$33,305,000	-\$18,813,300
	East Greenwich	156	76	-80	\$44,938,500	\$23,228,000	-\$21,710,500
	Warwick	1,443	740	-703	\$320,850,000	\$200,031,000	-\$120,819,000
	West Greenwich	13	5	-8	\$3,612,000	\$1,610,000	-\$2,002,000
	West Warwick	153	64	-89	\$31,126,500	\$16,723,000	-\$14,403,500
Newport	Jamestown	230	177	-53	\$67,839,300	\$53,697,000	-\$14,142,300
	Little Compton	147	96	-51	\$41,254,700	\$30,401,000	-\$10,853,700
	Middletown	115	66	-49	\$39,103,300	\$23,514,000	-\$15,589,300
	Newport	1,092	752	-340	\$269,365,800	\$193,027,000	-\$76,338,800
	Portsmouth	622	339	-283	\$149,248,300	\$90,368,000	-\$58,880,300
Providence	Tiverton	161	93	-68	\$39,847,100	\$25,576,000	-\$14,271,100
	Burrillville	53	31	-22	\$14,227,700	\$9,302,000	-\$4,925,700
	Central Falls	56	50	-6	\$9,998,800	\$6,813,000	-\$3,185,800
	Cranston	431	275	-156	\$122,633,000	\$79,634,000	-\$42,999,000
	Cumberland	58	43	-15	\$18,139,500	\$14,409,000	-\$3,730,500
	East Providence	246	160	-86	\$70,899,000	\$50,703,000	-\$20,196,000
	Foster	8	4	-4	\$2,432,000	\$1,240,000	-\$1,192,000
	Glocester	30	17	-13	\$7,640,500	\$4,353,000	-\$3,287,500
	Johnston	69	40	-29	\$20,339,400	\$13,657,000	-\$6,682,400
	Lincoln	233	215	-18	\$60,230,200	\$55,429,000	-\$4,801,200
	North Providence	62	47	-15	\$17,983,000	\$15,485,000	-\$2,498,000
	North Smithfield	34	19	-15	\$10,339,400	\$6,604,000	-\$3,735,400
	Pawtucket	90	82	-8	\$23,431,100	\$21,600,000	-\$1,831,100
	Providence	587	515	-72	\$161,164,600	\$143,221,000	-\$17,943,600
	Scituate	33	21	-12	\$8,103,900	\$5,429,000	-\$2,674,900
	Smithfield	153	122	-31	\$29,642,900	\$28,243,000	-\$1,399,900
	Woonsocket	166	155	-11	\$38,289,300	\$40,422,000	\$2,132,700
Washington	Charlestown	701	466	-235	\$196,417,500	\$138,402,000	-\$58,015,500
	Exeter	21	9	-12	\$5,706,900	\$2,344,000	-\$3,362,900
	Hopkinton	27	14	-13	\$6,273,500	\$3,383,000	-\$2,890,500
	Narragansett Tribe	1	1	0	\$100,000	\$215,000	\$115,000
	Narragansett	1,470	957	-513	\$411,426,300	\$280,661,000	-\$130,765,300
	New Shoreham	78	51	-27	\$23,658,200	\$17,998,000	-\$5,660,200
	North Kingstown	769	508	-261	\$218,073,000	\$154,746,000	-\$63,327,000
	Richmond	30	18	-12	\$9,469,000	\$7,738,000	-\$1,731,000
	South Kingstown	850	524	-326	\$223,818,500	\$153,634,000	-\$70,184,500
	Westerly	963	593	-370	\$259,636,900	\$172,511,000	-\$87,125,900
State Total		13,592	8,740	-4,852	\$3,578,245,800	\$2,483,762,000	-\$1,094,483,800

Source: FEMA Flood Insurance Data and Analytics

Risk Mapping, Assessment, and Planning Program

RIEMA works closely with FEMA, tribal, and local partners to identify flood risk and promote informed planning and development practices through the Risk MAP program. Risk MAP is the process used to make FIRMs which both map flood risk and provide informational datasets. Mapping occurs in four phases:

- Discovery: An initial investigation into a community's flood risk, challenges, and goals.

- **Analysis and Mapping:** A complete engineering analysis is performed that leads to the initial updates to the flood maps. Work is completed with technical experts in each community to make sure the drafts line up with community knowledge.
- **Preliminary Flood Map Release:** A preliminary flood map and supporting preliminary flood hazard data is generated for review and comment.
- **Map Adoption:** Community takes full ownership of the updated flood maps and data.

RIEMA works with FEMA during the map update process from discovery to map adoption. Since the last update, the State of Rhode Island has assisted communities during the Quinebaug map update, with the updated maps becoming effective on July 19, 2023. Currently the Blackstone Watershed Map update in Providence County is being completed. There was an initial preliminary map release on February 10, 2023, and a revised preliminary on August 1, 2023. The effective date is expected to be in late 2025. The table below shows the dates of the mapping process for the Blackstone Watershed:

Table 198: Mapping Process for the Blackstone Watershed

Metric	Date
Discovery Meeting Date	3/26/2017
Discovery Report Date	3/31/2018
Work Map Meeting Date	2/15/2022
Prelim Date	2/10/2023
Revised Prelim Date	8/1/2023
CCO Meeting Date	6/27/2023
Estimated Appeal Period Begin	3/1/2024

Source: US Census Bureau

In addition, RIEMA and the State of Rhode Island provide any available data to FEMA as requested. However, data is most shared from the following two tools:

- **RIEMA's Floodplain Mapping Tool:** An online mapping resource that provides flood risk information relative to residences, businesses, or other structures statewide. The tool provides an approximate flood zone determination relative to the flood hazard areas delineated on FEMA's FIRMs.
- **Rhode Island's STORMTOOLS:** An online system to illustrate and display storm inundation, with and without sea level rise, for different types of storms that could occur along Rhode Island's coastline. At the simplest level the system can be used to access flooding estimates for coastal areas. At a more sophisticated level the system can be used to perform studies in support coastal planning and engineering design. The impact of variations in coastal topography and implementation of potential shoreline protection studies can be evaluated for varying return period events, with or without sea level rise.

U.S. Army Corps of Engineers (USACE)

The New England District of the USACE is responsible for managing civil works responsibilities in a 66,000-square-mile region encompassing six New England states, including Rhode Island. The missions of the New England District include flood risk management, streambank and shoreline protection, disaster and emergency assistance, and engineering support.

The USACE and Rhode Island worked closely on the 2022 Rhode Island Coastline Coastal Storm Risk Management Project. The study area covers more than 457 miles of coastline from Point Judith to the Massachusetts line and includes the majority of Narragansett Bay and Block Island. All or part of 19 municipalities (home to approximately 650,000 people) are included in the study area. The study was conducted due to frequent flooding from coastal tides and storms in an effort to:

- Identify a plan to reduce the risk of coastal storm damage along a large portion of the Rhode Island coastline
- Contribute to the resilience of communities, important infrastructure, and the natural environment

The study area includes significant critical infrastructure at risk of damage from future flooding and coastal storms, and a range of measures were considered to manage the risk including:

- Structural
 - Floodwalls
 - river closure structures
 - living shorelines
- Nonstructural measures
 - wet/dry flood proofing
 - elevation of residential structures

Rhode Island Silver Jackets

The Rhode Island Silver Jackets is comprised of representatives from Federal and State agencies which support comprehensive and sustainable actions that reduce flood risk. In general, the Silver Jackets:

- Promote Rhode Island's capacity to implement projects, programs, plans, policies, and legislative actions to reduce vulnerability to flooding.
- Foster and facilitate statewide coordination with organizations, agencies, and stakeholders to achieve flood risk reduction.
- Establish or supplement mechanisms to collaboratively identify issues and implement or recommend solutions.
- Increase and improve flood risk communication and outreach.
- Advise and assist Rhode Island communities in their efforts to become more resilient and less vulnerable to flood hazards.

The Rhode Island Silver Jackets Team, in coordination with a number of partners, developed a comprehensive handbook for Rhode Island floodplain professionals in 2019. This document combines sound floodplain management techniques, federal requirements, and state regulations into a single concise resource.

FEMA National Safety of Dams Program

The Rhode Island Dam Safety Program, managed by RIDEM, is responsible for regulating the safety of dams and supports the National Safety of Dams Program.

5.8 Regional Initiatives

The following detail regional initiatives that Rhode Island currently participates in related to hazard mitigation:

Regional Greenhouse Gas Initiative

The Regional Greenhouse Gas Initiative is a cooperative effort among the states of Rhode Island, Connecticut, Delaware, Maine, Maryland, Massachusetts, New Hampshire, New Jersey, New York, Vermont, and Virginia to cap and reduce power sector carbon dioxide emissions. The program establishes a regional cap for Carbon Dioxide emissions and each state's allocation of Carbon Dioxide allowances under the budget. RGGI is composed of individual Carbon Dioxide Budget Trading Programs in each participating state and these programs require electric power generators with a capacity equal to or greater than 25 MWe (megawatt electrical), to surrender a tradable Carbon Dioxide allowance for each ton of Carbon Dioxide they emit. Through quarterly allowance auctions, allowances are purchased, and States invest the proceeds from the auctions into energy efficiency, renewables, and clean energy programs. Additional information on the investment of proceeds can be found on the Investment of Proceeds webpage. In Rhode Island, the Office of Energy Resources with guidance from RIDEM determines the allocation and distribution of Rhode Islands' auction proceeds.

Transportation and Climate Initiative

The Transportation and Climate Initiative is a regional collaboration of 13 Northeast and Mid-Atlantic states and the District of Columbia that seeks to improve transportation, develop the clean energy economy, and reduce carbon

emissions from the transportation sector. The Transportation and Climate Initiative is directed by state and district agencies.

5.9 State Emergency Response Teams

RIEMA, in conjunction with local jurisdictions, maintains a variety of emergency response teams who can rapidly identify, evaluate, and react to a wide spectrum of emergencies and provide support via the Emergency Management Assistance Compact. The following is a brief summary of these teams:

Rhode Island Heavy Rescue Task Force

This Task Force is organized into the East Bay (Middletown, Newport, and Portsmouth Fire Departments) and West Bay (Warwick Fire Department) Heavy Rescue Squad. When the two squads come together, they form the Rhode Island Heavy Rescue Task Force that is a Type-4 Urban Search and Rescue asset.

Hazardous Materials and Mass Decontamination Team

This Team is made up of six Hazardous Material Teams (Cranston, East Providence, Hope Valley, Providence, Warwick, and Woonsocket), and four Decontamination Teams (Hopkins Hill, Kingston, North Kingstown, and North Providence). The teams are used to assess and manage the consequences of a hazardous materials/WMD release, either accidental or as part of a terrorist attack.

Rhode Island Critical Stress Management Team

This team is designed to be a comprehensive, multi-component, multi-discipline program providing stress and crisis management to any fire, police, or ambulance department personnel responding to a critical incident, multi-casualty event, or disaster.

Rhode Island Disaster Medical Assistance Team

This Team recruits and trains medical health professionals to respond to public health emergencies and is currently comprised of over 800 volunteers.

Rhode Island Medical Reserve Corps

The Medical Reserve Corps has over 1,200 registered health volunteers who are called upon to assist in public health and emergency situations.

Wilderness Search and Rescue Team

This Team formalizes search and rescue capabilities and enhances efforts with the utilization of civilian volunteers. While the operations of all missing person searches shall remain a law enforcement function, there is a need to integrate the services of both trained and untrained civilian volunteers into a large-scale search effort.

Weapons of Mass Destruction Tactical Team

This Team is made up of members from the Rhode Island State Police and is used to contain, control, and de-escalate hostile situations where the use of weapons of mass destruction has been, or threatened to be, introduced.

Flammable Liquids Task Force

This Task Force (East Providence, Johnston, Providence, Warwick, Valley Falls, and Woonsocket Fire Departments) protects life, property, and critical infrastructure by using its assets to contain, control, and extinguish fires, or releases and spills involving flammable or combustible liquids or gasses. The Task Force uses five Williams Foam Trailers with 660 gallons of Aqueous Film Forming Foam-Alcohol Resistant Thunderstorm Foam Concentrate in 330-gallon totes to extinguish liquid and gaseous fires.

Port of Providence Marine Strike Team

This Team (Providence, East Providence, Cranston, and Warwick) is a waterside asset to mitigate chemical, incendiary, environmental and life safety incidents that occur in the port of Providence or contiguous waters. Each team has one boat, with the capability to conduct nighttime Search and Rescue, fire suppression, hazmat detection, weapons of mass destruction detection, and can support emergency medical services and dive operations.

5.10 Evaluation of State Mitigation Capabilities

The hazard mitigation capabilities of the State of Rhode Island continue to strengthen over time. Along with the litany of capabilities enumerated above, notable advances include:

- More robust climate change policies and actions.
- Excellent local hazard mitigation plan compliance from local jurisdictions.
- Enhanced GIS capabilities and increased assistance to jurisdictional planners.
- Strong retention of staff and no loss of positions due to budget constraints.
- Strengthened outreach efforts on mitigation grant availability, along with grant writing assistance.
- Identifying public-private partnerships and investigation and obtaining non-standard funding mechanisms would help alleviate these challenges.

5.11 Local Jurisdiction Capabilities Assessment

Currently, all State of Rhode Island jurisdictions have an emergency management program that has the primary responsibility for local response, recovery, hazard mitigation planning. During a local or state disaster, the head of the local government and/or the director of the local emergency management program can activate their local emergency operation center and establish direct communications with both RIEMA and State political leadership. However, the capability of each emergency management program varies based largely on the size and financial capabilities of the jurisdiction. While all jurisdictions have the capability needed to conduct mitigation planning, many rely on the technical expertise of RIEMA to apply for grant funding and oversee mitigation projects. Augmenting local emergency management capabilities, RIEMA aids with state and federal mitigation and emergency management initiatives and available funding opportunities. Other local personnel capabilities vary widely, and again are often tied to jurisdiction size and financial capabilities. In general, more urban, or larger jurisdictions have a greater range of full-time personnel dedicated to planning, engineering, mapping, and response, while smaller jurisdictions lack these capabilities.

Technical capabilities for each jurisdiction vary widely and are generally based on financial capabilities. As with personnel, and in general, more urban, or larger jurisdictions have a greater range of technical capabilities related to planning, engineering, mapping, and response, while smaller jurisdictions lack these capabilities. It should be noted that RIEMA offers a variety of programs to provide local jurisdictions with technical expertise, including mapping and planning.

A review of local level hazard mitigation plans indicates that most local jurisdictions have limited funding sources for mitigation initiatives and rely on available state or federal grant programs. However, tax generated funding (often generated from property and/or sales tax) can be allocated to infrastructure improvements for schools, public works, and other government functions. In addition, some larger jurisdictions have dedicated transportation or capital improvements sales or use taxes that can be obligated to fund mitigation projects.

On a yearly basis, many jurisdictions throughout the State of Rhode Island fully allocated their tax revenue to basic services and programs. As a result, funding for mitigation projects is often unavailable or severely limited. While the capability to assess special taxes or issue bonds does exist, historically it has been shown that passing these measures is extremely difficult. As a result, many needed mitigation projects throughout the state are not completed due to lack of funding. It should be noted that RIEMA offers a variety of programs to provide local jurisdictions with financial expertise, including grant application and determining potential funding streams.

The following table details local departments and positions and their roles in supporting hazard mitigation planning:

Table 199: Local Jurisdiction Department and Positions Supporting Mitigation Planning		
Department or Position	Description	Role in Mitigation
Building Officials	Implements and enforces the Rhode Island Building Code and locally zoning and subdivision adopted ordinances	Ensures construction standards are consistently applied.

Table 199: Local Jurisdiction Department and Positions Supporting Mitigation Planning

Department or Position	Description	Role in Mitigation
Conservation Commissions	Oversees issues relating to natural resources areas, critical areas of concern, other environmentally sensitive areas, and assists with the implementation and enforcement control ordinances.	Works for open space acquisition, and wetlands, buffer zone and other SFHAs preservation,
Emergency Management Director	Directs local response, recovery, and mitigation programs.	Develops Local Emergency Operations Plan, Continuity Plans, and Hazard Mitigation Plans helping to minimize loss of life and property damage and the local level.
NFIP/CRS Coordinators	Oversees compliance with the NFIP and CRS and addresses flood determinations, mapping issues, and construction standards within SFHAs.	Reviews floodplain/building permits for structures within floodplains and inspects developments to determine compliance with the community development standards and NFIP requirements. Explains floodplain development requirements to community leaders, citizens, and the general public.
Planning Boards	Per the State Enabling Act, implements local subdivision regulations, recommends land use regulations	Coordinates with the NFIP Coordinator and the Local Hazard Mitigation Committee through the mitigation planning process and the implementation of the plans.
Public Works Departments	Responsible for municipal drainage and storm water management systems.	Provides for the ongoing maintenance and upgrading of local storm water systems to help reduce flood risks.
Town/City Council	Approves subdivision, zoning and land ordinances and bylaws and facilitates capital improvements budget and plan.	Provide leadership and approval for local hazard mitigation plans, projects, grants, and programs.

A select set of local jurisdictional capabilities has been pulled from local hazard mitigation plans to illustrate the abilities of Rhode Island jurisdictions. The capabilities presented in the local hazard mitigation plans have been grouped into the following categories:

- Available plans, to include hazard mitigation, debris management, evacuation and emergency operations plans
- Building codes
- Land use restrictions
- Mapping
- Public Outreach

The following table illustrates these capabilities for each jurisdiction:

Table 200: Local Jurisdiction Capabilities Supporting Risk Reduction, Resiliency, and Mitigation

County	Jurisdiction	Approve Hazard Mitigation Plan	Comprehensive Plan	Response and Recovery Plans	Building Codes	Land Use Policies and Ordinances	NFIP Community	GIS Mapping Program	Public Outreach
Bristol	Barrington	x	x	x	x	x	x	x	x
Bristol	Bristol		x	x	x	x	x	x	x

Table 200: Local Jurisdiction Capabilities Supporting Risk Reduction, Resiliency, and Mitigation

County	Jurisdiction	Approve Hazard Mitigation Plan	Comprehensive Plan	Response and Recovery Plans	Building Codes	Land Use Policies and Ordinances	NFIP Community	GIS Mapping Program	Public Outreach
Bristol	Warren	x	x	x	x	x	x	x	
Kent	Coventry		x	x	x	x	x	x	x
Kent	East Greenwich	x	x	x	x	x	x	x	x
Kent	Warwick	x	x	x	x	x	x		
Kent	West Greenwich		x	x	x	x	x		x
Kent	West Warwick	x	x	x	x	x	x	x	x
Newport	Jamestown		x	x	x	x	x	x	x
Newport	Little Compton	x	x	x	x	x	x		
Newport	Middletown	x	x	x	x	x	x	x	x
Newport	Newport	x	x	x	x	x	x	x	x
Newport	Portsmouth	x	x	x	x	x	x	x	x
Newport	Tiverton		x	x	x	x	x	x	x
Providence	Burrillville	x	x	x	x	x	x		
Providence	Central Falls	x		x	x	x	x	x	
Providence	Cranston	x	x	x	x	x	x	x	x
Providence	Cumberland		x	x	x	x	x	x	x
Providence	East Providence	x	x	x	x	x	x	x	x
Providence	Foster	x	x	x	x	x	x	x	x
Providence	Glocester	x	x	x	x	x	x	x	x
Providence	Johnston	x	x	x	x	x	x		x
Providence	Lincoln	x	x	x	x	x	x	x	x
Providence	North Providence	x	x	x	x	x	x	x	
Providence	North Smithfield	x	x	x	x	x	x		
Providence	Pawtucket	x	x	x	x	x	x		
Providence	Providence	x	x	x	x	x	x		x
Providence	Scituate		x	x	x	x	x		x
Providence	Smithfield		x	x	x	x	x	x	
Providence	Woonsocket	x	x	x	x	x	x	x	x
Washington	Charlestown		x	x	x	x	x	x	x
Washington	Exeter		x	x	x	x	x		
Washington	Hopkinton		x	x	x	x	x	x	x
Washington	Narragansett	x	x	x	x	x	x	x	x
Washington	New Shoreham		x	x	x	x	x	x	
Washington	North Kingstown	x	x	x	x	x	x	x	x
Washington	Richmond	x	x	x	x	x	x	x	x
Washington	South Kingstown	x	x	x	x	x	x	x	x
Washington	Westerly	x	x	x	x	x	x		x

Partnerships

The most common hazard mitigation and emergency management coordination between local jurisdictions are Local Emergency Planning Committees (LEPCs). LEPCs are required by the Emergency Planning and Community Right-to-Know Act of 1986, which was passed to encourage the coordination of planning efforts and information sharing at all levels of government, private industry, and the public. LEPCs include representatives from both public and private organizations as well as representatives from every facility in the jurisdiction subject to the emergency planning requirements of the act. The following table details each LEPC District and the current representative:

Table 201: Rhode Island LEPCs

LEPC District	Representative	Address
District 1	John Silva	1951 Mineral Spring Avenue, North Providence, RI 02904
District 2	David Aucoin	Narragansett Bay Commission, 1 Service Road, Providence, RI 02904
District 3	Currently Vacant	-
District 4	Jake Longolucco	Narragansett Fire Department, 40 Caswell Street, Narragansett, RI 02882
District 5	Michael Demello	4 Annawamscutt Drive, Bristol, RI 02809

The following map illustrates the jurisdictions covered by each LEPC District:

Map 77: LEPC Districts

Source: RIEMA

Planning Integration

All local jurisdictions with an active hazard mitigation plan have committed to integrating all planning efforts with hazard mitigation planning. Effective plan integration can allow for a more comprehensive approach to hazard mitigation, and for the institutionalization of hazard mitigation principles into daily governmental activities.

Education and Awareness

Hazard awareness programs, designed to inform citizens as to the nature and extent potential hazards, are an effective way to inform citizens on mitigation related topics. As citizens are made more aware of potential hazards and the local and regional process to mitigation against their impacts, it was believed that they would take a stronger role in making their homes, neighborhoods, schools, and businesses safer from the potential effects of natural hazards. Public outreach efforts were conducted as part of all local hazard mitigation planning programs.

Local Jurisdiction Hazard Mitigation Program Effectiveness

State of Rhode Island local jurisdictions have demonstrated an overall exceptional level of hazard mitigation programming. While reviewing local capabilities for this plan, the following the hallmarks of an effective hazard mitigation program were noted in the majority of jurisdictions:

- **Clear Goals and Objectives:** These programs have clear and specific goals and objectives that outline what they aim to achieve.
- **Community Engagement:** Involving the community in the planning process is crucial. Engaging with residents, businesses, and organizations ensures that the program addresses the needs and concerns of the community.
- **Collaboration:** Effective mitigation programs involve collaboration among various stakeholders, including local government agencies, emergency services, nonprofits, businesses, and neighboring communities.

- **Long-Term Focus:** Effective programs take a long-term perspective. Mitigation is an ongoing process, and measures should be implemented with a focus on future generations, not just short-term fixes.
- **Education and Outreach:** Public education and outreach efforts are essential to inform residents about the risks they face and how to prepare for and respond to disasters. Community awareness and education can help reduce vulnerabilities.
- **Leadership and Commitment:** Strong leadership at the local government level is crucial. Elected officials and community leaders must be committed to the mitigation program and its long-term goals.

Local jurisdictions within Rhode Island demonstrate that effective hazard mitigation programs are proactive, forward-thinking, and adaptable to the changing nature of hazards. However, opportunities for improvement always exist. The following opportunities for improvement have been identified for local hazard mitigation programs:

- **Staff turnover:** Frequent turnover of local staff and elected and appointed officials creates gaps and misunderstandings in the hazard mitigation planning process. To help address this problem, frequent communication between local emergency management agencies, their departments and elected and appointed officials, and RIEMA should be conducted over the life of a local hazard mitigation plan. This will help ensure that plans do not get completed at the last minute and information is carried forward with new staff.
- **Prioritization of mitigation planning:** Many communities do not see the benefit of mitigation planning, and therefore do not prioritize their completion and adoption. Success stories concerning completed mitigation projects should be widely disseminated by local emergency management agencies and RIEMA to illustrate the benefits of participation.
- **Grant application challenges:** Often the requirements to apply for grant funding through mitigation programs are beyond the capability of local jurisdictions. Federal and state level training, along with assistance, should be provided to those jurisdictions who lack the capacity to complete applications.

5.12 Hazard Mitigation Challenges

As always, challenges exist due to the day-to-day demands of the working environment, including limited staff time and limited budgets. These issues can, and do, impact the utilization and incorporation of the SHMP and the completion of identified hazard mitigation projects. Additionally, a lack of funding remains a challenge as tax revenues have been impacted by aging populations and, recently, the COVID-19 pandemic. Despite repeated attempts to get all jurisdictions to engage in hazard mitigation grant funding programs, many eligible jurisdictions are reticent to participate due to lack of perceived capabilities and required match expense. Conversations indicate that the requirements to obtain grant funding through available programs and the required matching funds are large barriers to small communities who do not have the necessary resources to compete with larger, better funded, and better staffed jurisdictions.

To overcome these identified challenges, RIEMA, the SIHMC, and participating stakeholders will work collaboratively using the following strategies as appropriate:

- **Innovation and Adaptation:** Foster a culture of innovation and adaptability. Encourage employees to think creatively, embrace change, and explore new ways of doing things to overcome challenges and stay competitive in a dynamic environment.
- **Training and Development:** Invest in training and development to enhance skills and knowledge.
- **Communication Improvement:** Enhance communications and provide clear and transparent communication when sharing information, aligning teams, and addressing concerns.
- **Collaboration and Teamwork:** Encourage collaboration and teamwork which allows for the pooling of diverse skills and perspectives, leading to more effective problem-solving.
- **Technology Adoption:** Embrace technology to streamline operations and enhance productivity.
- **Diversity and Inclusion:** Foster a diverse and inclusive workplace. Embracing diversity brings different perspectives and approaches to problem-solving, contributing to more innovative solutions.

- **Agile Project Management:** Implement agile project management methodologies to enhance flexibility and responsiveness to changing conditions. Agile approaches empower teams to adapt quickly to challenges and deliver incremental value.
- **Continuous Improvement:** Adopt a culture of continuous improvement. Encourage employees at all levels to identify opportunities for improvement and implement changes incrementally

As appropriate, these strategies will be tailored for specific circumstances. Additionally, a combination of these strategies is often more effective than relying on a single approach.

5.13 Opportunities for Improvement

As part of this plan update, the following opportunities for improvement were noted:

- **Mitigation Planning**
 - Notification and provision of guidance on FEMA mandated changes to the hazard mitigation planning process.
 - Encouraging better integration with jurisdictional comprehensive plans, capital improvement plans, and other long-term community plans.
 - Continuous tracking of planning data from local hazard mitigation plans for integration onto SHMP.
- **Fiscal**
 - Provision of successful examples of cost-effective mitigation measures when developing mitigation actions on a local level.
 - Discussion of the usage of in-kind funding methods to meet required grant funding match.
- **Land Use Planning and Regulations**
 - Continued encouragement of using land use planning to identify areas at risk to hazards and prohibit growth in identified areas.
- **Floodplain Management**
 - Increased promotion of the CRS program for greater local jurisdiction uptake.
 - Assisting with the creation of Emergency Action Plans for all high and significant hazard dams.
- **Hazard Specific Data Gaps**
 - Access to High and Significant hazard inundation mapping is very limited, making vulnerability analysis difficult.
 - A complete mapping of road and highway infrastructure has not been completed to aid in the understanding of vulnerability to flooding.

Section 6 – Coordination of Local Hazard Mitigation Planning

6.1 Introduction

It is the goal of the State of Rhode Island to provide all available assistance to local governments in developing and maintaining hazard mitigation plans and in successfully executing mitigation projects. Local hazard mitigation planning and projects help ensure that:

- Local variances are accounted for and considered.
- Local jurisdictions have a plan that can help inform political and planning decisions based on sound principles.
- Local vulnerabilities to identified hazards can be lessened through the application of locally derived mitigation projects.
- Land use planning can use local hazard data to inform zoning and building decisions.

RIEMA is invested in providing all available support to local jurisdictions concerning hazard mitigation planning. RIEMA, and the State Hazard Mitigation Officer provide consistent assistance to jurisdictional mitigation planners through:

- Attending and informing mitigation discussions at planning meetings.
- Providing training on new planning requirements.
- Awarding grant funding to eligible jurisdictions when applied for.
- Provision of technical assistance when requested.
- Advertising the availability of mitigation grants when available.
- Reviewing all submitted hazard mitigation plans, providing feedback for plan improvement, and submitted plans to FEMA Region I for their review and approval.
- Working with jurisdictions on amending plans as per FEMA recommendations.
- Provision of HMGP Applicant Briefings in coordination with FEMA Region I following declared events.

6.2 Local Hazard Mitigation Plan Status

While Rhode Island is subdivided geographically into five counties, county government was abolished in the state in 1842. All local government in the state is vested in its 39 jurisdictions. RIEMA is actively working with all jurisdictions to ensure they have an approved, current hazard mitigation plan. To date:

- 27 jurisdictions have a FEMA-approved hazard mitigation plans.
- 11 jurisdictions have expired plans
- One jurisdiction has no plan

Despite all efforts, these numbers match exactly the number of jurisdictions without an approved hazard mitigation plan for the 2019 SHMP.

The State of Rhode Island actively works with all applicable jurisdictions to ensure they have an approved, current hazard mitigation plan. Using available hazard mitigation plans RIEMA can effectively coordinate with local jurisdictions and assess how to distribute project funding and technical assistance most efficiently. The following table shows the status of local hazard mitigation plans as of this plan.

Table 202: 2023 Local Hazard Mitigation Plan Status

County	Jurisdiction	Status	Expiration
Bristol	Barrington	Approved	2/7/2028
Bristol	Bristol	Expired	1/5/2022
Bristol	Warren	Approved	9/14/2027
Kent	Coventry	Expired	2/14/2023
Kent	East Greenwich	Approved	12/13/2027
Kent	Warwick	Approved	11/11/2024

Table 202: 2023 Local Hazard Mitigation Plan Status

County	Jurisdiction	Status	Expiration
Kent	West Greenwich	Expired	8/25/2010
Kent	West Warwick	Approved	10/29/2024
Newport	Jamestown	Expired	3/23/2022
Newport	Little Compton	Approved	9/26/2023
Newport	Middletown	Approved	1/14/2025
Newport	Newport	Approved	12/13/2027
Newport	Portsmouth	Approved	1/31/2024
Newport	Tiverton	Expired	2/8/2023
Providence	Burrillville	Approved	5/23/2026
Providence	Central Falls and Pawtucket	Approved	10/21/2023
Providence	Cranston	Approved	4/16/2028
Providence	Cumberland	Expired	7/20/2022
Providence	East Providence	Approved	8/28/2027
Providence	Foster	Approved	9/17/2023
Providence	Glocester	Approved	9/24/2023
Providence	Johnston	Approved	6/16/2026
Providence	Lincoln	Approved	5/18/2027
Providence	North Providence	Approved	8/19/2023
Providence	North Smithfield	Approved	10/8/2024
Providence	Providence	Approved	8/4/2024
Providence	Scituate	Expired	9/4/2022
Providence	Smithfield	No Plan	-
Providence	Woonsocket	Approved	2/6/2024
Washington	Charlestown	Expired	7/24/2022
Washington	Exeter	Expired	7/10/2010
Washington	Hopkinton	Expired	3/12/2023
Washington	Narragansett	Approved	1/13/2024
Washington	Block Island and New Shoreham	Expired	7/24/2022
Washington	North Kingstown	Approved	8/22/2024
Washington	Richmond	Approved	9/13/2027
Washington	South Kingstown	Approved	6/9/2024
Washington	Westerly	Approved	3/21/2028

Source: RIEMA

6.3 Available Technical Assistance for Local Hazard Mitigation Planning

RIEMA continues to emphasize the completion and adoption of FEMA-approved plans for all Rhode Island jurisdictions. As funding for planning purposes is oftentimes limited, and RIEMA is unable to provide planning funds to every jurisdiction, technical assistance is a major method used to provide planning assistance. RIEMA has been actively supporting local jurisdictions and contractors by providing various forms of training and outreach on a regular basis. This available assistance includes:

- Planning workshops
- Trainings
- Planning consultation via phone and email
- On-site visits

Additionally, RIEMA makes all resources, data, and information used to update the SHMP available to local planners for use in understanding risk and vulnerability and for the creation of local Hazard Mitigation Plans.

6.4 Barriers to Local Hazard Mitigation Planning

RIEMA recognizes that creating local Hazard Mitigation Plans can present a burden to local jurisdictions. Potential barriers that have been identified include:

- **Resource Constraints:** Limited financial, personnel, and time resources can be a significant barrier. High turnover in staffing is also a concern that we see in Rhode Island. The state sees many of the same planners, but they seem to switch around within communities, so the lack of continuity is a concern. Also, the inconsistency of who is the responsible party for updating the local hazard mitigation plan varies in communities. Many communities think it belongs to the Emergency Management Agency Directors, which often means it falls on a part-time employee who doesn't have the adequate time to dedicate to an update and often misses an opportunity to apply for funding, resulting in an expired plan.
- **Political Leadership:** A lack of political leadership can hinder the development and adoption of mitigation plans or a lack of understanding the importance of a hazard mitigation plan.
- **Competing Priorities:** Communities often face various challenges and priorities, which can divert attention and resources away from hazard mitigation planning. Most communities in Rhode Island are understaffed and taking on job responsibilities that are out of their job scopes.
- **Short-Term Focus:** Communities may prioritize immediate needs over long-term planning. Many communities in Rhode Island are not preparing for the renewal of their hazard mitigation plan and the expiration date comes as a surprise and applying for funding is time consuming.
- **Institutional Barriers:** Bureaucratic barriers, lack of inter-agency coordination, or inefficiencies can hinder the mitigation planning process.
- **Funding and Grants:** While funding can be a resource constraint, communities may also struggle to secure grants and financial support for mitigation projects, especially if they lack the expertise or capacity to navigate the complex grant application process. The FEMA application and grant award process is extremely time-consuming. Applications for a plan update can take between four months to 11 months to be approved, which in turn delays securing a consultant and then plans expire in the meantime. FEMA contributes to these delays due to a lack of staff, which mirrors staffing challenges the state and local municipalities are facing. Unfortunately, these delays greatly impact the municipalities desire to apply for funding, which we have seen causes plans to expire.
- **Technical Expertise:** A lack of in-house expertise can be a barrier, especially for smaller communities. Many small communities do not have a town planner and their Emergency Management director is responsible for applying for a grant to update the plan while having several lifesaving community responsibilities.

Overcoming these barriers requires a concerted effort, commitment, and support from local, state, and federal agencies, as well as the community.

To help overcome these barriers, RIEMA continues to provide as much technical assistance to local planners as time and staffing allows. In addition, and in order to standardize in-state planning efforts and minimize planning variances, RIEMA has worked to develop a local hazard mitigation template for local planners. This template will provide both a recommended structure and recommended content which are matched to FEMA's local guidance. RIEMA, specifically the State Hazard Mitigation Officer, is tracking dates and reaching out to communities three years before their plan expires to explain the importance of applying for funds early. RIEMA is also helping communities by standardizing the components of an application for a plan update as much as possible, allowing use components as necessary. Recently RIEMA has asked to join the Governor's monthly municipal call to let the highest governing authority in the communities and the Governor know about funding opportunities and the importance of having an updated plan. RIEMA has also visited two municipalities that appeared on local news after flooding events to discuss BRIC and project scoping, and how it can help, providing they have an updated plan. RIEMA will continue to do presentations to inform local jurisdictions of available funding opportunities to those who have a current hazard mitigation plan. Lastly, RIEMA has adopted Element H (from the review tool), which requires local municipalities to provide a month when they are going to hold their annual Hazard Mitigation Plan meeting. RIEMA will be tracking these months and reaching out around the time they designated to follow up on their annual meeting.

6.5 Review and Approval of Local Hazard Mitigation Plans

When a plan is submitted to RIEMA for review it is logged and assigned to a reviewer. Any subsequent information regarding the plan is then tracked until the plan reaches final FEMA approval. The plans are reviewed in the order in which they are received. When possible, RIEMA will review plans within 30 days of their submission. If revisions are necessary, the period for resubmittal will vary depending on the request for information. RIEMA will then submit the plan to FEMA Region I where it should be reviewed within 45 days whenever possible.

If revisions are needed, FEMA will notify RIEMA and RIEMA will notify the jurisdiction within five business days. Depending on the request for information will determine how long the jurisdiction takes to resubmit to RIEMA. Once the revised plan is submitted to RIEMA, it will be logged and reviewed in the order that it was received and resubmitted to FEMA Region I.

If the plan is deemed acceptable, FEMA will notify the local municipality directly with a copy to RIEMA that the plan has reached approved pending adoption status or fully approved if adopted before submission. The jurisdiction then must obtain resolutions from the participating jurisdictions. Once final resolutions are received, the local municipality will then submit the final mitigation plan to FEMA Region I. FEMA then issues a final approval letter.

6.6 Local Hazard Mitigation Strategies

The analysis of the types of mitigation goals and activities provides a baseline assessment of local hazard mitigation priorities. It also demonstrates the recognition among local communities that an effective mitigation strategy is comprised of a wide array of activities and actions and provides areas of opportunity to strengthen local mitigation strategies. All mitigation goals, and objectives described in the local Hazard Mitigation Plans were reviewed and analyzed to identify trends and to help inform the SHMP. In general, mitigation goals identified fell into the following categories:

- Reduction of risk of injury or death.
- Protection of structures or property.
- Protection of critical infrastructure.
- Provision of public education on hazard mitigation.
- Protection of cultural, historical, natural, and economical environments.

A large majority of local plans include similar mitigation actions and categories of actions. The most popular mitigation actions are indicated below:

- Infrastructure improvements
- Public outreach and education
- Acquisition of repetitive loss structures
- Tree trimming programs
- Code improvements and enforcement

6.7 Prioritization of Local Assistance for Mitigation Planning and Projects

In general, federal and state funding for mitigation planning and projects is limited and often competitive. When reviewing initial requests for assistance with mitigation planning and projects, RIEMA considers:

- As a principal consideration, does this project maximize potential benefits
- Availability of funds
- A demonstrated desire and capability to complete the planning process
- The jurisdiction has the ability to complete proposed mitigation projects
- Expiration date of current plan, if applicable

When funding because available RIEMA advertises a notice of intent period. Notice of intents are received for a limited amount of time then graded on various criteria to determine which ones will be selected to develop planning applications. RIEMA determination criteria for planning projects include the following:

- Is this project for a mitigation plan development or update?

- Current plan expiration date?
- Will the plan be multi-jurisdictional or single?
- Has a notice of intent been submitted under previous grants that went unfunded?
- Has the jurisdiction participated in training directly related to hazard mitigation?

Available federal funds for pre-disaster and post-disaster mitigation grants will most likely not be sufficient to support all eligible project applications. None the less, an attempt will be made to award grants to the maximum number of eligible projects. When considering hazard mitigation projects, the State of Rhode Island attempts to align priorities with that of FEMA. As such, RIEMA uses the priorities outlined in the FEMA Notice of Funding Opportunity to prioritize project applications. When funding comes from the HMGP, priority is given to mitigation projects which are related to the hazard that necessitated the disaster declaration and those jurisdictions included in the disaster declaration. In the event that there are not enough eligible projects to use these criteria, RIEMA places a priority on local mitigation projects that:

- Involve non-structural and lower cost solutions such as updating and enforcing local flood ordinances, retrofitting high-risk structures, or the acquisition of repetitive loss structures.
- Exhibit: cost-effectiveness as established by a benefit cost ratio calculated using the FEMA Benefit-Cost Analysis software.
- Contribute the greater solutions to stated problems.
- Are in communities with the highest risk.
- Address repetitive flood loss.
- Are proposed by communities with the most intense development pressure.

In evaluating hazard mitigation applications for grant funding, a scoring system is used to prioritize projects according to both federal eligibility criteria and the State eligibility criteria as published in the grant application guidance. For each round of grant funding, RIEMA reviews the applications. Mitigation projects submitted for consideration for grant funding are evaluated using a scoring process emphasizing protection of life and property, reduction of risk, and cost-effectiveness. Proposed mitigation projects must also comply with the goals and objectives in both the local hazard mitigation plan and the SHMP. Additionally, for mitigation projects seeking federal funding, applicants must follow national priorities and requirements as detailed in the specific grant program guidance literature. RIEMA staff work with each grant applicant to ensure that proposed projects meet these criteria. When prioritizing grant applications, the seriousness of risk is emphasized when considering an applicant's response to the following federal and State eligibility criteria. Among the criteria receiving greatest weight in scoring are those dealing with reduction of risk posed by hazards, prevention of repetitive loss structures, and protection of critical areas including frequently flooded areas and geologically hazardous areas.

Project Eligibility

All projects seeking to receive either pre- or post-disaster grant funding must meet the following criteria as determined by RIEMA:

- Must be in conformance with a FEMA-approved local and/or multi-jurisdictional all hazards mitigation plan which meets the mitigation planning requirements per the Disaster Mitigation Act of 2000.
- Must be in conformance with the Rhode Island SHMP.
- Must be in, or have a beneficial impact upon, past declared disaster areas or in a high-risk area for potential impacts a natural hazards. The high-risk area should be identified in either the local, regional, or State mitigation plan.
- Must be in compliance with all existing Rhode Island Laws and Regulations for construction, land alterations, and natural resource protection.
- Must be in compliance with municipal ordinances and zoning regulations.
- Must be in conformance with 44 CFR, Part 9, Floodplain Management and Protection of Wetlands, and 44 CFR, Part 10, Environmental Considerations.
- Must provide a solution to a problem independently or provide a significant functional portion of a solution being addressed in a combined project. If the project constitutes a significant functional portion of a solution

being addressed, the status of any associated dependent or supporting projects must be given. There must be reasonable assurance that the total mitigation project will be completed. The identification or analysis of a problem does not automatically qualify for eligibility.

- Must meet FEMA's cost-effective criteria.

Recommendations for funding will be made to the FEMA Region I office by the SHMO in consultation with NFIP State Coordinator, under advisement by the SIHMC. FEMA will make the final selection of grants to be awarded. The mitigation measure proposed should not be intended to only replace what was damaged, but rather should provide more protection to life and property than what existed prior to the storm.

Due to the onset of COVID-19 early in 2019, many available local resources, funding, and capabilities were reassigned to help manage the pandemic. As such, the State of Rhode Island was able to complete only a small number of the mitigation actions that were identified in local level plans. The following table presents awarded mitigation projects during the past five-year period.

Table 203: RIEMA Hazard Mitigation Awarded Local Mitigation Projects

Year	Jurisdiction	Project	Project Amount	Grant Program
2019	Narragansett	WWTP Drainage Improvements	\$270,000	PDM
2019	Narragansett	Advanced Assistance - Foddering Farm/Harbour Island Causeway Concept Study	\$50,000	PDM
2019	Woonsocket	Advanced Assistance - Stormwater Project	\$128,320	PDM
2019	North Smithfield	Advanced Assistance - Cherry Brook Watershed Analysis	\$14,000	PDM
2019	Exeter	HMP Update	\$10,500	PDM
2019	Cranston	HMP Update	\$36,976	PDM
2019	East Greenwich	HMP Update	\$11,000	PDM
2019	Newport	HMP Update	\$41,090	PDM
2019	Lincoln	HMP Update	\$33,690	PDM
2020	Barrington	HMP Update	\$30,500	BRIC
2020	Bristol	HMP Update	\$18,600	BRIC
2020	Cumberland	HMP Update	\$27,933	BRIC
2020	East Providence	HMP Update	\$17,500	BRIC
2020	New Shoreham	HMP Update	\$14,160	BRIC
2020	Smithfield	HMP Update	\$10,000	BRIC
2020	Newport	Project Scoping - North and South Easton Pond Dam Mitigation Planning	\$178,000	BRIC
2020	Bonnet Shores Fire District	Project Scoping - Protect Little Beach Shore Line	\$69,800	BRIC
2021	Central Falls/Pawtucket	HMP Update	\$38,500	BRIC
2021	Hopkinton	HMP Update	\$41,500	BRIC
2021	Providence	HMP Update	\$61,980	BRIC
2021	Warwick	HMP Update	\$50,000	BRIC
2021	Westerly	HMP Update	\$75,000	BRIC
2022	Town of Lincoln	Sewer Pump Station Improvements	\$495,650	PDM
2022	Foster	HMP Update	-	BRIC
2022	Portsmouth, Middletown, Newport	HMP Update	-	BRIC
2022	Narragansett	Foddering Farm Road Causeway Project Scoping Study	-	BRIC

Source: RIEMA

-: Data unavailable

6.8 Utilization of Local Plan Data in 2024 SHMP

RIEMA provided all available local mitigation plans for data integration purposes into the SHMP. A comprehensive review was conducted of all currently approved local hazard mitigation plans to integrate the local risk assessments into the state's analysis. Collected data was then translated into a unified model that accounted for any variability in methodologies. The result of this assessment provides a larger scale snapshot of how local jurisdictions view risk and allows both local and state planners to integrate data into plans with a whole picture methodology. By integrating data across plans the overall planning efforts of the State become more robust. Additionally, all mitigation strategies were reviewed and incorporated into the revision of state strategies to ensure statewide unity of effort.

For this SHMP, each of the above referenced local hazard mitigation plans was reviewed for the following elements, with data incorporated into the state plan:

- Mitigation goals and objectives
- Mitigation strategies (used to inform state strategies)
- Prioritized hazards
- Risk assessment
- Jurisdictional capabilities
- Mitigation projects

Details of information utilized from local hazard mitigation plans can be found throughout this SHMP. As new or revised local hazard mitigation plans become available, they will be reviewed and incorporated into subsequent plan revisions. RIEMA will also make all data from the state hazard mitigation plan available to all local jurisdictions for review and inclusion in their planning efforts.

6.9 Dissemination of State-Level Hazard Mitigation Data

RIEMA takes a proactive and continuous approach on sharing all state-level hazard mitigation data and mitigation priorities with all local jurisdictions. As part of this process, RIEMA makes the latest version of the SHMP, along with all data, available in electronic format for data sharing and works to help integrate state level data into local plans during regular planning meetings and during the local Hazard Mitigation Plan submission process.

Section 7 – Mitigation Strategy

7.1 Introduction

As part of this planning effort, the State of Rhode Island worked to minimize the risk of future impacts from identified hazards to all citizens of the state. In an attempt to shape future regulations, ordinances and policy decisions on both a state and local basis, the SIHMC reviewed, revised, and developed a comprehensive hazard mitigation strategy. This comprehensive strategy includes:

- Goals to guide the selection of activities to mitigate and reduce potential loss.
- Objectives to be pursued in order to reach the goals.
- A discussion of state funding capabilities for hazard mitigation projects.
- Identification, evaluation, and prioritization of mitigation actions along with potential funding sources.

Rhode Island’s mitigation strategy promotes long-term hazard resilience that will have a positive impact on quality-of-life issues. By minimizing both the exposure to, and potential impacts from, identified hazards jurisdictions can expect minimize injuries and loss of life, reduce property damage, and minimize the day to day social and economic disruptions that follow hazard events.

7.2 Mitigation Problem Statements

As part of this planning process, the SIHMC worked to identify gaps and deficiencies both in State and local capabilities and hazard assessment. Resulting from this assessment is a series of problem statements, concise descriptions of issues or challenges that needs to be addressed:

- Local Hazard Mitigation Plans contain inconsistent data that is difficult to combine on a State level.
- Continued climate change is driving an increased incidence of major hazard occurrences, stressing the response, recovery, and mitigation capabilities of even the most prepared jurisdiction.
- Understanding and utilization of climate change models and predictions is underutilized in local plans.
- The expansion of hazard areas related to sea level rise and storm surge inundation will continue to increase the number of State and local facilities and assets at risk.
- The aging dam infrastructure and lack of analysis on vulnerability is compounded by staffing challenges, potentially decreasing inspections and data output.
- With the potential advent of greater periods of drought, and the large footprint of WUI areas in the State, Rhode Island is at increased risk for a large and destructive wildfire event.
- Despite attempts to engage all communities to ensure planning equity, little buy-in was achieved.
- The hardening of aging state assets against identified hazards will be costly and slow.
- The difficulties in applying for and managing hazard mitigation grants are beyond the capability of smaller jurisdictions.
- Engagement with the public on hazard mitigation, and actions they can take to protect themselves and their property, are lacking.
- Staffing at all levels is stretched thin, with many personnel wearing multiple hats, compromising planning, response, recovery, and mitigation capabilities.

To overcome these identified challenges, RIEMA, the SIHMC, and participating stakeholders will work collaboratively using the following strategies as appropriate:

- Innovation and Adaptation: Foster a culture of innovation and adaptability. Encourage employees to think creatively, embrace change, and explore new ways of doing things to overcome challenges and stay competitive in a dynamic environment.
- Training and Development: Invest in training and development to enhance skills and knowledge.
- Communication Improvement: Enhance communications and provide clear and transparent communication when sharing information, aligning teams, and addressing concerns.

- **Collaboration and Teamwork:** Encourage collaboration and teamwork which allows for the pooling of diverse skills and perspectives, leading to more effective problem-solving.
- **Technology Adoption:** Embrace technology to streamline operations and enhance productivity.
- **Diversity and Inclusion:** Foster a diverse and inclusive workplace. Embracing diversity brings different perspectives and approaches to problem-solving, contributing to more innovative solutions.
- **Agile Project Management:** Implement agile project management methodologies to enhance flexibility and responsiveness to changing conditions.
- **Continuous Improvement:** Adopt a culture of continuous improvement. Encourage employees at all levels to identify opportunities for improvement and implement changes incrementally,

As appropriate, these strategies will be tailored for specific circumstances. Additionally, a combination of these strategies is often more effective than relying on a single approach.

7.3 Goals and Objectives

The State of Rhode Island's overall mitigation goal is to minimize the protect lives and properties within the state from the impacts of hazards identified in this plan. To assist with this overarching goal, the SIHMC convened to discuss the hazard mitigation goals for this plan. As part of this effort, a vision statement (or long-term declaration of ambition) for hazard mitigation was developed for the State of Rhode Island:

- **Vision Statement:** Rhode Island is resilient to natural, technological, and human caused hazards and climate change.

Based on this vision statement, and the discussions by the SIHMC, it was determined that the goals (desired outcomes) and objectives (specific and measurable steps) identified in the 2019 SHMP remained viable and valid. These goals were developed using the SMART principle, specifically is the goal:

- Specific
- Measurable
- Ambitious
- Relevant
- Time-Bound

By utilizing SMART goals, the State of Rhode Island provides the SHMP with:

- Purpose
- Direction
- Focus

The following represent the identified goals and objectives for the 2024 SHMP:

- **Goal 1:** Rhode Island has the capacity to promote and implement projects, programs, plans, policies, and legislative actions to reduce vulnerability and repetitive loss to natural, technological, and human-caused hazards, under current and future conditions.
 - Objective 1.1: Maintain and implement a SHMP that fosters innovation, advances public support, and gains long-term commitments to action from all stakeholders.
 - Objective 1.2: Increase the capacity of and diversify representation on the SIHMC.
 - Objective 1.3: Increase the capability of Rhode Islanders to address natural, technological, and human-caused hazards in their daily lives and work.
- **Goal 2:** Statewide coordination of hazard mitigation with organizations, agencies, and stakeholders.
 - Objective 2.1: Leverage resources and expertise and capitalize on opportunities that will further hazard mitigation efforts.
 - Objective 2.2: Institutionalize hazard mitigation into the activities of all stakeholders.

- o Objective 2.3: Increase communications and streamline policies, procedures, and operations to eliminate conflicts and duplication of effort.
 - o Objective 2.4: Continue to integrate the SHMP with State climate goals and plans.
- **Goal 3:** Local communities address natural, technological, and human-caused hazards and long-term risk reduction in local decision making and planning for current and future conditions.
 - o Objective 3.1: Aid in the development and implementation of Local Hazard Mitigation Plans.
 - o Objective 3.2: Improve the quality of comprehensive plans in addressing natural, technological, and human-caused hazards.
 - o Objective 3.3: Increase the capabilities of local decision makers in hazard mitigation.
 - o Objective 3.4: Continue to engage the public in hazard mitigation planning and solutions.
- **Goal 4:** The public understands, supports, and acknowledges the need for hazard mitigation.
 - o Objective 4.1: Identify and implement opportunities to educate the public about hazard specific issues.
 - o Objective 4.2: Engage the private sector in hazard mitigation initiatives.
 - o Objective 4.3: Increase stakeholder, public sector, and private sector coordination.
 - o Objective 4.4: Implement and maintain an extensive planning outreach strategy in order to gain feedback from all stakeholders and the public throughout the hazard mitigation planning process.
- **Goal 5:** The built environment, infrastructure, high hazard dams, people, natural environment, and economy are resilient to the impacts of natural, technological, and human-caused hazards under current and future conditions (including repetitive loss [RL] and severe repetitive loss [SRL]).
 - o Objective 5.1: Implement measurable actions to reduce vulnerability, before and after disasters and during recovery.
 - o Objective 5.2: Increase and measure losses avoided as a result of mitigation actions through regular monitoring and evaluation.

RIEMA will continuously evaluate these identified goals and objectives against current capabilities and conditions. As part of this process, RIEMA and the SIHMC will utilize a monitoring and evaluation system to systematically track, assess, and measure the progress of activities and outcomes related to goals or objectives outlined in this SHMP. Key components to the monitoring and evaluation system include:

- Establishment of baseline data to quantify the starting point upon the approval of this plan. This will provide a reference against which progress can be measured.
- Enactment of a monitoring plan which outlines the specific activities, tasks, and responsibilities for regularly collecting, analyzing, and reporting data on the performance indicators.
- Identification and specification of the methods for collecting data, whether through surveys, interviews, focus groups, or observations.
- Definition of the criteria and methods for analyzing collected data. This includes determining how quantitative and qualitative data will be processed and interpreted to assess progress.
- Involvement of SIHMC stakeholders to ensure that all perspectives are considered, and that feedback on the progress of achieving the delineated goals is taken into account.

In addition, both RIEMA and the SIHMC will work with all State of Rhode Island agencies and policy makers to help integrate the goals delineated in the SHMP and goals and plans for combating climate change (Objective 2.4).

7.4 Review and Creation of Mitigation Actions

For this plan update members of the SIHMC were provided with a complete list of previously identified mitigation actions and asked to review them to determine their status. Previously identified mitigation status was reported using the following definitions:

- **Completed:** The action has been fully completed.
- **Not Completed:** The action was not started or has been started and is not completed.
- **Revised:** Action has been revised to reflect current planning environment or identified changes.

- **Cancelled:** The action has been removed from consideration due to either a lack of resources or changing mitigation priorities.
- **Ongoing:** The action is completed and has become an ongoing activity or capability of RIEMA or another state agency.

SIHMC members were also asked if any of the previously identified actions were considered as part of continuing programs or as part of an agency's standard operating procedures. Actions identified as such were removed from the action list and are detailed in a stand-alone table.

Additionally, SIHMC members and stakeholders were provided with forms to identify and incorporate newly identified actions based on the changing hazard environment or previously unidentified needs. A wide range of activities were considered based on the following factors:

- Updated state risk assessment and information from local risk assessments.
- Goals and objectives.
- Existing state actions.
- State and local capabilities.
- Actions identified in local plans.

In preparing a mitigation strategy, all reasonable and obtainable mitigation actions were considered to help achieve the general goals. Priorities were developed based on past damage, existing exposure to risk, and weaknesses identified by the State and local capability assessments. In identifying mitigation actions, the following activities were considered:

- The use of applicable building construction standards.
- Hazard avoidance through appropriate land-use practices.
- Relocation, retrofitting, or removal of structures at risk.
- Removal or elimination of the hazard.
- Reduction or limitation of the amount or size of the hazard.
- Segregation of the hazard from that which is to be protected.
- Modification of the basic characteristics of the hazard.
- Control of the rate of release of the hazard.
- Provision of protective systems or equipment for both cyber and physical risks.
- Establishment of hazard warning and communication procedures.
- Redundancy or duplication of essential personnel, critical systems, equipment, and information materials.

In general, all identified mitigation actions can be classified under one of the following broad categories:

- **Local plans and regulations:** Actions that create or update plans to reflect situational changes and/or actions that aid in the creation, revision, or adoption of regulations related to hazard mitigation and management.
- **Natural resource protection:** Actions that, in addition to minimizing hazard losses, also preserve or restore the functions of natural systems.
- **Preparedness and response:** Emergency response or operational preparedness actions.
- **Public education and awareness:** Actions to inform and educate citizens, elected officials, and property owners about the hazards and potential ways to mitigate them.
- **Structural:** Actions that the modification of existing buildings or structures or involve the construction of structures to reduce the impact of hazard.

7.5 Prioritization of Mitigation Actions

The SIHMC and subject matter experts worked together to prioritize both previously identified and newly identified hazard mitigation actions. The methodology used to determine and revise mitigation priorities was based upon the following:

- Review of the updated risk assessments.
- Review of revised goals and objectives.
- Review of State and local capabilities.

A multi-pronged and flexible analysis method was used for determining and prioritizing mitigation actions. An initial review of previously identified but not completed actions was conducted to ensure that, based on current condition and capabilities, the actions were still viable. Actions that were considered viable by the SIHMC were retained in this plan update, with minor revisions completed as necessary.

For identified actions that were retained, and for newly identified actions, the FEMA recommended Social, Technical, Administrative, Political, Legal, Economic, and Environmental (STAPLEE) criteria were used to assist with prioritization. The following table details the STAPLEE criteria:

Table 204: STAPLEE Review Criteria

Criteria	Discussion	Example Considerations
Social	There should be community acceptance and support for the mitigation action?	Does the action have community acceptance? Will the proposed action adversely affect one segment of the population?
Technical	The proposed mitigation action should be technically feasible and should provide a long-term reduction in losses.	How effective is the action in avoiding or reducing future losses? Does it solve a problem or only a symptom? Does the action create additional problems?
Administrative	Personnel and administrative capabilities should be available to administer all phases of the project.	Are the staffing and administrative capabilities to implement the action in place? Is there someone to coordinate and lead the effort?
Political	Political support for the mitigation action needs to be present.	Is the action politically acceptable? Have political leaders been involved in the planning process? Is there a political champion to help see the project to completion?
Legal	The legal authority to implement the actions need to be in place or possible with the passing of laws or regulations.	Does the legal authority to implement the proposed action exist? Are there potential legal repercussions?
Economic	The current budget (and/or general obligation bonds or other instruments) need to be in place to fully fund the mitigation action.	Do the potential benefits of this action exceed the potential costs? Has funding been secured for the proposed action? What are the potential funding sources (public, non-profit, and private)? How will this action affect the fiscal capability of the community(s)? Does the action contribute to other community goals, such as capital improvements or economic development?
Environmental	Actions should interface with the need for sustainable and environmentally healthy communities. Also, statutory considerations, such as the National Environmental Policy Act need to be considered for federal funds.	How will the action affect the environment? Will the action need environmental regulatory approvals? Will it meet federal, state, and local state regulatory requirements? Are endangered or threatened species likely to be affected?

Based on the prioritization review, the SIHMC assigned each action the following prioritized ranking:

- **High Priority:** Actions that provide substantial progress towards improving resiliency and are determined as potentially urgent in nature by the SIHMC. This would include actions that strongly support the reduction or elimination of high hazard risks and meet mitigation goals. Additionally, actions in this ranking may have imminent funding availability or strong community support.
- **Medium Priority:** Actions that provide reasonable progress towards improving resiliency and are determined as moderately urgent in nature by the SIHMC. This would include actions that would lessen impact hazard events, but not eliminate the impact completely.
- **Low Priority:** Actions that provide incremental progress towards improving resiliency and are determined as slightly urgent in nature by the SIHMC. This would include actions that are generally the responsibility of the local community, actions outside the normal authority of the State, or actions whose cost/benefit analysis returns a low yield.

7.6 Mitigation Action Progress

The onset of COVID-19 early in the life of the 2019 SHMP necessitated all available resources, funding, and capabilities to be reassigned to help manage the pandemic. Additionally, staff shortages and non-standard working arrangements were instituted for all agencies. As such, the State of Rhode Island only managed to complete a sub-set of previously identified mitigation action items since the completion of the last SHMP. The following table presents completed mitigation projects during the past five-year period:

Table 205: State of Rhode Island Completed Hazard Mitigation Action Summary

Previous SHMP Action Number	Description	Responsible Agency
8	Small Public Water System Emergency Preparedness Planning	RIDOH
19	Newport Climate Change Outreach for Vibrant Waterfronts	URI CRC and City of Newport
22	RIPTA Continuity of Operations	Rhode Island Public Transit Authority
25	State and Local Professional Development	RIEMA
44	Harbor of Refuge Wall Repair	RIDOT

7.7 Reclassified Mitigation Actions

As part of the mitigation action review process, the following previously identified actions were considered as part of continuing programs or as part of an agency's standard operating procedures. As such, they have been removed from the mitigation action list of this SHMP but will continue on as agency responsibilities.

Table 206: State of Rhode Island Hazard Mitigation Actions Reclassified as Standard Operating Procedure

Previous SHMP Action Number	Description	Responsible Agency
1	THIRA and HMP Integration	RIEMA
2	Flood modeling/mapping	Executive Climate Change Coordinating Council
3	Partnerships with Higher Education	RIEMA
4	State Interagency Hazard Mitigation Committee	RIEMA
6	Stormwater and Floodplain Management Coordination	RIDEM, RIDOT, and RIEMA
12	MEDS full-scale exercise	RIDOH
14	Develop a Petroleum Set-Aside Program	Office of Energy Resources
16	Emergency Planning for Patient Care Centers	RIDOH
20	Technological and human-caused Vulnerability Database	RIEMA

Table 206: State of Rhode Island Hazard Mitigation Actions Reclassified as Standard Operating Procedure

Previous SHMP Action Number	Description	Responsible Agency
22	Transportation and Infrastructure Based Climate Change Adaptation Planning for the Town of North Kingstown	North Kingstown Planning Department and URI CRC
24	Municipal disaster preparedness and resilience building activities incentives	RIEMA
26	Facility-level impacts storm predictions	RIEMA
27	Integrate Flood Hazard Information into Watershed Planning	RIDEM
28	Water Conservation	RIDOH
29	Disaster Evacuation Route and Shelter Location Outreach	RIEMA
32	Cyber Education	RIEMA
33	Partnerships with Private and Non-Profits	Rhode Island Division of Statewide Planning, State Drought Steering Committee
36	Dam Remediation Prioritization	RIDEM
40	Support of the Gas and Electric ISR program	Rhode Island Division of Public Utilities
41	Route 1A Structural Analysis and Maintenance	RIDOT
49	State-owned property retreat	RIDEM and CRMC
50	Beach Ecosystem Preservation	CRMC
51	Integrate climate resilience into the school construction process.	RIIB
53	Establish dedicated funding source for beach replenishment projects and/or property buyouts in high hazard areas	CRMC and Rhode Island Sea Grant

7.8 Mitigation Action Summary

The following table provides a summary of the State of Rhode Island mitigation actions that were not completed or were reclassified. Each of the actions can be implemented independently or in concert, and each activity will support the improvement of the overall resiliency of the State. It is important to note that discussions with the SIHMC indicate that due to the onset of COVID-19 early in the life of the 2019 SHMP, all available resources, funding, and capabilities were reassigned to help manage the pandemic. Additionally, staff shortages and non-standard working arrangements were instituted for all agencies. As such, these many of mitigation actions were generally shelved due to the critically of the COVID-19 response and the general lack of resources.

Table 207: State of Rhode Island Hazard Mitigation Action Summary

New Action Number	Previous SHMP Action Number	Description	Status
1	15	Rhode Island Mitigation Resources and Technical Assistance	Not completed, modified, carried over
2	23	Urban Tree Inventory and Urban Forest Master Plan	Not completed, modified, carried over
3	37	Green Stormwater Infrastructure	Not completed, modified, carried over
4	38	Stormwater Infrastructure Flood Mitigation	Not completed, modified, carried over
5	39	Repetitive Loss Property Mitigation	Not completed, modified, carried over
6	42	Power Vulnerability Analysis for Drinking Water Systems	Not completed, modified, carried over

Table 207: State of Rhode Island Hazard Mitigation Action Summary

New Action Number	Previous SHMP Action Number	Description	Status
7	43	Narragansett Seawall Stability Evaluation and Maintenance	Not completed, modified, carried over
8	45	Technical Assistance for Water Treatment System and Pumping Station Hardening Projects	On-going
9	9	Extreme Heat Resilience and Mitigation Plan	Not completed, modified, carried over
10	21	Comprehensive Planning Guidance Integration	On-going. Staff review municipal Comprehensive Plans for inclusion of required content under Planning Standard 9, "Planning for Natural Hazards and Climate Change"
11	34	Coordination Between Utilities and State Agencies	Not completed, modified, carried over
12	46	Fuel Terminal Hardening and Resilience Measures	Not completed, modified, carried over
13	47	Evacuation Route Reassessment	Not completed, modified, carried over
14	5	Facilities Vulnerability Database and Prioritization	Not completed, modified, carried over
15	11	RIWARN Education and Joint Vulnerability Reduction	Not completed, modified, carried over
16	13	Local Water Supplier Emergency Interconnection Programs	Not completed, modified, carried over
17	17	Spillway Management and Coordination	Not completed, modified, carried over
18	7	Revise RIDOH/Drinking Water Quality Emergency Operations Plan	On-going
19	18	Agency and Municipality Data Consistency	Not completed, modified, carried over
20	27	Resilient Coastal Communities	Not completed, modified, carried over
21	35	Public Hazard Mitigation Education	Not completed, modified, carried over
22	52	Coastal and Freshwater Wetland Habitat Preservation	Not completed, modified, carried over
23	54	Coastal Drinking Water Reservoir Assessment	Not completed, modified, carried over
24	16	Local Planning Technical Assistance for Livable and Resilient Communities	Not completed, modified, carried over
25	48	Reduction in Fuel Sources	Not completed, modified, carried over
26	28 and 30	Increased CRS Participation	Not completed, modified, carried over
27	-	Narragansett and Wickford Dock Mitigation	New
28	-	Construction of Public Health Laboratory	New, construction underway
29	-	State Road Resilience Improvement Plan	New, underway
30	-	State Asset High Hazard Dam Vulnerability	New, underway
31	-	Mitigation Plan Standardization	New
32	-	Flood Insurance Outreach	New
33	-	Commodity Flow Study	New

The SIHMC acknowledges that the adoption and approval of this plan does not obligate the State of Rhode Island to complete each identified action. Rather, the SIHMC understands that progress should be shown in statewide mitigation efforts which may include the completion of mitigation actions or other actions or progress in achieving the goals and objectives of the SHMP.

7.9 Mitigation Action Details

The following tables provide specific details concerning each identified mitigation action. Each identified Lead and Support agency is charged with championing the identified action, seeking available funding sources, providing expert oversight and management of the project, and working with RIEMA through the tracking and closeout process.

Prior to the implementation of any action further feasibility analysis will be performed, including confirmation that it meets the STAPLEE evaluation criteria. Additionally, a Benefit-Cost Analysis that determines the future risk reduction benefits of a hazard mitigation project and compares those benefits to its costs will be conducted as required. Applicants and sub-applicants must use FEMA approved methodologies and tools, such as the Benefit-Cost Analysis Toolkit, to demonstrate the cost-effectiveness of their projects. The result is a Benefit-Cost Ratio, and a project is considered cost-effective when the Benefit-Cost Ratio is 1.0 or greater. Depending on the project, either a full Benefit-Cost Analysis will be completed by entering documented values into the FEMA Benefit-Cost Analysis Toolkit, which calculates a benefit-cost ratio or, if the project meets specified criteria, a streamlined Benefit-Cost Analysis may be completed (FEMA's cost-effectiveness requirement is never waived).

Table 208: Mitigation Action Number 1

Action Title	Rhode Island Mitigation Resources and Technical Assistance
Priority	High
Description	Yearly update of RIEMA website to include mitigation resources such as available state and federal grant programs and technical resources for mitigation planning in Rhode Island. Identification of these resources will allow local jurisdictions to identify and apply for funding for local projects to reduce long-term vulnerability to identified hazards.
Hazard(s) Addressed	All
Vulnerability Analysis	The current RIEMA website does not adequately or fully detail mitigation resources.
Goal and Objective	2 and 4
Funding Sources	Agency fiscal budget
Benefit-Cost Analysis	No
Lead Agency	RIEMA
Support Agency	All State of Rhode Island departments
Completion Timeframe	Short-Term (1-2 years)
Current Status	Not completed, modified, carried over

Table 209: Mitigation Action Number 2

Action Title	Urban Tree Inventory and Urban Forest Master Plan
Priority	High
Description	Provide technical assistance, including mapping, to local jurisdictions in developing urban tree inventories and implementing urban forest master plans in order to mitigate against urban heat, drought, and wildfires. Urban trees help mitigate the impacts of drought, extreme heat, and wildfire. Encouraging localities to implement urban forest master plans identifies existing trees to be preserved, as well as locations that would benefit from planting vegetation.
Hazard(s) Addressed	Drought, Extreme Temperatures, and Wildfire
Vulnerability Analysis	A complete understanding of the conditions and locations of trees and forested areas is not available.
Goal and Objective	2 and 3
Funding Sources	Agency fiscal budget
Benefit-Cost Analysis	No
Lead Agency	RIDEM Forest Environment
Support Agency	-
Completion Timeframe	Medium-Term (2-3 years)
Current Status	Not completed, modified, carried over

Table 210: Mitigation Action Number 3

Action Title	Green Stormwater Infrastructure
Priority	High
Description	Enhance the service capability of local stormwater systems through the provision of technical expertise and guidance on the installation of green infrastructure, including rain gardens, impermeable surfaces, and green roofs. These systems provide a long-term increase in resiliency of stormwater systems due to a decrease in input.
Hazard(s) Addressed	Flood, Infrastructure Failure, and Sea Level Rise
Vulnerability Analysis	A potential increase in the frequency and severity of identified hazards, including severe rain events, is possible with climate change stressing current systems.
Goal and Objective	1 and 5
Funding Sources	BRIC
Benefit-Cost Analysis	No
Lead Agency	RIEMA
Support Agency	RIDOT, RIDEM
Completion Timeframe	Long-Term (5 years)
Current Status	Not completed, modified, carried over

Table 211: Mitigation Action Number 4

Action Title	Stormwater Infrastructure Flood Mitigation
Priority	High
Description	Mitigate impacts of coastal and riverine flooding on stormwater infrastructure and its performance. Identify at-risk structures, then enact projects and/or policies to address risks, and lastly establish process for long-term improvement of stormwater ponds.
Hazard(s) Addressed	Flood, Infrastructure Failure, and Sea Level Rise
Vulnerability Analysis	Identifying at-risk structures and appropriate mitigation projects ensures the protection of populations along the coast or in flooding-prone areas. Mitigative actions can reduce the risk of flooding, or mitigate the impact on people, property, and critical infrastructure.
Goal and Objective	5
Funding Sources	BRIC, HMGP
Benefit-Cost Analysis	No
Lead Agency	RIEMA
Support Agency	Rhode Island Office of Housing and Community Development, CRMC, RIDOT, local DPWs, NBC, RIDEM
Completion Timeframe	Long-Term (5 years)
Current Status	Completed

Table 212: Mitigation Action Number 5

Action Title	Repetitive Loss Property Mitigation
Priority	High
Description	Provide technical assistance to help local jurisdictions identify and devise a comprehensive mitigation strategy for repetitive loss and severe repetitive loss properties. Identification of properties has been completed by FEMA. Local jurisdictions will be helped in identify potential long-term actions such as elevation, buyout, floodproofing, and demolition
Hazard(s) Addressed	Flood and Sea Level Rise
Vulnerability Analysis	Repetitive loss and severe repetitive loss properties are at a higher risk of hazard impact. Identifying and developing a mitigation strategy to address these risks helps improve the resilience and sustainability of these structures and reduces the potential of future loss
Goal and Objective	3 and 5
Funding Sources	Agency fiscal budget

Table 212: Mitigation Action Number 5

Action Title	Repetitive Loss Property Mitigation
Benefit-Cost Analysis	Yes
Lead Agency	RIEMA
Support Agency	-
Completion Timeframe	Medium-Term (2-3 years)
Current Status	Not completed, modified, carried over

Table 213: Mitigation Action Number 6

Action Title	Power Vulnerability Analysis for Drinking Water Systems
Priority	High
Description	Identify gaps and vulnerabilities in emergency redundant power in Rhode Island's critical drinking water systems. This identification can help identify long-term projects to mitigate against the determined vulnerabilities.
Hazard(s) Addressed	Cybersecurity, Dam Failure, Earthquake, Extreme Temperatures, Flood, Infrastructure Failure, Sea Level Rise, Severe Thunderstorm, Severe Winter Weather, Tornado, Tropical and Extratropical Storms
Vulnerability Analysis	A potential increase in the frequency and severity of identified hazards, including severe rain events, is possible with climate change stressing current systems.
Goal and Objective	2 and 5
Funding Sources	RIDOH Division of Water Quality Budget; Federal grants
Benefit-Cost Analysis	Yes
Lead Agency	RIDOH
Support Agency	-
Completion Timeframe	Medium-Term (2-3 years)
Current Status	Not completed, modified, carried over

Table 214: Mitigation Action Number 7

Action Title	Narragansett Seawall Stability Evaluation and Maintenance
Priority	High
Description	The stability and effectiveness of the seawall should be evaluated to ensure that it is resilient to the 20-to 50-year storm. RIDOT will work with the jurisdictions to continue to evaluate the stability of the seawall and ensure continued maintenance and improvements.
Hazard(s) Addressed	Flood, Sea Level Rise, Severe Winter Weather, Tropical and Extratropical Storms
Vulnerability Analysis	Unstable seawalls have reduced capacity to protect against coastal hazards and leave communities vulnerable to impacts. Long-term improvement and regular maintenance will ensure seawall is in a stable condition and provide adequate protection.
Goal and Objective	5
Funding Sources	Agency fiscal budget
Benefit-Cost Analysis	No
Lead Agency	RIDOT
Support Agency	RIEMA
Completion Timeframe	Medium-Term (2-3 years)
Current Status	Not completed, modified, carried over

Table 215: Mitigation Action Number 8

Action Title	Technical Assistance for Water Treatment System and Pumping Station Hardening Projects
Priority	High
Description	Provide technical expertise to allow local jurisdictions to identify and implement water treatment and pumping station hardening, including floodproofing and potential relocation, projects throughout Rhode Island. The economic, environmental, and

Table 215: Mitigation Action Number 8

Action Title	Technical Assistance for Water Treatment System and Pumping Station Hardening Projects
	public health implications of damaged wastewater treatment and pumping stations are potentially severe.
Hazard(s) Addressed	Flood, Infrastructure Failure, Sea Level Rise, Severe Thunderstorms, Severe Winter Weather, Tornado, Tropical and Extratropical Storms
Vulnerability Analysis	A potential increase in the frequency and severity of identified hazards, including severe rain events, is possible with climate change stressing current systems.
Goal and Objective	1 and 5
Funding Sources	Rhode Island Division of Statewide Planning budget
Benefit-Cost Analysis	No
Lead Agency	Rhode Island Division of Statewide Planning
Support Agency	RIEMA, RIDEM
Completion Timeframe	Long-Term (5 years)
Current Status	On-going

Table 216: Mitigation Action Number 9

Action Title	Extreme Heat Resilience and Mitigation Plan
Priority	Medium
Description	Convene interagency and nongovernmental partners to develop and implement a state extreme heat resilience and mitigation plan, including state and local building design parameters, the development of potential heat and cold station locations, and the production of information for the public on the hazard.
Hazard(s) Addressed	Extreme Temperatures
Vulnerability Analysis	Climate change can exacerbate extreme temperature conditions. Without an initial action plan, citizens and vulnerable populations may be left at long-term risk.
Goal and Objective	2
Funding Sources	RIEMA fiscal budget
Benefit-Cost Analysis	No
Lead Agency	RIEMA
Support Agency	Department of Business Regulation, Red Cross
Completion Timeframe	Medium-Term (2-3 years)
Current Status	Not completed, modified, carried over

Table 217: Mitigation Action Number 10

Action Title	Comprehensive Planning Guidance Integration
Priority	Medium
Description	Promote continued and updated integration of natural and human-caused hazards from local and state hazard mitigation plans into local comprehensive planning guidance and provide workshops to disseminate guidance. Mitigation actions are bolstered when hazards are addressed and integrated into all municipal comprehensive plans as required by the Rhode Island Comprehensive Planning and Land Use Act RIGL § 45-22.2-6 (10).
Hazard(s) Addressed	All Hazards
Vulnerability Analysis	Excluding hazards from such plans can exacerbate hazard impacts and increase vulnerability of the locality.
Goal and Objective	3
Funding Sources	Agency fiscal budget
Benefit-Cost Analysis	No
Lead Agency	Rhode Island Division of Statewide Planning
Support Agency	-
Completion Timeframe	On-going

Table 217: Mitigation Action Number 10

Action Title	Comprehensive Planning Guidance Integration
Current Status	On-going. Staff review municipal Comprehensive Plans for inclusion of required content under Planning Standard 9, "Planning for Natural Hazards and Climate Change"

Table 218: Mitigation Action Number 11

Action Title	Coordination Between Utilities and State Agencies
Priority	Medium
Description	Enhance coordination between Rhode Island Energy and the Division of Public Utilities to help mitigate the impact of future hazards on the electrical grid. Initial steps include the determination of partners, the formation of a committee, the establishment of meeting timelines, and a setting of goals. Coordinating between Rhode Island Energy and the Division of Public Utilities creates a cohesive and comprehensive mitigation approach to reducing the vulnerability of utility operations.
Hazard(s) Addressed	All Hazards
Vulnerability Analysis	The electrical grid is vulnerable to all hazards, and failure often results in the cessation of vital government business.
Goal and Objective	2
Funding Sources	Agency fiscal budget
Benefit-Cost Analysis	No
Lead Agency	Rhode Island Division of Public Utilities, Rhode Island Energy
Support Agency	RIEMA
Completion Timeframe	Long-Term (5 years)
Current Status	Not completed, modified, carried over

Table 219: Mitigation Action Number 12

Action Title	Fuel Terminal Hardening and Resilience Measures
Priority	Medium
Description	Identify and implement fuel terminal long-term hardening and resilience measures in order reduce long-term vulnerability from future storms. Fuel terminals provide critical resources to all state operations. Implementing hardening and resilience measures, as well as restoration operations, reduce long-term storm impact
Hazard(s) Addressed	Flood, Infrastructure Failure, Sea Level Rise, Severe Thunderstorm, Severe Winter Weather, Tornado, Tropical and Extratropical Storms
Vulnerability Analysis	Fuel terminals are vulnerable to all hazards, and failure often results in the cessation of vital government business.
Goal and Objective	5
Funding Sources	BRIC
Benefit-Cost Analysis	No
Lead Agency	Rhode Island Office of Energy Resources
Support Agency	-
Completion Timeframe	Long-Term (5 years)
Current Status	Not completed, modified, carried over

Table 220: Mitigation Action Number 13

Action Title	Evacuation Route Reassessment
Priority	Medium
Description	Assess current evacuation routes and signage and update as needed in relation to changing hazard frequency and location due to climate change.
Hazard(s) Addressed	Flood, Sea Level Rise, Tropical and Extratropical Storms
Vulnerability Analysis	Evacuation routes may be increasingly vulnerable to impacts from all hazards, including flooding and sea level rise.

Table 220: Mitigation Action Number 13

Action Title	Evacuation Route Reassessment
Goal and Objective	1 and 3
Funding Sources	Agency fiscal budget
Benefit-Cost Analysis	No
Lead Agency	RIEMA
Support Agency	RIDOT
Completion Timeframe	Medium-Term (2-3 years)
Current Status	Not completed, modified, carried over

Table 221: Mitigation Action Number 14

Action Title	Facilities Vulnerability Database and Prioritization
Priority	Low
Description	Perform a more comprehensive examination of state and critical facility vulnerability to natural hazards to prioritize facility improvements or relocation.
Hazard(s) Addressed	All Hazards
Vulnerability Analysis	People, property, critical facilities, and infrastructure are more vulnerable to hazard impacts if high-risk areas are unknown or outdated. Regularly performing vulnerability assessments with best available data ensures communities understand high-risk locations and work to implement mitigative actions based on assessment results.
Goal and Objective	1
Funding Sources	Agency fiscal budget
Benefit-Cost Analysis	No
Lead Agency	RIEMA
Support Agency	Rhode Island Division of Public Utilities, Rhode Island Energy, RIDOT, RIDEM, Department of Administration
Completion Timeframe	Medium-Term (2-3 years)
Current Status	Not completed, modified, carried over

Table 222: Mitigation Action Number 15

Action Title	RIWARN Education and Joint Vulnerability Reduction
Priority	Low
Description	Educate key public drinking water authorities on the benefits of RIWARN. RIWARN will allow for drinking water authorities to identify and complete hazard vulnerability reduction projects as a unit rather than stand-alone projects.
Hazard(s) Addressed	All Hazards
Vulnerability Analysis	A potential increase in the frequency and severity of identified hazards is possible with climate change stressing current systems.
Goal and Objective	2
Funding Sources	RIDOH Division of Water Quality budget; FEMA Hazard Mitigation Assistance Grant
Benefit-Cost Analysis	No
Lead Agency	RIDOH – Drinking Water Quality
Support Agency	RIEMA
Completion Timeframe	Medium-Term (2-3 years)
Current Status	On-going

Table 223: Mitigation Action Number 16

Action Title	Local Water Supplier Emergency Interconnection Programs
Priority	Low
Description	Assist water suppliers in developing local Emergency Interconnection Programs. Identifying interconnections will allow for drinking water authorities to identify and complete hazard vulnerability reduction projects based on interconnection data and results as a unit rather than stand-alone projects.
Hazard(s) Addressed	All Hazards
Vulnerability Analysis	A potential increase in the frequency and severity of identified hazards is possible with climate change stressing current systems.
Goal and Objective	2
Funding Sources	Drinking water state revolving fund
Benefit-Cost Analysis	No
Lead Agency	RIDOH
Support Agency	Water suppliers, Rhode Island Water Resources Board
Completion Timeframe	Long-Term (5 years)
Current Status	Not completed, modified, carried over

Table 224: Mitigation Action Number 17

Action Title	Spillway Management and Coordination
Priority	Low
Description	Advance common goal setting and communication between water suppliers that manage reservoirs and downstream municipalities. Ensure downstream flood mitigation via proactive spillway management without adverse impacts on safe yield.
Hazard(s) Addressed	Dam Failure, Flood, Infrastructure Failure
Vulnerability Analysis	Spillway management coordination ensures flood mitigation measures are comprehensive, uniform, and effective by encouraging water suppliers to coordinate and advance common goals.
Goal and Objective	2 and 5
Funding Sources	Drinking water state revolving fund
Benefit-Cost Analysis	No
Lead Agency	RIDOH and Rhode Island Water Resources Board
Support Agency	University of Rhode Island, RIIB, and Providence Water Supply Board
Completion Timeframe	Short-Term (1-2 years)
Current Status	Not completed, modified, carried over

Table 225: Mitigation Action Number 18

Action Title	Revise RIDOH/Drinking Water Quality Incident Response Plan
Priority	High
Description	This revised Incidence Response Plan will work as a tool to assist Drinking Water Quality staff to implement its response and assistance to a Public Water System rapidly and effectively in an all-hazards environment. This plan will be revised to include steps to identify system vulnerabilities and allow for the proposal of long-term projects to reduce identified vulnerabilities.
Hazard(s) Addressed	All Hazards
Vulnerability Analysis	Aging water systems are increasingly vulnerable to hazard events. Additional potential changes in climate may increase the frequency and magnitude of future events further stressing the system without long-term improvements.
Goal and Objective	2 and 5
Funding Sources	RIDOH Division of Water Quality Budget
Benefit-Cost Analysis	No
Lead Agency	RIDOH - Drinking Water Quality
Support Agency	RIDOH (CEPR)

Table 225: Mitigation Action Number 18

Action Title	Revise RIDOH/Drinking Water Quality Incident Response Plan
Completion Timeframe	Medium-Term (6-24 months)
Current Status	On-going

Table 226: Mitigation Action Number 19

Action Title	Agency and Municipality Data Consistency
Priority	Low
Description	Ensure that consistent data is available to all state departments and local jurisdictions, including GIS layers (e.g., STORMTOOLS, critical infrastructure, precipitation projections) and related metadata, which will allow for more effective hazard mitigation planning and the more streamlined identification of long-term projects to reduce vulnerability.
Hazard(s) Addressed	All Hazards
Vulnerability Analysis	The changing nature of hazards make the availability of current hazard data a necessity.
Goal and Objective	2
Funding Sources	Agency fiscal budget
Benefit-Cost Analysis	No
Lead Agency	RIEMA
Support Agency	State of Rhode Island Departments
Completion Timeframe	Short-Term (1-2 years)
Current Status	Not completed, modified, carried over

Table 227: Mitigation Action Number 20

Action Title	Resilient Coastal Communities
Priority	Low
Description	Build the capacity of targeted sectors or community decision-makers to identify general strategies and implement long-term mitigation actions which they can put into practice to increase their resilience to sea level rise and future storms. This will be done through outreach, technical assistance and sharing of lessons learned.
Hazard(s) Addressed	Flood, Sea Level Rise, Tropical and Extratropical Storms
Vulnerability Analysis	Coastal communities are at increasing risk to sea level rise and coastal flooding due to continued climate change conditions.
Goal and Objective	2 and 3
Funding Sources	URI CRC budget
Benefit-Cost Analysis	No
Lead Agency	URI CRC
Support Agency	-
Completion Timeframe	Long-Term (5 years)
Current Status	Not completed, modified, carried over

Table 228: Mitigation Action Number 21

Action Title	Public Hazard Mitigation Education
Priority	Low
Description	Increase awareness and knowledge of hazard mitigation principles and practices. Develop programs (i.e., website, Public Service Announcement) to increase public awareness of the importance of mitigating the damage caused by natural hazards, through a coordinated effort with multiple stakeholders. Determine and schedule training focused on hazard mitigation education and encourage the public to participate through widely advertised notices.
Hazard(s) Addressed	All Hazards

Table 228: Mitigation Action Number 21

Action Title	Public Hazard Mitigation Education
Vulnerability Analysis	Without appropriate knowledge of hazard mitigation principles and practices, the public will be unable to effectively implement mitigation actions.
Goal and Objective	4
Funding Sources	Agency fiscal budget
Benefit-Cost Analysis	NO
Lead Agency	RIEMA
Support Agency	-
Completion Timeframe	On-going
Current Status	Not completed, modified, carried over

Table 229: Mitigation Action Number 22

Action Title	Coastal and Freshwater Wetland Habitat Preservation
Priority	Low
Description	Monitor and assess coastal wetland habitats and management practices to evaluate and prioritize future mitigation actions. Statewide models, such as the SLAMM, should be updated to identify opportunities for restoration and assist in planning for future wetland mitigation. Where possible, retreat rather than fortification should be emphasized as a coastal adaptation strategy.
Hazard(s) Addressed	Flood, Sea Level Rise, Tropical and Extratropical Storms
Vulnerability Analysis	As sea level rise and climate change worsen coastal hazards in Rhode Island, wetland habitats and resources are important natural assets to conserve and provide a sustainable mechanism to protect against these hazards. Implementing conservation management, developing buffers, and encouraging protection of these resources can increase the resilience of communities.
Goal and Objective	2, 3 and 5
Funding Sources	Agency fiscal budget
Benefit-Cost Analysis	No
Lead Agency	CRMC, RIDEM
Support Agency	University of Rhode Island Environmental Data Center
Completion Timeframe	Ongoing
Current Status	Not completed, modified, carried over

Table 230: Mitigation Action Number 23

Action Title	Coastal Drinking Water Reservoir Assessment
Priority	Low
Description	Assess the coastal drinking water reservoirs to determine if the reservoirs are vulnerable to storm surge, sea level rise, and saltwater intrusion.
Hazard(s) Addressed	Flood, Infrastructure Failure, Tropical and Extratropical Storms, Sea Level Rise
Vulnerability Analysis	Sea level rise is observed to be occurring along Rhode Island's coastline. Understanding the risk posed to critical coastal drinking water reservoirs can help identify appropriate mitigation actions to reduce potential impact.
Goal and Objective	1
Funding Sources	Drinking water state revolving fund
Benefit-Cost Analysis	No
Lead Agency	RIDOH
Support Agency	CRMC
Completion Timeframe	Short-Term (1-2 years)
Current Status	Not completed, modified, carried over

Table 231: Mitigation Action Number 24

Action Title	Local Planning Technical Assistance for Livable and Resilient Communities
Priority	Medium
Description	Provide technical assistance to local jurisdictions on the design and completion of multi-modal transportation projects. Multi-modal transportation can reduce greenhouse gas emissions and reduce the potential of climate change, thus reducing the potential increase in frequency and magnitude of identified hazards. First steps include planning, review, and revisions of local development regulations
Hazard(s) Addressed	All Hazards
Vulnerability Analysis	A potential increase in the frequency and severity of identified hazards is possible with climate change.
Goal and Objective Met	1 and 3
Funding Sources	Agency fiscal budget
Benefit-Cost Analysis	No
Lead Agency	Rhode Island Division of Statewide Planning
Support Agency	-
Completion Timeframe	Short-Term (1-2 years)
Current Status	Not completed, modified, carried over

Table 232: Mitigation Action Number 25

Action Title	Reduction in Fuel Sources
Priority	High
Description	Provide technical assistance and resources to local jurisdictions to conduct the reduction of fuel sources and vegetation management and clearance mitigation actions for wildfire to mitigate against future occurrences and reduce jurisdictional vulnerability.
Hazard(s) Addressed	Wildfire, Drought
Vulnerability Analysis	Future climate conditions, including an increase in drought and hot temperatures, may lead to an increase in wildfires.
Goal and Objective Met	1, 2, and 3
Funding Sources	RIDEM budget
Benefit-Cost Analysis	No
Lead Agency	RIDEM
Support Agency	-
Completion Timeframe	Medium-Term (6-24 months)
Current Status	Not completed, modified, carried over

Table 233: Mitigation Action Number 26

Action Title	Increased CRS Participation
Priority	High
Description	Educate and provide technical assistance to local jurisdictions on the benefit and application process for joining the CRS. Currently 11 out of 39 local jurisdictions are participants. CRS participation allows local jurisdictions to offer flood insurance discounts, potentially increasing uptake.
Hazard(s) Addressed	Flood
Vulnerability Analysis	A consistent year-over-year drop in both flood insurance policy and coverage has been noted throughout the State of Rhode Island.
Goal	2
Funding Sources	RIEMA budget
Benefit-Cost Analysis	No
Lead Agency	RIEMA

Table 233: Mitigation Action Number 26

Action Title	Increased CRS Participation
Support Agency	Local Emergency Management Agencies. Insurance Providers
Completion Timeframe	Three years
Current Status	Not completed, modified, carried over

Table 234: Mitigation Action Number 27

Action Title	Narragansett and Wickford Dock Mitigation
Priority	High
Description	RIEMA has applied for a \$17,000,000 grant (under COVID-19 disaster declaration) to raise the docks in both Narragansett and Wickford to an operational height of 5.06 above Mean High Water. This federally funded project will be constructed to provide the ability to further raise the docks as sea level continues to rise and will provide long-term protection to these important critical assets to continue to offer economic stability and growth.
Hazard(s) Addressed	Flood, Sea Level Rise, Severe Thunderstorms, Tropical and Extratropical Storms
Vulnerability Analysis	The current dock system is at risk to sea level rise and wave action from storms.
Goal	5
Funding Sources	COVID-19 disaster declaration
Benefit-Cost Analysis	Yes
Lead Agency	RIEMA
Support Agency	RIDEM
Completion Timeframe	Five years
Current Status	Grant applied for

Table 235: Mitigation Action Number 28

Action Title	Construction of Public Health Laboratory
Priority	High
Description	A new Public Health Laboratory is currently under construction in Providence. The building will also be home to a new Center for Biological Sciences that will function as a dedicated Genomics Sequencing Core Laboratory. The laboratory will enable the Rhode Island State Health Laboratory to continue providing long-term monitoring of infectious disease allowing for early identification and mitigation against transmissible disease.
Hazard(s) Addressed	Transmissible Disease
Vulnerability Analysis	The current public health infrastructure for testing is inadequate to meet the needs of the State of Rhode Island.
Goal	5
Funding Sources	Centers for Disease Control and Prevention
Benefit-Cost Analysis	Yes
Lead Agency	RIDOH
Support Agency	-
Completion Timeframe	Two years
Current Status	New, construction underway

Table 236: Mitigation Action Number 29

Action Title	State Road Resilience Improvement Plan
Priority	High
Description	Complete a Resilience Improvement Plan for all state roads to allow for the determination of long-term projects to reduce or eliminate risk to identified hazards.
Hazard(s) Addressed	All Hazards

Table 236: Mitigation Action Number 29

Action Title	State Road Resilience Improvement Plan
Vulnerability Analysis	Data concerning the vulnerability of State roads in hazard areas currently does not exist.
Goal	1
Funding Sources	RIDOT budget
Benefit-Cost Analysis	Yes
Lead Agency	RIDOT
Support Agency	RIEMA
Completion Timeframe	Two years
Current Status	New, underway

Table 237: Mitigation Action Number 30

Action Title	State Asset High Hazard Dam Vulnerability
Priority	High
Description	Conduct a GIS driven analysis of all state assets in identified high hazard dam inundation areas for the determination of long-term projects to reduce or eliminate the vulnerability of these identified assets.
Hazard(s) Addressed	Dam Failure, Flood
Vulnerability Analysis	Data concerning the vulnerability of State assets in high hazard dam inundation areas currently does not exist.
Goal	1
Funding Sources	RIDEM budget
Benefit-Cost Analysis	Yes
Lead Agency	RIDEM
Support Agency	RIEMA
Completion Timeframe	Two years
Current Status	New, underway

Table 238: Mitigation Action Number 31

Action Title	Mitigation Plan Standardization
Priority	High
Description	Complete and provide a standardized hazard assessment methodology for use in local plans to allow for a state level determination of most vulnerable assets and facilities to allow for the determination of long-term projects and the award of grant funding to reduce or eliminate risk to identified hazards.
Hazard(s) Addressed	All Hazards
Vulnerability Analysis	Data concerning hazard vulnerability varies amongst local level hazard mitigation plans making the synthesis and analysis of data a challenge.
Goal	1, 2 and 3
Funding Sources	RIEMA budget
Benefit-Cost Analysis	Yes
Lead Agency	RIEMA
Support Agency	Local Emergency Management Departments
Completion Timeframe	Three years
Current Status	New

Table 239: Mitigation Action Number 32

Action Title	Flood Insurance Outreach
Priority	High
Description	Collaborate with local jurisdictions to notify public of availability of flood insurance through both the NFIP and private providers through public outreach efforts.
Hazard(s) Addressed	Flood
Vulnerability Analysis	A consistent year over year drop in both flood insurance policy and coverage has been noted throughout the State of Rhode Island
Goal	4
Funding Sources	RIEMA budget
Benefit-Cost Analysis	No
Lead Agency	RIEMA
Support Agency	Local Emergency Management Agencies. Insurance Providers
Completion Timeframe	Three years
Current Status	New

Table 240: Mitigation Action Number 33

Action Title	Commodity Flow Study
Priority	High
Description	Conduct a commodity flow study to determine the nature and amount of all hazardous materials transported within and through the State to help identify high vulnerability areas and long-term projects to reduce vulnerability.
Hazard(s) Addressed	Hazardous Materials
Vulnerability Analysis	The transportation of hazardous materials through and within the state should be better understood to determine the full extent of hazard.
Goal	1
Funding Sources	Regional Catastrophic Preparedness Grant
Benefit-Cost Analysis	No
Lead Agency	RIEMA, RIDEM, RIDOT
Support Agency	RIDOH
Completion Timeframe	Three years
Current Status	New

In an effort to ensure that all identified hazards were addressed with mitigation actions, the following table details the action number and the hazard that action addresses.

Table 241: Mitigation Action Hazards Addressed

Action Number	Dam Failure	Drought	Earthquake	Extreme Temperatures	Flood	Sea Level Rise	Severe Thunderstorms	Severe Winter Weather	Tornado	Tropical/Extratropical Storms	Wildfire	Biological Incident	Civil Disorder	Cybersecurity Incident	Hazardous Materials Incident	Infrastructure Failure	Radiological Incident	Terrorism	Transmissible Disease
1	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
2		x		x							x								
3					x	x										x			
4					x	x										x			
5					x	x													
6	x		x	x	x	x	x	x	x	x				x					

Table 241: Mitigation Action Hazards Addressed

Action Number	Dam Failure	Drought	Earthquake	Extreme Temperatures	Flood	Sea Level Rise	Severe Thunderstorms	Severe Winter Weather	Tornado	Tropical/Extratropical Storms	Wildfire	Biological Incident	Civil Disorder	Cybersecurity Incident	Hazardous Materials Incident	Infrastructure Failure	Radiological Incident	Terrorism	Transmissible Disease
7					x	x		x		x						x			
8					x	x	x	x	x	x									
9				x															
10	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
11	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
12					x	x	x	x	x	x						x			
13					x	x				x									
14	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
15	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
16	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
17	x				x											x			
18	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
19	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
20					x	x				x									
21	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
22				x	x					x									
23					x			x		x						x			
24	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
25		x									x								
26					x														
27					x	x	x			x									
28																			x
29	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	
30	x				x														
31	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
32					x														
33															x				

7.10 Monitoring Mitigation Projects

The RIEMA Planning Branch is responsible for monitoring the progress of mitigation activities and projects throughout the state. RIEMA monitors the progress all hazard mitigation projects through quarterly reports, site visits, direct communications and questionnaires, the FEMA Grant Outcomes (FEMA GO) platform, mitigation project tracking database, and the State Grants Administrative Plan. Full details concerning monitoring procedures may be found in the 2021 RIEMA Sub-Recipient Grant Management Handbook.

Upon notification from FEMA that a HMA project has been approved for funding, the SHMO will notify the subrecipient and will arrange a meeting to provide the subrecipient with appropriate information on regulatory program requirements. As the recipient, RIEMA recognizes its role for project management, accountability of funds, and for ensuring that subrecipients meet all program and administrative requirements in accordance with 2 CFR Part 200. Approved applicants are considered subrecipients and are accountable to the recipient for funds awarded them.

RIEMA is committed to monitoring and providing technical assistance to all eligible and funded subrecipients. The SHMO, project manager, and/or technical support staff attend subgrantee meetings to ensure the policies and procedures

are explained correctly. Additionally, all subrecipients are required to review FEMA’s 2023 Hazard Mitigation Assistance Program and Policy Guide which simplifies and clarifies guidance and incorporates strategic mitigation priorities.

The following summarizes the various activities that RIEMA uses to monitor and evaluate the completion of mitigation projects, including non-federally funded projects:

Table 242: RIEMA Project Monitoring Activities

Monitoring Activity	Monitoring Actions	Timeframe
Site Visits	Ensure that the contracted work has been completed	Anytime during lifetime of grant
Questionnaire	Determine and document progress and gather information to evaluate successes/area of needed improvement	As required
Quarterly Reports	Each recipient of a mitigation grant must file quarterly reports with RIEMA	Quarterly
FEMA GO or RIEMA Tracking System	Update as appropriate.	Immediate upon plan approval
State Grant Administration Plan	Review current mitigation grant and project guidelines and make updates as appropriate	After every disaster declaration or every three years

Differing from the 2019 SHMP, RIEMA now monitors and tracks the progress of state and federally funded mitigation projects using the FEMA GO platform. RIEMA uses the FEMA GO platform to promote uniformity across all core grant management processes. The platform uses automated data collection tools and on-line forms which allow application submission and required reporting to be completed faster and more efficiently. Additionally, FEMA GO provides users with a system to house quarterly progress reporting data and programmatic information. By utilizing the FEMA GO platform, RIEMA achieves the following key objectives:

- Simplify grants management lifecycle processes
- Improve timeliness of funding to support survivors and facilitate community resilience
- Improve allocation of grant funds across the emergency management and homeland security communities
- Enhance FEMA’s ability to achieve equitable outcomes for our grant recipient communities
- Streamline and improve business performance through business process and decision-making improvements
- Improve business intelligence and decision-making by increasing access to data related to grant program performance and grant project status
- Facilitate compliance with regulations and statutes
- Reduce sustainment costs by consolidating legacy systems into a single Information Technology platform
- Execute a user-centric, business-driven approach to grants transformation founded on active engagement with all grant stakeholders

7.11 Conditions and Reporting Requirements Guidelines for Non-Disaster Grants

As a condition of receiving these types of grants from RIEMA, the recipient/sub-recipient must agree to comply with the most recent version of the Administrative Requirements, Cost Principles and Audit Requirements, as set forth in the current edition of the Office of Justice Programs Financial Guide. A summary of select regulations commonly applicable to these grants include:

Administrative Requirements

- 2 CFR §200, Uniform Administrative Requirements for Grants and Cooperative Agreements to State and Local Governments.
- 2 CFR §200, Uniform Administrative Requirements for Grants and Agreements with Institutions of Higher Education, Hospitals and Other Non-Profit Organizations.

Cost Principles

- 2 CFR §200 Subpart E, Cost Principles for State, Local and Indian tribal Governments. • Special Considerations (2 CFR §200.416-.417)
- 2 CFR §200 Subpart E, Cost Principles for Education Institutions. • Special Considerations (2 CFR §200.418-.419)
- 2 CFR §200 Subpart E, Cost Principles for Non- Profit Organizations. • Exemptions found in Appendix VIII to Part 200.
- Federal Acquisition Regulations (FAR), Part 31.2 Contract Cost Principles and Procedures, Contracts with Commercial Organizations.
- Financial and administrative requirements are set forth in the current edition of the Office of Justice Programs Financial Guide.

Audit Requirements

- Recipient/sub-recipient agrees to comply with the organizational audit requirements of Office of Management and Budget (OMB) Circular A-133, Audits of States, Local Governments and Non-Profit Organizations as further described in the current edition of the OJP Financial Guide, Chapter 3.19.
- Recipient/sub-recipient agrees to comply with all applicable Rhode Island State Purchases Laws pursuant to Rhode Island General Law Title 37 Chapter 2 et seq.; purchasing rules and regulations.
- Recipient/sub-recipient agrees to use federal funds granted under this award to supplement but not supplant state or local funds for homeland security preparedness.
- Recipient/sub-recipient understands and agrees that use of any federal funds, either directly or indirectly, in support of the enactment, repeal, modification or adoption of any law, regulation or policy, at any level of government, without express prior written approval of the Office of Domestic Preparedness is strictly prohibited.
- Recipient/sub-recipient is prohibited from transferring funds between programs (i.e., State Homeland Security Program, Law Enforcement Terrorism Prevention Program and Citizens Corps Program etc.).
- Recipients/sub-recipients receiving \$750,000 or more in federal funds during their fiscal year (FY), must have an audit conducted and submitted to RIEMA.

Reporting Requirements and Financial Guides

- Sub-recipients agree to meet reasonable fiscal and administrative requirements to account for its federal grant funds in accordance with OMB Circular A-102. Additionally, the Office of the Governor or RIEMA may require the submission of quarterly financial reports, quarterly progress reports and final financial reports. Quarterly reports must be signed by an Authorized Official and submitted to RIEMA within fifteen days after the close of each calendar quarter.
- The sub-recipient further agrees to establish fiscal control and fund accounting procedures which meet the minimum requirements of these guidelines to ensure proper disbursement of, and accounting for, grant funds. Such accounting procedures must provide for an accurate and timely recording of receipt of funds by source, of expenditures made from such funds, and of unexpended balances. Sub-recipients that are not a state or quasi-governmental entity are required to maintain a separate bank account for each grant.
- RIEMA recognizes that the costs associated with equipment lists that were submitted as part of the grant applications may only have been estimates. Therefore, should a cost savings be realized during an equipment acquisition process, the sub-recipient may return excess funds to RIEMA or petition RIEMA to change the quantity of a particular item or request that it be allowed to purchase an additional item. However, requests to purchase items not on the approved award must be submitted to RIEMA and ODP for final approval. 5.

Project Monitoring/Evaluation

- Sub-recipient agrees to fully cooperate with periodic programmatic, fiscal monitoring, records review and site visits conducted by RIEMA. The sub-recipient agrees to follow any other special conditions specified in the REC (Record of Environmental Consideration) and/or by the State and FEMA. Sub-recipient agrees to submit timely and accurate Program Evaluation Reports to RIEMA and to participate in RIEMA sponsored surveys

and all other required reports related to any RIEMA-administered grant program. RIEMA reserves the right to deny payment to any approved programs, for failure to comply with this provision.

Maintenance of Records

- All grant documents including but not limited to invoices, purchase orders, packing slips, equipment make, model and serial numbers, must be maintained for a minimum of three (3) years after RIEMA closeout date.

7.12 Mitigation Closeouts

Upon completion of a project, a member of the RIEMA Planning Branch will conduct a closeout site visit to:

- Review all files and all documents related to grant and state general revenue funds
- Review all procurement files and contracts to third parties
- Take photos of the completed project

Closeout packages will generally be submitted 90 days after a project has been completed, and will include the following:

- Summary of Documentation
- Pictures of completed project
- Materials, labor, and equipment forms, if required
- HMA Grant Close-Out Certification

7.13 Hazard Mitigation Funding

It is generally recognized that mitigation actions help realize long term savings by preventing future losses due to hazard events. However, many mitigation actions are beyond the budgetary capabilities of a single jurisdiction. This section provides a general description of some of the avenues available to defray the cost of implementing mitigation actions.

State Funding Programs

The Rhode Island Investment Bank's Municipal Resilience Program provides direct support to cities and towns to complete a municipal-driven workshop process that brings together climate change information and local knowledge to identify top hazards, current challenges, and community strengths. This process identifies priority projects and strategies to improve the municipality's resilience to all natural and climate-related hazards. Upon successful completion of a Municipal Resilience Program workshop, municipalities are designated as Resilient Rhody Municipalities and can apply for Municipal Resilience Program Action Grants for projects identified through the workshop process. The projects must improve climate resilience and result in design, engineering, and/or construction actions. Types of projects that can receive grant funding include:

- Dam retrofits or removal
- Road elevation
- Floodproofing or elevation of pump stations, berms and levies
- Culvert resizing
- Green stormwater infrastructure
- Solar and battery back-up power
- Energy efficiency
- Watershed restoration
- Urban tree planting
- Coastal and riparian resilience

Municipalities will be required to match grants with a 25% local project cost share. The Grants are not available for research-related activities.

The RI READY program is a statewide industrial site readiness initiative designed to prepare sites to achieve pre-permitted status. This program provides access to an array of services designed to create an inventory of pre-permitted properties ready for industrial development throughout the State. State investment is recaptured to the extent possible to make the initiative self-sustaining and extend the life of available funds. The RI READY program leverages the Quonset Development Corporation, a quasi-state agency which partners with both Rhode Island Commerce and municipalities and/or private parties to spur development activity throughout the state. Site-readiness awards are made in two categories, site-specific improvements, and municipal assistance. Site-specific improvements include site planning and pre-development activities and project improvements. Projects intended to catalyze growth in advanced industry clusters, regardless of project size, are prioritized. Municipal Assistance awards are for developing projects that will lead to further growth and support the municipality's development goals.

The Department of Environmental Management administers the Climate Resilience Fund to implement climate resilience projects for climate-driven challenges facing Rhode Island communities, both inland and coastal. Grants provide funding for projects that support restoring and improving the climate resilience of vulnerable coastal habitats, river and stream floodplains, and related habitat and are available to both local governments and non-profits. Grant recipients will be required to match grants with a 25% local project cost share.

The Department of Environmental Management Wastewater Treatment Facility Resilience Fund provides funding to municipalities and/or quasi-state entities to protect government-owned wastewater collection systems and treatment facilities against the effects of climate change and flooding. Wastewater Treatment Facility Resilience Fund projects harden, relocate, repair or replace, and/or provide redundancy infrastructure at these wastewater treatment facilities. Approximately \$5,000,000 has been awarded.

The Department of Environmental Management Each Division of Forest Environment manages the Volunteer Fire Assistance Grant Program. The program is funded through the US Department of Agriculture, Forest Service. The objective of the program is to improve the capacity and capability of fire departments in rural areas and rural communities (a population of 10,000 or less) to prevent and suppress wildfires. Applicants can request from \$1,000 - \$2,500 in grant funds, with a 50/50 match requirement.

The Office of Housing and Community Development administers the State Community Development Block Grant program, intended to develop viable communities by providing decent housing, expanding economic opportunities, and creating suitable living environments. The Office of Housing and Community Development manages the program for 33 Rhode Island municipalities, with the cities of Cranston, East Providence, Pawtucket, Providence, Warwick, and Woonsocket operating their own programs. Grant funding supports housing, economic development, facilities/improvements, services and planning projects, with a priority placed on housing, economic development, and neighborhood revitalization. By regulation, the State may only distribute funds to units of general local government. These communities may, however, distribute funds to non-profit and other entities to undertake eligible activities.

Rhode Island Historical Preservation and Heritage Commission manages the State Historic Preservation Grant Program. Projects under this program fund capital preservation for renovation projects at public historic sites, museums, and cultural art centers located in historic structures in the State of Rhode Island. The State Preservation Grants Program is a matching grant program that promotes careful planning for the restoration, rehabilitation, and preservation of a variety of historic resources around the state. Through its matching requirements, the program stimulates broader support and participation in historic preservation projects statewide.

Federal Programs

FEMA provides financial assistance to state, local, tribal, and territorial governments, as well as certain private non-profit organizations, to implement projects that help reduce the risk and impact of future disasters. These grant programs are designed to support initiatives aimed at mitigating hazards and improving resilience. The main grant program offered by FEMA for hazard mitigation is the Hazard Mitigation Assistance (HMA) program. The HMA program includes four subprograms, the Hazard Mitigation Grant Program (HMGP), the HMGP Post-Fire, Building Resilient Infrastructure and Communities (BRIC), and the Flood Mitigation Assistance (FMA) grant program. Applicants to these grant programs are required to submit project proposals that demonstrate the effectiveness of their proposed mitigation projects. The eligibility criteria, application process, and specific requirements for each program are outlined by FEMA in their guidelines and announcements, which are typically published on FEMA's website.

The following provides a general overview of major grant funding streams:

- **HMGP and HMGP Fire:** The HMGP grants assist in implementing long-term hazard mitigation measures following Presidential disaster declarations, including fire declarations. Funding is available to implement projects in accordance with State, Tribal, and local priorities.
- **BRIC:** BRIC supports states, local communities, tribes, and territories as they undertake hazard mitigation projects, reducing the risks they face from disasters and natural hazards. The BRIC program guiding principles are supporting communities through capability- and capacity-building; encouraging and enabling innovation; promoting partnerships; enabling large projects; maintaining flexibility; and providing consistency. Working in coordination with BRIC, the National Mitigation Investment Strategy is intended to provide a national, whole-community approach to investments in mitigation activities and risk management.
- **FMA Grant Program:** FMA is a competitive grant program that provides funding to states, local communities, federally recognized tribes, and territories. Funds can be used for projects that reduce or eliminate the risk of repetitive flood damage to buildings insured by the NFIP. FEMA chooses recipients based on the applicant's ranking of the project and the eligibility and cost-effectiveness of the project. FEMA requires state, local, tribal, and territorial governments to develop and adopt hazard mitigation plans as a condition for receiving certain types of non-emergency disaster assistance, including funding for hazard mitigation assistance projects.

The following figure summarizes HMA grants programs:

Figure 7: HMA Grant Program Summary

HMA Program Comparison	 HMGP	 HMGP Post Fire	 BRIC	 FMA
Program Type	Post-disaster	Post-disaster	Pre-disaster	Pre-disaster
Funding Availability	Presidentially declared disaster	FMAG-declared disaster	6% set aside from federal post-disaster grant funding	Annual appropriations
Competitive?	No	No	Yes	Yes
Eligible Applicants	States, federally recognized tribes, territories and the District of Columbia (DC)	States, federally recognized tribes, territories and DC	States, federally recognized tribes, territories and DC	States, federally recognized tribes, territories and DC
Eligible Subapplicants	State agencies, local governments, tribes and private nonprofit organizations	State agencies, local governments, tribes and private nonprofit organizations	State agencies, local governments and tribes	State agencies, local governments and tribes
Hazard Mitigation Plan Requirement	Yes	Yes	Yes	Yes
NFIP Participation	Communities with projects in Special Flood Hazard Areas (SFHAs)	Communities with projects in SFHAs	Communities with projects in SFHAs	Subapplicants and properties

Source: FEMA

Additionally, the following provide available grant funding avenues for hazard mitigation projects:

- **Rehabilitation Of High Hazard Potential Dam (HHPD) Grant Program:** HHPD awards provide technical, planning, design, and construction assistance in the form of grants for rehabilitation of eligible high hazard potential dams. A state or territory with an enacted dam safety program, the State Administrative Agency, or an equivalent state agency, is eligible for the grant.
- **Emergency Management Performance Grant:** Program provides state, local, tribal, and territorial emergency management agencies with the resources required for implementation of the National Preparedness System and works toward the National Preparedness Goal of a secure and resilient nation. Allowable costs support efforts to build and sustain core capabilities across the prevention, protection, mitigation, response and recovery mission areas.
- **State Homeland Security Program:** Program includes a suite of risk-based grants to assist state, local, tribal, and territorial efforts in preventing, protecting against, mitigating, responding to and recovering from acts of terrorism and other threats. This grant provides grantees with the resources required for implementation of the National Preparedness System and working toward the National Preparedness Goal of a secure and resilient nation.
- **Nonprofit Security Grant Program:** Program is one of three grant programs that support Department of Homeland Security/FEMA's focus on enhancing the ability of state, local, tribal, and territorial governments, as well as nonprofits, to prevent, protect against, prepare for, and respond to terrorist or other extremist attacks. These grant programs are part of a comprehensive set of measures authorized by Congress and implemented by DHS to help strengthen the nation's communities against potential terrorist or other extremist attacks. Among the five basic homeland security missions noted in the Department of Homeland Security Strategic Plan for Fiscal Years 2020-2024
- **Public Assistance Program:** The mission of FEMA's Public Assistance program is to provide assistance to State, Tribal and local governments, and certain types of Private Nonprofit organizations so that communities can quickly respond to and recover from major disasters or emergencies declared by the President. Through the Public Assistance program, FEMA provides supplemental Federal disaster grant assistance for debris removal, emergency protective measures, and the repair, replacement, or restoration of disaster-damaged, publicly owned facilities and the facilities of certain private non-profit organizations. The Public Assistance Program also encourages protection of these damaged facilities from future events by providing assistance for hazard mitigation measures during the recovery process. The Federal share of assistance is not less than 75% of the eligible cost for emergency measures and permanent restoration. The grantee determines how the non-Federal share (up to 25%) is split with the eligible applicants.
- **Individual Assistance Program:** After a disaster, the federal government determines if any county in the state meets the criteria for individual disaster assistance. The decision is based on damage related to the severity and magnitude of the event. When a county receives an Individual Assistance declaration from the President of the United States, anyone who lives in that county can apply for assistance.
- **Small Business Administration Disaster Loans:** The Small Business Administration provides low-interest disaster loans to homeowners, renters, businesses of all sizes, and most private nonprofit organizations. Small Business Administration disaster loans can be used to repair or replace the following items damaged or destroyed in a declared disaster: real estate, personal property, machinery and equipment, and inventory and business assets.
- **The Housing and Urban Development Agency:** Provides flexible grants to help cities, counties, and States recover from Presidentially declared disasters, especially in low-income areas, subject to availability of supplemental appropriations.
- **Community Development Block Grant Program:** This is a flexible program that provides communities with resources to address a wide range of unique community development needs. The program provides annual grants on a formula basis to general units of local government and States.
- **Individual and Households, Other Needs Assistance Program:** This program provides financial assistance to individuals or households who sustain damage or develop serious needs because of a natural or man-made disaster. The funding share is 75% federal funds and 25% state funds. The program provides grants for necessary expenses and serious needs that cannot be provided for by insurance, another federal program, or other source of assistance. The current maximum allowable amount for any one disaster to individuals or families is \$25,000. The program gives funds for disaster-related necessary expenses and serious needs, including personal property, transportation, medical and dental, funeral, essential tools, flood insurance, and moving and storage.

- **WUI Grants:** The 10-Year Comprehensive Strategy focuses on assisting people and communities in the WUI to moderate the threat of catastrophic fire through the four broad goals of improving prevention and suppression, reducing hazardous fuels, restoring fire-adapted ecosystems, and promoting community assistance. The WUI Grant may be used to apply for financial assistance towards hazardous fuels and educational projects within the four goals of: improved prevention, reduction of hazardous fuels, restoration of fire-adapted ecosystems and promotion of community assistance.
- **Bureau of Indian Affairs Aid to Tribal Governments:** This program provides funds to Indian Tribal governments to support general Tribal government operations, to maintain up-to-date Tribal enrollment, to conduct Tribal elections, and to develop appropriate Tribal policies, legislation, and regulations. Funds may be used in a variety of ways to strengthen the capabilities of Indian tribes in self-government, community planning, and maintenance of membership records.
- **Bureau of Indian Affairs Replacement and Repair of Indian Schools:** Providing safe, functional, code-compliant, economical, and energy efficient education facilities for American Indian students attending Bureau of Indian Affairs owned or funded primary and secondary schools or residing in Bureau owned or funded dormitories.
- **Bureau of Indian Affairs Wildland Fire Management:** Cooperative agreements for grants and reimbursable costs related to wildland fire management directly associated with programs contracted by tribes under the authority of the National Indian Forest Resources Management Act.

Small and impoverished communities that receive grants may receive a federal cost share of up to 90% of the total amount approved under the grant award. As defined in 44 CFR 201.2, a small and impoverished community is:

- A community of 3,000 or fewer individuals that is identified by the State as a rural community
- Is not a remote area within the corporate boundaries of a larger city
- Is economically disadvantaged, by having an average per capita annual income of residents not exceeding 80% of national, per capita income
- The local unemployment rate exceeds by one percentage point or more, the most recently reported, average yearly national unemployment rate
- Any other factors identified in the State Plan in which the community is located

7.14 Coronavirus Aid, Relief, and Economic Security Act

Due to the COVID-19 pandemic, and under the Coronavirus Aid, Relief, and Economic Security Act, Rhode Island received \$1,930,000,000 through the Coronavirus Relief Fund. These funds were used to pay for items that:

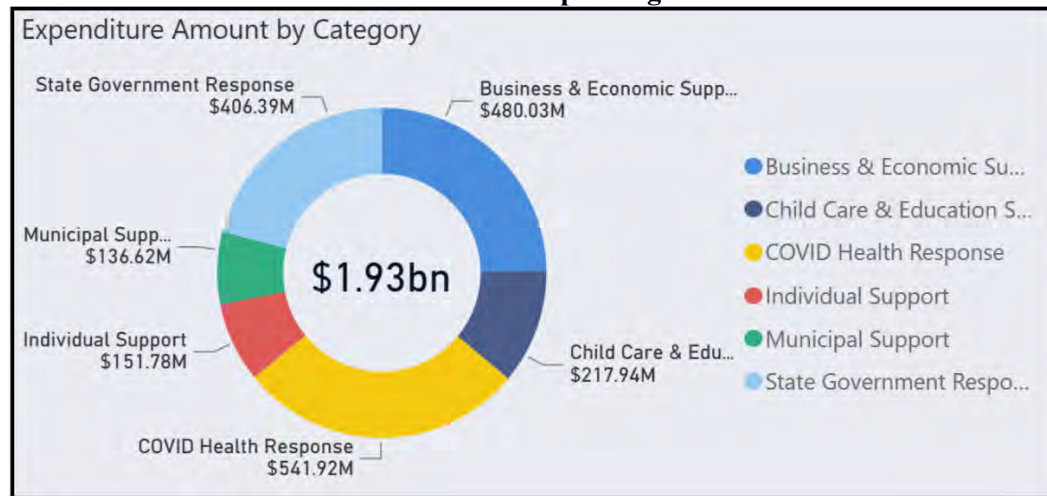
- Were necessary expenditures incurred due to the public health emergency with respect to COVID-19
- Were not accounted for in the budget most recently approved by the State before the public health emergency
- Were incurred from March 1, 2020, through December 31, 2021.

State expenditures and encumbrances were generally broken down into one of the following categories:

- **Business and Economic Support:** Costs to support businesses, workforce development, healthcare providers and hospitals.
- **Childcare and Education Support:** Costs to support childcare providers and assist school districts and institutions of higher education to adapt to the challenges posed by COVID-19.
- **COVID Health Response:** Costs associated with testing, contact tracing, personal protective equipment, the alternative hospital sites, community mitigation, quarantine and isolation, and other emergency procurements to protect the health of Rhode Islanders.
- **Individual Support:** Cash assistance, food support, and housing and rental assistance to help Rhode Islanders have access to essentials during the pandemic.
- **Municipal Support:** Support to municipalities for costs incurred responding to COVID-19.
- **State Government Response:** Costs related to state employees who have been critical in the pandemic response.

Available data from the Rhode Island COVID-19 Transparency Portal indicates that spending was allocated as follows:

Chart 43: COVID-19 Spending Allocation



Source: Rhode Island COVID-19 Transparency Portal

A full breakdown of expenditures cross referenced to the State of Rhode Island Department making the expenditure may be found at <http://www.transparency.ri.gov/COVID-19/payees.php>. Please note the data provided is a snapshot in time and reflects a preliminary assessment of spending that the State believes is eligible for federal reimbursement. As the State reviews this data and receives additional guidance from the federal government about COVID-19 spending, some expenditures may be recategorized.

Additionally, on August 5, 2021, FEMA announced every state, tribe, and territory that received a major disaster declaration in response to the COVID-19 pandemic will be eligible to receive 4% of those disaster costs to invest in mitigation projects that reduce risks from natural disasters.

7.15 Repetitive and Severe Repetitive Loss Mitigation Strategies

Mitigating the State's Repetitive Loss (RL) and Severe Repetitive Loss (SRL) properties remains a high priority for the State of Rhode Island. Mitigating these properties will require a multi-pronged approach and will require the combined efforts of agencies and organizations beyond the hazard mitigation program staff housed at RIEMA. Some identified remediation projects are coordinated and directed at the state level, while others require the support of the local governments and emergency management agencies. RIEMA will continue to seek assistance to implement this strategy through close cooperation with its public and private sector partners.

In acknowledgement of the importance of mitigating RL and SRL structures, the following mitigation actions will continue through the life of this plan:

- Annually perform updates of the PIVOT NFIP database in coordination with FEMA Region I.
- Provide RL and SRL datasets to local governments.
- Continue contact with local jurisdictions to promote grant programs to mitigate listed SRL and RL properties.
- Encourage jurisdictions to notify listed SRL property owners promoting mitigation at no cost through grant programs.
- Prioritize mitigation funding to RL and SRL properties for acquisition and demolition projects.

Additionally, RIEMA staff will continue to manage the tracking and management of RL and SRL properties. RIEMA will:

- Maintain access to PIVOT NFIP database of RL and SRL properties.

- Review potential acquisition projects to determine if new BCA analysis will all these structures to be eligible for HMA grants.
- Continue to complete NFIP Repetitive Loss Update Form (formerly AW-501). CRS communities use this form to submit corrections and updates to the list of repetitive flood loss properties within their jurisdictions. Ensure that all needed data is gathered during the full project close-out period.

Creating a competitive FEMA HMA grant application can be challenging for local officials. Local government-to-local government mentoring can be highly effective; RIEMA and FEMA will help to facilitate this process and match experienced grant participants with those that have not participated in HMA programs. In addition to mentoring local governments, the data analysis performed for the Risk Assessment in the 2019 SHMP update will be provided to localities to assist in targeting future mitigation opportunities. This will be done by:

- Providing State direction that each jurisdictional plan must include the targeting and mitigation of RL and SRL structures in the mitigation strategies section of every local §322 plan with RL and SRL properties.
- Examining the FEMA RL and SRL data sets to seek candidate properties that could potentially be mitigated through the FEMA HMA funding programs or any other available funding sources on an annual basis, or more frequently as required by disaster experience or available staffing resources. Include targeting of RL and SRL structures for mitigation in the mitigation strategies section of every §322 plan with RL and SRL properties.
- Developing and conducting education efforts that increase residential and business owners' knowledge and awareness of mitigation grants by conducting various outreach activities, especially to SRL property owners.
- After the distribution of a RL and SRL letter, RIEMA staff will meet with municipalities if requested for follow up.

RIEMA will use all available programs to fund projects that mitigation RL and SRL properties including:

- FEMA's HMA grant programs
- U.S. Department of Housing and Urban Development Community Development Block Grant – Disaster Recovery
- Other state, local, and privately funded efforts

Section 8 – Plan Maintenance

8.1 Introduction

The SHMP is a living document that will be updated and submitted to FEMA for approval every five years as required by 44 CRF 201.4. During the five-year cycle, the plan will undergo continuous monitoring and evaluation to ensure that the policies, procedures, priorities, and state environment established in the plan reflect current conditions. RIEMA has developed a method to ensure monitoring, evaluation, and updating of this SHMP. RIEMA will utilize the SIHMC to provide plan updates, revisions, and data collection for future HMP planning purposes.

8.2 Plan Update Responsibilities

RIEMA serves as the lead coordinating agency for emergency management in Rhode Island, including plan maintenance. Additional participants in the plan maintenance process include members of the SIHMC, subject matter experts, and representatives of local jurisdictions.

The SIHMC will facilitate the review and revision of the SHMP every five years. The review and revision will be an ongoing process. This process will incorporate all of the revisions made during the annual plan review, especially new data obtained from the local hazard mitigation plans. As these plans are completed, new information relative to hazards and threats will be incorporated into the formal five-year update.

Rhode Island will seek to expand participation in the mitigation planning process through expansion of the committee to represent municipal planners, other State agencies, and nongovernmental organizations. This expansion will focus on implementation of this plan update as well as to inform the next plan update.

8.3 Plan Review Meetings

The SIHMC will meet bi-annually for the first two years after plan approval. If needed, more meetings will take place during this timeframe. The RIEMA SHMO will determine the meeting dates and location and is responsible for sending invitations. RIEMA will be the main point of contact for these meetings and will maintain attendance and meeting minutes.

The purpose of these meetings is to discuss agency capability changes, the status of proposed projects, and any new studies or mapping that may inform the SHMP. Should a specific plan element or section require revision or amendment prior to the subsequent plan revision due to state or federal legislation or policy change, RIEMA staff will meet with all appropriate stakeholders and propose the change or addendum to FEMA as quickly as is practicable.

During these meetings, and in order to monitor SHMP progress, following information will be tracked by the SIHMC:

- How the actions from the mitigation strategy are being pursued and completed
 - Are actions being prioritized
- How the plan goals and objectives are being carried out
- How mitigation funding mechanisms are being utilized
- How local jurisdictions are receiving technical assistance

Additionally, the SIHMC will monitor the following elements to ensure the SHMP is current and correct:

- Reviewing the hazards and determining if any of them have changed
- Determining if there are new hazards that pose a risk to the state
- Ensuring goals and objectives are still relevant
- Determining if any actions have been completed or are deemed irrelevant
- Determining if new actions should be added
- Determining if capabilities have changed

In addition to these meetings, the Mitigation Branch at RIEMA will monitor and evaluate the progress of mitigation projects via quarterly reports, site visits, correspondence, and reimbursements. Completed projects will be evaluated for loss avoidance and alignment with local development plans.

8.4 Plan Update Schedule

At a minimum of two years before the expiration of the SHMP is the SIHMC will meet to begin preparation for the next SHMP update process. The SIHMC will review the procedures used for this plan update and recommend any changes to facilitate the update process. Additionally, a determination will be made concerning the use of contracting resources to assist with plan creation. The following indicates the tasks to be performed during this plan update period:

- **2026 Fall Meeting:** The SIHMC will begin updating the risk assessment portion of the plan. Hazards will be analyzed to determine if they are still relevant, if location should be updated, and if new hazards should be added. Previous occurrences will be reviewed to help determine the probability of future events.
- **2027 Spring Meeting:** The SIHMC will begin updating the vulnerability assessment. The SIHMC will update the vulnerability assessment portion of the plan. Data will need to be gathered for state assets, critical facilities, building stock values, jurisdictional damages, etc.
- **2027 Fall Meeting:** The SIHMC will review information received and determine if the goals and objectives are still relevant and if new ones should be added. Actions will be reviewed to determine if they should remain in the plan, have been completed, or are no longer relevant. The SIHMC will review the potential funding sources for each action.
- **2028 Spring Meeting:** The SIHMC will evaluate the policies, programs, capabilities, and funding sources from the previous plan to determine if they are still accurate and if any new items should be added. During this timeframe the RIEMA Mitigation Branch will begin reviewing approved local mitigation plans and incorporating their programs and capabilities into the plan. Additionally, RIEMA will compile a list of grant recipients over the five-year timeframe and incorporate it into the plan.
- **2028 Fall Meeting:** The SIHMC will review the draft copy of the mitigation plan and make comments and updates if necessary. Formal submittal to FEMA for re-approval will follow.

In general, the following steps will be taken to complete the next SHMP revision:

Table 243: State of Rhode Island SHMP Update Task List

Task	Action
1	Begin tracking communications associated with the plan update.
2	Review existing plan and completed crosswalks and identify needed updates.
3	Identify who will be responsible for updating the plan and the timeframe for the update.
4	Secure any necessary funding sources.
5	Develop a request for hazard mitigation project proposals.
6	Begin tracking significant plan changes.
7	Evaluate and update the planning process.
8	Review the stakeholder contact list, make necessary changes, and identify new stakeholders.
9	Initiate plan outreach and discussion, including a stakeholder meeting.
10	Consider the addition, removal, or modification of hazards identified in the plan.
11	Update and revise membership of the mitigation planning committees.
12	Evaluate risk assessment methodologies and data sources.
13	Evaluate and update state inventory information.
14	Evaluate and update the hazard profiles.
15	Evaluate and update the risk assessment summary.
16	Evaluate and update the mitigation strategy.
17	Evaluate and update the mitigation implementation system.
18	Evaluate and update the plan maintenance.
19	Develop the necessary annual mitigation reports.
20	Integrate new and updated local and tribal mitigation plans.
21	Integrate new and updated related state plans.
22	Evaluate and update other plans sections.
23	Identify and add any additional sections or information needed.
24	Review updated plan in its entirety.

Table 243: State of Rhode Island SHMP Update Task List

Task	Action
25	Conduct updated plan outreach, including public information, comment period, and meetings.
26	Integrate additional comments received.
27	Finalize plan document.
28	Complete crosswalk and submit final plan to FEMA for review and approval.
29	Make additional modifications as required.
30	Obtain signed letter from the Governor adopting the plan.

8.5 Post-Disaster Review

After each Presidential disaster declaration, and in coordination with FEMA, RIEMA will convene the SIHMC to document impacts on the State of Rhode Island and to determine if any mitigation actions should be considered to reduce future risk. This will allow for the development of hazard mitigation recommendations to FEMA during the disaster operation as well as to update the Mitigation Strategy as needed. The post-disaster review may coincide with established SIHMC or may be convened as separate events. In addition, following every declared disaster the SHMO may instigate an update of the HMGP Administrative Plan as required.

8.6 Continued Hazard Mitigation Plan Incorporation

The hazard mitigation plan is an overarching document that is both comprised of, and contributes to, various State, regional, and local plans. Integrated planning helps build resiliency and consistency with other local, state, regional, and national/federal mitigation programs. Additionally, the Threat Hazard Identification and Risk Assessment and Stakeholder Preparedness Review find their basis in natural hazards and risks described and detailed within the SHMP.

Each member of the SIHMC and each stakeholder agency has committed to actively encourage all employees and partners to use this plan when creating or updating plans, policies, procedures, and programs throughout the life of this plan.

8.7 Continued Public Involvement

The State of Rhode Island is dedicated to involving the public in the continual shaping of the SHMP and the development of its mitigation projects and activities.

RIEMA and the SIHMC will continue to keep the public informed about its hazard mitigation projects and activities through RIEMA's website. The public will also be invited to participate in regular SIHMC meetings to review and discuss the mitigation-related events.

A copy of the SHMP will be available online at RIEMA's website.

Appendix A – State of Rhode Island Adoption Documentation and FEMA Region I Approval Documentation



STATE OF RHODE ISLAND
OFFICE OF GOVERNOR DANIEL J. McKEE

February 8, 2024

Lori Ehrlich, Region 1 Administrator
FEMA Region 1
220 Binney Street
Cambridge MA 02142

Re: Rhode Island State Hazard Mitigation Plan Adoption

Dear Ms. Ehrlich:

Please find enclosed, for your review and formal approval, the updated Rhode Island State Hazard Mitigation Plan (SHMP). Through this letter, as Governor, I hereby adopt this plan on behalf of the State of Rhode Island. The updated plan is consistent with the mission of our state hazard mitigation program and has been prepared pursuant to the Disaster Mitigation Act (DMA) of 2000.

This update was led by the Rhode Island Emergency Management Agency and representatives from the State Interagency Hazard Mitigation Committee. The coordinated effort represents a commitment by numerous state agencies, public and private stakeholders and local governments involved in pre-disaster mitigation, emergency response efforts, and post-disaster recovery following a major natural disaster. The SHMP is a comprehensive document describing the assessed risks associated with natural disasters in Rhode Island and the existing mitigation measures our state has taken, and is planning to undertake in the future to minimize the damage and disruption caused by these natural disasters. The adoption of the plan authorizes the State to implement the strategies outlined in this update.

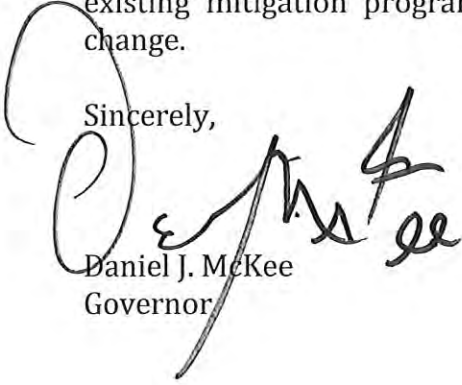
The SHMP is in full compliance with the planning requirements of the DMA 2000 and the Final Rule, Title 44 of the Code of Federal Regulations (CFR), parts 201 and 206, thus keeping the State of Rhode Island qualified to receive funding under all FEMA disaster assistance and hazard mitigation programs. Please accept our assurances that the State of Rhode Island will comply with all applicable Federal statutes and regulations in effect with respect to the periods in which it receives grant funding, in compliance with 44 CFR 13.11(c). Furthermore,

Lori Ehrlich
February 8, 2024
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the State of Rhode Island will amend its plan whenever necessary to reflect changes in State and Federal laws and statutes as required in 44 CFR 13.11(d).

In closing, the implementation of Rhode Island's SHMP will further complement our State's existing mitigation programs and encourage resiliency to natural hazards and climate change.

Sincerely,



Daniel J. McKee
Governor



FEMA

February 13, 2024

Daniel J. McKee
Office of the Governor
State of Rhode Island
82 Smith Street
Providence, RI 02903

Reference: Approval of the *2024 State of Rhode Island Hazard Mitigation Plan*

Governor McKee:

The U.S. Department of Homeland Security, Federal Emergency Management Agency (FEMA) Region 1 Mitigation Division has approved the *2024 State of Rhode Island Hazard Mitigation Plan* effective **February 13, 2024** through **February 12, 2029**. This plan is approved in accordance with applicable mitigation planning regulations and policy requirements.¹

In addition, this plan **has not met** the requirements of the High Hazard Potential Dams (HHPD) Grant Program to address all dam risks and the Fire Management Assistance Grant Program (FMAG) to address wildfire risks and mitigation measures.

A FEMA-approved State Mitigation Plan is a condition of receiving certain FEMA non-emergency assistance and mitigation grants from the following programs:

- Public Assistance Categories C-G (PA C-G)
- Pre-Disaster Mitigation Grant Program (PDM)
- Building Resilient Infrastructure and Communities (BRIC)
- Hazard Mitigation Grant Program (HMGP)
- Hazard Mitigation Grant Program – Post Fire (HMGP – Post Fire)
- Hazard Mitigation Grant Program – Post Fire (HMGP – Post Fire)
- Flood Mitigation Assistance (FMA)
- Safeguarding Tomorrow Revolving Loan Fund Program
- Fire Management Assistance Grants (FMAG)
- Rehabilitation of High Hazard Potential Dams Program (HHPD)

¹ Robert T. Stafford Disaster Relief and Emergency Assistance Act (Stafford Act), as amended; the National Flood Insurance Act of 1968, as amended; Title 44 Code of Federal Regulations (CFR) Part 201; and the “Water Infrastructure Improvements for the Nation Act,” or the “WIIN Act,” on December 16, 2016, which amends the National Dam Safety Program Act (Pub. L. 92-367).

Governor McKee

Page 2

Approval of a mitigation plan does not guarantee funding under any FEMA program. Please refer to the individual FEMA non-emergency assistance and mitigation grant program policy and/or annual Notice of Funding Opportunity for specific application and eligibility requirements for the FEMA programs listed above.

The State Mitigation Plan must be reviewed and revised to reflect changes in development, progress in statewide mitigation efforts, and changes in priorities and resubmitted for approval to the FEMA Region 1 Mitigation Division prior to the expiration date. Please allow sufficient time for FEMA's 45-day review, State adoption, formal approval, including any revisions that may be needed. FEMA will provide a reminder at least 12 months before the plan expiration date of the consequences of not having an approved mitigation plan, which is required to apply for and receive funding for FEMA non-emergency assistance and mitigation grant programs.

If at any time over the plan approval period, FEMA determines that the State is not complying with all applicable Federal statutes and regulations in effect during the periods for which it receives funding or is unable to fulfill mitigation commitments, FEMA may take action to correct the noncompliance (44 CFR §§ 201.3[b][5] and 201.4[c][7]). State and FEMA responsibilities are outlined in Section 2, Roles and Responsibilities of the *State Mitigation Planning Policy Guide*.

Thank you for your continued commitment to hazard mitigation planning in support of a resilient State of Rhode Island. We look forward to working with you to discuss the implementation of the State mitigation program each year over the approval period. If we can be of assistance, please contact Alexis Meehan, at (202) 394-6439 and alexis.meehan@fema.dhs.gov.

Sincerely,

Lori Ehrlich
Regional Administrator
DHS, FEMA Region 1

LE:am

Attachment: Final State Mitigation Plan Review Tool

cc: Marc R. Pappas, RIEMA, Director
Melinda Hopkins, RIEMA, Planning Branch Chief
Rae-Anne Culp, RIEMA, State Hazard Mitigation Officer
Dean Savramis, FEMA Region 1, Mitigation Division Director

Appendix B – FEMA State Plan Review Guide

State Mitigation Plan Review Tool

The State Mitigation Plan Review Tool (**Plan Review Tool**) demonstrates and documents how the state mitigation plan meets the regulations set forth in 44 CFR Part 201 and offers FEMA mitigation planners an opportunity to provide feedback to the state.

The **Regulation Checklist** must be completed by FEMA. The FEMA Plan Approver must reference the State Mitigation Planning Policy Guide when completing the Plan Review Tool. The purpose of the checklist is to identify the location of relevant or applicable content in the plan by element/sub-element and to determine if each requirement has been “Met” or “Not Met.”

The **Required Revisions** summary at the bottom of each element must clearly explain the revisions that are required for plan approval. Required revisions must be explained for each plan sub-element that is “Not Met.” Sub-elements should be referenced by the appropriate number, where applicable (e.g., S2-a, S2-b). Requirements for each element and sub-element are described in detail in Sections 3 and 4 of the State Mitigation Planning Policy Guide.

The **HHPD section and FMAG sub-elements** only need to be completed if the state is pursuing eligibility for those grant programs.

The **Plan Assessment** must be completed by FEMA. This assessment provides more comprehensive feedback to the state to acknowledge where the plan exceeds minimum requirements and provides suggestions for improvements. FEMA will describe the strengths that are demonstrated and highlight examples of best practices. FEMA’s suggestions for improvement are not required to be made for plan approval.

For greater clarification of the elements in the regulation checklist, please see Sections 3 and 4 in the State Mitigation Planning Policy Guide. This document defines terms and phrases used within this review tool.

1. Plan and Review Information

Plan Information	
State	Rhode Island
Title and Date of Plan	State of Rhode Island Hazard Mitigation Plan February 1, 2024
Plan Update Version	Update
State Point of Contact Name	Rae-Anne Culp
Title	RI State Hazard Mitigation Officer
Agency	Rhode Island Emergency Management Agency
Address	645 New London Avenue, Cranston, Rhode Island 02920
Phone Number	401-462-7048
Email	raeanne.culp@ema.ri.gov
Meets mitigation planning requirements for HHPD?	No
Meets mitigation planning requirements for FMAG?	No

Review Information	
Date Received by FEMA region	November 21, 2023
FEMA Reviewer (Planning – Name / Title)	Alexis Meehan, (R1 MIT, Risk Analysis Branch – Community Planner)
FEMA Reviewer (Planning – Name / Title)	Nan Johnson, (R1 MIT, Risk Analysis Branch – Senior Community Planner)
Plan Status (Not Approved, Approvable Pending Adoption, Approved)	Not Approved (January 3, 2024) – Requires Revisions
Review Information	
Date Received by FEMA region	January 31, 2024
FEMA Reviewer (Name / Title)	Alexis Meehan, (R1 MIT, Risk Analysis Branch – Community Planner)
FEMA Reviewer (Name / Title)	Nan Johnson, (R1 MIT, Risk Analysis Branch – Senior Community Planner)
Plan Status (Not Approved, Approvable Pending Adoption, Approved)	Approvable Pending Adoption (February 7, 2024)
Review Information	
Date Received by FEMA region	February 12, 2024
FEMA Reviewer (Name / Title)	Alexis Meehan, (R1 MIT, Risk Analysis Branch – Community Planner)
FEMA Reviewer (Name / Title)	Nan Johnson, (R1 MIT, Risk Analysis Branch – Senior Community Planner)
Plan Status (Not Approved, Approvable Pending Adoption, Approved)	Approved (February 13, 2024)

SUMMARY	YES	NO
STANDARD STATE MITIGATION PLAN		
Does the plan meet the standard state mitigation plan requirements?	☒	☐
ENHANCED STATE MITIGATION PLAN		
Does the plan meet the enhanced state mitigation plan requirements?	☐	☒

2. Standard State Mitigation Plan Regulation Checklist

PLANNING PROCESS

Requirements	Location in Plan (section and/or page number)	Met / Not Met
S1. Does the plan include a description of the process used to develop the plan? [44 CFR §§ 201.4(b) and 201.4(c)(1)]		
S1-a. Does the plan describe the current process used to update the plan, including how the plan was prepared, the schedule or time frame, specific milestones and activities, the agencies and stakeholders who were involved in the process, and if the mitigation planning process was integrated to the maximum extent possible with other state planning efforts?	Section 2.0, pg. 3-11	Met
S2. Does the plan describe how the state coordinated with other agencies and stakeholders? [44 CFR §§ 201.4(b) and 201.4(c)(1)]		
S2-a. Does the plan describe how the state coordinated with other state agencies, appropriate federal agencies, and other stakeholders, and how they were involved in the process?	Section 2.4, pg. 5-6; Section 2.5, pg. 7; Section 5.3, pg. 276	Met
Planning Process Required Revisions: Click or tap here to enter text.		

HAZARD IDENTIFICATION AND RISK ASSESSMENT

Requirements	Location in Plan (section and/or page number)	Met / Not Met
S3. Does the risk assessment include an overview of the type and location of all natural hazards that can affect the state? [44 CFR § 201.4(c)(2)(i)]		
S3-a. Does the plan include a current overview of all natural hazards that can affect the state, including the type, location and previous occurrences?	Section 4.0, pg. 48-274; Appendix D	Met
S4. Does the risk assessment provide an overview of the probabilities of future hazard events? [44 CFR § 201.4(c)(2)(i)]		
S4-a. Does the risk assessment provide an overview of the probability of future hazard events that includes projected changes in the location, range of anticipated intensities, frequency, and/or duration of each natural hazard?	Section 4.9 to 4.27, pg. 59-274	Met

Requirements	Location in Plan (section and/or page number)	Met / Not Met
S4-b. Does the probability include considerations of changing future conditions, including climate change (e.g., long-term weather patterns, average temperature, and sea levels) on the type, location and range of anticipated intensities of identified hazards?	Section 4.9 to 4.27, pg. 59-274	Met
S5. Does the risk assessment address the vulnerability of state assets located in hazard areas and estimate the potential dollar losses to these assets? [44 CFR §§ 201.4(c)(2)(ii) and 201.4(c)(2)(iii)]		
S5-a. Does the risk assessment include an overview and analysis of the vulnerability to state assets from the identified hazards as well as a summary of the most vulnerable assets?	Section 4.7, pg. 55; Section 4.9 to 4.27, pg. 59-274	Met
S5-b. Does the risk assessment estimate potential dollar losses to state assets located in identified hazard areas?	Section 4.7, pg. 55; Section 4.9 to 4.27, pg. 59-274	Met
S6. Does the risk assessment include an overview and analysis of jurisdictions' vulnerability to the identified hazards and the potential losses? [44 CFR §§ 201.4(c)(2)(ii) and 201.4(c)(2)(iii)]		
S6-a. Does the risk assessment provide an overview and analysis of vulnerable jurisdictions based on the state and local government risk assessments?	Section 3.3, pg. 22; Section 3.7, pg. 38; Section 4.3, pg. 49; Section 4.4, pg. 50-51; Section 4.9 to 4.27, pg. 53-274	Met
S6-b. Does the risk assessment include an overview and analysis of the potential losses to the identified vulnerable structures based on estimates in the local risk assessments as well as the state risk assessment?	Section 4.3, pg. 49; Section 4.9 to 4.27, pg. 53-274	Met
S7. Was the risk assessment revised to reflect changes in development? [44 CFR § 201.4(d)]		
S7-a. Does the plan provide a summary of recent development and potential or projected development in hazard-prone areas based on state and local government risk assessments?	Section 3.2, pg. 12–14; Section 3.3, pg. 14-22; Section 3.7, pg. 27-39	Met

Hazard Identification and Risk Assessment Required Revisions:

Click or tap here to enter text.

STATE MITIGATION CAPABILITIES

Requirements	Location in Plan (section and/or page number)	Met / Not Met
S8. Does the plan discuss the evaluation of the state's hazard management policies, programs, capabilities, and funding sources to mitigate the hazards identified in the risk assessment? [44 CFR § 201.4(c)(3)(ii)]		
S8-a. Does the plan include an evaluation of state laws, regulations, policies and programs related to hazards that improve or impede resilience to future natural hazard events and other future conditions, including the effects of climate change?	Section 3.11, pg. 44-47; Section 4.13.8-4.13.10, pg. 130-134; Section 5.3, pg. 277; Section 5.4, pg. 279-282; Section 5.7, pg. 293-300; Section 7.3, Goal #5 pg. 317; Section 7.9, Action #26 pg. 332-333 Section 7.15, pg. 344-345;	Met
S8-b. Does the plan include a general discussion of state funding capabilities for hazard mitigation actions and projects?	Section 3.7, pg. 28-38; Section 5.5, pg. 282; Section 7.13, pg. 339	Met
S8-c. Does the plan include a summary of obstacles, challenges and proposed solutions related to any state capabilities, including a brief discussion of potential strategies for overcoming any challenges related to implementing and enforcing hazard-resistant building codes statewide, as applicable, and changes since the previous plan approval?	Section 5.1, pg. 275; Section 5.3, pg. 278; Section 5.10, pg. 302; Section 5.12, pg. 306; Section 7.2, pg. 315	Met

State Mitigation Capabilities Required Revisions:

Click or tap here to enter text.

MITIGATION STRATEGY

Requirements	Location in Plan (section and/or page number)	Met / Not Met
S9. Does the mitigation strategy include goals to reduce long-term vulnerabilities from the identified hazards? [44 CFR § 201.4(c)(3)(i)]		
S9-a. Does the plan identify hazard mitigation goals representing what the state seeks to accomplish through mitigation plan implementation using a wide range of funding, including non-FEMA funding?	Section 7.3, pg. 316	Met
S9-b. Are the goals consistent with the hazards and vulnerabilities identified in the risk assessment?	Section 7.2, pg. 315; Section 7.3, pg. 316	Met
S10. Does the plan prioritize mitigation actions to reduce vulnerabilities identified in the risk assessment? [44 CFR §§ 201.4(c)(3)(i), 201.4(c)(3)(ii) and 201.4(c)(3)(iii)]		
S10-a. Does the plan identify actions based on the current risk assessment to reduce the vulnerability of jurisdictions within the state, as well as the vulnerability of state assets as described in Elements S5 and S6?	Section 7.8, pg. 321 and Section 7.9, pg. 323-336	Met
S10-b. Does the plan describe the process used by the state to evaluate and prioritize actions that are cost-effective, environmentally sound, and technically feasible?	Section 7.8, pg. 321 and Section 7.9, pg. 323-336	Met
S10-c. Does the plan describe how each action contributes to the hazard mitigation goals?	Section 7.9, pg. 323-336	Met
S10-d. Does the plan describe how local government mitigation strategies link to the state mitigation strategy?	Section 6.8, pg. 314 and Section 6.6, pg. 311	Met
S11. Does the plan identify current and potential sources of funding to implement mitigation actions and activities? [44 CFR § 201.4(c)(3)(iv)]		
S11-a. Do mitigation activities include the identification of current and/or potential sources of federal, state, local or private funding for implementation?	Section 7.9, pg. 323	Met

Mitigation Strategy Required Revisions:

Requirements	Location in Plan (section and/or page number)	Met / Not Met
S11-b. Does the plan identify FEMA mitigation funding sources (if applicable), including, but not limited to: HMGP, BRIC, FMA and PA Mitigation, at a minimum?	Section 7.13, pg. 339	Met
S12. Was the plan updated to reflect progress in statewide mitigation efforts and changes in priorities? [44 CFR § 201.4(d)]		
S12-a. Does the plan provide a narrative of the status of each mitigation action in the previous plan?	Section 7.7, pg. 320, Section 7.8, pg. 321 and Section 7.9, pg. 323	Met
S12-b. Was the prioritization of mitigation actions and activities updated based on the updated analysis of risks, capabilities and progress?	Section 7.5, pg. 318	Met

LOCAL PLANNING COORDINATION AND CAPABILITY BUILDING

Requirements	Location in Plan (section and/or page number)	Met / Not Met
S13. Does the plan generally describe and analyze the effectiveness of local government mitigation policies, programs, and capabilities? [44 CFR § 201.4(c)(3)(ii)]		
S13-a. Does the plan provide a summary of current local government policies, programs and capabilities of jurisdictions to accomplish hazard mitigation?	Section 5.11, pg. 302	Met
S13-b. Does the plan describe the effectiveness of local government mitigation policies, programs and capabilities?	Section 5.11, pg. 302	Met
S14. Does the plan describe the process to support the development of approvable local government mitigation plans? [44 CFR §§ 201.3(c)(5) and 201.4(c)(4)(i)]		
S14-a. Does the plan describe how the state supports developing or updating FEMA-approvable mitigation plans?	Section 6.1, pg. 308; Section 6.3, pg. 309	Met
S14-b. Does the plan provide a brief summary of barriers to developing or updating, adopting, and implementing FEMA-approved local government mitigation plans based on an analysis of plan and jurisdiction coverage data and trends across the state and steps to remove barriers to help local governments advance mitigation planning, including how plan and jurisdiction coverage data and trends inform those steps?	Section 6.2, pg. 308; Section 6.4, pg. 310	Met
S15. Does the plan describe the criteria for prioritizing funding? [44 CFR § 201.4(c)(4)(iii)]		
S15-a. Does the plan describe criteria for prioritizing jurisdictions to receive planning and project grants under available federal and non-federal programs?	Section 6.7, pg. 311	Met
S16. Does the plan describe the process and time frame to review, coordinate, and link local and tribal mitigation plans with the state mitigation plan? [44 CFR §§ 201.3(c)(6), 201.4(c)(2)(ii), 201.4(c)(3)(iii), and 201.4(c)(4)(ii)]		
S16-a. Does the plan describe the state's process and time frame to review and submit approvable local and tribal mitigation plans to FEMA?	Section 6.5, pg. 311	Met

Requirements	Location in Plan (section and/or page number)	Met / Not Met
S16-b. Does the plan describe the state's process and time frame to share risk assessment data and mitigation priorities with local governments for their plan updates, as well as integrate local risk assessment and mitigation actions into the state mitigation plan updates?	Section 6.8-6.9, pg. 314	Met
Local Planning Coordination and Capability Building Required Revisions: Click or tap here to enter text.		

REVIEW, EVALUATION, AND IMPLEMENTATION

Requirements	Location in Plan (section and/or page number)	Met / Not Met
S17. Is there a description of the method and schedule for keeping the plan current? [44 CFR §§ 201.4(c)(5)(i) and 201.4(d)]		
S17-a. Does the plan describe the agency/office responsible for monitoring, evaluating and updating the plan?	Section 7.3, pg. 316; Section 8.2, pg. 346; Section 8.3, pg. 346	Met
S17-b. Does the plan describe the schedule for monitoring, evaluating, and updating the plan?	Section 8.3, pg. 346; Section 8.4, pg. 347	Met
S18. Does the plan describe the systems for monitoring implementation and reviewing progress? [44 CFR §§ 201.4(c)(5)(ii) and 201.4(c)(5)(iii)]		
S18-a. Does the plan describe the system for tracking the implementation of the mitigation activities and projects identified in the mitigation strategy, including all mitigation activities and not just those funded by FEMA?	Section 7.10, pg. 336	Met
S18-b. Does the system include the schedule, the agency/office responsible for coordination, and the role of the agencies/offices identified in the mitigation strategy as responsible for implementation of actions?	Section 7.9, pg. 323	Met
S18-c. Does the plan describe a system for reviewing progress on achieving the mitigation strategy's goals that includes the criteria and process for evaluating progress?	Section 7.3, pg. 316; Section 8.3, pg. 346	Met

Review, Evaluation, and Implementation Required Revisions:

Click or tap here to enter text.

ADOPTION AND ASSURANCES

Requirements	Location in Plan (section and/or page number)	Met / Not Met
S19. Did the state provide documentation that the plan has been formally adopted? [44 CFR § 201.4(c)(6)]		
S19-a. Did the state provide documentation of formal adoption by the highest elected official or designee prior to FEMA approval?	Signed copy located in Appendix A	Met
S20. Did the state provide assurances? [44 CFR § 201.4(c)(7)]		
S20-a. Does the plan include assurances that the state will manage and administer FEMA funding in accordance with applicable federal statutes and regulations?	Section 1.2, pg. 1; Section 1.4, pg. 2; Appendix A	Met
S20-b. Does the plan include assurances that the state will update its plan whenever necessary to reflect changes in state or federal laws and statutes?	-Section 1.2, pg. 1; Section 1.4, pg. 2; Appendix A	Met
Adoption and Assurances Required Revisions:		
Click or tap here to enter text.		

HIGH HAZARD POTENTIAL DAMS

Requirements	Location in Plan (section and/or page number)	Met / Not Met
HHPD1. Did Element S2 (planning process) describe how the state dam safety agency, other agencies, and stakeholders participated in the planning process and contributed expertise, data, studies, information, etc. relative to high hazard potential dams?		
HHPD1-a. Does the plan describe how the state dam safety agency, other agencies, and stakeholders were involved in the planning process?	Section 2.4, pg. 7; Section 2.8, pg. 8	Met
HHPD1-b. Does the plan describe the types of data contributed?	Section 2.4, pg. 5; Section 2.8, pg. 8-9	Met

Requirements	Location in Plan (section and/or page number)	Met / Not Met
HHPD2. Did Element S6 (risk assessment) address all dam risk for high hazard potential dams in the risk assessment?		
HHPD2-a. Does the plan provide a list of high hazard potential dams that have been identified by the state with their names, National Inventory of Dams identification numbers, locations by jurisdiction, and other relevant information, as well as maps?	Section 4.9.2, pg. 64; Section 4.9.7, pg. 71; Appendix D,	Met
HHPD2-b. Does the plan summarize statewide vulnerabilities to/from high hazard potential dams from hazards and the potential consequences associated with dam incidents?	Section 4.9.5, pg. 62-63	Not Met
HHPD2-c. Does the plan document limitations and describe the approach to address deficiencies?	Section 4.9.5, pg. 64; Section 7.2, pg. 315	Met
HHPD3. Did Element S9 (mitigation goals) include mitigation goals to reduce long-term vulnerabilities from high hazard potential dams?		
HHPD3-a. Does the plan address a reduction in vulnerabilities to/from high hazard potential dams from hazards and the potential consequences associated with dam incidents as part of their own goals or with other long-term strategies?	Section 7.3, pg. 316 (Goal 5); Section 7.9, pg. 323	Met
HHPD3-b. Does the plan link the proposed actions to reduce long-term vulnerabilities consistent with the goals?	Section 7.4, pg. 317; Section 7.5, pg. 318; Section 7.9, pg. 323	Not Met
HHPD4. Did Element S10 (mitigation actions) prioritize mitigation actions and activities to reduce vulnerabilities from high hazard potential dams?		
HHPD4-a. Does the plan include actions to reduce vulnerabilities to/from high hazard potential dams?	Section 7.9, pg. 323	Not Met
HHPD4-b. Does the plan describe the process to evaluate and prioritize actions related to high hazard potential dams that are cost-effective, environmentally sound and technically feasible?	Section 7.5, pg. 318	Not Met
HHPD4-c. Does the plan describe how each action to reduce risks related to high hazard potential dams contributes to the goals and describe how strategies are linked to the state mitigation strategy?	Section 7.9, pg. 323	Not Met

Requirements	Location in Plan (section and/or page number)	Met / Not Met
HHPD5. Did Element S11 (funding sources) identify current and potential sources of funding to implement mitigation actions and activities for high hazard potential dams?		
HHPD5-a. Does the plan include various funding sources to mitigate vulnerabilities to and from high hazard potential dams from hazards and the potential consequences associated with dam incidents, as well as funding sources to rehabilitate or remove high hazard potential dams?	Section 7.13, pg. 339	Not Met
HHPD6. Did Element S13 (local coordination) generally describe and analyze the effectiveness of local mitigation policies, programs, and capabilities that address high hazard potential dams?		
HHPD6-a. Does the plan provide a summary of the local policies, programs, and capabilities to implement mitigation actions and reduce vulnerabilities from high hazard potential dams from hazards and the potential consequences associated with dam incidents?	Section 4.9, pg. 59; Section 5.3 pg. 275	Not Met
HHPD6-b. Does the plan describe challenges to implementing local mitigation policies, programs and capabilities to reduce vulnerabilities to and from high hazard potential dams and the approach to overcome these challenges?	Section 5.3, pg. 275; Section 7.2, pg. 315	Not Met
HHPD6-c. Does the plan describe opportunities for implementing mitigation actions to reduce risks to and from high hazard potential dams through local capabilities?	Section 5.3, pg. 275; Section 5.6, pg. 286	Not Met
HHPD7. Did Element S15 (prioritizing funding) describe the criteria for prioritizing funding for high hazard potential dams?		
HHPD7-a. Does the plan describe the method for funding actions to reduce vulnerabilities to and from high hazard potential dams if these actions were prioritized differently than mitigation actions for other hazards?	Section 7.13, pg. 339	Not Met
HHPD7-b. Does the plan document limitations and describe the approach to addressing deficiencies?	Section 4.9, page 59, and Section 7.2, page 315	Not Met

HHPD Required Revisions:

Note: The State HMP Update can receive formal FEMA approval without fulfilling all of the HHPD requirements at the time of the initial approval. If needed, the State can complete the required HHPD revisions after the plan has been formally approved by FEMA. FEMA would then review the plan again to verify that the HHPD requirements have been met.

HHPD2-b. “To meet this requirement with a specific focus on high hazard potential dams, the plan must: Summarize statewide vulnerabilities to/from high hazard potential dams from hazards and the potential consequences associated with dam incidents, including: (1) Potential cascading impacts of storms seismic events, landslides, wildfires, etc. on dams that might affect upstream and downstream flooding potential. (2) Potential significant economic, environmental, or social impacts, as well as multi-jurisdictional impacts from a dam incident. (3) Location and size of PARs from high hazard potential dams, as well as potential impacts to institutions and critical infrastructure/facilities/community lifelines. (4) Methodology and/or assumptions for risk data and inundation modeling should be noted.”

- Summary of vulnerabilities to/from high hazard potential dams is incomplete.
 - o The location and size of PARs from high hazard potential dams were not addressed.

HHPD3-b. “Link the proposed actions to reducing long-term vulnerabilities consistent with the goals.” To be determined, pending completion of HHPD4 (no mitigation actions for HHPD were provided).

HHPD4-a. No mitigation actions were provided that met this requirement. Provide the required actions to reduce vulnerabilities to/from high hazard potential dams as described by the State Mitigation Planning Policy Guide under requirement HHPD-a. Also, refer to S10-a.

HHPD4-b. Provide a statement that confirms the process for prioritizing mitigation actions (as described in Section 7.5, pg. 318-320) also applies to actions related to high hazard potential dams.

HHPD4-c. No mitigation actions related to high hazard potential dams were provided. To be determined, pending completion of HHPD3-b and HHPD4-a.

HHPD5. Rehabilitation of High Hazard Potential Dam (HHPD) Grant Program is the only funding source included. “To meet the requirement, the mitigation **plan must include various funding sources to mitigate vulnerabilities to and from high hazard potential dams from hazards and the potential consequences associated with dam incidents**, as well as funding sources to rehabilitate or remove high hazard potential dams. Funding may include sources other than FEMA for activities other than rehabilitation of high hazard potential dams.”

HHPD6-a-c. Requirements a through c were not addressed in Section 5.11 (pg. 302-306) regarding the local policies, programs, and capabilities to implement mitigation actions and reduce vulnerabilities from high hazard potential dams from hazards and the potential consequences associated with dam incidents.

HHPD7-a. State whether prioritization of funding for actions to reduce vulnerabilities to and from high hazard potential dams are different than how actions were prioritized for funding mitigation actions for other hazards. If different, describe the method for prioritizing this funding (See State Mitigation Planning Policy Guide HHPD7-a for an example).

HHPD7-b. Could not locate where this requirement was addressed in the plan.

FIRE MANAGEMENT ASSISTANCE GRANTS

Requirements	Location in Plan (section and/or page number)	Met / Not Met
FMAG1. Does the plan address wildfire risks? [44 CFR 201.4(c)(2); 44 CFR § 204.51(d)(2)]		
FMAG1-a. Does the risk assessment provide an overview of the location and previous occurrences of wildfire hazards in the state?	Section 4.19.2, pg. 205; Section 4.19.3, pg. 210; Section 4.19.4, pg. 212;	Met
FMAG1-b. Does the risk assessment provide an overview of the probability of future wildfire events that includes projected changes in the location, intensity, frequency and/or duration of wildfire hazards?	Section 4.19.2, pg. 205; Section 4.19.4, pg. 212; Section 4.19.5, pg. 212	Met
FMAG1-c. Does the risk assessment address the vulnerability of state assets located in wildfire hazard areas and estimate the potential dollar losses to those assets?	Section 4.19.6, pg. 215-217; Appendix D	Met
FMAG1-d. Does the risk assessment include an overview and analysis of local governments' vulnerability to wildfires and the potential losses to vulnerable structures?	Section 4.19.6, pg. 216-217; Section 4.19.7, pg. 218	Not Met
FMAG2. Does the plan's mitigation strategy contain wildfire-related mitigation initiatives? [44 CFR 201.4(c)(3); 44 CFR § 204.51(d)(2)]		
FMAG2-a. Does the mitigation strategy identify mitigation actions and activities to reduce the vulnerability of jurisdictions within the state as well as the vulnerability of state-owned assets as described in Elements S5 and S6?	Section 5.6, pg. 286; Section 7.8, pg. 321; Section 7.9, pg. 323	Not Met

FMAG Required Revisions:

Note: The State HMP Update can receive formal FEMA approval without fulfilling all of the FMAG requirements at the time of the initial approval. If needed, the State can complete the required FMAG revisions after the plan has been formally approved by FEMA. FEMA would then review the plan again to verify that the FMAG requirements have been met.

FMAG1-d. “The risk assessment must include an overview and analysis of local governments’ vulnerability of jurisdictions within the state as well as the vulnerability of state-owned assets as described in Elements S5 and S6.”

- The local community wildfire protection plans (CWPP) were not mentioned in this plan update, the plan must include an overview and analysis of local governments’ vulnerability to wildfires. Address whether the CWPPs are developed for local jurisdictions within Rhode Island with respect to local mitigation planning and hazard mitigation plans. Confirm if these were reviewed and incorporated in this update for the state’s analysis of the local governments’ vulnerability to wildfires.

FMAG2-a. “The mitigation strategy must identify mitigation actions and activities to reduce the vulnerability of jurisdictions within the state as well as the vulnerability of state-owned assets as described in Elements S5 and S6.”

- The action for wildfire hazards is not specific to the vulnerabilities of the states’ owned and operated assets identified in the plan’s updated risk assessment. Mitigation Action Number 26 does not identify actions and activities to reduce the vulnerability of state owned and operated assets. Also, the action provided does not directly reduce or eliminate the vulnerability to the identified vulnerable state assets (both existing and future assets) such as state structures, facilities, systems; and infrastructure and historic structures exposed to wildfire hazards.
- The statements are too broad and must be further developed to connect with the local and state vulnerabilities including more specific vulnerable assets.
- There is no clear indication of what the mitigation action(s) is/are.

3. Enhanced State Mitigation Plan Regulation Checklist

ENHANCED STATE PREREQUISITES: *NOT APPLICABLE, THIS IS A STANDARD STATE PLAN*

Requirements	Location in Plan (section and/or page number)	Met / Not Met
E1. Does the enhanced plan include all elements of the standard state mitigation plan? [44 CFR § 201.5(b)]		
E1-a. Does the enhanced plan meet all the required elements of the standard state mitigation plan?	Click or tap here to enter text.	Choose an item.
E2. Regarding HMA, is the state maintaining the capability to meet application time frames and submitting complete project applications? [44 CFR § 201.5(b)(2)(iii)(A)]		
E2-a. Are all applications complete and submitted by the end of each program's respective application period?	Click or tap here to enter text.	Choose an item.
E2-b. Are all applications entered into FEMA's electronic data systems (i.e., NEMIS, eGrants, and/or FEMA GO)?	Click or tap here to enter text.	Choose an item.
E2-c. Is a complete Minimum Criteria Checklist for Project Subapplicants or equivalent documentation prepared for all subapplications?	Click or tap here to enter text.	Choose an item.
E2-d. Are all applications determined to be complete by FEMA within 90 days of submittal or selection for further review, or after the first request for information response?	Click or tap here to enter text.	Choose an item.
E3. Regarding HMA, is the state maintaining the capability to prepare and submit accurate environmental reviews and benefit-cost analyses? [44 CFR § 201.5(b)(2)(iii)(B)]		
E3-a. Are all applications and amendments determined to be complete by FEMA within 90 days of submittal or selection for further review, or after the first request for information response, including all data requested by FEMA to support cost-effectiveness determinations and EHP compliance reviews?	Click or tap here to enter text.	Choose an item.
E4. Regarding HMA, is the state maintaining the capability to submit complete and accurate quarterly progress and financial reports on time? [44 CFR § 201.5(b)(2)(iii)(C)]		
E4-a. Are all progress reports complete and submitted on time?	Click or tap here to enter text.	Choose an item.
E4-b. Are all FFR SF-425s submitted on time?	Click or tap here to enter text.	Choose an item.

Requirements	Location in Plan (section and/or page number)	Met / Not Met
E4-c. Does the state consistently comply with the Financial Management Standard requirements described in 2 CFR §§ 200.300 to 200.309?	Click or tap here to enter text.	Choose an item.
E5. Regarding HMA, is the state maintaining the capability to complete HMA projects within established performance periods, including financial reconciliation? [44 CFR § 201.5(b)(2)(iii)(D)]		
E5-a. Is all work as part of HMA subawards completed by the end of the period of performance, as described in the HMA Guidance?	Click or tap here to enter text.	Choose an item.
E5-b. Have there been no major findings on the last single audit obtained by the state related to HMA programs?	Click or tap here to enter text.	Choose an item.
E5-c. Are all grant closeout activities, including financial reconciliation, completed within 120 days from the end of the performance period as outlined in 2 CFR 200.344?	Click or tap here to enter text.	Choose an item.
E5-d. Have actual expenditures been documented and are they consistent with SF-424A or SF-424C?	Click or tap here to enter text.	Choose an item.
Enhanced State Prerequisites Required Revisions: Click or tap here to enter text.		

INTEGRATED PLANNING

Requirements	Location in Plan (section and/or page number)	Met / Not Met
E6. Does the plan demonstrate integration, to the extent practicable, with other state and/or regional planning initiatives and FEMA mitigation programs and initiatives? [44 CFR § 201.5(b)(1)]		
E6-a. Does the enhanced plan demonstrate integration with other state and/or regional planning initiatives?	Click or tap here to enter text.	Choose an item.
E6-b. Does the enhanced plan demonstrate integration of FEMA mitigation programs and initiatives?	Click or tap here to enter text.	Choose an item.

Integrated Planning Required Revisions:

Click or tap here to enter text.

DEMONSTRATING COMMITMENT TO A COMPREHENSIVE STATE MITIGATION PROGRAM

Requirements	Location in Plan (section and/or page number)	Met / Not Met
E7. Does the state demonstrate commitment to a comprehensive mitigation program? [44 CFR §§ 201.3(c), 201.5(b)(4) and 201.6(d)]		
E7-a. Does the state demonstrate commitment to statewide programs, initiatives and plans that advance mitigation and resilience?	Click or tap here to enter text.	Choose an item.
E7-b. Does the state demonstrate a commitment to mitigation training and capability building?	Click or tap here to enter text.	Choose an item.
E7-c. Does the state demonstrate a commitment to its mitigation planning responsibilities by helping local governments update and adopt their plans before they expire?	Click or tap here to enter text.	Choose an item.

Demonstrating a Commitment to a Comprehensive State Mitigation Program Required Revisions:

Click or tap here to enter text.

EFFECTIVE USE OF EXISTING MITIGATION PROGRAMS TO ACHIEVE MITIGATION GOALS

Requirements	Location in Plan (section and/or page number)	Met / Not Met
E8. Is the state effectively using existing mitigation programs to achieve mitigation goals? [44 CFR §§ 201.5(a) and 201.5(b)(3)]		
E8-a. Does the state demonstrate and document the full and effective use of existing FEMA programs (if funding is available)?	Click or tap here to enter text.	Choose an item.
E8-b. Does the state demonstrate and document the full and effective use of non-FEMA programs?	Click or tap here to enter text.	Choose an item.

Effective Use of Existing Mitigation Programs to Achieve Mitigation Goals Required Revisions:

Click or tap here to enter text.

DOCUMENTATION OF THE STATE'S IMPLEMENTATION CAPABILITY

Requirements	Location in Plan (section and/or page number)	Met / Not Met
E9. Does the enhanced plan document capability to implement mitigation actions? [44 CFR §§ 201.5(b)(2)(i), 201.5(b)(2)(ii), and 201.5(b)(2)(iv)]		
E9-a. Does the enhanced plan describe the system to rank the mitigation measures according to established eligibility criteria, including a process to prioritize between funding programs, jurisdictions, and proposals that address different or multiple hazards?	Click or tap here to enter text.	Choose an item.
E9-b. Does the enhanced plan describe how the state will assess the effectiveness of mitigation actions, mitigation the agencies that are involved as well as the timeline, and use the results to inform the mitigation strategy?	Click or tap here to enter text.	Choose an item.
Documentation of the State's Implementation Capability Required Revisions: Click or tap here to enter text.		

4. Plan Assessment

The Plan Assessment comments can be used to help guide the ongoing maintenance and update of your mitigation plan.

Standard State Mitigation Plan Requirements

PLANNING PROCESS

Strengths

- The Rhode Island Hazard Mitigation Plan Update stakeholder survey for the public yielded over 100 community member responses and comments during the period of time that it was available through the online survey portal. Input from this survey was included in Appendix C of the plan and input provided the SIHMC with a clearer understanding of local concerns.
- All established local plans submitted to the State were used as a source to inform the 2024 Rhode Island SHMP Update and were utilized for information on identified hazards, vulnerability and risk assessment methodologies, jurisdictional capabilities, mitigation goals, and identified mitigation actions (both proposed and completed).
- The plan incorporated information from a variety of sources.
- It was great to see PPL Rhode Island Energy participate in the planning process representing the energy sector.

Opportunities for Improvement

- The plan would benefit from a better understanding of the coordination between the state agencies and appropriate federal agencies. Document and describe the decision-making, discussions, etc. in the analysis that led to the findings of the outcomes of each of the plan elements. How were things worked through to arrive at analysis and plan development? Additionally, describe coordination with appropriate federal agencies that were involved in the process rather than listing them as technical resources.
- The State is encouraged to revisit and expand the lack of information on the federally recognized Tribal Nation, the Narragansett Indian Tribe, in Rhode Island. The Tribal Nation is federally recognized by the U.S. Bureau of Indian Affairs and has jurisdiction over their reservation lands. The State HMP should include a map and a discussion of the tribe. The local and Tribal governments are key to successful implementation of the identified actions and can directly contribute to risk mitigation efforts. Tribal partnerships and engagement help to support external capacity building.
- The Plan states that, "Repeated outreach efforts were made to identified equity partners... extending opportunities to have a representative on the SIHMC." Continue to seek collaboration with equity partners and groups that support underserved communities and socially vulnerable populations. Describe how the state will break down those barriers in the future. Consider how they can support the plan and how the planning process can also help them. Common engagement methods include meetings, surveys, data calls, and material review. Develop a plan to gather input and provide changes to review throughout the process. Set meeting dates, outreach methods, and other activities early.

- Continue to diversify and increase the number of state agencies and other stakeholders that participate in the planning process for future SHMP updates - beyond the core agencies of RIEMA, RI DOT, RI Department of Housing, and a handful of other agencies. Bringing multiple partners together on common goals can leverage others' successes and combine efforts. Reference **Table 1. Sectors and their connected lifelines** of the [State Mitigation Planning Key Topics Bulletins: Planning Process \(fema.gov\)](#) to see the links between each sector with the community lifelines identified by the [National Response Framework \(fema.gov\)](#) and their connections to the plan.
- Supporting documentation, such as meeting sign-in sheets, preparation slides, handouts, completed forms with ideas from attendees, meeting minutes, sample invitations, a list of invitees, and announcements in traditional and digital media along with other supporting documentation from the planning process are a good record of how decisions were made and who was involved.

HAZARD IDENTIFICATION AND RISK ASSESSMENT

Strengths

- Effective use of the FEMA National Risk Index (NRI) as a data resource throughout the plan, the National Risk Index is great an online tool and database to visualize risk metrics on expected annual losses.
- Estimated potential dollar losses to state assets located in identified hazard areas considers factors affecting vulnerability such as construction type, age or intended use. For example, Wood Framed Facilities valuations for Extreme Temperatures or Manufactured Facilities and Assets valuations for Severe Thunderstorms.
- Projected Changes in Location, Intensity, Frequency, and Duration subsections under each of the hazard profiles effectively describe the effect of changing future conditions and highlight the impact of climate change.
- In the Earthquake hazard profile, FEMA's HAZUS-MH software was utilized to provide projections on Hazus Estimated Annualized Earthquake Losses that can be expected in Rhode Island as well as annualized losses for each county.
- Each hazard profile in the plan includes discussion on vulnerability in the categories of People and Communities, Facilities and Critical Infrastructure, Environment, and State Operations and Assets.

Opportunities for Improvement

- A 4.7% increase in population from 2000-2020 in the State of Rhode Island is not an insignificant change. Describe this change in population more and in terms of changes in vulnerability.
- Make a clearer connection between the changes in land use and development and risk. Where is the development and growth occurring? Is development moving out of cities and into more suburbs/rural areas? State agencies responsible for planning, economic, and environmental policies may know where growth is expected based on past trends. They may also know areas that are targeted for future growth.

- Refer to pages 6 and 7 of the [State Mitigation Planning Key Topics Bulletin: Risk Assessment \(fema.gov\)](https://www.fema.gov/state-mitigation-planning-key-topics-bulletin-risk-assessment) for additional federal data sources to use in the risk assessment. The data sources listed cover a variety of hazards. Academic institutions, state natural resources, environmental protection, economic development, or community development agencies and departments also have data on hazards and assets.
- Recommend using NIDIS National Integrated Drought Information System for the drought hazard profile.
- Continue to monitor development trends in the state and how that development intersects with known or potential hazard areas – affecting overall vulnerability of populations, structures, and infrastructure in the state’s communities.
- When the plan says there has not been “significant” or “notable” changes in development since the last plan update, it is strongly suggested to define what significant or notable is. For example, it could be over a certain acreage or size, etc. Otherwise, what is identified in the plan (or not) as significant or notable changes in development could be perceived as a matter of personal opinion.
- Continue to evaluate and highlight which State owned and operated assets are most vulnerable. State assets include state-owned or operated critical facilities, buildings, infrastructure, and community lifelines. Factors affecting vulnerability may include asset use and function as well as construction type, age, or intended use.
- Evaluate other State-owned-and-operated assets that may warrant integration into the plan as well, including colleges and universities, if they have not developed their own hazard mitigation plans. The State may also consider assets that are not state-owned and operated that, if compromised, could have a profound impact on the State’s economy and its ability to recover from disasters (e.g.: utilities).

STATE MITIGATION CAPABILITIES

Strengths

- The State of Rhode Island adopts and enforces the International Building Code. Thorough information was provided on the statewide building code (IBC 2018) and specific sections within that help create more disaster-resistant development.
- The state identifies, reviews, and analyzes various authorities, policies, programs, staff, funding and other resources in the evaluation and description of capabilities to mitigate the hazard identified in the risk assessment.
- The section on how the state works to administer National Flood Insurance Program (NFIP) pre- and post-disaster and Community Rating System (CRS) effectively describes and evaluates the capability through narrative summary and tables with specific community data related to the NFIP and CRS. Trends in changes in policies and coverage are captured in the narrative as well as in *Table 197: Changes in State of Rhode Island NFIP Coverage*.

Opportunities for Improvement

- The capability assessment is not simply a list or report of existing programs but rather a comprehensive evaluation based on the state’s existing capabilities that demonstrates the state’s commitment to mitigation. Identify whether each capability supports, actively facilitates,

or is in conflict with risk reduction. The state planning team should consider how capabilities that are in conflict with hazard mitigation could be amended to better support risk reduction.

- Expand Section 5.10 Evaluation of State Mitigation Capabilities to address any other changes with capabilities since the last update of the plan. Have they lost capabilities? Were there any changes to building codes, did they build support, etc. What, if any, were the changes/improvements on capabilities since the last plan update.
- Add information about generally how state funding capabilities have changed since the last plan approval. For example, identify if there is greater or less funding available overall, any new programs that are being utilized, or if any new programs are being anticipated in the future.
- Add to the highlights of some of the programs that RIEMA operates and oversees more information about State participation in FEMA's Flood Hazard Mapping Program (Risk MAP). Discuss how the state supports the Risk MAP program. How does the state encourage the use of Risk MAP products in local hazard mitigation plans (thus growing the capabilities of local communities to communicate risk)? How does the state teach its population about flood risk in the state through Risk MAP?
- In addition to the evaluation of Rhode Island Agencies in *Table 182: Rhode Island Agency Capability Review*, within each of the writeups of each of the state agencies, identify areas where the state needs to strengthen its mitigation capabilities as well as areas to target improvements. Include RIEMA in this capability review in addition to the summary description in Section 5.5 RIEMA Profile.

MITIGATION STRATEGY

Strengths

- The plan clearly identifies how each mitigation action aligns with the plan's stated goals and objectives.
- Gaps and challenges identified in various parts of the plan are addressed as future actions in the mitigation strategy. Actions address data deficiencies noted in the risk assessment or gaps identified in the state mitigation capabilities assessment.
- The plan clearly describes how each action relates to the goals. It also showcases the goal(s) that each action will address when it is implemented.
- The status of proposed mitigation actions from the previously approved state hazard mitigation plan is thoroughly documented and described; whether the action has been completed, partially completed, reclassified as a capability or standard operating procedure, or carried over and included in the updated mitigation strategy is clear.

Opportunities for Improvement

- Mitigation actions should clearly connect to the risk assessment. The risk assessment will identify new or changed vulnerabilities, or it may identify areas where the probability of future hazard events or other changing conditions could increase the vulnerability. Where these changes have occurred or have high probability of occurring in the future without mitigation, states may need to add new mitigation actions.
- FEMA defined four primary kinds of mitigation actions to reduce long-term vulnerabilities: plans and regulations, structure and infrastructure, natural systems protection projects, and education

and awareness programs. Consider including more structure and infrastructure projects that involve modifying existing structures and infrastructure or constructing new structures to reduce the impact of hazard in future updates of the state hazard mitigation plans.

- Leverage the planning team and stakeholder during the review, evaluation, and update of mitigation actions. The planning team should assist in determining which aspects of the previous mitigation strategy are still relevant, and what should be included in the plan update. Collaboration during this process not only allows for the development of these relationship, but also identifies possibilities for integrating mitigation strategies across plans and initiatives.

LOCAL PLANNING COORDINATION AND CAPABILITY BUILDING

Strengths

- The evaluation of local jurisdiction capabilities is well documented in Section 5.11. Each jurisdiction's available plans, building codes, land use restrictions, mapping and public outreach were reviewed to demonstrate the existing local jurisdictions capabilities support risk reductions, resiliency, and mitigation in the State of Rhode Island.
- The description of the local governments' effectiveness in mitigation is thorough. The plan includes the challenges of applying the local policies, programs, and capabilities due to frequent local staff turnover, lack of success stories being widely disseminated to illustrate the benefits of participation, and grant application challenges.

Opportunities for Improvement

- Clarify what the timeframe is for when and how local HMP information is being incorporated into State planning efforts.
- Continue to highlight and showcase mitigation successes in the SHMP, including call out boxes and graphics. This could not only include State successes, but local ones too. This may increase buy in and interest at the local level on how important hazard mitigation planning is at the State and local levels.
- Continue to find ways to further integrate information from the LHMPs into the State HMP. Consider adding LHMP "roll up" information into the SHMP's plan appendices for reference. This could be accomplished by providing this information within an Excel spreadsheet.
- Discussion of challenges should include any gaps in aiding underserved communities and challenges that come from the impacts of climate change. Do any capabilities disproportionately benefit wealthy communities? Do any capabilities actively increase the vulnerability or risk exposure of underserved and socially vulnerable communities? Are the capabilities able to address the effects of climate change on local populations?

REVIEW, EVALUATION, AND IMPLEMENTATION

Strengths

- The plan discusses the monitoring, evaluating, and updating the plan separately. It is clear the state sees these plan maintenance processes as three distinct activities.

- The plan includes monitoring tool(s) that were made to be used over its lifespan. This will ensure the plan is routinely monitored on the schedule it identifies. The results can show progress in hazard mitigation through the plan's 5-year lifespan.
- The update schedule allows for a thorough planning process within the timeline, as well as for FEMA's review. This will ensure the plan coverage for the state will not lapse.

Opportunities for Improvement

- Submit the next draft plan for FEMA review twelve months in advance of the plan expiration. Suggest starting the update years in advance and account for 45-day FEMA review periods, potential required revisions, and additional time considerations. Getting FEMA input as early as possible in the planning process on full plan drafts will facilitate the review process until official approval.
- Once the SHMP is formally approved, implementation should begin immediately. During FEMA's Annual Hazard Mitigation Planning consultation meetings, the state should be prepared to provide a timeline and progress on where the state is regarding updating the next SHMP.
- It is important to discuss the successes of and obstacles to the previous plan's maintenance strategies. This will clearly note going forward how the plan will be monitored, evaluated, and updated. Using these successes and learning from challenges of maintaining the previous plan is a strength.
- Determine the opportunities and challenges, including barriers to implementation and approaches to overcoming them—funding, capacity, permitting, technical, community support, or other.

ADOPTION AND ASSURANCES

Strengths

- Assurances language was added into the plan as well (not just in the adoption letter).
- The plan will be adopted by the state's highest elected official. This signals that the state's investment in the process and provides support secure and rally resources and investment to implement the plan.

Opportunities for Improvement

- None identified

HIGH HAZARD POTENTIAL DAMS

Strengths

- The plan identifies gap and limitations in data and addresses and describes the approach to address these deficiencies in the mitigation strategy through *Mitigation Action #30 State Asset High Hazard Dam Vulnerability*.

Opportunities for Improvement

- Continue to document limitations and describe the approach to address deficiencies and include various funding potential sources to mitigate vulnerabilities to and from high hazard potential dams from hazards and the potential consequences associated with dam incidents.

- In the future update of the plan, ensure that all groups in the state related to dams and dam safety are invited to the planning process. They can offer the state valuable data, vulnerability information, and resources to use in the plan. A robust participation from these stakeholders for the dam safety program adds greater value to the risk assessment.
- As listed in the State Mitigation Planning Policy Guide (2022), potential actions to reduce vulnerabilities to/from high hazard potential dams include:
 - o Proposing, enacting, and/or delegating authority for local land use regulations, ordinances, and/or other standards to protect life and property from high hazard potential dams.
 - o Working with dam owners to create/update and share EAPs or dam incident annexes to emergency operations plans.
 - o Delegating authority to local governments to adopt and enforce land use ordinances in inundation zones.
 - o Acquiring structures both upstream and downstream of high hazard potential dams.
 - o Rehabilitating and/or removing high hazard potential dams.

FIRE MANAGEMENT ASSISTANCE GRANTS

Strengths

- The plan includes a complete history of the state's wildfires. This shows the general trends, as well as the events with the largest magnitude. It also describes the impacts of historical events.

Opportunities for Improvement

- Analyze local government wildfire risk assessments including those from the Community Wildfire Protection Plans where they exist by jurisdictions, to further build out the wildfire analysis within the plan and to see the vulnerability of the state's local governments to wildfire. How many communities in the state have Community Wildfire Protection Plans?
- Be sure to address the identified vulnerable state-owned or operated assets identified in the risk assessment with new developed actions in the mitigation strategy. Using the risk assessment to guide the mitigation strategy will ensure that the state is taking action directed at addressing the most vulnerable areas.

Miscellaneous Resources

- [Climate and Economic Justice Screening Tool \(CEJST\)](#), White House. This interactive tool and map shows information about the climate, environmental, health, and socioeconomic burdens that communities are facing throughout the nation.
- [Climate Mapping for Resilience and Adaptation \(CMRA\)](#). Data and maps available in this tool are downscaled results from global climate models. Results for selected geographies indicate how local exposure to five common climate-related hazards is projected to change through this century.

- [Climate Risk and Resilience Portal](#) (ClimRR). ClimRR's Local Climate Projections allow users to generate and export a report of climate projections at a given point on the map.
- [Community Resilience Estimates](#) (CRE), Census Bureau. The Community Resilience Estimates (CRE) provide an easily understood metric for how at-risk every neighborhood in the United States is to the impacts of disasters, including COVID-19.
- [Environmental Justice Screening & Map Tool](#) (EJScreen), EPA. EJScreen is the EPA's environmental justice mapping and screening tool that provides EPA with a nationally consistent dataset and approach for combining environmental and demographic socioeconomic indicators.
- [Risk Assessment Bulletin](#), FEMA, Oct. 2022. Lays out the five steps to conduct the state risk assessment. It includes advice on how to use climate projections to support the plan. It also provides ideas for updating the risk assessment with changes in development.
- [HAZUS](#). FEMA's HAZUS Program provides standardized tools and data for estimating risk from earthquakes, floods, tsunamis, and hurricanes. HAZUS models combine expertise from many disciplines to create actionable risk information that increases community resilience. HAZUS software is distributed as a GIS-based desktop application with a growing collection of simplified open-source tools. Risk assessment resources from the program are always freely available and transparently developed.
- [Low-Income Energy Affordability Data \(LEAD\) Tool](#), DOE. The Low-Income Energy Affordability Data (LEAD) Tool was created to help stakeholders understand housing and energy characteristics for low- and moderate-income households. Using data, maps, and graphs from the LEAD Tool, stakeholders can make data-driven decisions when planning for their energy goals.
- [National Risk Index](#), FEMA. The National Risk Index is a dataset and online tool to help illustrate the United States communities most at risk for 18 natural hazards at the county and census tract level. The site has information on Total Building Exposure Values by hazard and Expected Annual Building Loss Values by hazard.
- [Neighborhoods at Risk](#) – Headwaters Economics. Neighborhoods at Risk is an easy-to-use website with interactive maps, charts, and resources to help communities identify neighborhoods that may be more impacted by climate change. It shows where people may experience unequal impacts from flooding and extreme heat.
- [Northeast-Midwest Wildfire Risk Explorer](#). Consists of a variety of data, user-friendly applications and planning tools for 20 Northeast and Midwest states that can be used to help prioritize areas where tactical analyses, community interaction and education, or mitigation treatments might be necessary to reduce risk from wildfires. It includes a [Community Exposure Dashboard](#) that identifies the Top 5 communities at risk from wildfire in a state, based on the number and percentage of housing units exposed – and more.

- [Resilience Analysis and Planning Tool](#) (RAPT), FEMA. RAPT includes over 100 preloaded layers including community resilience indicators from peer-reviewed research, the most current census demographic data, infrastructure data, and data on weather, hazards, and risk.
- [Social Vulnerability Index \(SVI\)](#), CDC. The CDC/ATSDR Social Vulnerability Index (CDC/ATSDR SVI) uses 16 U.S. census variables to help local officials identify communities that may need support before, during, or after disasters.
- [The Climate Explorer](#). Explore how climate is projected to change in any U.S. County.
- [Wildfire Risk to Communities](#), USDA/USFS. This is a website with interactive maps, charts, and resources to help communities understand, explore, and reduce wildfire risk. The website was created by the USDA Forest Service under the direction of Congress and is designed to be a starting point to help community leaders, such as elected officials, community planners, and fire managers, to be able to assess and reduce risk to homes, businesses, and other valued resources.
- [The Path to Achieving Justice40 | OMB | The White House](#) President Biden has made historic commitments to use every lever at his disposal to advance environmental justice and spur economic opportunity for disadvantaged communities. And within his first weeks in office, he established the [Justice40 Initiative](#).

Standard State Plan Review Checklist (44 CFR § 201.4)

Planning Process

Requirements	Location in Plan (section and/or page number)
S1. Does the plan include a description of the process used to develop the plan? [44 CFR §§ 201.4(b) and 201.4(c)(1)]	
S1-a. Does the plan describe the current process used to update the plan, including how the plan was prepared, the schedule or time frame, specific milestones and activities, the agencies and stakeholders who were involved in the process, and if the mitigation planning process was integrated to the maximum extent possible with other state planning efforts?	Section 2.0, pgs. 3-11
S2. Does the plan describe how the state coordinated with other agencies and stakeholders? [44 CFR §§ 201.4(b) and 201.4(c)(1)]	
S2-a. Does the plan describe how the state coordinated with other state agencies, appropriate federal agencies, and other stakeholders, and how they were involved in the process?	Section 2.4, pg. 5 and 6, and Section 2.5, page 7

Hazard Identification and Risk Assessment

Requirements	Location in Plan (section and/or page number)
S3. Does the risk assessment include an overview of the type and location of all natural hazards that can affect the state? [44 CFR § 201.4(c)(2)(i)]	
S3-a. Does the plan include a current overview of all natural hazards that can affect the state, including the type, location and previous occurrences?	Section 4.0, pgs. 48 – 274 and Appendix D
S4. Does the risk assessment provide an overview of the probabilities of future hazard events? [44 CFR § 201.4(c)(2)(i)]	
S4-a. Does the risk assessment provide an overview of the probability of future hazard events that includes projected changes in the location, range of anticipated intensities, frequency, and/or duration of each natural hazard?	Section 4.9 to 4.27, pg. 59-274
S4-b. Does the probability include considerations of changing future conditions, including climate change (e.g., long-term weather patterns, average temperature, and sea levels) on the type, location and range of anticipated intensities of identified hazards?	Sections 4.9 to 4.27, pgs. 59-274
S5. Does the risk assessment address the vulnerability of state assets located in hazard areas and estimate the potential dollar losses to these assets? [44 CFR §§ 201.4(c)(2)(ii) and 201.4(c)(2)(iii)]	
S5-a. Does the risk assessment include an overview and analysis of the vulnerability to state assets from the identified hazards as well as a summary of the most vulnerable assets?	Section 4.7, pg. 55; Section 4.9 to 4.27, pg. 59-274
S5-b. Does the risk assessment estimate potential dollar losses to state assets located in identified hazard areas?	Section 4.7, pg. 55; Section 4.9 to 4.27, pg. 59-274
S6. Does the risk assessment include an overview and analysis of jurisdictions' vulnerability to the identified hazards and the potential losses? [44 CFR §§ 201.4(c)(2)(ii) and 201.4(c)(2)(iii)]	
S6-a. Does the risk assessment provide an overview and analysis of vulnerable jurisdictions based on the state and local government risk assessments?	Section 4.3, pg. 49; Section 4.9 to 4.27, pg. 53-274
S6-b. Does the risk assessment include an overview and analysis of the potential losses to the identified vulnerable structures based on estimates in the local risk assessments as well as the state risk assessment?	Section 4.7, pg. 55; Section 4.9 to 4.27, pg. 53-274
S7. Was the risk assessment revised to reflect changes in development? [44 CFR § 201.4(d)]	
S7-a. Does the plan provide a summary of recent development and potential or projected development in hazard-prone areas based on state and local government risk assessments?	Section 3.7, pg. 21-39

State Mitigation Capabilities

Requirements	Location in Plan (section and/or page number)
S8. Does the plan discuss the evaluation of the state's hazard management policies, programs, capabilities, and funding sources to mitigate the hazards identified in the risk assessment? [44 CFR § 201.4(c)(3)(ii)]	
S8-a. Does the plan include an evaluation of state laws, regulations, policies and programs related to hazards that improve or impede resilience to future natural hazard events and other future conditions, including the effects of climate change?	Section 3.11, pg. 44; Section 5.2, pg. 275-; Section 5.4, pg. 280; Section 5.7, pg. 293
S8-b. Does the plan include a general discussion of state funding capabilities for hazard mitigation actions and projects?	Section 5.5, pg. 282; Section 7.13, pg. 339
S8-c. Does the plan include a summary of obstacles, challenges and proposed solutions related to any state capabilities, including a brief discussion of potential strategies for overcoming any challenges related to implementing and enforcing hazard-resistant building codes statewide, as applicable, and changes since the previous plan approval?	Section 5.10, pg. 302; Section 5.12, pg. 306; Section 7.2, pg. 315

Mitigation Strategy

Requirements	Location in Plan (section and/or page number)
S9. Does the mitigation strategy include goals to reduce long-term vulnerabilities from the identified hazards? [44 CFR § 201.4(c)(3)(i)]	
S9-a. Does the plan identify hazard mitigation goals representing what the state seeks to accomplish through mitigation plan implementation using a wide range of funding, including non-FEMA funding?	Section 7.3, pg. 316
S9-b. Are the goals consistent with the hazards and vulnerabilities identified in the risk assessment?	Section 7.2, pg. 315; Section 7.3, pg. 316
S10. Does the plan prioritize mitigation actions to reduce vulnerabilities identified in the risk assessment? [44 CFR §§ 201.4(c)(3)(i), 201.4(c)(3)(ii) and 201.4(c)(3)(iii)]	
S10-a. Does the plan identify actions based on the current risk assessment to reduce the vulnerability of jurisdictions within the state, as well as the vulnerability of state assets as described in Elements S5 and S6?	Section 7.8, pg. 321 and Section 7.9, pg. 323
S10-b. Does the plan describe the process used by the state to evaluate and prioritize actions that are cost-effective, environmentally sound, and technically feasible?	Section 7.5, pg. 318; Section 7.9, pg. 323
S10-c. Does the plan describe how each action contributes to the hazard mitigation goals?	Section 7.9, pg. 323
S10-d. Does the plan describe how local government mitigation strategies link to the state mitigation strategy?	Section 6.8, page 286 and Section 6.6, page 283
S11. Does the plan identify current and potential sources of funding to implement mitigation actions and activities? [44 CFR § 201.4(c)(3)(iv)]	
S11-a. Do mitigation activities include the identification of current and/or potential sources of federal, state, local or private funding for implementation?	Section 7.9, pg. 323
S11-b. Does the plan identify FEMA mitigation funding sources (if applicable), including, but not limited to: HMGP, BRIC, FMA and PA Mitigation, at a minimum??	Section 7.13, pg. 339
S12. Was the plan updated to reflect progress in statewide mitigation efforts and changes in priorities? [44 CFR § 201.4(d)]	
S12-a. Does the plan provide a narrative of the status of each mitigation action in the previous plan?	Section 7.7, pg. 320, Section 7.8, pg. 321 and Section 7.9, pg. 323
S12-b. Was the prioritization of mitigation actions and activities updated based on the updated analysis of risks, capabilities and progress?	Section 7.5, pg. 318

Local Planning Coordination and Capability Building

Requirements	Location in Plan (section and/or page number)
S13. Does the plan generally describe and analyze the effectiveness of local government mitigation policies, programs, and capabilities? [44 CFR § 201.4(c)(3)(ii)]	
S13-a. Does the plan provide a summary of current local government policies, programs and capabilities of jurisdictions to accomplish hazard mitigation?	Section 5.11, pg. 320
S13-b. Does the plan describe the effectiveness of local government mitigation policies, programs and capabilities?	Section 5.11, pg. 302
S14. Does the plan describe the process to support the development of approvable local government mitigation plans? [44 CFR §§ 201.3(c)(5) and 201.4(c)(4)(i)]	
S14-a. Does the plan describe how the state supports developing or updating FEMA-approvable mitigation plans?	Section 6.1, pg. 308; Section 6.3, pg. 309
S14-b. Does the plan provide a brief summary of barriers to developing or updating, adopting, and implementing FEMA approved local government mitigation plans based on an analysis of plan and jurisdiction coverage data and trends across the state and steps to remove barriers to help local governments advance mitigation planning, including how plan and jurisdiction coverage data and trends inform those steps?	Section 6.4, pg. 310
S15. Does the plan describe the criteria for prioritizing funding? [44 CFR § 201.4(c)(4)(iii)]	
S15-a. Does the plan describe criteria for prioritizing jurisdictions to receive planning and project grants under available federal and non-federal programs?	Section 6.7, pg. 311
S16. Does the plan describe the process and time frame to review, coordinate, and link local and tribal mitigation plans with the state mitigation plan? [44 CFR §§ 201.3(c)(6), 201.4(c)(2)(ii), 201.4(c)(3)(iii), and 201.4(c)(4)(ii)]	
S16-a. Does the plan describe the state's process and time frame to review and submit approvable local and tribal mitigation plans to FEMA?	Section 6.5, pg. 311
S16-b. Does the plan describe the state's process and time frame to share risk assessment data and mitigation priorities with local governments for their plan updates, as well as integrate local risk assessment and mitigation actions into the state mitigation plan updates?	Section 6.9, pg. 314

Review, Evaluation, and Implementation

Requirements	Location in Plan (section and/or page number)
S17. Is there a description of the method and schedule for keeping the plan current? [44 CFR §§ 201.4(c)(5)(i) and 201.4(d)]	
S17-a. Does the plan describe the agency/office responsible for monitoring, evaluating and updating the plan?	Section 8.2, pg. 346; Section 8.3, pg. 346
S17-b. Does the plan describe the schedule for monitoring, evaluating, and updating the plan?	Section 8.3, pg. 346; Section 8.4, pg. 347
S18. Does the plan describe the systems for monitoring implementation and reviewing progress? [44 CFR §§ 201.4(c)(5)(ii) and 201.4(c)(5)(iii)]	
S18-a. Does the plan describe the system for tracking the implementation of the mitigation activities and projects identified in the mitigation strategy, including all mitigation activities and not just those funded by FEMA?	Section 7.10, pg. 336
S18-b. Does the system include the schedule, the agency/office responsible for coordination, and the role of the agencies/offices identified in the mitigation strategy as responsible for implementation of actions?	Section 7.9, pg. 323
S18-c. Does the plan describe a system for reviewing progress on achieving the mitigation strategy's goals that includes the criteria and process for evaluating progress?	Section 7.3, pg. 316; Section 8.3, pg. 346

Adoption and Assurances

Requirements	Location in Plan (section and/or page number)
S19. Did the state provide documentation that the plan has been formally adopted? [44 CFR § 201.4(c)(6)]	
S19-a. Did the state provide documentation of formal adoption by the highest elected official or designee prior to FEMA approval?	To be submitted upon APA; location to be inserted. Section 1.4, pg. 2; Appendix A
S20. Did the state provide assurances? [44 CFR § 201.4(c)(7)]	
S20-a. Does the plan include assurances that the state will manage and administer FEMA funding in accordance with applicable federal statutes and regulations?	To be submitted upon APA; location to be inserted. Section 1.2, pg. 1
S20-b. Does the plan include assurances that the state will update its plan whenever necessary to reflect changes in state or federal laws and statutes?	To be submitted upon APA; location to be inserted. Section 1.2, pg. 1

Appendix C – Community Feedback

2024 Rhode Island Hazard Mitigation Plan Survey

MOVE 

FILTER BY SEGMENT
Project Engagement

OPTIONS 



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VIEWS

330

PARTICIPANTS

100

RESPONSES

1,649

COMMENTS

244

Where in Rhode Island do you live?

West Greenwich

one month ago

Johnston

one month ago

South Kingstown

one month ago

Narragansett

one month ago

Barrington

one month ago

Cumberland

one month ago

Coventrt

one month ago

Little Compton

one month ago

West Greenwich

one month ago

North kingstown

one month ago

Providence

one month ago

Woonsocket

one month ago

Charlestown

one month ago

Charlestown

one month ago

charlestown

one month ago

Charlestown

one month ago

Greene

one month ago

Riverside

one month ago

North Smithfield

one month ago

South Kingstown

one month ago

South County

one month ago

Hope Valley

one month ago

East Providence

one month ago

North Providence

one month ago

West Greenwich

one month ago

Barrington

one month ago

Westerly

one month ago

Newport

one month ago

Narragansett

one month ago

Charlestown

one month ago

Rumford

one month ago

Pawtucket

one month ago

Exeter

one month ago

Warwick

one month ago

Hope Valley

one month ago

Warwick

one month ago

Pawtucket

one month ago

West Greenwich

one month ago

South Kingstown

one month ago

Smithfield

one month ago

Narragansett

one month ago

West Warwick

one month ago

Pascoag

one month ago

Coventry (Western)

one month ago

West Greenwich

one month ago

Jamestown

one month ago

Charlestown

one month ago

Warwick

one month ago

Warwick

one month ago

Woonsocket

one month ago

Smithfield

one month ago

Providence

one month ago

West Greenwich

one month ago

Narragansett

one month ago

Pawtucket

one month ago

Cranston

one month ago

Newport

one month ago

Tiverton

2 months ago

Pawtucket (Oak Hill), RI

2 months ago

Barrington

2 months ago

Woonsocket

2 months ago

Johnston

2 months ago

Portsmouth

2 months ago

Johnston

2 months ago

Johnston

2 months ago

Providence

2 months ago

North Kingstown

2 months ago

Warwick

2 months ago

Tiverton

2 months ago

Burrillville

2 months ago

West Warwick

2 months ago

North Kingstown

2 months ago

Providence

2 months ago

Providence

2 months ago

Providence

2 months ago

Johnston

2 months ago

Warwick

2 months ago

Cranston

2 months ago

Woonsocket

2 months ago

Cranston

2 months ago

Exeter

2 months ago

East Providence

2 months ago

Cumberland

2 months ago

Warwick

2 months ago

Warwick

2 months ago

North Kingstown

2 months ago

East Greenwich

2 months ago

Burrillville

2 months ago

North Kingstown

2 months ago

The following natural hazards have been identified as having the potential to impact Rhode Island.
Please tell us about your concerns about the identified hazards.

	No Concern	Some Concern	Moderate Concern	Significant Concern
Dam Failure	39% No Concern	33% Some Concern	19% Moderate Concern	9% Significant Concern
Drought	15%	44%	31%	10%

	No Concern	Some Concern	Moderate Concern	Significant Concern
Earthquake	78% No Concern	21% Some Concern	20% Moderate Concern	10% Significant Concern
Extreme Temperatures	13% No Concern	29% Some Concern	31% Moderate Concern	27% Significant Concern
Flash Flood / Flood	5% No Concern	29% Some Concern	31% Moderate Concern	35% Significant Concern
Sea Level Rise	20% No Concern	27% Some Concern	25% Moderate Concern	27% Significant Concern
Severe Storm (including Lightning, Thunderstorms, Hail, and Wind)	3% No Concern	20% Some Concern	34% Moderate Concern	42% Significant Concern
Severe Winter Weather	5% No Concern	23% Some Concern	42% Moderate Concern	30% Significant Concern
Tornado	17% No Concern	51% Some Concern	24% Moderate Concern	8% Significant Concern
Tropical Storms (Hurricanes) and Extratropical Storms (Nor'easters)	- No Concern	23% Some Concern	32% Moderate Concern	45% Significant Concern
Wildfires	20% No Concern	47% Some Concern	22% Moderate Concern	12% Significant Concern

95 respondents

The following human caused and technological hazards have been identified as having the potential to impact Rhode Island. Please tell us about your concerns about the identified hazards.

	No Concern	Some Concern	Moderate Concern	Significant Concern
Biological Incident	13% No Concern	52% Some Concern	17% Moderate Concern	18% Significant Concern
Civil Disorder	16% No Concern	42% Some Concern	26% Moderate Concern	17% Significant Concern
Cybersecurity Incident	3% No Concern	18% Some Concern	39% Moderate Concern	40% Significant Concern
Infrastructure Failure	4% No Concern	25% Some Concern	37% Moderate Concern	34% Significant Concern
Novel Transmissible Disease	11% No Concern	28% Some Concern	32% Moderate Concern	29% Significant Concern
Radiological Incident	33% No Concern	42% Some Concern	10% Moderate Concern	16% Significant Concern
Terrorism	13% No Concern	48% Some Concern	18% Moderate Concern	21% Significant Concern

90 respondents

Have you been impacted by any of the identified hazards in the last five years? If yes, please tell us about what happened.

Flash flood

one month ago

Flooding and wind

one month ago

Smoke from West Greenwich wildfires that came within 1 mile of our house.

one month ago

No

one month ago

No

one month ago

As a prior antiterrorism specialist and emergency manager, it is just a matter of time before most of the man-made threats become realities.

one month ago

No

one month ago

There was a severe storm somewhat like a microburst. My neighbors 70+ ft pine tree came down pulling down the wires on the street & there were live wires on my neighbors lawn that started little fires on several lawns. Even the rain didn't fully help stopping the fire.

one month ago

COVID 19

one month ago

No

one month ago

None

one month ago

Cybersecurity. Bank account hacked

one month ago

High wind power outages and trees down causing loss of electricity and operation of well pumps for drinking water.
Local flooding on roads.
Bridge repairs leading to road closures.
Increased verbal discord on local social and print media which could incite violence in the future.
Loss of forest to development increasing affects of climate changes.

one month ago

Not really

one month ago

Repair of dam, severe weather damage, financial impact of COVID-19

one month ago

I live in Oakland Beach and there has not been a flood since Hurricane Carol. Lower flood zone flood insurance is ridiculous

one month ago

Terrible water in Pawtucket. Lots of iodine in water to prevent rust yet water is some times unpalatable to drink or feel safe to cook with.

one month ago

Wildfires in the completely unmanaged state lands. Fuel loads are very high on both State and private land. This poses SIGNIFICANT hazard to communities and our fire services in RI. Both of the large fires this year displayed an increased potential for catastrophic wildfires to happen in RI. DEM Forestry is woefully understaffed and local fire departments are ill-equipped to combat a larger fire. I commend the current staff at DEM and the recent push to increase the effectiveness of firefighters with additional equipment. I fear that this was only a reactive stopgap and that State leaders will not prioritize further proactive measures as time passes. It should be know that all firefighters, on both fires, were incredibly lucky to not be overrun by fire. Solid tactical decision making, and slightly cooperative weather, undoubtedly saved this State from significant loss of life in our fire service. To be clear, we had ZERO means to survive being overrun by fire. No emergency fire shelters at all! This is simply unacceptable to expect men and women to combat fires without the most basic gear necessary to potentially survive being overrun.

one month ago

Covid my wife and hated the lockdown and believed it unnecessary

one month ago

Flash flooding in Warwick has led to car damage, being trapped in flooded waters.

one month ago

COVID 19, flood and tornado warnings.

one month ago

No

one month ago

Poorly managed infrustructure resulting in power outages and poor roads

one month ago

The terrible erosion of Scarborough Beach is an imminent disaster waiting to happen. The state should be replenishing the entire beach before it is totally gone. There has NEVER been a replenishment in spite of the loss of dozens and dozens of feet of beach especially at the north and south ends.

one month ago

Folding due to high water and heavy rains

one month ago

No

one month ago

no

one month ago

Flooding

2 months ago

Covid affecteded us all

2 months ago

Extreme temperatures, flash flooding, cybersecurity incidents... all have happened in increasing frequency over the past 5 years. I live right on the Blackstone River past the Woonsocket Falls Dam. Luckily we have the Army Corp handling that but if it were to fail, plus sea level rising, extreme rain events... it would be very scary.

2 months ago

No

2 months ago

My basement flooded with these past flash floods. It had never done this before. I was also located closer to the tornado that touched down.

2 months ago

Hurricane, tornado, flooding, wildfire

2 months ago

power outage after storm

street flooding after heavy rain

2 months ago

Forest fire

Water quality

Area identified as industrial should be 50% + preserved for open space.

More protection for Stafford Pond. Increase water shed area and open space on West, north & south sided. No industrial allowed within 1 mile.

2 months ago

Weather related hazards that normally happen.

2 months ago

Virus (COVID-19), severe weather both in wind storms and winter snow events

2 months ago

COVID-19 affected everyone, have watched RI houses destroyed during storms due to sea level rise and storm surge, malicious actors have "hacked" several large organizations in the State already and held data/functionality hostage.

2 months ago

No

2 months ago

no

2 months ago

Flooding, civil disobedience, virus/pandemic, tornado, extreme temps, severe storm

2 months ago

Affected by Covid- both personally and professionally.

2 months ago

Yes, house damage

2 months ago

No

2 months ago

Flooding, wild fires

2 months ago

Yes storm damage as a result of wind

2 months ago

Obviously a novel transmittable disease (Covid) and flooding in particular have affected my life enormously.

2 months ago

No

2 months ago

Civil disorder in Providence (2020), power outage due to winter weather and tropical storms in recent years

2 months ago

No

2 months ago

Yes, storm dropped tree on garsge

2 months ago

Are there any other hazards that were not identified that may be worth consideration for inclusion in the Hazard Mitigation Plan?

Tsunami

one month ago

Anything to the west that prevailing winds could carry in here.

one month ago

No

one month ago

School lock downs, active shooter, and chemical shelter in place plans are not exercised properly. Terrifying optics of what could happen in a traumatic event prevents realistic training. This strategy is the road to failure.

one month ago

No

one month ago

Not that I can think of.

one month ago

Storm Surge

one month ago

Increased crime

one month ago

Hazard waste

one month ago

Robo phone calls and Junk mail are a nuisance.

one month ago

Antivaxxer movement impacting immunity from many diseases. Need more public education and physician advocacy.

one month ago

Nitrogen level in drinking water and coastal ponds from overdevelopment, fertilizer and septic discharge.

Coastal erosion issues.

These have impact but not affecting me directly at the moment as being 3-5 miles inland from coast but affect taxes.

one month ago

EMP or Coronal Mass Ejection

one month ago

Environmental impact of destruction of green space for development

one month ago

Aircraft incident

one month ago

Stop the solar farms! They are ruining our state, our towns environments

one month ago

The potential for extreme lose of life in large pubic venues. All large public venues lack "hardening" measures meant to decrease the ability of individuals intent on causing harm. To my knowledge, no state entity has prioritized the security issues of these venues. I strongly believe that government officials and leaders are failing the citizens of our State in this matter.

one month ago

Too heavy a drain on the aquafier, causing no access to water for drinking, bathing, bathroom, etc.

Use of pesticides causing reduction in pollinator population and gutting of farming etc

one month ago

Gun violence, in schools and elsewhere.

one month ago

State government overreach in the name of safety. Excessive state spending and useless bureaucracy that makes hardworking taxpayers have to work more and live less safe lives.

one month ago

Burying powerlines especially in high wind areas like Narragansett.

one month ago

No

one month ago

Constant roadway flooding city and highway

one month ago

Stupidity

2 months ago

Protection of tidal areas and marshes.

2 months ago

I would consider pollution of our air and water to be a hazard. I live near the waste water treatment center in Woonsocket and the air quality, especially on hot days is not great. There have been discharges of partially treated waste in the river as well...

2 months ago

No

2 months ago

Chemical and biological hazards associated with storms and flooding.

2 months ago

no

2 months ago

Nothing hazardous in a rural small town. Not equipped for it.

2 months ago

no

2 months ago

Communication of disasters to all citizens including those who do not use English eg Spanish speakers, Deaf people (ASL). Portuguese, Chinese etc

2 months ago

n/a

2 months ago

none that i can see

2 months ago

No

2 months ago

No

2 months ago

None I can think or

2 months ago

No

2 months ago

No

2 months ago

Are there any projects that could help make safer communities in Rhode Island that you would like to be considered for the Hazard Mitigation Plan? (Example: construction of stormwater retention ponds or severe weather safe rooms)

Government funded insurance

one month ago

Bury power lines that are most exposed to storm damage and prolonged power outages.

one month ago

Any intervention that could improve stormwater drainage and reduce flash flooding. Continue improving infrastructure.

one month ago

I'll leave most of the natural events to the thousands of people who work on them daily.

one month ago

Early warning responses, better use of cell phone alerts, identifiable shelter/safe locations

one month ago

Honestly, the most hazard I've encountered are our roads. Water pooling in potholes do incredible damage to cars and are quite likely to cause accidents involving more than one vehicle.

one month ago

A warming signal to sound so people can seek shelter quickly and go to a safe place

The severe weather safe rooms are a great idea! I would like to know how to prepare & what should be stored in a safe room to be prepared in case we need to stay there for hours/days?

one month ago

Storm drain maintenance

one month ago

Conservation design development

Retention of forest

Energy reduction

Not rebuilding on barrier beaches after storm destruction.

Not building in flood plains.

Enforcement of wetland and wetland buffers violations.

Increase DEM and CRMC oversight of building in sensitive areas.

Access to freshwater during major storm events.

one month ago

Hardening of infrastructure

one month ago

Since first responders are in high demand, perhaps the formation of neighborhood chosen reporters work hand in hand with the PD or FD's.

one month ago

The NFPA Fire Wise program is HIGHLY recommended to start with.

one month ago

Outlaw basement apartments

one month ago

More Stormwater BMPS in areas where runoff becomes too extreme for highways to handle would be very useful

one month ago

Safe rooms for storms and biological incidents

one month ago

Thinning the herd

one month ago

drinking water safety---old pipes, etc.

one month ago

No

one month ago

Term limits

2 months ago

Stop voting for democrats, they have not solutions

2 months ago

All of the above

2 months ago

Severe weather safe rooms is a great idea, especially when people are going to need places to use electricity, heat, access to clean water during storms.

2 months ago

Flood plain maps update to make it user friendly and easier to understand.

2 months ago

Every reasonable preventative measure should be taken to protect RI from the inevitable effects of climate induced changes in weather patterns. Protecting RI from flash floods and hurricane related flooding seems like it should be the priority given our sea level position and proximity to the ocean. Both physical and educational preparation is necessary. I hope that RI will be as proactive as possible and not wait until there is a severe disaster to take action.

2 months ago

update evacuation map. In Warwick for example the building in Oakland Beach identified as a fire station has been a bar for many years. The evacuation route on Cowesett Road seems to take people onto Rt 95 north but there is no on ramp there.

2 months ago

Forest fire safety. No fireworks by individuals. No hazardous businesses allowed. Don't build out or over develop certain areas of town. Already too much development in North Tiverton and near Stafford Pond. All conservation efforts seem to be made in south tiverton. Spread protections around town. Protect other area's esp stafford pond. Provide more open space around the pond & ensure NO development on open space. No wind turbines. Protect water dept owned land from development.

2 months ago

resilient infrastructure

2 months ago

Protecting or bringing back natural barriers to sea level rise (e.g marshland, salt water pond environments, etc.).
Requiring cyber security precautions at any organization that provides a critical good/service (e.g. healthcare).
Reducing reliance on fossil fuels and by extension decreasing our contribution to climate change.

2 months ago

I dont know enough about how to contain or curtain recent flooding but I would like to see experts working on this immediately

2 months ago

barriers to preserve the salt marshes from sea level rise

2 months ago

None at this time.

2 months ago

Electrical infrastructure and investment

2 months ago

Construction of Big River Reservoir, detention ponds and creating additional capture areas for storm water

2 months ago

RIDOT current Resilience Improvement Plan effort is identifying infrastructure projects of highest vulnerability to sea level rise.

2 months ago

natural and nature-based shoreline resilience measures

2 months ago

Storm water mitigation projects around the state to prevent flooding on evacuation routes

2 months ago

No

2 months ago

Culvert studies in all communities

2 months ago

Is there anything else concerning Hazard Mitigation that you would like us to know?

Continuously exercise and rewrite all hazards plans to move forward.

one month ago

Yes as to all the land being destroyed for sola as well as damage will farms are doing to out fish

one month ago

Plan now before emergency events occur. Some problems may even be preventable if the infrastructure is improved.

one month ago

I'd like to know how to pre plan for an emergency/dangerous event before hand to be prepared.

one month ago

Yes, stopping windfarms which are killing our ocean life. Littering land with solar farms. Concern for outbreaks of diseases that have nit been an issue in our Country for years now returning due to the influx of several million illegal immigrants.

one month ago

Pre planning rather than waiting for imminent danger

one month ago

See above

one month ago

Not knowing the current state of hazard mitigation I don't feel comfortable commenting

one month ago

What is going to happen to all the waste from the solar farms when they are no longer profitable

one month ago

Yes. Way too many, deep pot holes these days that could impact driving in RI and a behind the times infrastructure. Another is the big push for solar and the digging to put turbines in the ocean which are causing upset with whales and dolphins way of communicating through sound.

one month ago

Stop the solar building in our state

one month ago

Stop beating the climate change drum and actually do something tangible to decrease the current hazards.

one month ago

Update flood zones for climate change impacts while biting the bullet on gerrymandered flood zones!!! Increase housing code violation penalties.

one month ago

No

one month ago

No

one month ago

No

one month ago

Democrats from reproducing

2 months ago

Education across populations as to how to mitigate dangers and protect communities

2 months ago

I think that a lot of conversation is focused on the coast and sea level rising, a lot of communities that are along rivers also have their share of problems and poorer communities are left out of conversations. They are most in need of services during these severe events, the elderly, disabled, etc.

2 months ago

I am concerned about the transmission of diseases like measles, rubella, etc given vaccine hesitancy in sub populations of the state and surrounding areas,

2 months ago

climate change continues to negatively affect our state.

2 months ago

General info on cybersecurity to all citizens

2 months ago

no

2 months ago

No

2 months ago

Nope

2 months ago

Cybersecurity should include issues of identity theft

2 months ago

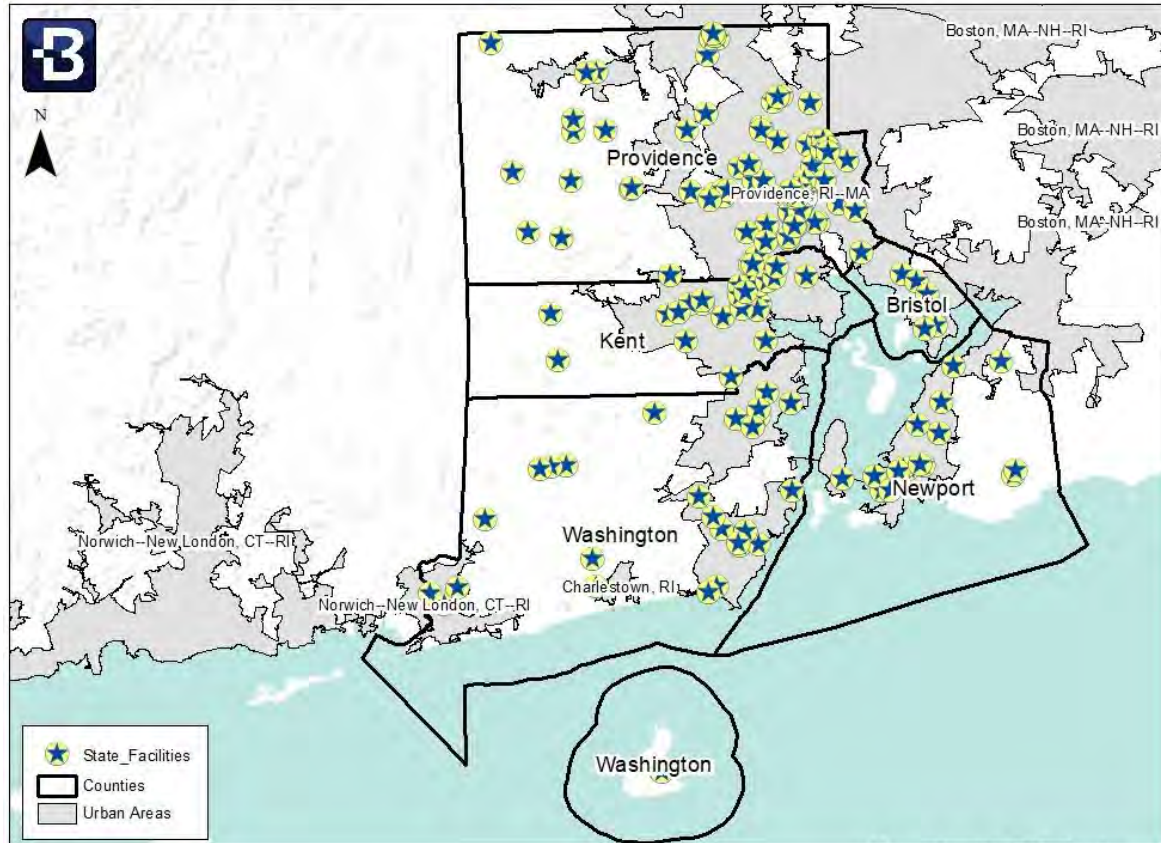
No

2 months ago

Appendix D – State Facility, Community Lifeline, and Critical Facility Vulnerability Maps

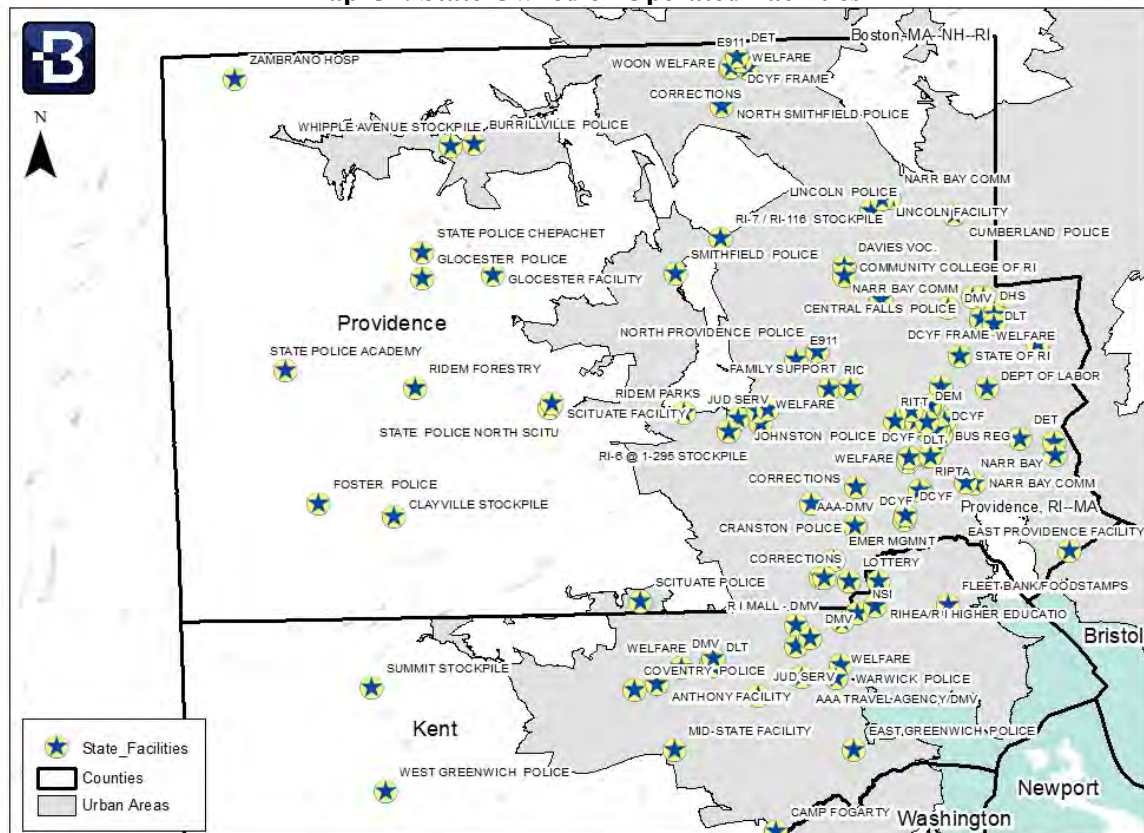
Appendix C1 State Owned or Operated Facility Maps

Map C1: State Owned or Operated Facilities



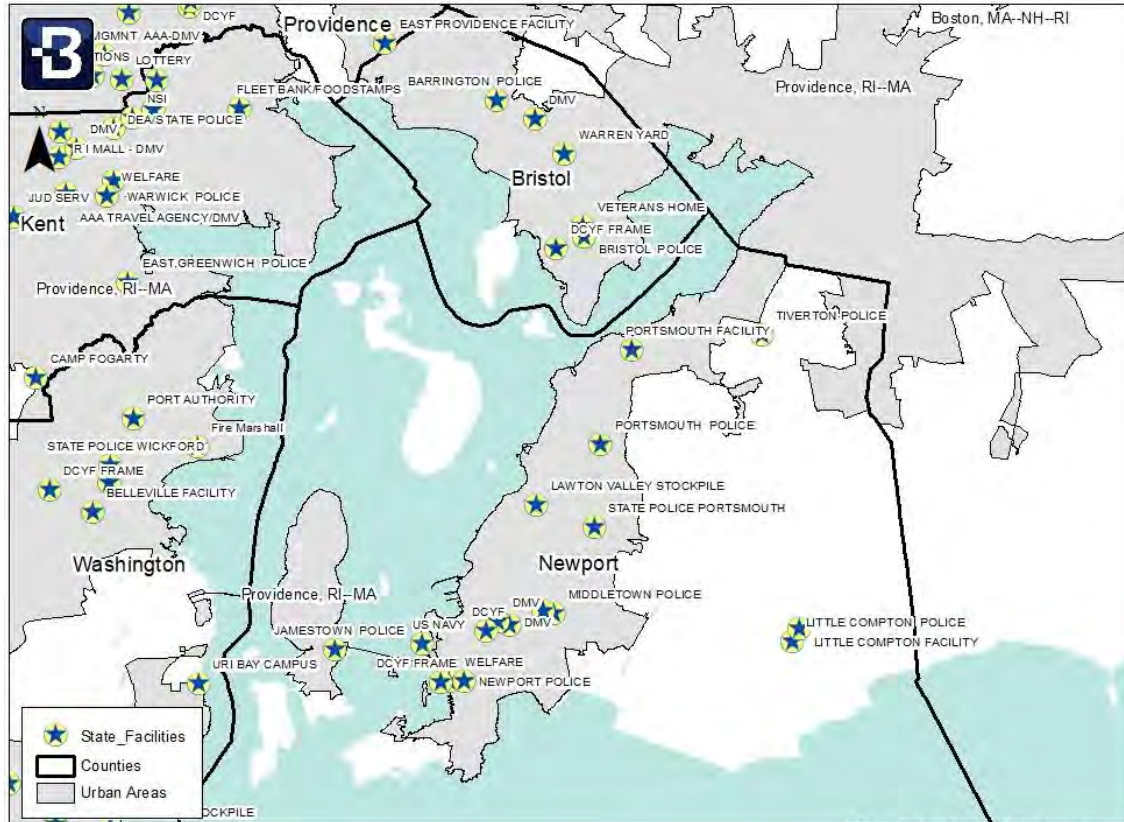
Map produced by BOLD planning.

Map C2: State Owned or Operated Facilities



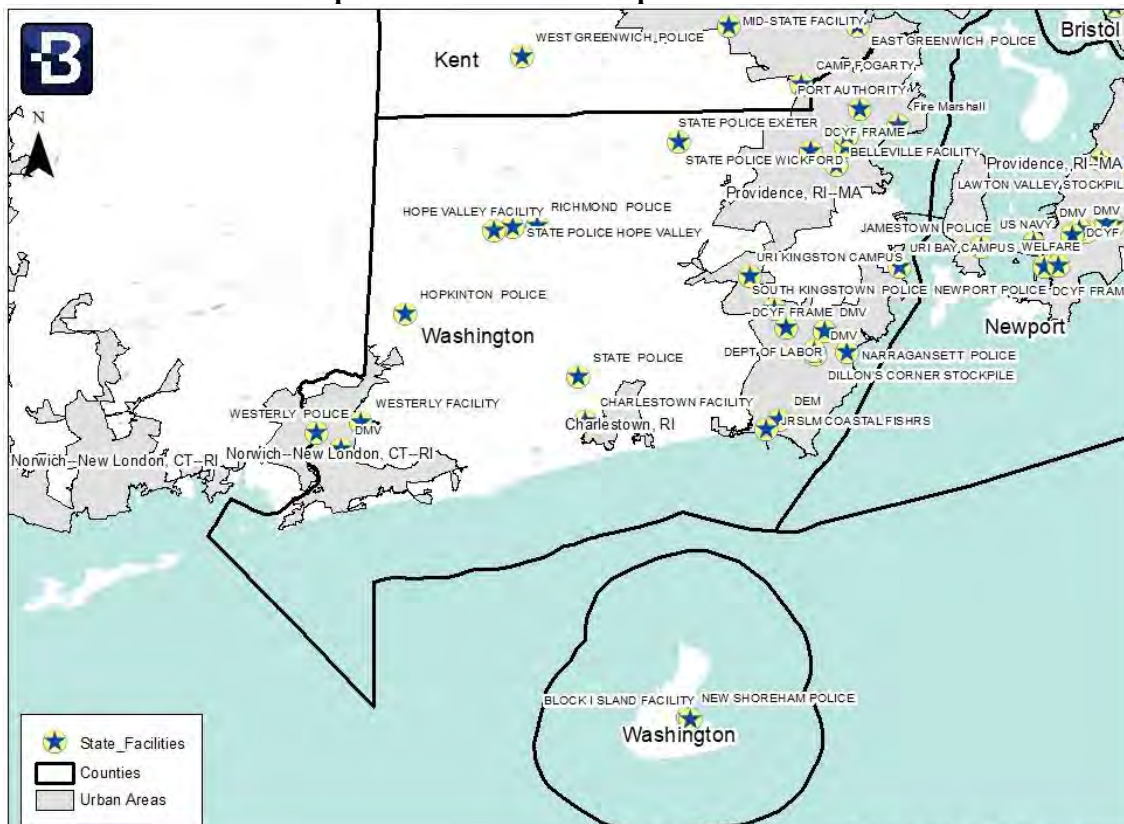
Map produced by BOLD planning.

Map C3: State Owned or Operated Facilities



Map produced by BOLD planning.

Map C4: State Owned or Operated Facilities



Map produced by BOLD planning.

Map of Rhode Island showing State Facilities, Counties, and Urban Areas.

Legend:

- State Facilities (Yellow Star)
- Counties (Black Outline)
- Urban Areas (Grey Shaded Area)

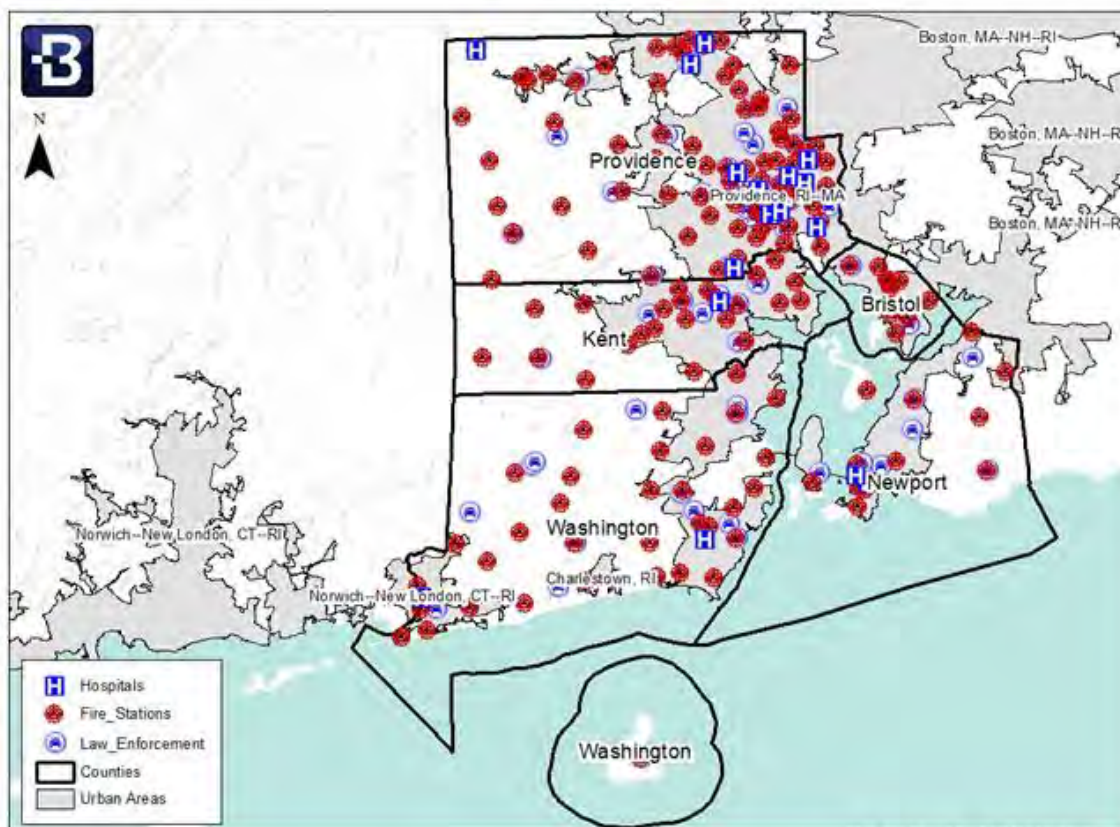
Key Locations and Facilities:

- Providence:** Providence, RI-MA, Providence Police, Foster Police, Clayville Stockpile, Scituate Facility, State Police North Scituate, Jud Serv, Welfare, Ried, Dem, Ritt, DCYF, Bus Reg, Det, Corrections, Welfare, RFTA, Narr Bay, Narr Bay Comm, East Providence Facility, Fleet Bank Foodstamps, Community College of RI, DMV, DEA State Police, Welfare, DMV, DLT, Coventry Police, Anthony Facility, Jud Serv, Welfare, Warwick Police, AAA Travel Agency/DMV, East Greenwich Police, MD State Facility, Camp Fogarty, Port Authority, Fire Marshall, State Police Wickford, DCYF Frame, Belleville Facility, North Kingstown Police, URI Kingston Campus, URI Bay Campus, Jamestown Police, Hope Valley Facility, Richmond Police, State Police Hope Valley.
- Kent:** Kent, West Greenwich Police, Summit Stockpile.
- Washington:** Washington, State Police Exeter.
- Newport:** Newport, Fire Marshall.

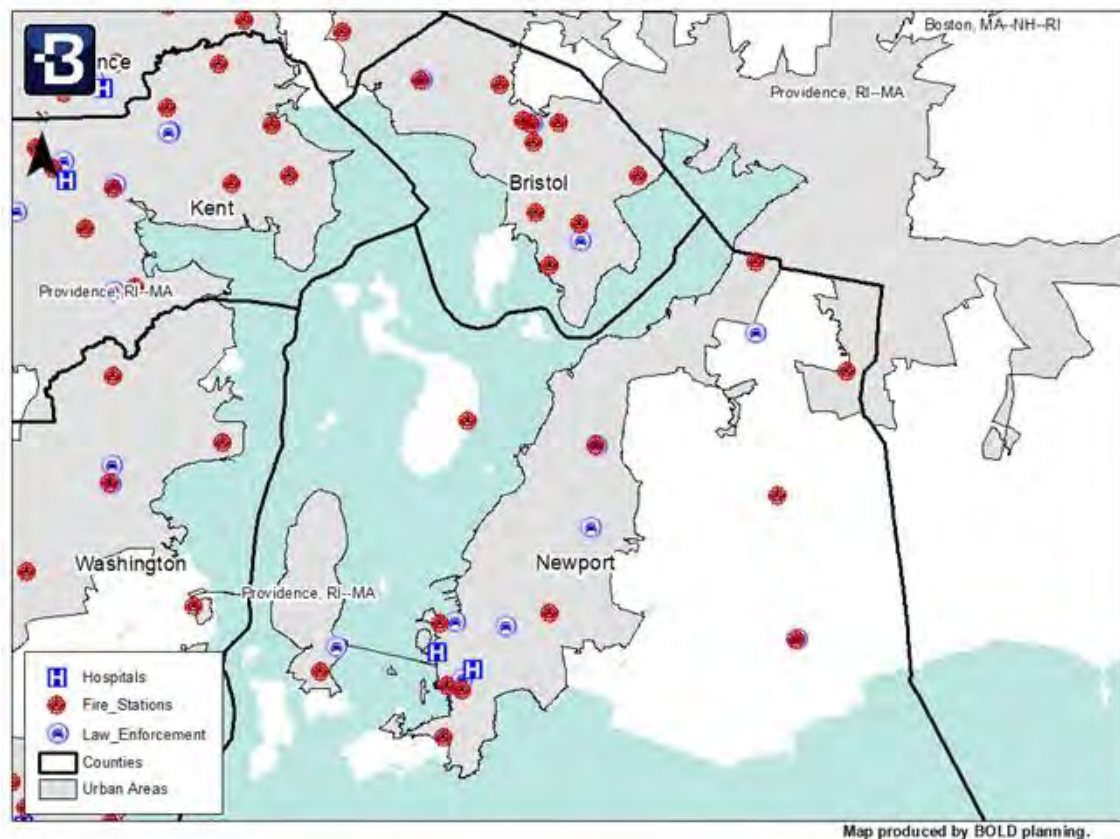
Map produced by BOLD planning.

Appendix C2 Critical Facility Maps

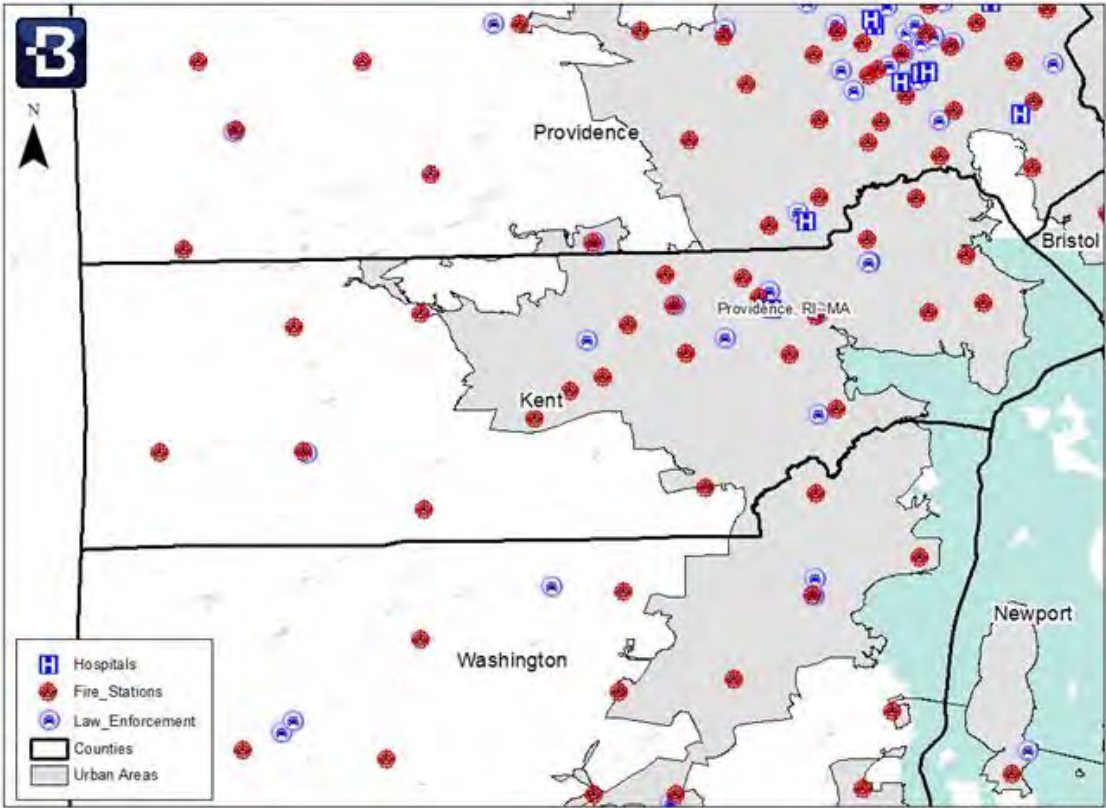
Map C6: Critical Facilities, Hospitals, Fire Stations, and Law Enforcement Centers



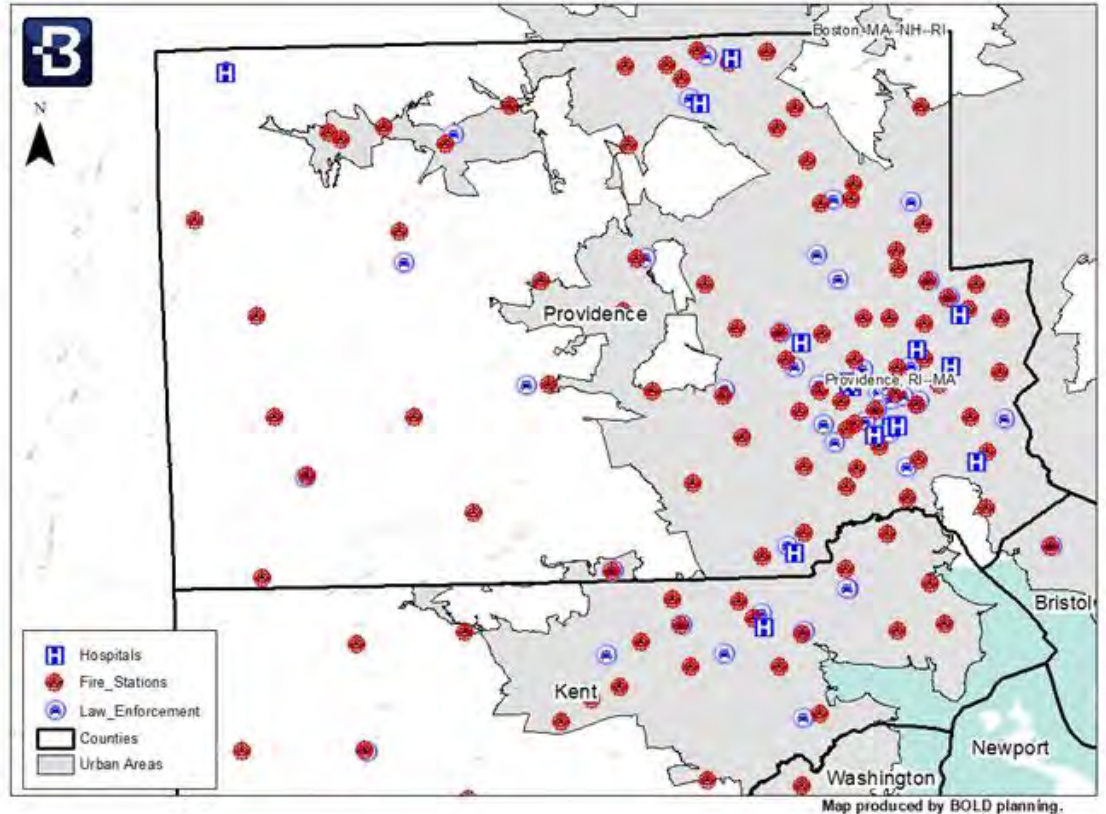
Map C7: Critical Facilities, Hospitals, Fire Stations, and Law Enforcement Centers



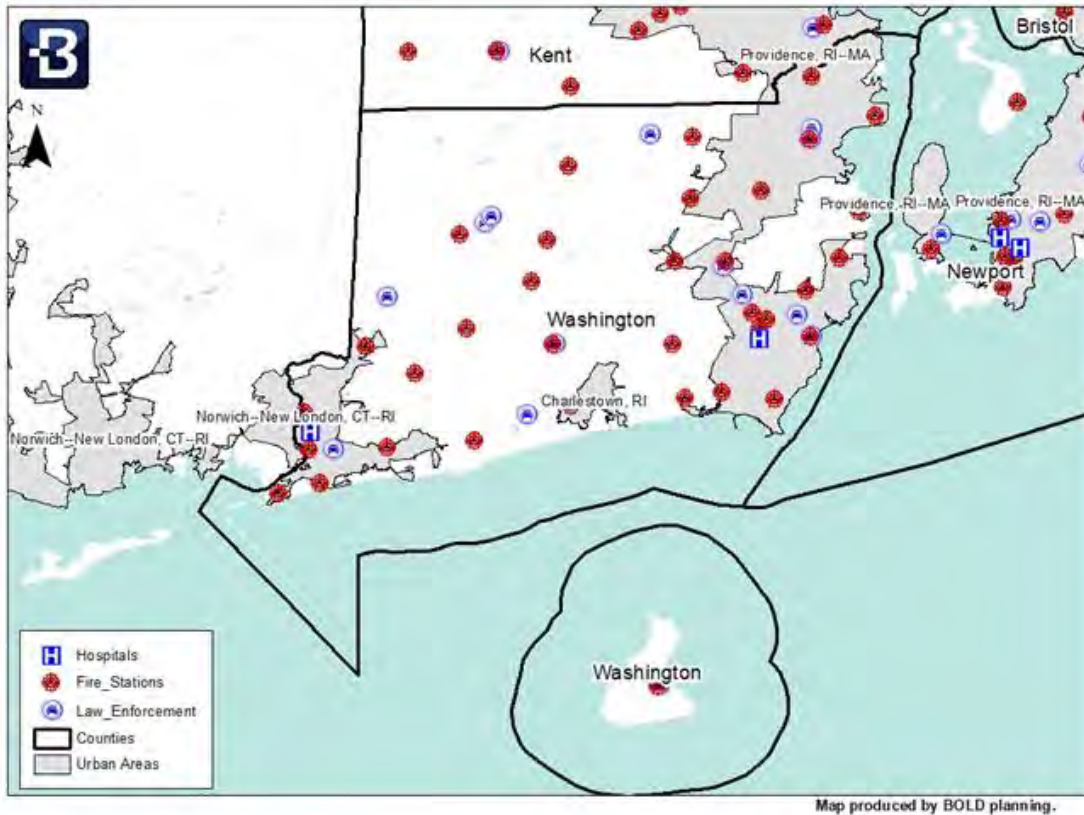
Map C8: Critical Facilities, Hospitals, Fire Stations, and Law Enforcement Centers



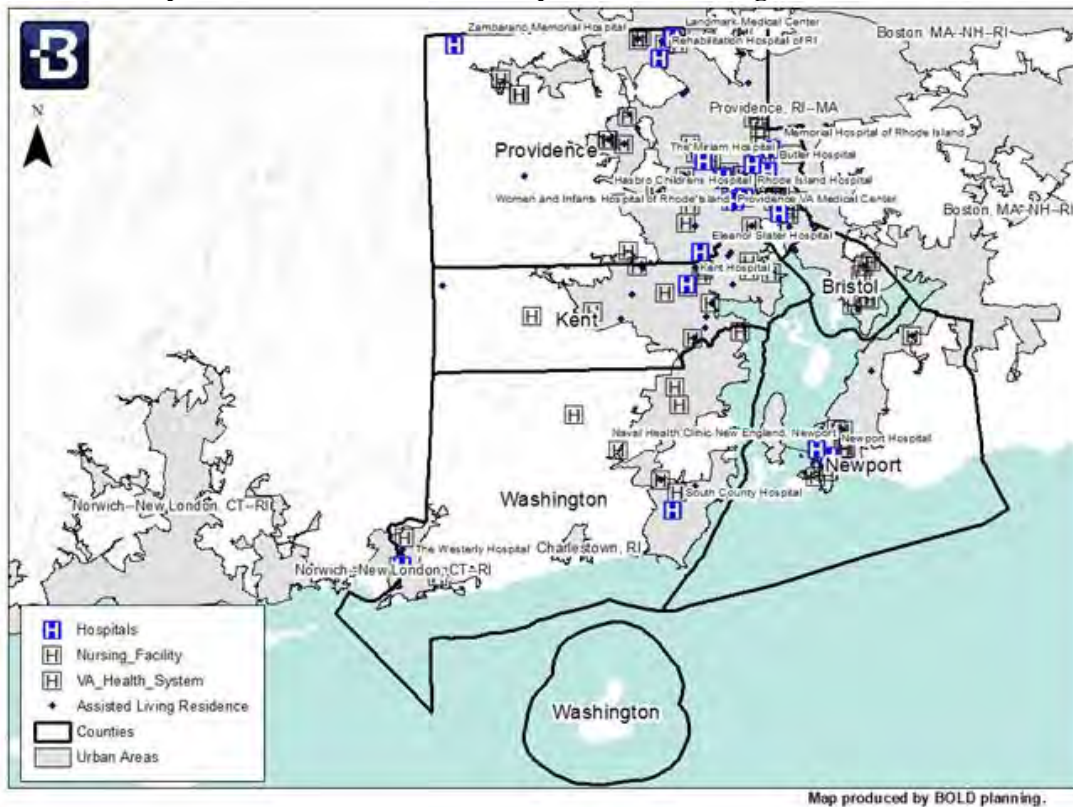
Map C: Critical Facilities, Hospitals, Fire Stations, and Law Enforcement Centers



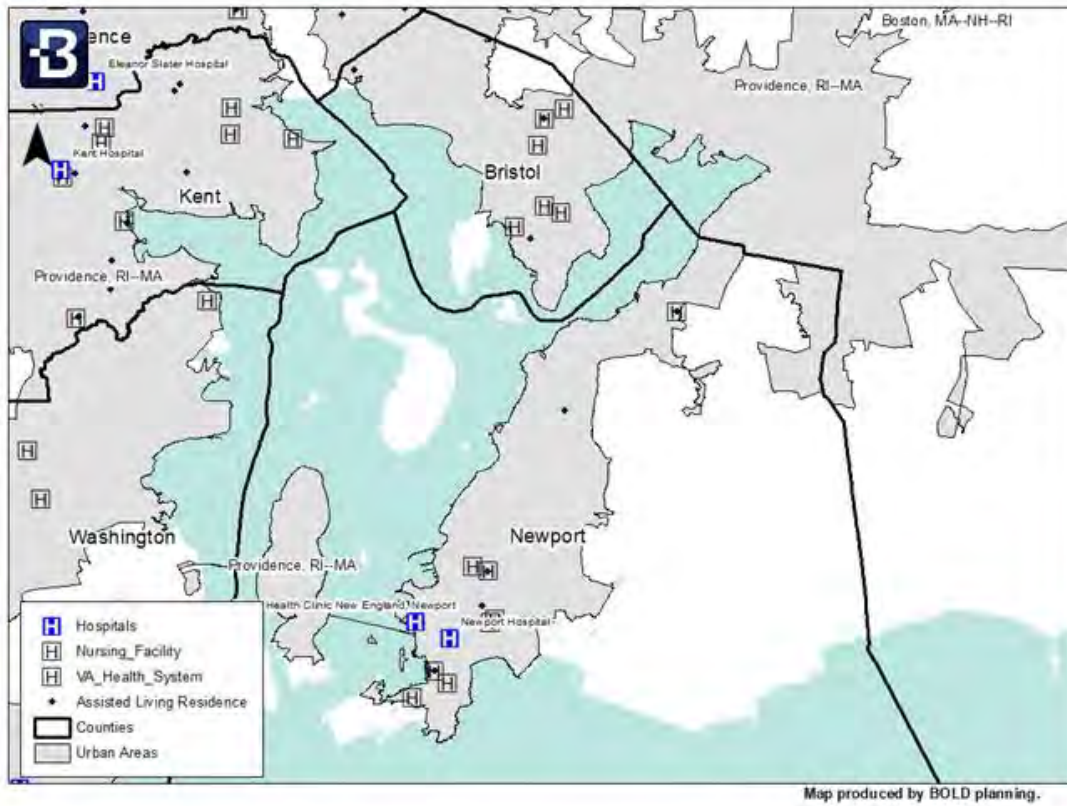
Map C10: Critical Facilities, Hospitals, Fire Stations, and Law Enforcement Centers



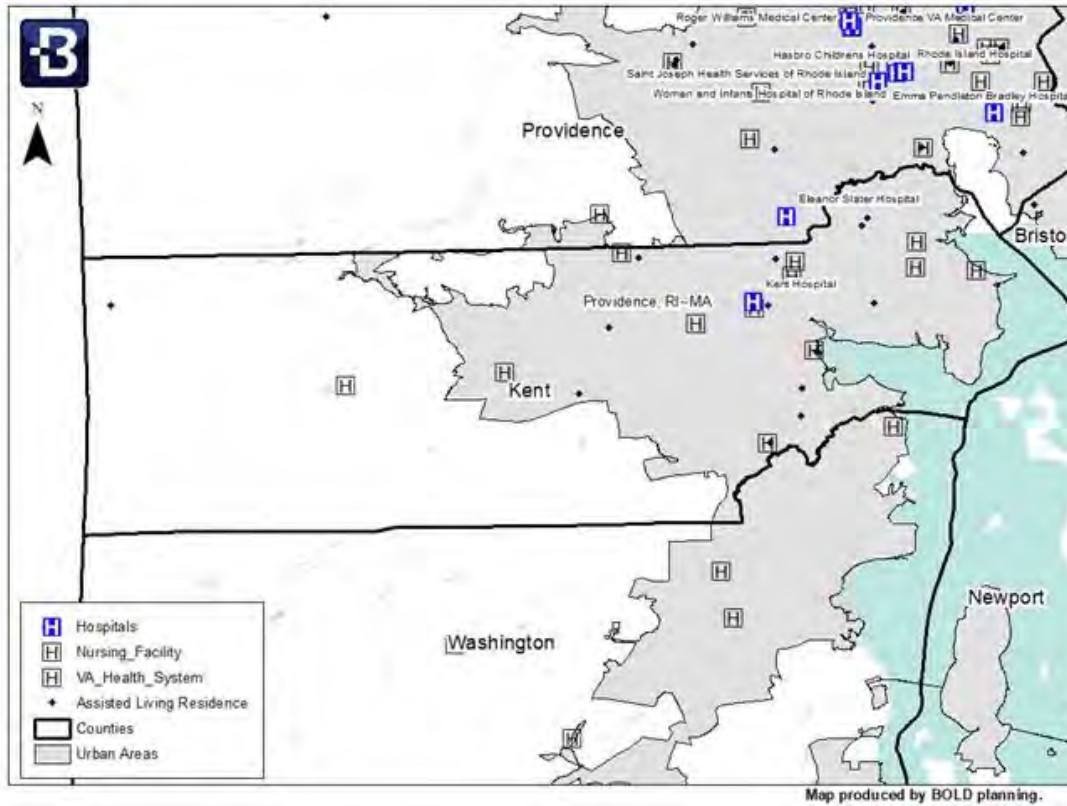
Map C11: Critical Facilities, Hospitals and Nursing Home Facilities



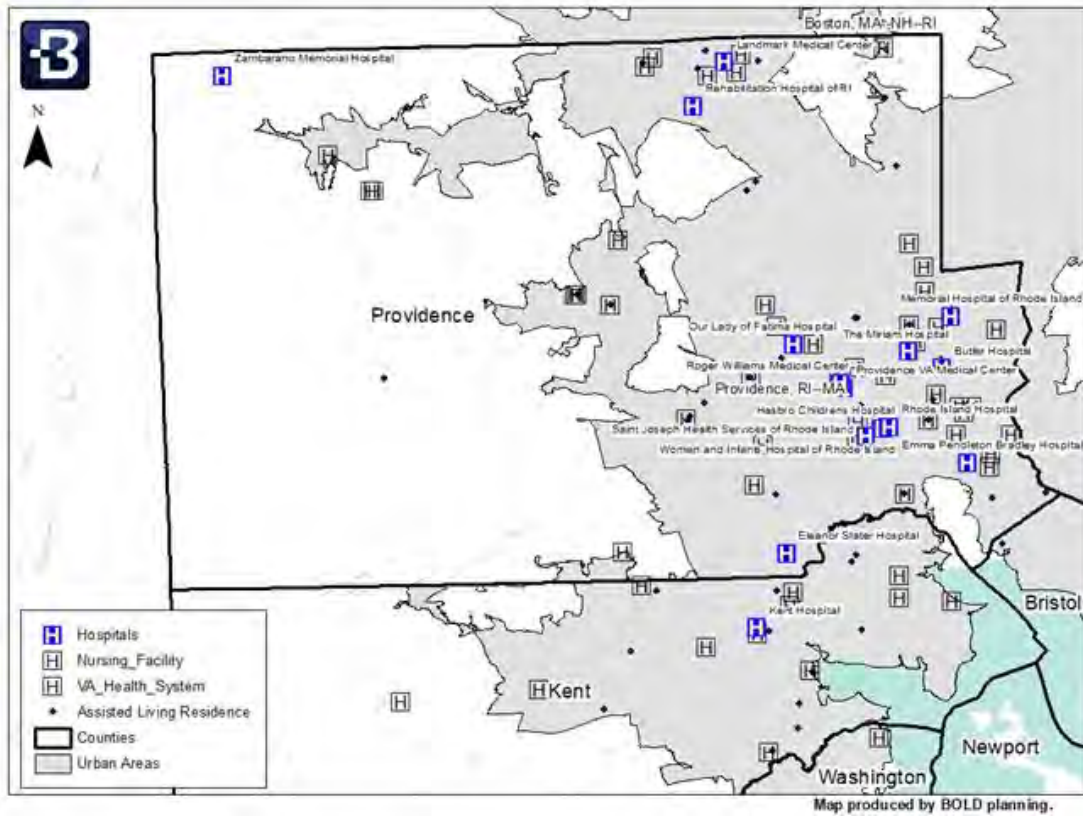
Map C11: Critical Facilities, Hospitals and Nursing Home Facilities



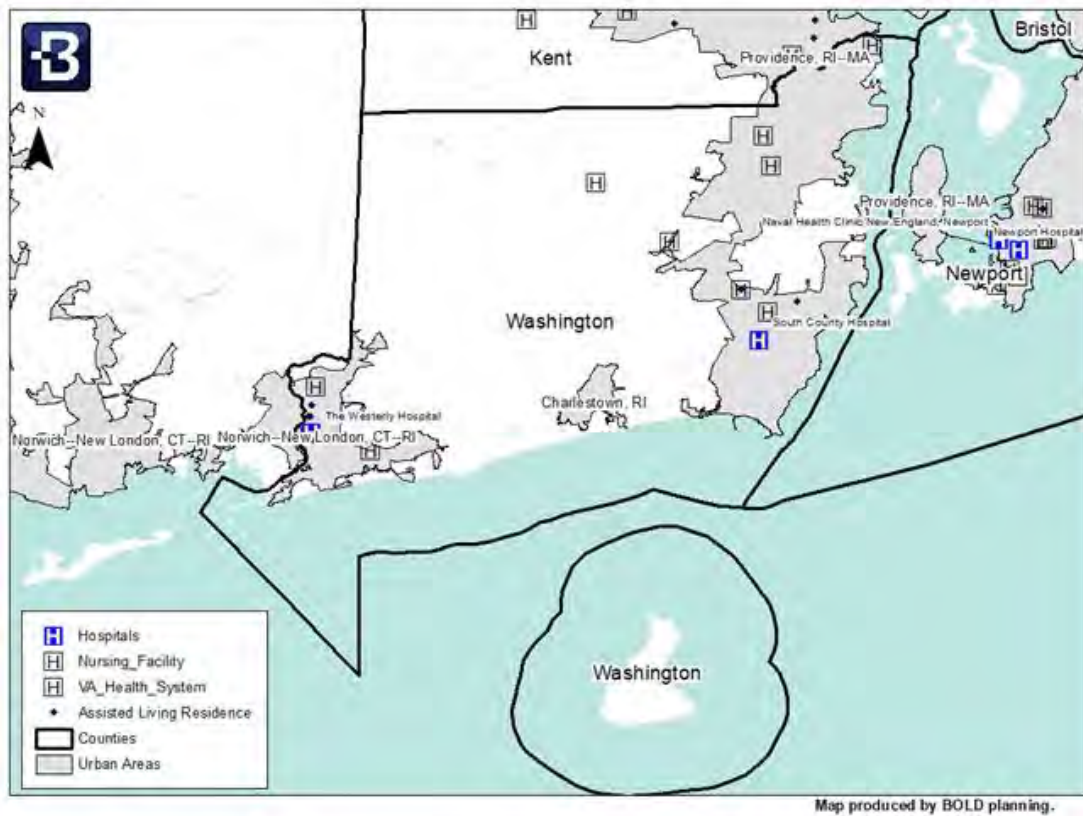
Map C12: Critical Facilities, Hospitals and Nursing Home Facilities



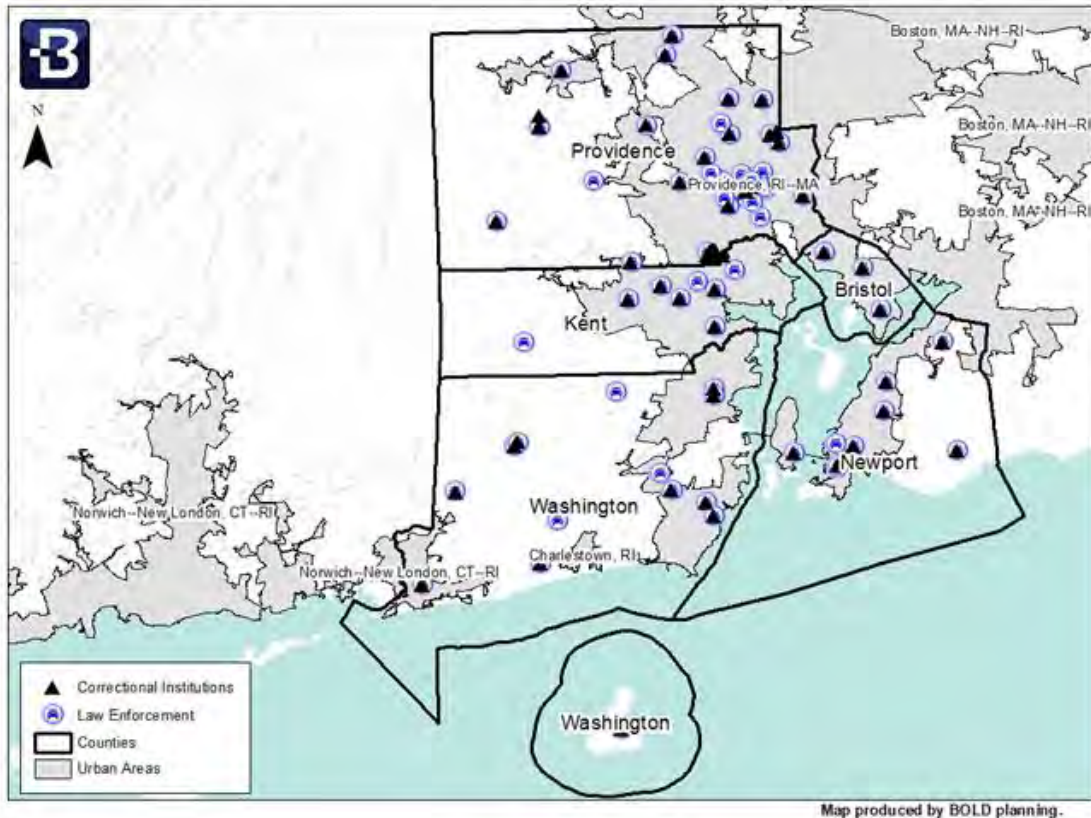
Map C13: Critical Facilities, Hospitals and Nursing Home Facilities



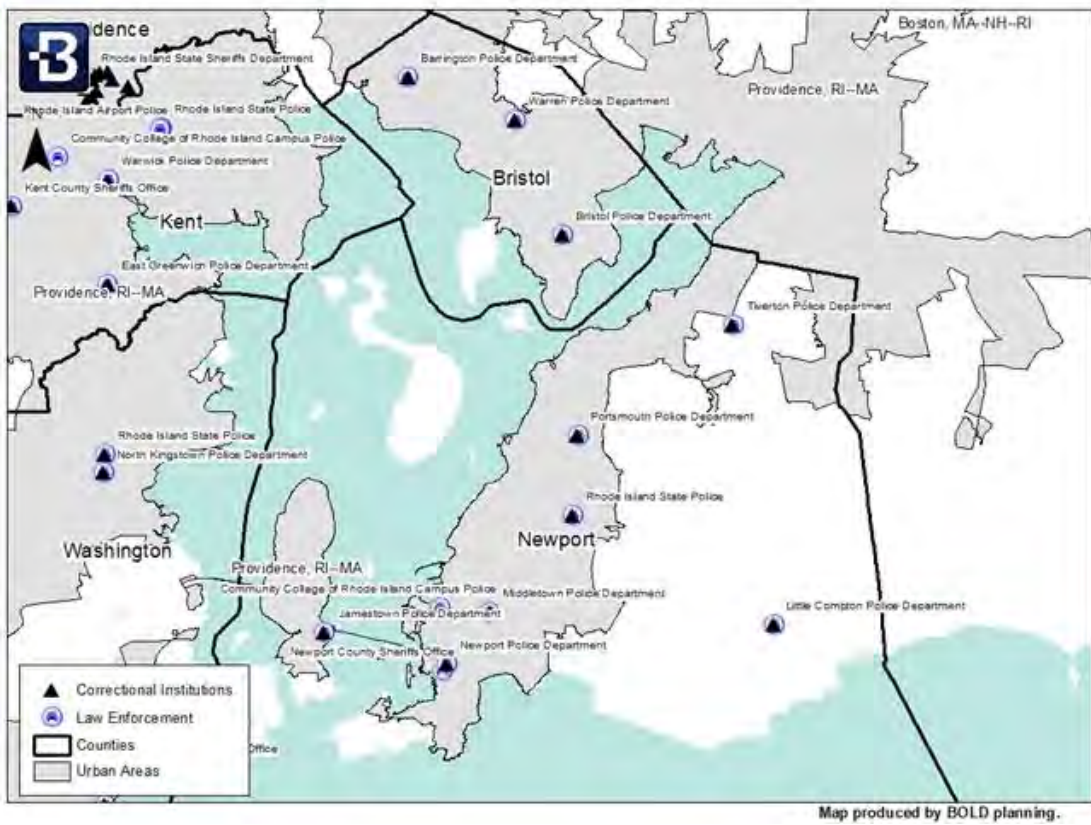
Map C14: Critical Facilities, Hospitals and Nursing Home Facilities



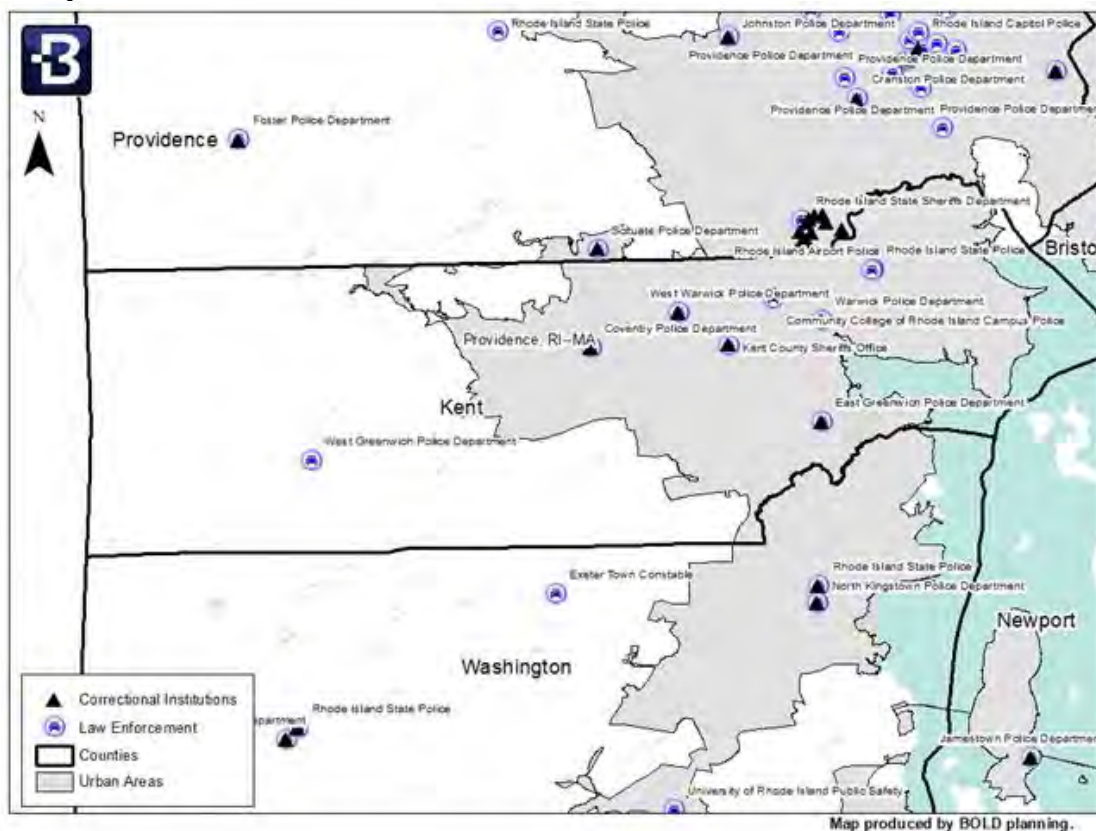
Map C15: Critical Facilities, Law Enforcement Centers and Correctional Institutions



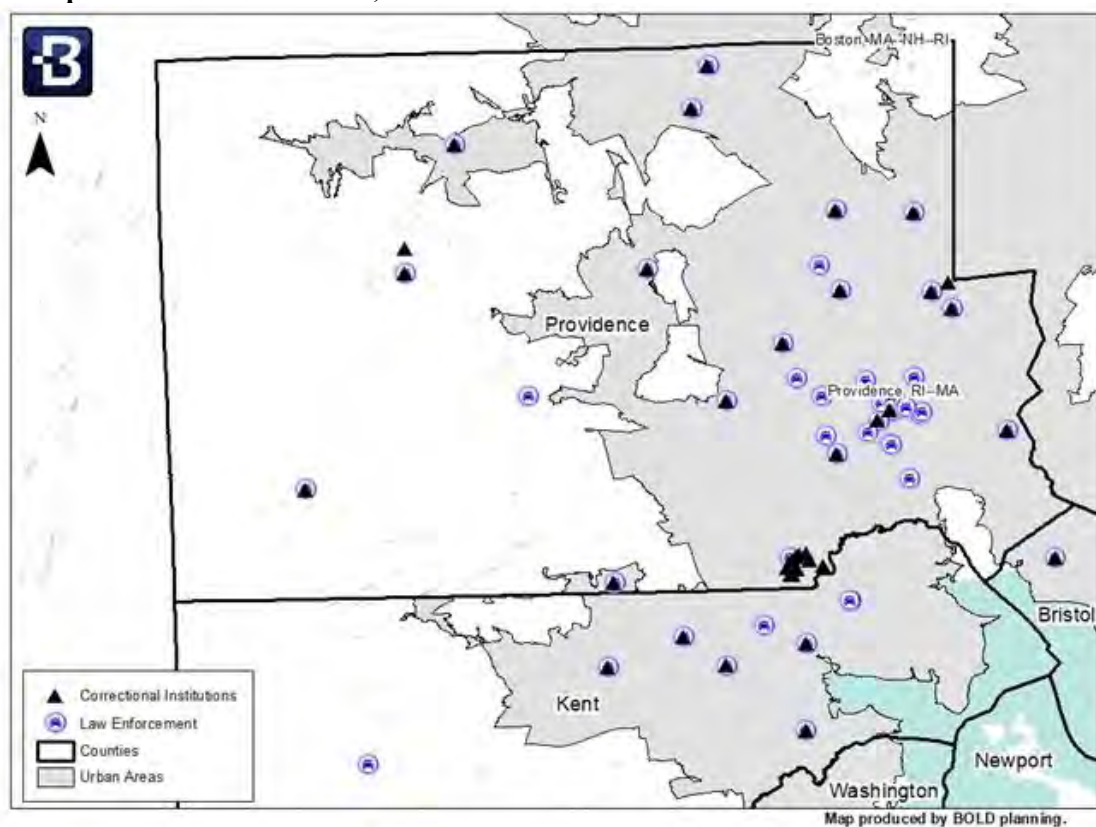
Map C16: Critical Facilities, Law Enforcement Centers and Correctional Institutions



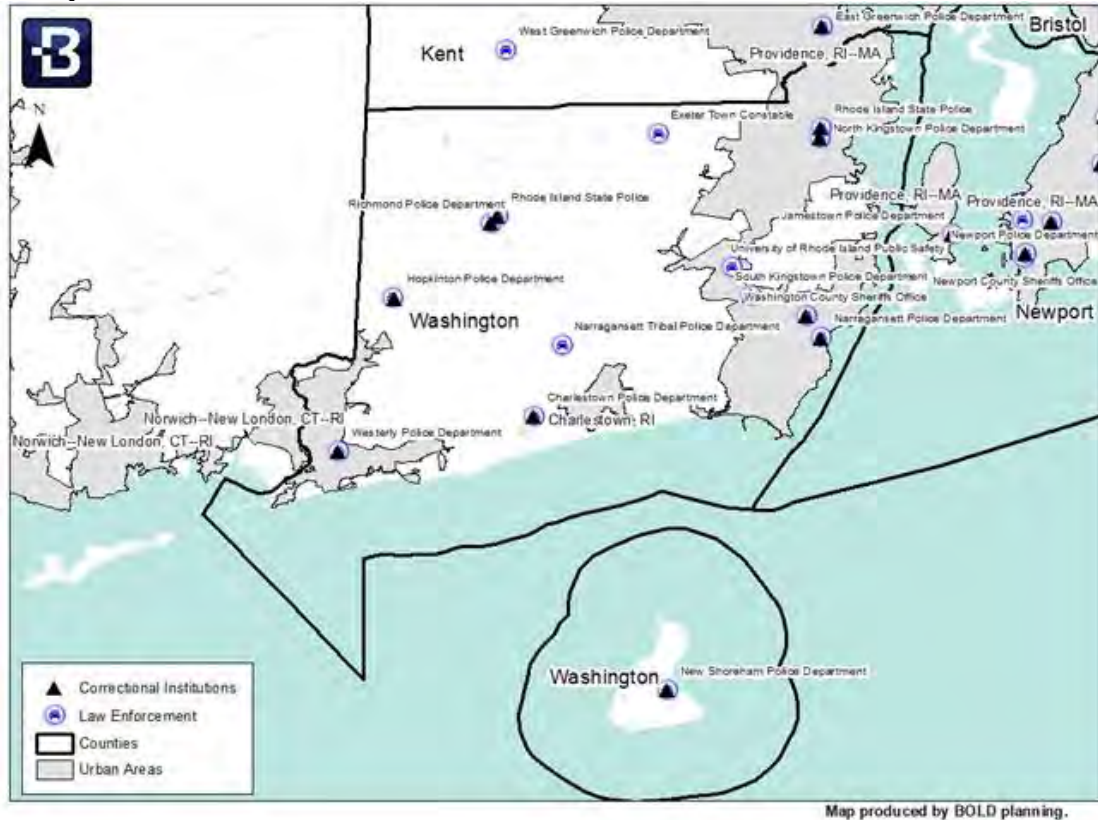
Map C17: Critical Facilities, Law Enforcement Centers and Correctional Institutions



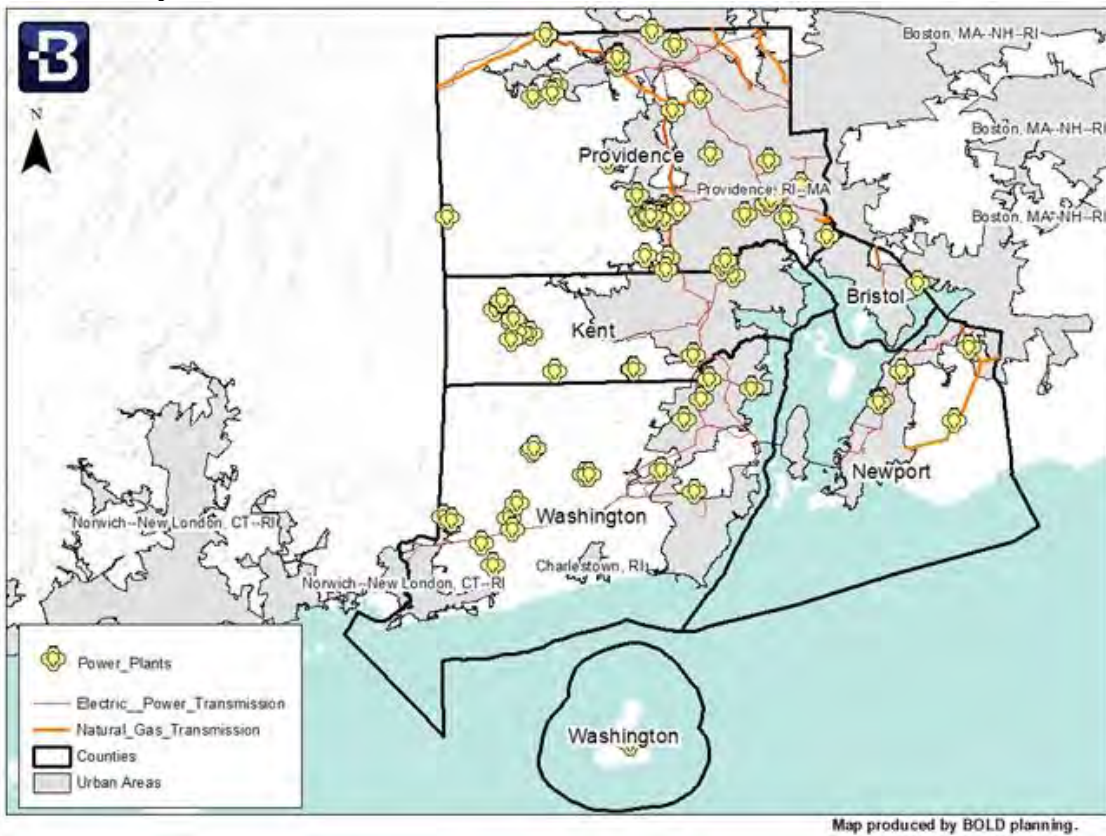
Map C18: Critical Facilities, Law Enforcement Centers and Correctional Institutions



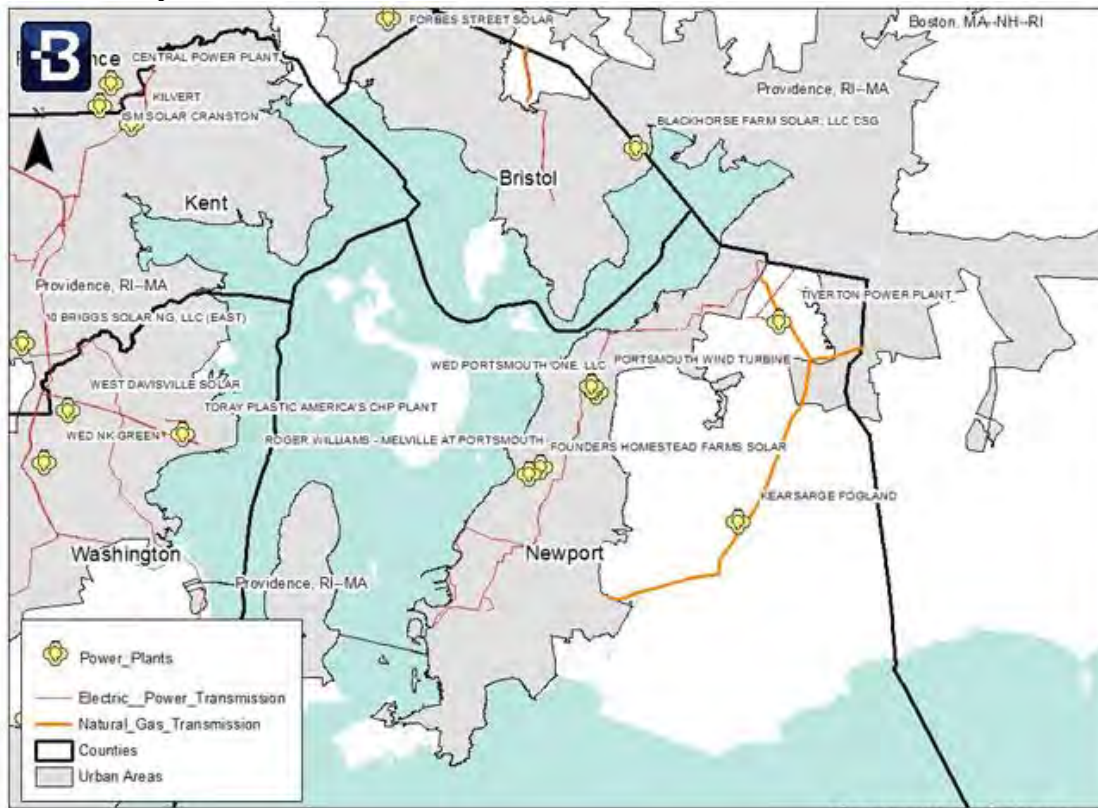
Map C19: Critical Facilities, Law Enforcement Centers and Correctional Institutions



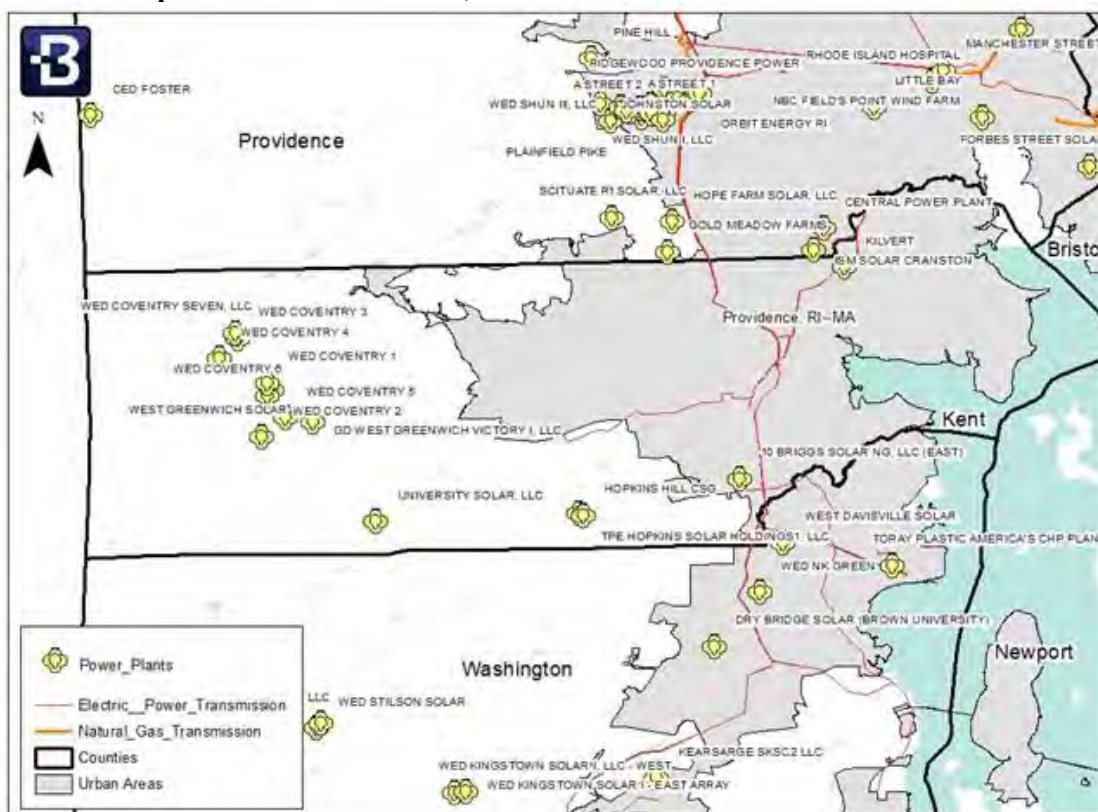
Map C20: Critical Facilities, Electrical and Natural Gas Infrastructure



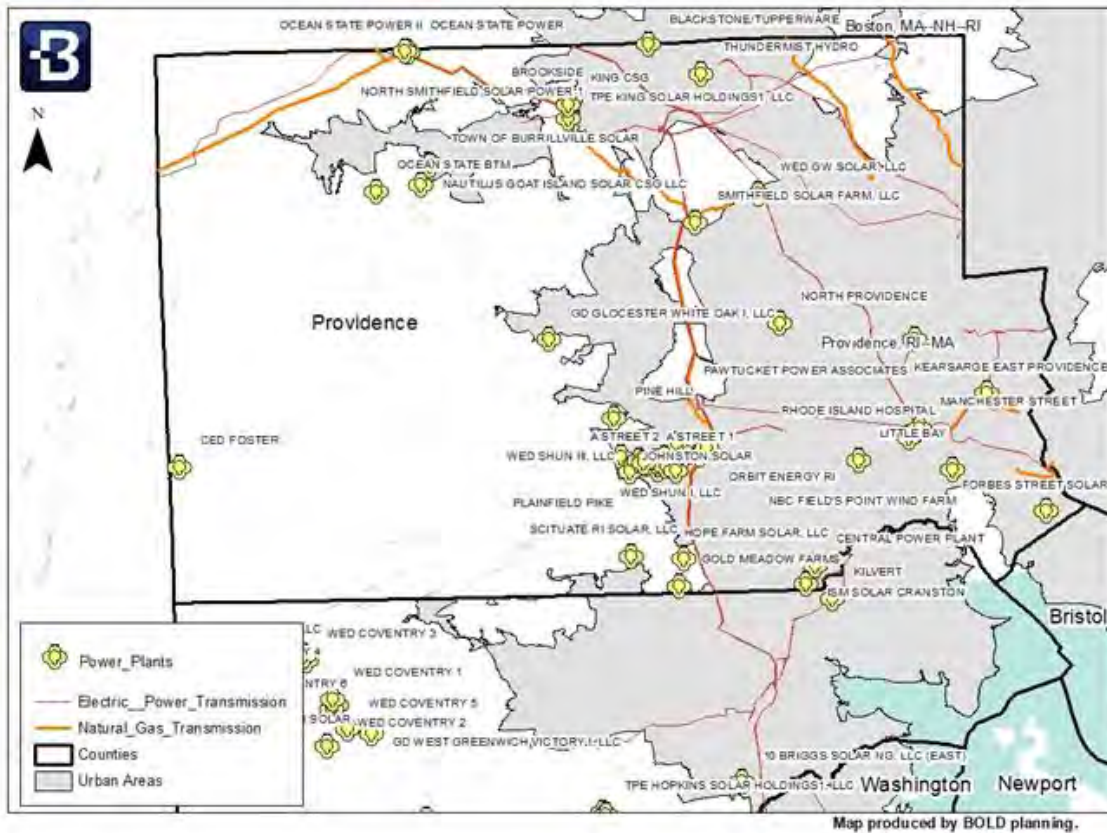
Map C21: Critical Facilities, Electrical and Natural Gas Infrastructure



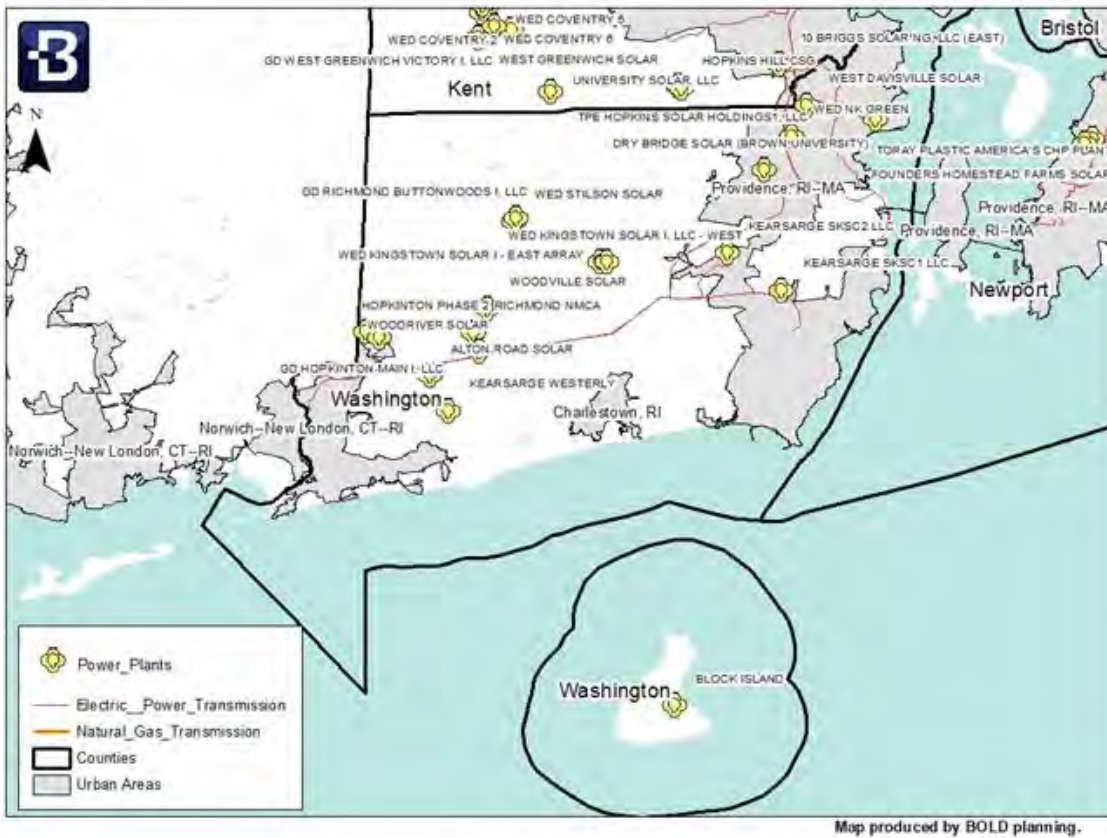
Map C22: Critical Facilities, Electrical and Natural Gas Infrastructure



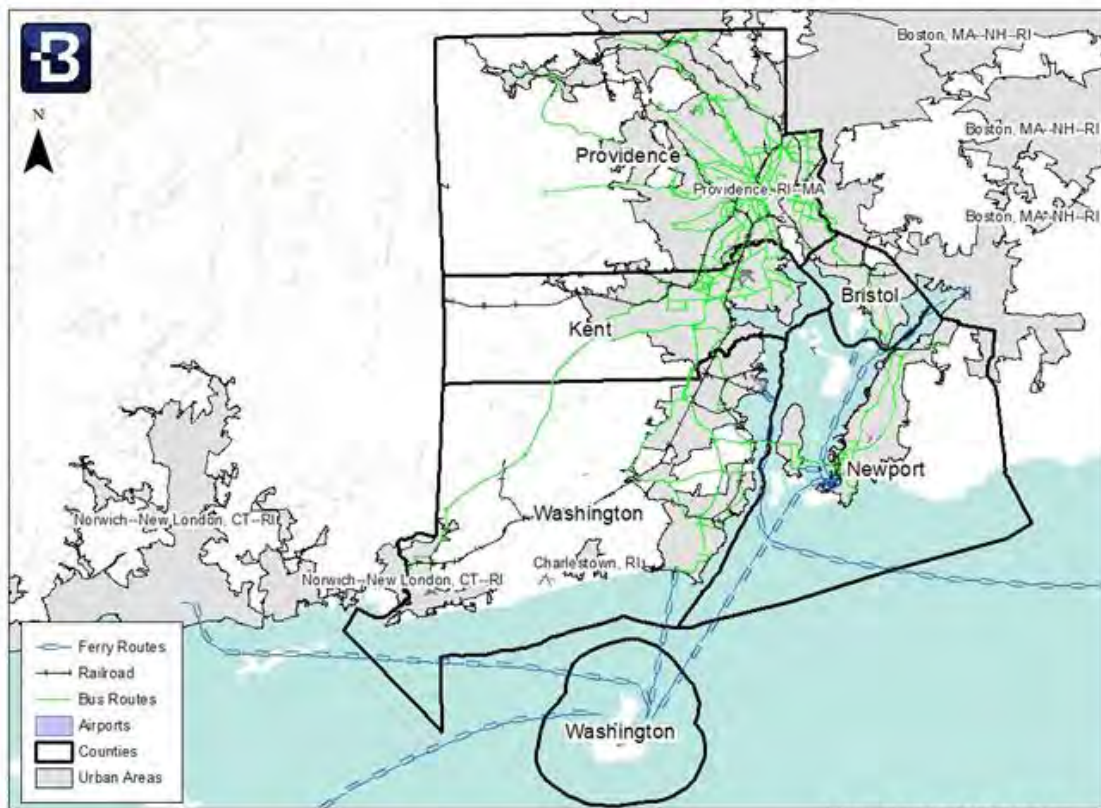
Map C23: Critical Facilities, Electrical and Natural Gas Infrastructure



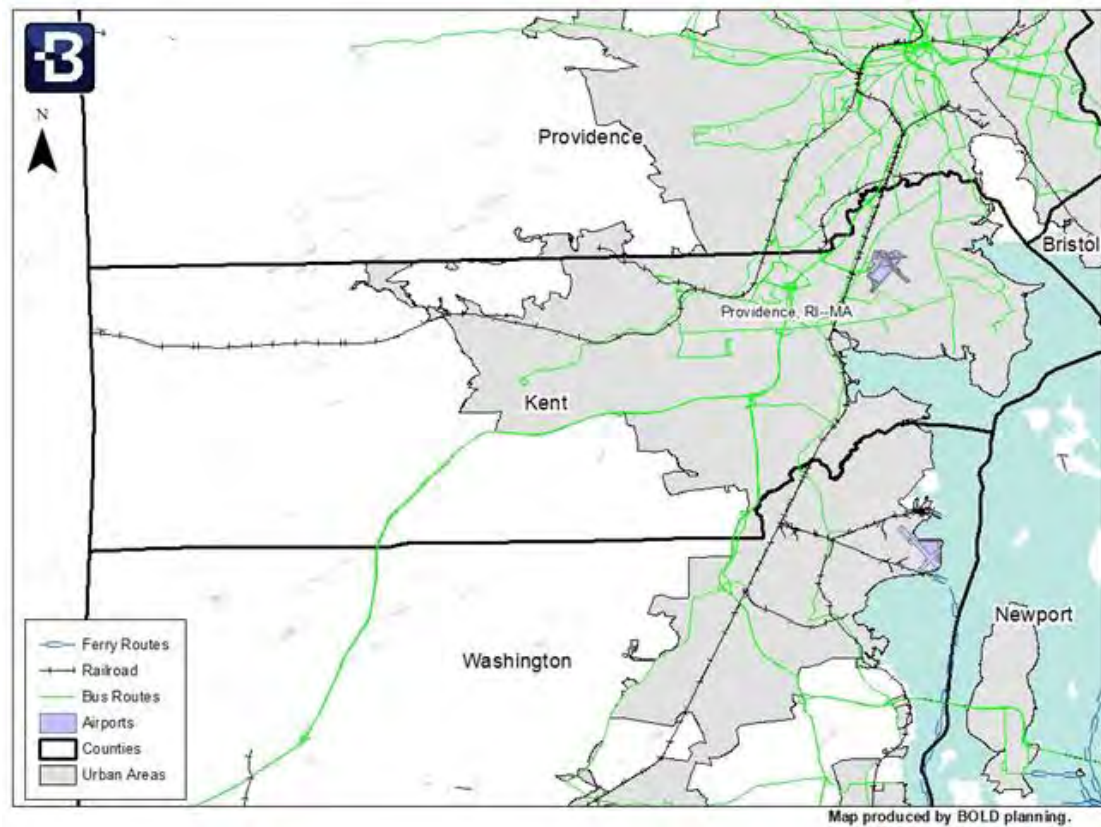
Map C24: Critical Facilities, Electrical and Natural Gas Infrastructure



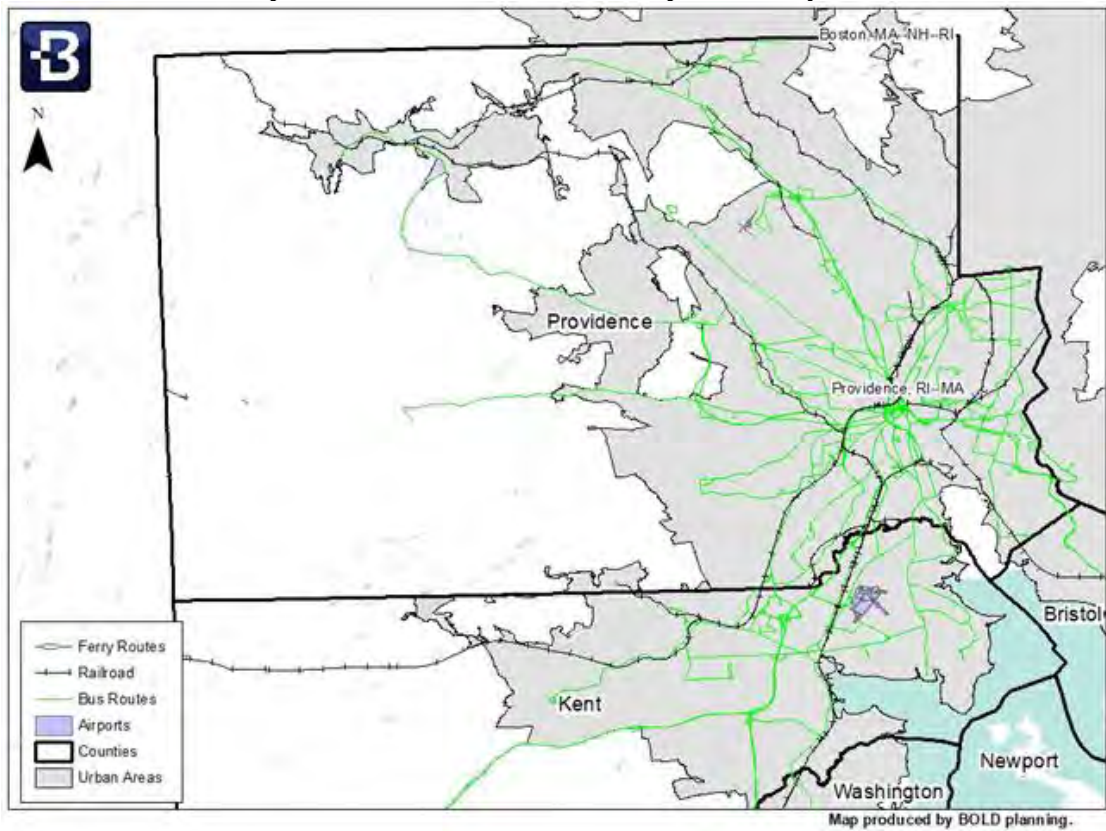
Map C25: Critical Facilities, Transportation Systems



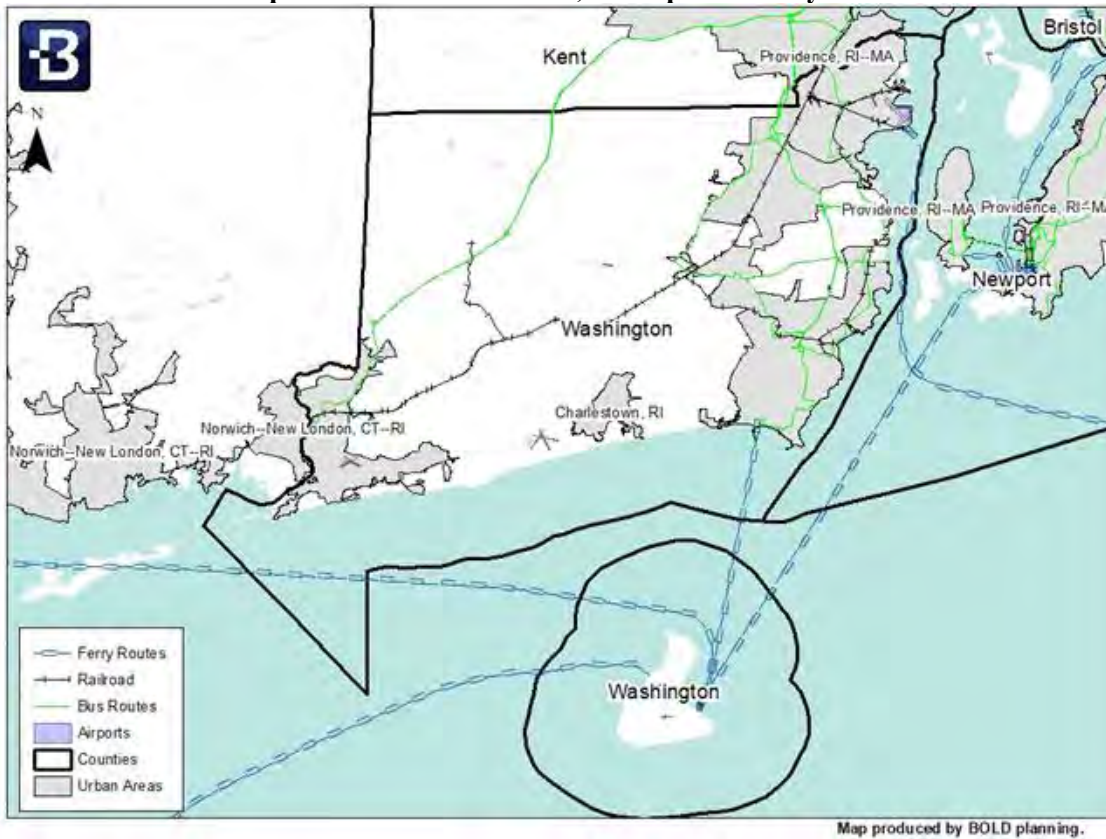
Map C26: Critical Facilities, Transportation Systems



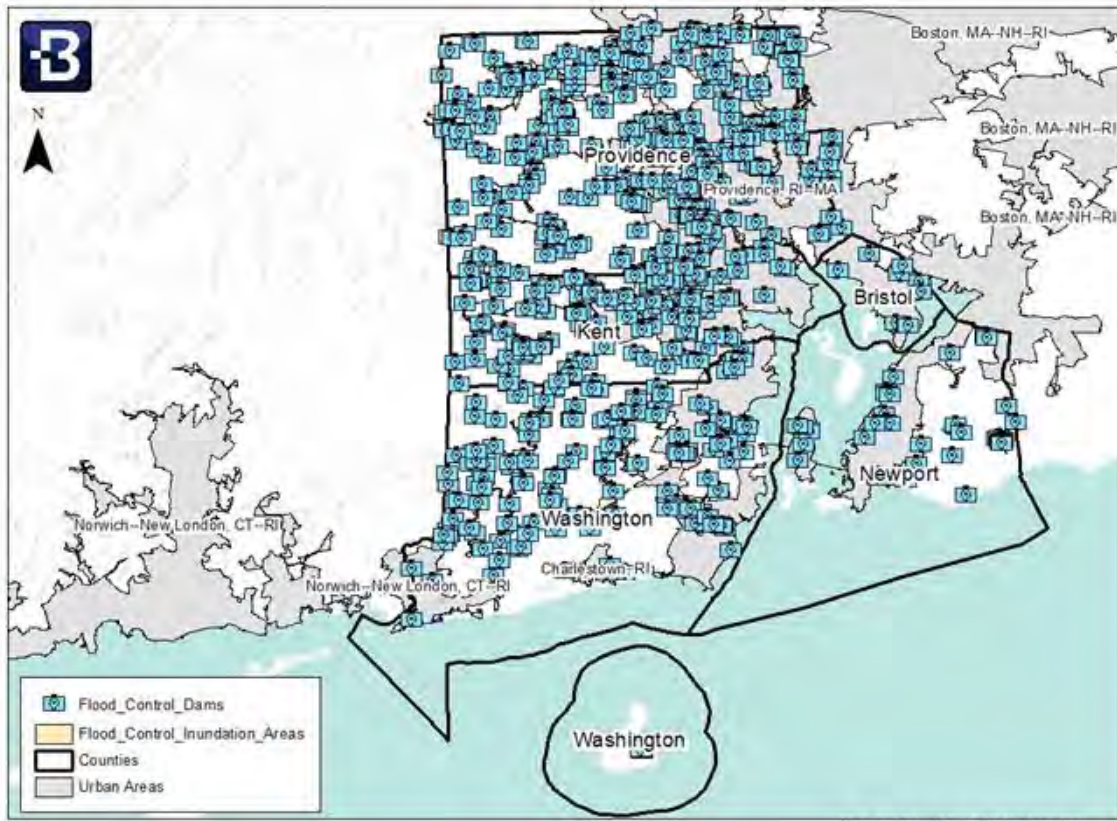
Map C27: Critical Facilities, Transportation Systems



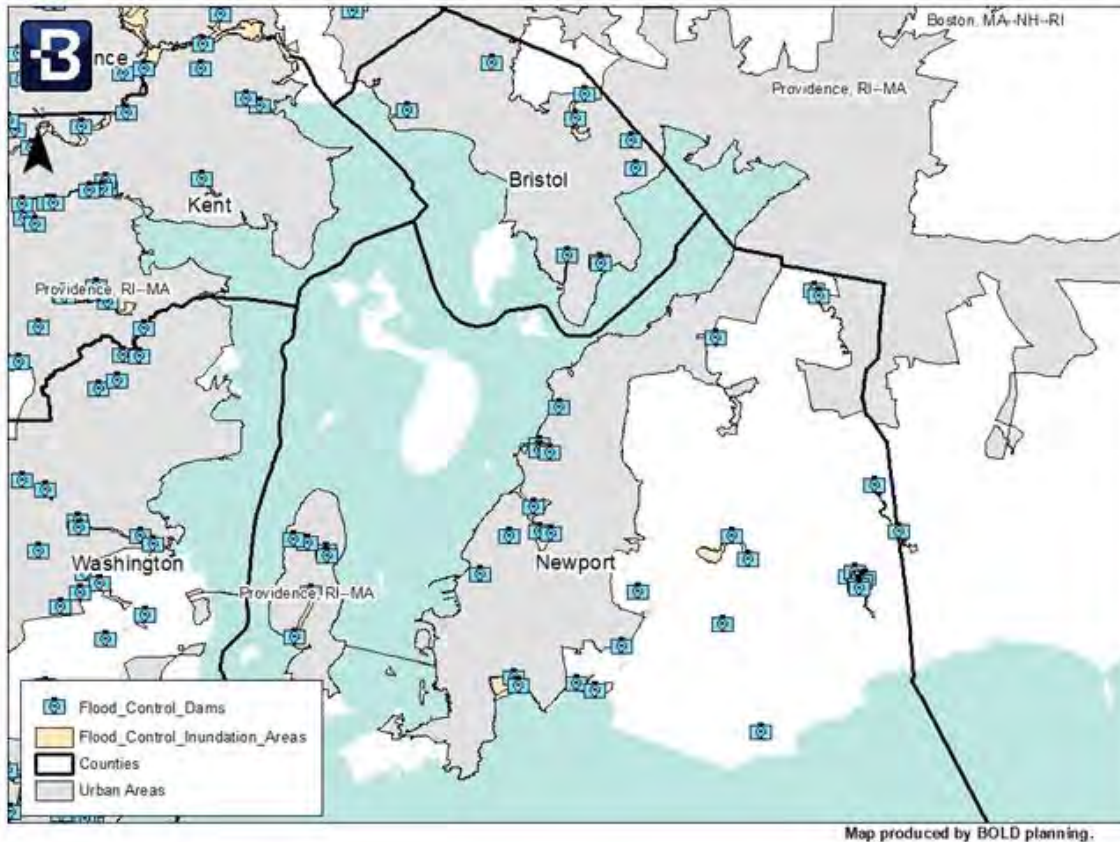
Map C28: Critical Facilities, Transportation Systems



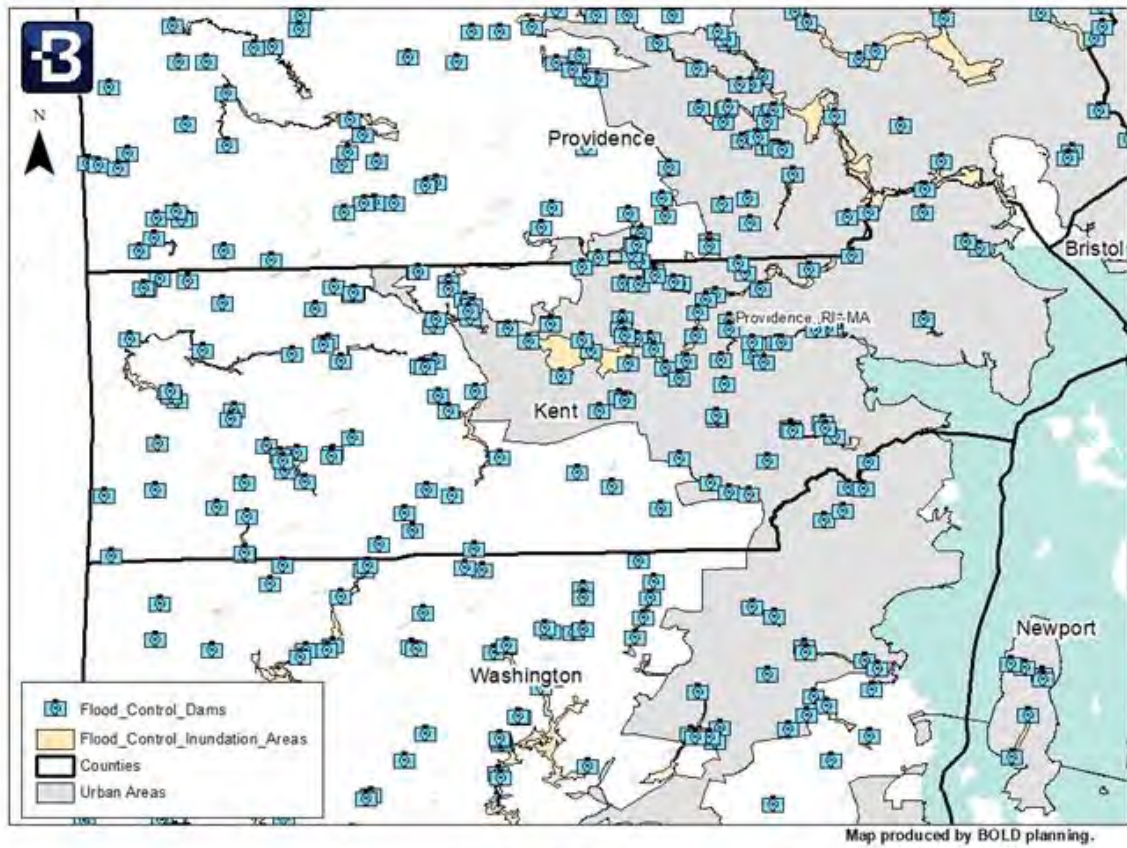
Map C29: Critical Facilities, Flood Control Dams



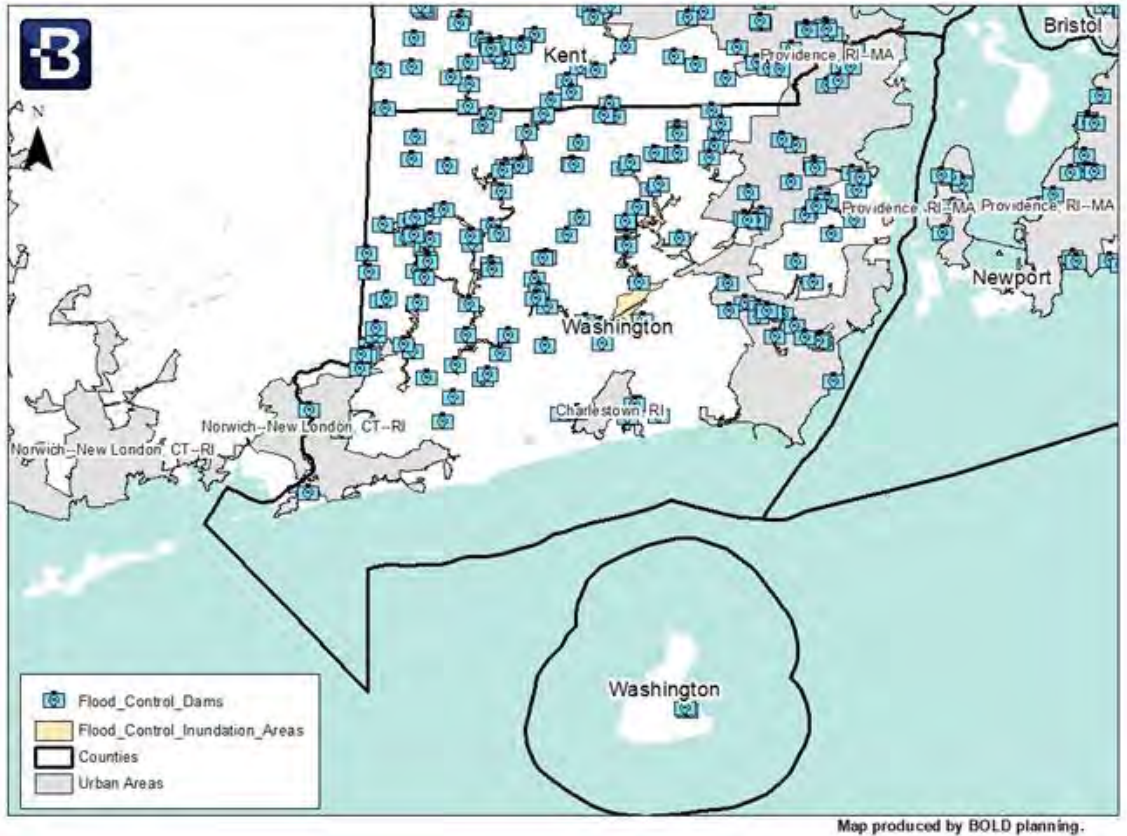
Map C30: Critical Facilities, Flood Control Dams



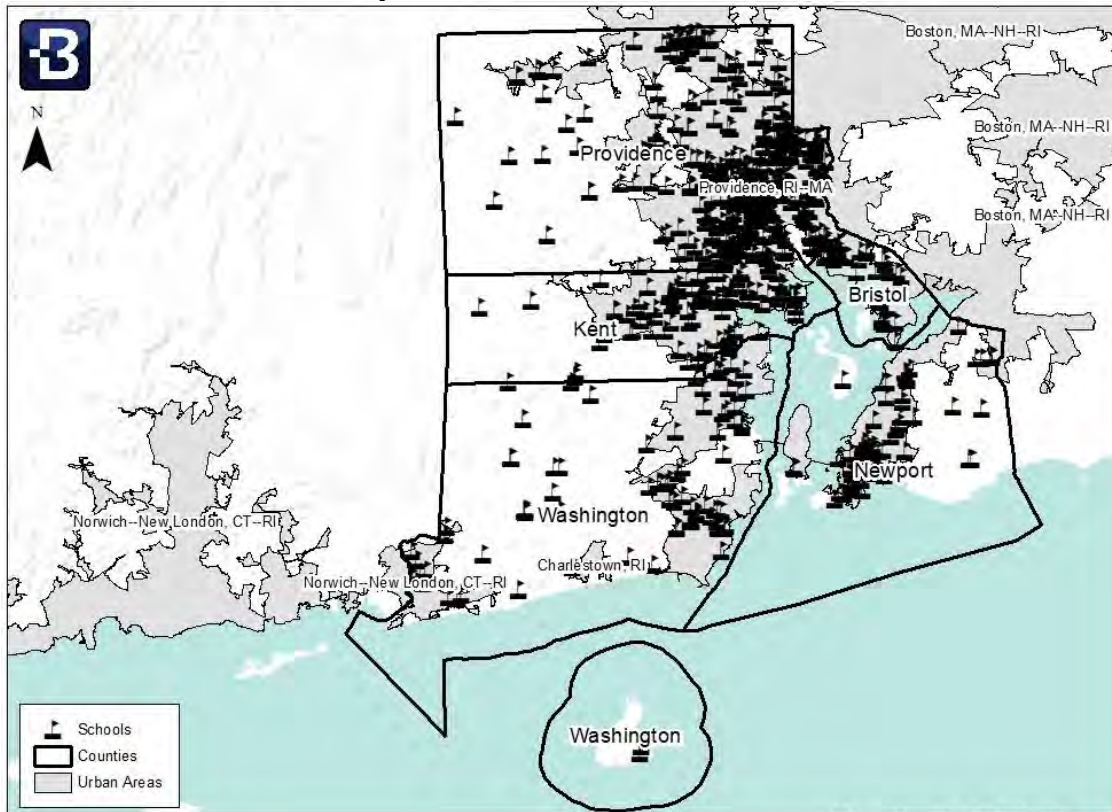
Map C31: Critical Facilities, Flood Control Dams



Map C32: Critical Facilities, Flood Control Dams

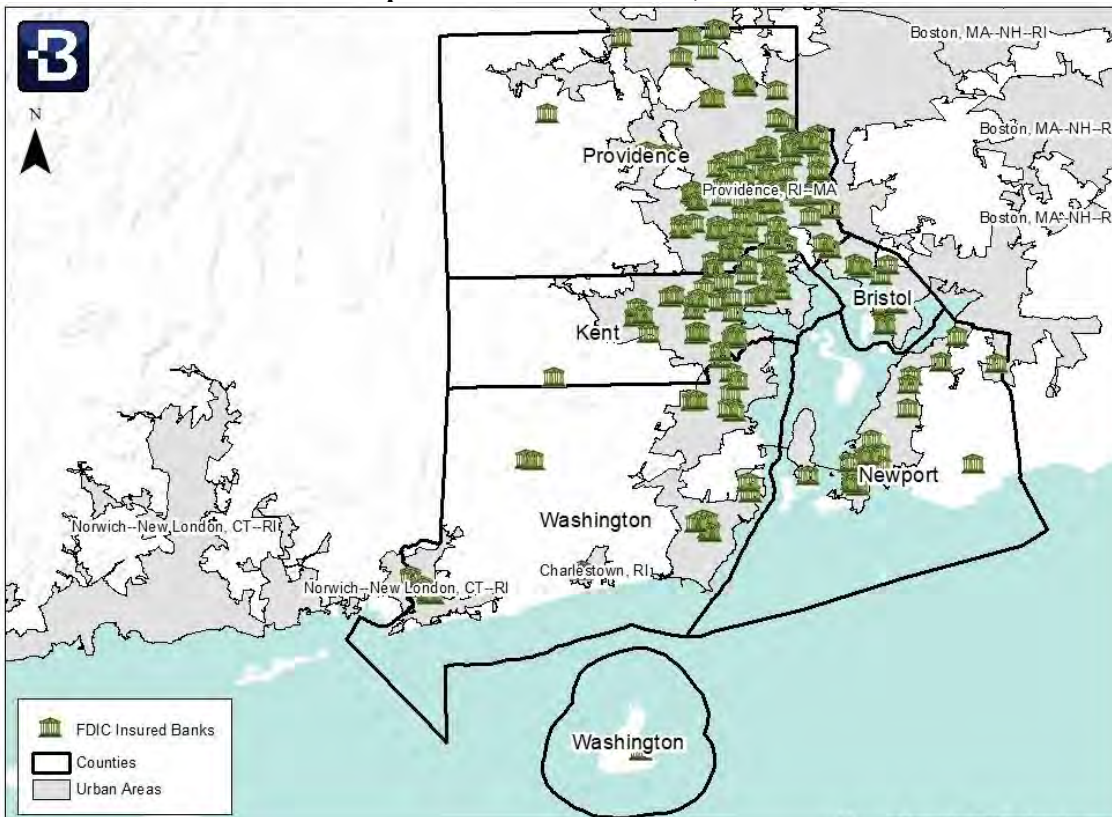


Map C33: Critical Facilities, Schools



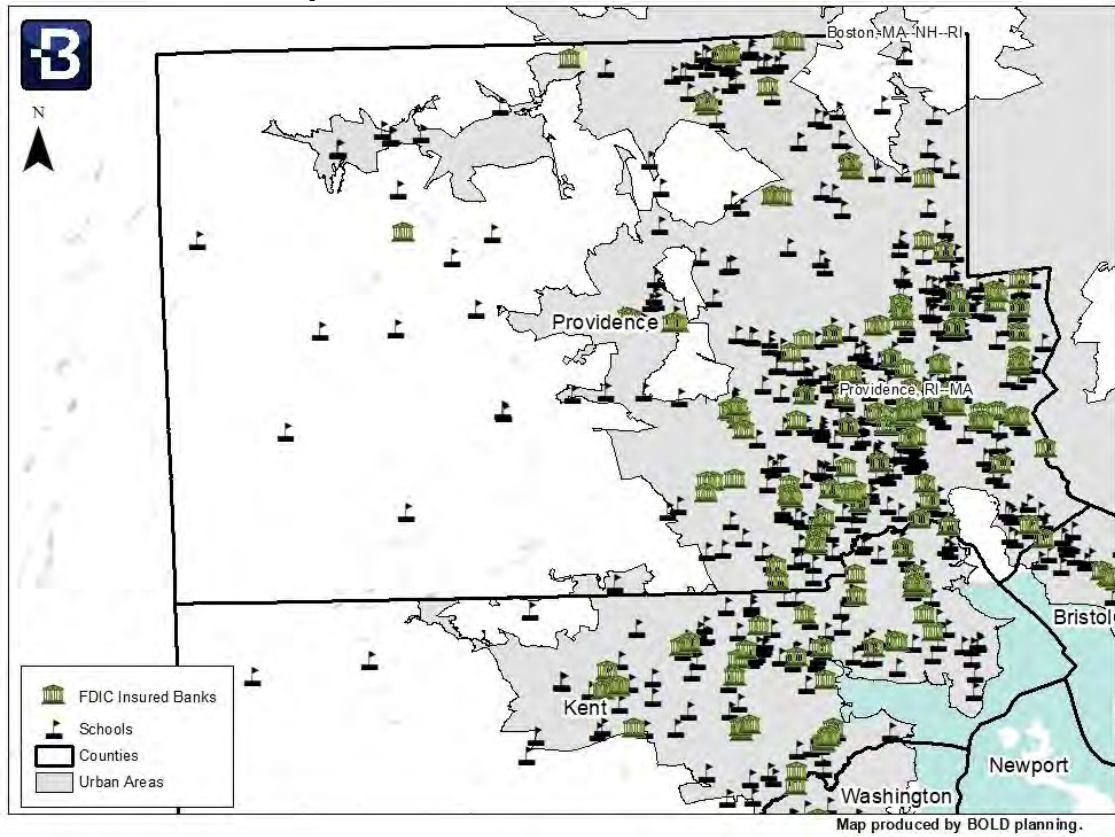
Map produced by BOLD planning.

Map C34: Critical Facilities, Banks

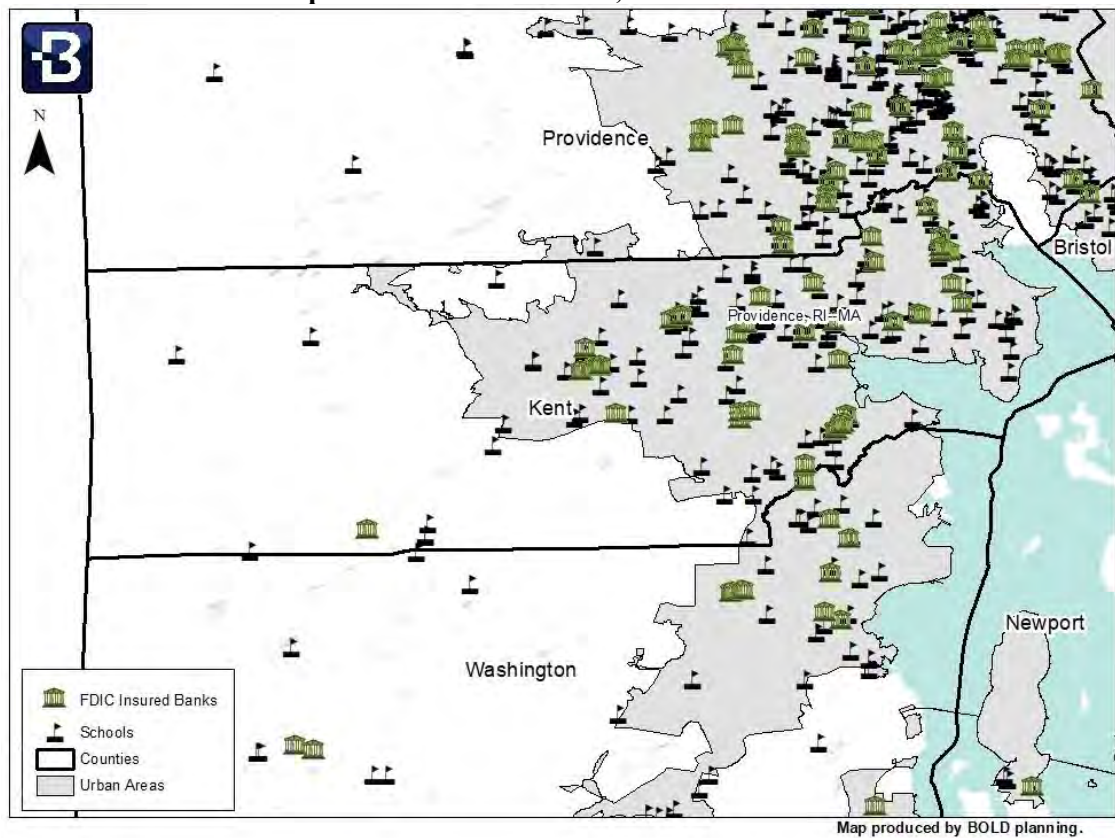


Map produced by BOLD planning.

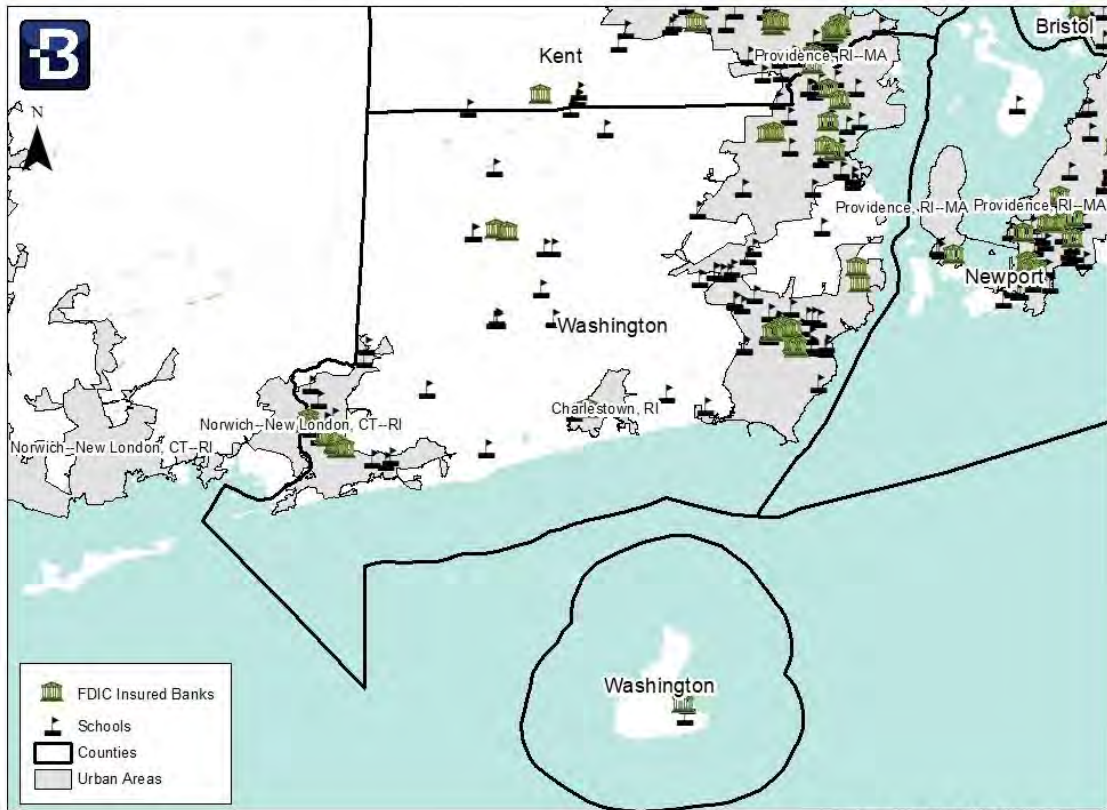
Map C35: Critical Facilities, Schools and Banks



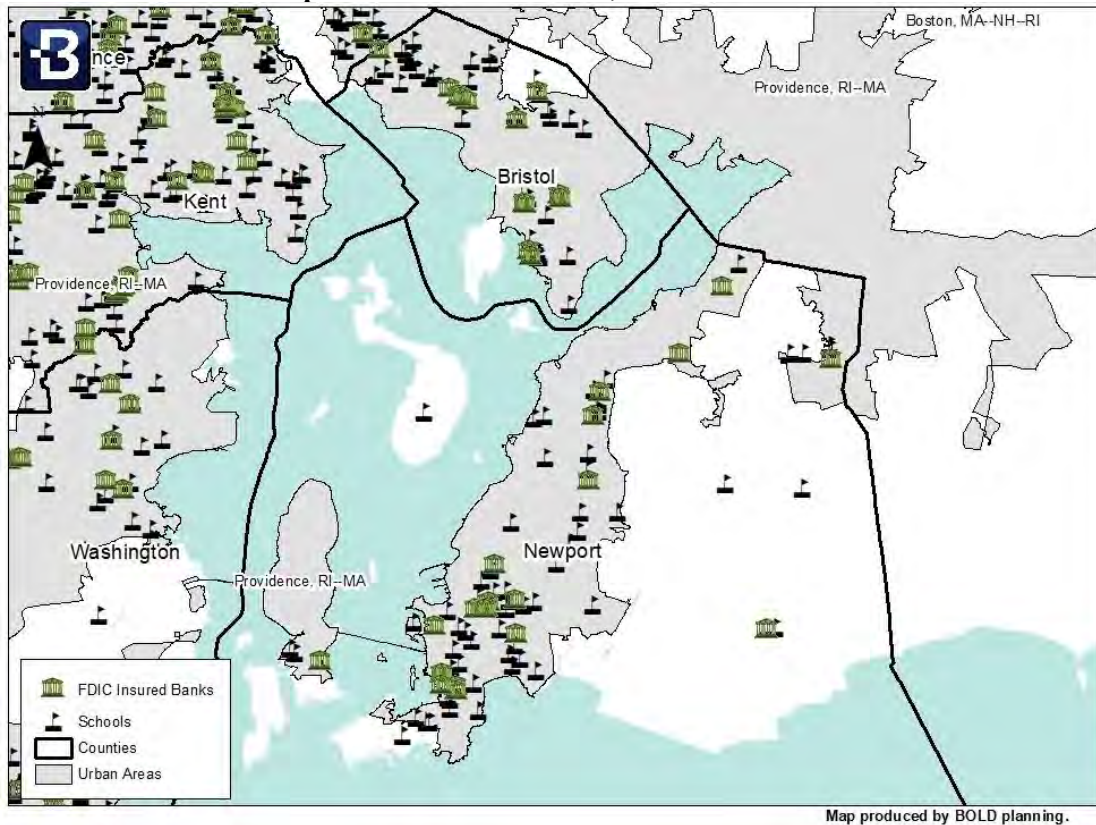
Map C36: Critical Facilities, Schools and Banks



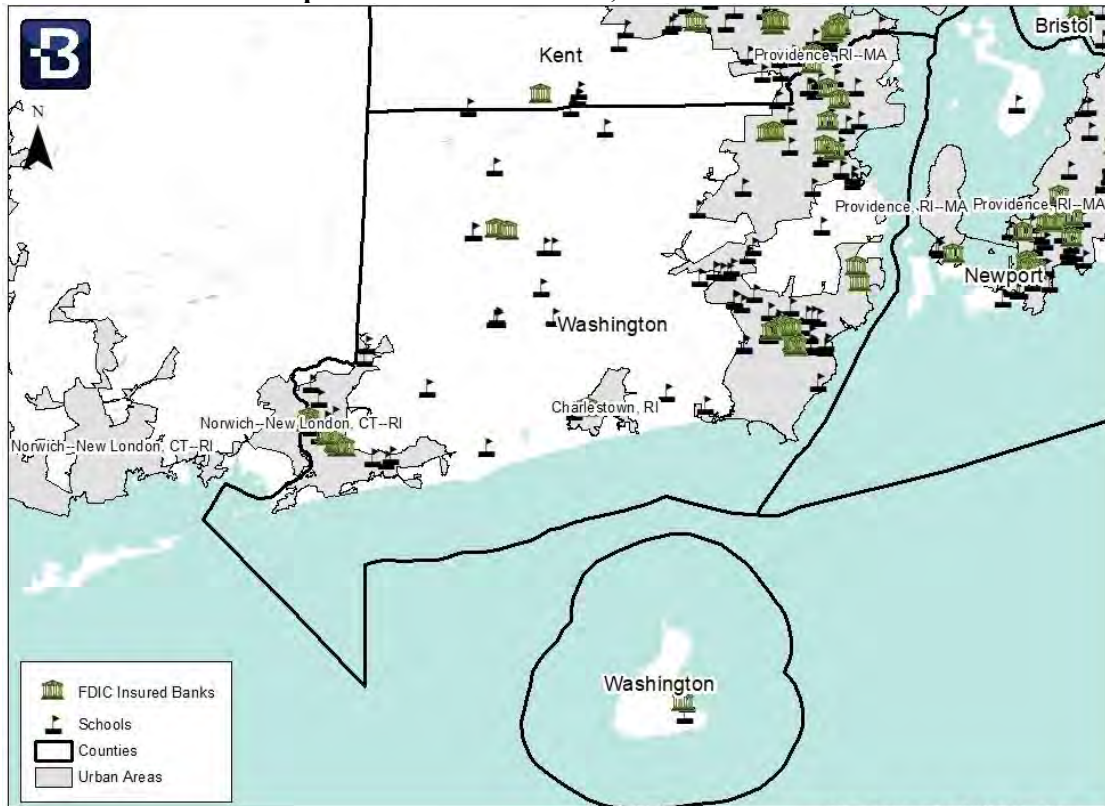
Map C37: Critical Facilities, Schools and Banks



Map C38: Critical Facilities, Schools and Banks

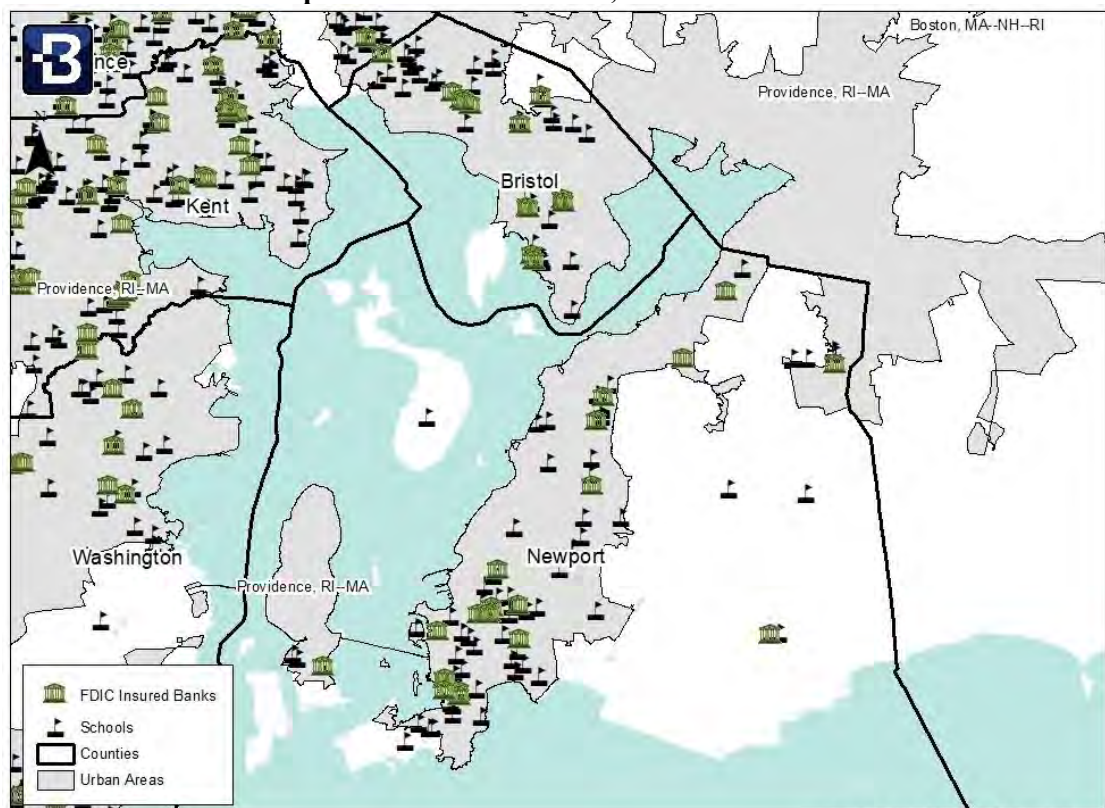


Map C39: Critical Facilities, Schools and Banks



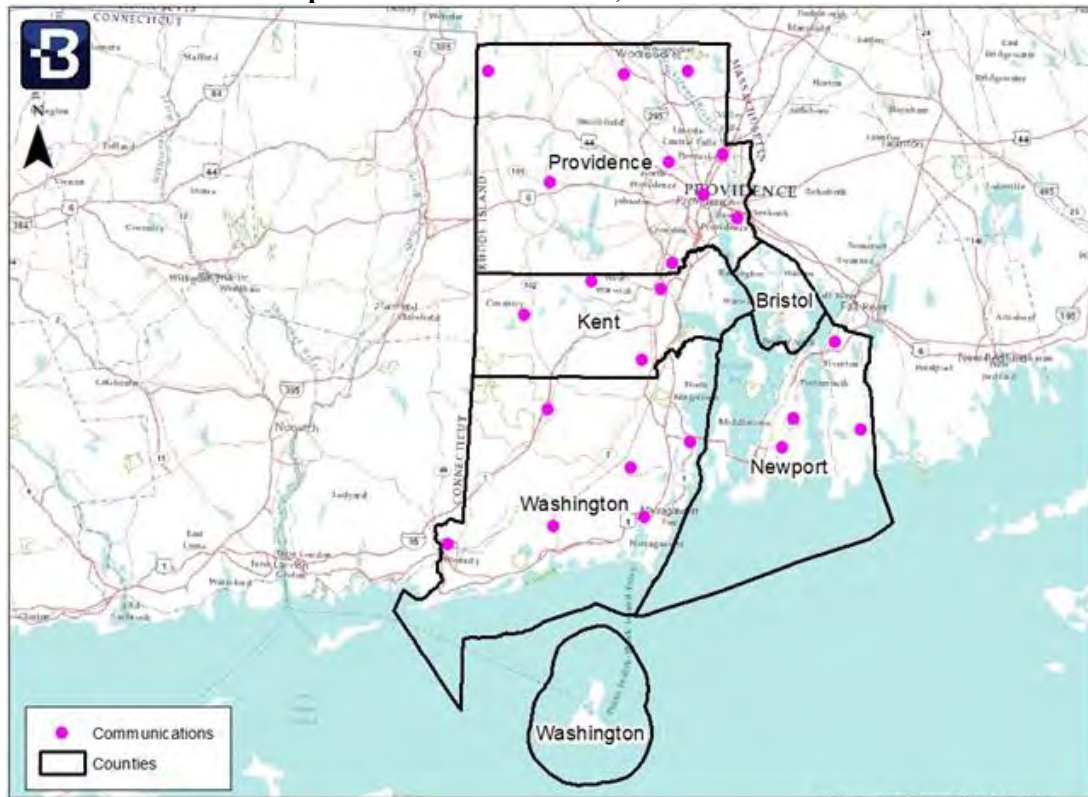
Map produced by BOLD planning.

Map C40: Critical Facilities, Schools and Banks



Map produced by BOLD planning.

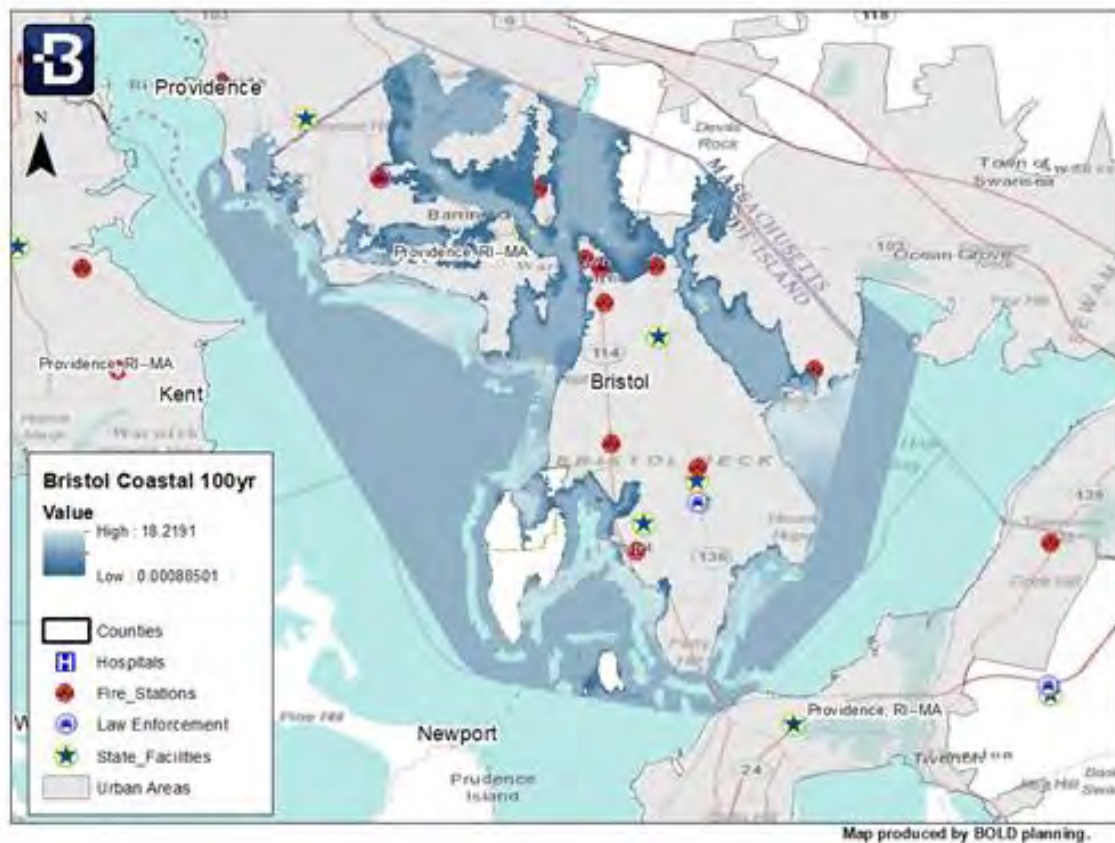
Map C41: Critical Facilities, Communications



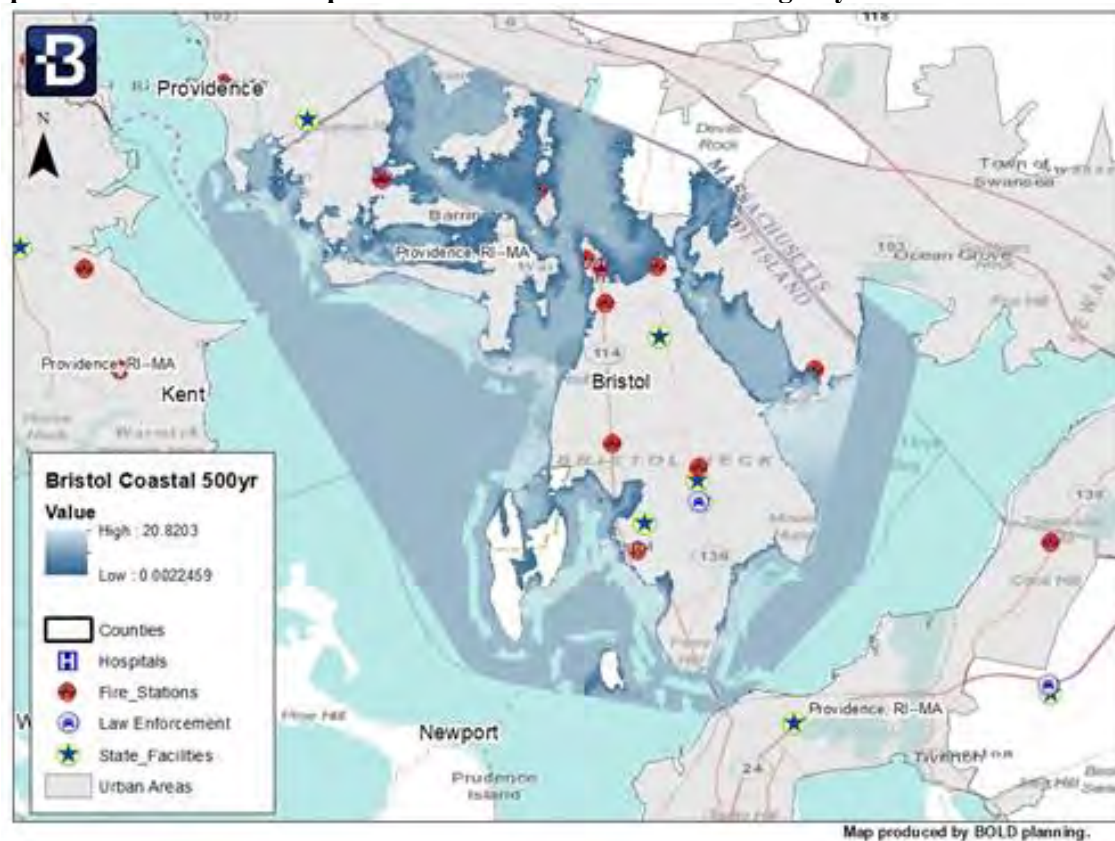
Map produced by BOLD planning.

Appendix C3 State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones

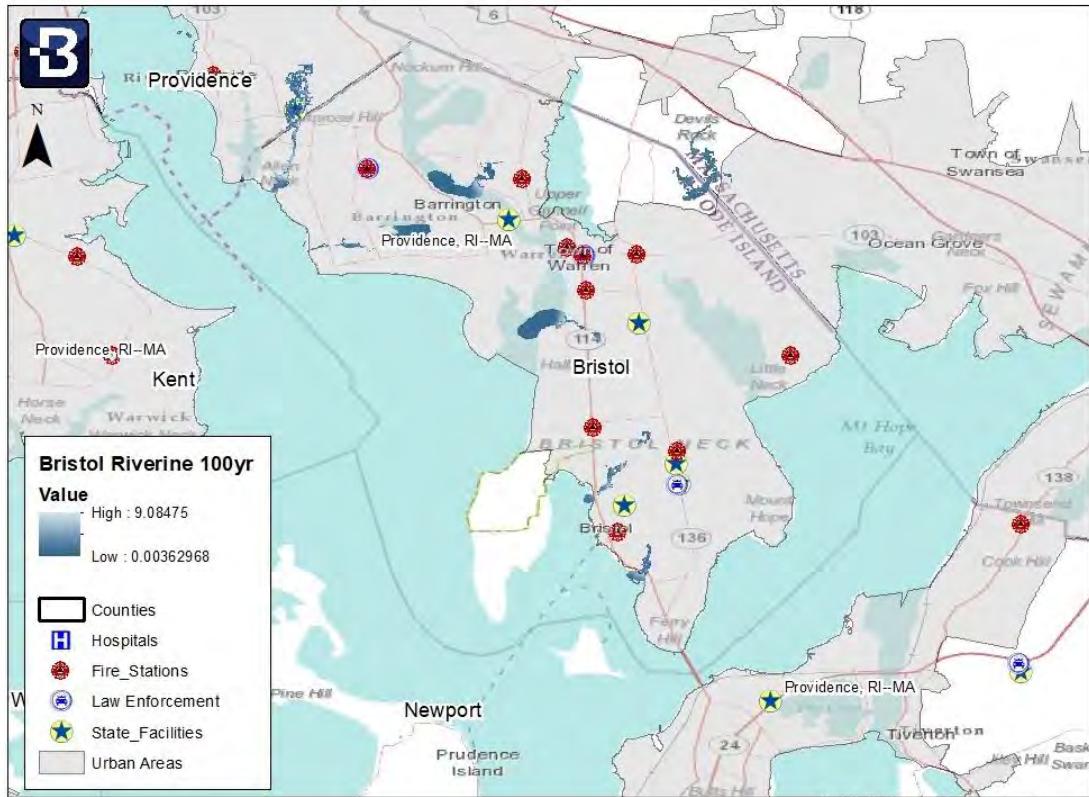
Map C42: State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones



Map C43: State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones

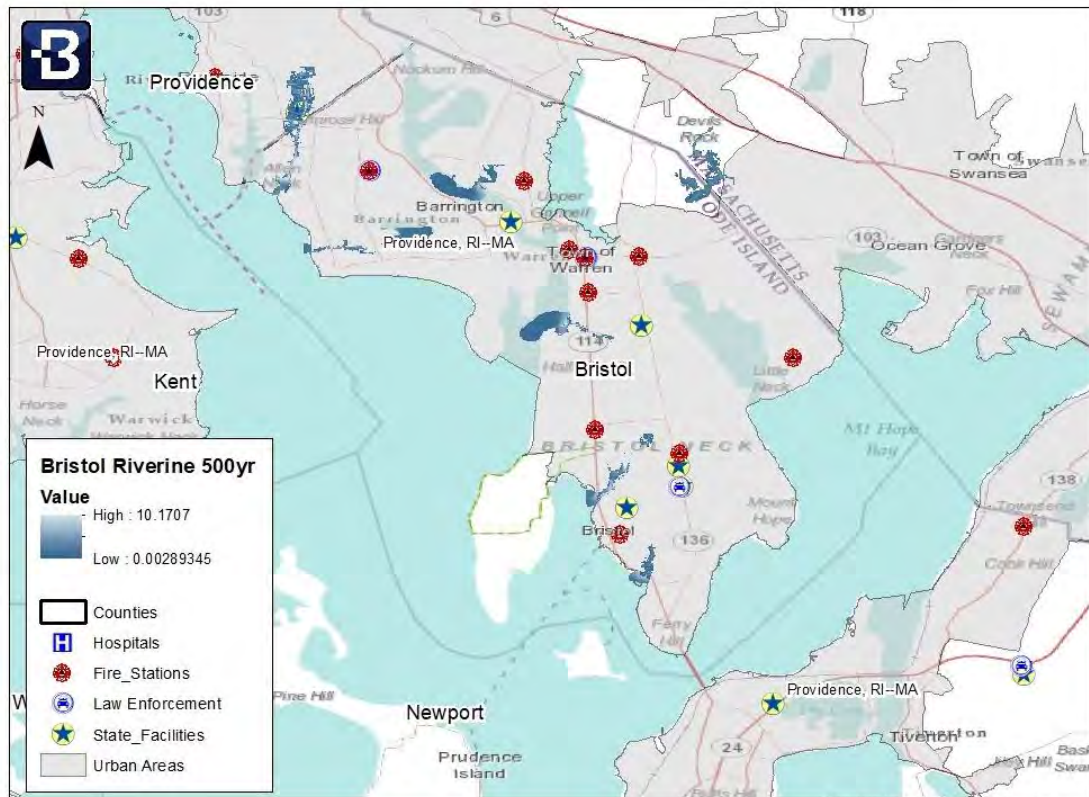


Map C44: State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones



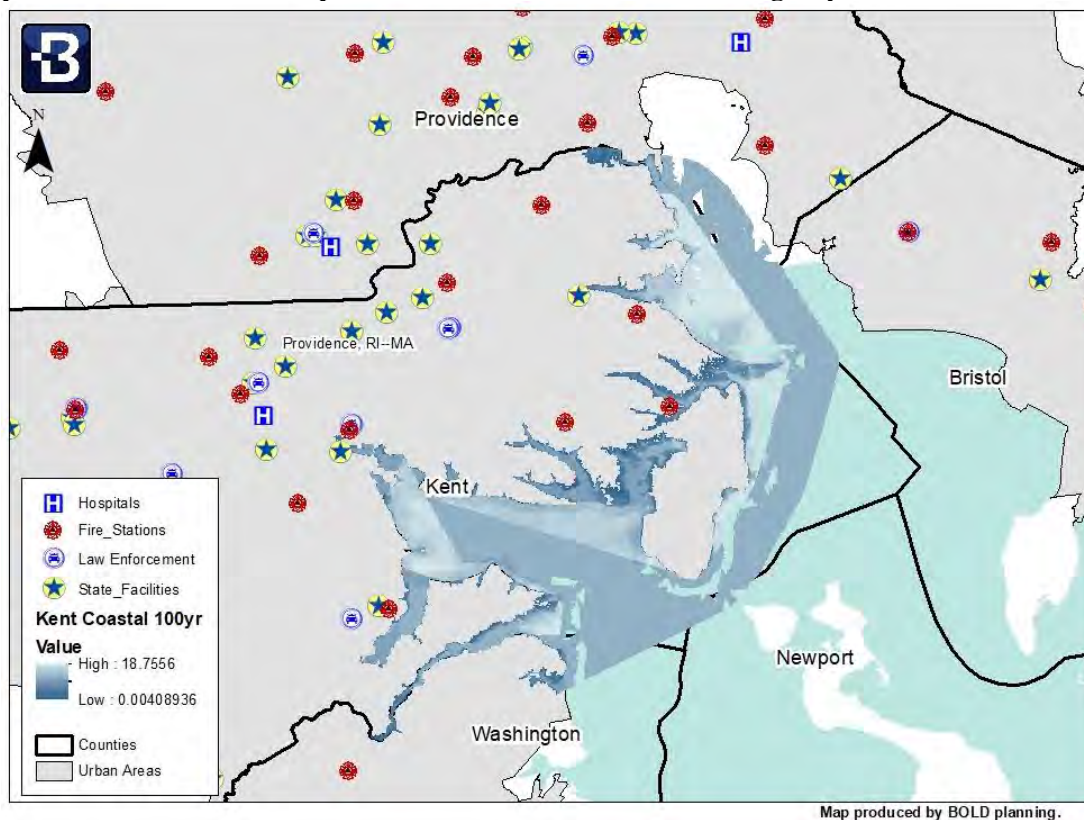
Map produced by BOLD planning.

Map C45: State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones

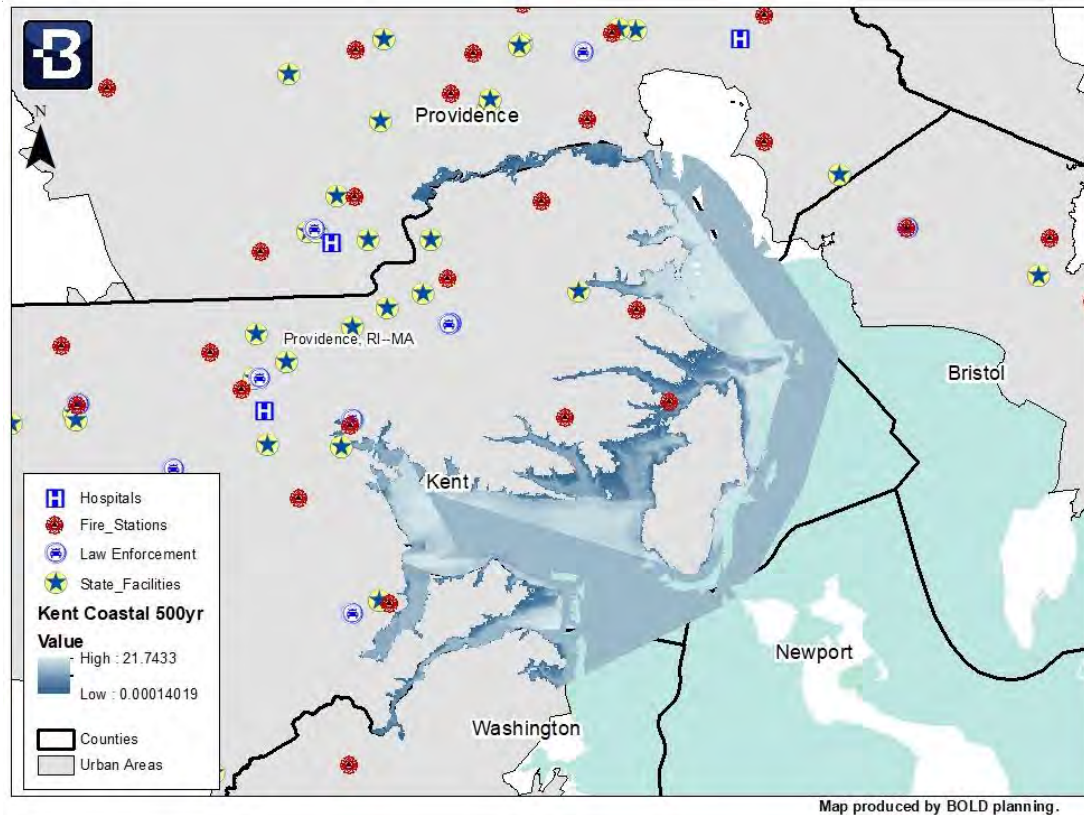


Map produced by BOLD planning.

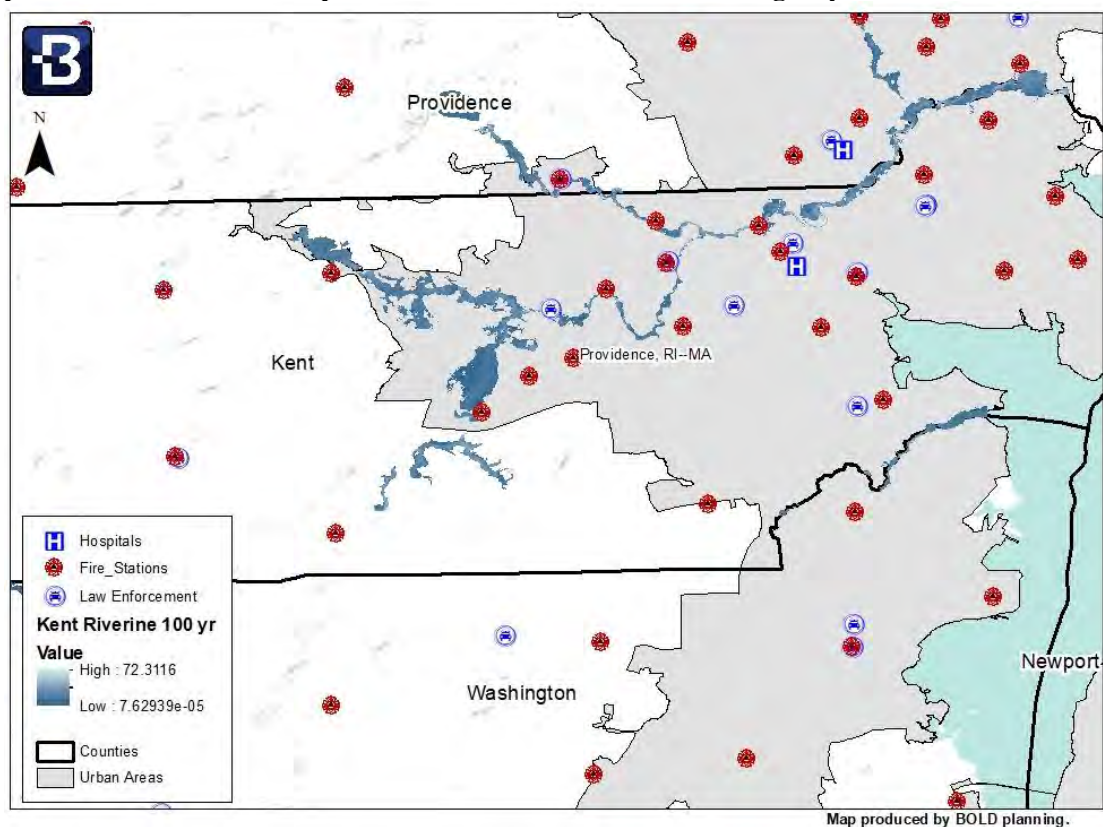
Map C46: State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones



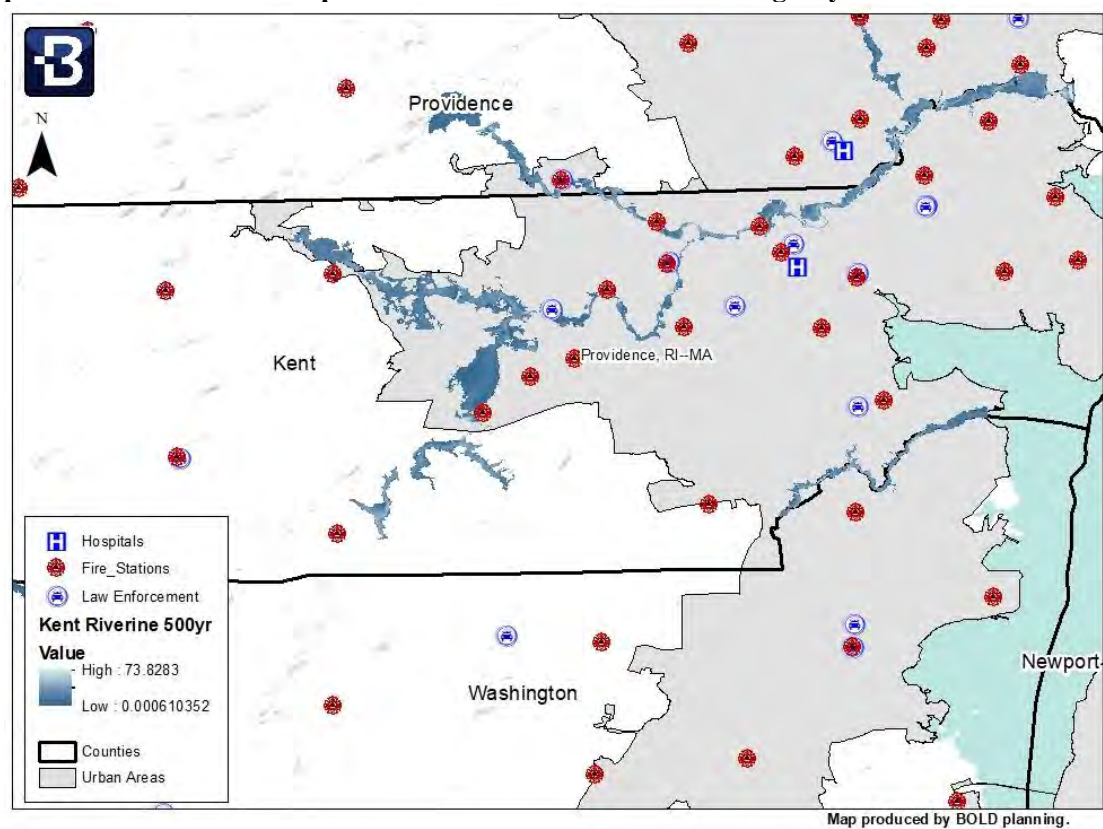
Map C47: State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones



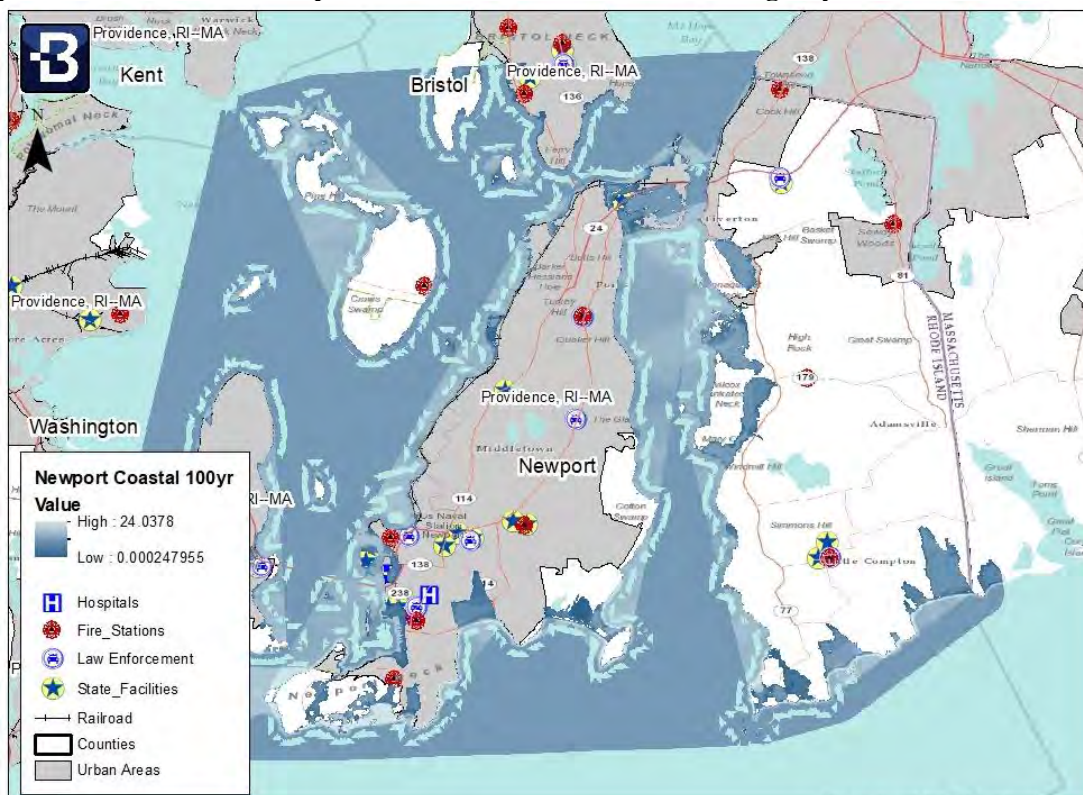
Map C48: State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones



Map C49: State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones

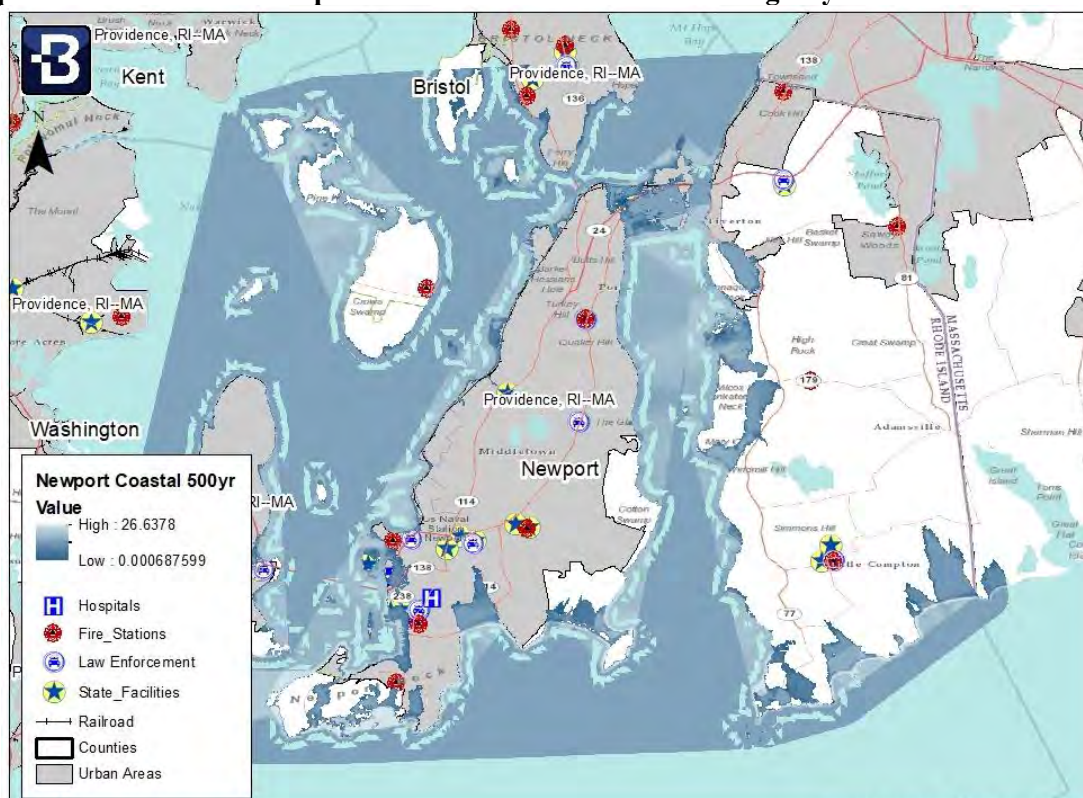


Map C50: State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones



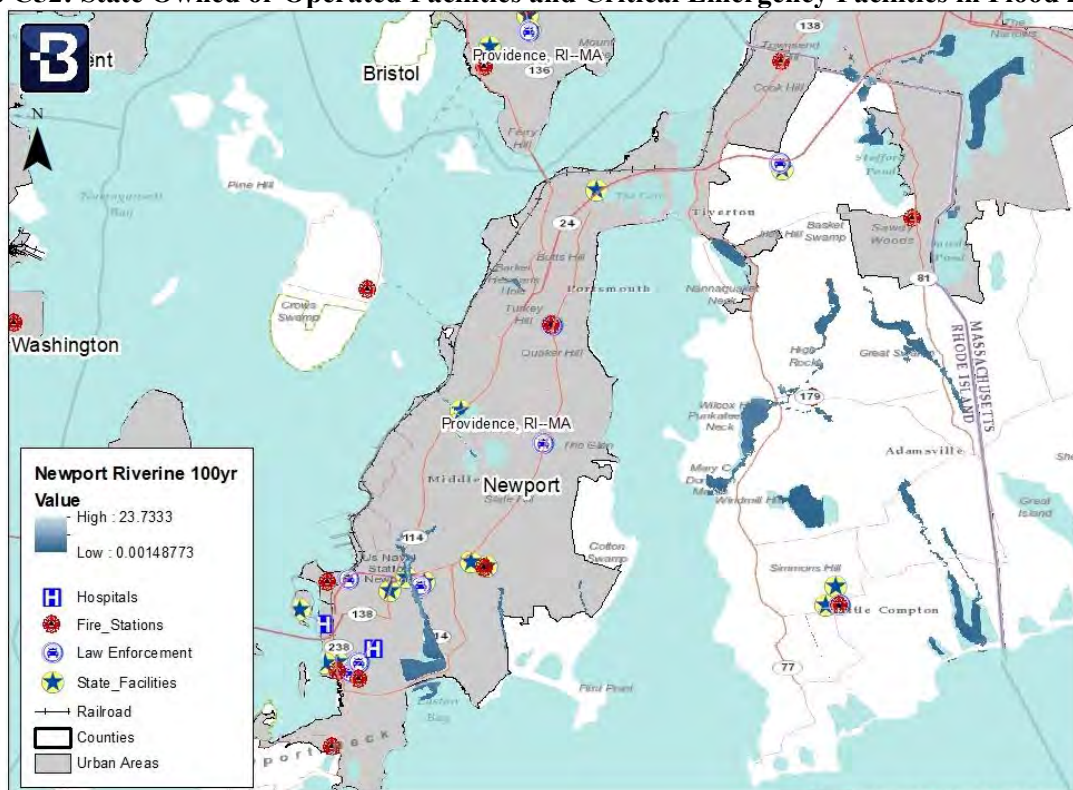
Map produced by BOLD planning.

Map C51: State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones



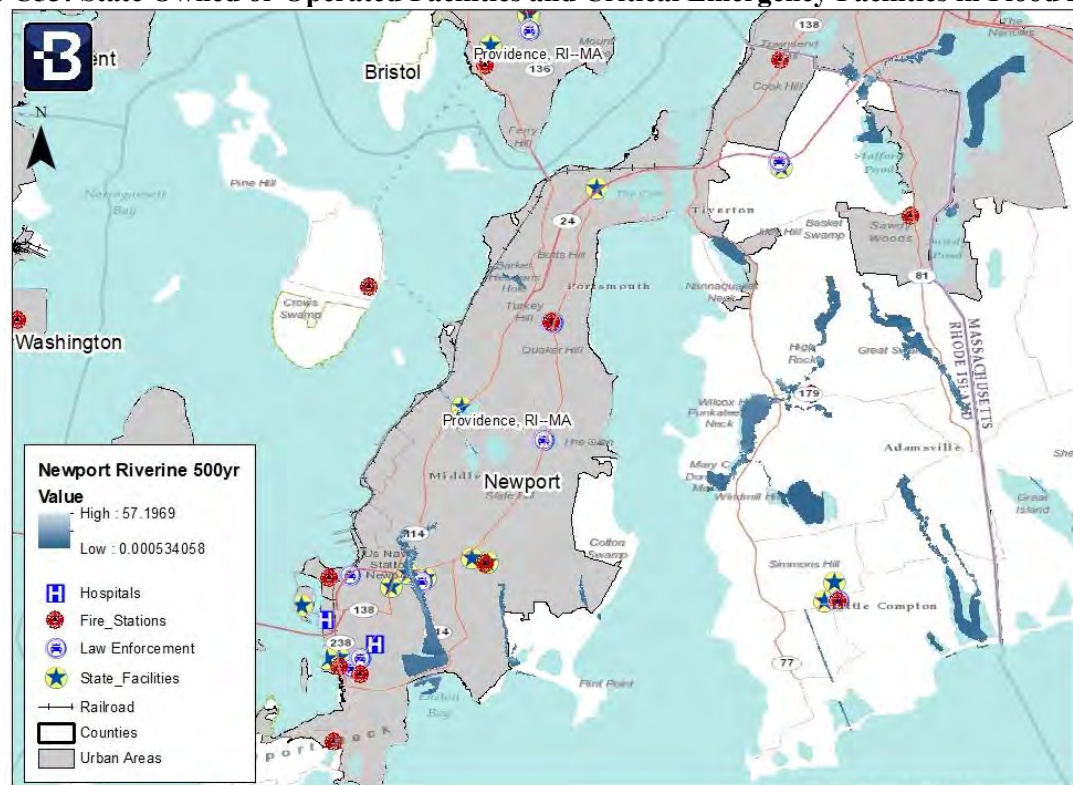
Map produced by BOLD planning.

Map C52: State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones



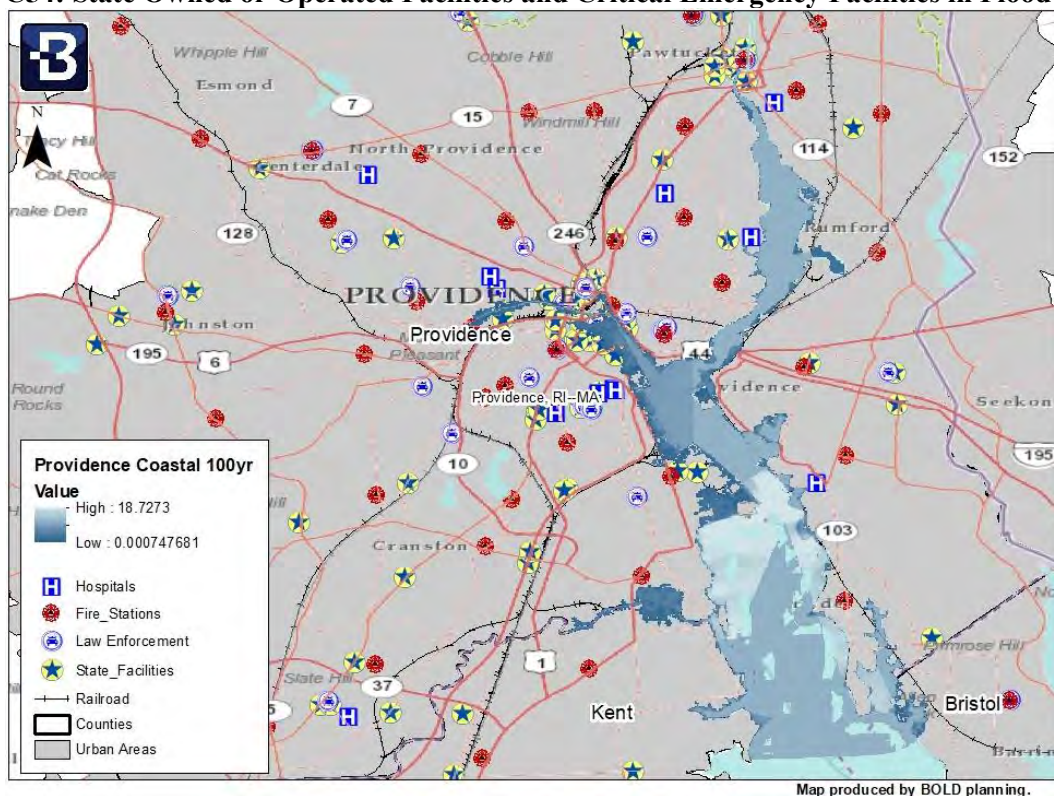
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Map C53: State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones

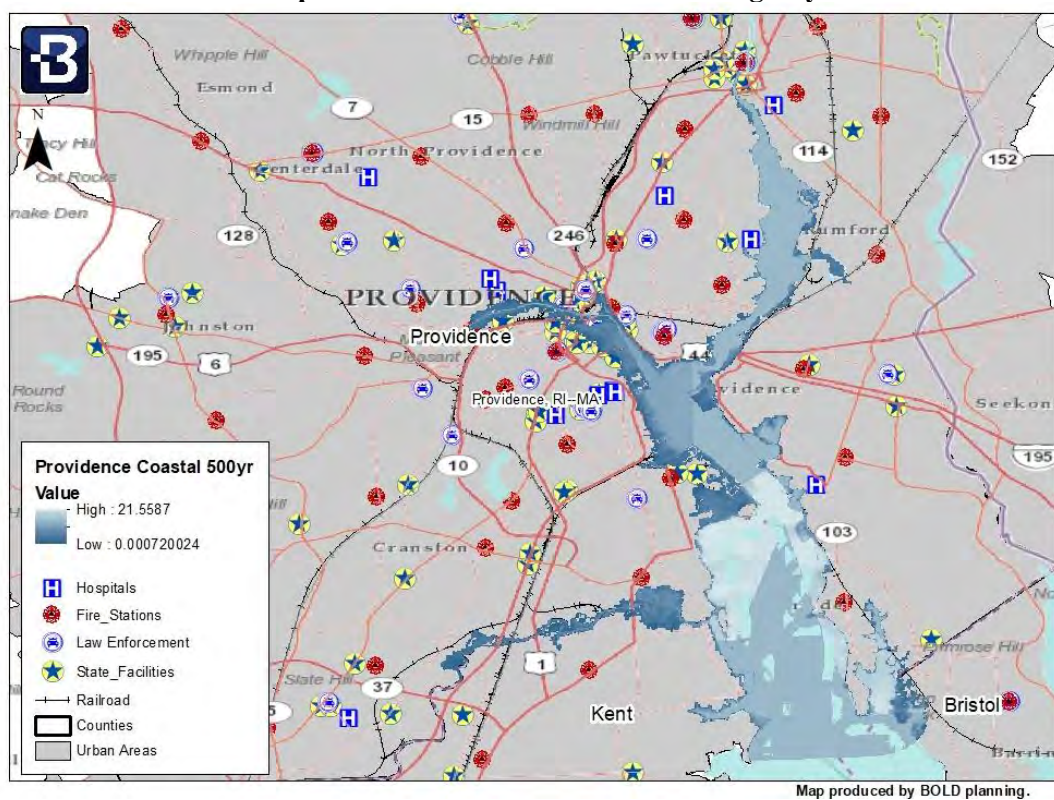


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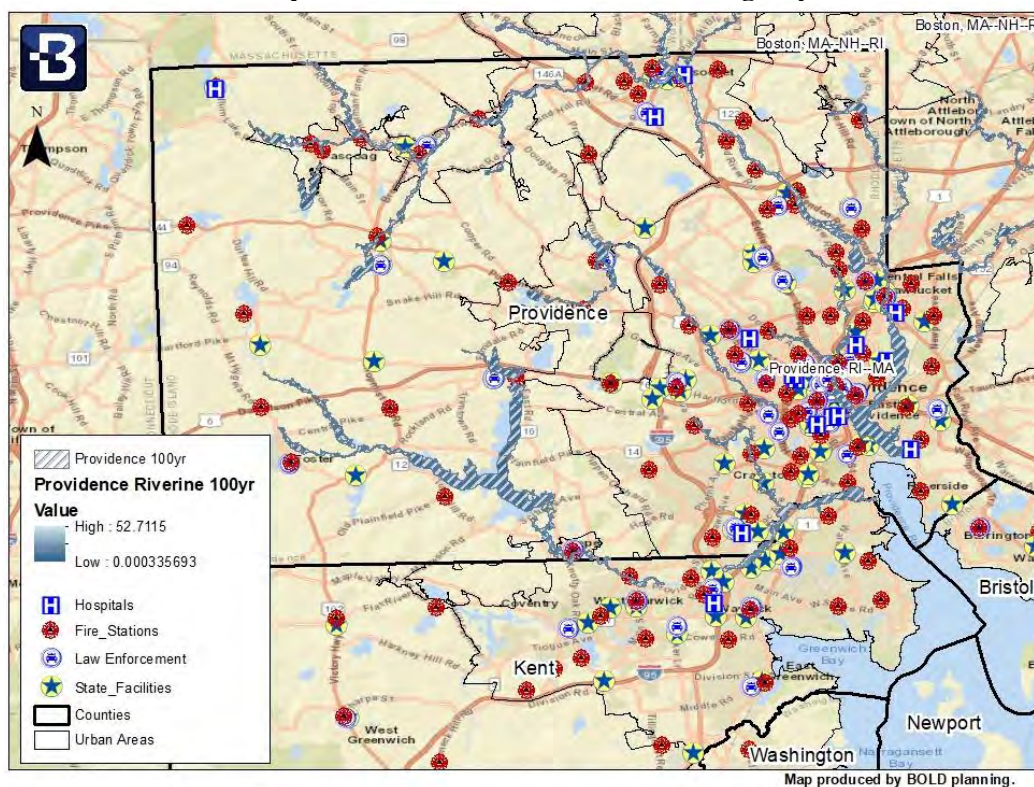
Map C54: State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones



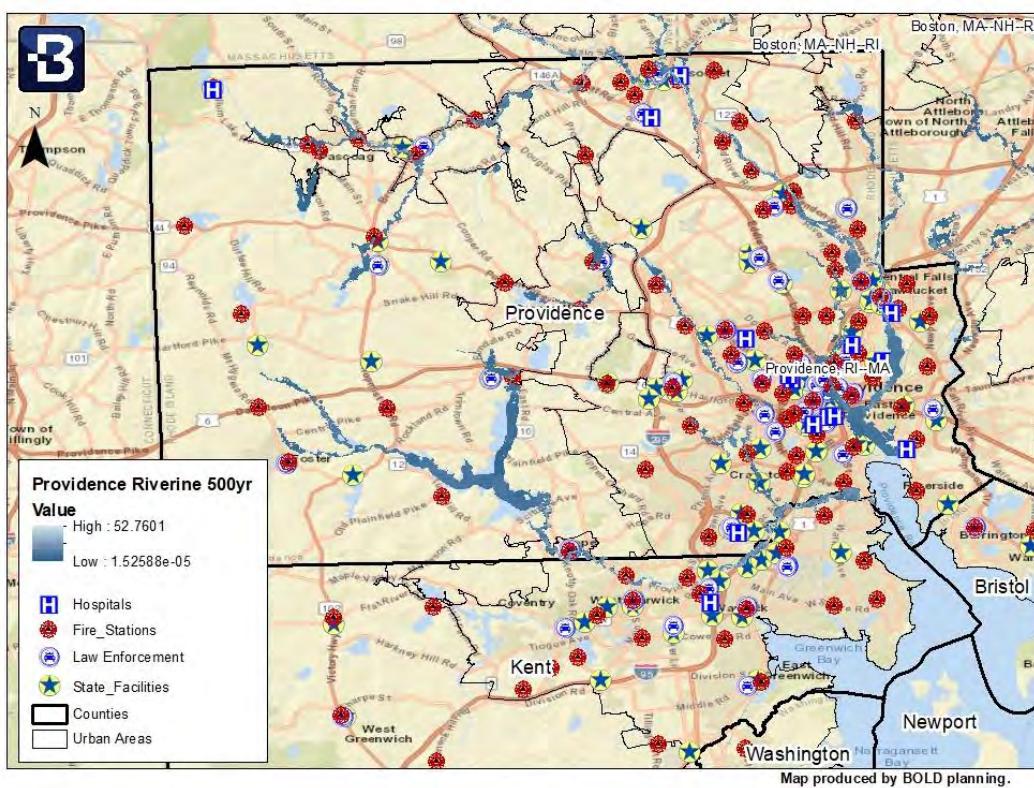
Map C55: State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones



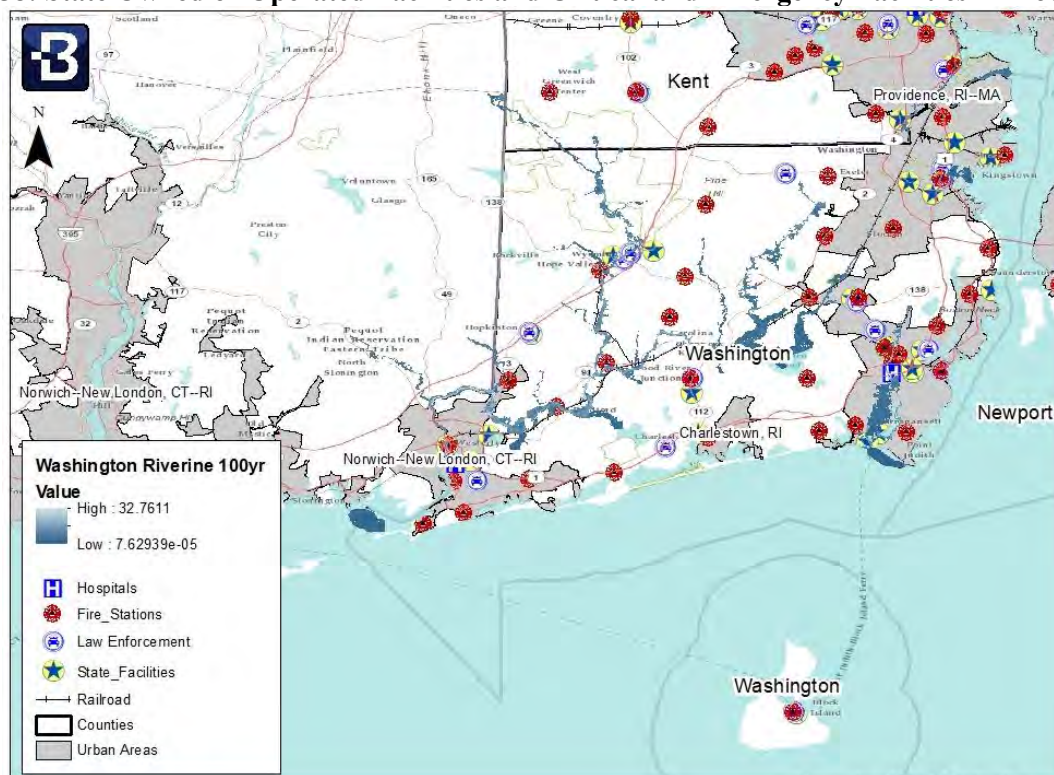
Map C56: State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones



Map C57: State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones

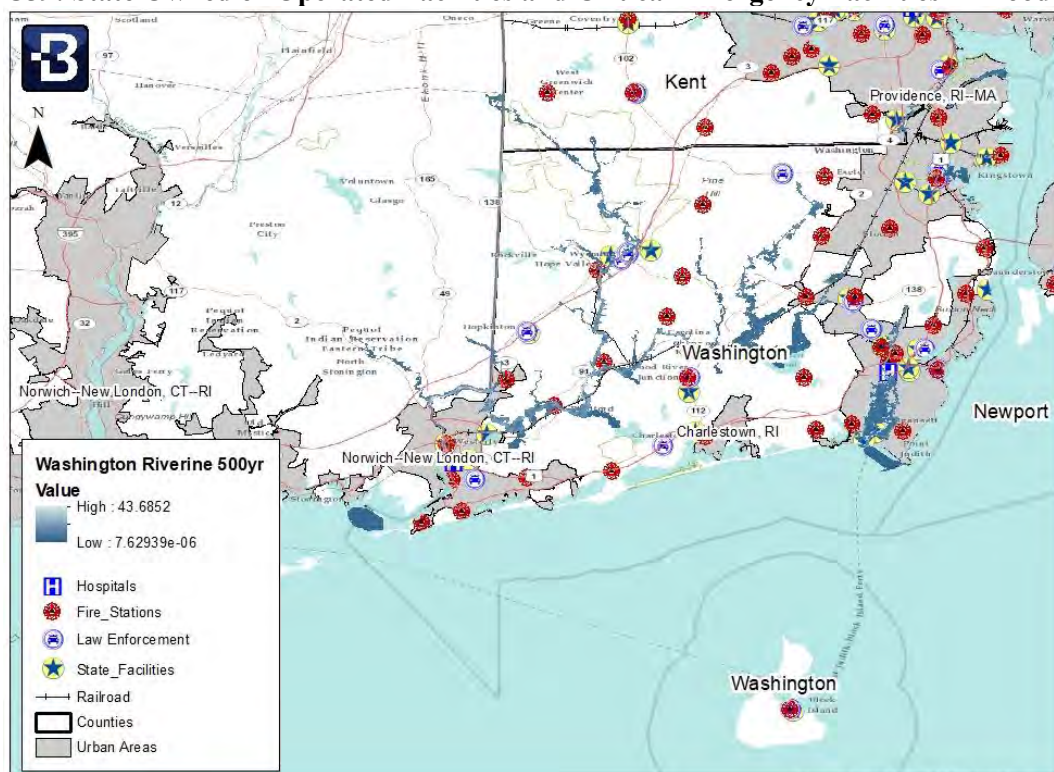


Map C58: State Owned or Operated Facilities and Critical and Emergency Facilities in Flood Zones



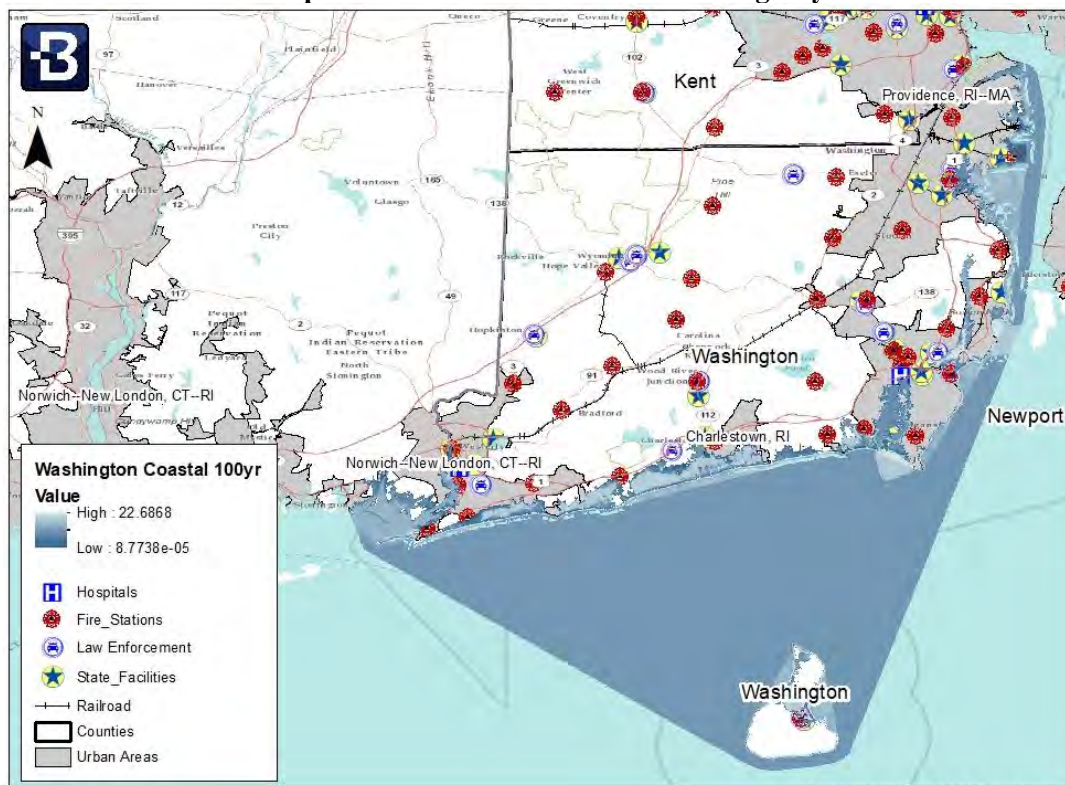
Map produced by BOLD planning.

Map C59: State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones



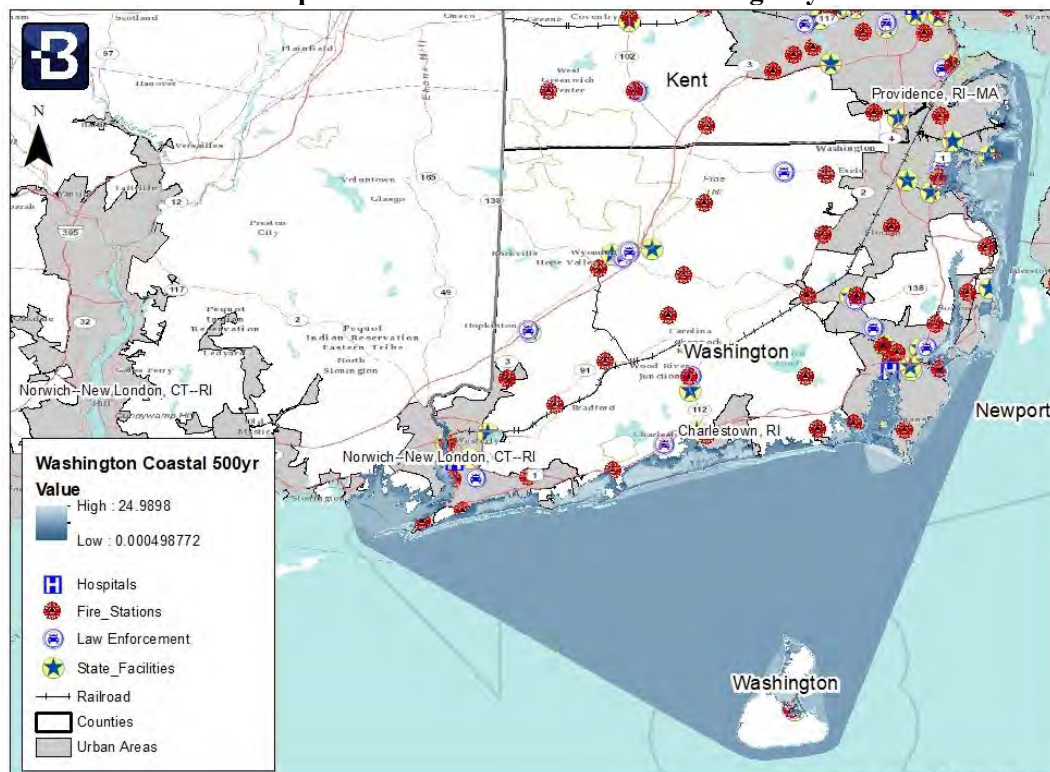
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Map C60: State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones



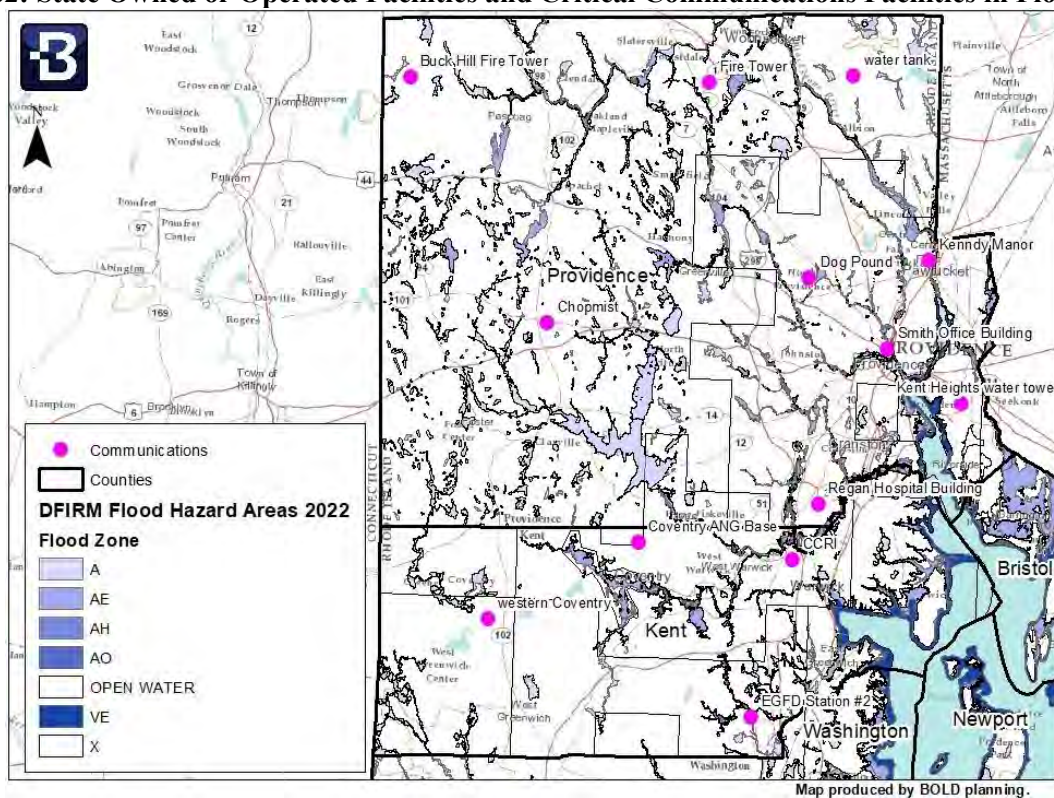
Map produced by BOLD planning.

Map C61: State Owned or Operated Facilities and Critical Emergency Facilities in Flood Zones

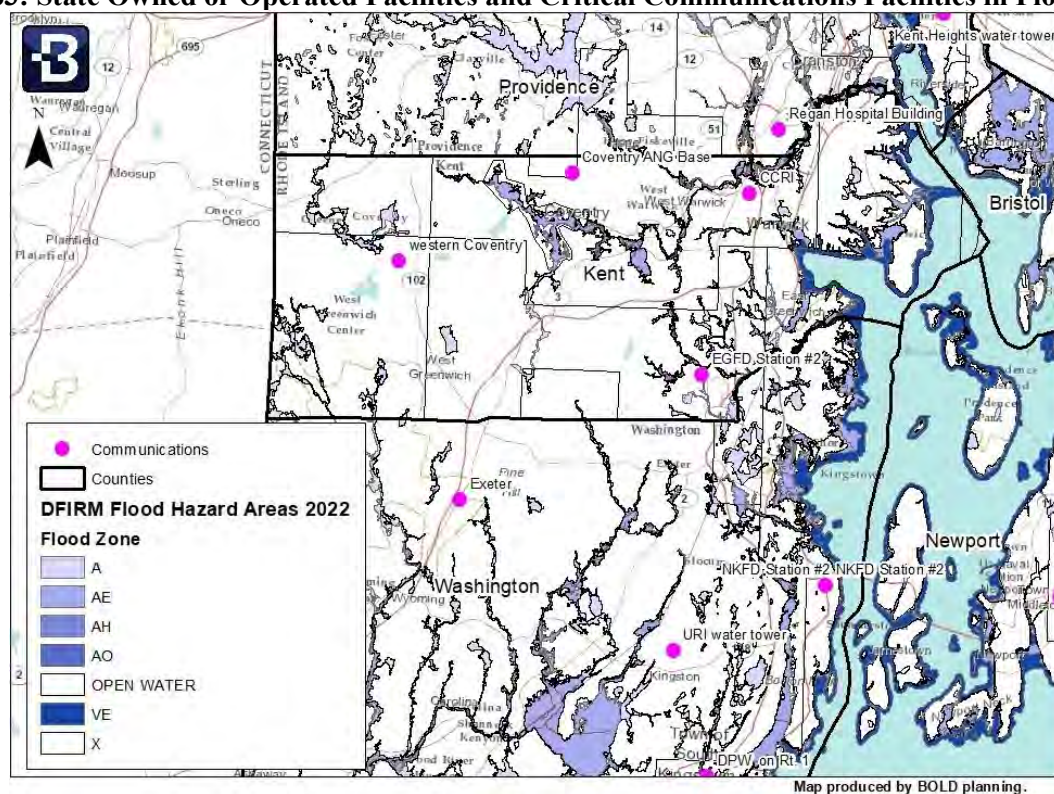


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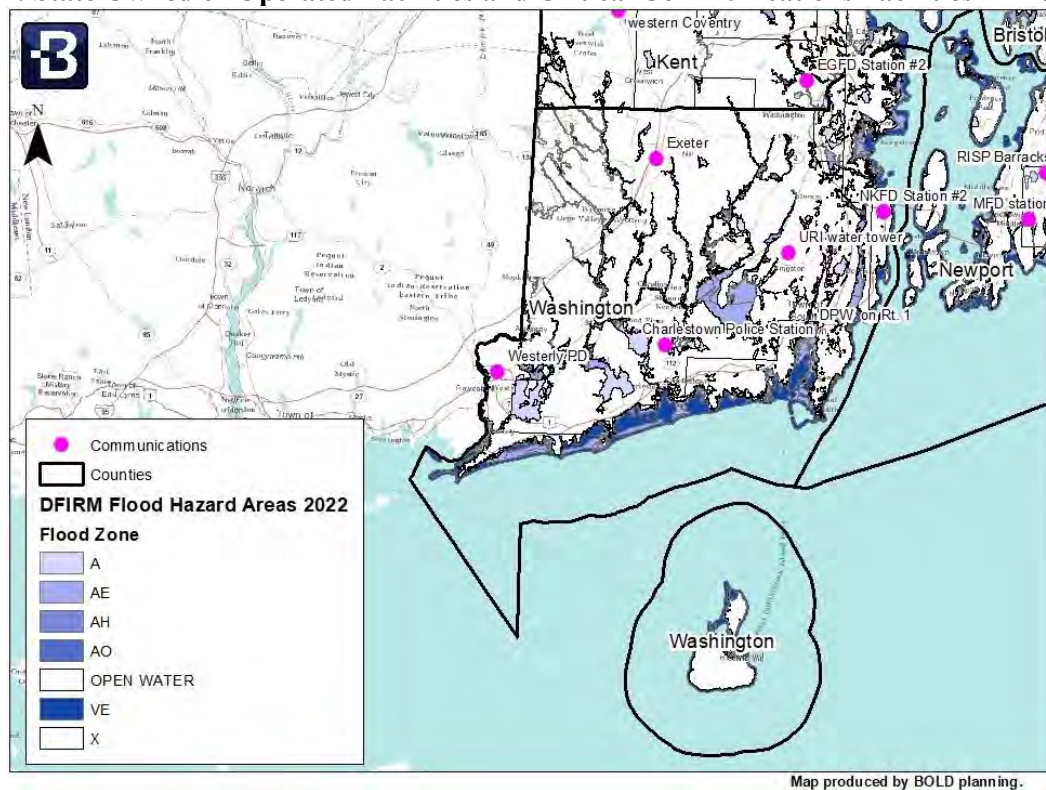
Map C62: State Owned or Operated Facilities and Critical Communications Facilities in Flood Zones



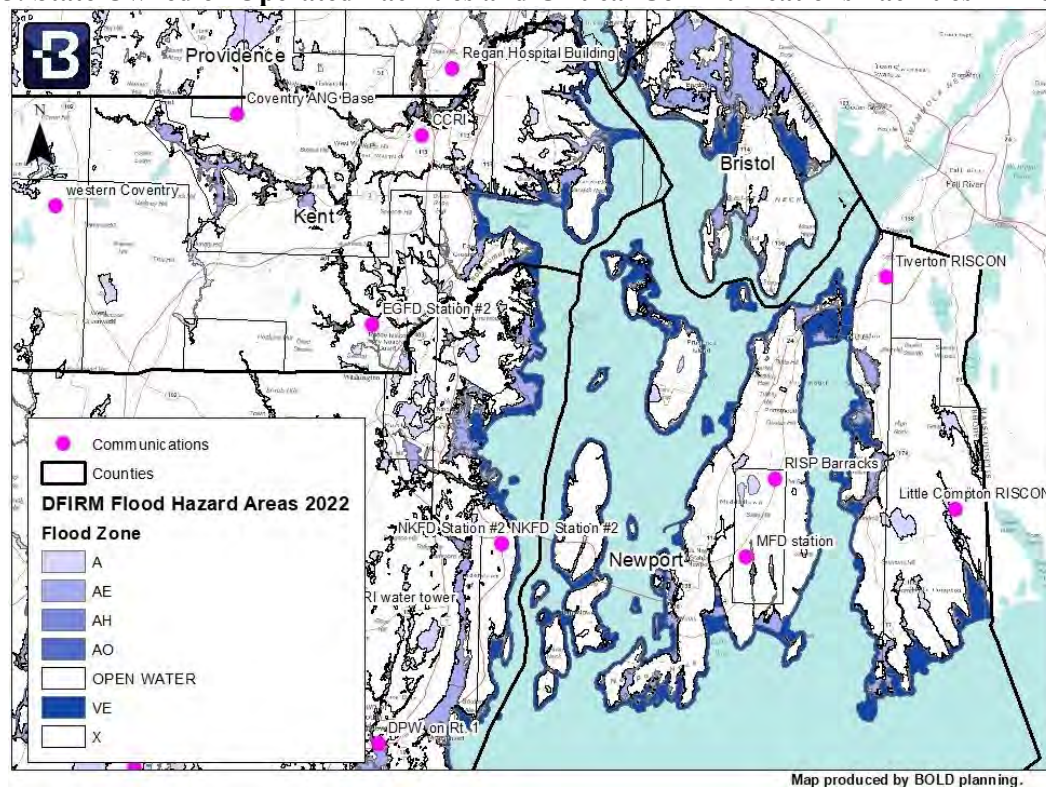
Map C63: State Owned or Operated Facilities and Critical Communications Facilities in Flood Zones



Map C64: State Owned or Operated Facilities and Critical Communications Facilities in Flood Zones

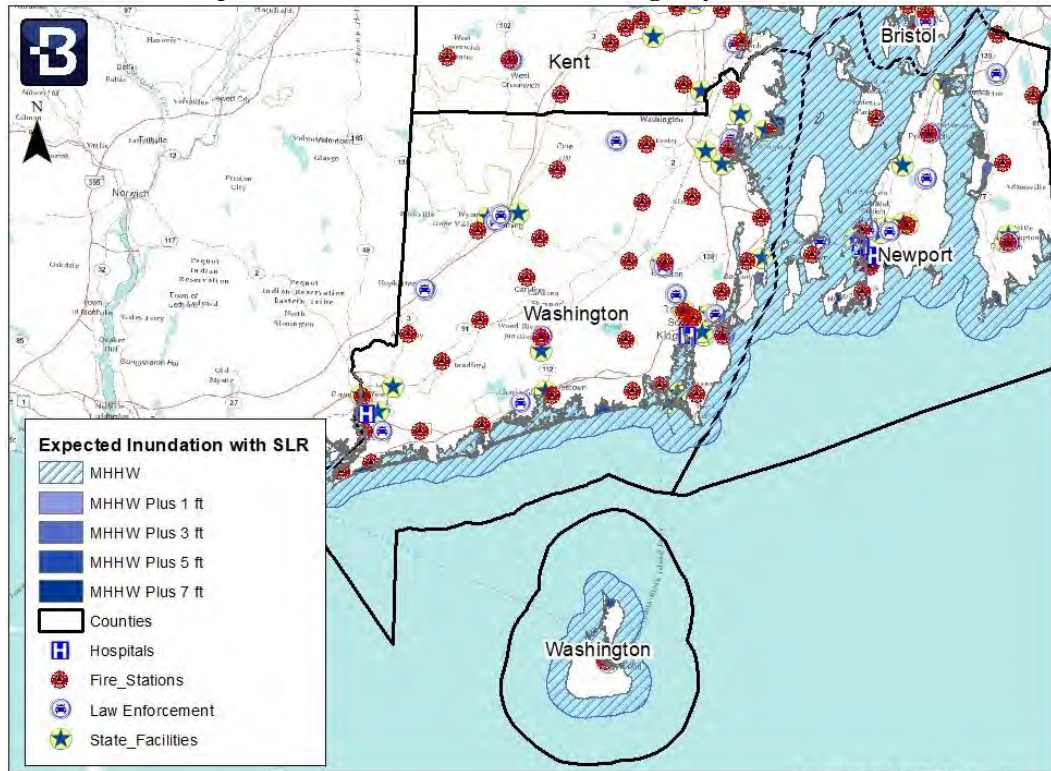


Map C65: State Owned or Operated Facilities and Critical Communications Facilities in Flood Zones

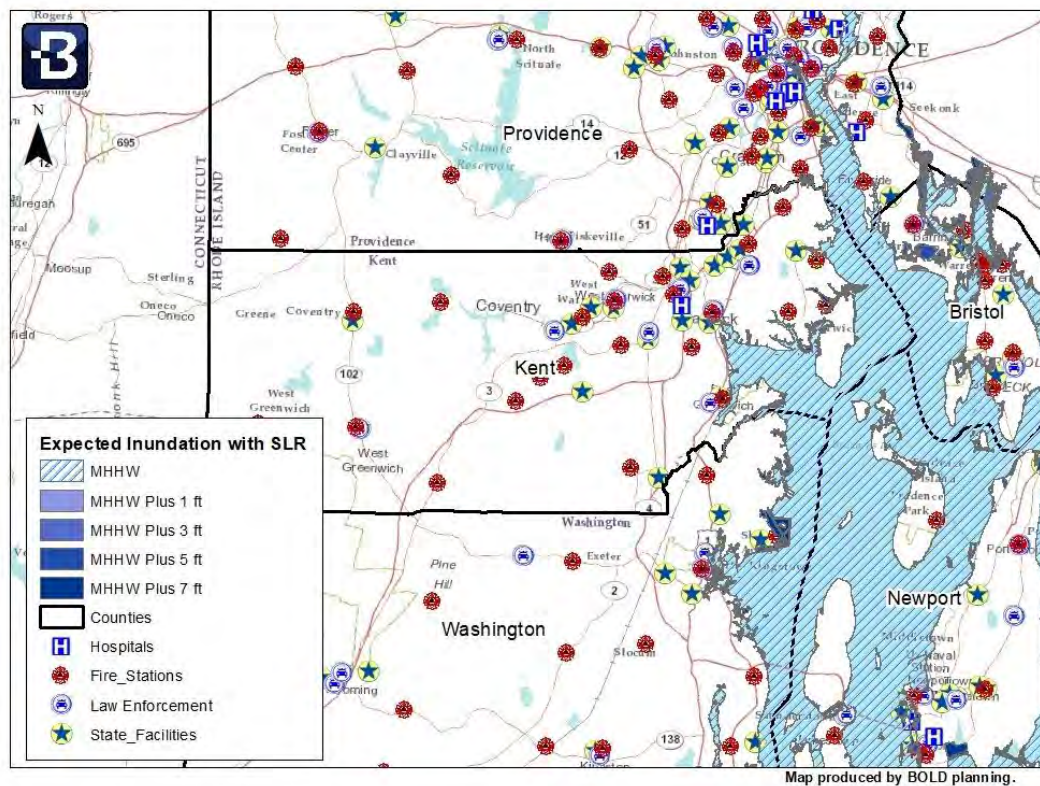


Appendix C4 State Owned or Operated Facilities and Critical Facilities in Sea Level Inundation Areas

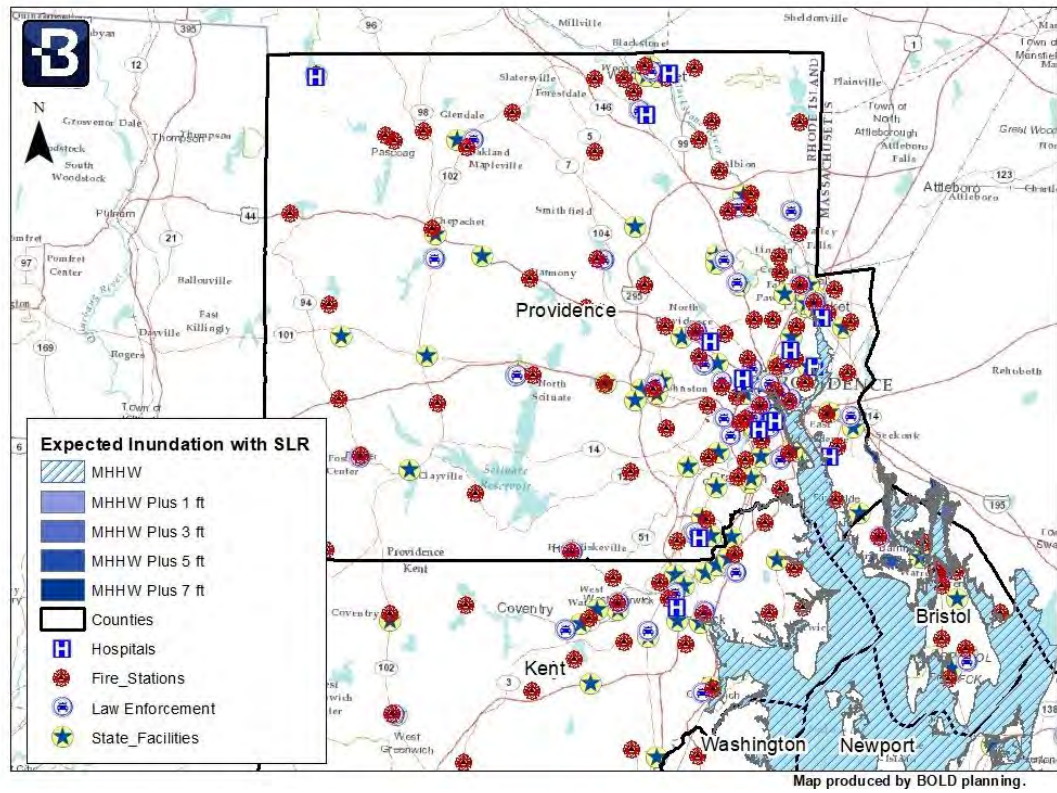
Map C66: State Owned or Operated Facilities and Critical Emergency Facilities in Sea Level Inundation Areas



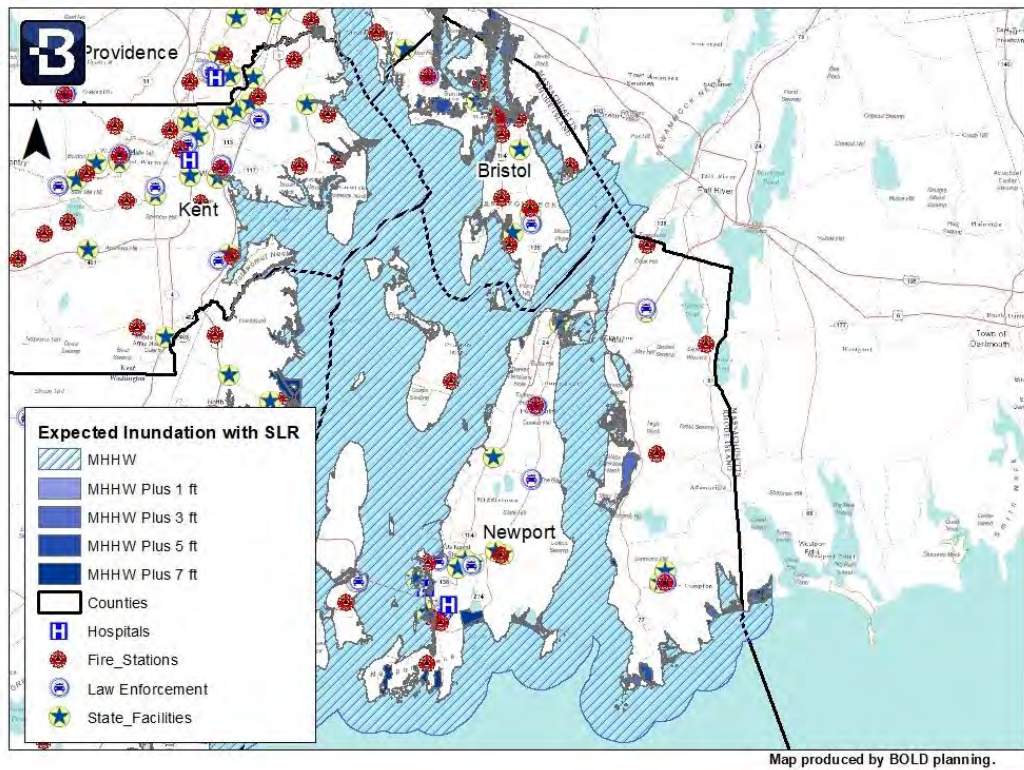
Map C67: State Owned or Operated Facilities and Critical Emergency Facilities in Sea Level Inundation Areas



Map C68: State Owned or Operated Facilities and Critical Emergency Facilities in Sea Level Inundation Areas

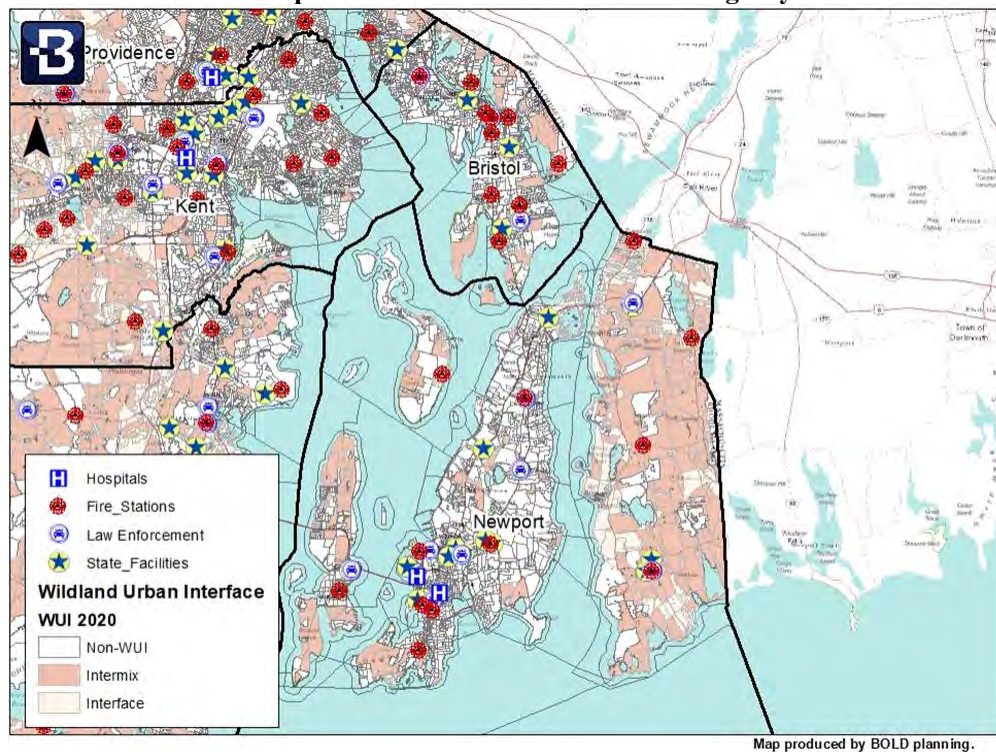


Map C69: State Owned or Operated Facilities and Critical Emergency Facilities in Sea Level Inundation Areas

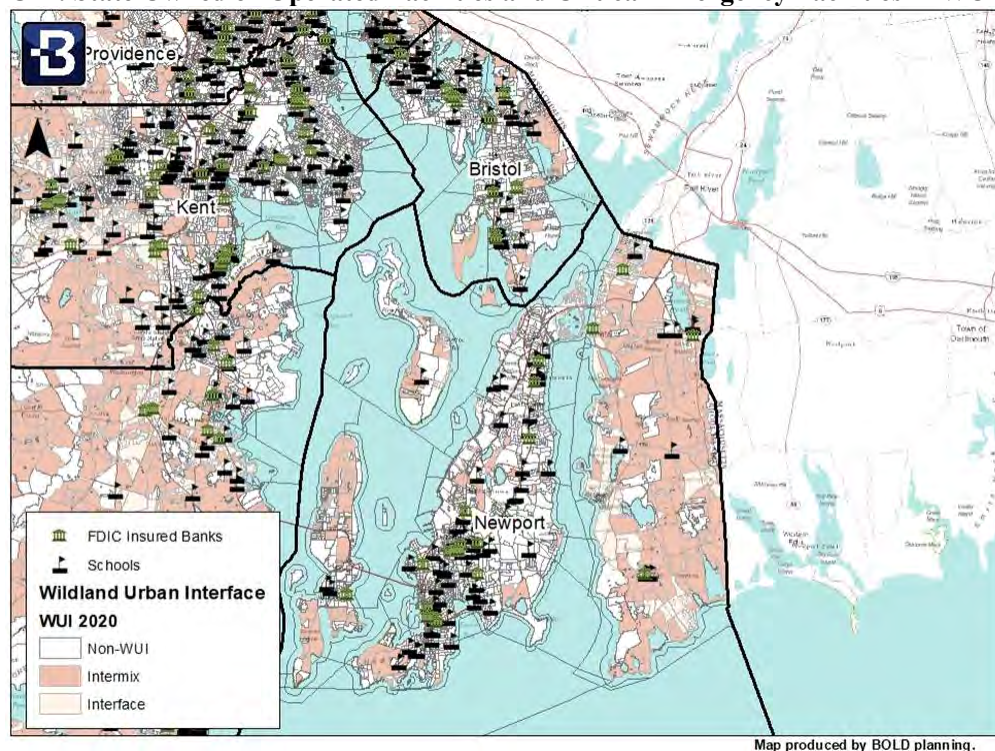


Appendix C5 State Owned or Operated Facilities and Critical Facilities in WUI Areas

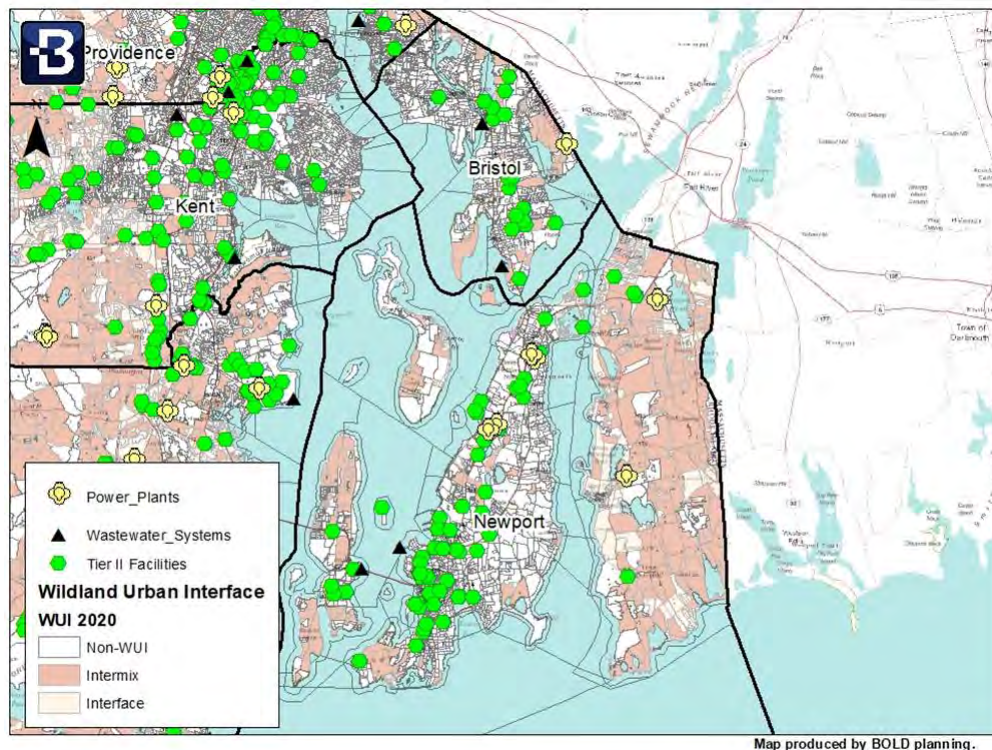
Map C70: State Owned or Operated Facilities and Critical Emergency Facilities in WUI Areas



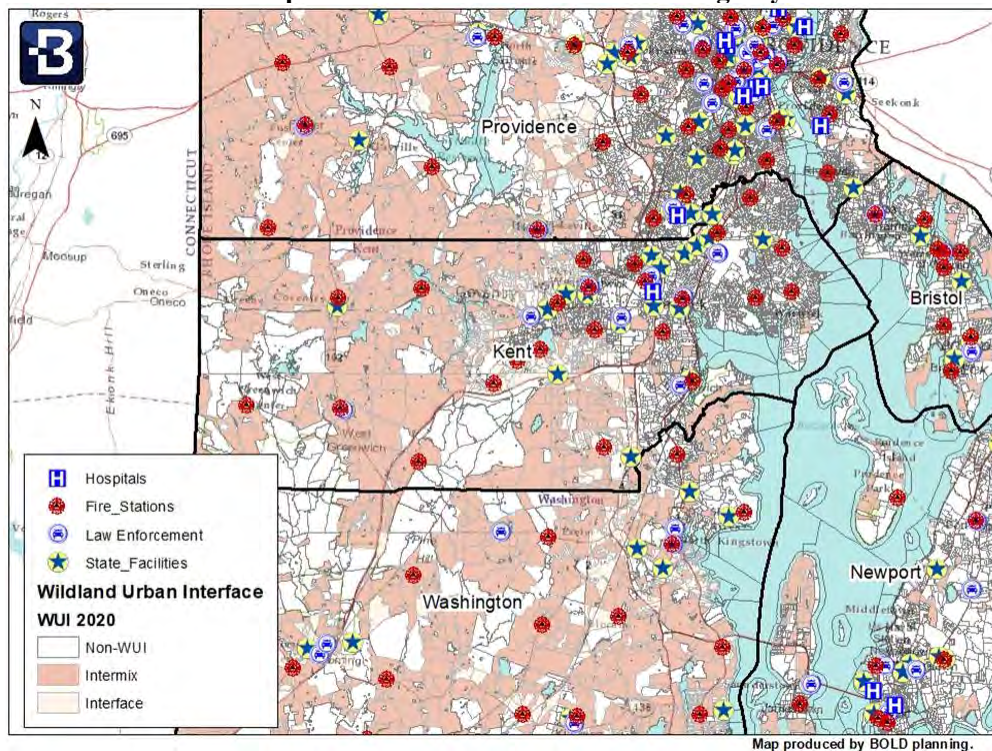
Map C71: State Owned or Operated Facilities and Critical Emergency Facilities in WUI Areas



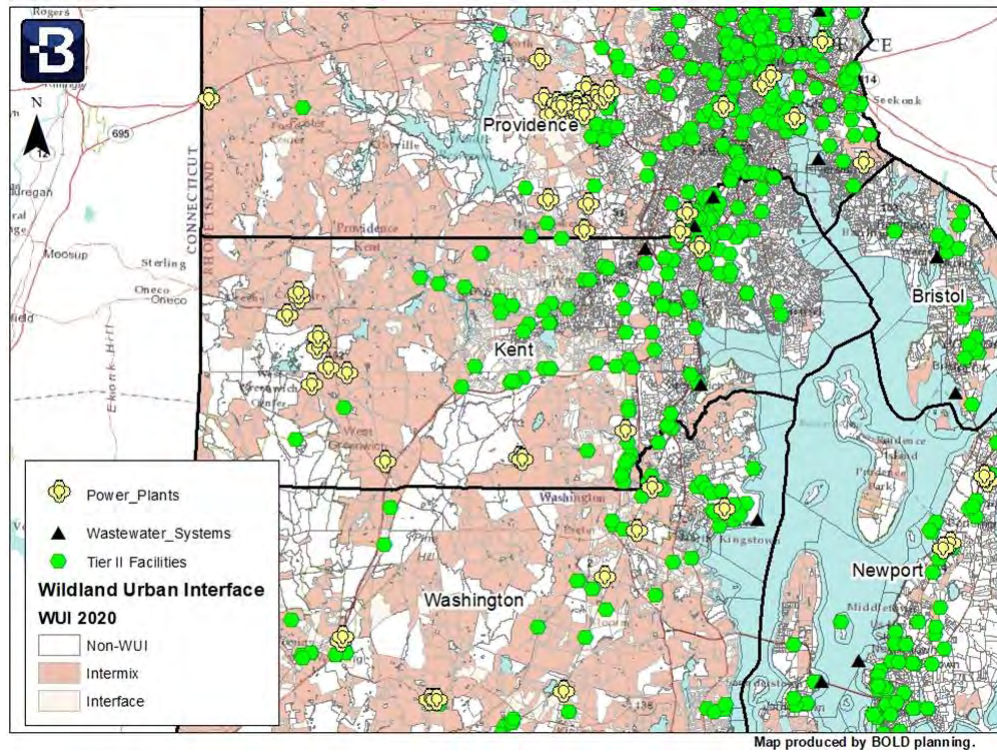
Map C72: State Owned or Operated Facilities and Critical Emergency Facilities in WUI Areas



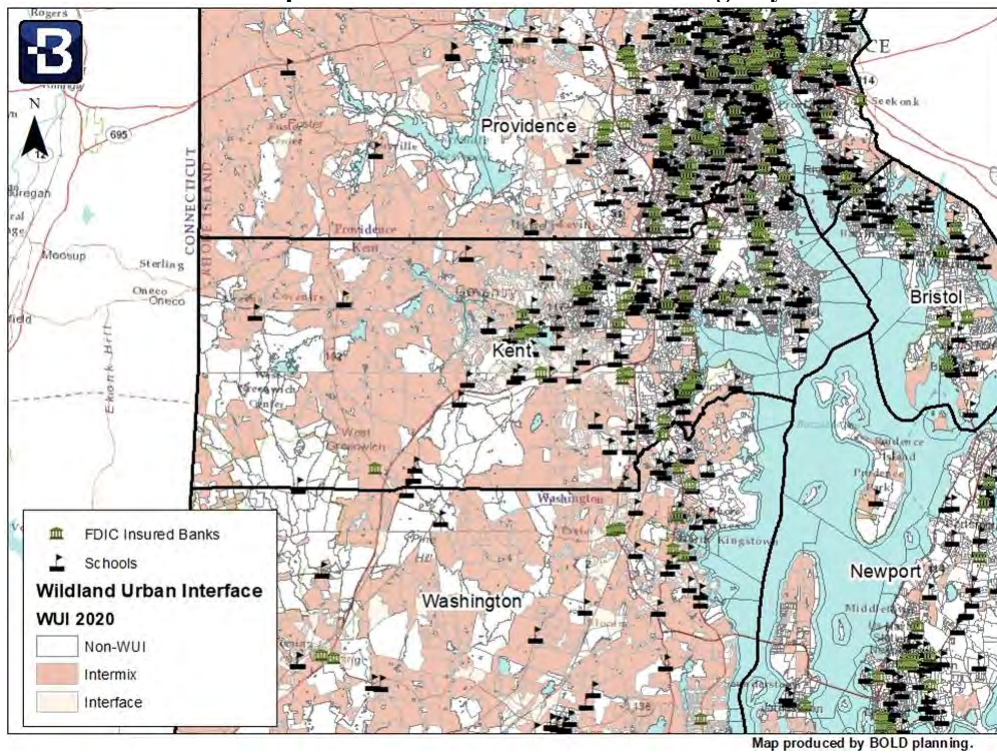
Map C73: State Owned or Operated Facilities and Critical Emergency Facilities in WUI Areas



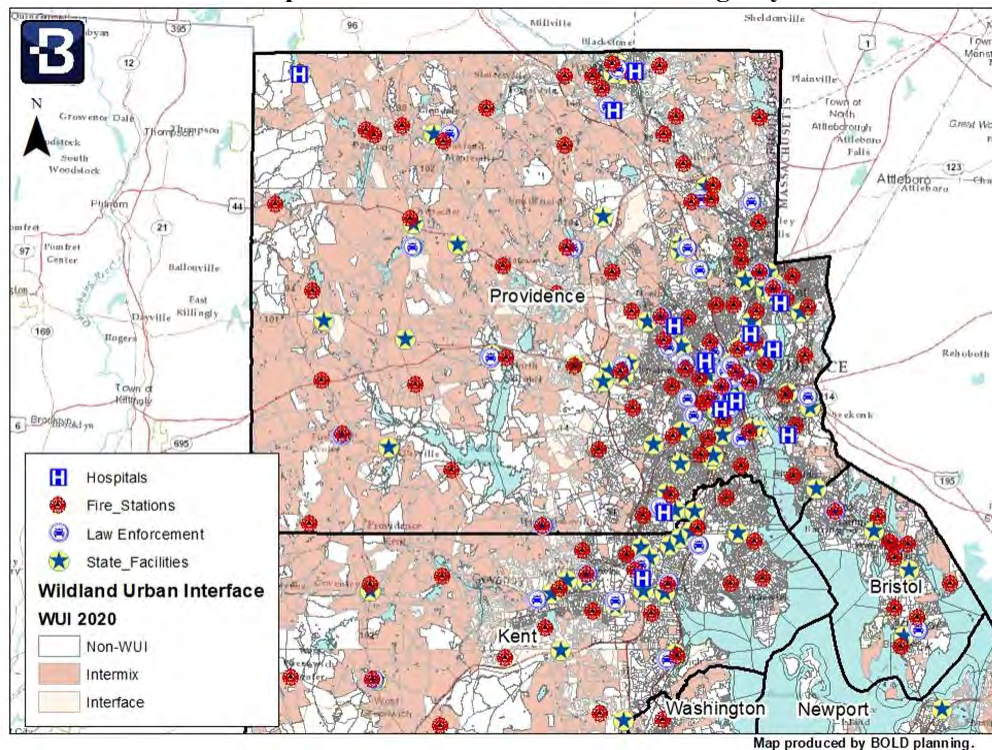
Map C74: State Owned or Operated Facilities and Critical Emergency Facilities in WUI Areas



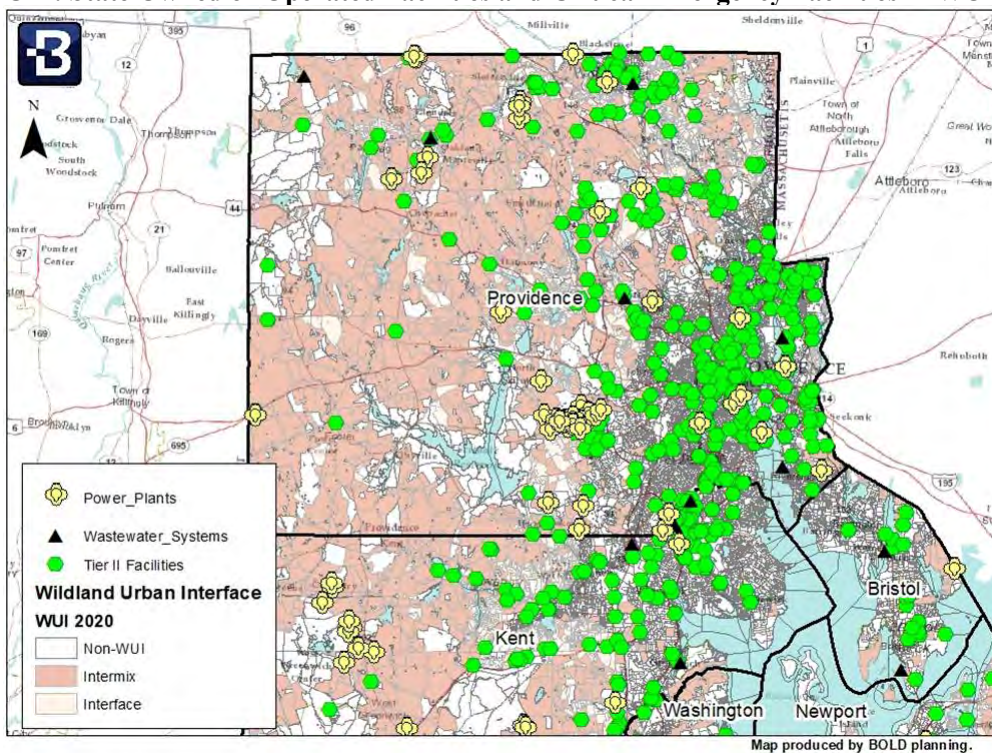
Map C75: State Owned or Operated Facilities and Critical Emergency Facilities in WUI Areas



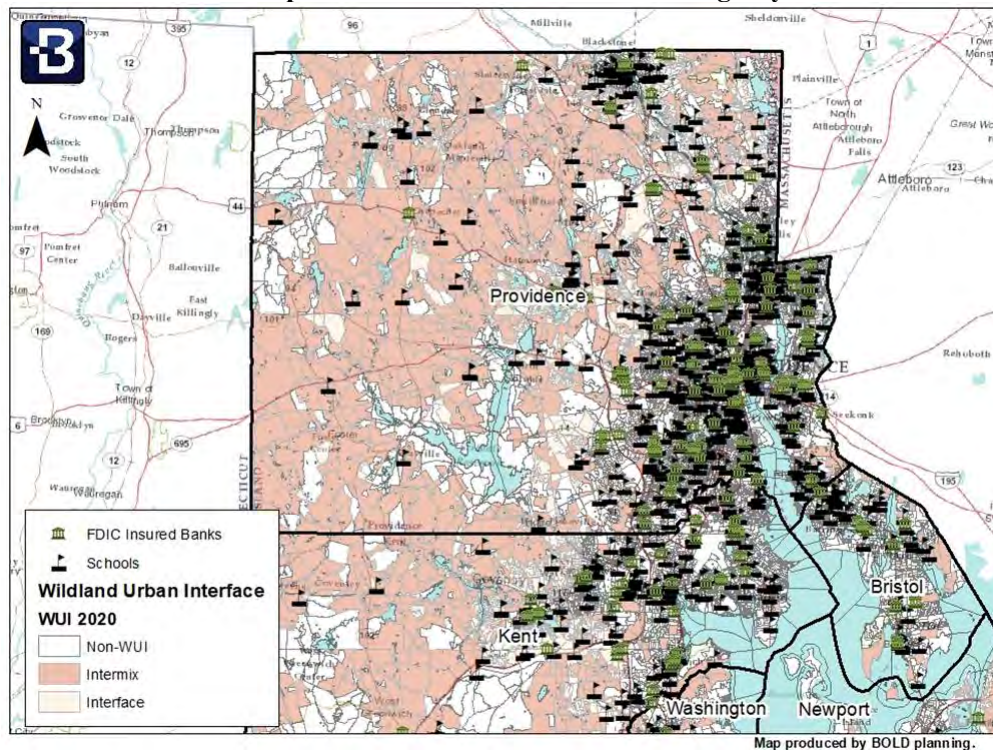
Map C76: State Owned or Operated Facilities and Critical Emergency Facilities in WUI Areas



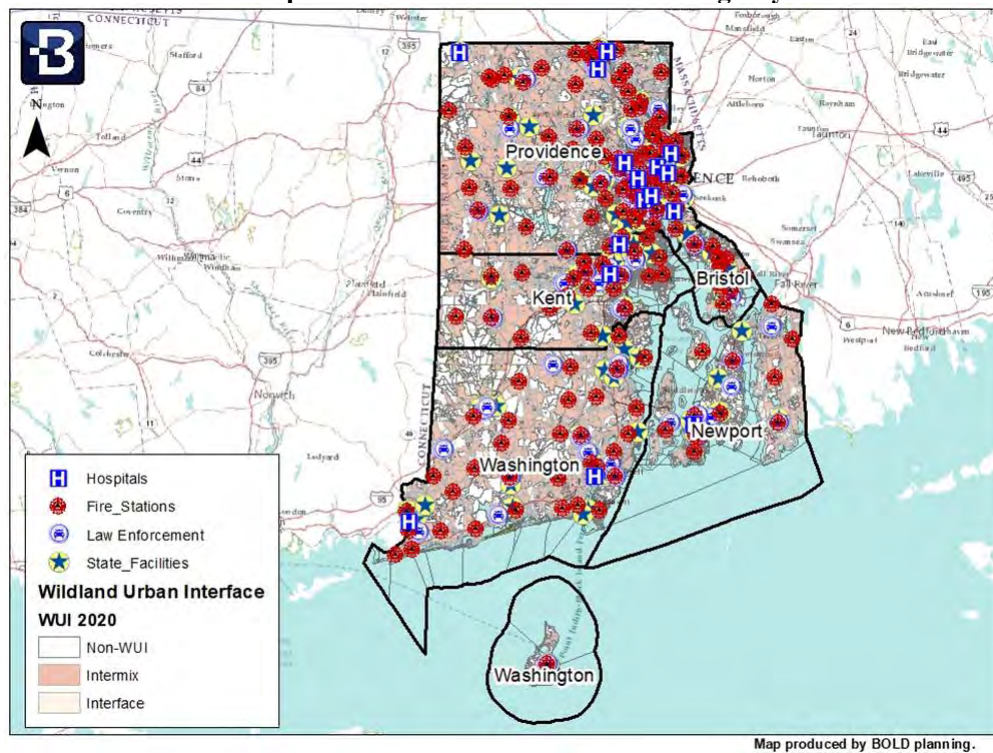
Map C77: State Owned or Operated Facilities and Critical Emergency Facilities in WUI Areas



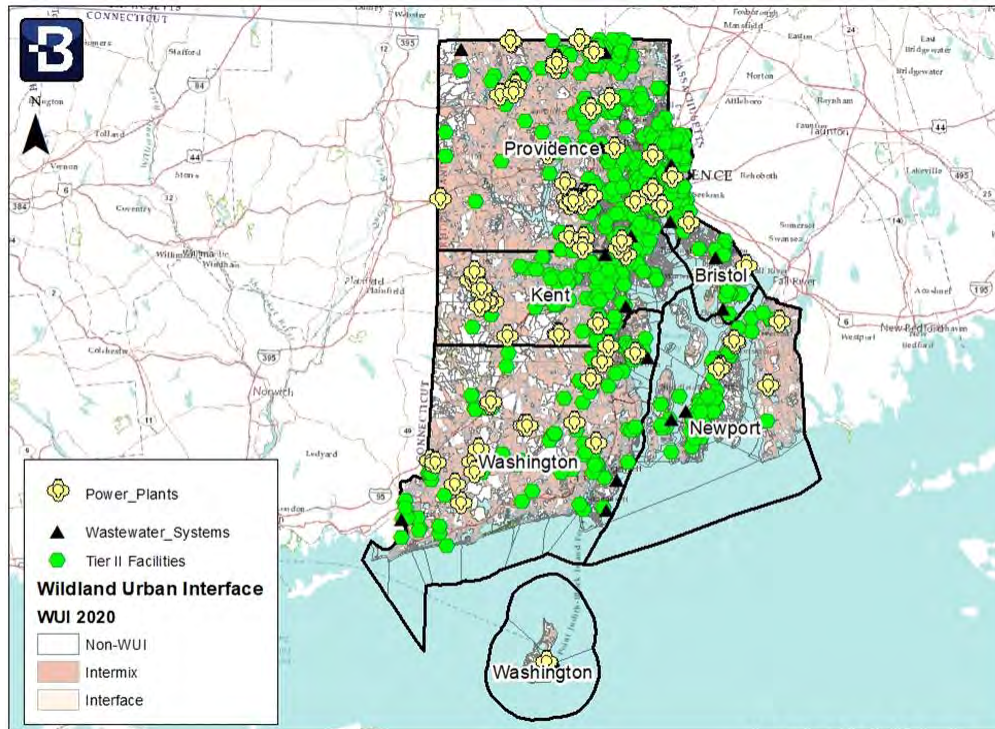
Map C78: State Owned or Operated Facilities and Critical Emergency Facilities in WUI Areas



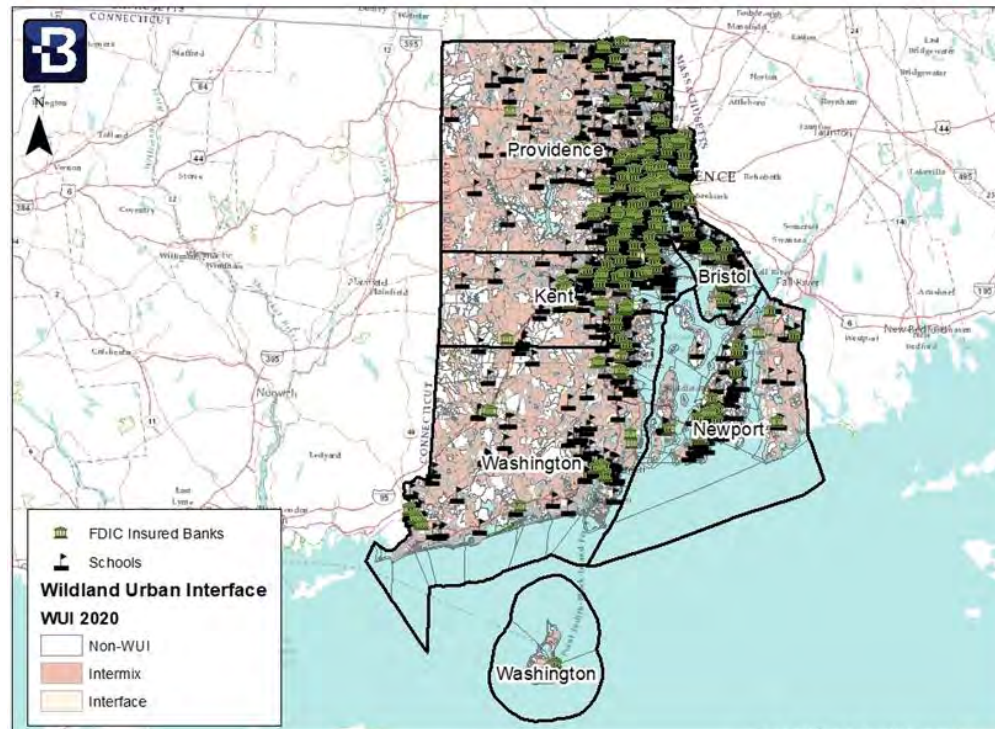
Map C79: State Owned or Operated Facilities and Critical Emergency Facilities in WUI Areas



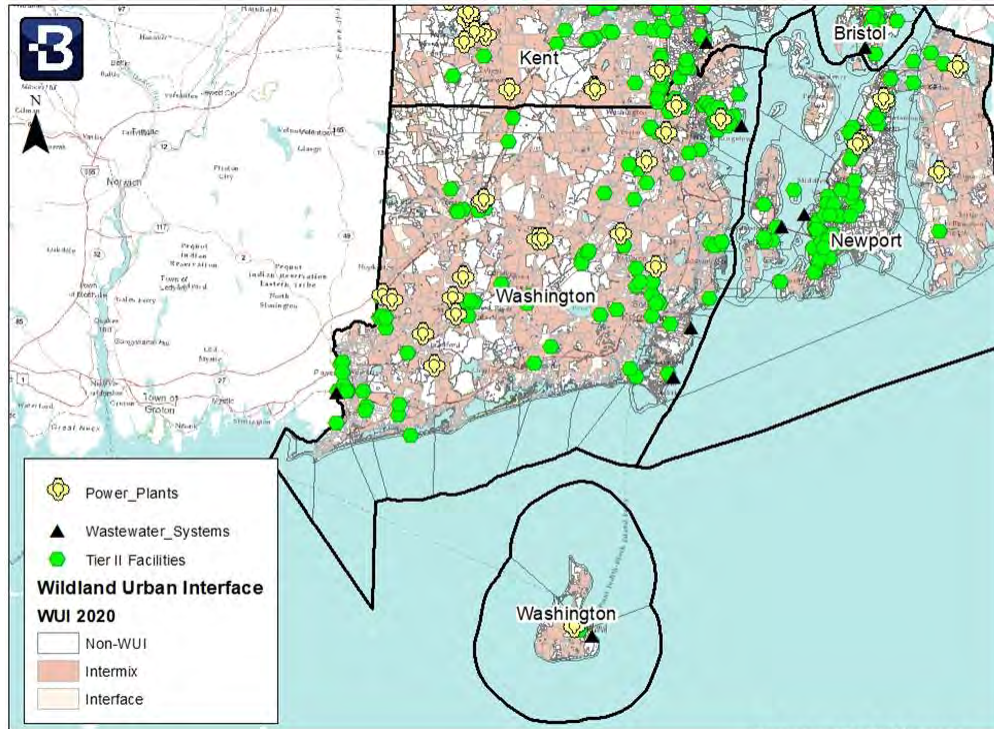
Map C80: State Owned or Operated Facilities and Critical Emergency Facilities in WUI Areas



Map C81: State Owned or Operated Facilities and Critical Emergency Facilities in WUI Areas



Map C82: State Owned or Operated Facilities and Critical Emergency Facilities in WUI Areas



Map C69: State Owned or Operated Facilities and Critical Emergency Facilities in WUI Areas

